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Dr. Martin Sternstein, Professor Emeritus at Ithaca College, was honored by Princeton Review as one of the nation's "300 Best College Professors." He is a long-time College Board consultant and has been a Reader and Table Leader for the AP Statistics exam for many years. He has strong interests in national educational and social issues concerning equal access to math education for all. For two years, he was a Fulbright Professor in Liberia, West Africa, after which he developed a popular "Math in Africa" course, and he is the only mathematician to have given a presentation at the annual Conference on African Linguistics. He also taught the first U.S. course for college credit in chess theory.

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FULL-LENGTH PRACTICE TESTS.**

How to Use This Book

This book provides comprehensive review and extensive practice for the latest AP Statistics course and exam.

About the Exam

Start with the Exam Overview, which outlines the exam format. Familiarize yourself with all of the units covered on this test, review the different question types, and learn how the exam will be scored.

Review and Practice

Study all five units in [Part 2](#), which are organized according to the five units of AP Statistics, and cover the topics recommended by the AP Statistics Development Committee. Every unit includes Learning Objectives that will be covered, a review of each topic, dozens of examples that illustrate key concepts, and end-of-unit summaries. Interspersed among these units are quizzes (mini-AP exams with both multiple-choice and free-response questions), which should be used as progress checks.

Then, consult [Part 3](#), the Final Review, which has several invaluable sections.

- “Selecting an Appropriate Inference Procedure” offers hints on inference recognition followed by two quizzes on naming the procedure to use, defining parameters, listing conditions to be checked, and stating hypotheses, if appropriate.

- “Statistical Insights into Social Issues” includes two quizzes of comprehensive review questions that cover the whole AP Statistics curriculum. These quizzes aim to give an appreciation of the power of statistics and show how this subject gives insights into some of society’s most pressing issues.
- “50 Misconceptions,” “50 Common Errors on the AP Exam,” and “50 AP Exam Hints, Advice, and Reminders” provide tips to remember and pitfalls to avoid on test day.

Diagnostic Test

When you are ready for final review, take the full-length diagnostic test in [Part 1](#) to determine which topics you know well and which ones you may want to brush up on. Complete the entire test, and then check all of the answer explanations, especially for any questions you may have missed. Then, consult the Study Guide for units in the book that you should focus on in your review.

Practice Tests

There are four full-length practice tests in the book that mirror the actual exam in format, content, and level of difficulty. Each test is followed by detailed answers and explanations for all questions.

Appendices

The end of this book consists of a series of helpful appendices, including the answers and explanations for all quizzes, important formulas to know, templates, a guide to inference, and much more. Be sure to go over these sections before completing your final review.

Online Practice

There are also four additional full-length practice tests online where all questions are answered and explained. You may take these tests in practice (untimed) mode or in timed mode.

For Students

This book is intended both as a topical review during the year and as a final review in the weeks before the AP exam. Study the text and illustrative examples carefully, and try to complete the practice quiz problems before referring to the solutions. Simply reading the detailed explanations without first striving to work through the questions on your own is not the best approach. Remember, *mathematics is not a spectator sport!* Use the practice quizzes at appropriate times throughout the school year, and you will develop confidence and a deeper understanding of the material.

A good piece of advice is to develop critical practices (like checking assumptions and conditions), acquire strong technical skills, and always write clear and thorough, yet to the point, interpretations and conclusions in context. Final answers to most problems should not be numbers but, rather, sentences explaining and analyzing numerical results. To help develop skills and insights to tackle AP free-response questions (which often choose contexts students haven't seen before), read newspapers and magazines, and figure out how to apply what you are learning to better understand articles that reference numbers, graphs, and studies.

On the day of the exam, eat a healthy meal before the test. Bring a watch (not a cell phone or smartwatch) to help pace yourself. Do not bring a ruler, white out, or highlighters. Bring extra batteries for your calculator, pens, and more than one sharp pencil with good erasers. Know that scoring at least 40% correct on the exam should be enough for at least a 3 or higher, so don't panic if you can't answer a question. And scoring 70% should earn you a 5! Furthermore, no matter what your score, plan to take more statistics classes in college.

The AP Statistics course is one of the fastest growing and most important courses offered in the high school curriculum. If you work with your teacher and study hard, you will find this to be an enjoyable course, you will do well on the AP exam, and you will develop into a more thoughtful citizen of this world! One day, when you're a data scientist, an

engineer, a doctor or nurse, a manager, or whatever it is you do, you may look back on the AP exam as only a distant memory, but you'll remember the truly important things you learned in your AP Statistics class. You'll remember how to think critically about the world around you and how to communicate your knowledge, insights, and discoveries; these are skills you'll carry with you for the rest of your life.

For Teachers

This book is fully aligned with the five units and exam format outlined in the AP Statistics Course and Exam Description (the CED) and elaborated upon in AP Classroom. Ideally, each individual unit review, paired with practice quiz problems, should be assigned after the unit has been covered in class. The full-length diagnostic test and practice tests should be reserved for final review shortly before the AP exam. These tests are at the same level of difficulty as the actual exam, have the number of multiple-choice questions for each topic as specified by the College Board, and have the topic specific free-response questions as itemized by the College Board. While this review book is not designed to substitute for an in-class experience, students have found it to be a valuable resource both for learning the essential concepts and for preparing for the AP exam.

Barron's Essential 5

As you review the content in this book and work toward earning that **5** on your AP STATISTICS exam, here are five things that you **MUST** know:

- 1** **Graders want to give you credit—help them!** Make them understand *what* you are doing, *why* you are doing it, and *how* you are doing it. Don't make the reader guess at what you are doing.
 - **Communication** is just as important as statistical knowledge!
 - Be sure you understand **exactly what you are being asked to do or find or explain** and **underline key points**.
 - Approach each problem systematically. Some problems look scary on first reading but are not overly difficult and are surprisingly straightforward if you work step-by-step through each part.
 - **Use proper terminology!** Words like *bias*, *correlation*, *normal*, *range*, *skew*, and even *statistic* have specific statistical meanings and should not be used colloquially.
 - *Naked* or *bald* answers will receive little or **no** credit. You must show where answers come from.
 - On the other hand, don't give more than one solution to the same problem—you will receive credit only for the weaker one.

-
- 2** **Describing and comparing distributions is fundamental in descriptive statistics.**
 - Reference shape (such as bell-shaped, skewed, uniform, unimodal, or bimodal), center (mean or median), variability (range, IQR, or standard deviation), and unusual features (such as outliers, gaps, and clusters) when describing one-variable quantitative data. Don't forget context!
 - **Use nondeterministic language** like *roughly* symmetric and *approximately* normal.

- Use **comparative language**, rather than simply making separate lists, when comparing two distributions. Don't forget context!
 - Be able to analyze displays such as dotplots, histograms, boxplots, and stemplots.
 - Be able to analyze parallel boxplots and back-to-back stemplots.
 - Reference form (linear or nonlinear), direction (positive or negative), strength (weak, moderate, or strong), and unusual features (such as outliers, influential points, and clusters) when describing bivariate data. Don't forget context!
-

3

Data collection is the first step in the data analysis process.

- Understand the difference between observational studies and experiments, and know the strengths and weaknesses of each.
 - Understand that random sampling, the use of chance in selecting a sample from a population, is critical in being able to generalize from a sample to a population.
 - **Be able to describe how to implement sampling methods**, including simple random sampling, stratified random sampling, cluster random sampling, and systematic random sampling.
 - Understand that random assignment of subjects to treatments in experiments is critical in minimizing the effect of possible confounding variables.
 - **Be able to describe how to set up an experiment** using random assignment and possibly blinding or blocking.
-

4

Distributions describe variability, and variability is the single most fundamental concept in statistics. Understand the terminology:

- *population distribution* (variability in an entire population),
- *sample distribution* (variability within a particular sample), and
- *sampling distribution* (variability between samples).

- The larger the sample size, the more the **sample distribution** looks like the population distribution.
 - Central limit theorem: the larger the sample size, the more the **sampling distribution** (probability distribution of the sample means) looks like a normal distribution.
-

5

Choosing the correct procedure and performing proper checks are critical.

- Categorical variables lead to proportions or chi-square procedures, while quantitative variables lead to means or linear regression.
 - Estimating a quantity indicates a confidence interval, while looking for evidence to test a claim indicates a hypothesis test.
 - **Know the proper checks for each procedure** and state them correctly. (Listing wrong conditions will lose points.)
 - **Understand the three fundamental conditions:** the randomization condition, the 10% condition, and the normality condition.
 - Verifying assumptions and conditions means more than simply listing them with little check marks—you must show work or give some reason to confirm verification.
-

Exam Overview

Exam Format

The exam consists of two parts: a 90-minute section with 42 multiple-choice questions and a 90-minute free-response section with four open-ended questions to complete. During grading, the two sections of the exam are given equal weight. Students have remarked that the first section involves “lots of reading” while the second section involves “lots of writing.” The percentage of questions from each content area is approximately 25% data analysis, 15% experimental design, 25% probability, and 35% inference.

In 2025 the AP Statistics exam was given as a hybrid digital exam. The multiple-choice questions were viewed and answered in the Bluebook testing app. The free-response questions were viewed in the app; however, the free-response answers were written in paper exam booklets that were returned for scoring. Beginning in 2026, the Desmos graphing calculator will be embedded in the Bluebook testing platform for AP Statistics. Of course, you will still be allowed to use a handheld graphing calculator on the exam.

Multiple-Choice Section

In the multiple-choice section, the questions are much more conceptual than computational, and thus use of the calculator is minimal. The multiple-choice section can be broken down by units of instruction as follows:

- Unit 1: Exploring One-Variable Data and Collecting Data (8–13 questions)
- Unit 2: Probability, Random Variables, and Probability Distributions (6–10 questions)
- Unit 3: Inference for Categorical Data: Proportions (6–10 questions)
- Unit 4: Inference for Quantitative Data: Means (4–8 questions)
- Unit 5: Regression Analysis (4–8 questions)

The multiple-choice section can be broken down by statistical practices as follows:

- Practice 1: Formulate Questions (2–4 questions)
- Practice 2: Collect Data (8–13 questions)
- Practice 3: Analyze Data (10–15 questions)
- Practice 4: Interpret Results (10–15 questions)

Good strategies for working multiple-choice questions include the following:

- Read carefully so you are clear what is being asked for.
- Within the Bluebook app:
 - Underline or circle key words or numbers in the question.
 - Make notes in the margin as to what numbers represent.
 - Cross out answer choices you know are incorrect.
 - Cross out answer choices that may be true but don't relate to the specific question asked.
- Use this process of elimination and then narrow down to the BEST answer.
- Answer every question (you will not be penalized for wrong answers).

Free-Response Section

In the free-response section, the four open-ended questions are weighted equally and scored on 10-point scales. The free-response section can be

broken down as follows, including the exact order the four questions will appear on the exam:

- 1 multipart question with a primary focus on formulating questions and collecting data
- 1 multipart question with a primary focus on analyzing data and interpreting results
- 1 multipart question with a primary focus on inference
- 1 multipart question focusing on multiple content areas

Good strategies and guidance for working free-response questions include the following:

- Be sure you understand exactly what you are being asked to do, find, or explain.
- Within Bluebook, underline or highlight key words, phrases, and numbers, and make notes in the margin as to what numbers represent.
- Use proper terminology: words like *bias*, *correlation*, *normal*, *power*, *range*, *skew*, and even *statistic* have specific statistical meanings.
- Indicate your methods clearly, as problems will be graded on the correctness of the methods as well as the accuracy of the results and explanations.
- One-variable distributions: don't forget context as well as shape, center, spread, and unusual features.
- Linear regression: be able to interpret the slope, the y-intercept, and the coefficient of determination in context.
- Sampling methods or experimental design: be able to explain how you will pick a random sample or randomly assign subjects to treatment groups.
- Probability: name the distribution, give the parameters, note boundary values, and give the calculated answer.
- Inference: name procedures, define parameters, include confirmation of underlying assumptions, perform calculations (use software if quicker), and state conclusions in context, not just as numbers.
- Describe both an advantage of one thing and a disadvantage of the other if asked why one thing is better than another.

Calculator Usage

On the AP Statistics exam, you will be furnished with a list of formulas (from (I) Descriptive Statistics, (II) Probability and Distributions, and (III) Inferential Statistics) and tables (including standard normal probabilities, t -distribution critical values, and χ^2 critical values). While you can use the embedded Desmos graphing calculator or bring a graphing calculator with statistics capabilities to the exam, you should not leave answers in terms of calculator syntax. Furthermore, many students have commented that calculator usage was less than they had anticipated. However, even though the calculator is a tool, to be used sparingly and as needed, you need to be proficient with this technology. You also must be comfortable with reading generic computer output.

Scoring

The score on the multiple-choice section is based on the number of correct answers, with no points deducted for incorrect answers. So don't leave any blank answers! Starting in 2027, free-response questions will be scored on a 0 to 10 scale.

Each completed AP exam will receive a grade based on a 5-point scale, with 5 being the highest score and 1 being the lowest score. Most colleges and universities accept a grade of 3 or better for credit, advanced placement, or both. Over the years, average cut scores, together with the approximate percent of students receiving each score, are as in the following table.

AP Score	Total Points	Students (%)
5	70–100	15%
4	55–69	20%
3	40–54	25%
2	30–39	20%
1	0–29	20%

Part 1

Diagnostic Test

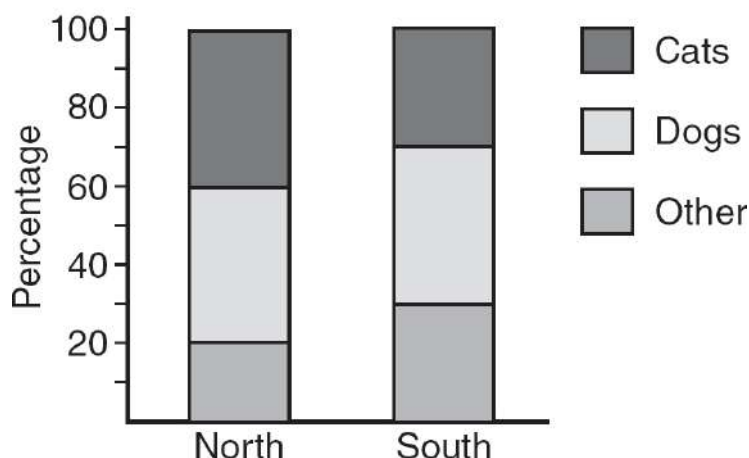
Diagnostic Test

Section I: Questions 1–42

SPEND 90 MINUTES ON THIS PART OF THE EXAM.

DIRECTIONS: The questions or incomplete statements that follow are each followed by four suggested answers or completions. Choose the response that best answers the question or completes the statement.

1. It is estimated that 70 percent of Americans have pets. However, favorite pet of choice differs by geographic location. A random sample of pet owners is cross-classified by geographic location and pet of choice. The results are summarized in the following segmented bar chart.



Which of the following is an *incorrect* conclusion?

- (A) More pet owners in the South than in the North answered “Other.”
- (B) Twice as many pet owners in the North answered “Dogs” than answered “Other.”
- (C) The same number of pet owners in the South answered “Cats” as answered “Other.”
- (D) A greater proportion of pet owners in the North than in the South answered “Cats.”

Questions 2-4 refer to the following:

A nutritionist is researching a possible linear relationship between calories and sodium content (mg) in beef hot dogs. A random sample of 20 beef hot dogs gives the following regression output:

Dependent variable is: Sodium				
Predictor	Coef	SE Coef	T	P
Constant	-228.33	77.97	-2.93	0.009
Calories	4.0133	0.4922	8.15	0.000
S = 48.5799 R-Sq = 78.7% R-Sq(adj) = 77.5%				

[Follow for extended description](#)

2. Which of the following statements is a correct interpretation of the value 4.0133 in the output?
- (A) There is a predicted increase of 4.0133 calories for every 1 mg increase in sodium.
 - (B) There is a predicted increase of 4.0133 mg in sodium for every 1 unit increase in calories.
 - (C) Approximately 4.0133% of the variability in sodium is explained by the linear model with calories.
 - (D) The predicted sodium is 4.0133 mg if a hot dog has 0 calories.
3. Which of the following has no logical interpretation in context?
- (A) The predicted sodium content for a beef hot dog with over 228.33 calories
 - (B) The true sodium content for a beef hot dog with over 228.33 calories
 - (C) A beef hot dog with a predicted sodium content of 228.33 mg
 - (D) The y-intercept of the regression line
4. Which of the following statements is a correct interpretation of the value 78.7% in the output?

- (A) Approximately 78.7% of the variability in predicted sodium is explained by the linear model with calories.
 - (B) Approximately 78.7% of the variability in sodium is explained by the linear model with calories.
 - (C) Approximately 78.7% of the sodium is explained by the calories.
 - (D) Approximately 78.7% of the sodium is explained by the variability in calories.
-

5. In tossing a fair coin, which of the following sequences is most likely to appear?
- (A) HHHHHH
 - (B) HTHTHT
 - (C) HTHHTTH
 - (D) All are equally likely.
6. There have been growing numbers of news stories about White Americans calling the police on people of color whose behavior is completely normal. A criminologist hypothesizes that the mean number of such incidents across the country is 2 per day. A sociologist believes the true mean is greater than 2 per day and plans a hypothesis test at the 5% significance level on a random sample of 50 days. For which of the following possible true values of μ will the power of the test be greatest?
- (A) 1.5
 - (B) 1.85
 - (C) 2.0
 - (D) 2.4
7. A simple random sample is defined by
- (A) the method of selection.
 - (B) how representative the sample is of the population.
 - (C) whether or not a random number generator is used.
 - (D) examination of the outcome.

8. A delivery company keeps track of truckloads with two variables: number of parcels and total weight. Are these variables discrete or continuous?

- (A) Both are discrete.
- (B) Both are continuous.
- (C) Number of parcels is discrete, and weight is continuous.
- (D) Number of parcels is continuous, and weight is discrete.

Questions 9–10 refer to the following:

A researcher would like to show that a new oral diabetes medication she developed helps control blood sugar level better than insulin injection. She plans to run a hypothesis test at the 5% significance level.

9. What would be a Type I error?

- (A) The researcher concludes she has sufficient evidence that her new medication helps more than insulin injection, and her medication really is better than insulin injection.
- (B) The researcher concludes she has sufficient evidence that her new medication helps more than insulin injection when in reality her medication is not better than insulin injection.
- (C) The researcher concludes she does not have sufficient evidence that her new medication helps more than insulin injection, and her medication really is not better than insulin injection.
- (D) The researcher concludes she does not have sufficient evidence that her new medication helps more than insulin injection when in reality her medication is better than insulin injection.

10. What would be a Type II error?

- (A) The researcher concludes she has sufficient evidence that her new medication helps more than insulin injection, and her medication really is better than insulin injection.
- (B) The researcher concludes she has sufficient evidence that her new medication helps more than insulin injection when in reality her medication is not better than insulin injection.
- (C) The researcher concludes she does not have sufficient evidence that her new medication helps more than insulin injection, and her medication really is not better than insulin injection.

- (D) The researcher concludes she does not have sufficient evidence that her new medication helps more than insulin injection when in reality her medication is better than insulin injection.
-

11. A financial analyst determines the yearly research and development investments for 50 blue chip companies. She notes that the distribution is distinctly not bell-shaped. If the 50 dollar amounts are converted to z -scores, what can be said about the standard deviation of the 50 z -scores?
- (A) It is less than the standard deviation of the raw scores.
(B) It is greater than the standard deviation of the raw scores.
(C) It equals $\frac{\sigma}{\sqrt{50}}$ where s is the population standard deviation of the raw scores.
(D) It equals 1.
12. A coin is weighted so that the probability of heads is 0.6. The coin is tossed 20 times, and the number of heads is noted. This procedure is repeated a total of 200 times, and the number of heads is recorded each time. What kind of distribution has been simulated?
- (A) The sampling distribution of the sample proportion with $n = 20$ and $p = 0.6$
(B) The sampling distribution of the sample proportion with $n = 200$ and $p = 0.6$
(C) The binomial distribution with $n = 20$ and $p = 0.6$
(D) The binomial distribution with $n = 200$ and $p = 0.6$
13. A random sample of 120 students at a high school are to be selected, and each will be asked if their parents have ever given them an alcoholic drink. Which of the following is a reason for increasing the size of this study from 120 to 220 students?
- (A) To reduce response bias
(B) To reduce nonresponse bias
(C) To more likely obtain an estimate that is closer to the population proportion
(D) All of the above are valid reasons to increase the sample size.

14. Of the 423 seniors graduating this year from a city high school, 322 plan to go on to college. When the principal asks an AP student to calculate a 95% confidence interval for the proportion of this year's graduates who plan to go to college, the student says that this would be inappropriate. Why?
- (A) The independence assumption may have been violated (students tend to do what their friends do).
 - (B) There is no evidence that the data come from a normal or nearly normal population (GPAs help determine college admission and may be skewed).
 - (C) Randomization was not used.
 - (D) The population proportion is known, so a confidence interval has no meaning.
15. An AP Statistics student in a large high school plans to survey his fellow students with regard to their preferences between using a laptop or using a tablet. Which of the following survey methods is unbiased?
- (A) None of the below sampling methods are unbiased.
 - (B) The student passes a survey card to every student with instructions to fill it out at home and drop the filled-out card in a box by the school entrance the next day.
 - (C) The student creates an online survey and asks everyone to respond.
 - (D) The student goes to all of the high school sports events for a week, hands out the survey, and waits for each student to fill it out and hand it back.
16. In a random sample of 500 students, it was reported that test grades went up an average of at least 10 points for 70 percent of the students when usage of cell phones was banned during the school day. What was the degree of confidence if the margin of error was ± 2.5 percent?
- (A) $P(-0.025 < z < 0.025)$
 - (B) $P(0.675 < z < 0.725)$
 - (C) $P\left(-\frac{0.025}{\sqrt{(0.5)(0.5)/500}} < z < \frac{0.025}{\sqrt{(0.5)(0.5)/500}}\right)$
 - (D) $P\left(-\frac{0.025}{\sqrt{(0.7)(0.3)/500}} < z < \frac{0.025}{\sqrt{(0.7)(0.3)/500}}\right)$
17. In a random sample of 10 insects of a newly discovered species, an entomologist measures an average life expectancy of 17.3 days with a standard deviation of 2.3 days. Assuming all conditions for inference are met, what is a

95% confidence interval for the mean life expectancy for insects of this species?

- (A) $17.3 \pm 1.96 \left(\frac{2.3}{\sqrt{9}} \right)$
- (B) $17.3 \pm 1.96 \left(\frac{2.3}{\sqrt{10}} \right)$
- (C) $17.3 \pm 2.228 \left(\frac{2.3}{\sqrt{9}} \right)$
- (D) $17.3 \pm 2.262 \left(\frac{2.3}{\sqrt{10}} \right)$

18. Time management and procrastination are difficult problems for college students. The distribution for the percentage of study time that occurs in the 24 hours prior to a final exam is approximately normal with mean 44 and standard deviation 21. Consider two different random samples taken from the population of college students, one of size 10 and one of size 100. Which of the following is true about the sampling distributions of the sample mean for the two sample sizes?

- (A) Both distributions are approximately normal. Both the mean and the standard deviation for the $n = 10$ sampling distribution are greater than for the $n = 100$ distribution.
- (B) Both distributions are approximately normal with the same mean. The standard deviation for the $n = 10$ sampling distribution is greater than for the $n = 100$ distribution.
- (C) Only the $n = 100$ sampling distribution is approximately normal. Both distributions have mean 44 and standard deviation 10.
- (D) Only the $n = 100$ sampling distribution is approximately normal. Both distributions have the same mean. The standard deviation for the $n = 10$ sampling distribution is greater than for the $n = 100$ distribution.

19. Do male and female high school seniors choose colleges for the same reasons? A random sample of male high school seniors and a random sample of female high school seniors are given questionnaires asking which of the following is their primary reason for choosing a particular college: academics, sports teams, location, or where friends are going. Among the following, which is the best statistical test to use in analyzing these data?

- (A) One-sample z-test for a population proportion
- (B) Two-sample z-test for a difference of two population proportions
- (C) Chi-square test of independence

(D) Chi-square test of homogeneity

20. Suppose that a study was carried out in which each student in a random sample of students at a particular high school was asked whether he or she thought that male and female sports teams should be equally funded. Which of the following is a correct understanding of the type of investigative question this study might address?

- (A) The data could be used to estimate a population mean since the data collected are numerical.
- (B) The data could be used to estimate a population proportion since the data collected are numerical.
- (C) The data could be used to estimate a population mean since the data collected are categorical.
- (D) The data could be used to estimate a population proportion since the data collected are categorical.

Questions 21–23 refer to the following:

A campus has 55% male and 45% female students. Suppose 30% of the male students pick basketball as their favorite sport compared with 20% of the females.

21. What is the probability that a randomly chosen student is male and picks basketball as his favorite sport?

- (A) 0.30
- (B) 0.55
- (C) $(0.55)(0.30)$
- (D) $(0.55)(0.30) - (0.45)(0.20)$

22. What is the probability that a randomly chosen student picks basketball as the student's favorite sport?

- (A) $0.30 + 0.20$
- (B) $(0.55)(0.30) + (0.45)(0.20)$
- (C) $\frac{0.30 + 0.20}{(0.55)(0.30) + (0.45)(0.20)}$
- (D) $\frac{(0.30)(0.20)}{(0.55)(0.30) + (0.45)(0.20)}$

23. If a randomly chosen student picks basketball as the student's favorite sport, what is the probability the student is male?

- (A) $\frac{0.30}{0.30 + 0.20}$
 - (B) $\frac{(0.55)(0.30)}{(0.55)(0.30) + (0.45)(0.20)}$
 - (C) $\frac{0.30}{(0.55)(0.30) + (0.45)(0.20)}$
 - (D) $\frac{0.55}{(0.55)(0.30) + (0.45)(0.20)}$
-

24. The kelvin is a unit of measurement for temperature; 0 K is absolute zero, the temperature at which all thermal motion ceases. Conversion from Fahrenheit to Kelvin is given by $K = \frac{5}{9} \times (F - 32) + 273$. The average daily temperature in Monrovia, Liberia, is 78.35°F with a standard deviation of 6.3°F. If a scientist converts Monrovia daily temperatures to the Kelvin scale, what will be the new mean and standard deviation?

- (A) Mean, 231.75 K; standard deviation, 3.5 K
- (B) Mean, 298.75 K; standard deviation, 3.5 K
- (C) Mean, 298.75 K; standard deviation, 258.72 K
- (D) Mean, 298.75 K; standard deviation, 276.5 K

25. A cattle veterinarian is considering two experimental designs to compare two sources of bovine growth hormone, or BVH, to spur increased milk production in Guernsey cattle. Design 1 involves flipping a coin as each cow enters the stockade, and if *heads*, giving it BVH from bovine cadavers, and if *tails*, giving it BVH from engineered *E. coli*. Design 2 involves flipping a coin as each cow enters the stockade, and if *heads*, giving it BVH from bovine cadavers for a specified period of time and then switching to BVH from engineered *E. coli* for the same period of time, and if *tails*, the order is reversed. With both designs, daily milk production is noted. Which of the following is accurate?

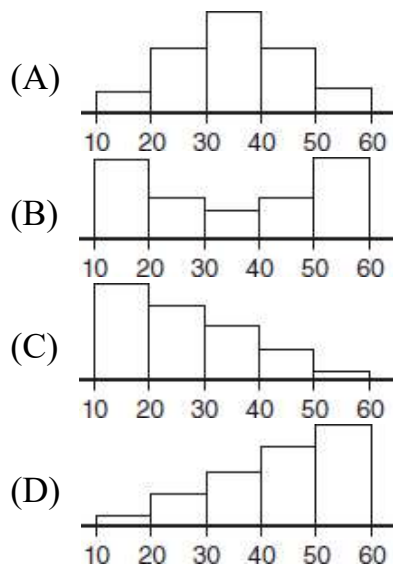
- (A) Neither design uses randomization since there is no indication that cows will be randomly picked from the population of all Guernsey cattle.
- (B) Design 1 is a completely randomized design, while Design 2 is a block design.
- (C) In the second design, BVH from bovine cadavers and BVH from engineered *E. coli* are confounded.

- (D) One of the two designs is actually an observational study, while the other is an experiment.
26. Recent studies have determined with 94% confidence that the mean number of trees cut down every 24 hours to make toilet paper is between 25,000 and 29,000. What is meant by the confidence *level* in this context?
- (A) A confidence interval of the true mean number of trees cut down every 24 hours to make toilet paper was calculated using z -scores of ± 1.881 .
 - (B) A confidence interval of the true mean number of trees cut down every 24 hours to make toilet paper was calculated using t -scores consistent with $df = n - 1$ and tail probabilities of 0.03.
 - (C) We are 94% confident that the true mean number of trees cut down every 24 hours to make toilet paper is between 25,000 and 29,000.
 - (D) If all possible random samples of data are obtained by this method, approximately 94% will yield confidence intervals that capture the true mean number of trees cut down every 24 hours to make toilet paper.
27. Which of the following is a true statement about sampling?
- (A) If the sample is random, the size of the sample usually doesn't matter.
 - (B) If the sample is random, the size of the population usually doesn't matter.
 - (C) A sample of less than 1% of the population is too small for statistical inference.
 - (D) A sample of more than 10% of the population is too large for statistical inference.
28. Suppose, in a study of mated pairs of soldier beetles, it is found that the measure of the elytron (hardened forewing) length is always 0.5 millimeters longer in the female. What is the correlation between elytron lengths of mated females and males?
- (A) -0.5
 - (B) 0
 - (C) 0.5
 - (D) 1
29. A chi-square test of homogeneity was conducted to determine whether different racial groups have different distributions in the size of their social networks. The chi-square test statistic and the p -value of the test were 17.43

and 0.037, respectively. Which of the following is the correct interpretation of the p -value?

- (A) Assuming that each racial group has the same distribution in size of social network, the probability of finding a test statistic of 17.43 or smaller is 0.037.
- (B) Assuming that each racial group has the same distribution in size of social network, the probability of finding a test statistic of 17.43 or larger is 0.037.
- (C) There is a 0.037 probability that the different racial groups have the same distribution of social networks.
- (D) There is a 0.037 probability that the different racial groups have different distributions of social networks.

30. The age distribution for a particular debilitating disease has a mean greater than the median. Which of the following graphs most likely illustrates this distribution?



31. In a random sample of 10 people, the probability that k people show a particular genetic abnormality is given by $\binom{10}{k} (0.38)^k (0.62)^{10-k}$ for $k = 0, \dots, 10$.

What is the mean of the associated random variable?

- (A) 0.38
- (B) 3.8

- (C) 5.0
(D) 6.2

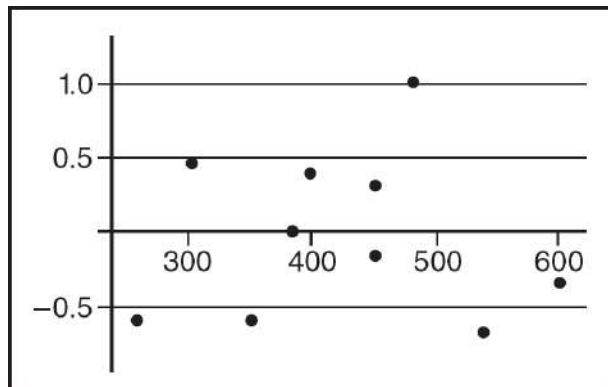
32. A soccer coach believes that soccer players run an average of 7 miles per match and tests this hypothesis on a random sample of 100 players. The sample is found to have a mean of $\bar{x} = 6.8$ with a standard deviation of 1.3. Which of the following is the correct test statistic?

- (A) $t = \frac{6.8 - 7}{1.3/10}$
(B) $t = \frac{6.8 - 7}{1.3/100}$
(C) $t = \frac{7 - 6.8}{1.3/10}$
(D) $t = \frac{7 - 6.8}{1.3/100}$

33. A recent study noted prices and battery lives of 10 top-selling tablet computers. The data follow:

	1	2	3	4	5	6	7	8	9	10
Cost	303	450	260	480	540	390	350	400	600	450
Battery life (hr)	8.5	10	7	11	10	9	8	9.5	11	9.5

The residual plot of the least squares model is



What is the model's predicted battery life for the tablet computer costing \$480?

- (A) 10 hr
(B) 10.5 hr

- (C) 11 hr
- (D) 12 hr

34. Should college athletes be required to give their coaches their social media account user names and passwords? A survey of student-athletes is to be taken. The statistician believes that Division I, II, and III players may differ in their views, so she selects a random sample of athletes from each division to survey. This is a

- (A) simple random sample.
- (B) stratified random sample.
- (C) cluster random sample.
- (D) convenience sample.

35. In a random sample of 1,500 college students, a pollster found 45% prefer a female president and 42% prefer a male president. To calculate a 95 percent confidence interval for the difference in the proportion of college students who prefer a female president over a male president, he uses

$$(0.45 - 0.42) \pm 1.96 \sqrt{\frac{(0.45)(0.55)}{1,500} + \frac{(0.42)(0.58)}{1,500}}. \text{ Were conditions for inference met?}$$

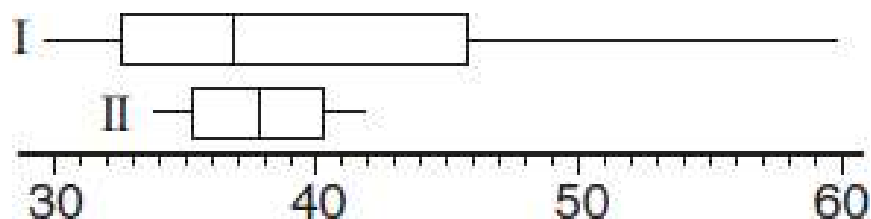
- (A) Yes, because there was a random sample, 1,500 is less than 10% of all college students, and $1,500(0.45)$, $1,500(0.55)$, $1,500(0.42)$, and $1,500(0.58)$ are all ≥ 10 .
- (B) No, because there was no random assignment between female and male college students.
- (C) No, because the independence assumption is violated.
- (D) No, because the random sample may not be truly representative of the population.

36. The population of the Greater Tokyo area is 34,400,000 and of Karachi is 17,200,000. A random sample of citizens is to be taken in each city, and 95% confidence intervals for the mean age in each city will be calculated. Assuming roughly equal sample standard deviations, to obtain the same margin of error for each confidence interval,

- (A) the sample sizes should be the same.
- (B) the sample in Greater Tokyo should be twice the size of the sample in Karachi.

- (C) the sample in Karachi should be twice the size of the sample in Greater Tokyo.
- (D) the sample in Greater Tokyo should be four times the size of the sample in Karachi.
37. For a class project, AP Statistics students set up a race for all students at their high school. The distance was 1.96 miles with prizes for the fastest time, the 75th percentile time, the median time, the 25th percentile time, and the slowest time. The prize-winning times in minutes were 11.1, 13.7, 15.1, 16.1, and 19.4. Which of the following statements about outliers is true?
- (A) Both the fastest and slowest times were outliers.
- (B) The fastest time was an outlier. The slowest time was not an outlier.
- (C) The fastest time was not an outlier. The slowest time was an outlier.
- (D) Neither the fastest nor slowest times were outliers.
38. Which of the following is an *incorrect* statement?
- (A) Statistics are random variables with their own probability distributions.
- (B) Bias means that, on average, our estimate of a parameter is different from the true value of the parameter.
- (C) There are some statistics for which the sampling distribution is not approximately normal, no matter how large the sample size.
- (D) The larger the sample size, the closer the sample distribution is to a normal distribution.
39. For Air Force cadets, the recommended fitness level with regard to the number of push-ups is 34. In a test whether or not current classes of recruits can meet this standard, a t -test of $H_0: \mu = 34$ against $H_a: \mu < 34$ gives a p -value of 0.068. Using these data, among the following, which is the largest level of confidence for a two-sided confidence interval that does not contain 34?
- (A) 85%
- (B) 90%
- (C) 92%
- (D) 95%
40. A particular car is tested for stopping distance in feet on wet pavement at 30 mph using tires with one tread design and then tires with another tread design.

For each set of tires, the test is repeated 30 times, and the following parallel boxplots give a comparison of the resulting five-number summaries.



Which of the following is a reasonable conclusion?

- (A) Distribution I is skewed right, while distribution II is bell-shaped.
 - (B) The mean of distribution I is greater than the mean of distribution II.
 - (C) The range of distribution I is approximately $46 - 33 = 13$.
 - (D) The upper 50% of the values in distribution I are all greater than the lower 50% of the values in distribution II.
41. E and F are events on the same probability space with $P(E) = 0.3$ and $P(F) = 0.4$. What is the relationship between $P(E \cap F)$ and 0.12?
- (A) $P(E \cap F) > 0.12$
 - (B) $P(E \cap F) < 0.12$
 - (C) $P(E \cap F) = 0.12$
 - (D) There is not enough information to determine a relationship between $P(E \cap F)$ and 0.12.
42. Do middle school and high school students have different views on what makes someone popular? Random samples of 100 middle school and 100 high school students yield the following counts with regard to three choices: lots of money, good at sports, and good looks:

What Makes a Student Popular			
	Money	Sports	Looks
Middle school students	22	48	30
High school students	36	24	40

A chi-square test of homogeneity yields which of the following test statistics?

- (A) $\frac{(22-29)^2}{22} + \frac{(48-36)^2}{48} + \frac{(30-35)^2}{30} + \frac{(36-29)^2}{36} + \frac{(24-36)^2}{24} + \frac{(40-35)^2}{40}$
- (B) $\frac{(22-29)^2}{29} + \frac{(48-36)^2}{36} + \frac{(30-35)^2}{35} + \frac{(36-29)^2}{29} + \frac{(24-36)^2}{36} + \frac{(40-35)^2}{35}$
- (C) $\left(\frac{22}{29}\right)^2 + \left(\frac{48}{36}\right)^2 + \left(\frac{30}{35}\right)^2 + \left(\frac{36}{29}\right)^2 + \left(\frac{24}{36}\right)^2 + \left(\frac{40}{35}\right)^2$
- (D) $\frac{(22)(29)}{58} + \frac{(48)(36)}{72} + \frac{(30)(35)}{70} + \frac{(36)(29)}{58} + \frac{(24)(36)}{72} + \frac{(40)(35)}{70}$

STOP

If there is still time remaining, you may check your work on this section.

Section II: Free-Response

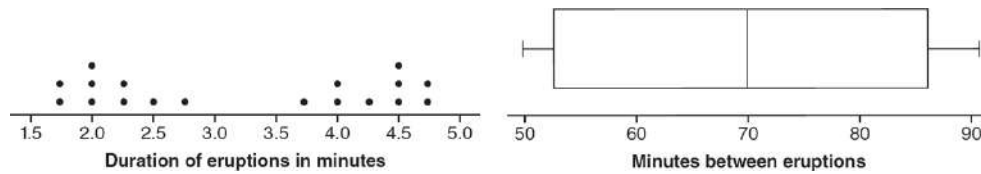
DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your results and explanations.

1. A horticulturist plans a study on the use of compost tea for plant disease management. She obtains 18 identical beds, each containing a random selection of five minipink rose plants. Some bed or beds will not receive compost tea. For those beds receiving compost tea, she plans to use two different composting times (2 and 5 days), two different compost preparations (aerobic and anaerobic), and two different spraying techniques (with and without adjuvants). Midway into the growing season she will check all plants for rose powdery mildew disease.
 - A.
 - i. **Identify** the experimental units.
 - ii. **Identify** the treatments.
 - B. **Explain** why the horticulturist included a group that does not receive compost tea?
 - C.
 - i. **Explain** the advantage of using only minipink roses in this study.
 - ii. **Explain** a disadvantage of using only minipink roses in this study.
 - D.
 - i. Using the proportion of plants in each bed that are infected with rose powdery mildew disease, **identify** a graphical representation that is appropriate to use to display these proportions.
 - ii. **Explain** why a histogram would not be appropriate to display these proportions.
 - E. The horticulturist would like to perform a study on proportions of plants that are infected with rose powdery mildew disease in beds using aerobic and

anaerobic preparations. **Determine** a valid investigative question. Ensure all three components of a valid investigative question are included in your response.

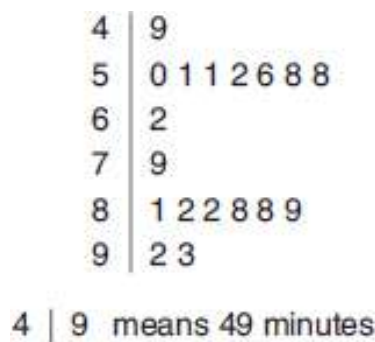
2. A National Park employee measures times between eruptions and eruption durations for the Old Faithful geyser during a 24-hour period.

A dotplot of eruption duration times and a boxplot of the times between eruptions are as follows:



[Follow for extended description](#)

- A.
 - i. **Identify** the type of study the National Park employee conducted.
 - ii. **Identify** the variables in the study.
- B. **Describe** the distribution of duration times of Old Faithful geyser eruptions.
- C. **Determine** what the boxplot illustrates about the center and variability of the distribution of times between eruptions?
- D. A stemplot of the times between eruptions is as follows:



[Follow for extended description](#)

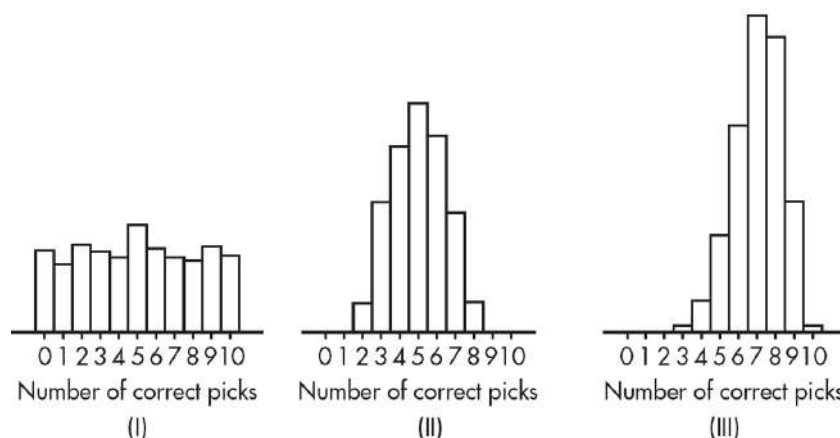
- i. **Determine** the median time between eruptions. Show your work.
 - ii. **Describe** what the stemplot shows about shape that is missed in the boxplot?
- E. **Determine** the population for valid investigative questions for the National Park employee's study.

3. A top-100, 7.0-rated tennis pro wishes to compare a new racket against his current model. He is interested in whether his hitting speed with the new racket is different from that with his old racket. He strings the new racket with the same type of strings at 60 pounds tension that he uses on his old racket. From past testing, the tennis pro knows that the average forehand crosscourt volley with his old racket is 82 miles per hour (mph). On an indoor court, using a ball machine set at 70 mph, which is the same speed he had his old racket tested against, he takes 47 swings with the new racket. (Assume that swings are independent of each other.) An associate with a speed gun records an average of 83.5 mph with a standard deviation of 3.4 mph. At the 0.01 level of significance, complete the appropriate inference procedure to determine if there is convincing statistical evidence that his mean hitting speed with the new racket is different from that with his old racket.

- A. **Identify** and **classify** the variable(s) in the study.
- B. **Identify** the appropriate inference procedure.
- C. **Complete** the inference procedure, including **calculating** the appropriate statistics.
- D. **Justify** a conclusion in context.

4. Researchers want to study whether a capuchin monkey named Rafiki can correctly predict (better than guesswork) whether the Dow Jones Industrial Average (DJIA) will go up or down on any given day. In a random sample of 10 days, Rafiki correctly predicted the rise or fall of the DJIA on 7 of the 10 days. (He did this by choosing to eat a banana from either a box with an “up” arrow or a box with a “down” arrow.)

- A.
 - i. **Identify** the parameter of interest.
 - ii. **Identify** the null and alternative hypotheses.
- B. Assuming Rafiki was guessing, **calculate** the probability he guessed correctly at least 7 out of 10 times.
- C. Suppose you conduct a simulation to investigate whether the observed result provides convincing evidence that Rafiki can correctly predict the rise or fall of the DJIA.
 - i. **Identify** what you would use for the probability of success.
 - ii. **Identify** what you would use for the sample size.
- D.
 - i. **Identify** which of the following graphs could reasonably have come from the simulation. **Explain**.



ii. **Identify** which of 6.5, 0.65, 0.15, and 0.015 is closest to the p -value. **Interpret** it in context.

- E. Based on the simulation and p -value, **determine** whether there is sufficient evidence that Rafiki can correctly predict (better than guesswork) whether the Dow Jones Industrial Average (DJIA) will go up or down on any given day. **Explain** your reasoning.

STOP

If there is still time remaining, you may check your work on this section.

Answer Explanations

Section I

1. **(A)** Without knowing the actual number of the sample pet owners from each location, only proportions, not numbers of pet owners, can be compared between locations.
2. **(B)** The slope gives the *predicted* or *average* increase or decrease in the dependent variable, sodium, for each 1 unit increase in the independent variable, calories.
3. **(D)** The y -intercept is -228.33 , and a negative sodium content does not have a logical interpretation.
4. **(B)** The coefficient of determination, 78.7%, gives the percent of the variability in the dependent variable, sodium (not the predicted sodium), that is explained by the linear model with the independent variable, calories.

5. **(A)** The shortest sequence has a greater probability than any longer sequence.
6. **(D)** Power, the probability of rejecting a false null hypothesis, will be the greatest for true values furthest from the hypothesized value in the direction of the alternative hypothesis. Here, $H_a: \mu > 2.0$, and 2.4 is furthest from 2.0 among the value choices that are greater than 2.0.
7. **(A)** A simple random sample may or may not be representative of the population. It is a method of selection in which every possible sample of the desired size has an equal chance of being selected.
8. **(C)** The number of parcels is an integer, a countable variable, thus this is discrete. Weight can take on an infinite number of values between any two values, thus it is continuous.
9. **(B)** The null hypothesis is that the new medication is no better than insulin injection, while the alternative hypothesis is that the new medication is better. A Type I error means a mistaken rejection of a true null hypothesis.
10. **(D)** A Type II error means a mistaken failure to reject a false null hypothesis. A false null hypothesis here means that her medication really is better than insulin injection, and failure to realize this means she does not have sufficient evidence that it is better.
11. **(D)** No matter what the distribution of raw scores, the set of z-scores always has mean 0 and standard deviation 1.
12. **(C)** There are two possible outcomes (heads and tails), with the probability of heads always 0.6 (independent of what happened on the previous toss), and we are interested in the number of heads in 20 tosses. Thus, this is a binomial model with $n = 20$ and $p = 0.6$. Repeating this over and over (in this case 200 times) simulates the resulting binomial distribution.
13. **(C)** Increasing sample size increases accuracy but has no effect on any bias that is present.
14. **(D)** When a complete census is taken (all 423 seniors were in the study), the population proportion is known and a confidence interval has no meaning.
15. **(A)** (B) and (C) are voluntary response surveys, and (D) suffers from undercoverage bias.

16. **(D)** $SE(\hat{p}) = \sqrt{(0.7)(0.3)/500}$. Then $z SE(\hat{p}) = 0.025$ gives critical z -score of $\pm \frac{0.025}{SE(\hat{p})}$. The degree of confidence is the probability between these two values.
17. **(D)** With $df = n - 1 = 10 - 1 = 9$ and 95% confidence, the critical t -values are $\pm_{invT} = \pm 2.262$. $\pm \frac{0.025}{SE(\hat{p})}$. Then $z SE(\hat{p}) = 0.025$ gives critical z -score of $\pm \frac{0.025}{SE(\hat{p})}$. The confidence interval is $\bar{x} \pm t^* SE(\bar{x}) = 17.3 \pm 2.262 \left(\frac{2.3}{\sqrt{10}} \right)$.
18. **(B)** Because the population is approximately normal, normality can be assumed for both sampling distributions. Both will have mean 44. The standard deviation for the $n = 10$ sampling distribution is $\frac{21}{\sqrt{10}}$ which is greater than the standard deviation for the $n = 100$ sampling distribution, which is $\frac{21}{\sqrt{100}}$.
19. **(D)** When we pick samples from each of two or more populations to compare the distribution of some variable among the different populations, the proper test is a chi-square test of homogeneity.
20. **(D)** The data are categorical as each student responds yes or no. This leads to estimating a population proportion—that is, the proportion of students who feel that male and female sports teams should be equally funded.
21. **(C)** $P(\text{male} \cap \text{basketball}) = P(\text{male})P(\text{basketball} | \text{male}) = (0.55)(0.30)$
22. **(B)** $P(\text{basketball}) = P(\text{male} \cap \text{basketball}) + P(\text{female} \cap \text{basketball}) = (0.55)(0.30) + (0.45)(0.20)$

23. **(B)**



$$P(\text{basketball}) = (0.55)(0.30) + (0.45)(0.20)$$

$$P(\text{male} | \text{basketball}) = \frac{P(\text{male} \cap \text{basketball})}{P(\text{basketball})} = \frac{(0.55)(0.30)}{(0.55)(0.30) + (0.45)(0.20)}$$

24. **(B)** Adding the same constant to every value in a set adds the same constant to the mean but leaves the standard deviation unchanged. Multiplying every value in a set by the same constant multiplies the mean and standard deviation by that constant. So the new mean is $\frac{5}{9} \times (78.35 - 32) + 273 = 298.75$, and the new standard deviation is $\frac{5}{9} \times 6.3 = 3.5$.
25. **(B)** Design 2 is an example of a matched pairs design, a special case of a block design; here, each subject is compared with itself with respect to the two treatments. Both designs definitely use randomization with regard to assignment of treatments, but since they do not use randomization in selecting subjects from the general population, care must be taken in generalizing any conclusions. It's not clear whether or not the researchers who do the observations and measurements know which treatment individual cows are receiving, so there is no way to conclude if there is or is not blinding. The two sources of BVH are different treatments, and so they are not being confounded. In both designs, treatments are randomly applied, so neither is an observational study.
26. **(D)** Confidence level (94% in this example) is the percentage of times repeated applications of this method (finding numbers of trees cut down in 24-hour periods) would capture the true population parameter (mean in this example). (C) is a correct interpretation of confidence *interval*, not confidence *level*.
27. **(B)** The size of the sample always matters: the larger the sample, the greater the power of statistical tests. One percent of a large population is large. Larger samples are better, but if the sample is greater than 10% of the population, the best statistical techniques are not those covered in the AP curriculum. While not in the AP curriculum, if the population is small, its size may matter as shown by the finite population correction factor.
28. **(D)** The points on the scatterplot all fall on the straight line:

$$\text{Female length} = \text{Male length} + 0.5$$

29. **(B)** The null hypothesis is that there is no difference in the distributions in size of social networks among different racial groups. Under the assumption that the null hypothesis is true, the p -value gives the probability of obtaining a test statistic as extreme as or more extreme than what was obtained in the sample.
30. **(C)** The distributions in (A) and (B) appear roughly symmetric, so the mean and median will be roughly the same. The distribution in (C) is skewed to the

right, so the mean will most likely be greater than the median, while the distribution in (D) is skewed to the left, so the mean will most likely be less than the median.

31. **(B)** This is a binomial with $n = 10$ and $p = 0.38$, so the mean is $np = 10(0.38) = 3.8$.
32. **(A)** The numerator of the test statistic is the sample mean minus the assumed population mean, while the denominator is the sample standard deviation divided by the square root of the sample size. $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{6.8 - 7}{1.3/\sqrt{100}} = \frac{6.8 - 7}{1.3/10}$
33. **(A)** Residual = observed – predicted, so $1.0 = 11 - \text{predicted}$ and predicted = 10.
34. **(B)** Stratified random sampling is when the population is divided into homogeneous groups (the three divisions in this example) and a random sample of individuals is chosen from each group.
35. **(C)** The two-sample z -interval for a difference in proportions is inappropriate here, because the two proportions do not come from *independent* samples.
36. **(A)** The margin of error, $\pm t^* \frac{s}{\sqrt{n}}$, depends on n , the sample size, not the population size.
37. **(D)** Outliers are values less than $Q_1 - 1.5(\text{IQR}) = 13.7 - 1.5(16.1 - 13.7) = 10.1$ or greater than $Q_3 + 1.5(\text{IQR}) = 16.1 + 1.5(16.1 - 13.7) = 19.7$. Both $11.1 > 10.1$ and $19.4 < 19.7$, so neither the fastest nor the slowest times are outliers.
38. **(D)** The larger the sample size, the closer the sample distribution is to the population distribution. The central limit theorem roughly says that if multiple samples of size n are drawn randomly and independently from a population, the histogram of the means of those samples will be approximately normal. Statistics have probability distributions called sampling distributions. The central limit theorem does not apply to all statistics as it does to sample means. Many sampling distributions are not normal; for example, the sampling distribution of the sample max is not a normal distribution. An estimator of a parameter is unbiased if we have a method that, through repeated samples, is on average the same value as the parameter.
39. **(A)** With 0.068 in a tail, the confidence interval with 34 at one end would have a confidence level of $1 - 2(0.068) = 0.864$, so anything higher than 86.4% confidence will contain 34.

40. **(B)** From a boxplot there is no way of telling if a distribution is bell-shaped. (Very different distributions can have the same five-number summary.) Distribution I appears strongly skewed right, and so its mean is probably much greater than its median, while distribution II appears roughly symmetric, and so its mean is probably close to its median. The interquartile range, not the range, in distribution I is 13.
41. **(D)** If E and F are independent, then it would follow that $P(E \cap F) = (0.3)(0.4) = 0.12$, but without further information, all that can be said is that $P(E \cap F)$ is between 0 and 0.3.
42. **(B)** $\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$ and cell calculations [expected value of a cell equals (row total)(column total)/(table total)] or χ^2 -test on a calculator such as the TI-84, Casio Prizm, or HP Prime will yield expected cells of 29, 36, 35, 29, 36, and 35.

Section II

1. Model Solution:

- A.
- i. The treatments are applied to the beds. So the experimental units are the 18 beds, not the individual rose plants.
 - ii. There is one treatment with no compost tea and $2 \times 2 \times 2 = 8$ treatments involving compost tea.
 - No compost tea
 - Two-day, aerobic, with adjuvant
 - Two-day, aerobic, without adjuvant
 - Two-day, anaerobic, with adjuvant
 - Two-day, anaerobic, without adjuvant
 - Five-day, aerobic, with adjuvant
 - Five-day, aerobic, without adjuvant
 - Five-day, anaerobic, with adjuvant
 - Five-day, anaerobic, without adjuvant
- B. The purpose of the group receiving no compost tea is to determine whether using no compost tea is better or worse than using any of the 8 particular applications of compost tea.

- C. i. Using only minipink roses gives reduced variability and increases the likelihood of determining differences among the treatments.
 ii. Using only minipink roses limits the scope of the experiment and makes generalizing to other species difficult.
- D. i. A bar chart (bar graph) is useful to display relative frequencies (proportions). In this context, the height of each bar would correspond to the proportion of plants in a bed that are infected with rose powdery mildew disease.
 ii. Histograms provide a visual representation of the distribution of a quantitative variable, not of relative frequencies corresponding to a categorical variable (the beds).
- E. Is there convincing statistical evidence to suggest that the proportion of minipink rose plants that are infected with rose powdery mildew disease in beds using aerobic preparations is different from the proportion of minipink rose plants that are infected with rose powdery mildew disease in beds using anaerobic preparations?

A.	i.	Identify the experimental units.	The experimental units are the 18 beds.	1 point
	ii.	Identify the treatments.	No compost tea Two-day, aerobic, with adjuvant Two-day, aerobic, without adjuvant Two-day, anaerobic, with adjuvant Two-day, anaerobic, without adjuvant Five-day, aerobic, with adjuvant Five-day, aerobic, without adjuvant Five-day, anaerobic, with adjuvant Five-day, anaerobic, without adjuvant	1 point
B.		Explain the purpose of a control group in this context.	To determine if using no compost tea is better or worse than using any of the 8 particular applications of compost tea	1 point
C.	i.	Explain the advantage of using only minipink roses.	To reduce variability and increase the likelihood of determining differences among the treatments	1 point
		Explain the disadvantage of using only minipink roses.	Makes generalizing to other species difficult	1 point
D.	i.	Identify an appropriate graph.	A bar chart (bar graph)	1 point
	ii.	Explain why a histogram is not appropriate.	Histograms are for the distribution of a quantitative variable, not a categorical variable like the beds.	1 point

E.	First component (identify variables of interest)	The variables are the beds treated with the aerobic preparation and the beds treated with the anaerobic preparation.	1 point
	Second component (identify the parameter and direction of the alternative hypothesis)	The proportion of plants that are infected with rose powdery mildew disease is different in beds using aerobic preparations and in beds using anaerobic preparations.	1 point
	Third component (identify the population to which conclusion is applicable)	All minipink rose plants in beds being treated with aerobic preparations and in beds being treated with anaerobic preparations.	1 point

2. Model Solution:

- A. i. Observational study
 ii. Duration of eruptions in minutes and minutes between eruptions
- B. The distribution of eruption time durations is roughly symmetric with two clusters and a gap, has a center between 3.0 and 3.5 minutes, and has a range of $4.75 - 1.75 = 3.0$ minutes.
- C. The boxplot of times between eruptions shows a median slightly greater than 70 minutes and a range slightly greater than 40 minutes.
- D. i. With an even number of values, the median is the average of the two middle ones: $(62 + 79)/2 = 70.5$
 ii. The stemplot shows that the distribution of times between eruptions has two roughly symmetric clusters and a gap between 62 and 79.
- E. The population for valid investigative questions for the National Park employee's study would be all Old Faithful geyser eruptions.

A.	i.	Identify the type of study.	Observational	1 point
	ii.	Identify the variables.	Duration of eruptions Minutes between eruptions	1 point
B.		Describe a distribution.	Distribution of eruption time durations	1 point
			Roughly symmetric with two clusters and a gap	1 point
			Center between 3.0 and 3.5 minutes, and a range of $4.75 - 1.75 = 3.0$ minutes	1 point
C.		Determine the center.	Median slightly greater than 70 minutes	1 point
		Determine the variability.	Range slightly greater than 40 minutes	1 point

D.	i.	Determine the median.	$(62 + 79)/2 = 70.5$	1 point
	ii.	Describe what the stemplot shows that the boxplot misses.	Two roughly symmetric clusters and a gap between 62 and 79	1 point
E.		Determine the population for an investigative question.	All Old Faithful geyser eruptions	1 point

3. Model Solution:

- A. The type of racket, current model or new model, is categorical. Speed of hit ball is numerical.
- B. An appropriate inference procedure is a one-sample t -test for a population mean.
- C. The null hypothesis is $H_0: \mu = 82$, and the alternative hypothesis is $H_a: \mu \neq 82$.
For new racket swings similar to those in the study, let μ = true mean hitting speed with the new racket. The randomization condition is satisfied as we are given that the swings are independent. The normality condition is satisfied because the sample size $n = 47 \geq 30$. This is sufficiently large for the central limit theorem to apply. The value of the test statistic is $t = 3.0246$ with a p -value of 0.00406 and $df = 46$.
- D. With this small a p -value, $0.00406 < 0.01$, there is sufficient evidence to reject the null hypothesis. There is convincing evidence that the tennis pro's hitting speed with the new racket is different from that with his old racket.

A.	Identify the variables.	Type of racket, whether current model or new model; speed of hit ball	1 point
	Classify the variables.	Racket type is a categorical variable; speed of hit ball is a numerical variable.	1 point
B.	Identify the appropriate procedure.	One-sample t -test for a population mean	1 point
C.	Identify H_0 and H_a .	$H_0: \mu = 82$ and $H_a: \mu \neq 82$	1 point
	Identify the parameter.	μ = the true mean hitting speed with the new racket	1 point
	Justify whether the randomization condition is met.	We are given that the swings are independent.	1 point
	Justify whether the normality condition is met.	The sample size $n = 47 \geq 30$ is sufficiently large for the central limit theorem to apply.	1 point

	Calculate the test statistic and p -value.	$t = 3.0246$ with a p -value of 0.00406 and $df = 46$	1 point
D.	Compare the p -value to the significance level.	$0.00406 < 0.01$ so the null hypothesis should be rejected.	1 point
	Justify whether the claim is or is not supported.	There is convincing evidence that the tennis pro's hitting speed with the new racket is different from that with his old racket	1 point

4. Model Solution:

- A. i. The parameter of interest is p = the true proportion of times Rafiki can correctly predict whether the DJIA will go up or down on any given day.
 ii. $H_0 : p = 0.5$ and $H_a : p > 0.5$
- B. Binomial distribution with $n = 10$ and $p = 0.5$. $P(x \geq 7) = 0.172$
- C. i. Probability of success $p = 0.5$
 ii. Sample size $n = 10$
- D. i. Graph (II). The simulation should be centered at the value based on the null hypothesis probability, not the observed value.
 ii. Looking at areas under (II), which is the correct graph, among the given choices, 7 or greater picks seems to have a probability closest to 0.15. If the null hypothesis were true—that is, if Rafiki can do no better than guesswork—the estimated probability of Rafiki correctly predicting 7 or more out of 10 rises or falls of the DJIA is about 0.15.
- E. With this large a p -value, $0.15 > 0.05$, there is not sufficient evidence to reject H_0 . There is not convincing evidence that Rafiki can pick the rise and fall of the DJIA better than guesswork.

A.	i.	Identify the parameter.	p = the true proportion of times Rafiki can correctly predict whether the DJIA will go up or down on any given day	1 point
	ii.	Identify the hypotheses.	$H_0 : p = 0.5$ and $H_a : p > 0.5$	1 point
B.		Identify the distribution and parameters.	Binomial distribution with $n = 10$ and $p = 0.5$	1 point
		Calculate the probability.	$P(x \geq 7) = 0.172$	1 point
C.	i.	Identify the probability of success.	$p = 0.5$	1 point
	ii.	Identify the sample size.	Sample size $n = 10$	1 point

D.	i.	Identify the graph.	Graph (II). The simulation should be centered at the value based on the null hypothesis probability, not the observed value.	1 point
	ii.	Calculate the p -value.	From area under Graph (II), $p \approx 0.15$	1 point
E.		Determine whether a claim is supported.	There is not convincing evidence that Rifiki can pick the rise and fall of the DJIA better than guesswork.	1 point
		Explain the reasoning.	With this large a p -value, $0.15 > 0.05$, there is not sufficient evidence to reject H_0 .	1 point

AP Score for the Diagnostic Test

Multiple-choice section (42 questions)

Number correct $\times 1.19 =$ _____

Free-response section (4 open-ended questions plus an investigative task)

Question 1 _____ $\times 1.25 =$ _____
out of 10

Question 2 _____ $\times 1.25 =$ _____
out of 10

Question 3 _____ $\times 1.25 =$ _____
out of 10

Question 4 _____ $\times 1.25 =$ _____
out of 10

[Follow for extended description](#)

Total points from multiple-choice and free-response sections = _____

Conversion chart based on a recent AP exam:

Total Points	AP Score
70–100	5

Total Points	AP Score
55–69	4
40–54	3
30–39	2
0–29	1

Study Guide for the Diagnostic Test Multiple-Choice Questions

Circle all of the questions you answered incorrectly, and focus your review on the unit or units that contain the most circled questions.

UNIT 1: Exploring One-Variable Data and Collecting Data

Questions: 1, 7, 8, 11, 13, 15, 24, 25, 30, 34, 37, 40

UNIT 2: Probability, Random Variables, and Probability Distributions

Questions: 5, 12, 21, 22, 23, 31, 38, 41

UNIT 3: Inference for Categorical Data: Proportions

Questions: 9, 10, 12, 14, 16, 19, 20, 27, 29, 35, 42

UNIT 4: Inference for Quantitative Data: Means

Questions: 6, 17, 18, 26, 32, 36, 39

UNIT 5: Regression Analysis

Questions: 2, 3, 4, 28, 33

Part 2

Units Review

1

Exploring One-Variable Data and Collecting Data

(8–13 multiple-choice questions on the AP exam)

Learning Objectives

In this unit, you will learn how

- To establish a research question in a statistical study.
- To be able to distinguish between categorical variables and quantitative variables.
- To be able to distinguish between discrete and continuous quantitative variables.
- To use counts and relative frequencies to describe categorical variables.
- To be able to create and interpret bar charts and pie charts for categorical data given as frequency or relative frequency tables.
- To be able to calculate summary statistics for univariate (one-variable) quantitative data including center (mean and median), variability (range, interquartile range, variance, and standard deviation), and the five-number summary (minimum, Q_1 , median, Q_3 , and maximum).
- To be able to describe univariate quantitative data including shape (such as bell-shaped, symmetric, right or left skewed, uniform, unimodal, or bimodal), center, variability (spread), and unusual features (such as gaps, clusters, modes, and outliers), always remembering to do so in context (including units).
- To be able to read and interpret specific information from dotplots, stemplots, histograms, and boxplots.
- To be able to compare two distributions, being sure to use comparative terminology rather than simply making two separate lists, and not forgetting context.
- To be able to calculate z-scores and to use z-scores to compare relative positions of individual values.
- To be able to distinguish between a sample and a population and between a statistic and a parameter.
- To be able to explain why a particular study is observational or an experiment.
- To be able to explain why or why not the results of a study can be generalized to a larger population.
- To be able to explain whether an observational study is retrospective or prospective.
- To be able to identify, describe, and implement different sampling methods, including a simple random sample, a cluster random sample, a stratified random sample, and a

systematic random sample.

- To be able to describe the benefits and drawbacks of various sampling methods.
- To recognize different sources and types of bias, including voluntary response bias, undercoverage bias, nonresponse bias, and response bias.
- To be able to explain how a particular bias is likely to result in the sample underestimating or overestimating a population parameter.
- To identify elements of a well-designed experiment, including comparison, random assignment, replication, and control.
- To be able to design an experiment using one or more of the following: different levels of an explanatory variable, a control group possibly using a placebo, blinding or double-blinding, blocking, and matched pairs design.
- To be able to explain in context why a particular variable might lead to confounding.
- To be able to explain why or why not a cause-and-effect conclusion is justified.
- To understand the difference between, and how to perform, a random sampling and a random assignment.

Statistical Studies

Statistical studies typically involve the following steps:

- Asking an investigative question
- Noting the population of interest
- Deciding what unknown population parameter is of interest
- Selecting a sample from the population from which to obtain data
- Summarizing the sample data numerically and graphically
- Analyzing the sample statistics from the data
- Making conclusions in context

The field of statistics is all about collecting, describing, analyzing, and drawing conclusions from data.

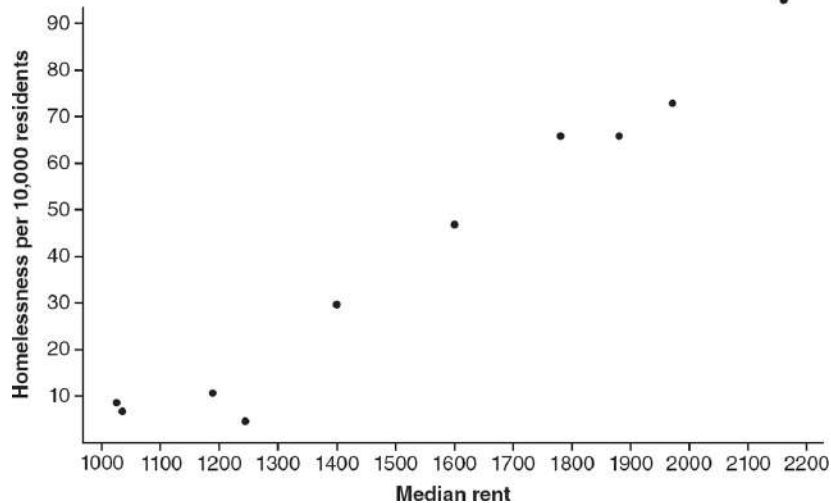
- A **population** consists of the entire set of individuals or objects that you want to learn about.
- A **sample** is a subset of the population that is selected for study. From the sample, data are collected and analyzed in order to answer investigative questions about the population.
- The population size is often designated by N , while the sample size is often designated by n .
- A **parameter** is a numerical summary of a population, while a **statistic** is a numerical summary of sample data.
- We want to estimate an unknown population characteristic (parameter) by considering a sample measurement (statistic).

- Conclusions should be given **in context**, that is, should refer back to the population in a real-world frame of reference.

As an introduction to what is expected on the exam, consider the following two examples.

➤ Example 1.1

Homelessness affects millions of people across the United States, and understanding the causes is essential to addressing the problem. A primary cause of homelessness in America is thought to be lack of affordable housing, especially in urban areas. Researchers randomly selected 10 American cities. For each city, they calculated the median rent and the number of homeless per 10,000 residents. The data are displayed in the following scatterplot:



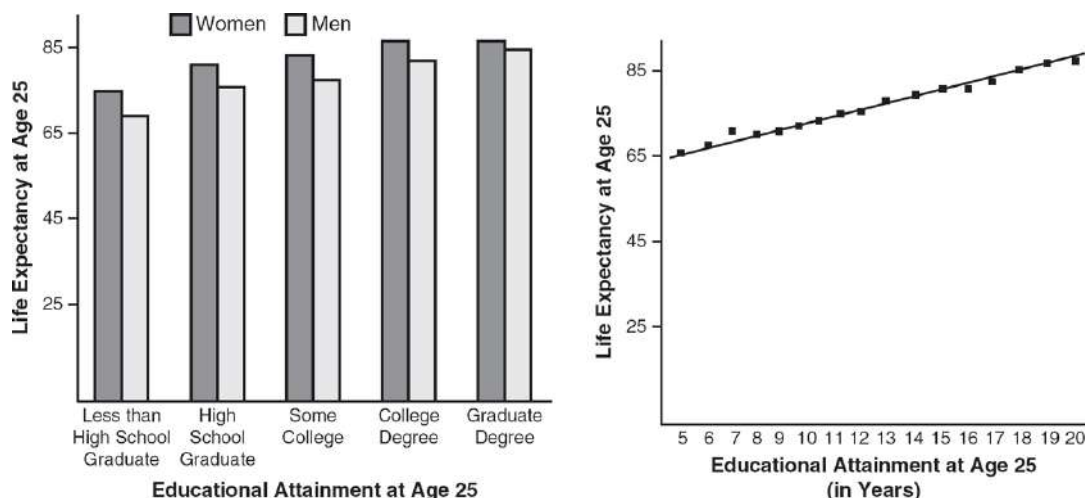
- (a) What is the investigative question of interest?
- (b) What is the population of interest?
- (c) What was the selected sample?
- (d) What do these data indicate about median rent versus number of homeless per 10,000 city residents?
- (e) Do the data prove that high median rent causes high homelessness?
- (f) The slope of a best-fitting straight line to these data is 0.079. Interpret this in context.

✓ Solutions

- (a) Is there a relationship between median rent and homelessness in American cities?
- (b) The population of interest is the set of all American cities.
- (c) The selected sample was the set of 10 randomly selected American cities.
- (d) As median rent goes up, the number of homeless per 10,000 residents tends to go up.
- (e) The data do not prove a cause-and-effect relationship. Instead, they show only an association. For example, it could be that landlords raise rents in cities with greater substance abuse problems hoping to discourage undesirable renters. Furthermore, people with substance abuse problems have difficulty maintaining employment and housing. If this was the case, then greater substance abuse problems, not higher median rent, would be the root cause of higher homelessness.
- (f) This predicts a rise of 0.079 in homelessness per 10,000 residents for every \$1 increase in median rent. In other words, this is a prediction of 7.9 more homeless per 10,000 residents for every \$100 increase in median rent.

➤ Example 1.2

An epidemiologist is interested in a possible association between life expectancy and educational attainment and if it relates to gender in American adults. She gathers data from a random sample of $n = 2,000$ obituaries of American adults and notes gender, educational attainment, and age at death. The data are summarized in the following two graphs.



- (a) What is the investigative question of interest?
- (b) What is the population of interest?

- (c) What was the selected sample?
- (d) What do the data say about women's versus men's life expectancies? Explain.
- (e) Do the data say that attaining a higher educational level will give a person greater life expectancy? Explain.
- (f) Could a pie chart be used to display these data? Explain.
- (g) Do the points in the second graph seem to follow a pattern that is linear or nonlinear? Is this pattern in the positive or negative direction? Is it strongly, moderately, or weakly aligned?
- (h) Two points on the second graph are (12, 72) and (20, 84). They correspond to 25-year-olds who graduated from high school and graduate school, respectively. What information does the slope of a line between these points predict?

Solutions

- (a) The epidemiologist's investigative question of interest is whether or not there is convincing statistical evidence of an association between life expectancy and educational attainment among all American adults and if it relates to gender.
- (b) The population of interest is the set of all American adults.
- (c) The selected sample was the set of 2,000 American adults whose obituaries were available.
- (d) At each educational attainment level, women's life expectancy is higher than men's.
- (e) This does not indicate a cause-and-effect relationship. Instead, it only shows an association. For example, it could be that more intelligent people attain higher educational levels and also take better care of their health and so have greater life expectancies. If this was the case, intelligence, not educational level, would lead to greater life expectancy.
- (f) No, a pie chart could not be used to display these data because the life expectancy for each educational attainment level does not represent parts of a single whole.
- (g) Visually, the points appear to follow a linear, positive, strong pattern because the points are tightly clustered around an upward-sloping, straight line.
- (h) The slope is $(84 - 72)/(20 - 12) = 12/8 = 1.5$. This predicts a 1.5-year rise in life expectancy for each additional year of education.

Variables

Data are pieces of information about **observational units** (items or individuals).

Variables are characteristics that may change from one observational unit to another.

A **categorical variable** (or **qualitative variable**) places individuals into groups called categories. For example, gender, musical genre, blood type, and political party are categorical variables. Categorical variables often lead to research questions involving proportions.

A **quantitative variable** (or **numerical variable**) takes on values for a measurable quantity. For example, height, weight, speed, and radius are numerical variables. Quantitative variables can be classified as **discrete** (taking on values with gaps between them) or **continuous** (taking on all possible values within intervals). For example, the numbers of students in classrooms, the salaries of employees, and how many text messages are sent are discrete quantitative variables, while backpack weights, wind speeds, and times spent doing homework are continuous quantitative variables. Quantitative variables often lead to research questions involving means.

Example 1.3

Classify each of the following variables as categorical, discrete quantitative, or continuous quantitative.

- (a) Heights of students in your classroom
- (b) Amounts spent by customers in a grocery store
- (c) Favorite movies
- (d) Numbers of people attending high school football games on given weekends
- (e) Brands of cereal eaten by students at your high school
- (f) Lengths of customer service calls

Solutions

- (a) Continuous quantitative
- (b) Discrete quantitative
- (c) Categorical
- (d) Discrete quantitative
- (e) Categorical
- (f) Continuous quantitative

One Categorical Variable (Tables, Graphs, Summary Statistics)

The values taken on by categorical variables can be organized into frequency tables or relative frequency tables. They can be represented graphically by displays such as bar charts (bar graphs) and pie charts.

> Example 1.4

During the first week of 2026, the results of a survey revealed that 1,100 parents wanted to keep the school year to the current 180 days, 300 wanted to shorten it to 160 days, 500 wanted to extend it to 200 days, and 100 expressed no opinion. (Noting that there were 2,000 parents surveyed, proportions and percentages can be calculated.)

A table showing frequencies (counts) and relative frequencies (proportions and percentages):

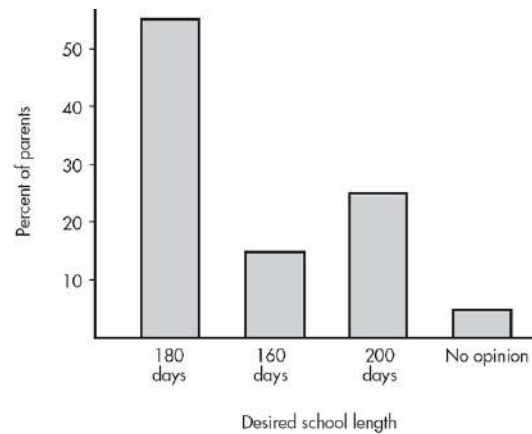
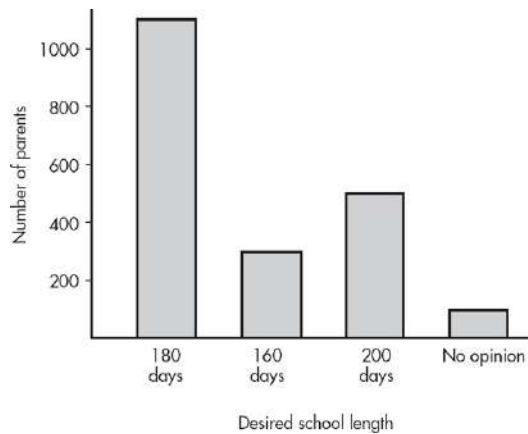
NOTE

Frequency gives the number of cases falling into each category, while relative frequency gives the proportion or percent of cases falling into each category.

Desired School Length	Number of Parents (frequency)	Relative Frequency	Percent of Parents
180 days	1,100	$1,100/2,000 = 0.55$	55%
160 days	300	$300/2,000 = 0.15$	15%
200 days	500	$500/2,000 = 0.25$	25%
No opinion	100	$100/2,000 = 0.05$	5%

Bar charts (bar graphs) showing frequencies and relative frequencies:

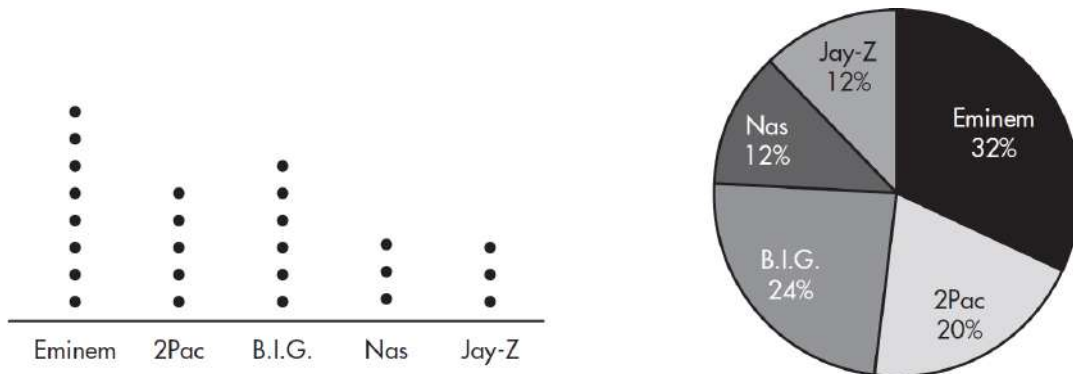
The distribution of a variable describes the possible values the variable could assume and how often each of those values occurs.



Note that, as above, bar charts (bar graphs) should always have appropriate labeling and scaling.

Example 1.5

When asked to choose their favorite hip-hop artist, 8 students chose Eminem, 5 picked 2Pac, 6 picked Notorious B.I.G., 3 picked Nas, and 3 picked Jay-Z. These data can be displayed in the following dotplot and pie chart.



NOTE

$8 + 5 + 6 + 3 + 3 = 25$, and $8/25 = 32\%$, $5/25 = 20\%$, $6/25 = 24\%$, and $3/25 = 12\%$.

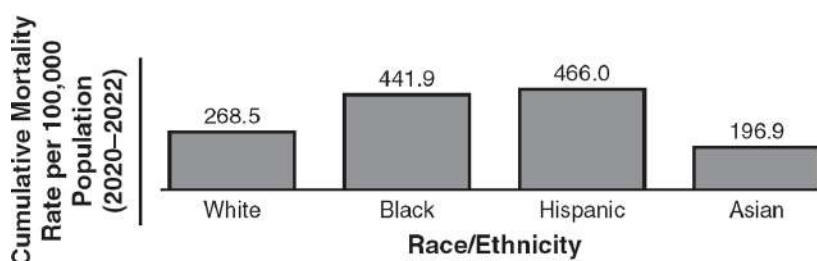
- Among those students who did not choose Eminem, what proportion chose Jay-Z?
- What ratio compares the Eminem choices to the 2Pac choices?

Solutions

- (a) Because $25 - 8 = 17$ students did not choose Eminem and 3 of these 17 students chose Jay-Z, the proportion is $\frac{3}{17}$.
- (b) The ratio comparing the Eminem choices to the 2Pac choices is 8:5.

➤ Example 1.6

The first years of the Covid-19 pandemic illustrated stark disparities in the health outcomes for people of color. The following bar chart shows 2020–2022 mortality data (cumulative mortality rates per 100,000 population) by race/ethnicity.



- (a) Can a pie chart be used to display the above data?
- (b) Is the shape of the above graphical display meaningful?

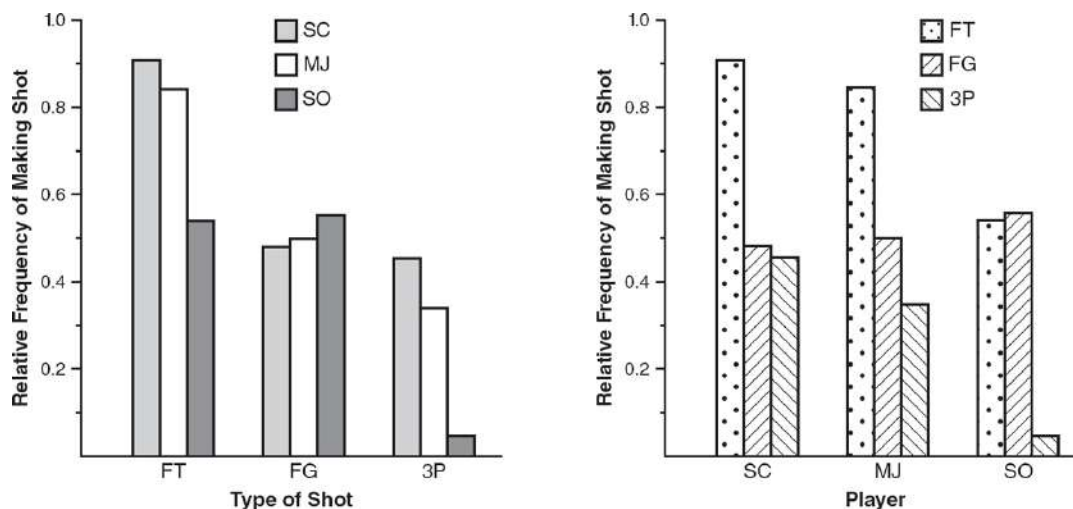
✓ Solutions

- (a) It would not be correct to use a pie chart to display these data because the mortality rate of each race/ethnicity does not represent parts of a single whole.
- (b) The order of race/ethnicity along the horizontal axis does not need to be in any specific order. Therefore, we do not talk about the “shape” of such a bar chart.

Bar charts are useful when comparing categorical data.

➤ Example 1.7

How do Stephen Curry (SC), Michael Jordan (MJ), and Shaquille O’Neal (SO) compare with regard to free throw percentage (FT%), field goal percentage (FG%), and three-point percentage (3P%) shooting? This can be illustrated by using either of the following side-by-side bar charts. The first graph groups by “type of shot,” and the second graph groups by “player” along the respective horizontal axes.



- Is there evidence to support the claim that Shaquille O’Neal had a higher field goal percentage than Jordan or Curry?
- Is there evidence to support the claim that Jordan’s free throw percentage was more than twice his three-point percentage?

✓ Solutions

- Yes, the first comparative bar chart shows that Shaquille O’Neal had a higher field goal percentage than Jordan or Curry. (The simple reason was that most of O’Neal’s shots were dunks.)
- Yes, the second comparative bar chart shows the ratio of Jordan’s free throw percentage to his three-point percentage to be approximately $0.85:0.35 = 17:7$, which is greater than $2:1$.

One Quantitative Variable (Tables and Graphs)

A quantitative variable takes on numerical values for a measured or counted quantity. The values can be organized into frequency tables or relative frequency tables or can be represented graphically by displays such as dotplots, histograms, stemplots, cumulative relative frequency plots, or boxplots.

A quantitative variable can be categorized as either *discrete* or *continuous*. A discrete quantitative variable takes on a finite or countable number of values. There are “gaps” between each of the values. For example, the number of teachers in different schools throughout a state is a countable or finite amount in each school. No school has 100.7 or $252\frac{1}{2}$ teachers. There is a “gap” between 100 and 101 and between 252 and 253. Other examples of discrete quantitative variables include the SAT scores of students in a school or the amount of money

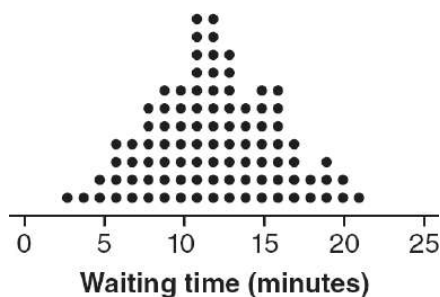
in bank accounts in a city. A continuous quantitative variable can take on uncountable or infinite values with no gaps, such as heights and weights of students. Time is a continuous quantitative variable because, for example, you can measure someone's age as 18 years, 6 months, 17 days, 4 hours, 19 seconds, 39 milliseconds, . . . and so on.

TIP

To help decide if a variable is a quantitative variable, ask if it makes sense to find an average. For example, you can find the average “height of students” but not the average “eye color.”

➤ Example 1.8

The *dotplot* below shows the waiting time lengths (in minutes) for the 95 people who showed up at a hospital emergency room on a randomly chosen day.



[Follow for extended description](#)

TIP

Typically, discrete quantitative variables represent something that is being counted, whereas continuous quantitative variables represent something that is being measured.

- (a) What were the shortest and longest waiting times?
- (b) What proportion of the people waited at least 20 minutes?

✓ Solutions

- (a) The shortest waiting time was 3 minutes, and the longest was 21 minutes.

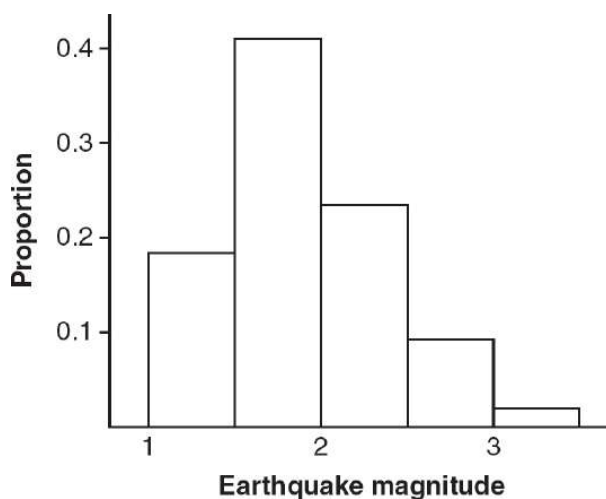
- (b) Three people waited at least 20 minutes, so the proportion is $\frac{3}{95} \approx 0.032$ or 3.2%.

Histograms, which are useful for large data sets involving quantitative variables, show frequencies or relative frequencies with bars over intervals called bins. Relative frequencies correspond to relative areas. On the AP exam, there often are questions that involve interpreting given histograms.

Note: A histogram displays the distribution of a quantitative variable with a horizontal axis identifying intervals of values that the variable takes. In contrast, a bar chart displays the distribution of a categorical variable with a horizontal axis identifying the categories. Histograms appear with no space between the bars (unless no data have values over that bin), while bar charts appear with spaces to separate the categories.

➤ Example 1.9

The following histogram shows magnitudes of the 2,650 earthquakes that occurred in a certain locale over the past 50 years.



[Follow for extended description](#)

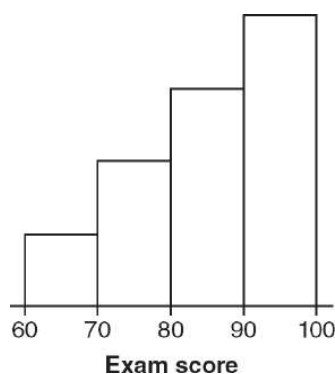
- (a) Approximately what proportion of the earthquakes had magnitudes less than 2.0?
- (b) Approximately how many earthquakes had magnitudes over 2.5?

✓ Solutions

- (a) Approximately $0.18 + 0.40 = 0.58$
 (b) Approximately $(0.10 + 0.02) \cdot 2,650 = 318$

Example 1.10

Consider the following histogram of exam scores, where the vertical axis has not been labeled.

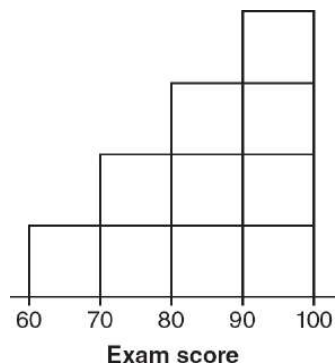


[Follow for extended description](#)

What can we learn from this histogram?

Solution

It is impossible to determine the actual frequencies; that is, we have no idea if there were 25 students, 100 students, or any particular number of students who took the exam. However, relative frequencies correspond to relative areas, so we can determine the relative frequencies by noting the fraction of the total area that is over any interval.



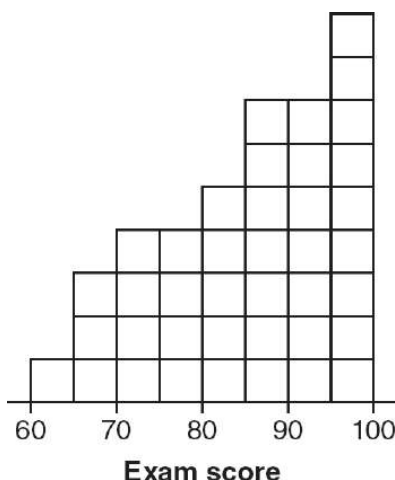
[Follow for extended description](#)

We can divide this area into 10 equal portions and then note that $\frac{1}{10}$, or 10%, of the area is between 60 and 70, so 10% of the students scored between 60 and 70. Similarly, 20% scored between 70 and 80, 30% scored between 80 and 90, and 40% scored between 90 and 100.

TIP

Relative frequencies are the usual choice when comparing distributions of different-size populations.

Suppose we have further information as given in the following graph.



[Follow for extended description](#)

- (a) Now what percentage of the students scored between 60 and 70?
- (b) Now what percentage of the students scored between 90 and 100?
- (c) What was the lowest exam score?
- (d) What was the highest exam score?

✓ Solutions

- (a) Still 10%. We could calculate $\frac{4}{40} = 0.10$ or simply realize that the proportion of area hasn't changed.
- (b) Still 40%. We could calculate $\frac{16}{40} = 0.40$, but again the proportion of area remains the same.

- (c) It is impossible to say exactly what the lowest score was, only that it was between 60 and 65.
- (d) It is impossible to say exactly what the highest score was, only that it was between 95 and 100.

Although a histogram may show how many scores fall into each grouping or interval, the exact values of individual scores are lost. An alternative pictorial display, called a *stemplot* (also called a stem and leaf display), retains this individual information and is useful for giving a quick overview of a distribution, displaying the relative density and shape of the data. Each data value is split into a “leaf” (the last digit) and a “stem” (the first digit or digits). A stemplot contains two columns separated by a vertical line. The left column contains the stems, and the right column contains the leaves. Both stems and leaves are ordered from smallest to largest.

► Example 1.11

How many nonstop push-ups can a teenager between the ages of 15 and 18 do? In one study in a mixed-gender high school gym class of 25 students, the numbers of push-ups were {2, 5, 7, 10, 12, 12, 14, 16, 16, 18, 19, 20, 21, 29, 32, 34, 35, 37, 37, 38, 39, 39, 42, 44, 50}.



Number of push-ups
(5|0 means 50 push-ups)

[Follow for extended description](#)

TIP

All stemplots must have keys!

NOTE

Stemplots must include all stems between the lowest and highest stem, even if that stem has no leaves in it. This is important to show any gaps in the data.

- (a) What was the minimum number of nonstop push-ups by teenagers in this class?
- (b) What was the maximum number of nonstop push-ups by teenagers in this class?
- (c) What was the middle number (median) of nonstop push-ups by teenagers in this class?
- (d) Give a possible explanation for the bimodal nature of these data.

✓ Solutions

- (a) The minimum number of nonstop push-ups by teenagers in this class was 2.
- (b) The maximum number of nonstop push-ups by teenagers in this class was 50.
- (c) The middle number (median) of nonstop push-ups by teenagers in this class was 21.
- (d) The class was mixed gender. It is possible that the peak centered around 15 nonstop pushups was mostly from female students, while the peak centered around 37 was mostly from male students.

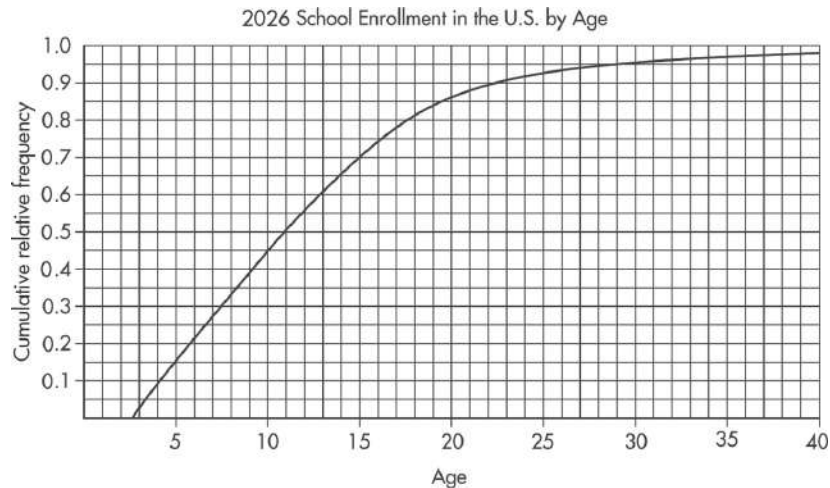
Stemplots allow for easy visualization of the distribution of data:

- Because a stemplot preserves all the original data values, this makes it impractical with large data sets.
- Stemplots must include all stems between the lowest and highest stem, even if that stem has no leaves in it. This is important to show any gaps in the data.
- All stemplots must have keys!
- Note how stemplots look something like bar charts.
- Stemplots are useful in highlighting outliers and modes.

Sometimes we sum frequencies and show the result visually in a cumulative relative frequency plot (also known as an *ogive*). While an ogive can help you visualize data, you will not be tested on it.

➤ Example 1.12

The following graph shows 2026 school enrollment in the United States by age.



- (a) Interpret the point (5, 0.15) in context.
- (b) 50% of school enrollment is above and below what age?
- (c) What proportion of school enrollment is below age 29?
- (d) The middle 50% of school enrollment is between what two ages?

✓ Solutions

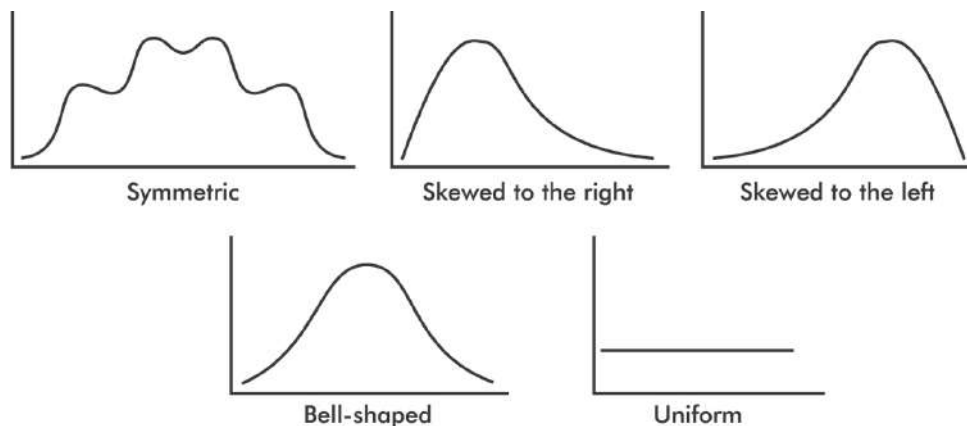
- (a) 15% of school enrollment is below age 5.
- (b) By going over to the graph from 0.5 on the vertical axis, we see that the middle age is 11.
- (c) By going up to the graph from 29, we see that 95% of the school enrollment is below 29.
- (d) By going over to the graph from both 0.25 and 0.75, we see that the middle 50% of school enrollment is between ages 6 and 7 at the lower end and age 16 at the upper end.

One Quantitative Variable (Describing a Distribution)

When describing a distribution, always comment on Shape, Outliers, Center, and Spread (SOCS). Alternatively, you can use the acronym Center, Unusual features, Shape, and Spread (CUSS). Always mention context somewhere in the description. Important aspects that should be addressed include the following.

1. *Shape*: Although distributions come in an endless variety of shapes, certain common patterns deserve special mention.

- A *symmetric* distribution is one in which the two halves are mirror images of each other. For example, the weights of all people in some organizations fall into an approximately symmetric distribution with two mirror image bumps, one for men's weights and one for women's weights.
- A distribution is *skewed to the right* if it spreads far and thinly toward the higher values. For example, ages of nonagenarians (people in their 90s) is a distribution with sharply decreasing numbers as one moves from 90-year-olds to 99-year-olds.
- A distribution is *skewed to the left* if it spreads far and thinly toward the lower values. For example, track meet times often show a distribution bunched at a higher end with a few low values.
- A *bell-shaped* distribution is symmetric with a center mound and two sloping tails. For example, the distribution of IQ scores across the general population is approximately symmetric with a center mound at 100 and two sloping tails.
- A distribution is *uniform* if its histogram is a horizontal line. For example, tossing a fair die and noting how many dots (pips) appear on top yields an approximately uniform distribution because 1 through 6 are all equally likely.
- A distribution with one main peak is called *unimodal* and with two distinct peaks is called *bimodal*. For example, the number of siblings of each student in most high schools tends to be unimodal with a peak at 1 sibling, while the distribution of heights in fifth-grade classrooms tends to be bimodal because girls usually have earlier growth spurts.



In the real world, distributions are rarely perfectly symmetric, perfectly bell-shaped, or perfectly uniform. So we usually say “roughly” or “approximately” symmetric, bell-shaped, or uniform.

2. *Center*: The center either separates the values (or the area under the curve in a histogram) roughly in half (the *median*) or is calculated as an average (the *mean*).
3. *Spread*: The spread or variability is the scope of the values from smallest to largest. More specifically, the *range*, *interquartile range*, *variance*, and *standard deviation* are discussed depending on the distribution.
4. *Unusual features*: Distinctive distribution attributes should always be commented upon.
 - *Outliers*: Outliers are values that are unusually small or large relative to the rest of the data. (For example, most people can hold their breath between 20 and 60 seconds; 5 seconds for a person with emphysema and 5 minutes for a free diver would be outliers in a breath-holding distribution.) Two specific mathematical definitions of outliers will be discussed later.
 - *Gaps*: Gaps show holes where no values fall. (For example, the Office of the Dean sends letters to students being put on the honor roll and to those being put on academic warning for low grades. Thus, the GPA distribution of students receiving letters from the dean has a huge gap in the middle.)
 - *Clusters*: Clusters show natural subgroups into which the values fall. (For example, the salaries of teachers in Ithaca, NY, fall into three overlapping clusters: one for public school teachers, a higher one for Ithaca College professors, and an even higher one for Cornell University professors.) Although clusters may overlap, they are often separated by gaps.

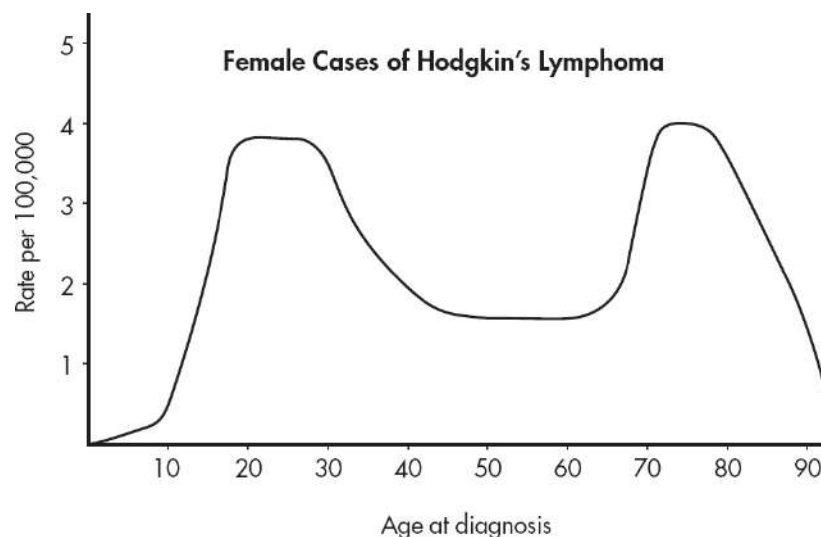
You have already seen several of these on the previous pages:

- The dotplot of [Example 1.8](#) illustrates that the distribution of waiting times at the hospital emergency room is roughly bell-shaped with a center around 12 and a spread from 3 to 21.
- The histogram of [Example 1.9](#) illustrates that the distribution of the magnitudes of the earthquakes is skewed right with a center between 1.5 and 2 and a spread from between 1 and 1.5 on the left to between 3 and 3.5 on the right.

- The histogram of [Example 1.10](#) illustrates that the distribution of exam scores is skewed left with a center between 85 and 90 (counting 20 boxes from each end in the third graph) and a spread from between 60 and 65 on the left to between 95 and 100 on the right.
- The stemplot of [Example 1.11](#) illustrates a bimodal distribution with a center of 21 push-ups, a spread from 2 to 50 push-ups, and a gap between 21 and 29 push-ups.

➤ Example 1.13

Hodgkin's lymphoma is a cancer of the lymphatic system, the system that drains excess fluid from the blood and protects against infection. Consider the following histogram:



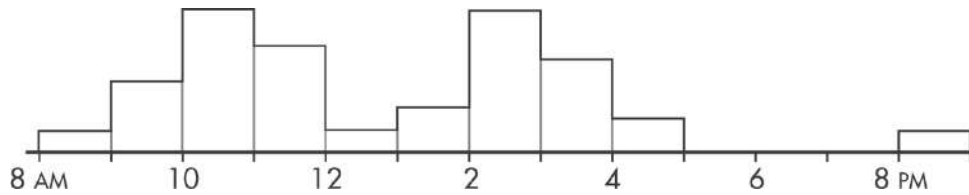
Describe the distribution in context.

✓ Solution

The distribution of ages of diagnosis of female cases of Hodgkin's lymphoma is bimodal, roughly symmetric, with a center around 50, a spread from 0 to around 93, and two distinct clusters centered around 25 and 75 years old.

➤ Example 1.14

The histogram below shows employee computer usage (number accessing the Internet) during given times at a company's main office.



NOTE

In plots such as this, there is no way to know the exact minimum or maximum values, only intervals in which these values fall.

- (a) Describe the distribution in context.
- (b) Suggest a possible explanation for the outlier.

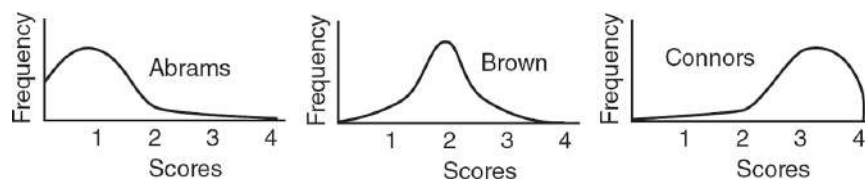
✓ Solutions

- (a) The distribution of employee computer usage (number accessing the Internet) during given times is bimodal, with a center around 1 P.M., a spread from around 8 A.M. to around 9 P.M., a gap between 5 P.M. and 8 P.M., and an outlier or outliers between 8 P.M. and 9 P.M. Computer usage at this company appears heaviest at midmorning and midafternoon with a dip in usage during the noon lunch hour.
- (b) The evening usage could indicate employees returning after dinner or possibly a hacker using company computer power for free!

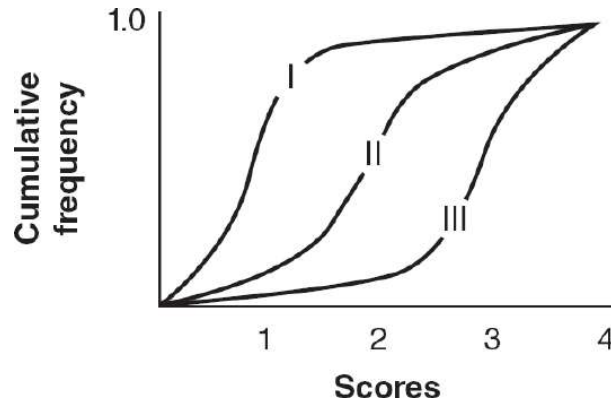
A distribution skewed to the left has a cumulative frequency plot that rises slowly at first and then steeply later. In contrast, a distribution skewed to the right has a cumulative frequency plot that rises steeply at first and then slowly later.

➤ Example 1.15

Consider the essay-grading policies of three teachers: Mr. Abrams gives very low scores; Mrs. Brown gives equal numbers of low and high scores; and Ms. Connors gives very high scores. Histograms of the grades (with 1 representing the lowest score and 4 representing the highest score) are as follows:



- (a) Compare the three distributions.
- (b) Which of the distributions corresponds to each of the following three cumulative frequency plots?



[Follow for extended description](#)

✓ Solutions

- (a) The distribution of Mr. Abrams's scores is skewed right, of Mrs. Brown's scores is roughly bell-shaped, and of Ms. Connors's scores is skewed left. All three distributions are unimodal and have roughly the same spread from around 0 to around 4. The center of Mr. Abrams's scores is the lowest, while the center of Ms. Connors's scores is the highest.
- (b) The Abrams distribution with many low scores corresponds to cumulative frequency graph I, which rises sharply over the low scores. The Brown distribution with many middle scores corresponds to cumulative frequency graph II, which rises slowly to start, then rises more sharply over the middle scores, and finally rises slowly over the higher scores. The Connors distribution with many high scores corresponds to cumulative frequency graph III, which rises sharply over the high scores.

Quiz 1

Multiple-Choice Questions

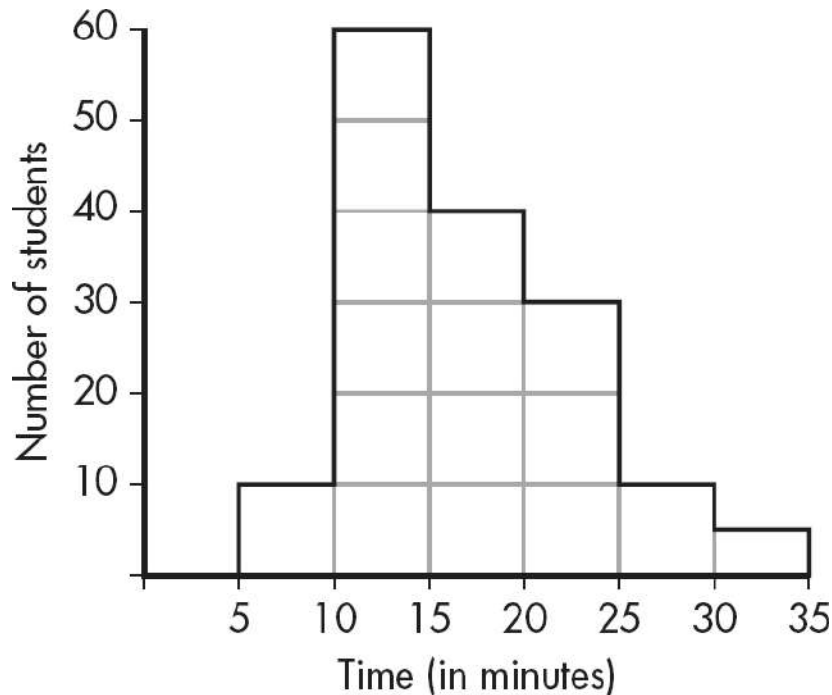
DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. Data were collected on 100 stamps from around the world. Two variables were primary color and width of stamp. Are these variables categorical or quantitative?
 - (A) Both are categorical.
 - (B) Both are quantitative.
 - (C) Primary color is categorical and width is quantitative.
 - (D) Primary color is quantitative and width is categorical.

2. An elevator manufacturing company rates each elevator type with two variables: maximum number of passengers and maximum weight. Are these variables discrete or continuous?
 - (A) Both are discrete.
 - (B) Number of passengers is discrete and weight is continuous.
 - (C) Number of passengers is continuous and weight is discrete.
 - (D) These are categorical, not quantitative, variables, so neither is discrete or continuous.

3. Suppose that a study was carried out in which each student in a random sample of students at a particular high school was asked whether he or she thought too much homework was assigned in most classes. Which of the following is a correct understanding of the type of investigative question this study might address?
 - (A) The data could be used to estimate a population mean since the data collected are numerical.
 - (B) The data could be used to estimate a population proportion since the data collected are numerical.
 - (C) The data could be used to estimate a population mean since the data collected are categorical.
 - (D) The data could be used to estimate a population proportion since the data collected are categorical.

4. The times, in minutes, taken by 155 high school students to complete a level of the video game *Angry Birds* are summarized in the histogram below.



Based on the histogram, which of the following must be true?

- (A) The minimum time taken by any of these students was 5 minutes.
- (B) The maximum time taken by any of these students was 35 minutes.
- (C) If the times are arranged in order, the middle time would be between 10 and 15 minutes.
- (D) If the times are arranged in order, the middle time would be between 15 and 20 minutes.

5. Which of the following is a true statement?

- (A) Stemplots are useful for both quantitative and categorical data sets.
- (B) Stemplots are equally useful for small and very large data sets.
- (C) Stemplots can show symmetry, gaps, clusters, and outliers.
- (D) Stemplots may or may not show individual values.

6. Which of the following is an *incorrect* statement?

- (A) In histograms, relative areas correspond to relative frequencies.
- (B) In histograms, frequencies can be determined from relative heights.

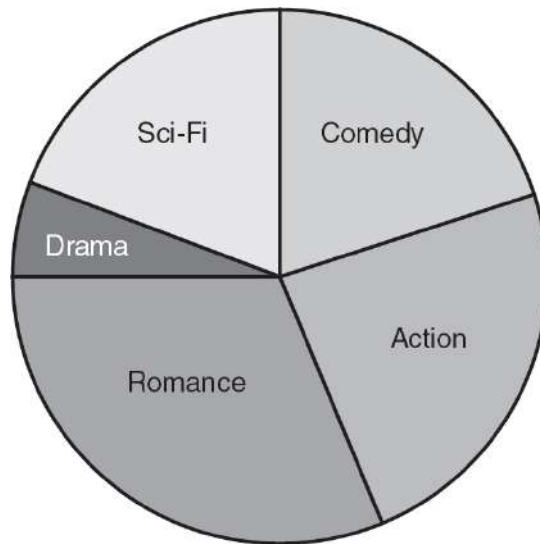
- (C) Symmetric histograms may have multiple peaks.
- (D) Two students working with the same set of data may come up with histograms that look different.

7. A sales rep asked 50 Generation X, 75 Millennial, and 100 Generation Z respondents their preference between Company A and Company B with regard to sound quality of their smart speakers. The results are shown in the following table.

	Company A	Company B
Generation X	19	31
Millennials	43	32
Generation Z	68	32

Which of the following statements is supported by the table?

- (A) The same percentage of Millennials and Generation Z preferred Company B.
 - (B) The percentage of Generation X who preferred Company B is greater than the percentage of Millennials who preferred Company B.
 - (C) Of those who preferred Company A, more than half were either Generation X or Millennials.
 - (D) Of those who preferred Company B, less than half were not Generation Z.
8. In a favorite movie survey of 20 AP Statistics students, 4 chose Comedy, 5 chose Action, 6 chose Romance, 1 chose Drama, and 4 chose Sci-Fi. These data are illustrated in the following pie chart.



Favorite Type of Movie

Among those not choosing Comedy or Romance, what proportion chose Sci-Fi or Action?

- (A) $\frac{4+5}{4+6}$
 (B) $\frac{4+5}{20-(4+6)}$
 (C) $\frac{4+5}{20-(4+5)}$
 (D) $\frac{4+5}{20}$

Free-Response Questions

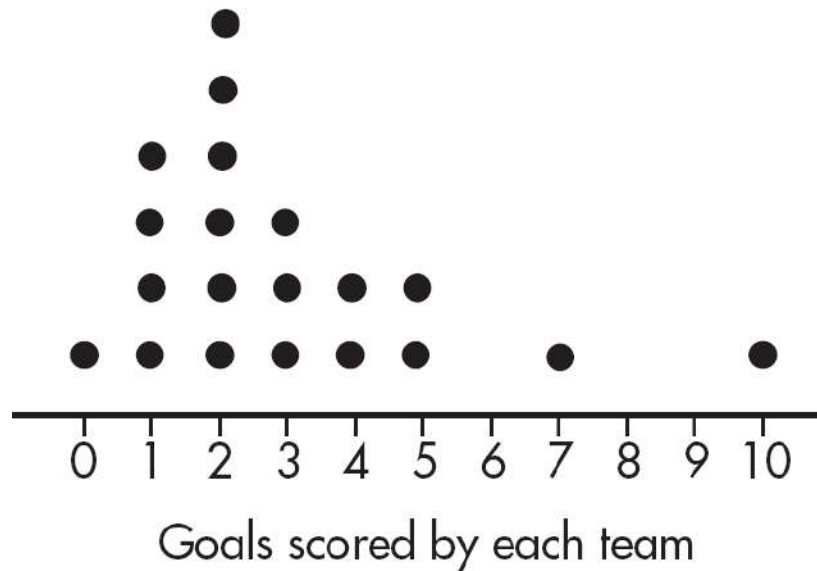
DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. The miles run by 25 randomly selected runners during a given week are given below.

Miles run (x)	$5 \leq x < 10$	$10 \leq x < 15$	$15 \leq x < 20$	$20 \leq x < 25$
Frequency	3	11	7	4

- (a) Which is more appropriate for displaying these data, a dotplot, a stemplot, or a histogram? Explain.
- (b) What percent of the runners ran under 15 miles?
- (c) If each of the four intervals is divided in half and the corresponding numbers given, what percent would be under 15 miles?

2. The dotplot below shows the numbers of goals scored by the 20 teams playing in a city's high school soccer games on a particular day.



- (a) Describe the distribution.
 - (b) One superstar scored six goals, but his team still lost. What are all possible final scores for that game? Explain.
 - (c) Is it possible that each of the teams that scored exactly two goals were the winners of that game? Explain.
3. The winning percentages for a major league baseball team over the past 22 years are shown in the following stemplot:

46		0 8
47		1 4 7 9
48		5 8 8 9
49		3 4 7
50		
51		
52		5 8
53		2 5 6
54		4 8 9
55		6
(55		6 means 55.6%)

[Follow for extended description](#)

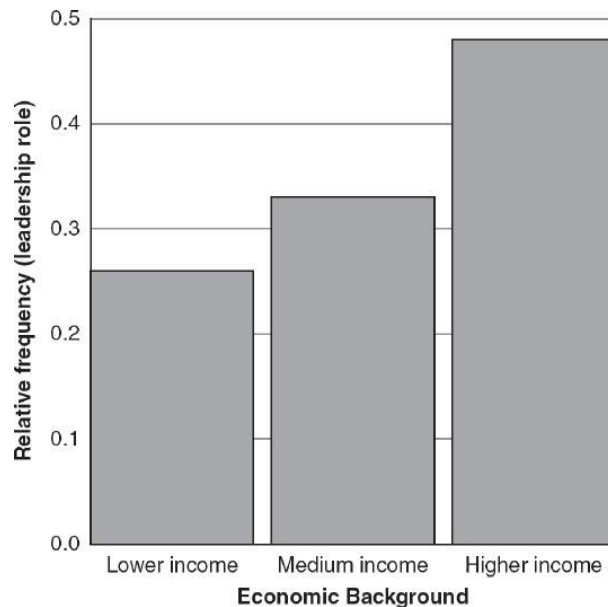
- Interpret the lowest value.
- Describe the distribution.
- Give a reason that one might argue that the team is more likely to lose a given game than win it.
- Give a reason that one might argue that the team is more likely to win a given game than lose it.

- Data collected from over 10,000 graduating college seniors asked about economic backgrounds and leadership position experiences. The data are summarized in the following table:

	Economic background		
	Lower income	Medium income	Higher income
Number of students	1,450	8,280	2,975
Number claiming leadership role	377	2,760	1,428

- What is the investigative question of interest?
- Are the data categorical or numerical? Explain.
- Explain why a relative frequency distribution is more appropriate than a frequency distribution in analyzing the data.

- (d) The following bar chart is constructed from the data. What does it indicate about the relationship between college students' economic backgrounds and leadership position experience?



[Follow for extended description](#)

The answers for this quiz can be found beginning on [page 468](#).

Quiz 2

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. The following relative frequency table shows choices of reading materials for a random sample of high school students.

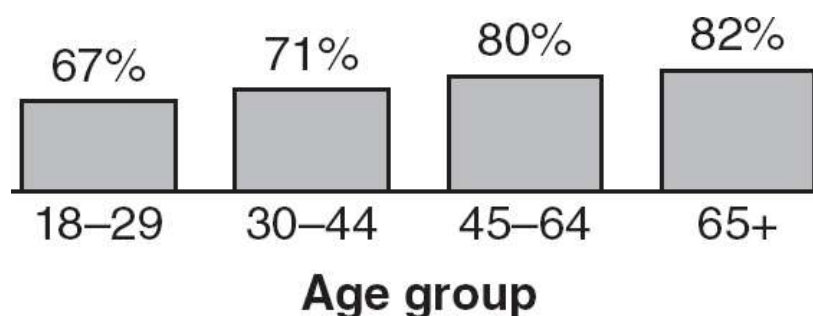
Book	Relative frequency
Classic	0.23

Contemporary fiction	0.51
Nonfiction	0.26

Which of the following statements is not supported by the relative frequency table?

- (A) Only 23 students chose “Classic.”
- (B) More than half the students chose “Contemporary fiction.”
- (C) More than one-quarter of the students chose “Nonfiction.”
- (D) The number of students choosing “Contemporary fiction” minus the number choosing “Classic” was greater than the number choosing “Nonfiction.”

2. The following bar chart shows the relative frequencies of adults from different age groups who agree that formerly incarcerated people who have served their prison sentence deserve to be considered for jobs.



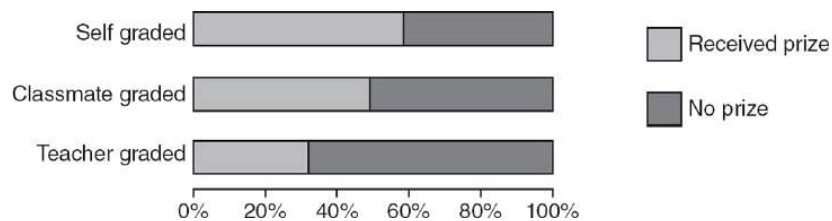
[Follow for extended description](#)

Which of the following statements is supported by the bar chart?

- (A) The greatest number of adults participating in this survey were in the 65+ age group.
- (B) The number of adults participating in this survey was 4% greater in the 30–44 age group than in the 18–29 age group.
- (C) If the 45–64 and 65+ age groups were combined, the relative frequency of those who agree that formerly incarcerated people who have served their prison sentence deserve to be considered for jobs would be 81%.

(D) Age group 18–29 had the least relative frequency of any age group with regard to agreeing that formerly incarcerated people who have served their prison sentence deserve to be considered for jobs.

3. A college psychology professor has three introductory classes. One day, she hands out a surprise quiz in all three classes and announces that anyone achieving a certain score will receive a prize, a large chocolate chip cookie. In one class, she grades the quizzes herself. In another class, she has the students grade their own quizzes. In the third class, she has students hand their quizzes to classmates to grade. The graph below summarizes the data.



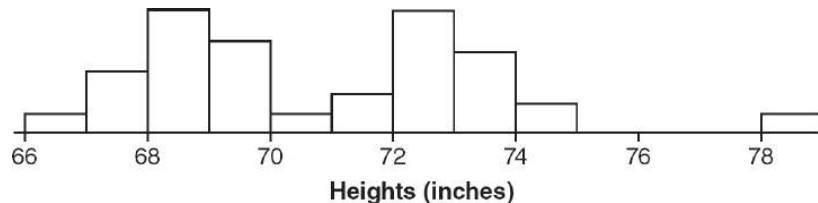
[Follow for extended description](#)

Which statement is supported by the above graph?

- (A) The number of students who self-graded and received a prize is greater than the number of students who were teacher graded who received a prize.
 - (B) When allowed to self-grade, students will cheat.
 - (C) Teacher-graded quizzes had the lowest relative frequency of receiving a prize compared with self-graded or classmate-graded quizzes.
 - (D) Classmate-graded quizzes seem to be the fairest way of grading.
4. A random sample of weights of student backpacks gives a skewed right graph. Which of the following statements is most reasonable based on this graph?
- (A) The mean weight is less than the median weight.
 - (B) The mean weight is the average of the first and third quartiles.
 - (C) There are more weights between the first quartile and median than there are between the median and the third quartile.

(D) The value of the maximum weight minus the median weight is greater than the median weight minus the minimum weight.

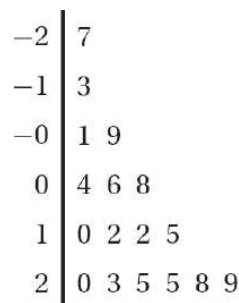
5. The heights of high school students trying out for the baseball and basketball teams is summarized in the following histogram.



[Follow for extended description](#)

Which of the following is the best description of the distribution?

- (A) There are two clusters, a gap, an outlier between 78 and 79, and a range between 7 and 9.
- (B) There are two clusters, a gap, an outlier between 78 and 79, and a range between 11 and 13.
- (C) The distribution is bimodal, is skewed right, and has a gap between 75 and 78.
- (D) The distribution is bimodal, is roughly symmetric, and has outliers below 66 and above 75.
6. The following stemplot shows the difference in heights (husband minus wife) of 17 randomly chosen couples.



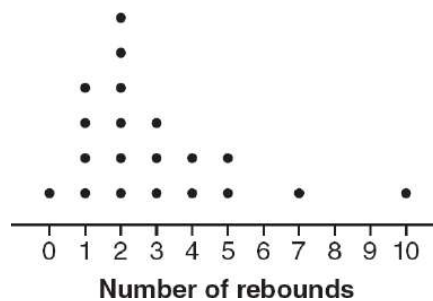
Key: 2 | 0 means 2.0 inches

[Follow for extended description](#)

Suppose a second stemplot is drawn showing the difference in heights (wife minus husband) using the same data. Which of the following is a true statement?

- (A) Both stemplots illustrate skewed left distributions.
- (B) The original stemplot illustrates a skewed left distribution, and the second stemplot illustrates a skewed right distribution.
- (C) The original stemplot illustrates a skewed right distribution, and the second stemplot illustrates a skewed left distribution.
- (D) Both stemplots illustrate skewed right distributions.

7. The following dotplot shows the number of rebounds grabbed by the 20 players participating in a randomly picked high school basketball game.

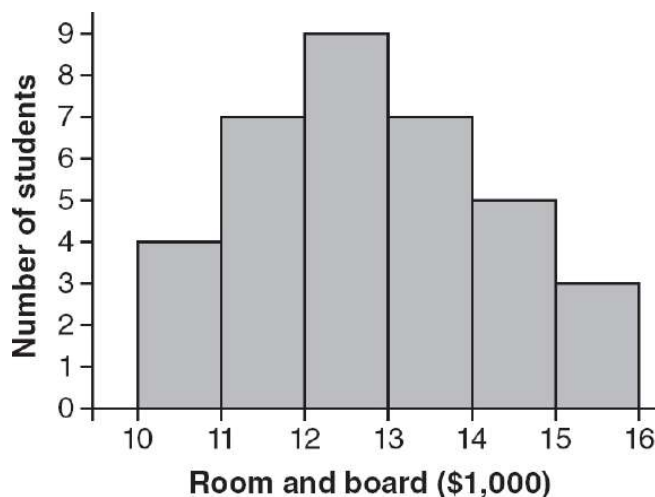


[Follow for extended description](#)

Which of the following best describes the type of variable represented in the dotplot?

- (A) Categorical and discrete
- (B) Categorical and continuous
- (C) Quantitative and discrete
- (D) Quantitative and continuous

8. The following histogram summarizes the amount spent on room and board per year (in \$1,000) by a random sample of 35 college students.



[Follow for extended description](#)

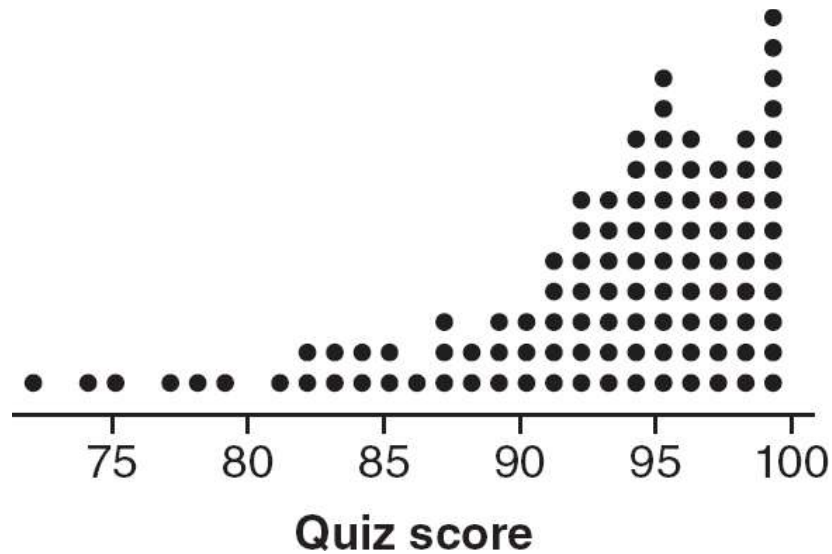
If the interval size is decreased from \$1,000 to \$500, which of the following will remain the same on the new histogram?

- (A) The shape of the distribution
- (B) The sum of the frequencies
- (C) The heights of the bars
- (D) The widths of the bars

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Ms. D has 3 AP Statistics sections with a total of 105 students. The scores on one quiz are summarized in the following dotplot.



[Follow for extended description](#)

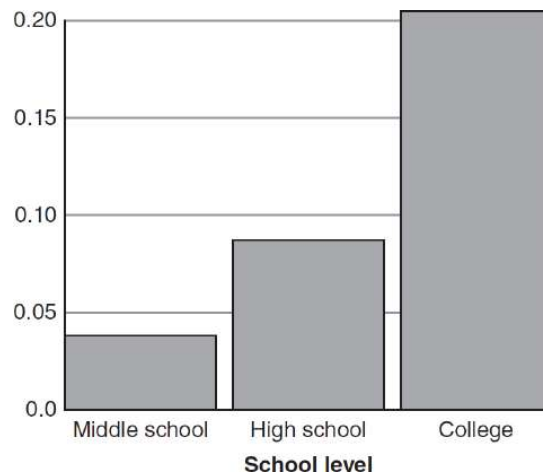
- Describe the distribution.
- Would you expect the median to be less than or greater than the mean?

- Many students perceive e-cigarettes as less harmful than traditional cigarettes, but research shows that both cigarettes and e-cigarettes can lead to nicotine addiction, respiratory issues, and other health problems. Data collected from over 5,000 students asking about e-cigarette use are summarized in the following table.

	Current school level		
	Middle school	High school	College
Number of students	1,400	2,900	1,800
Number admitting to using e-cigarettes	51	241	371

- What is the investigative question of interest?
- Are the data categorical or numerical? Explain.
- Explain why a relative frequency distribution is more appropriate than a frequency distribution in analyzing the data.
- The following bar chart showing relative frequencies of e-cigarette use by school level is constructed from the data. What does the bar

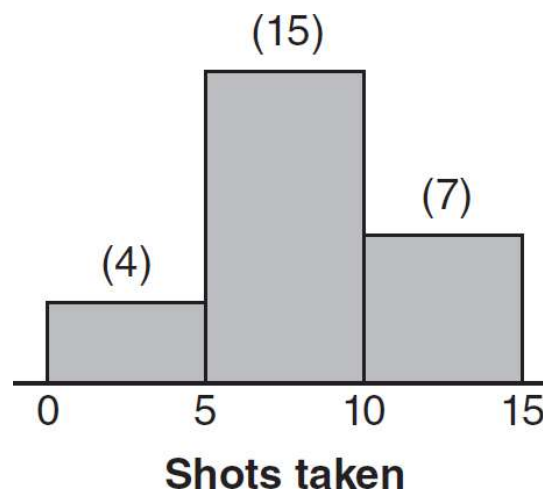
chart indicate about the relationship between school level and e-cigarette use?



[Follow for extended description](#)

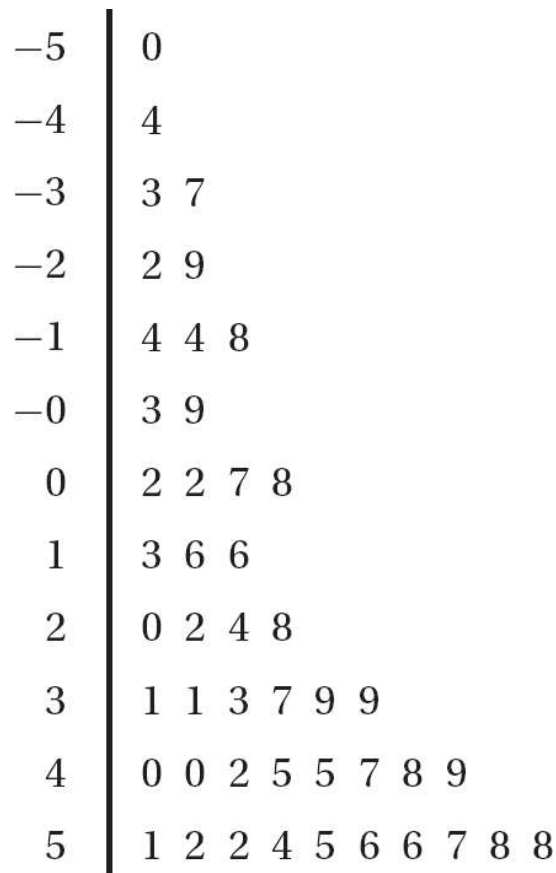
3. Twenty-six professional basketball players took shots from midcourt until they made a basket. The number of shots taken by the players is summarized in the following table and histogram.

Shots taken	$1 \leq x < 5$	$5 \leq x < 10$	$10 \leq x < 15$
Number of players	4	15	7



[Follow for extended description](#)

- (a) Given the table, why was a histogram more appropriate than a dotplot or a stemplot for displaying these data?
- (b) What percentage of players took over 4 attempts before making a half-court basket?
- (c) If each of the intervals is divided in half and the corresponding numbers given, what percent would be over 4?
4. A medical researcher performs a year-long test to compare two new weight loss medications. Each of 46 volunteers takes Medication *A* for 6 months and Medication *B* for 6 months. Twenty-three of the participants are randomly chosen to take *A* first while the remaining 23 participants take *B* first. The difference in weight loss (under Medication *A* minus under Medication *B*) for the participants is summarized in the following stemplot.



Key: 5|1 = 5.1 pounds

[Follow for extended description](#)

- (a) Explain what a negative difference indicates in the context of this study.
- (b) Describe the distribution given by the stemplot.
- (c) What conclusion is reasonable from this set of data? Explain.

The answers for this quiz can be found beginning on [page 469](#).

One Quantitative Variable (Summary Statistics and Graphs)

There are two primary ways of denoting the “center” of a distribution:

1. The *median*, which is the middle number of a set of numbers arranged in numerical order.
2. The *mean*, which is found by summing items in a set and dividing by the number of items.

Example 1.16

Consider the following set of home run distances (in feet) to center field in 13 ballparks: {387, 400, 400, 410, 410, 410, 414, 415, 420, 420, 421, 457, 461}. What are the median and mean home run distances to center field for this set of 13 ballparks?

Solution

The median is 414 feet (there are six values below 414 and six values above), while the mean is

$$\frac{387 + 400 + 400 + 410 + 410 + \cdots + 457 + 461}{13} = 417.3 \text{ feet}$$

REMEMBER

Don't forget to put the data in order before finding the median.

The word *median* is derived from the Latin *medius*, which means “middle.” The values under consideration are arranged in ascending or descending order. If there is an odd number of values, the median is the middle one. If there is an even number of values, the median is found by adding the two middle values and

dividing by 2. Thus the median of a set has the same number of elements above it as below it.

The median is not affected by exactly how large the larger values are or by exactly how small the smaller values are. Thus, it is a particularly useful measurement when the extreme values, *outliers*, are in some way suspicious or when you want to diminish their effect. For example, if 10 mice try to solve a maze and nine succeed in less than 15 minutes while one is still trying after 24 hours, the most representative value is the median (not the mean, which is over 2 hours). Similarly, if the salaries of four executives are each between \$240,000 and \$245,000 while a fifth is paid less than \$20,000, again the most representative value is the median (the mean is under \$200,000). The median is “resistant” to extreme values.

NOTE

A parameter, such as μ , is a numerical summary of a population, while a statistic, such as $\bar{x}$, is a numerical summary of sample data.

Although the median is often useful in descriptive statistics, the *mean*, or more accurately, the *arithmetic mean*, is most important for statistical inference and analysis. Also, for the layperson, the “average” is usually understood to be the mean.

The mean of a *whole population* (the complete set of items of interest) is denoted by the Greek letter μ (mu), while the mean of a *sample* (a part of a population) is denoted by $\bar{x}$. For example, the mean value of the set of all houses in the United States might be $\mu = \$156,400$, while the mean value of 100 randomly chosen houses might be $\bar{x} = \$152,100$ or perhaps $\bar{x} = \$163,800$ or even $\bar{x} = \$224,000$. (Each sample will probably contain different dollar amounts than other samples and therefore will likely have a different mean than other samples.) In statistics we learn how to estimate a population mean from a sample mean.

Mathematically, the mean $= \frac{\sum x}{n}$, where $\sum x$ represents the sum of all the elements of the set under consideration and n is the actual number of elements.



Example 1.17

Suppose the salaries of the six part-time employees at a computer security company are \$3,000; \$7,000; \$15,000; \$22,000; \$23,000; and \$38,000.

(a) What are the mean and median salaries?

- (b) What will the new mean and median salaries be if everyone receives a \$3,000 increase?
- (c) What will the new mean and median salaries be if only the person with the highest salary receives a \$3,000 increase?
- (d) What will be the new mean and median if everyone receives a 10% raise?

✓ Solutions

(a) The mean is $\frac{3,000 + 7,000 + 15,000 + 22,000 + 23,000 + 38,000}{6} = \$18,000.$

The median is $\frac{15,000 + 22,000}{2} = \$18,500.$

(b) The mean is $\frac{6,000 + 10,000 + 18,000 + 25,000 + 26,000 + 41,000}{6} = \$21,000.$

The median is $\frac{18,000 + 25,000}{2} = \$21,500.$

Note that adding the same constant to each value increases both the mean and the median by that same constant ($\$18,000 + \$3,000 = \$21,000$ and $\$18,500 + \$3,000 = \$21,500$).

(c) The mean is $\frac{3,000 + 7,000 + 15,000 + 22,000 + 23,000 + 41,000}{6} = \$18,500.$

The median is $\frac{15,000 + 22,000}{2} = \$18,500.$

Note that the mean, unlike the median, is sensitive to a change in any value.

(d) The mean is $\frac{3,300 + 7,700 + 16,500 + 24,200 + 25,300 + 41,800}{6} = \$19,800.$

The median is $\frac{16,500 + 24,200}{2} = \$20,350.$

Note that multiplying each value by the same constant multiplies both the mean and the median by that same constant (110% of \$18,000 is \$19,800 and 110% of \$18,500 is \$20,350).

In describing a set of numbers, not only is it useful to designate a central value (mean or median), but also it is important to be able to indicate the *variability* of the measurements. An explosion engineer in mining operations aims for small variability—it would not be good for the 30-minute fuses actually to have a range of 10–50 minutes before detonation. On the other hand, a teacher interested in distinguishing better students from those who have difficulty with the material aims to design exams with large variability in results—it would not

be helpful if all the students scored exactly the same. The players on two basketball teams may have the same average height, but this observation doesn't tell the whole story. If the dispersions are quite different, one team may have a 7-foot player, whereas the other has no one over 6 feet tall. Two Mediterranean holiday cruises may advertise the same average age for their passengers. One, however, may have passengers only between 20 and 25 years old, while the other has only middle-aged parents in their forties together with their children under age 10.

TIP

Variability is the single most fundamental concept in statistics and is the key to understanding statistics.

To understand variability, we either calculate the mean and standard deviation (defined below) or calculate the following *five-number summary* of a data set:

- The minimum.
- The 25th percentile, Q_1 , is the median of the lower half of the data set. In other words, about 25% of the numbers in the data set lie below Q_1 and about 75% lie above Q_1 .
- The median.
- The 75th percentile, Q_3 , is the median of the upper half of the data set. In other words, about 75% of the numbers in the data set lie below Q_3 and about 25% lie above Q_3 .
- The maximum.

There are four primary ways of describing variability (or spread):

1. *Range*: the difference between the largest and smallest values (maximum minus minimum).
 - Although the range gives a measure of spread, it is entirely dependent on the two extreme values and is insensitive to the values in the middle.
 - Some quality control techniques involve taking periodic small samples and basing further action on the range found in such samples.
2. *Interquartile range (IQR)*: the difference between the largest and smallest values after removing the lower and upper quartiles.

- IQR = range of middle 50% = $Q_3 - Q_1$ = 75th percentile minus 25th percentile.
- This removes the influence of extreme values.
- 3. *Variance*: the average of the squared differences from the mean.
 - The variance of a complete population is $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.
 - The variance of a sample is $s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$.
- 4. *Standard deviation (SD)*: the square root of the variance.
 - The standard deviation of a complete population is $\sigma = \sqrt{\frac{\sum(x - \mu)^2}{n}} = \sqrt{\frac{1}{n} \sum(x - \mu)^2}$.
 - The standard deviation of a sample is $s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{1}{n - 1} \sum(x - \bar{x})^2}$.
 - The standard deviation gives a typical distance that each value is away from the mean. In other words, it is a representative value for the variability of a population or sample.
 - Although variance is measured in square units, the standard deviation is measured in the same units as are the data.

There are two numerical rules for designating outliers:

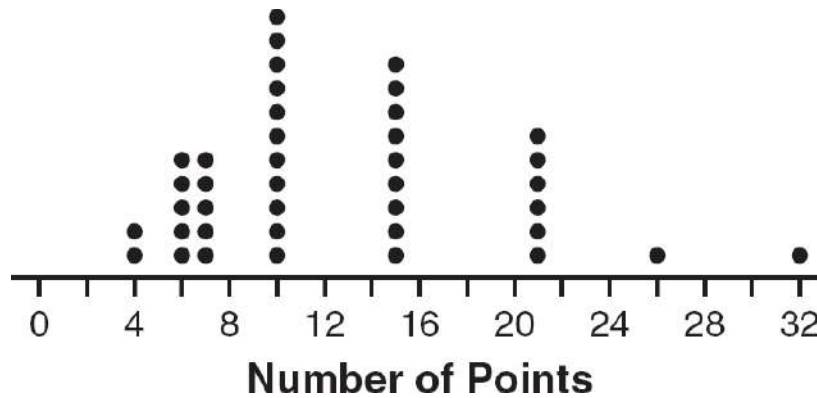
1. The more common rule is that outliers are any value less than $Q_1 - 1.5(\text{IQR})$ or greater than $Q_3 + 1.5(\text{IQR})$.
2. An acceptable alternative rule is that outliers are all values located more than two standard deviations above, or below, the mean.

The mean, range, variance, and standard deviation are called *nonresistant* or *nonrobust* measures because outliers can greatly affect their values. The median and IQR are called *resistant* or *robust* measures because outliers do not greatly (if at all) affect their values.

➤ Example 1.18

In a random sample of 40 NBA players, the points scored by each player on a particular game day are summarized in the following table and are shown in the dotplot.

Number of points	4	6	7	10	15	21	26	32
Number of players	2	5	5	11	9	6	1	1



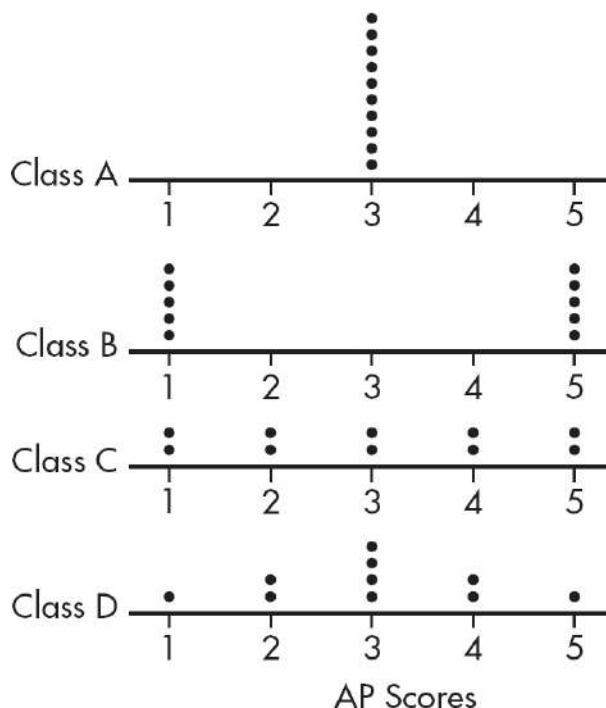
- What is the five-number summary?
- What are the range and interquartile range? Show your work.
- Using the $1.5(\text{IQR})$ rule, what values are outliers, if any? Show your work.
- The mean number of points scored is 12.55 with a standard deviation of 6.46. Using the two standard deviations rule, what values are outliers, if any? Show your work.

✓ Solutions

- The minimum number of points scored is 4. With $n = 40$, count 10 values to obtain $Q_1 = 7$. Counting 20, 21 values shows that the median is 10. Counting 10 values from the high side gives $Q_3 = 15$. The maximum is 32. The five-number summary is $\{4, 7, 10, 15, 32\}$.
- The range is $\text{Max} - \text{Min} = 32 - 4 = 28$.
The interquartile range is $\text{IQR} = Q_3 - Q_1 = 15 - 7 = 8$.
- $Q_1 - 1.5(\text{IQR}) = 7 - 1.5(8) = -5$ and no values are below this.
 $Q_3 + 1.5(\text{IQR}) = 15 + 1.5(8) = 27$ and there is one value, 32, above this.
Using the $1.5(\text{IQR})$ rule, there is one outlier, 32.
- $\bar{x} - 2s = 12.55 - 2(6.46) = -0.37$ and no values are below this.
 $\bar{x} + 2s = 12.55 + 2(6.46) = 25.47$ and there are two values, 26 and 32, above this.
Using the two standard deviations rule, there are two outliers, 26 and 32.

➤ Example 1.19

Consider the following dotplots of student scores on an AP exam of four AP Calculus classes, each with 10 students. Without making any calculations, arrange the four classes in order from smallest SD to largest SD.



✓ Solution

Although all four classes have a mean score of 3, variability is the issue, as it is different for each class. The scores in Class A have no variability at all, so the SD is smallest for Class A. The scores in Class B are all as far as possible away from the mean, so the SD is greatest for Class B. Note that Class D has more scores near the middle, while Class C has more scores further from the middle. So the SD for Class D is less than the SD for Class C. The correct ordering of the classes from smallest SD to largest SD is: A-D-C-B.

We have seen ways of choosing a value to represent the center of a distribution. We also need to be able to talk about the *position* of other values. There are several important, recognized procedures for designating position:

1. *Simple ranking*, which involves arranging the elements in some order and noting where in that order a particular value falls.
 - Simple ranking is easily calculated and easily understood. For example, we know what it means for someone to graduate second in their class of

435 or for someone to have the seventh-best batting average on a team of 30 players.

- Simple ranking is useful even when there are no numerical values. For example, ice cream brands can be ranked according to taste without any numerical value of taste.
2. *Percentile ranking*, which indicates what percentage of all values fall at or below the value under consideration.
- Percentile ranking is helpful when comparing positions with different bases. For example, we can compare a rank of 176 out of 704 with a rank of 187 out of 935 by noting that the first has a rank of 75% and the second a rank of 80%.
 - Percentile ranking is also useful when the exact population size is unknown or is irrelevant. For example, it is more meaningful to say that Jennifer scored in the 90th percentile on a national exam rather than trying to determine her exact ranking among some large number of test takers.
3. The *z-score*, which states very specifically the number of standard deviations a particular value is above or below the mean.
- The *z-score* of a raw score x is given by $z = \frac{x - \mu}{\sigma}$, where μ is the mean and σ is the standard deviation.
 - A positive *z-score* indicates that a value is greater than the mean, while a negative *z-score* indicates that a value is less than the mean.
 - When given a *z-score*, we can find the corresponding raw score by calculating $x = \mu + z\sigma$.
 - No matter what the distribution of raw scores, the set of *z-scores* always has mean 0 and standard deviation 1.
 - *z-scores* can be useful as a common “measuring stick” with which to compare values from different distributions. For example, a high school senior might have a lower GPA than a high school sophomore, but the senior’s GPA might actually be far above the average senior GPA while the sophomore’s GPA might be only slightly above the average sophomore GPA. We can make a comparison using *z-scores*.
 - When we come to *normal* distributions, software will enable us to quickly go between *z-scores* and probabilities.
4. Values that divide a data set into equal parts.
- Quartiles split data into four equal parts. The first quartile, Q_1 , lies one-quarter of the way up a list, and the third quartile, Q_3 , lies three-quarters

of the way up a list. Their percentile ranks are 25% and 75%, respectively.

- Deciles split data into 10 equal parts. The first decile, D_1 , marks the point below which 10% of the data fall; the second decile, D_2 , marks the point below which 20% of the data fall; and so on. This continues up to the ninth decile, D_9 , which marks the point below which 90% of the data fall.

➤ Example 1.20

It is recommended that the “good cholesterol,” high-density lipoprotein (HDL), be present in the blood at levels of at least 40 mg/dL. Suppose all 50 members of a high school football team are tested, and the resulting HDL levels are as follows: {53, 26, 45, 33, 64, 29, 73, 29, 21, 58, 70, 41, 48, 55, 55, 39, 57, 48, 9, 59, 56, 39, 68, 50, 65, 30, 38, 54, 49, 35, 56, 70, 43, 86, 52, 40, 28, 40, 67, 50, 47, 54, 59, 29, 29, 42, 45, 37, 51, 40}. The mean of this list is 47.22 mg/dL with a standard deviation of 15.05 mg/dL. Given that there are 31 higher HDL levels than 41, what is the position of the HDL score of 41 as measured in simple ranking, percentile ranking, and z-score?

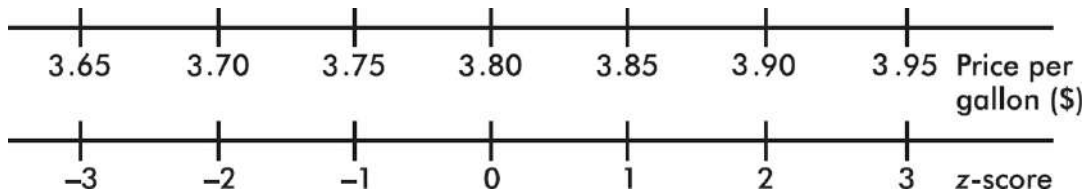
✓ Solution

Since there are 31 higher HDL levels on the list, the 41 has a simple ranking of 32 out of 50. Of all the HDL levels, 19 are at or lower than 41, so the percentile ranking is $\frac{19}{50} = 38\%$. The HDL score of 41 has a z-score of $\frac{41 - 47.22}{15.05} = -0.413$.

➤ Example 1.21

Because of hurricanes, refinery outages, and Middle East tensions, the mean price of gasoline in one northeast city hit \$3.80 per gallon with a standard deviation of \$0.05. Then \$3.90 has a z-score of $\frac{3.90 - 3.80}{0.05} = +2$, while \$3.65 has a z-score of $\frac{3.65 - 3.80}{0.05} = -3$. Alternatively, a z-score of +2.2 corresponds to a raw score of $3.80 + 2.2(0.05) = 3.80 + 0.11 = 3.91$, while a z-score of -1.6 corresponds to $3.80 - 1.6(0.05) = 3.72$.

It is often useful to portray integer z-scores and the corresponding raw scores as follows:



Changing units, for example, from dollars to rubles or from miles to kilometers, is common in a world that seems to become smaller all the time. It is instructive to note how measures of center, variability, and, by extension, position are affected by such changes.

- When the same constant is added to every value:
 - The mean and median increase by that same constant.
 - The range and standard deviation remain unchanged (because the distances between the increased values stay the same as before).
- When every value is multiplied by the same constant:
 - The mean, median, range, and standard deviation are all multiplied by that same constant.
- Neither adding the same constant to every value nor multiplying every value by the same constant will affect a simple ranking, a percentile ranking, or even a z -score.

► Example 1.22

A set of experimental measurements of the freezing point of an unknown liquid yields a mean of 25.32 degrees Celsius with a standard deviation of 1.47 degrees Celsius.

- (a) If all the measurements are converted to the Kelvin scale, what are the new mean and standard deviation? (Kelvins are equivalent to degrees Celsius plus 273.16.)
- (b) What is the z -score of 27 degrees Celsius?

✓ Solutions

- (a) The new mean is $25.32 + 273.16 = 298.48$ Kelvins. The standard deviation, however, remains numerically the same, 1.47 Kelvins. Graphically, you should picture the whole distribution moving over by the constant 273.16; the mean moves, but the standard deviation (which measures variability) doesn't change.

- (b) For 27 degrees Celsius, $z = (27 - 25.32)/1.47 = 1.14$. It is instructive to note that 27 degrees Celsius corresponds to $27 + 273.16 = 300.16$ Kelvins with again a z -score of $z = (300.16 - 298.48)/1.47 = 1.14$.

➤ Example 1.23

Measurements of the sizes of farms in an upstate New York county yield a mean of 59.2 hectares with a standard deviation of 11.2 hectares.

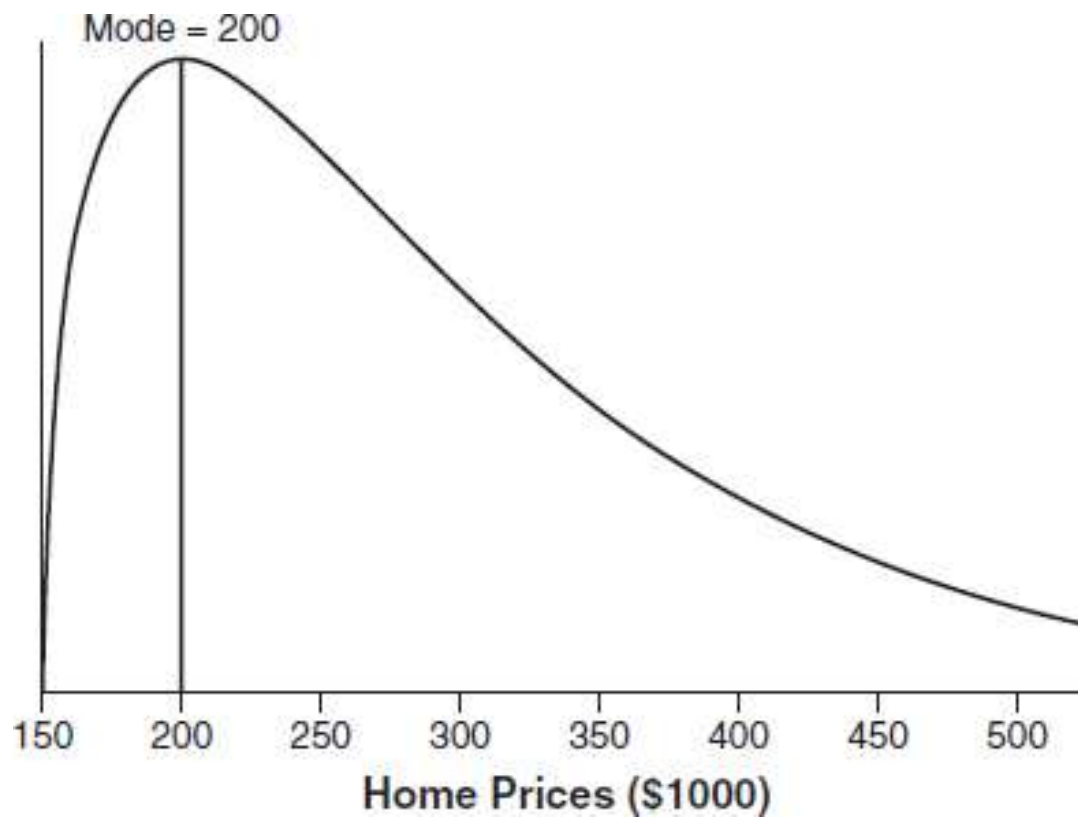
- (a) If all the measurements are converted from hectares (metric system) to acres, what are the new mean and standard deviation? (One hectare is equivalent to 2.471 acres.)
- (b) What is the z -score of 55 hectares?

✓ Solutions

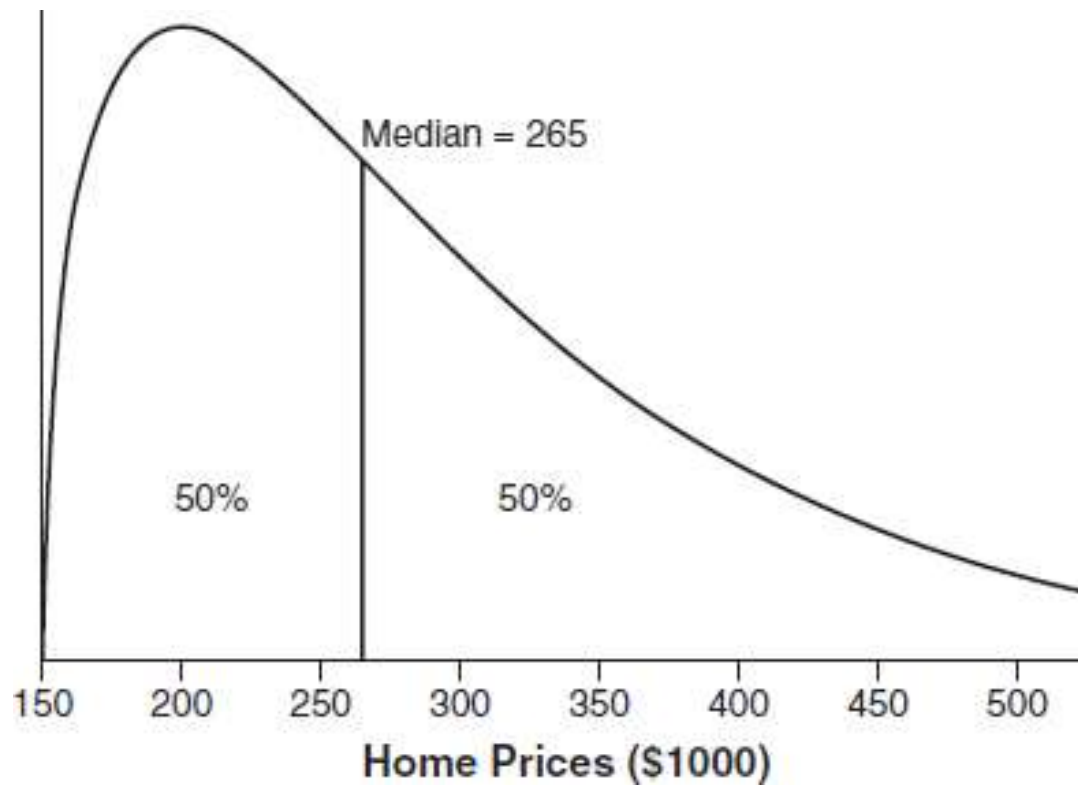
- (a) The new mean is $2.471 \times 59.2 = 146.3$ acres with a standard deviation of $2.471 \times 11.2 = 27.7$ acres. Graphically, multiplying each value by the constant 2.471 both moves and spreads out the distribution.
- (b) $z = (55 - 59.2)/11.2 = -0.375$. It is instructive to note that 55 hectares equals $2.471(55) = 135.905$ acres, and again $z = (135.905 - 146.3)/27.7 = -0.375$.

➤ Example 1.24

Below is a histogram of the home prices (in \$1,000) of a Midwestern urban city.



The distribution of home prices is unimodal, is skewed right, has a mode of \$200,000, and varies from \$150,000 to about \$525,000.



The median, \$265,000, divides the distribution in half, so it can be illustrated by a line that divides the area of the histogram in half.



The mean, \$300,000, is affected by the spacing of all the values. If the histogram is considered a solid region, the mean corresponds to a line passing through the center of gravity, or balance point.

➤ Example 1.25

Suppose that the faculty salaries at a college have a median of \$82,500 and a mean of \$88,700. What does this indicate about the shape of the distribution of the salaries?

✓ Solution

The median is less than the mean, and so the salaries are probably skewed to the right. There are a few highly paid professors, with the bulk of the faculty at the lower end of the pay scale.

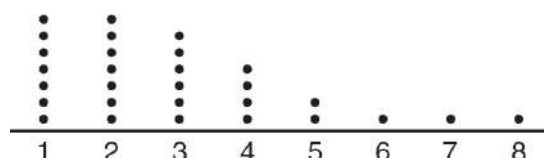
There are some general principles:

- If a distribution is approximately symmetric, the mean and median will be close to each other.
- If a distribution is skewed right, the mean is usually greater than the median.
- If a distribution is skewed left, the mean is usually less than the median.

It should be noted that the above principle is a useful, but not hard-and-fast, rule.

► Example 1.26

The set given by the dotplot below is skewed to the right; however, its median (3) is greater than its mean (2.97).



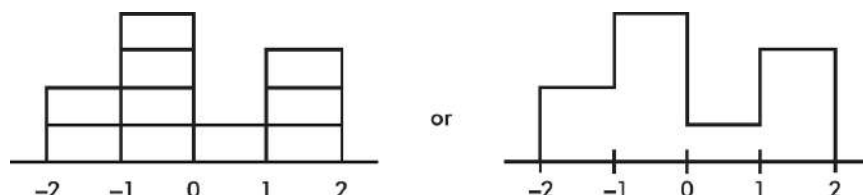
We have seen that in histograms, relative frequencies are represented by relative areas, and so labeling the vertical axis is not crucial. If we know the standard deviation, the horizontal axis can be labeled in terms of z -scores. In fact, if we are given the percentile rankings of various z -scores, we can construct a histogram.

► Example 1.27

Suppose we are asked to construct a histogram from these data:

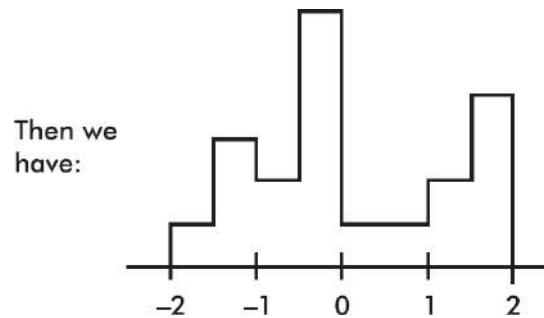
z -score	-2	-1	0	1	2
Percentile ranking	0	20	60	70	100

We note that the entire area is less than z -score +2 and greater than z -score -2. Also, 20% of the area is between z -scores -2 and -1, 40% is between -1 and 0, 10% is between 0 and 1, and 30% is between 1 and 2. Thus the histogram is as follows.

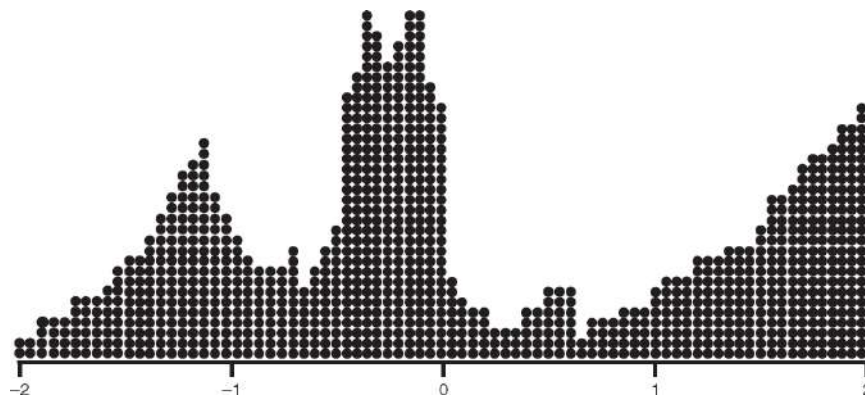


Now suppose we are given four in-between z -scores as well:

z-score	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0
Percentile ranking	0	5	20	30	60	65	70	80	100



With 1,000 z-scores perhaps the histogram would look like:



The height at any point is meaningless; what is important is relative areas.

- In the final diagram above, what percentage of the area is between z-scores of +1 and +2?
- What percent is to the left of 0?

✓ Solutions

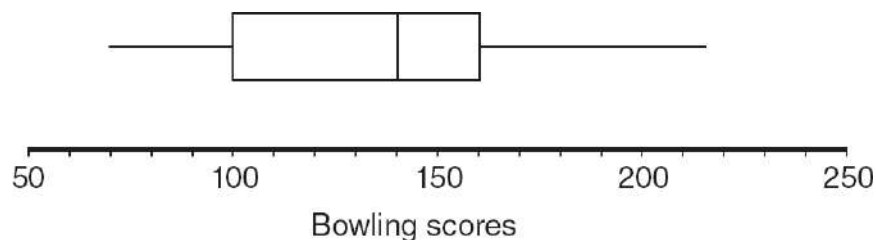
- Still 30%.
- Still 60%.

A *boxplot* (also called a *box and whisker display*) is a visual representation displaying the distribution of data that shows the smallest value, the largest value, the middle (median), the middle of the bottom half of the set (Q_1), and the

middle of the top half of the set (Q_3). The five numbers that are identified in a boxplot are called the five-number summary. In addition, values more than or less than $1.5 \times \text{IQR}$ (1.5 times the interquartile range) outside the box are plotted separately as outliers.

► Example 1.28

After an AP Statistics teacher hears that every one of her 27 students received a 3 or higher on the AP exam, she treats the class to a game of bowling. The individual student bowling scores are 210, 130, 150, 140, 150, 210, 150, 125, 85, 200, 70, 150, 75, 90, 150, 115, 120, 125, 160, 140, 100, 95, 100, 215, 130, 160, and 200. The students note that the greatest score is 215, the smallest is 70, the middle is 140, the middle of the top half is 160, and the middle of the bottom half is 100. A boxplot of these five numbers is



Note that the display consists of a box together with two “whiskers”—hence the alternative name. The box shows the spread of the two middle quarters; the whiskers (together with any outliers) show the spread of the two outer quarters. This relatively simple display conveys information not immediately available from histograms or stem and leaf displays.

Putting the above data into a list, for example, L1, on the TI-84, not only gives the five-number summary

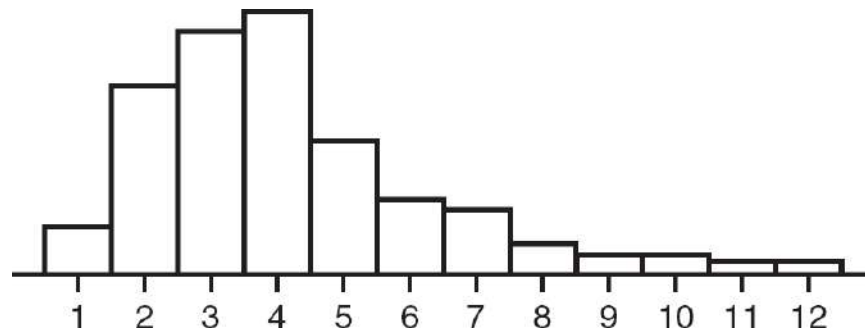
```
1-Var Stats
minX=70
Q1=100
Med=140
Q3=160
MaxX=215
```

but also gives the boxplot itself using STAT PLOT, choosing the boxplot from among the six type choices, and then using ZoomStat or in WINDOW, letting Xmin=0 and Xmax=225.

Example 1.29

Inputting the lengths of words in a selection of Shakespeare's plays results in the following summary statistics and histogram:

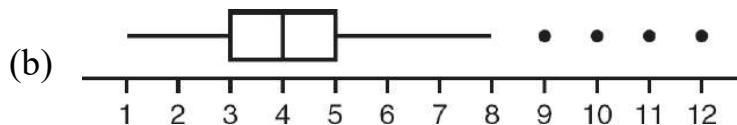
n	mean	SD	min	Q_1	med	Q_3	max
498	4.114	2.085	1	3	4	5	12



- (a) Find all outliers, if any, using the $1.5(IQR)$ rule.
- (b) Draw a boxplot.

Solutions

- (a) Outliers consist of any word lengths less than $Q_1 - 1.5(IQR) = 3 - 1.5(5 - 3) = 0$ or greater than $Q_3 + 1.5(IQR) = 5 + 1.5(5 - 3) = 8$. Word lengths of 9, 10, 11, and 12 are outliers.

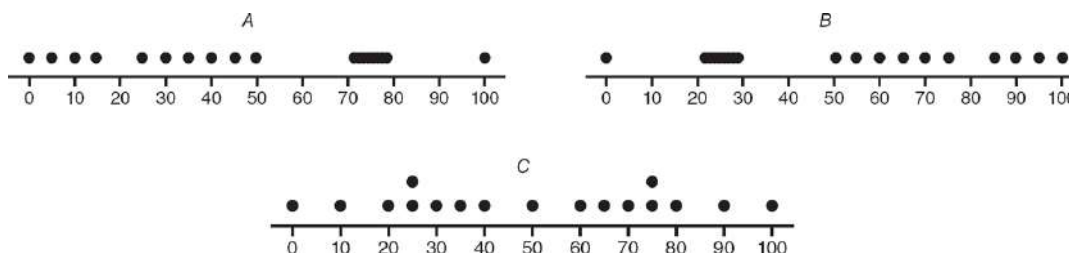


[Follow for extended description](#)

Example 1.30

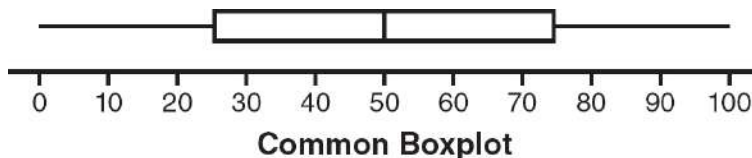
Let $A = \{0, 5, 10, 15, 25, 30, 35, 40, 45, 50, 71, 72, 73, 74, 75, 76, 77, 78, 100\}$, $B = \{0, 22, 23, 24, 25, 26, 27, 28, 29, 50, 55, 60, 65, 70, 75, 85, 90, 95, 100\}$, and $C = \{0, 10, 20, 25, 25, 30, 35, 40, 50, 60, 65, 70, 75, 75, 80, 90, 100\}$. Simple inspection indicates very different distributions; however, the TI-84 gives

identical boxplots with $\text{Min} = 0$, $Q_1 = 25$, $\text{Med} = 50$, $Q_3 = 75$, and $\text{Max} = 100$ for each. The dotplots are shown below.



Draw the common boxplot.

✓ Solution



Boxplots enable you to do a quick graphical examination of a data set:

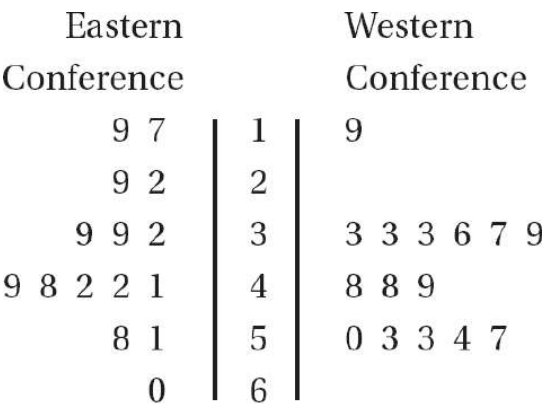
- A boxplot gives one measure of center (the median), gives two measures of variability (the range and the IQR), and shows outliers.
- Be careful! When outliers are present, the whiskers are drawn to the smallest and largest values that are not outliers; they are *not* drawn to $Q_1 - 1.5(\text{IQR})$ and $Q_3 + 1.5(\text{IQR})$ unless these are actual values.
- Concepts like symmetry, gaps, and overall shape are generally very difficult to conclude from just a boxplot. Although boxplots cannot show overall shape, they may suggest skewness or symmetry.
- Different sets can have the same five-number summary, and thus the same boxplots, but can have dramatically different distributions.
- Be careful of language! The IQR is a length, the length of the box; IQR is not the box itself. So, although the median is in the box, or is between Q_1 and Q_3 , it is wrong to say that the median is in the IQR.

One Quantitative Variable (Distribution Comparisons)

Many studies involve comparing groups. Following are three examples showing comparisons involving back-to-back stemplots, side-by-side histograms, and parallel boxplots.

➤ Example 1.31

The numbers of wins for the 30 NBA teams at the end of a recent season are shown in the following *back-to-back stemplot*.



7 | 1 | 9 represents Eastern and Western Conference teams with 17 and 19 wins, respectively.

[Follow for extended description](#)

NOTE

The leaves radiate out from a common stem.

NOTE

Although both sets in this example are of the same size, in general you do not have to have the same number of data points on each side of the plot.

Compare the two distributions.

✓ Solution

Important

When asked for a comparison, don't forget to address Shape, Unusual features (Outliers, gaps, clusters), Center, and Spread (SOCS or CUSS) and to refer to context. You must use *comparative* words—that is, you must state which center and which spread is larger (or if they are approximately the same). Simply describing each distribution separately is not enough and will be penalized.

Shape: The distribution of wins in the Eastern Conference (EC) is roughly bell-shaped, while the distribution of wins in the Western Conference (WC) is roughly uniform with a low outlier.

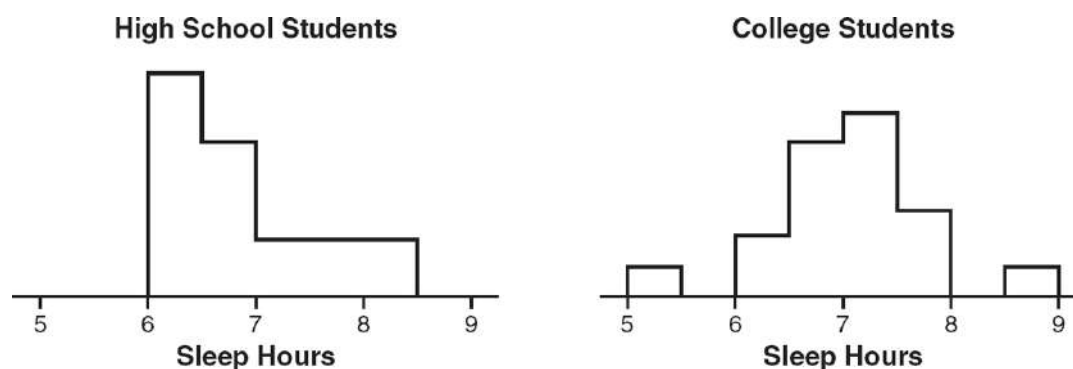
Center: Counting values (8th out of 15) gives medians of $m_{EC} = 41$ and $m_{WC} = 48$. Thus, the WC distribution of wins has the greater center.

Spread: The range of the EC distribution of wins is $60 - 17 = 43$, while the range of the WC distribution of wins is $57 - 19 = 38$. Thus, the EC distribution of wins has the greater spread.

Unusual features: The WC distribution has an apparent outlier at 19 and a gap between 19 and 33, which is different than the EC distribution that has no apparent outliers or gaps.

► Example 1.32

Two surveys, one of high school students and one of college students, asked students how many hours they sleep per night. The following histograms summarize the distributions.



Compare the two distributions.

✓ Solution

Shape: The distribution of sleep hours in the high school student distribution is skewed right, while the distribution of sleep hours in the college student distribution is unimodal and roughly symmetric.

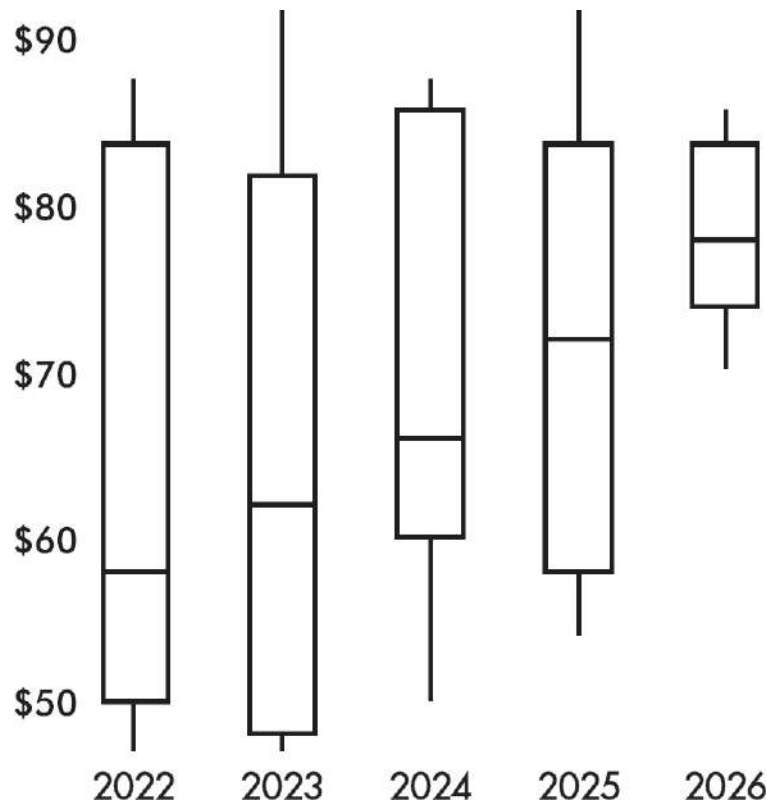
Center: The median sleep hours for the high school students (between 6.5 and 7) is less than the median sleep hours for the college students (between 7 and 7.5).

Spread: The range of the college student sleep hour distribution is greater than the range of the high school student sleep hour distribution.

Unusual features: The college student sleep hour distribution has two distinct gaps, 5.5 to 6 and 8 to 8.5, and possible low and high outliers, while the high school student sleep hour distribution doesn't clearly show possible gaps or outliers.

➤ Example 1.33

The following are parallel boxplots showing the daily price fluctuations of a particular common stock over the course of 5 years. What trends do the boxplots show?



✓ Solution

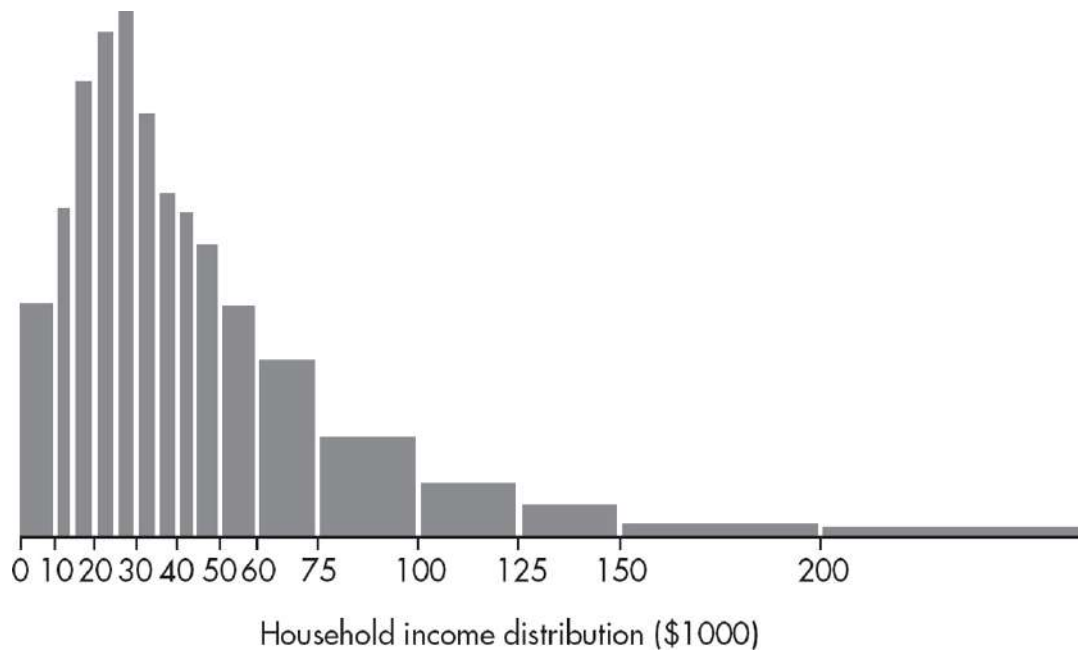
The parallel boxplots show that from year to year the median daily stock price has steadily risen 20 points from about \$58 to about \$78, the third quartile value has been roughly stable at about \$84, the yearly low has never decreased from that of the previous year, and the interquartile range has never increased from one year to the next. Note that the lowest median stock price was in 2022, and the highest was in 2026. The smallest spread (as measured by the range) in stock prices was in 2026, and the largest was in 2023. None of the price distributions shows an outlier.

Quiz 3

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. The graph below shows household income in Laguna Woods, California.



What can be said about the ratio $\frac{\text{Mean household income}}{\text{Median household income}}$?

- (A) Approximately zero
 - (B) Less than one, but definitely above zero
 - (C) Approximately one
 - (D) Greater than one
2. Students in an algebra class were timed in seconds while solving a series of mathematical brainteasers. One student's time had a standardized score of $z = 2.40$. If the times are all changed to minutes, what will then be the student's standardized score?
- (A) $z = 0.04$
 - (B) $z = 0.4$
 - (C) $z = 1.80$
 - (D) $z = 2.40$
3. Dieticians are concerned about sugar consumption in teenagers' diets (a 12-ounce can of soft drink typically has 10 teaspoons of sugar). In a random

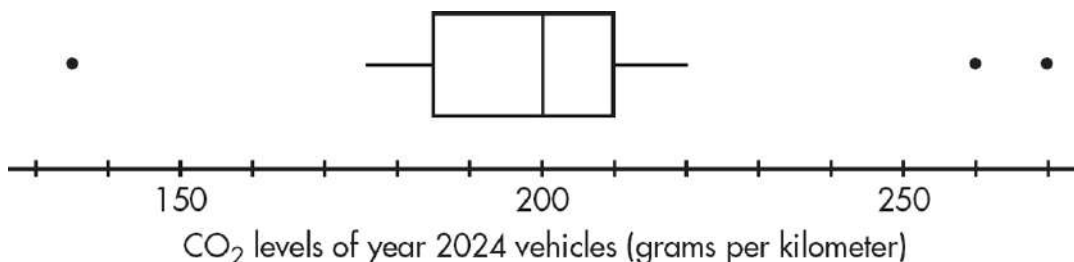
sample of 55 students, the number of teaspoons of sugar consumed for each student on a randomly selected day is tabulated. Summary statistics are noted below:

Min = 10	Max = 60	First quartile = 25	Third quartile = 38
Median = 31	Mean = 31.4	$n = 55$ $s = 11.6$	

Which of the following is a true statement?

- (A) The value 10 is an outlier, and there can be no others.
- (B) The value 60 is an outlier, and there can be no others.
- (C) Both 10 and 60 are outliers, and there may be others.
- (D) The value 60 is an outlier, and there may be others at the high end of the data set.

4. Below is a boxplot of CO₂ levels (in grams per kilometer) for a sampling of year 2024 vehicles.



Suppose follow-up testing determines that the low outlier should be 10 grams per kilometer less and the two high outliers should each be 5 grams per kilometer greater. What effect, if any, will these changes have on the mean and median CO₂ levels?

- (A) Both the mean and median will be unchanged.
- (B) The median will be unchanged, but the mean will increase.
- (C) The mean will be unchanged, but the median will increase.
- (D) Both the mean and median will change.

5. Which of the following statements are true?
- I. If two data sets have different ranges, the data set with the larger range will always have the larger standard deviation.
 - II. The standard deviation cannot be larger than the mean.
 - III. The standard deviation cannot be larger than the interquartile range.

- (A) I only
- (B) II only
- (C) III only
- (D) None of the statements are true.

6. Can either the standard deviation or the interquartile range ever be negative?

- (A) Standard deviation: never
Interquartile range: never
- (B) Standard deviation: never
Interquartile range: sometimes
- (C) Standard deviation: sometimes
Interquartile range: never
- (D) Standard deviation: sometimes
Interquartile range: sometimes

7. A teacher is teaching two AP Statistics classes. On the final exam, the 20 students in the first class averaged 92, while the 25 students in the second class averaged only 83. If the teacher combines the classes, what will the average final exam score be?

- (A) 87
- (B) 87.5
- (C) 88
- (D) More information is needed to make this calculation.

8. Allan famously quipped that when he moved from Pennsylvania to California, the average IQ dropped in both states. What would have had to been true for this to happen?

- (A) Allan's IQ was greater than the average IQ in both Pennsylvania and California.
- (B) Allan's IQ was greater than the average IQ in Pennsylvania and less than the average IQ in California.
- (C) Allan's IQ was less than the average IQ in both Pennsylvania and California.
- (D) There is no way for Allan's statement to be true.

Questions 9 and 10 refer to the following: 250 students are taking a college history course. There is one class of size 150 and four classes of size 25.

9. What is the average class size among the five classes?

- (A) 50
- (B) 100
- (C) 125
- (D) 137.5

10. Among the 250 students, what is the average size, per student, of their history class?

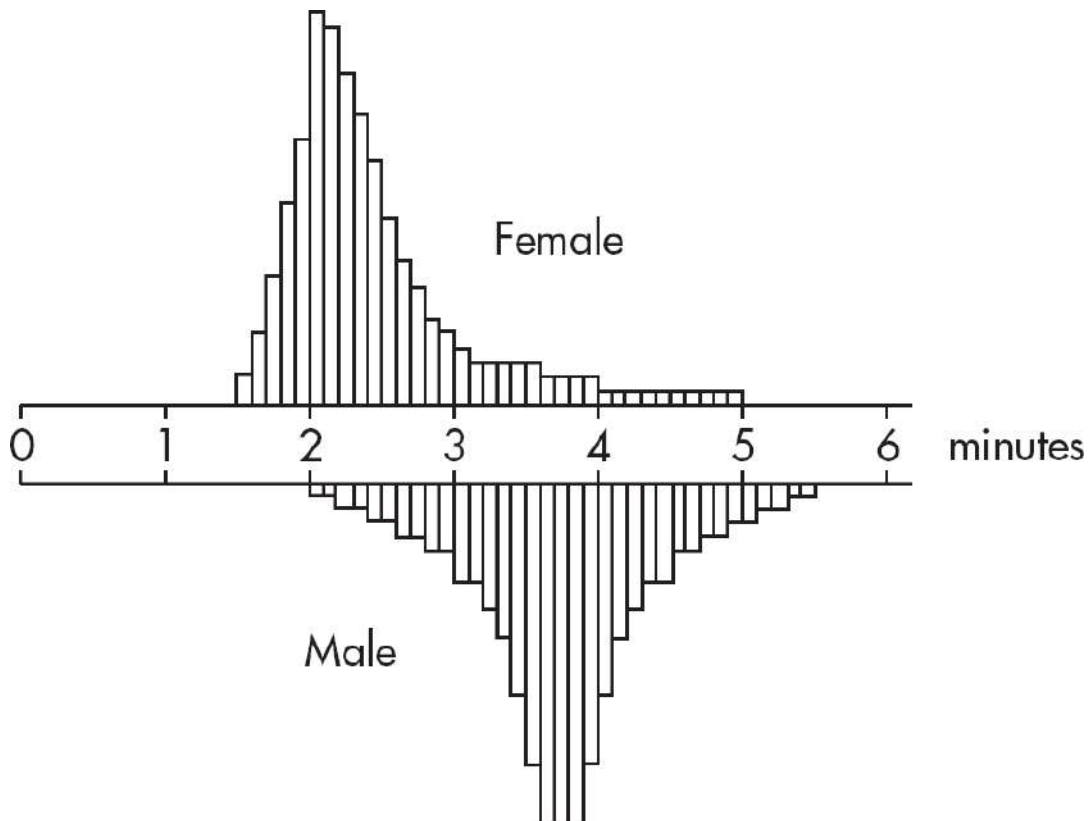
- (A) 35
- (B) 50
- (C) 100
- (D) 125

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

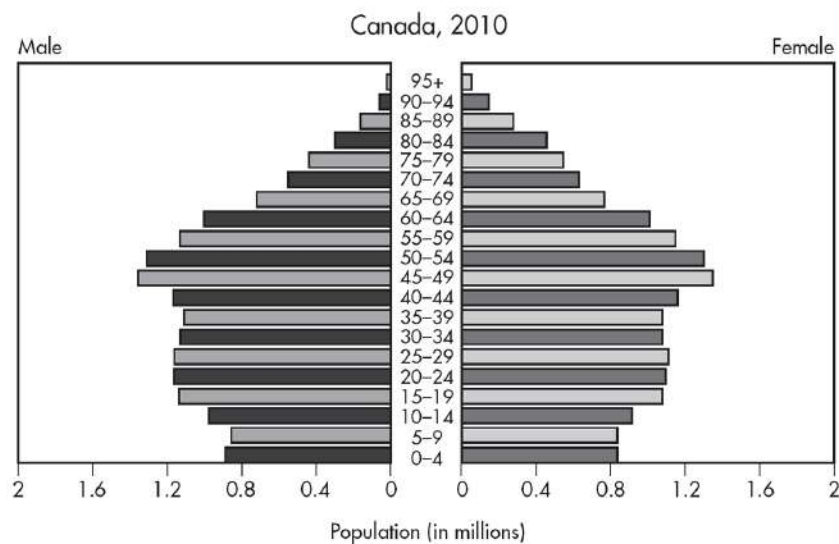
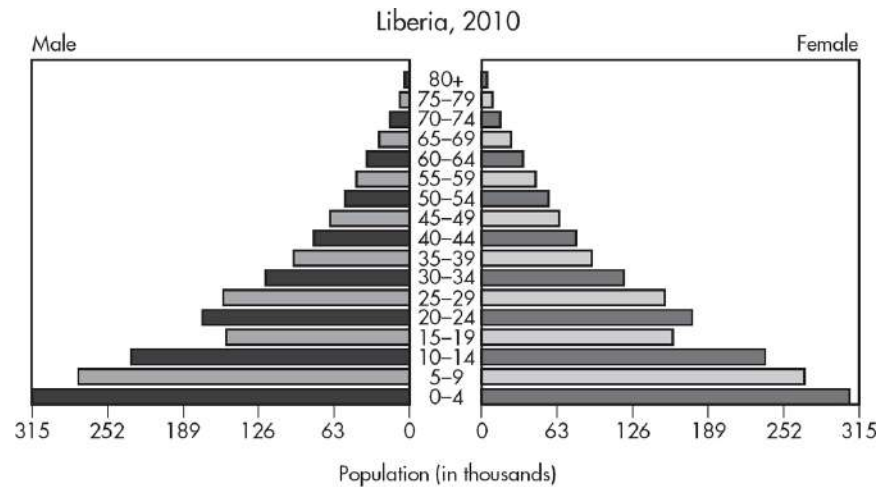
1. Victims spend from 5 to 5,840 hours repairing the damage caused by identity theft with a mean of 330 hours and a standard deviation of 245 hours.

- (a) What would be the mean, range, standard deviation, and variance for hours spent repairing the damage caused by identity theft if each of the victims spent an additional 10 hours?
- (b) What would be the mean, range, standard deviation, and variance for hours spent repairing the damage caused by identity theft if each of the victims' hours spent increased by 10%?
2. Are women better than men at multitasking? Suppose in one study of multitasking a random sample of 200 female and 200 male high school students were assigned several tasks at the same time, such as solving simple mathematics problems, reading maps, and answering simple questions while talking on a telephone. Total times taken to complete all the tasks are given in the histograms below.



Write a few sentences comparing the distributions of times to complete all tasks by females and by males.

3. Suppose a distribution has mean 300 and standard deviation 25. If the z -score of Q_1 is -0.7 and the z -score of Q_3 is 0.7 , what values would be considered outliers?
4. Below are population pyramids from Liberia and Canada.



- (a) The approximate median age of the Liberian population falls in which of these intervals: 0–4, 15–19, 30–34, 40–44? Explain.
- (b) Explain why it is impossible to calculate the mean age of either population.
- (c) Which country has more children younger than 10 years of age? Explain.

- (d) Does the population pyramid indicate that Canadian men or Canadian women live longer? Explain.
- (e) In 2010, Liberia had recently come out of a civil war with the extensive use of child soldiers. How is this visible in the population pyramid?

The answers for this quiz can be found beginning on [page 470](#).

Quiz 4

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. A school nutritionist surveys random samples of underclassmen and upperclassmen as to the number of minutes it takes to eat a cafeteria lunch. The data are shown in the following back-to-back stemplot.

Underclassmen		Upperclassmen
9 8	0	9
7 6 2	1	4 4 8
7 7 4 3 0	2	2 3 5 9
8 6 6 5 3 3 2 1	3	0 1 1 5 8 9 9
3 3 2 1 1 0 0	4	1 2 2 3 6 7 7 8 9

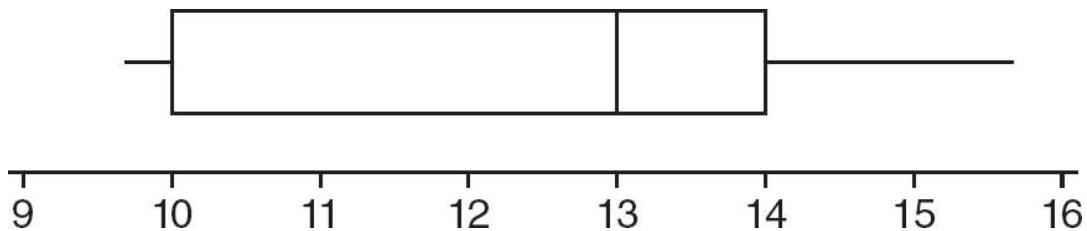
Key: 0|9 = 9 minutes

[Follow for extended description](#)

Which of the following statements is true about the distributions of eating times?

- (A) The distributions of both underclassmen and upperclassmen times are skewed left.

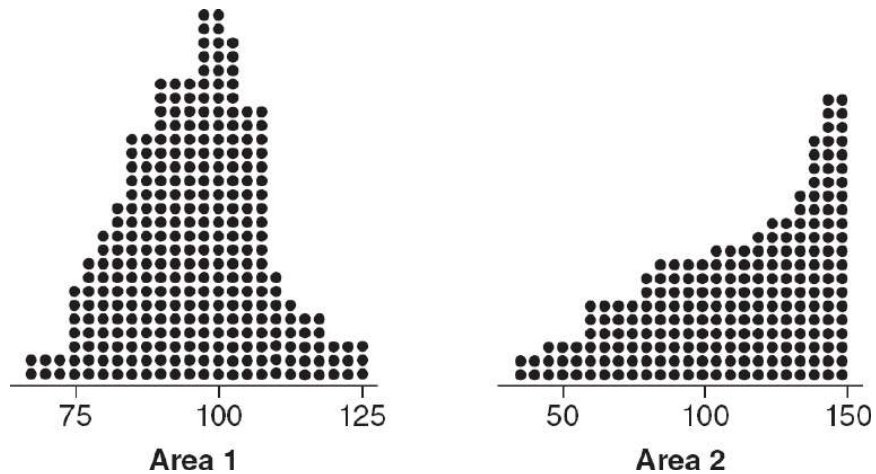
- (B) The distributions of both underclassmen and upperclassmen times are skewed right.
 - (C) The distribution of underclassmen times is skewed left, and the distribution of upperclassmen times is skewed right.
 - (D) The distribution of underclassmen times is skewed right, and the distribution of upperclassmen times is skewed left.
2. The following boxplot summarizes the times (in minutes) that it took a random sample of teachers to walk/run a mile.



[Follow for extended description](#)

Which of the following statements is supported by the boxplot?

- (A) The mean time was 13 minutes.
 - (B) More teachers took at least 14 minutes than took at most 10 minutes.
 - (C) Approximately 50% of the teachers took between 10 and 14 minutes.
 - (D) 25% of the teachers took over 10 minutes.
3. The kilowatt-hours (kWh) generated in a random sample of days from solar panel setups in two areas are shown below.



[Follow for extended description](#)

Based on the dotplots, which of the following is the best comparison of the ranges and means of the two distributions?

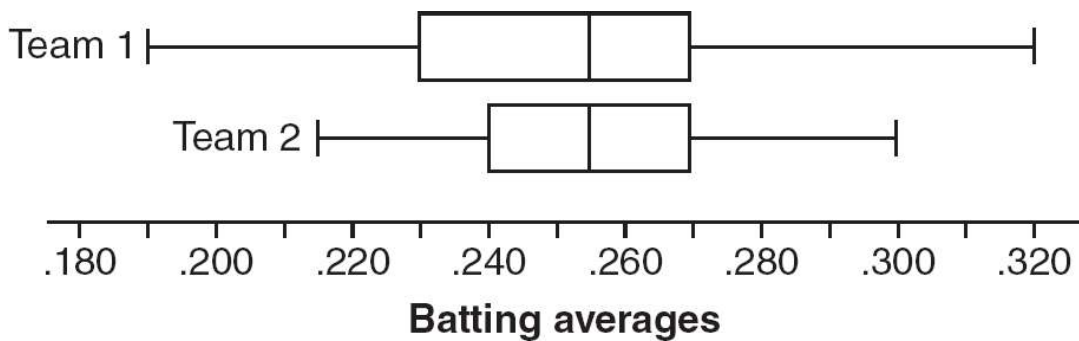
- (A) The ranges are approximately equal, and the mean of the Area 1 distribution is less than the mean of the Area 2 distribution.
 - (B) The ranges are approximately equal, and the mean of the Area 1 distribution is greater than the mean of the Area 2 distribution.
 - (C) The range of the Area 1 distribution is less than that of Area 2, and the mean of the Area 1 distribution is greater than the mean of the Area 2 distribution.
 - (D) The range of the Area 1 distribution is less than that of Area 2, and the mean of the Area 1 distribution is less than the mean of the Area 2 distribution.
4. An employee's salary of \$80,000 has a standardized score of $z = 1.3$ among all employees in the company. If everyone receives a 10% raise, what will be the standardized score of the employee's new salary of \$88,000?
- (A) 1.3
 - (B) 1.4
 - (C) 1.43
 - (D) This depends on knowing the mean and standard deviation of the company salaries.

5. The distribution of temperatures of 25 spring break vacation destinations has mean 75 degrees and standard deviation 4 degrees. Suppose an Alaskan glacier destination is added to the set with a temperature of 35 degrees. Which of the following claims about the new distribution is supported?

- I. The median temperature is not likely to change.
- II. The mean is likely to decrease.
- III. The range is not likely to change.

- (A) I only
- (B) II only
- (C) I and III only
- (D) I and II only

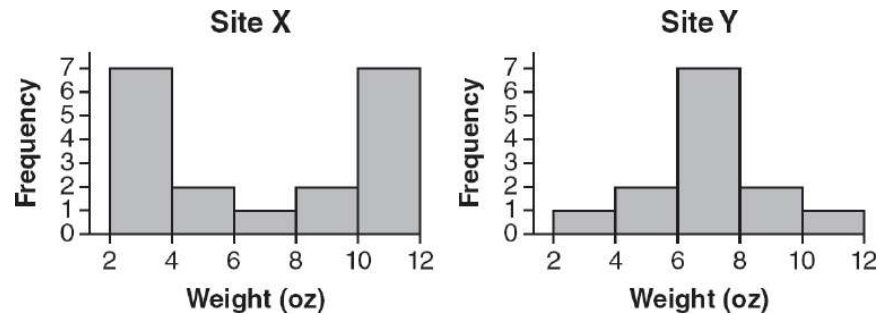
6. The boxplots below summarize the batting averages for the players on two college baseball teams.



[Follow for extended description](#)

Which of the following statements must be true based on the boxplots?

- (A) Team 1 has more players than Team 2.
 - (B) On both teams, at least half their players have an average over .250.
 - (C) On Team 2, more players have an average over .255 than under .255.
 - (D) Both team batting average distributions are roughly bell-shaped.
7. The histograms below show the distribution of weights of small broken pieces of pottery found at two different archeological sites.

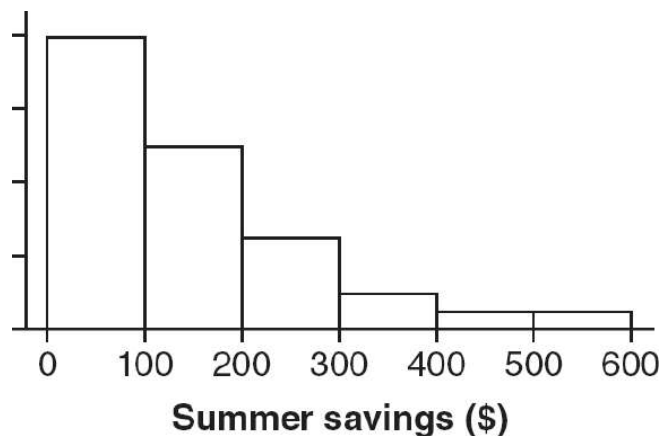


[Follow for extended description](#)

Which of the following statements must be true based on the histograms above?

- (A) The medians of the two distributions must be the same.
- (B) The means of the two distributions must be the same.
- (C) The standard deviation of the distribution from Site X is less than the standard deviation of the distribution from Site Y.
- (D) The standard deviation of the distribution from Site X is greater than the standard deviation of the distribution from Site Y.

8. The histogram below summarizes summer savings for a random sample of high school students.



[Follow for extended description](#)

Which of the following could be the median and mean of the summer savings data?

- (A) Median = \$90, Mean = \$75
- (B) Median = \$90, Mean = \$110
- (C) Median = \$160, Mean = \$110
- (D) Median = \$160, Mean = \$190

9. At an international swimming event, the times for the men's and women's 100-meter freestyle event are summarized in the following table where all values are in seconds.

	Minimum	First quartile	Median	Third quartile	Maximum
Men	47.0	47.6	47.7	47.9	48.3
Women	51.7	52.8	53.2	53.6	54.7

Which of the following statements is a correct comparison of the two distributions with regard to outliers?

- (A) Both distributions have at least one low outlier and at least one high outlier.
 - (B) The men's distribution has at least one low outlier, and the women's distribution has no outliers.
 - (C) The women's distribution has at least one low outlier, and the men's distribution has no outliers.
 - (D) The men's distribution has at least one low outlier, and the women's distribution has at least one high outlier.
10. The set of SAT Math scores for students attending a charter STEM high school has summary statistics reported in the following table.

Minimum	Q ₁	Median	Q ₃	Maximum	Mean	Standard deviation
620	650	695	735	775	710	25

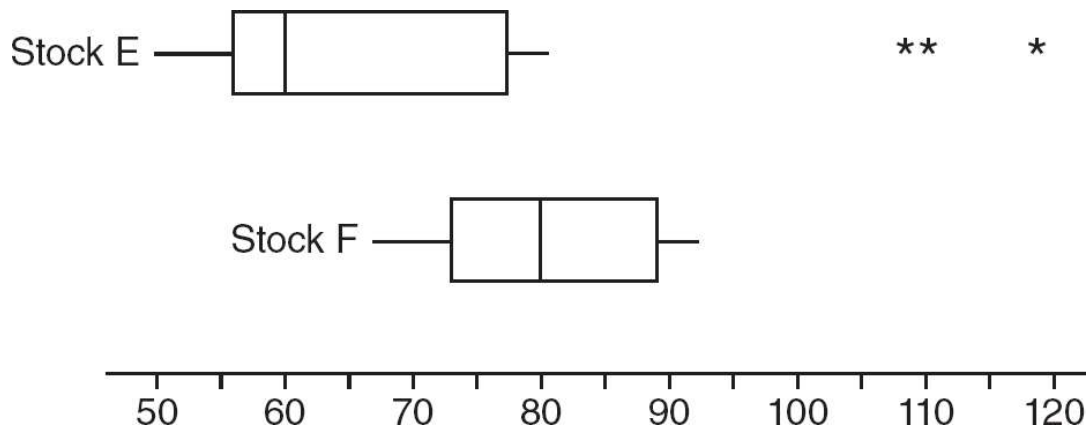
It was later reported that an error was made in the initial reporting of the scores and that all scores were 25 greater than what was initially reported. What are the interquartile range (IQR) and the standard deviation (SD) of the corrected scores?

- (A) IQR = 85 and SD = 25
- (B) IQR = 85 and SD = 50
- (C) IQR = 110 and SD = 25
- (D) IQR = 110 and SD = 50

Free-Response Questions

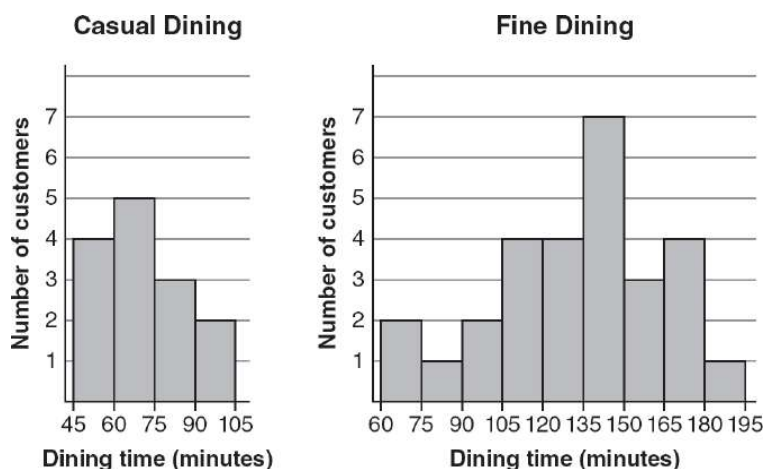
DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. The following parallel boxplots give the daily fluctuations (in dollars) of two stocks over the course of a year.



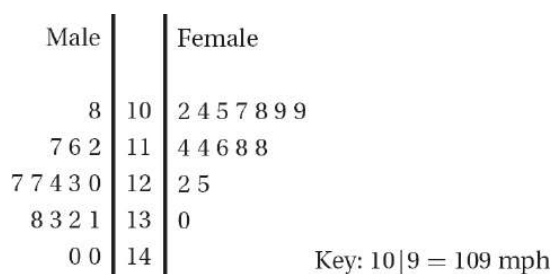
[Follow for extended description](#)

- (a) Compare the two distributions.
 - (b) Based on the boxplots, give a reason for preferring Stock E.
 - (c) Based on the boxplots, give a reason for preferring Stock F.
2. An entrepreneur owns two restaurants, one advertised as casual dining and the other advertised as fine dining. To estimate time spent in his restaurants, the entrepreneur randomly samples 42 customers, 14 with reservations at the casual dining site and 28 with reservations at the fine dining site. The following histograms summarize the times spent. (Each interval includes values from the left endpoint up to but not including the right endpoint.)



[Follow for extended description](#)

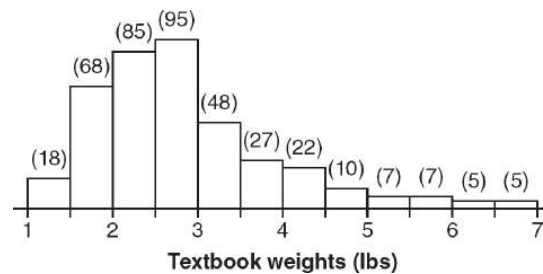
- (a) Compare the two distributions of dining times.
 - (b) How many of the dining times were under 90 minutes?
 - (c) Based on the histograms, what is a reasonable estimate for the median dining time for this random sample of 42 customers?
3. The following back-to-back stemplot summarizes the average first-serve speeds for random samples of 15 male and 15 female professional tennis players.



[Follow for extended description](#)

- (a) Compare the distributions.
- (b) How many average first serves were over 125 mph among the 30 players?
- (c) What was the median first-serve speed among the 30 players?

4. In a random sample of 397 textbooks in a college bookstore, the weights of the textbooks are summarized in the following histogram.



[Follow for extended description](#)

- What would be a statistical advantage to using the sample median rather than the sample mean as an estimate for the typical weight of the textbooks sold in that bookstore?
- Explain a method to find the median weight, and then use your method to find the median weight.
- Based on the histogram, the interquartile range is between 0.5 and 1.5 pounds. Which of these two IQR values would potentially identify more textbook weights as outliers? Explain.

The answers for this quiz can be found beginning on [page 472](#).

The Investigative Question

There are three principal components to the investigative question:

- Identifying the variable(s) of interest
- Identifying the parameter(s) and identifying the appropriate inference procedure:
 - Estimating a quantity indicates a confidence interval
 - Testing a claim indicates a hypotheses test; the direction of the alternative hypothesis must be indicated.
- Justifying a conclusion in context:
 - Determining the population to which a conclusion is applicable
 - Justifying a cause-and-effect conclusion when appropriate

> Example 1.34

A sociologist plans to interview 500 randomly selected adults over the age of 65 living in retirement homes in her home state of Florida as to whether or not they have been victimized by scams. What is a valid investigative question, and what are the components of this question?

 Solution

A valid investigative question:

What is a confidence interval of plausible values for the proportion of all American adults over the age of 65 living in retirement homes in Florida who have been victimized by scams?

The components of the investigative question:

1. The variable of interest is whether or not an American adult over the age of 65 living in a retirement home in Florida has been victimized by a scam.
2. The parameter of interest is the true proportion of all American adults over the age of 65 living in retirement homes in Florida who have been victimized by scams.
3. This can be generalized to all adults over the age of 65 living in retirement homes in Florida.

 Example 1.35

A school administrator wants to perform a study about the average first period class grade received by students who are granted parking privileges and those who are denied parking privileges. She believes that those students granted parking privileges do better than those students denied parking privileges. From among 100 applicants for parking privileges, she plans to randomly select 50 to grant privileges to and to deny privileges to the other 50 applicants. Grades in first-period classes will be compared between the two groups at the end of the semester. What is a valid investigative question, and what are the components of this question?

 Solution

A valid investigative question:

Is there convincing statistical evidence to suggest that the mean first period class grade for all students who are granted parking privileges is greater than the mean

first period class grade for all students who are denied parking privileges?

The components of the investigative question:

1. The variables of interest are whether or not a student has been granted parking privileges, and students' first period class grades.
2. The parameters of interest are the mean first period class grade for all high school students who are granted parking privileges and the mean first period class grade for all high school students who are denied parking privileges. The alternative hypothesis is that the mean grade for high school students who are granted parking privileges is greater than the mean grade for students who are denied parking privileges.
3. If convincing evidence is found, a cause-and-effect conclusion would be appropriate; that is, a conclusion would follow that granting parking privileges will result in higher first period class grades for the population of all students who apply for parking privileges at the administrator's high school.

Data Collection

There is a key question that you will begin to reflect upon in this unit and that will be more fully answered in the later units on inference. That is, suppose we measure some difference between two groups: what are the possible explanations? The difference could have resulted simply by chance, or there could be some unknown confounding variable that is causing the difference in measure, or there could be a causal association whereby something in one group causes the difference in measure.

In the real world, time and cost considerations usually make it impossible to analyze an entire population. Do companies like Apple question every potential consumer before designing a new smartphone? Does a television producer check every household's viewing preferences before deciding whether a pilot program will be continued? By studying statistics, we learn how to estimate a population characteristic (called a population *parameter*) by considering a sample measurement (called a sample *statistic*). Later in this review book we will see how to estimate population proportions and means by looking at sample proportions and means.

To derive conclusions about the larger population, we need to be confident that the sample we have chosen represents the population fairly. Analyzing the data with computers is often easier than gathering the data, but the frequently quoted "garbage in, garbage out" applies here. If data are poorly collected, then

the measurements we calculate will be meaningless. Nothing can help if the data are collected poorly. Unfortunately, many of the statistics that we are bombarded with in newspapers, on the radio, and on television are based on poorly designed data collection procedures.

The following questions should always be asked when collecting data:

- How are the data being collected?
- Does the collection method reasonably lead to a sample representative of the population or not?
- How might the individuals in our sample differ from those not in our sample?
- If our sample is not representative of the intended population, how might this lead to an underestimate or overestimate of the measurement of interest?

A sample is considered random when all observational units in the sample are selected from the population using some type of random mechanism. With random sampling, it is appropriate to make generalizations from the sample to the population. However, the following myth and fact hold true:

- Myth: A random sample will *always* be representative of the population.
- Fact: A random sample, *simply by chance*, might turn out not to be very representative.

There are four important principles in picking a representative sample in order to be able to generalize our findings to the whole population:

1. The use of *randomization* in sample selection increases the chance that the resulting sample exhibits all the known and unknown features of the population.
2. The *sample size*, not the fraction of the population surveyed, is key. Larger samples give better information about the population than smaller samples.
3. A sample is generalizable only to the population from which the sample was selected.
4. It is not possible to conclude a cause-and-effect relationship between variables using data collected from an observational study.

There are four important principles in designing an experiment in order to be able to make a cause-and-effect conclusion:

1. There should be at least two treatment groups, one of which can be a control group.
2. The use of *randomization* in assignment to treatment groups increases the chance that treatment groups are as similar as possible other than which treatment each group receives.
3. The size of treatment groups—that is, the number of subjects in each treatment group—is key. Larger numbers of subjects give better information than smaller numbers.
4. If there is an important difference among the subjects, similar subjects should be grouped into blocks, and treatments should be randomized within the blocks.

In an experiment, we impose some change or treatment and measure the response. In an observational study, we simply observe and measure something that has taken place or is taking place while trying not to cause any changes by our presence. Understanding the differences between observational studies and experiments is one of the most important ideas to come away with from your AP Statistics class. The table below summarizes and previews the terminology and concepts that will be explained throughout the remainder of this unit.

Observational Studies	Experiments
Observe and measure without influencing	Impose <i>treatments</i> and measure responses
Can only show associations	Can suggest <i>cause-and-effect</i> relationships
Use of <i>random sampling</i> in order to be able to generalize to a population	Use of <i>random assignment</i> to minimize the effect of confounding variables
Usually less expensive and less time-consuming	Can be expensive and time-consuming
Use of <i>strata</i> , and randomization within strata, for greater accuracy and to reduce variability	Use of <i>blocks</i> , and randomization within blocks, to control for certain variables
	Possible ethical concerns over imposing certain treatments
	Use of a control group for a baseline comparison
	Use of blinding and double-blinding

➤ Example 1.36

Identify each of the following as an observational study or an experiment.

- (a) Researchers measured the systolic blood pressures for a random sample of 100 dog owners and compared these values with the systolic blood pressures for a random sample of 100 subjects who did not own dogs.
- (b) Resting times before full recovery from vigorous exercise was tabulated for 25 athletes who chose to drink chocolate milk after exercising and for 25 athletes who chose to drink Gatorade after exercising. Resting times were then compared between the two sets of athletes.
- (c) Researchers randomly assigned 38 students, each to one of two groups. The students in one group were instructed to take an exam where all the questions were displayed on a small screen in portrait orientation, while the students in the other group were instructed to take the same exam but where all the questions were displayed on a small screen in landscape orientation. Results on the exam were compared.
- (d) Data were gathered on the rate of fatal heart attacks per 100,000 people living in the South versus living in the North. Rates were compared and indicated a greater percentage of Southerners than Northerners had heart attacks.
- (e) Patients recovering from knee replacement surgery were randomly assigned to two groups. The patients in one of the groups were asked to play video games for at least five hours per day for the first three days after surgery. After three days, a comparison was made on how much pain relief medication was needed for patients in each group.

Solutions

- (a) This was an observational study as people in the study either were pet owners or were not. No one was assigned to have or not have a pet.
- (b) This was an observational study as the athletes themselves chose to drink either chocolate milk or Gatorade after exercising. No one was assigned a particular drink.
- (c) This was an experiment where two treatments were applied. Students were randomly assigned to two groups and then were instructed as to which display to use.
- (d) This was an observational study as no one was made to live in the South or the North.
- (e) This was an experiment because the patients were assigned to two groups. One group received the “treatment” of playing video games, while the other group was a control group.

Example 1.37

A study is to be designed to determine whether a particular commercial review course actually helps raise SAT scores among students attending a particular high school.

- (a) How could an observational study be performed?
- (b) How could an experiment be performed?
- (c) What is an argument in favor of the experimental approach over the observational study approach?
- (d) What is an argument in favor of the observational study approach over the experimental approach?

Solutions

- (a) An observational study would interview students at the school who have taken the SAT exam, asking whether or not they took the review course, and then the SAT results of those who have and have not taken the review course would be compared.
- (b) An experiment performed on students at the school who are planning to take the SAT exam would use chance to randomly assign some students to take the review course while others to not take the review course and would then compare the SAT results of those who have and have not taken the review course.
- (c) An argument in favor of the experimental approach is that with the observational study, there could be many explanations for any SAT score difference noted between those who took the course and those who didn't. For example, students who took the course might be the more serious high school students. Higher score results might be due to their more serious outlook and have nothing to do with taking the course. The experiment tries to control for confounding variables by using random assignment to make the groups taking and not taking the course as similar as possible except for whether or not the individuals take the course.
- (d) An argument in favor of the observational study approach is that there are ethical considerations in forcing some students to take the review course who might not have wanted to and in withholding the review course from some students who might have wanted to take it.

TIP

It is important to identify the *response variable* (SAT scores in this example) right away. This is critical in answering any experimental design question.

Observational Studies

Observational studies aim to gather information about a population without disturbing the population. *Retrospective studies* look backward, examining existing data, while *prospective studies* watch for outcomes, tracking individuals into the future.

Example 1.38

Retrospective studies of the 2014–2016 Ebola epidemic in West Africa have looked at the timing, numbers, and locations of reported cases and have tried to understand how the outbreak spread. There is now much better understanding of transmission through contact with bodily fluids of infected people. Several prospective studies involve ongoing surveillance to see how experience and tools to rapidly identify cases will now limit future epidemics.

Retrospective studies tend to be smaller scale, quicker to complete, and less expensive. In cases such as addressing diseases with low incidence, the study begins right from the start with people who have already been identified. However, researchers have much less control, usually having to rely on past recordkeeping of others. Furthermore, the existing data will often have been gathered for other purposes than the topic of interest. Then there is the problem of subjects' inaccurate memories and possible biases.

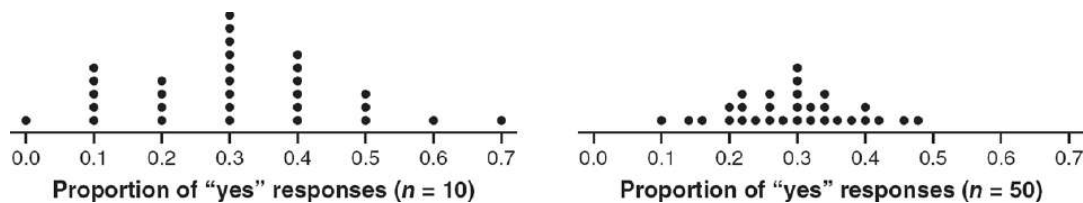
Prospective studies usually have greater accuracy in data collection and are less susceptible to recall error from subjects. Researchers do their own recordkeeping and can monitor exactly what variables they are interested in. However, these studies can be very expensive and time-consuming, as they often follow large numbers of subjects for a long time.

No matter how well designed and well conducted a survey is, it still gives a sample statistic as an estimate for a population parameter. Different samples give different statistics, all of which are estimates for the same population parameter. So *sampling variability* is naturally present.

Example 1.39

Pesticide residue on store-bought fruit can be a health problem. Ms. D has two AP Statistics classes with 30 students in each. For a study asking high school students whether or not they wash fruit before eating them, each student in the first class obtained a random sample of size 10 to question, while each student in

the second class obtained a random sample of size 50 to question. The data obtained are displayed in two dotplots:



[Follow for extended description](#)

Note the relationship between sampling variability and sample size. Estimates from larger samples produce less sampling variability; that is, they are more tightly clustered around the presumed true parameter value.

Collecting data from every individual in a population, called a *census*, might seem to provide complete information, but unless the population is small, this method would be prohibitively expensive and time-consuming. Furthermore, it could be ridiculous, as, for example, tasting every single pizza a company makes would leave nothing left to sell. Finally, if a questionnaire is poorly worded, answers, even from an entire population, might be meaningless.

Given that it is almost always impractical, if not impossible, to examine a whole population, we must settle for examining a sample. The list of all individuals from which the sample is drawn is called the *sampling frame*. Of primary importance is how to choose a representative sample from the sampling frame.

A *simple random sample* (SRS) is one in which **every possible sample of the desired size has an equal chance of being selected**. (In an SRS, it is also true that every individual will have an equal chance of being selected.) One technique to create an SRS is to write the name of each member of the population on a card, thoroughly mix the cards, and pull out a specified number of cards. This method is often too time-consuming. A more practical method is to assign a number to everyone in the population and have a computer generate random numbers to indicate those chosen.

Important

To receive full credit, there are 3 steps:

1. assigning integers,
2. using a computer random number generator to generate distinct integers in a given range, and

3. linking selected integers with corresponding individuals selected for the sample.

Example 1.40

Suppose 80 students are taking an AP Statistics course and the teacher wants to pick a sample of 10 students randomly to try out a practice exam. She can make use of a random number generator on a computer as follows. Assign the students numbers 1, 2, 3, . . . , 80. Use a computer to generate 10 random integers between 1 and 80 without replacement—that is, throw out repeats. The sample consists of the students with assigned numbers corresponding to the 10 unique computer-generated numbers.

Advantages of simple random sampling include the following:

- The simplicity of simple random sampling makes it relatively easy to interpret data collected.
- This method requires minimal advance knowledge of the population other than knowing the complete sampling frame.
- Simple random sampling allows us to make generalizations (i.e., statistical inferences) from the sample to the population.
- Among the major parameters with which we will work, simple random sampling tends to be *unbiased*; that is, when repeated many times, it gives sample statistics that are centered around the true parameter value.

Disadvantages of simple random sampling include the following:

- The need for a list of all potential subjects can be a formidable task.
- Although simple random sampling is a straightforward procedure to understand, it can be difficult to execute, especially if the population is large. For example, how would you obtain a simple random sample of students from the population of all high school students in the United States?
- The need to repeatedly contact nonrespondents can be very time-consuming.
- Important groups may be inadvertently left out of the sample when using this method.

There are other sampling methods available. Each has its own set of advantages and disadvantages.

Stratified random sampling involves dividing the population into non-overlapping homogeneous groups called *strata*, then picking random samples from each of the strata, and finally combining these individual samples into what is called a *stratified random sample*. For example, we can stratify by age, gender, income level, or race/ethnicity; pick a sample of people from each stratum; and combine to form the final sample. For instance, a political pollster might want to be sure to include respondents from various minority groups. Stratifying by religion or race/ethnicity and including people from each group would ensure this diversity in the final sample.

NOTE

All individuals in a given stratum have a characteristic in common.

TIP

You should be able to explain why you think that stratification should be used in a given problem.

Advantages of stratified random sampling include the following:

- This ensures that every stratum is adequately represented and removes the chance of any significant segment of the population being entirely neglected.
- Looking at subgroups within a population can make the work of collecting data more manageable.
- This method may speed up the time it takes to conduct a survey and so is more cost-effective.
- Samples taken within a stratum have reduced variability, which means the resulting estimates are more precise than when using other sampling methods.
- More information, insights, and important differences among groups can become apparent.

Disadvantages of stratified random sampling include the following:

- It necessitates prior knowledge about the population and the variables used to stratify it, which may not be always available or accurate.

- It may be difficult to classify every member of a population into one of the strata.
- It may necessitate knowing the size of the population in each stratum.
- Evaluating the results may be more difficult and more complex than with simple random sampling.
- Like an SRS, this method might be difficult to implement with large populations.
- Significant planning may be necessary to ensure that the subdivisions are logical in the context of the study. Forcing subdivisions when none really exist is meaningless.

We could further do *proportional sampling*, where the sizes of the random samples from each stratum depend on the proportion of the total population represented by the stratum.

Cluster random sampling involves dividing the population into non-overlapping heterogeneous groups called *clusters*, each similar to each other, and then picking everyone or everything in a random selection of one or more of the clusters. For example, to survey high school seniors, we could randomly pick several senior class homerooms in which to conduct our study and sample all students in those selected homerooms.

NOTE

Each cluster should resemble the entire population.

Advantages of cluster random sampling include the following:

- Clusters are often chosen for ease, convenience, and quickness.
- With limited fixed funds, cluster random sampling usually allows for a larger sample size than do other sampling methods.
- This method is especially appropriate when the construction of a complete list of the population is impossible, expensive, or too difficult to organize.

Disadvantages of cluster random sampling include the following:

- Although clusters should mirror the full diversity of the population, realistically this is difficult to accomplish.
- Difficulties in analysis can result if the clusters are of very different sizes.

- With a given sample size, cluster random sampling usually provides less precision than either an SRS or a stratified random sample provides.
- If the population doesn't have natural clusters and the designated clusters are not representative of the population, selection could easily result in a biased sample.

The differences between strata and clusters include the following:

- Although all strata are represented in the sample, only a subset of clusters are in the sample.
- Stratified random sampling is accurate when each stratum consists of *homogeneous* elements, while cluster random sampling is accurate when each cluster consists of *heterogeneous* elements.
- Randomization plays a key role in both methods! With stratified random sampling, the randomization takes place within each stratum. With cluster random sampling, the randomization takes place when choosing which clusters to include.

TIP

Depending upon how the random starting point is chosen, it may be necessary to wind the list around to the beginning.

Systematic random sampling is a relatively simple and quick method. It involves listing the population in some order (for example, alphabetically), choosing a random point to start, and then picking every tenth (or hundredth, or thousandth, or *k*th) person from the list. This gives a reasonable sample as long as the original order of the list is not in any way related to the variables under consideration.

Advantages of systematic random sampling include the following:

- Systematic random sampling is simpler and quicker to implement than the other methods discussed. Subjects don't have to be numbered. For example, this is a nice option for situations like exit polling or surveying people as they leave a stadium.
- It is a practical, straightforward, reasonably random method especially useful for investigators unfamiliar with advanced statistics.
- If similar members in the list are grouped together, we can end up with a kind of stratified random sample, only more easily implemented.

Disadvantages of systematic random sampling include the following:

- If randomization is a top priority, this method is weak as the only randomization in systematic random sampling occurs in choosing the initial subject.
- If the population size isn't roughly known, determining the fixed interval for sample selection is problematic and can lead to inaccurate results.
- If the list happens to have a periodic structure similar to the period of the sampling, a very biased sample could result. For example, picking every 12th seat in an airplane when there are 6 seats in a row would result in all aisle or all middle or all window seats in the sample.

Example 1.41

Suppose a sample of 100 high school students from a Chicago school of size 5,000 is to be chosen to determine their views on whether they think the Cubs will win another World Series this century. One method would be to have each student write his or her name on a card, put the cards into a box, have the principal reach in and pull out 100 cards, and then choose the names on those cards to be the sample. However, questions could arise regarding how well the cards are mixed. For example, how might the outcome be affected if all students in one PE class toss their names in at the same time so that their cards are clumped together? A second method would be to number the students 1 through 5,000 and use a random number generator to pick 100 unique (throw out repeats) integers between 1 and 5,000. The sample is then the students whose numbers correspond to the 100 generated numbers. What are alternative procedures?

Solution

- *Systematic random sampling:* From a list of the students, the surveyor could choose a random starting name from among the first 50 names and then include every 50th name from that point into the sample.
- *Stratified random sampling:* Since students in each class have certain characteristics in common, the surveyor could use a random selection method to pick 25 students from each of the separate lists of freshmen, sophomores, juniors, and seniors.
- *Cluster random sampling:* If each homeroom has a random mix of 20 students from across all grade levels, the surveyor could randomly pick five homerooms with the sample consisting of all the students in these five homerooms.

Sometimes we randomly pick a single sample, cross-categorize on two variables, and then test to see if there is an association between the variables. However, sometimes we instead randomly pick a sample from each of two or more populations and then compare the distribution of some variable among the different populations. This second method is called *sampling for homogeneity*.

The one thing that most quickly invalidates a sample and makes obtaining useful information and drawing meaningful conclusions impossible is *bias*. **A sampling method is biased if in some critical way it consistently results in samples that do not represent the population.** This typically leads to certain responses being repeatedly favored over others, and results in a statistic being consistently smaller or larger than the parameter the statistic is used to estimate.

The way data are obtained is crucial. A large sample size cannot make up for a poor survey design.

Voluntary response surveys are based on individuals who choose to participate:

Sampling bias is a property of the sampling method, not of any one sample generated by the method. Sampling methods that do not use random chance to select individuals are particularly susceptible to bias. Two examples of nonrandom sampling methods are voluntary response surveys and convenience surveys.

TIP

Think about potential biases before collecting data!

- Such surveys typically give too much emphasis to people with strong opinions and undersample people who don't care much about a topic.
- Radio call-in programs about controversial topics do not produce meaningful data on what proportion of the population favors or opposes related issues.
- Online surveys posted to websites miss people without Internet access and tend to have many unanswered questions if the survey is long.
- Although a voluntary response survey may be easy and inexpensive to conduct, the sample is likely to be composed of strongly opinionated people, especially those with negative opinions on a subject.

Convenience surveys are based on choosing individuals who are easy to reach:

- These surveys tend to produce data highly unrepresentative of the entire population.
- Mall surveys miss those who do not frequent malls and are affected by differences in shopper demographics during different times of day or different seasons.
- Door-to-door household surveys miss people who do not happen to be at home and typically miss groups such as college students and prison inmates.
- Although convenience samples may allow a researcher to conduct a survey quickly and inexpensively, generalizing from the sample to a population is almost impossible.

It is important to recognize when a sample design is flawed, explain why it is flawed, and understand the direction of the bias. You will need to be able to decide if the bias will likely cause an underestimate or overestimate of a population parameter.

The following are biases that you should recognize by name:

NOTE

Undercoverage bias and *voluntary response bias* happen when choosing a sample, while *response bias*, *nonresponse bias*, and *wording bias* happen after choosing a sample.

- *Undercoverage bias* happens when there is inadequate representation, and thus some groups in the population are left out of the process of choosing the sample. For example, telephone surveys ignore those who don't have telephones. Studies planning to conclude something about all residents in a city often do not adequately take into account the homeless when choosing a sample. Voluntary response surveys often suffer from undercoverage bias because those who choose to participate may differ from those who do not choose to participate. Similarly, convenience surveys often suffer from undercoverage bias because hard-to-reach people may have different views from those easy to reach.
- *Response bias* occurs when the question itself can lead to misleading results because people don't want to be perceived as having unpopular or

unsavory views or don't want to admit to having committed crimes. Nonneutral (leading) or poorly worded questions can result in very unrepresentative responses. Even the order in which questions are asked can make a difference.

- *Nonresponse bias*, where there are low response rates, occurs when individuals chosen for the sample can't be contacted or refuse to participate, and it is often unclear which part of the population is responding. As with undercoverage bias, nonresponse bias can misrepresent results if those who respond differ in some systematic way from those who do not respond. To minimize nonresponse bias, every effort should be made to follow up with those who do not respond to initial requests for information.

➤ Example 1.42

The *Military Times*, in collaboration with the Institute for Veterans and Military Families at Syracuse University, conducted a voluntary and confidential online survey of U.S. service members who were readers of the *Military Times*. Their military status was verified through official Defense Department email addresses. What were possible sources of bias?

✓ Solution

First, *voluntary online surveys* are very suspect because they typically overcount strongly opinionated people. Second, *undercoverage bias* is likely because only readers of the *Military Times* took part in the survey. Note that *response bias* was probably not a problem because the survey was confidential.

When given a data collection method and asked to describe a bias that could occur, you need to address the following:

TIP

Don't confuse *bias* with *precision*. Increasing the sample size will not reduce bias but may simply give a more precise estimate of a wrong answer.

- Why is there bias? (For example, "The survey was voluntary response," or "This was a convenience survey," or "Only certain people had a chance to

be selected,” or “People might be untruthful because they don’t want to be perceived as having an unpopular view.”)

- How is it biased? (For example, “Those choosing to answer might feel stronger about the issue than the general population,” or “Those picked for the sample might be less likely than the general population to have full-time jobs.”)
- What will this do to the estimate of the population parameter of interest? (For example, “This would likely result in an underestimate of the population proportion of ...,” or “This would likely result in an overestimate of the population mean of”)

In some questions, you can receive full credit arguing either direction (a likely underestimate or overestimate) as long as there is a clear direction and a reasonable explanation tied to the data collection method that supports the argument.

Example 1.43

For each of the following, identify a possible question of interest and the potential problem.

- (a) Before buying a package of strawberries, you carefully look at the visible top rows of strawberries.
- (b) In a presidential survey, registered voters were asked, “Many people believe this candidate is weak on foreign policy. What do you think?”
- (c) A nutritionist asks adults entering a fast-food restaurant what they plan on ordering.
- (d) An online survey asks if drug companies should be held accountable for the number of overdose deaths reported using drugs they manufacture.
- (e) A teacher asks her students if they use AI in writing term papers.
- (f) A college administrator who is considering recommending a change in food vendor interviews students eating lunch in the college cafeteria.

Solutions

- (a) Question of interest: Are the strawberries fresh?
Problem: This is a convenience sample. The freshest strawberries may have been placed on top, leading you to overestimate the true quality of your purchase.
- (b) Question of interest: What proportion of registered voters believe the candidate is weak on foreign policy?

Problem: The wording, which first states what some others have said, may lead to response bias and an overestimate of the true proportion of registered voters who believe the candidate is weak on foreign policy.

- (c) Question of interest: Do adults who frequent fast-food restaurants pick unhealthy options?

Problem: There could be response bias because adults may not want to admit they plan to order unhealthy foods. There could also be nonresponse bias because adults planning to order unhealthy foods may refuse to take part in the survey. Both cases would lead to an underestimate of the true proportion of adults who order unhealthy foods at fast-food restaurants.

- (d) Question of interest: What proportion of people think that drug companies should be held accountable for the number of overdose deaths reported using drugs they manufacture?

Problem: This is a voluntary response survey and probably has undercoverage bias. People who feel most strongly are likely to respond, particularly if they have lost family or friends to drug overdoses. This would lead to an overestimate of the true proportion of people who think that drug companies should be held accountable for the number of overdose deaths reported using drugs they manufacture.

- (e) Question of interest: What proportion of the teacher's students use AI in writing term papers?

Problem: Many students would not want to admit to using AI, leading to response bias and an underestimate of the true proportion of the teacher's students who use AI in writing term papers.

- (f) Question of interest: What proportion of students are satisfied with the cafeteria food?

Problem: This is a convenience sample of those students who do use the cafeteria. Students not using the cafeteria, possibly because they don't like the food offered, are not being sampled, leading to undercoverage bias and an overestimate of the true proportion of students who are satisfied with the cafeteria food.

Quiz 5

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. In September 2020, medical researchers analyzed the summer records of hospitalized Covid-19 patients, collecting data on disease severity and whether or not the patients had been taking statin medications for high cholesterol within 30 days prior to hospitalization. The researchers found that statin use was associated with a lesser severity of the disease. What kind of research was this?
 - (A) A retrospective observational study
 - (B) A prospective observational study
 - (C) A voluntary response survey
 - (D) A convenience survey
2. An increasing number of U.S. public schools are adopting school uniforms. Two possible wordings for a question on whether or not students should have to wear school uniforms are as follows:
 - I. Many educators believe in creating a level playing field to reduce socioeconomic disparities. Do you believe that students should have to wear school uniforms?
 - II. Many sociologists believe that students have a right to express their individuality. Do you believe that students should have to wear school uniforms?

One of these questions showed that 18% of the population favors school uniforms, while the other question showed that 23% of the population favors school uniforms. Which question probably produced which result and why?

- (A) The first question probably showed 23% of the population favors school uniforms, and the second question probably showed 18% because of the lack of randomization in the choice of pro-uniform and anti-uniform arguments as evidenced by the wording of the questions.
- (B) The first question probably showed 18% and the second question probably showed 23% because of stratification in the wording of the questions.
- (C) The first question probably showed 18% and the second question probably showed 23% because of response bias due to the wording of the questions.

- (D) The first question probably showed 23% and the second question probably showed 18% because of response bias due to the wording of the questions.

3. Which of the following is a true statement?

- (A) If bias is present in a sampling procedure, it can be overcome by dramatically increasing the sample size.
- (B) Sampling techniques that use probability techniques effectively eliminate bias.
- (C) Convenience samples often lead to undercoverage bias.
- (D) Voluntary response samples often underrepresent people with strong opinions.

4. In a study of successes and failures in adopting new academic standards, a random sample of high school principals will be selected from each of the 50 states. Selected individuals will be asked a series of evaluative questions. Why is stratification used here?

- (A) To minimize response bias
- (B) To minimize nonresponse bias
- (C) Because each state is roughly representative of the U.S. population as a whole
- (D) To obtain higher statistical precision because variability of responses within a state is less likely than variability of responses found in the overall population

5. A random sample of 100 students at a high school are to be selected, and each will be asked if they ever tried edible cannabis. Which of the following is a reason for increasing the size of this study from 100 to 200 students?

- (A) To reduce response bias
- (B) To reduce nonresponse bias
- (C) To reduce undercoverage bias
- (D) To more likely obtain an estimate that is closer to the population proportion

6. Each of the 30 MLB teams has 25 active-roster players. A sample of 60 players is to be chosen as follows. Each team will be asked to place 25 cards with its players' names into a hat and randomly draw out two names. The two names from each team will be combined to make up the sample. Will this method result in a simple random sample of the 750 baseball players?
- (A) Yes, because each player has the same chance of being selected.
 - (B) Yes, because each team is equally represented.
 - (C) Yes, because this is an example of stratified random sampling, which is a special case of simple random sampling.
 - (D) No, because not each group of 60 players has the same chance of being selected.
7. A researcher plans a study to examine the depth of belief in God among the adult population. He obtains a simple random sample of 100 adults as they leave church one Sunday morning. All but one of them agree to participate in the survey. Which of the following is a true statement?
- (A) Proper use of chance as evidenced by the simple random sample makes this a well-designed survey.
 - (B) The high response rate makes this a well-designed survey.
 - (C) Undercoverage bias makes this a poorly designed survey.
 - (D) The validity of this survey depends on whether or not the adults attending this church are representative of all churches.
8. A sports statistician plans to survey Division 1 college football programs as to graduation rates of their student athletes. The statistician will use a cluster random sample with the NCAA Division 1 football conferences as clusters. A random sample of two conferences will be selected and their football programs analyzed. What is one disadvantage to selecting a cluster random sample to investigate Division 1 football program graduation rates?
- (A) With two full conference football programs, the sample might be too large to effectively analyze.
 - (B) The conferences in the sample might not be representative of the Division 1 population.

- (C) The graduation rates of student athletes might not be homogeneous among schools in a given conference.
- (D) There is no disadvantage to using the proposed cluster random sample.

9. Political advisors want to survey registered voters as to their top concerns. Which of the following sampling methods does NOT have voluntary response bias as a potential source of error? In each case, assume all 100 chosen people answer the survey.

- (A) Give the survey to 100 randomly chosen people from among callers to a radio station who identify themselves as registered voters.
- (B) Give the survey to 100 randomly chosen people from official voter registration lists.
- (C) Give the survey to 100 randomly chosen people from among those who approach and agree to answer questions of a surveyor standing on a busy street corner.
- (D) Give the survey to 100 randomly chosen people from among those giving feedback to the question posted on a neutral online website.

10. A survey is to be taken as to whether or not people believe the moon landing conspiracy theory, which claims the six crewed landings (1969–1972) were faked. Which of the following describes a design in which *nonresponse bias* is likely present?

- (A) Surveys were completed by the first 100 people responding to the question posed on a radio call-in program.
- (B) Surveys were completed by 100 of the 200 people to whom the survey was mailed.
- (C) Surveys were completed by 100 people chosen at random.
- (D) Surveys were completed by all 100 students who were sent the survey at a small liberal arts college with 850 total students.

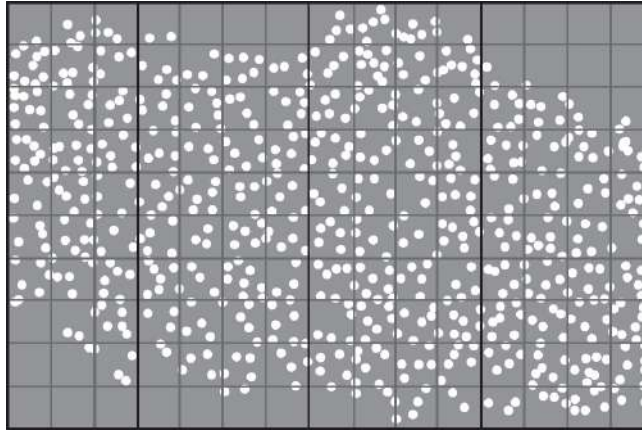
Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. A student is interested in estimating the average length of words in a 600-page textbook and plans the following three-stage sampling procedure:
 - I. After noting that each of the 12 chapters has a different author, the student decides to obtain a sample of words from each chapter.
 - II. Each chapter is approximately 50 pages long. The student uses a random number generator to pick three pages from each chapter and use all words from these pages to find the average length of words for that author.
 - III. On each chosen page, the student notes the length of every tenth word.
 - (a) The first stage above represents what kind of sampling procedure? Give an advantage in using it in this context.
 - (b) The second stage above represents what kind of sampling procedure? Give an advantage in using it in this context.
 - (c) The third stage above represents what kind of sampling procedure? Give a *disadvantage* to using it dependent upon an author's writing style.

2. Pelicans are considered an endangered species, and there is a need to protect them from becoming extinct. Unfortunately, the U.S. Fish & Wildlife Service is unable to protect all pelican colonies; however, it plans to protect those colonies identified as most important. One criterion in determining the importance of a geographical region is the density of nesting sites within it.

Given an aerial view of the nests in a geographical region used for pelican nesting, the service plans to divide the rectangular picture into 150 equal-size plots as below:



The service will randomly pick 15 plots (10% of the total), count the nests in each of those plots, sum and multiply by 10 to estimate the total number of nests in the region, and then calculate density. Two methods for counting the number of nests are under consideration.

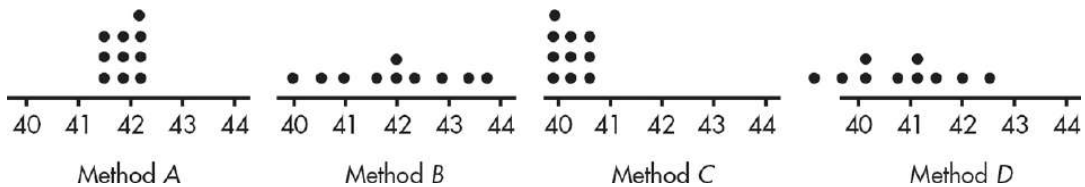
Method I: Randomly pick one of the above rows and use all the 15 plots in that row.

Method II: Randomly pick one of the plots in each of the 15 columns to obtain the 15 plots.

- (a) What sampling procedure is each method striving for, and how can each method be implemented?
 - (b) Which of these two methods is preferable here? Explain.
3. For each of the following survey designs, identify the problem and the effect it would have on the estimate of interest.
 - (a) Should spending be increased for the athletic program? A school official decides to sample spectators randomly at the next home basketball game.
 - (b) Do people believe that eating organic will lower the risk of cancer? A health magazine sends a questionnaire to all their subscribers.
 - (c) A supermarket quality control inspector opens boxes of strawberries. If no mold is seen among the visible strawberries, he accepts the box.
 - (d) Are the patients following their diet instructions? A family physician asks each of her patients during their office visits.
 - (e) Parents are asked to write to a newspaper with a yes or no response to whether or not they are happy with the education their children are

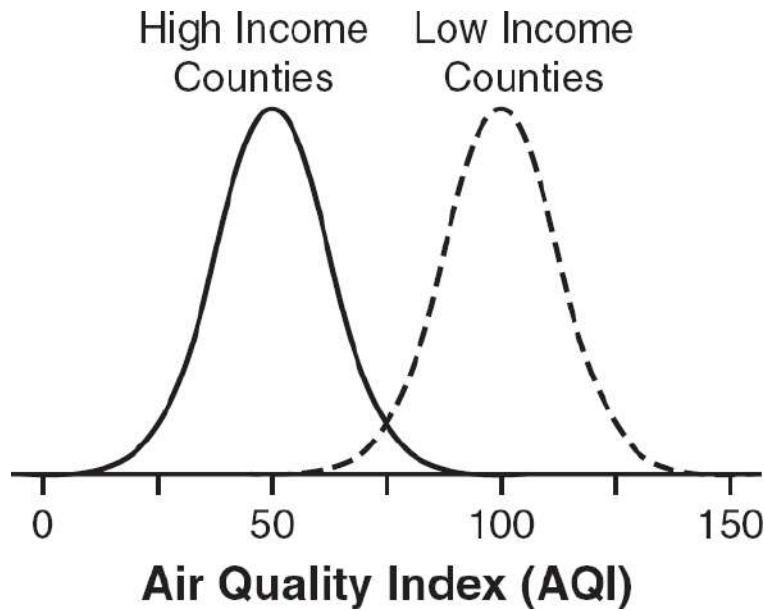
receiving in the local public schools.

4. Suppose we are trying to estimate the mean age of high school teachers and have four methods of choosing samples. We choose 10 samples using each method. Later we find the true mean, $\mu = 42$. The following are plots of the results of each sampling method.



[Follow for extended description](#)

- (a) Which method or methods indicate high precision? Explain.
- (b) Which method or methods indicate probable bias? Explain.
5. Air Quality Index (AQI) is the EPA's index for reporting air quality. An AQI value of 50 or below represents good air quality, while an AQI value over 150 represents unhealthy air quality. Suppose that the distribution of AQI scores in the 500 highest-income counties is roughly normal with a mean of 50, while the distribution of AQI scores in the 500 lowest-income counties is roughly normal with a mean of 100. The figure below represents the distributions of AQI scores in these high- and low-income counties.

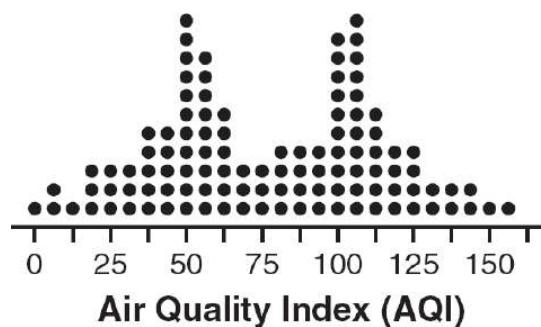


For an environmental study, 110 of these 1,000 counties will be randomly selected. Two methods are under consideration.

Method I: Take an SRS from among the 1,000 low- and high-income counties.

Method II: Randomly select either the low- or high-income counties and take an SRS of those 500 counties.

- Will a sample obtained using Method II be representative of the 1,000 counties? Explain.
- Which of the two sampling methods will result in a sample with less variability in AQI? Explain.
- The figure below represents the AQI of 110 counties obtained by using one of the two methods. Explain which method was most likely used.



[Follow for extended description](#)

The answers for this quiz can be found beginning on [page 473](#).

Quiz 6

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. Twenty obese men volunteer for a 2-month weight loss study. For one trial, while eating dinner the volunteers watched television episodes about morbidly obese individuals confronting lifelong emotional and physical struggles. For a second trial, while eating dinner the volunteers listened to heavy metal music. Each trial lasted for 1 month, and each volunteer completed both trials in a randomized order. The researchers noted each volunteer's weight loss or weight gain after each trial. Which of the following best describes this study?
 - (A) A completely randomized experiment
 - (B) A randomized matched pairs experiment
 - (C) A randomized placebo-controlled experiment
 - (D) An observational study

2. Which of the following is best when trying to minimize *response bias*?
 - (A) Picking a smaller sample
 - (B) Picking a larger sample
 - (C) Carefully wording the survey questions
 - (D) Increasing the number of survey questions

3. When is it preferable to use stratified random sampling rather than simple random sampling?
 - (A) The population can be divided into heterogeneous strata where each stratum looks roughly like the entire population.

- (B) The population can be divided into homogeneous strata where all individuals in a given stratum have a characteristic in common.
 - (C) The population can be divided into equal-size strata and every individual has an equal chance of being put in any strata.
 - (D) The population can be divided into a small number of strata, each containing a large number of individuals.
4. A nutritionist believes that overeating before an exam leads to lower test scores. He conducts an observational study in which students taking a first-period exam self-report on what they had for breakfast. He finds a negative correlation of $r = -0.21$ between calories eaten for breakfast and exam scores, and the nutritionist concludes that overeating causes lower test scores. Why is such a conclusion incorrect?
- (A) The correlation is too far from -1 to support such a conclusion.
 - (B) There is no indication that this was a random sample of students.
 - (C) Only $(-0.21)^2 = 4.41\%$ of the variability in test score is explained by the linear model.
 - (D) Causation cannot be determined from an observational study.
5. Which of the following is a correct statement about random sampling and random assignment?
- (A) Both random sampling and random assignment allow for generalization from samples to populations.
 - (B) Random sampling allows for generalization from a sample to a population, while random assignment minimizes the effect of possible confounding variables.
 - (C) Random sampling minimizes the effect of possible confounding variables, while random assignment allows for generalization from samples to populations.
 - (D) Both random sampling and random assignment minimize the effect of possible confounding variables.
6. At a certain college, students can select whether to live in a dormitory or to live off campus in a rented apartment. A college administrator randomly picks two dormitories and then selects a random sample of students leaving

each of those dormitories one morning to survey. The resulting sample of students are asked whether or not they were happy with the college dormitories. The administrator later publishes a report saying that 93% of the students attending the college are happy with the college dormitories. Which of the following statements is true?

- (A) This study suffers from undercoverage bias.
- (B) Because of the use of randomization, the 93% statistic is an accurate estimate of the population parameter.
- (C) Because no treatment was imposed in this study, no conclusion is possible about the percentage of college students who are happy with the dormitories.
- (D) The conclusion would have been more accurate if students from all the dormitories were surveyed.

7. A survey is to be conducted about alcohol use. Which of these questions is most likely to contribute to response bias?

- (A) How often do you drink beer?
- (B) Do you ever have two glasses of wine at a meal?
- (C) Do you ever drive after becoming intoxicated?
- (D) Have you ever called a ride share service after drinking?

8. A sociologist plans to survey 30 college tennis players about whether or not they plan to turn pro at some point. On a randomly selected day, there were 50 teams participating, each with 6 players for a total of 300 players. The 50 teams come from 3 different conferences. Which of the following describes a cluster random sampling method?

- (A) Number the players 1 through 300, use a random number generator to pick 30 unique numbers, and survey the players with numbers corresponding to the chosen integers.
- (B) Randomly select 5 teams out of the 50, and choose all 6 players from each of the 5 teams to participate in the survey.
- (C) Randomly pick 10 players from each of the 3 conferences to participate in the survey.

- (D) Number the players 1 through 300, and randomly pick a number between 1 and 10 to start with. Choose the corresponding player and every 10th player from that point forward to participate in the survey.
9. Five soil samples were taken within 50 feet of a dump site, and 5 more soil samples were taken over 100 feet away from the dump site. Measurements were taken of the concentration of a particularly dangerous contaminant. What type of study was conducted, and what was the response variable?
- (A) This was an observational study, and the response variable was the concentration of the contaminant.
- (B) This was an observational study, and the response variable was the distance from the dump site.
- (C) This was an experiment, and the response variable was the concentration of the contaminant.
- (D) This was an experiment, and the response variable was the distance from the dump site.
10. A survey was designed to collect information about employees' satisfaction with regard to overtime pay. Which of the following describes a procedure in which nonresponse bias is likely?
- (A) The survey question asked, "Do you feel employees are greedy and unrealistic with regard to overtime pay?"
- (B) Surveys were handed out to and completed by all the employees in the 8-to-5 work schedule.
- (C) Surveys were completed by a random sample taken of roughly 10% of the employee base.
- (D) Surveys were mailed to all employees, and 50% were returned.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. A company has 50 employees, 40 blue-collar workers and 10 white-collar workers, and all workers belong to a company union. The union plans to form a 5-person committee to represent the union in negotiations for a new contract.
 - (a) For sake of convenience, the union is considering picking an SRS. Describe a process for choosing a simple random sample of 5 people from among the 50 employees.
 - (b) An alternative sampling method would be to select a stratified random sample where the two strata are the blue-collar workers and the white-collar workers. In the context of this situation, give a statistical advantage of selecting a stratified random sample as opposed to an SRS.
2. Researchers performed a study to determine if increasing daily protein ingestion contributes to gaining lean body mass, muscle strength, and physical test performance in healthy student athletes at the college level. Thirty student athlete volunteers were randomly assigned to 3 groups, each group following the identical set diet. Each day for 5 weeks, participants in group 1 were given a supplement of 1 gram of protein per kilogram of body weight, participants in group 2 were given a supplement of 2 grams of protein per kilogram of body weight, and participants in group 3 received no protein supplement.
 - (a) Was this an experiment or an observational study? Explain.
 - (b) Why was a group included that received no supplement?
 - (c) Is it reasonable to generalize conclusions from this study beyond the 30 participants? Explain.
3. A grocery store manager decides to do a survey to see if passing out a free piece of candy to the first 50 customers is more of an inducement to purchase candy than offering a storewide sale on candy. On a randomly chosen day, he surveys the first 100 customers who enter the store by asking the following question, “Some managers believe that customers arriving early at the store deserve a reward. Would you rather receive a free piece of candy for arriving early, or would you rather the store offer a storewide sale on candy?”

- (a) Explain how bias may have been introduced based on the way the convenience sample was selected, and suggest a way of selecting the sample to avoid this bias.
- (b) Explain how bias may have been introduced based on the way the question was worded, and explain how this bias could have been avoided.

4. There are approximately 1,800 public colleges and 4,000 private colleges in the United States. A survey is to be taken of the students of U.S. colleges, and two sampling methods have been proposed.

Method I: Randomly select 25 colleges from among the 5,800 colleges, and survey all the students at those colleges.

Method II: Randomly select 2,000 students from the public colleges and 4,500 students from the private colleges to survey.

- (a) Identify the method that results in a stratified random sample, and identify the strata.
- (b) Identify the method that results in a cluster random sample, and identify the clusters.
- (c) Give one statistical advantage and one statistical disadvantage of using Method I.
- (d) Give one statistical advantage and one statistical disadvantage of using Method II.

5. A large company is investigating ways of promoting greater productivity. Two programs they are considering are to provide at the beginning of the workday either short professional-led exercise sessions or short professional-led meditation sessions. One hundred employees volunteer to participate. Their productivity will be measured before and after a 2-month period of sessions.

- (a) Some of the volunteers who are friends ask to be placed into the same program. Give an example of why this would not be a good idea as far as measuring the success of one or the other program.
- (b) What advantage would there be to include a control group?
- (c) Will it be reasonable to generalize any conclusion from this study?

The answers for this quiz can be found beginning on [page 475](#).

Experimental Design

An experiment is performed on objects called *experimental units*. If the units are people, they are called *subjects*. Experiments involve *explanatory variables*, called *factors*, that are believed to have an effect on *response variables*. A group is intentionally treated with some *level* of the explanatory variable, and the outcome of the response variable is measured.

Example 1.44

In an experiment to test exercise and blood pressure reduction, volunteers are randomly assigned to participate in 0, 1, or 2 hours of exercise per day for 5 days per week over the next 6 months.

TIP

If asked to list all the treatments, don't create ones that aren't there!

NOTE

When there is more than one factor, the number of treatments is the product of the number of levels for each factor.

- (a) What is the explanatory variable with corresponding levels, and what is the response variable?
- (b) Suppose the volunteers were further randomly assigned to follow either the DASH (Dietary Approaches to Stop Hypertension) or the TLC (Therapeutic Lifestyle Changes) diet for the 6 months. What would be the factors, the levels, and the treatments?

Solutions

- (a) The explanatory variable, hours of exercise, is being implemented at three levels: 0, 1, and 2 hours per day. The response variable is not specified but could be the measured change in either systolic or diastolic blood pressure readings after 6 months.
- (b) There would be two factors, hours of exercise with three levels and diet plan with two levels, and a total of six treatments (DASH diet with 0 hours daily exercise, DASH with 1 hour exercise, DASH with 2 hours exercise, TLC

diet with 0 hours daily exercise, TLC with 1 hour exercise, and TLC with 2 hours exercise).

The following are the four principles of experimental design.

1. In *comparison*, we wish to compare a response variable with regard to two or more experimental conditions (treatments):
 - If the purpose of an experiment is to determine whether some treatment has an effect, we compare to a control group that doesn't receive that treatment.
 - Depending on the experiment, a control group may be given an inactive treatment (a placebo), an established active treatment, or no treatment at all.
 - *Blinding* occurs when the subjects don't know which of the different treatments (such as placebos) they are receiving. When blinding is used, subjects in the treatment and control groups experience the placebo effect equally.
 - *Double-blinding* is when neither the subjects nor the response evaluators know who is receiving which treatment.
2. In *random assignment*, we use chance to randomly assign treatments to experimental units:
 - The goal is to create treatment groups, including the control group, that are similar in all respects except for the treatment imposed.
 - This potentially allows for cause-and-effect conclusions. On the exam, it is not enough to say "use random assignment."
 - You must be able to carefully explain how the random assignment of treatments to subjects is to be accomplished. Don't forget to name which treatment is given to which randomly assigned group.
3. In *replication*, the larger the sample is, the more significant the observation:
 - The treatment should be repeated on a sufficient number of subjects so that real response differences are more apparent.
 - Natural variation in outcomes can be lessened by using more subjects. We want to rule out that observed differences can be due just to random chance.
 - Replication refers to having more than one experimental unit in each treatment group, not multiple trials of the same experiment.
4. *Control* allows us to minimize the sources of variation other than the treatments by making conditions as similar as possible for all treatment groups:

- Controlling extraneous sources of variation makes it easier to identify true differences among the treatment groups.
- We want to minimize the possibility of variables we are not interested in from affecting the response variable.
- Blocking (randomly assigning treatments within homogeneous blocks) allows for a comparison of treatments within each block without worrying about how different responses might be due more to characteristics of different groups rather than to the different treatments.

Example 1.45

A dermatologist conducted an experiment to test whether a particular medication could reduce hair loss in men. From over 1,000 men volunteering for the study, 240 were over the age of 60, and these men were chosen as the test subjects. The men were randomly assigned to 2 groups of 120 each. One group received the medication daily for 6 months, while the other group received a placebo for 6 months. An independent expert looked at before and after photos of every subject and visually determined the change in hair density.

- What question is the experiment trying to answer?
- What are the experimental units?
- Explain how the random assignment could have been performed.
- What were the treatments?
- What was the response variable?
- Explain how control with regard to gender and age played a role in this experiment.
- Did the experiment use blinding, and if so, why?
- Did the experiment use double-blinding? Whether it did or did not, would double-blinding have been appropriate? Explain.

Solutions

- Does a particular medication reduce hair loss in men over the age of 60?
- The experimental units are the 240 chosen men.
- Number the chosen men from 1 to 240. Use a random number generator to pick integers between 1 and 240, throwing out repeats, until 120 unique integers have been picked. The men corresponding to these 120 integers will be the group given the medication, and the remaining 120 men will be the group given the placebo.
- The two treatments were the medication and the placebo.
- The response variable was the change in hair density.

- (f) The medication may work differently for women or for men under the age of 60. These extraneous variables might have affected the response variable. So, for control, the experiment was limited to men over the age of 60.
- (g) The use of a placebo is an example of blinding. The purpose here is to guard against the placebo effect, which refers to the fact that many people respond to any kind of perceived treatment.
- (h) It's not stated whether or not the independent expert knew which of the men were receiving the medication and which were receiving the placebo. It would be best if this was a double-blind experiment. If the independent expert knew who did and who did not receive the medication, that could have subconsciously influenced his judgment about hair density.

Two variables are *confounded* when both are associated with a response variable and are also associated with each other. Their effects on the response variable cannot be distinguished, and it may be impossible to tell which explanatory variable truly causes the response.

Example 1.46

In a random survey of adults, it is noted that those who regularly brush their teeth have fewer cavities. Explain how diet could be a confounding variable.

Solution

Those who regularly brush their teeth may also choose better diets. So, it might be unclear whether regular brushing or a better diet leads to fewer cavities. In this case, brushing and diet would be confounded.

Example 1.47

In an experiment to determine if two different fertilizers cause differences in plant growth, a horticulturist used many test plots containing either the first or the second fertilizer, with equal numbers of sunny and shady plots for each fertilizer, so that fertilizer and sunlight would not be confounded. Explain how watering could be a confounding variable.

Solution

If the two fertilizers require different amounts of watering, it might be difficult to determine if the difference in fertilizers or the difference in watering is the real

cause of an observed difference in plant growth. In this case, fertilizer and water would be confounded.

Example 1.48

Many are concerned that vaping produces the so-called gateway effect. Studies have shown that young people who vape are more likely to go on to smoke cigarettes than young people who do not vape.

- (a) What are the explanatory and response variables?
- (b) Are these observational studies or experiments?

Some researchers suggest that a confounding variable might be that many people feel that smoking in general is “cool.”

- (c) Explain what this means to be a confounding variable in the above context.
- (d) Explain how this confounding variable might lead researchers to conclude a higher or lower proportion of people become cigarette smokers because they vape than actually become cigarette smokers because they vape.

Solutions

- (a) The explanatory variable is whether or not a young adult vapes, and the response variable is whether or not the person becomes a cigarette smoker later in life.
- (b) These are observational studies. It would have been highly unethical to randomly select young adults and instruct them to vape as part of an experiment to see if they go on to cigarette smoking.
- (c) People who think smoking is “cool” are more likely to vape as young adults and are more likely to smoke cigarettes later in life. So, it may be that it’s not vaping that leads to cigarette smoking but, rather, that thinking smoking is “cool” leads both to vaping as a young adult and to smoking cigarettes later in life.
- (d) This confounding variable would lead researchers to conclude a higher proportion of people become cigarette smokers because they vape than actually become cigarette smokers because they vape, because some become cigarette smokers because they feel that smoking is “cool,” not because they vape.

In an experiment, there is often a *control group* to determine if the treatment of interest has an effect. There are several types of control groups. A control group can be a collection of experimental units not given any treatment, or given

the current treatment, or given a treatment with an inactive substance (a placebo). When a control group receives the current treatment or a placebo, these count as treatments if you are asked to enumerate the treatments.

Random assignment to the various treatment groups, including to the control group, can help reduce the problem posed by confounding variables. The beauty of random assignment is that it works on variables that you “know” to be confounders, variables that you “suspect” to be confounders, and variables you are clueless about or don’t even know they exist. Random assignment to treatments, called *completely randomized design*, is critical in good experimental design. This is especially true if the subjects are not randomly selected, as is the case in most medical and drug experiments.

Example 1.49

Sixty patients, ages 5 to 12, all with common warts are enrolled in a study to determine if application of duct tape is as effective as cryotherapy in the treatment of warts. Subjects will receive either cryotherapy (liquid nitrogen applied to each wart for 10 seconds every 2 weeks) for 6 treatments or duct tape occlusion (applied directly to the wart) for 2 months. Describe a completely randomized design.

Solution

Assign each patient an integer from 1 to 60. Use a random integer generator on a calculator to pick integers between 1 and 60, throwing away repeats, until 30 unique integers have been selected. The 30 patients corresponding to the 30 selected integers will be given the cryotherapy treatment. (A second design would be to put the 60 names on identical slips of paper, put the slips in a hat, mix them well, and then pick out 30 slips, without replacement, with the corresponding names given cryotherapy.) The remaining 30 patients will receive the duct tape treatment. At the end of the treatment schedules, compare the proportion of each group that had complete resolution of the warts being studied.

The goal is to control every variable we can so that all treatment groups have the same experience, save the one variable we are manipulating. Blinding and double-blinding are useful tactics to accomplish that, and a placebo is a useful tool in blinding.

A *matched pairs design* (also called a *paired comparison design*) is when two treatments are compared based on the responses of paired subjects, one of whom receives one treatment while the other receives the second treatment. Often the

paired subjects are really single subjects who are given both treatments, one at a time in random order.

Example 1.50

Does seeing pictures of accidents caused by drunk drivers influence one's opinion on penalties for drunk drivers? A comparison test is to be designed for 100 subjects.

- (a) One experimental design would be to ask all 100 subjects about drunk driving penalties before and then again after seeing the pictures, noting any change in answers. Give a weakness of this design.
- (b) Describe how to implement a completely randomized design.
- (c) Describe how to implement a matched pairs design using 50 sets of biological twins as the subjects.

Solutions

- (a) This would be a poor design because there is no control group, there is no use of randomization, and subjects might well change their answers because they realize that that is what is expected of them after seeing the pictures.
- (b) Use randomization to split the subjects into two groups, half of whom simply answer the questions while the other half first see the pictures and then answer the questions. For example, put the 100 names on identical slips of paper, put the slips in a hat, mix them well, and then pick out 50 slips (without replacement) with the corresponding names designating the subjects who see the pictures.
- (c) One of each set of twins is randomly picked (e.g., based on choosing an odd or even digit from a random number table) to answer the questions without seeing the pictures, while the other first sees the pictures and then answers the questions. The answers could be compared from each set of twins. This is a paired comparison test that might help minimize confounding variables due to family environment, heredity, and so on.

TIP

Use proper terminology! The language of experiments is different from the language of sampling. Don't mix up *blocking* and *stratification*.

Just as stratification, in sampling, first divides the population into representative groups called strata, *blocking*, in experimental design, first divides the subjects into representative groups called *blocks* based on similar characteristics. Subjects within *each* block are randomly assigned treatments. One can think of blocking as running parallel experiments before combining the results. This technique helps reduce variability as well as helps control certain possible confounding variables by bringing them directly into the picture. The paired comparison (matched pairs) design is a special case of blocking in which each pair can be considered a block of size 2.

Example 1.51

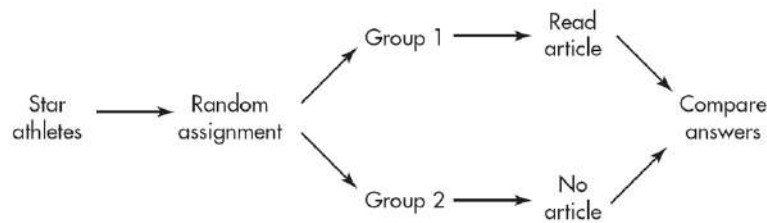
There is a rising trend for star college athletes to turn professional without finishing their degrees. A study is performed to assess whether reading an article about professional salaries has an impact on such decisions. Randomization can be used to split the subjects into two groups, and those in one group are given the article before answering questions. Suppose it is believed that freshman and sophomore athletes would be impacted differently than junior and senior athletes. How can a block design be incorporated into the design of this experiment?

Solution

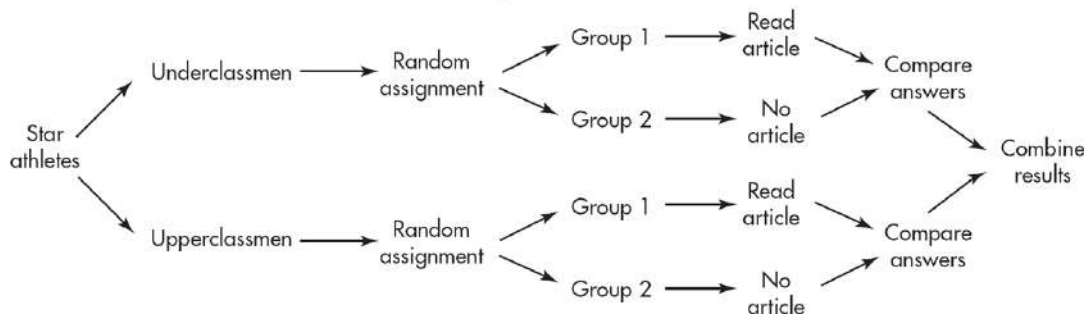
The subjects can be split into two blocks, underclass and upperclass, before using randomization to assign some to read the article before questioning. With this design, the impact of the salary article on freshmen and sophomores can be distinguished from the impact on juniors and seniors.

Similarly, blocking can be used to separately analyze men and women, those with high GPAs and those with low GPAs, those in different sports, those with different majors, and so on.

Completely Randomized Design



Block Design



TIP

Most experiments are not performed on random samples from a population. So, don't use *random sample* terminology when you mean *random assignment*.

NOTE

When the results are combined, this estimates the effect, separate from blocks, of the independent variable of interest.

What kind of conclusions result from well-designed experiments? Can we conclude that observed differences in the treatment groups are statistically significant; that is, are the differences so large as to be unlikely to have occurred by chance? How can we decide whether the differences are large enough?

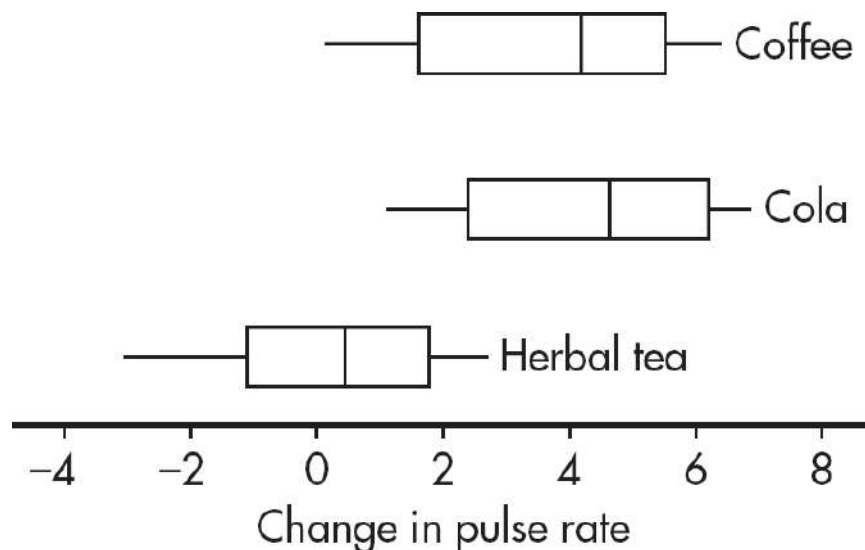
If the treatments make no difference, just by chance there probably will be some variation. In later units, you will learn mathematically how to decide if something is different enough to be considered statistically significant. However, for now, you should understand that something is statistically significant if the probability of it happening just by chance is so small that you're convinced there must be another explanation.

➤ Example 1.52

For a fair pair of dice, the probability of throwing “7 or 11” is approximately 22.2 percent. We would expect for the sum to come up 7 or 11 an average of 22.2 times for every 100 tosses if the dice are fair. Suppose we throw a pair of dice 100 times, and 7 or 11 comes up 26 times. This would appear to be within the bounds of random fluctuations, and we would have no reason to suspect the dice of being unfair. However, if 7 or 11 came up 55 times, that would raise our suspicions. While it would still be possible that the dice were fair, we could comfortably say that we had convincing evidence to believe that they were unfair. However, what if 7 or 11 came up 30 times? Is that different enough from 22.2 to raise our suspicions? We’ll learn how to answer this question later after studying probability distributions.

➤ Example 1.53

Sixty students volunteered to participate in an experiment comparing the effects of coffee, caffeinated cola, and herbal tea on pulse rates. Twenty students are randomly assigned to each of the three treatments. For each student, the change in pulse rate was measured after consuming eight ounces of the treatment beverage. The results are summarized with the parallel boxplots below.



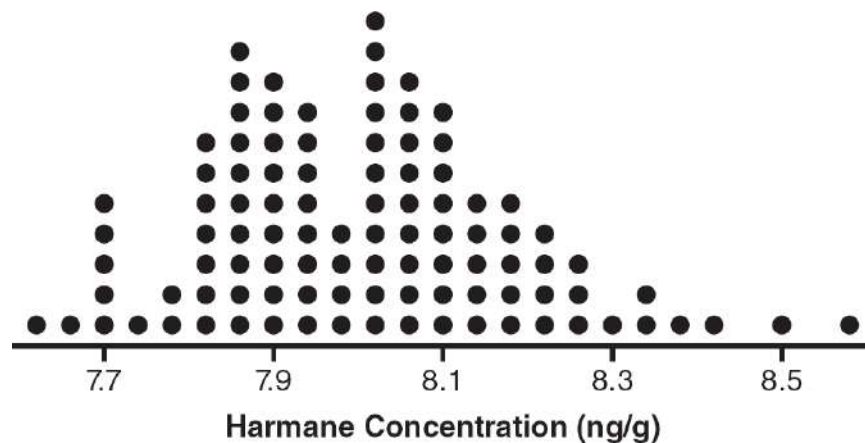
What are reasonable conclusions?

✓ Solution

The median change in pulse rate for the cola drinkers was higher than that for the coffee drinkers; however, looking at the overall spreads, that observed difference does not seem significant. The difference between the coffee and caffeinated cola drinkers with respect to change in pulse rate is likely due to random chance. Now compare the coffee and caffeinated cola drinkers' results to that of the herbal tea drinkers. Although there is some overlap, there is not much. It seems reasonable to conclude the difference is statistically significant; that is, drinking coffee or caffeinated cola results in a greater rise in pulse rate than drinking herbal tea.

➤ Example 1.54

Harmane is a potent neurotoxin linked to human disease and is found in a variety of cooked meats. Chicken in fast-food restaurants appears to be a particular problem. A fast-food chain claims that their cooked chicken has an 8 ng/g (nanograms per gram) concentration, but a nutritionist believes the true figure is greater. She randomly samples chicken nuggets on ten different days and obtains a mean harmane concentration of 8.48 ng/g. Is this statistically significant? The chain claims that this is possible just by chance sample variability. The chain's scientists simulate 100 such samples of size 10 from a laboratory setting where the known mean concentration is 8 ng/g in order to show that 8.48 ng/g is possible and not unusual. The resulting 100 means are summarized in the following dotplot:



Does the simulation support the nutritionist or the chain?

✓ Solution

The simulation shows that although a sample with a mean 8.48 ng/g concentration is possible if the true mean is 8 ng/g, it is highly unlikely. The simulation supports the nutritionist.

Quiz 7

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. Which of the following is not a primary element of experimental design?
 - (A) Random sampling
 - (B) Random assignment
 - (C) Control
 - (D) Replication

2. Which name describes an experiment where neither the subjects nor the response evaluators know which subjects are in the treatment group and which are in the control group?
 - (A) Double-sided
 - (B) Double-sample
 - (C) Double-blind
 - (D) Double-origin

3. What is the name of the beneficial effect experienced by some patients due to the patients' belief in the treatment?
 - (A) Ambiguous effect
 - (B) Mistaken effect
 - (C) Placebo effect
 - (D) Psychological effect

4. Randomized experiments allow for cause-and-effect conclusions between ____ and ____.

- (A) subjects; treatments
 - (B) subjects; response variables
 - (C) statistical significance; statistical confidence
 - (D) explanatory variables; response variables
5. Suppose you wish to compare the average class size of mathematics classes to the average class size of English classes in your high school. Which is the most appropriate technique for gathering the needed data?
- (A) Census
 - (B) Sample survey
 - (C) Experiment
 - (D) Observational study
6. Two studies are run to compare the experiences of families living in high-rise public housing to those of families living in townhouse subsidized rentals. The first study interviews 25 families who have been in each government program for at least 1 year, while the second randomly assigns 25 families to each program and interviews them after 1 year. Which of the following is a true statement?
- (A) Both studies are observational studies because of the time period involved.
 - (B) Both studies are observational studies because there are no control groups.
 - (C) The first study is an observational study, while the second is an experiment.
 - (D) The first study is an experiment, while the second is an observational study.
7. A consumer product agency tests miles per gallon for a sample of automobiles using each of four different octanes of gasoline. Which of the following is true?
- (A) There are four explanatory variables and one response variable.
 - (B) There is one explanatory variable with four levels of response.
 - (C) Miles per gallon is the only explanatory variable, but there are four response variables corresponding to the different octanes.
 - (D) There are four levels of a single explanatory variable.

8. A study is made to determine whether taking AP Statistics in high school helps students achieve higher GPAs when they go to college. In comparing records of 200 college students, half of whom took AP Statistics in high school, it is noted that the average college GPA is higher for those 100 students who took AP Statistics than for those who did not. Based on this study, guidance counselors begin recommending AP Statistics for college-bound students. Which of the following is *incorrect*?
- (A) Although this study indicates a relation, it does not prove causation.
 - (B) There could well be a confounding variable responsible for the seeming relationship.
 - (C) Self-selection here makes drawing the counselors' conclusion difficult.
 - (D) A more meaningful study would be to compare a simple random sample (SRS) from each of the two groups of 100 students.
9. A cardiologist surveyed a random sample of adults aged 21 to 75, looking at whether or not they reported exercising regularly and then used an echocardiogram to measure heart strength of those surveyed. She found a strong positive association. Which of the following is a proper conclusion?
- (A) This was a randomized experiment, which gave strong evidence that regular exercise causes greater heart strength.
 - (B) This was an observational study, which proved that regular exercise causes greater heart strength.
 - (C) This was an observational study, which gave strong evidence that regular exercise causes greater heart strength.
 - (D) It is likely that there is a confounding variable.
10. In designing an experiment, blocking is used
- (A) to reduce bias.
 - (B) to reduce variation.
 - (C) as a substitute for a control group.
 - (D) as a first step in randomization.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. For each of the following, would you recommend using random sampling, random assignment, both, or neither? Explain.
 - (a) A movie critic wants to investigate whether a greater proportion of teenagers in public or private schools have seen all three of the *Back to the Future* films.
 - (b) A company CEO wants to investigate whether giving hourly pay raises or end-of-the-year bonuses leads to greater productivity on an assembly line.
 - (c) A sportscaster wants to investigate whether the Chicago Cubs or the Chicago White Sox won a greater number of interleague games during the shortened 2020 season.

2. Rock and pop fame is associated with risk-taking and premature mortality. A study following the lives of 100 rock and pop stars periodically asked them about their frequency of substance abuse and followed their lives until death. Researchers noted the age of death and noticed that a greater amount of substance abuse was associated with an earlier age at death.
 - (a) What are the explanatory and response variables?
 - (b) Is this a prospective observational study, a retrospective observational study, or an experiment? Explain.
 - (c) Does this study show that greater frequency of substance abuse leads to early death?

3. Much of the profit in movie theaters comes from sales at the concession stands. A theater manager wants to run an experiment to evaluate whether subliminal advertising will help to increase sales. He has software that flashes “BUY POPCORN” on the screen for a fraction of a second.
 - (a) What are the explanatory and response variables?
 - (b) Is this a prospective observational study, a retrospective observational study, or an experiment? Explain.
 - (c) Should blinding and/or double-blinding or neither be incorporated? Explain.
 - (d) Should random sampling or random assignment be incorporated? Explain how.
 - (e) Why is it important to have some kind of control with regard to which days the subliminal message takes place?

(f) In context, explain how a cause-and-effect conclusion will or will not be justified.

4. A six-month study looked at 800 adult volunteers who were randomly divided into four groups.
- The first group wore a clip-on fitness tracker and received a cash incentive if they walked more than 50,000 steps a week.
 - The next group also wore an activity tracker and received a cash incentive for walking more than 50,000 steps a week. However, they were required to donate the cash to charity.
 - A third group wore the activity tracker but did not receive a cash incentive.
 - The last group did not wear a fitness tracker or get a cash incentive.

The measured outcome was “change in blood pressure over a six-month period.”

- (a) Was this an observational study or an experiment? Explain.
- (b) Identify the treatments, experimental units, and response variable.
- (c) What is the fourth group called, and what is the purpose of including it?
- (d) Describe an appropriate method for assigning the subjects to the four groups so that each group will have an equal number of subjects.
- (e) Is it reasonable to generalize the results of the study beyond the 800 participants? Explain why or why not.

The answers for this quiz can be found beginning on [page 477](#).

Quiz 8

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. Some diseases have developed resistance to medications that were once commonly used to treat them. A dermatologist conducted an experiment on a form of chronic eczema that had become increasingly difficult to treat.

From a sample of 20 teenagers with the disease, he treated 10 with a new calcineurin inhibitor for 5 days and treated the remaining people with a standard treatment of topical corticosteroids for 5 days. He had all 20 teenagers rate their itch intensity on a scale of 0 to 10 before and after their respective treatments. What were the experimental units of this study?

- (A) All teenagers with chronic eczema
- (B) The 5 days of treatment
- (C) The sample of 20 teenagers
- (D) The new calcineurin inhibitor and the topical corticosteroid

2. A new strain of flower is being tested for THC content. Fifty flowers of the strain are randomly assigned to be analyzed when either dried or fresh. Which of the following would provide the best evidence that the average THC content of the flower is different when dried or fresh?

- (A) The variation in THC content among the dried flowers is different from the variation in the THC content among the fresh flowers.
- (B) The variation in the THC content within each group is small compared with the variation in the THC content between the two groups.
- (C) The variation in the THC content within each group is large compared with the variation in the THC content between the two groups.
- (D) The variation in the THC content within each group is the same as the variation in the THC content between the two groups.

3. An experiment was conducted to investigate whether teenagers can stand longer on one foot with their eyes open or closed. Thirty students from a teacher's math classes volunteered to participate. Drawing names out of a hat resulted in two groups of equal size, one to stand with eyes open and the other with eyes closed. The mean number of seconds volunteers could stand on one foot with their eyes closed was calculated to be shorter than with eyes open. Which of the following elements of a well-designed experiment is missing?

- I. Random assignment of treatments to experimental units
- II. Comparison of at least two treatment groups
- III. Replication

- (A) I only
- (B) II only

- (C) III only
- (D) None of the above elements are missing.

4. A game designer plans to test whether any of three different background music genres leads to improved game enjoyment. Sixty gamers who actively play Minecraft volunteer for the study and will be randomly assigned to 3 groups of 20 volunteers each. Three musical genres will be randomly assigned to be piped in through headphones to the volunteers so that each group of 20 volunteers receive a different genre for 1 hour while playing Minecraft. Before and after the session, everyone rates their enjoyment of playing Minecraft, and changes are recorded. Why is this considered to be a completely randomized design?
- (A) Each of the music genres is a block, and the volunteers were randomly assigned to the blocks.
 - (B) Each of the volunteers is a block with randomly assigned music piped in through his or her headphones.
 - (C) There is no blocking variable, and the music genres will be randomly assigned to the volunteers.
 - (D) There is no blocking variable, and the volunteers were selected at random.
5. A study was conducted to evaluate the effectiveness of a topical cream for relief of arthritis. Sixteen volunteers suffering from moderate arthritis rubbed the cream on painful joints for 10 days. Twelve of the volunteers reported some pain relief. The company concluded that the topical cream was more effective than no treatment for moderate arthritis 75% of the time. Which of the following best explains why the company's conclusion is not supported by this study?
- (A) The study had no control.
 - (B) The study used volunteers.
 - (C) The sample size was too small.
 - (D) The volunteers self-reported pain relief.
6. An experiment will be conducted to determine whether first graders do better on a standardized aptitude test if a parent is sitting beside them. Participating children will be randomly assigned to take the test alone or with a parent sitting beside them. Because girls generally mature earlier

than boys, it is felt that gender might influence the outcome, so blocking by gender will be used. Explain why it will be impossible to incorporate blinding in this experiment.

- (A) The child will know whether he or she is a boy or a girl.
- (B) The child will know whether or not a parent is sitting beside him or her.
- (C) The person running the experiment will know whether a child is a boy or a girl.
- (D) The person running the experiment will know whether or not a parent is sitting beside a child.

7. A study is conducted to determine which of two seamstresses can sew a better dress. They each sew a dress using their own pattern, material, and sewing machine. Both seamstresses then use the same questionnaire passed out to a random sample of their customers who view and evaluate their final product. Answers on the questionnaires were then compared. Which of the following is not true about this study?

- (A) Because the seamstresses used their own patterns, choice of pattern is confounded with who can sew a better dress.
- (B) Because the seamstresses used their own machines, machine quality is confounded with who can sew a better dress.
- (C) Because the seamstresses have different populations of customers, customer population is confounded with who can sew a better dress.
- (D) Because the seamstresses used the same questionnaire, the conclusions can be generalized to the population of their combined customer bases.

8. A school is trying to evaluate two SAT prep review books. Two-hundred twenty of the school's students are signed up to take the SAT exam in 2 months. The school randomly assigns 110 of these students to receive a free copy of one of the review books and gives the other 110 students a free copy of the other review book. After the exam, the school notes that the students given the first review book averaged 1470 for the combined SAT score, while those given the second review book averaged 1250. Based on this, the school orders copies of the first review book for all of its students planning to take the next SAT exam. One of those students ends up scoring 1255. Which of the following is the best explanation of the difference

between that student's score and the score reported in the study for those using that review book?

- (A) The reported 1470 result is an average; there could have been a wide spread of exam scores for students using either of the review books.
- (B) The student probably used the review book for the other group either intentionally or by mistake.
- (C) This indicates some form of bias in the original experiment.
- (D) This indicates a confounding variable in the original experiment.

9. In search for the perfect chocolate chip cookie, 16 separate batches were prepared using white or brown sugar, butter or vegetable oil, all-purpose or bread flour, and semisweet or milk chocolate chips. Each of the batches was divided into three parts, with the three parts randomly assigned to be baked immediately, chilled for 1 hour before baking, or chilled for 24 hours before baking. The cookies were then judged on a 1 to 10 scale as to whether they were chewy and soft with a crisp exterior. What were the treatments in this experiment?

- (A) The 18 batches
- (B) The choices of sugar, oil, flour, and chips
- (C) Being baked immediately, chilled for 1 hour before baking, or chilled for 24 hours before baking
- (D) Chewy interior, soft interior, and crisp exterior

10. A randomized block design tests the effect of three teaching methods on student performance. Which of the following should guide the formation of blocks?

- (A) Students in each block should be as similar as possible, for example, similar IQ scores.
- (B) Students in each block should be as different as possible, for example, different IQ scores.
- (C) Students should be randomly assigned to the blocks.
- (D) Students in the same block should receive the same teaching method.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. A medical researcher plans to test a new medication to improve focus and reduce impulsivity in teens with ADHD (attention deficit hyperactivity disorder). The researcher has recruited 25 pairs of teens, each pair with the same age, same gender, same ADHD diagnosis (inattentive, hyperactive-impulsive, or combined), and same severity of symptoms. Each teen will receive either the new medication or a placebo for 1 month. At the end of the month, each teen will be evaluated as to lessening of symptoms.
 - (a) Identify the treatments, experimental units, and response variable.
 - (b) Describe a statistical advantage of using a matched pairs design rather than using a completely randomized design.
 - (c) Describe how the treatments can be randomly assigned using a matched pairs design.

2. A bike-riding school teaches children to ride on bikes with training wheels before teaching them to ride on bikes without training wheels. A study was conducted with 25 children to determine if first using a balance bike (remove the pedals) before using training wheels would be quicker. Each of 15 children participating in the study first trained on a balance bike for 1 lesson followed by using training wheels until they could ride without the training wheels. Each of the other 10 children participating in the study first trained on a balance bike for 2 lessons followed by using training wheels until they could ride without the training wheels.
 - (a) Was this an observational study or an experiment? Explain.
 - (b) When the study ended, it was clear that first training on a balance bike for 2 lessons followed by using training wheels was more successful than training on a balance bike for 1 lesson followed by using training wheels. Would the researchers be justified in saying that first training on a balance bike for 2 lessons followed by using training wheels is a quicker method of learning to ride a bike than just using lessons with training wheels? Explain.
 - (c) Does it matter how the 25 children were separated into the two groups? Explain.

3. For each of the following, would you recommend using random sampling, random assignment, both, or neither? Explain.
 - (a) A teacher wants to investigate whether the promise of bonus points or a chocolate chip cookie motivates students to study more for a test.

- (b) A sports statistician wants to investigate whether a greater proportion of high school students would rather attend a football game or a basketball game.
 - (c) A student wants to know if the average age of administrators is greater than the average age of teachers at her school.
4. There is a pressure point on the wrist that some doctors believe can be used to help control the nausea experienced following certain medical procedures. The idea is to place a band containing a small marble firmly on a patient's wrist so that the marble is located directly over the pressure point. A certain procedure involves a 2-day hospital stay because of severe nausea complications. A doctor plans to test the correct pressure point placement versus an improper marble placement on 50 patients. He will use a matched pair test where each patient is a block. Each day, patients will rate their nausea level on a scale of 1 to 10.
- (a) Identify the explanatory variable in this study and classify it as categorical or quantitative.
 - (b) Identify the response variable in this study and classify it as categorical or quantitative.
 - (c) Is this a prospective observational study, a retrospective observational study, or an experiment? Explain.
 - (d) Should blinding be incorporated? Explain.
 - (e) Should random sampling or random assignment be incorporated? Explain how.
 - (f) Why is it important for there to be some kind of control with regard to which days the correct pressure point is used?
 - (g) In context, explain how a cause-and-effect conclusion will or will not be justified.

The answers for this quiz can be found beginning on [page 479](#).

USE CORRECT TERMINOLOGY!

Bias: when a study systematically favors certain outcomes

Blinding: refers to subjects not knowing which treatment they are receiving

Block design: an experimental design where subjects are divided into representative groups to bring certain differences into the picture and reduce

variation (for example, blocking by gender, age, or race) and randomization then takes place within each block

Census: an attempt to include the entire population

Cluster random sample: involves dividing the population into heterogeneous groups called clusters and then picking everyone in a random sample of the clusters

Completely randomized design: an experimental design in which everyone has an equal chance of receiving any treatment

Confounding: when there is uncertainty as to which variable is causing an effect

Control group: a group given no treatment or a sham treatment

Double-blinding: refers to subjects and those evaluating their responses not knowing who received which treatments

Experimental study: involves applying a treatment to one or more groups and observing the responses

Nonresponse bias: when a large fraction of those sampled do not respond

Observational study: researchers merely observe (no treatments are applied)

Parameter: a numerical measurement describing some characteristic of the population

Placebo: a dummy or sham treatment such as a sugar pill

Population: the entire set of items, events, people, objects, and so on that are of interest

Random assignment: in experiments, when subjects are randomly assigned to treatments (evens out effects over which we have no control)

Random sampling: use of chance in selecting a sample from a population (allows for generalization of conclusions)

Response bias: when the question itself leads to misleading results (for example, people don't want to be perceived as having unpopular, unsavory, or illegal views)

Sample: the part of the population actually examined

Simple random sample (SRS): a sample selected in such a way that every possible sample of the desired size has an equal chance of being selected (each element of the population will also have an equal chance of being selected)

Statistic: a numerical measurement describing some characteristic of the sample

Statistical significance: a resulting difference among treatment groups too large to be attributed to chance

Stratified random sample: involves dividing the population into homogeneous groups called strata and then picking random samples from each of the strata

Systematic random sample: involves ordering the population, choosing a random point to start, and then picking every n th person for some n

Undercoverage bias: when part of the population is ignored (for example, some telephone surveys miss all those who only have cell phones)

Voluntary response bias: when individuals choose whether or not to respond (for example, radio call-in shows and Internet surveys)

Summary

- The four keys to describing a distribution are shape, center, spread (variability), and unusual features (such as outliers, clusters, gaps, and modes).
- Always provide context.
- Look for reasons behind any unusual features.
- A few common shapes that arise are symmetric, skewed to the right, skewed to the left, bell-shaped, uniform, unimodal, and bimodal distributions.
- Categorical variables are summarized by counts (frequency) and proportions (relative frequency) and visually displayed by bar charts (bar graphs) and pie charts.
- For quantitative data, dotplots, histograms, cumulative relative frequency plots (ogives), stemplots, and boxplots, give useful displays.
- In a histogram, relative area corresponds to relative frequency.
- The two principal measurements of the center of a distribution are the mean and the median.
- The principal measurements of the variability (spread) of a distribution are the range (maximum value minus minimum value), the interquartile range ($IQR = Q_3 - Q_1$), the variance, and the standard deviation.
- Adding the same constant to every value in a set adds the same constant to the mean and median but leaves all the aforementioned measures of

variability unchanged.

- Multiplying every value in a set by the same constant multiplies the mean, median, range, IQR, and standard deviation by that constant.
- The median and interquartile range are called resistant (or robust) measures because they are not sensitive to extreme values. The mean, range, variance, and standard deviation are called nonresistant (or nonrobust) measures because they are sensitive to extreme values.
- The principal measurements of position are simple ranking, percentile ranking, and the z -score (which measures the number of standard deviations a particular value is above or below the mean).
- No matter what the distribution of raw scores, the set of z -scores always has mean 0 and standard deviation 1.
- In skewed left data, the mean is usually less than the median, while in skewed right data, the mean is usually greater than the median.
- Boxplots show the five-number summary: the minimum value, the first quartile (Q_1), the median, the third quartile (Q_3), and the maximum value; they indicate outliers as distinct points. In general, boxplots cannot show overall shape; however, they may suggest symmetry and skewness.
- Two sets can have the same five-number summary and thus the same boxplots but have dramatically different distributions.
- Outliers are any values less than $Q_1 - 1.5(IQR)$ or greater than $Q_3 + 1.5(IQR)$. (An accepted second method of calculating outliers is to designate as outliers all values located more than two standard deviations above, or below, the mean.)
- When asked for a comparison of two data sets, in addition to comparing shape and unusual features, you must state which center and which spread is greater (two separate calculations are not a comparison).
- Advantages and disadvantages of four principal plots or graphical displays:
 - *Dotplot*: Advantages—shows every value, and it is easy to see the shape of the distribution. Disadvantage—difficult to plot for very large data sets.
 - *Stemplot*: Advantages—shows every value, and it is easy to see the shape of the distribution. Disadvantage—difficult to plot for very large data sets.
 - *Histogram*: Advantages—easy to make for large data sets, and it is easy to see the shape of the distribution. Disadvantage—does not show individual values.

- *Boxplot*: Advantages—quickly shows the five-number summary and outliers while splitting the data into quartiles. Disadvantages—does not show individual values and can hide features of the distribution, such as clusters, gaps, and even the overall shape.
- Retrospective studies look backward, examining existing data, while prospective studies watch for outcomes, tracking individuals into the future.
- Bias is the tendency to favor the selection of certain members of a population:
 - Convenience surveys (for example, interviews at shopping malls) are based on choosing individuals who are easy to reach. These surveys tend to produce data highly unrepresentative of the population.
 - Voluntary response bias occurs when individuals choose whether to respond (for example, Internet surveys) and typically give too much emphasis to people with strong opinions (particularly strong negative opinions).
 - Undercoverage bias is when part of the population is ignored (for example, some telephone surveys miss the homeless).
 - Response bias is when the question itself leads to misleading results (for example, people don't want to be perceived as having unpopular, unsavory, or illegal views).
 - Nonresponse bias is when a large fraction of those sampled do not respond.
- Use of randomization is absolutely critical in selecting an unbiased sample:
 - A simple random sample (SRS) is one in which every possible sample of the desired size has an equal chance of being selected.
 - Stratified random sampling involves dividing the population into homogeneous groups called strata and then picking random samples from each of the strata.
 - Cluster random sampling involves dividing the population into heterogeneous groups called clusters, randomly selecting one or more of the clusters, and then using everyone in the selected clusters.
 - Systematic random sampling involves ordering the population, randomly choosing a point to start, and then picking every n th person for some n .
- Use of randomization in selecting a sample is crucial in being able to generalize from a sample to the population.

- While observational studies involve observing without causing changes, experiments involve actively imposing treatments and observing responses.
- Differences in observed responses are statistically significant when it is unlikely that they can be explained by chance variation.
- Random assignment of subjects to treatment groups is extremely important in handling unknown and uncontrollable differences.
- Random assignment refers to what is done with subjects after they've been picked for a study, whereas random sampling refers to how subjects are selected for a study.
- Variables are said to be confounded when there is uncertainty as to which variable is causing an effect.
- The placebo effect refers to the fact that many people respond to any kind of perceived treatment.
- Blinding refers to subjects not knowing which treatment they are receiving.
- Double-blinding refers to subjects and those evaluating their responses not knowing who received which treatments.
- Completely randomized designs refer to experiments in which everyone has an equal chance of receiving any treatment.
- Blocking is the process of dividing the subjects into representative groups to bring certain differences into the picture (for example, blocking by gender, age, or race/ethnicity).
- Randomized block designs refer to experiments in which the randomization occurs only within blocks.
- Randomized paired comparison designs refer to experiments in which subjects are paired and randomization is used to decide who in each pair receives what treatment.

2

Probability, Random Variables, and Probability Distributions

(6–10 multiple-choice questions on the AP exam)

Learning Objectives

In this unit, you will learn how

- To be able to calculate frequencies and conditional frequencies from a two-way table.
- To be able to interpret side-by-side bar charts, segmented bar charts, and mosaic plots.
- To understand how the law of large numbers relates to relative frequencies.
- To be able to make basic probability calculations involving the complement, union, and intersection of events, including the use of the general addition rule.
- To understand when events are mutually exclusive and when they are independent and how that applies to probability calculations.
- To be able to calculate conditional probabilities.
- To be able to calculate reverse conditional probabilities and use a tree diagram if appropriate (multistage probability calculations).
- To be able to calculate and interpret the mean (expected value) and standard deviation of a random variable.
- To be able to calculate the probability of exactly x successes in n trials in a binomial scenario.
- To be able to calculate the probability of at least or at most x successes in n trials in a binomial scenario.
- To be able to calculate the mean (expected value) and standard deviation of a binomial probability distribution.
- To be able to calculate using the empirical rule applied to approximately normal distributions.
- To be able to find probabilities and percentiles using the normal distribution.
- To be able to find the value that corresponds to a given percentile when the distribution is approximately normal.

- To understand the concept of a sampling distribution.
- To understand the content and importance of the central limit theorem.

Two Categorical Variables

Qualitative data often encompass two categorical variables that may or may not have a dependent relationship. These data can be displayed in a *two-way table* (also called a *contingency table*).

▶ Example 2.1

The Cuteness Factor: A Japanese study had 250 volunteers look at pictures of cute baby animals, adult animals, or tasty-looking foods before testing their level of focus in solving puzzles.

		Level of focus			
		Low	Medium	High	Total
Pictures viewed	Baby animals	5	20	40	65
	Adult animals	30	40	15	85
	Tasty foods	55	35	10	100
Total		90	95	65	250

The row sums and column sums above are called *marginal frequencies*.

Instead of frequencies (counts), the entries in two-way tables are sometimes given as relative frequencies (proportions).

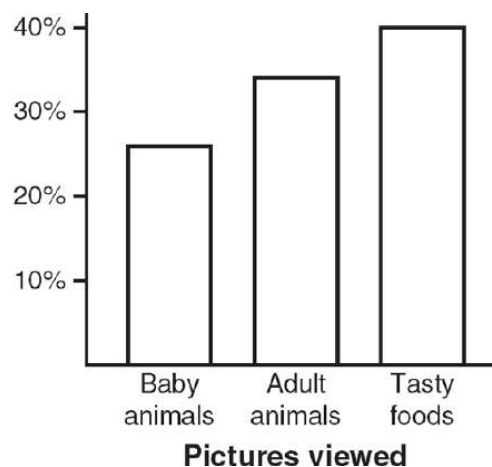
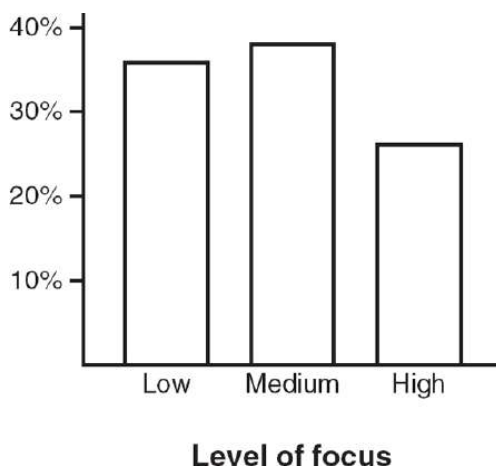
Determine the table of relative frequencies.

✓ Solution

Divide each cell's frequency by 250, the table total, to obtain:

		<i>Level of focus</i>			
		Low	Medium	High	<i>Total</i>
<i>Pictures viewed</i>	Baby animals	0.02	0.08	0.16	0.26
	Adult animals	0.12	0.16	0.06	0.34
	Tasty animals	0.22	0.14	0.04	0.40
<i>Total</i>		0.36	0.38	0.26	1.00

The row sums and column sums above are called *marginal relative frequencies*. The distributions of these marginal relative frequencies can be displayed in bar charts (bar graphs):



[Follow for extended description](#)

The relative frequency distributions described and calculated above do not describe or measure the relationship between the two categorical variables. For this we must consider the information in the body of the table, not just the sums in the margins.

➤ Example 2.2

We are interested in predicting the level of focus from the pictures viewed, and so we look at *conditional relative frequencies* for each row separately. In [Example 2.1](#), what proportion or percentage of the participants who viewed baby animals then had each of the levels of focus?

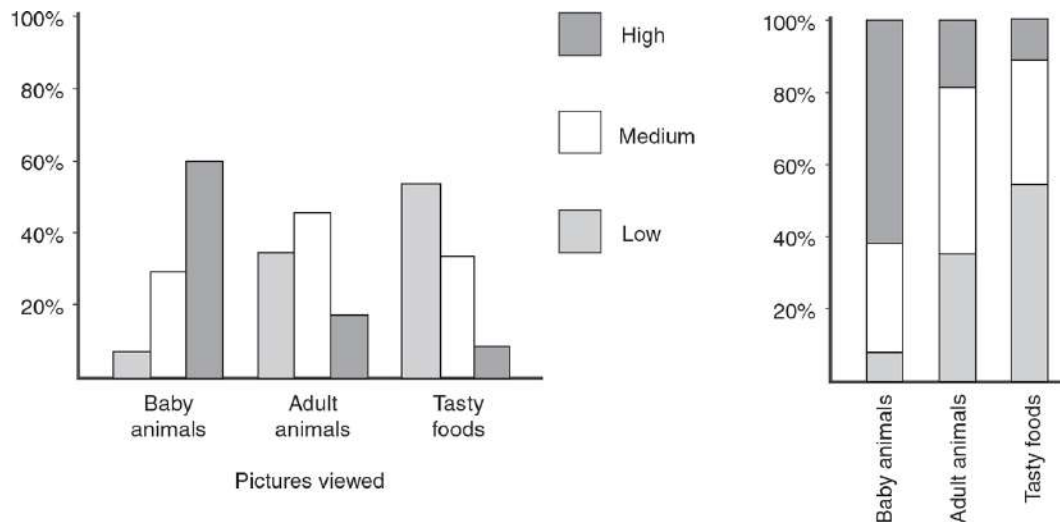
NOTE

Conditional relative frequencies are the relative frequencies for a specific part of the table, for example, the cell frequencies in a row divided by the total for that row or the cell frequencies in a column divided by the total for that column.

✓ Solutions

Low:	$\frac{5}{65} = 0.077 = 7.7\%$
Medium:	$\frac{20}{65} = 0.308 = 30.8\%$
High:	$\frac{40}{65} = 0.615 = 61.5\%$

This *conditional relative frequency distribution* can be displayed either with a side-by-side bar chart or by a *segmented bar chart* where each segment has a length corresponding to its relative frequency:



NOTE

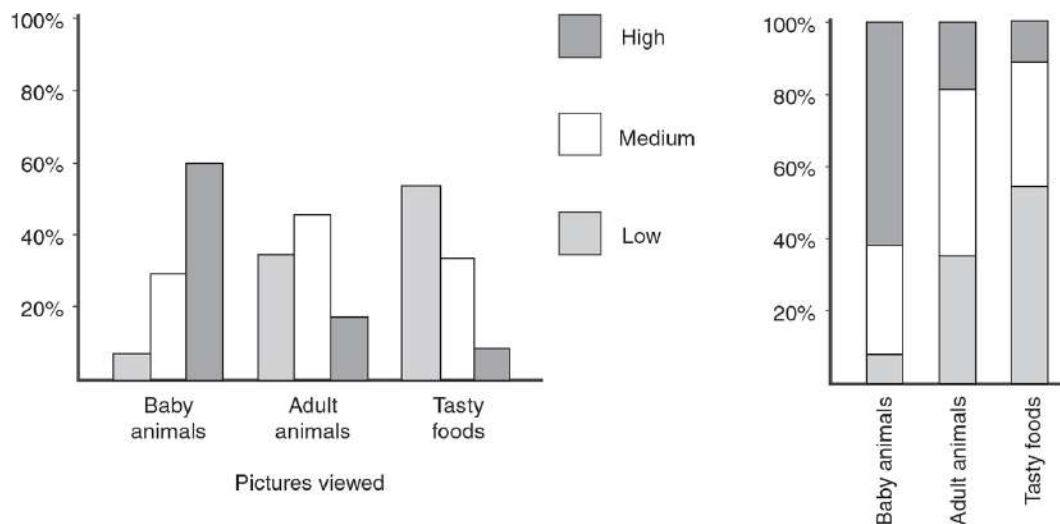
Segmented bar charts are like pie charts, except that the data set is represented by a rectangular bar rather than a circle.

Calculate the level of focus conditional relative frequency distributions for the participants who viewed adult animals and who viewed tasty foods.

✓ Solutions

Level of focus					
Participants who viewed adult animals	Low:	$\frac{30}{85} = 0.353 = 35.3\%$	Participants who viewed tasty foods	Low:	$\frac{55}{100} = 0.55 = 55\%$
	Medium:	$\frac{40}{85} = 0.471 = 47.1\%$		Medium:	$\frac{35}{100} = 0.35 = 35\%$
	High:	$\frac{15}{85} = 0.176 = 17.6\%$		High:	$\frac{10}{100} = 0.10 = 10\%$

Both of the following bar charts give good visual representations of the data:



NOTE

Because the distributions of focus levels are different for each picture viewed, we say that the two variables, focus level and pictures viewed, are **associated**.

What do the side-by-side bar charts and segmented bar charts indicate about an association between viewing pictures and levels of focus?

✓ Solution

The side-by-side bar charts and the segmented bar charts both clearly indicate that viewing pictures of baby animals tends to lead to higher levels of focus, viewing tasty foods tends to lead to lower levels of focus, and viewing adult animals tends to lead to medium levels of focus.

In summary:

- *Frequency* refers to *counts*.
- *Relative frequency* refers to *percentages*.
- *Marginal* refers to row or column *totals*.
- *Conditional* refers to a single row or column *category*.

➤ Example 2.3

As texting becomes a bigger part of our lives, emojis are an increasingly important part of our daily conversations. Are there generational differences? In one study, random samples of Baby Boomers (born 1946–1964), Generation X (born 1965–1980), Millennials (born 1981–1996), and Generation Z (born 1997–2012) were asked, “Your friend just sent you a hilarious message. How would you respond?” The data are tabulated in the following table:

	LOL			<i>Totals</i>
Baby Boomers	190	390	20	600
Generation X	210	750	40	1,000
Millennials	195	600	105	900
Generation Z	420	620	460	1,500
<i>Totals</i>	1,015	2,360	625	4,000

- What is the *marginal frequency distribution* of the generations?
- What is the *marginal frequency distribution* of the responses?
- What is the *marginal relative frequency distribution* of the generations?
- What is the *marginal relative frequency distribution* of the responses?
- What is the *conditional frequency distribution* among those choosing the  response?
- What is the *conditional frequency distribution* of responses among Generation Z?
- What is the *conditional relative frequency distribution* among those choosing the  response?
- What is the *conditional relative frequency distribution* of responses among Millennials?

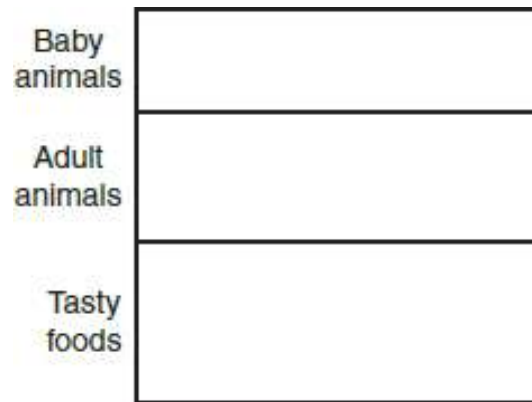
✓ Solutions

- This is the *counts* of the generations: 600 Baby Boomers, 1,000 Generation X, 900 Millennials, and 1,500 Generation Z.

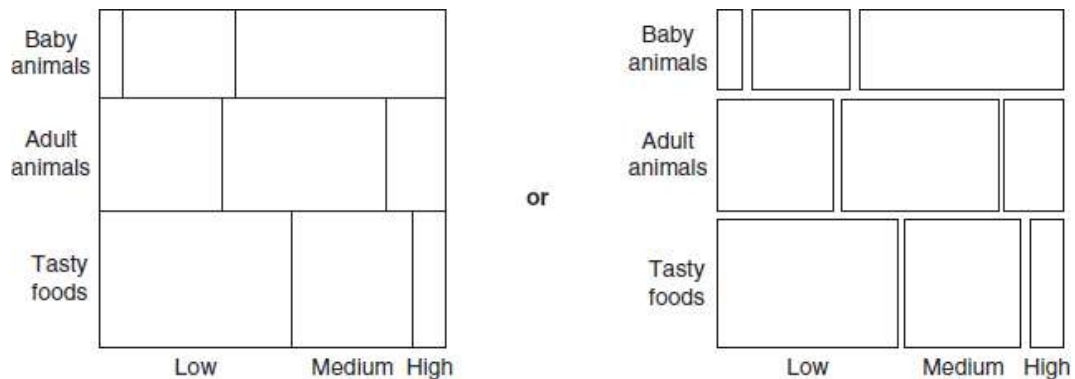
- (b) This is the *counts* of the responses: 1,015 LOL, 2,360 😊, and 625 🤔.
- (c) This is the *percentage* of the sample for each generation:
 $\frac{600}{4,000} = 15\%$ Baby Boomers, $\frac{1,000}{4,000} = 25\%$ Generation X, $\frac{900}{4,000} = 22.5\%$ Millennials, and $\frac{1,500}{4,000} = 37.5\%$ Generation Z.
- (d) This is the *percentage* of the sample for each response:
 $\frac{1,015}{4,000} = 25.4\%$ LOL, $\frac{2,360}{4,000} = 59\%$ 😊, and $\frac{625}{4,000} = 15.6\%$ 🤔.
- (e) This is the *counts* of the generations among those choosing the 😊 response: 390 Baby Boomers, 750 Generation X, 600 Millennials, and 620 Generation Z.
- (f) This is the *counts* of responses among Generation Z: 420 LOL, 620 😊, and 460 🤔.
- (g) Among those choosing the 🤔 response, this is what *percentage* are each generation: $\frac{20}{625} = 3.2\%$ Baby Boomers, $\frac{40}{625} = 6.4\%$ Generation X, $\frac{105}{625} = 16.8\%$ Millennials, and $\frac{460}{625} = 73.6\%$ Generation Z.
- (g) Among the Millennials, this is what *percentage* chose each response: $\frac{195}{900} = 21.7\%$ LOL, $\frac{600}{900} = 67.7\%$ 😊 and $\frac{105}{900} = 11.7\%$ 🤔.

➤ Example 2.4

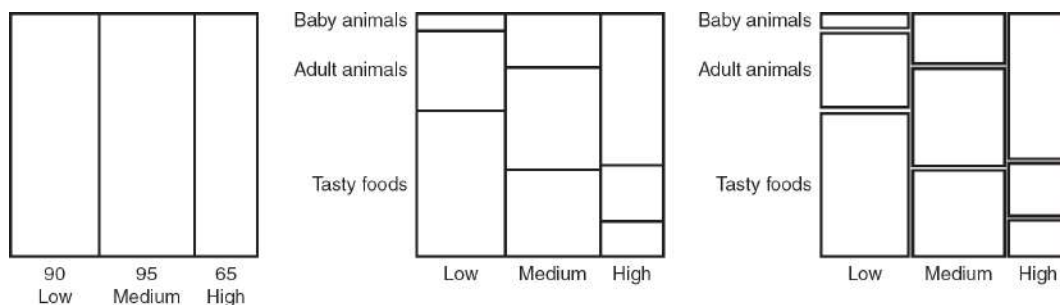
Mosaic plots are another graphical display that show the cell frequencies in a two-way table. They look like segmented bar charts in which the width of each rectangle is proportional to the number of individuals in the corresponding group. In [Examples 2.1](#) and [2.2](#), we can start with horizontal sections based on pictures viewed. The area of each rectangle is equal to the marginal relative frequency of that group (26%, 34%, and 40%).



Then, each horizontal section is split into sections based on focus level. The area of each smaller rectangle is equal to the conditional relative frequency for each specific group as calculated in [Example 2.2](#) (for example, 7.7%, 30.8%, and 61.5% for baby animals).



Alternatively, we could have first started with vertical sections based on level of focus.



➤ Example 2.5

Suppose you need heart surgery and are trying to decide between two surgeons, Dr. Fixit and Dr. Patch. You find out that each operated 250 times last year with the following results:

	Dr. F	Dr. P
Died	60	50
Survived	190	200

Which surgeon would you choose? Among Dr. Fixit's 250 patients, 190 survived, for a survival rate of $\frac{190}{250} = 0.76$ or 76%, while among Dr. Patch's 250 patients, 200 survived, for a survival rate of $\frac{200}{250} = 0.80$ or 80%. Your choice seems clear.

However, everything may not be so clear-cut. Suppose that on further investigation you determine that the surgeons operated on patients who were in either good or poor condition with the following results:

	Good condition			Poor condition	
	Dr. F	Dr. P		Dr. F	Dr. P
Died	8	17	Died	52	33
Survived	60	120	Survived	130	80

[Follow for extended description](#)

Note that adding corresponding boxes from these two tables gives the original table above.

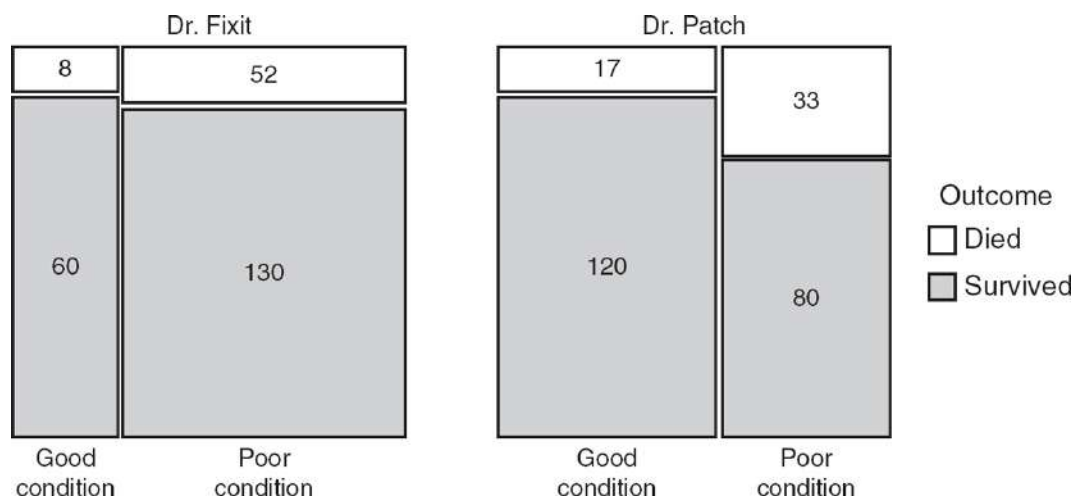
How do the surgeons compare when operating on patients in good health? In poor health?

Dr. Fixit's 68 patients in good condition have a survival rate of $\frac{60}{68} = 0.882$, or 88.2%, while Dr. Patch's 137 patients in good condition have

a survival rate of $\frac{120}{137} = 0.876$, or 87.6%. Similarly, we note that Dr. Fixit's 182 patients in poor condition have a survival rate of $\frac{130}{182} = 0.714$ or 71.4%, while Dr. Patch's 113 patients in poor condition have a survival rate of $\frac{80}{113} = 0.708$ or 70.8%.

Thus, Dr. Fixit does better with patients in good condition (88.2% versus Dr. Patch's 87.6%) and also does better with patients in poor condition (71.4% versus Dr. Patch's 70.8%). However, Dr. Fixit has a lower overall patient survival rate (76% versus Dr. Patch's 80%)! How can this be?

This problem is an example of *Simpson's paradox*, where a conjecture can be reversed when several different groups of data are combined to form a single group. The effect of another variable is masked when the groups are combined. In this particular example, closer scrutiny reveals that Dr. Fixit operates on many more patients in poor condition than Dr. Patch, and these patients in poor condition are precisely the ones with lower survival rates. Thus, even though Dr. Fixit does better with each group of patients, his overall rating is lower. Our original table hid the effect of the variable related to the condition of the patients. Mosaic plots help clarify the paradox:



Note how the above visually illustrates that Dr. Fixit operates on more patients in poor condition than in good condition, while Dr. Patch operates on more patients in good condition than in poor condition. It also illustrates that Dr. Fixit's patients, in both good condition and poor condition, survive at higher rates than Dr. Patch's patients.

Quiz 9

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

Questions 1–5 are based on the following: Students, teachers, and administrators were asked which of three teacher characteristics (*challenging, enthusiastic, strict*) they considered most important for a successful classroom experience. Five hundred people in a high school community were surveyed with the following results:

	Challenging	Enthusiastic	Strict	Total
Student	50	150	50	250
Teacher	125	50	25	200
Administrator	15	10	25	50
Total	190	210	100	500

1. What percentage of those surveyed were students?
(A) 10%
(B) 30%
(C) 40%
(D) 50%
2. What percentage of those surveyed picked *challenging* as most important and were teachers?
(A) 25%
(B) 38%
(C) 40%

(D) 62.5%

3. What percentage of administrators picked *strict* as most important?

- (A) 5%
- (B) 10%
- (C) 25%
- (D) 50%

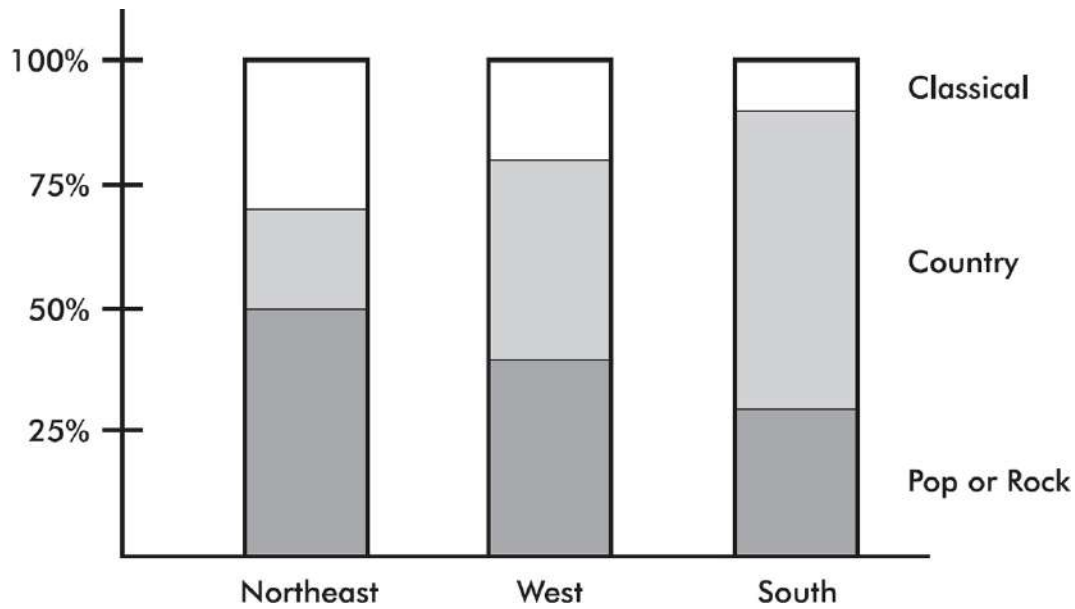
4. What percentage of those picking *enthusiastic* as most important were students?

- (A) 30%
- (B) 42%
- (C) 60%
- (D) 71.4%

5. Which group of people were most likely to pick *strict* as most important?

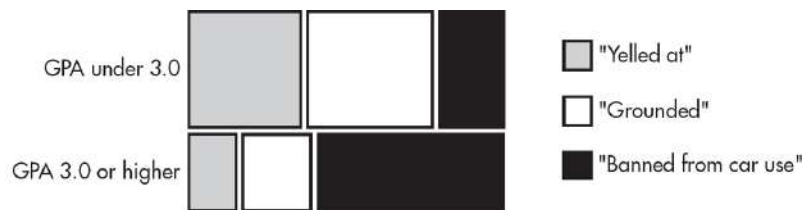
- (A) Student
- (B) Teacher
- (C) Administrator
- (D) Teacher and administrator, equally

Questions 6–7 are based on the following. A study of music preferences in three geographic locations resulted in the following segmented bar chart:



6. Which of the following is greatest?
- (A) The percentage of those from the Northeast who prefer classical is greatest.
 - (B) The percentage of those from the West who prefer country is greatest.
 - (C) The percentage of those from the South who prefer pop or rock is greatest.
 - (D) It is impossible to determine the answer without knowing the actual numbers of people involved.
7. Which of the following is greatest?
- (A) The number of people in the Northeast who prefer pop or rock is greatest.
 - (B) The number of people in the West who prefer classical is greatest.
 - (C) The number of people in the South who prefer country is greatest.
 - (D) It is impossible to determine the answer without knowing the actual numbers of people involved.
-

8. High school students were surveyed as to whether or not their GPA was 3.0 or higher and what was the most severe punishment they had ever received from their parents. The results are displayed in the following mosaic plot.



Based on the above plot, which of the following is a true statement?

- (A) The number of students with GPAs under 3.0 who were banned from car use is greater than the number of students with GPAs 3.0 or higher who were banned from car use.
- (B) More students had a GPA 3.0 or higher and were either yelled at or grounded than had a GPA under 3.0 and were yelled at.
- (C) Of the students who were grounded, a greater proportion had GPAs 3.0 or higher than GPAs under 3.0.
- (D) More students had a GPA under 3.0 than a GPA 3.0 or higher.
9. The average speed of a first serve in men's professional tennis is 120 mph. Assume that the server aims either for the corner or down the line. Assume the receiver moves toward the corner or the line depending on where he thinks the serve will go. The following table summarizes the directions taken by the server and the receiver over 500 serves.

		Server aims		
		For the corner	Down the line	Total
Receiver moves	Toward the corner	300	50	350
	Toward the line			

Toward the line	100	50	150
<i>Total</i>	400	100	500

Which of the following indicates an association between the server's choice and the receiver's choice?

- (A) Given a receiver's choice, the conditional relative frequencies of that choice are the same whichever the server's choice.
- (B) Given a receiver's choice, the conditional relative frequencies of that choice are different depending on the server's choice.
- (C) The marginal relative frequencies for the server and receiver are equal.
- (D) The marginal relative frequencies for the server and receiver are not equal.

10. The following table summarizes the relative frequencies from the data collected from a group of middle school and high school students about their favorite type of pet.

	Dogs	Cats	Other
Middle school students	0.20	0.10	0.05
High school students	0.30	0.25	0.10

Which of the following indicates an association between the school level and pet choice?

- (A) The conditional relative frequency of middle school students who preferred dogs is different from the conditional relative frequency of high school students who preferred dogs.
- (B) The conditional relative frequency of middle school students who preferred dogs is different from the relative frequency of middle school students who preferred cats.
- (C) The percent of the group who were high school students who preferred dogs is greater than the percent who were middle

school students who preferred dogs.

- (D) The percent of the group who were high school students who preferred a pet other than dogs or cats is twice the percent of the group who were middle school students who preferred a pet other than dogs or cats.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. For one study of racial disparities in traffic stops, 225 random records for drivers pulled over for speeding at night (when it is difficult to distinguish race/ethnicity at a distance) were analyzed. The following two-way table summarizes one aspect of the data.

	White drivers	Black drivers	Latino drivers	<i>Total</i>
Ticket	36	30	27	93
Warning	84	30	18	132
<i>Total</i>	120	60	45	225

- (a) Of the drivers pulled over for speeding who were given a ticket, what proportion were Latino?
- (b) Of the drivers pulled over for speeding who were White, what percent were given a warning?
- (c) The mosaic plot displays the distribution of tickets versus warnings given by race/ethnicity. Describe what this graph reveals about the association between these two variables for the 225 drivers pulled over for speeding.



2. In 1973, a question arose concerning possible gender discrimination in admissions to a particular graduate school. The graduate school analyzed admissions to the largest graduate programs, and below are partial results of that study:

Program	Men Accepted	Men Rejected	Women Accepted	Women Rejected
A	511	314	89	19
E	53	138	95	298

- Find the percentage of men and the percentage of women accepted by each of the two programs. Comment on any pattern or bias you see.
 - Find the percentage of men and the percentage of women accepted overall by these two programs. Does this appear to contradict the results from part (a)?
 - If you worked in the graduate admissions office, what would you say to an inquiring reporter who is investigating gender bias in graduate admissions?
3. To compare two review books with regard to success rates on the AP Statistics exam, researchers randomly sampled students who had purchased one of the two books before the exam. A score of 3 or higher generally results in college credit at most institutions. The following table summarizes the data.

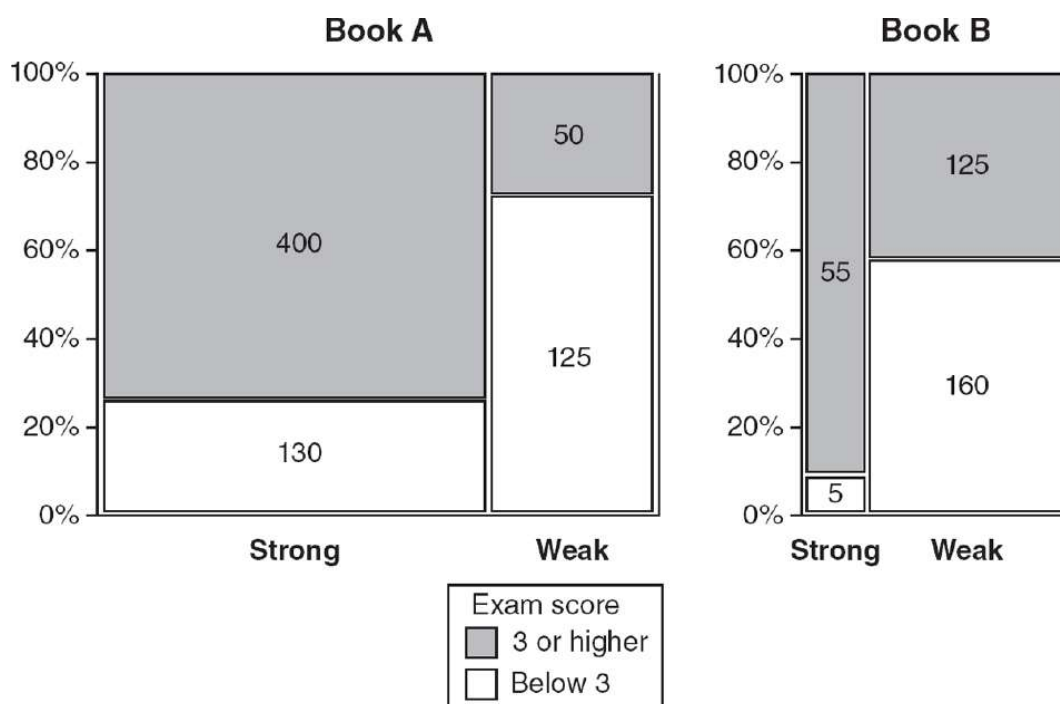
	Review book A	Review book B	<i>Total</i>
Scored 3 or higher	450	180	630
Scored below 3	255	165	420
<i>Total</i>	705	345	1,050

- (a) Complete the following table showing *relative frequencies* of scoring 3 or higher and of scoring below 3 for students using each of the review books.

	Review book A	Review book B
Scored 3 or higher		
Scored below 3		

- (b) Based on the above relative frequency table, using which book seems to lead to better results? Explain.
- (c) Can the researchers conclude that using the book you chose in (b) causes a better exam result? Explain.

An AP Statistics teacher believes that separating the data into results obtained by stronger and weaker students would lead to a more meaningful conclusion. She examines the data and constructs the following two mosaic plots (with Weak and Strong to note the student strength).



[Follow for extended description](#)

Based on the mosaic plots, the teacher concluded that book B leads to better results with both stronger and weaker students.

- (d) Give statistical calculations to support the teacher's conclusions.
- (e) Explain the apparent contradiction between the answers in (b) and (d).

The answers for this quiz can be found beginning on [page 481](#).

[The Law of Large Numbers](#)

In the world around us, unlikely events sometimes take place. At other times, events that seem inevitable do not occur. Although we are aware of the possible outcomes, we have no idea what an individual outcome will be. Chance is everywhere! The cards you are dealt in a poker game, the particular genes you inherit from your parents, and the coin toss at the beginning of a tennis match to determine who serves first are examples of chance behavior that mathematics can help us understand. Even though we may not be able to foretell a specific result, we can sometimes assign what

is called a *probability* to indicate the likelihood that a particular event will occur.

TIP

Calculators express very small probabilities in scientific notation such as 3.4211073E-6. Know what this means, and remember that probabilities are never greater than 1.

NOTE

An *event* is a collection of one or more outcomes.

- Probabilities are always between 0 and 1.
- A probability close to 0 means that the event is unlikely to occur.
- A probability close to 1 means that the event is likely to occur.
- The sum of the probabilities of all the separate outcomes of an experiment is always 1.

The *relative frequency* of an event is the proportion of times the event happened—that is, the number of times the event happened divided by the total number of trials. Relative frequencies may change every time an experiment is performed. The *law of large numbers* states that when an experiment is performed a large number of times, the relative frequency of an event tends to become closer to the true probability of the event; that is, *probability is long-run relative frequency*. There is a sense of predictability about the long run. The law of large numbers has two conditions:

The idea that a few outcomes will immediately be balanced by other outcomes is also called the *gambler's fallacy*.

1. The chance event under consideration does not change from trial to trial.
2. Any conclusion must be based on a large (a very large!) number of observations.

The law of large numbers says nothing at all about short-run behavior. There is no such thing as a law of small numbers or a law of averages. Gamblers might say that “red” is due on a roulette table, a basketball player is due to make a shot, a player has a hot hand at the craps table, or a certain number is due to come up in a lottery. However, if events are independent, the probability of the outcome of the next trial has nothing to do with what happened in previous trials. Even though casinos and life insurance companies might lose money in the short run, they make long-term profits because of their understanding of the law of large numbers.

Example 2.6

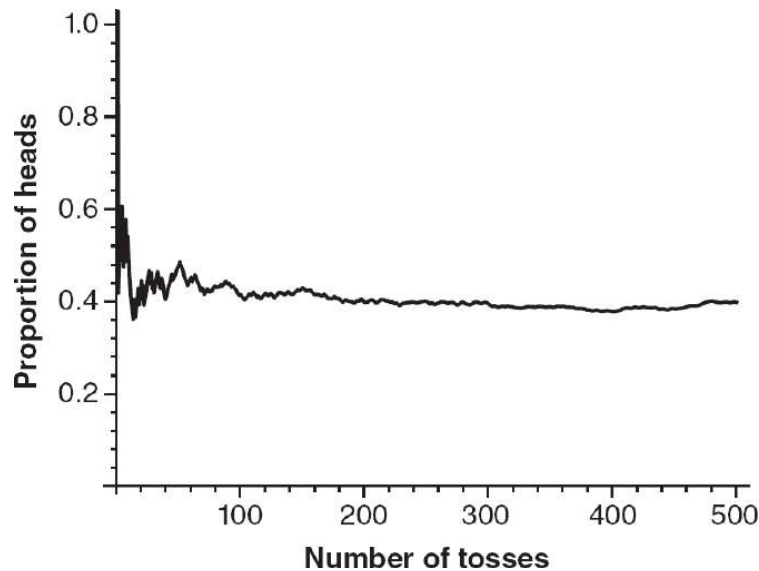
There are two games involving flipping a fair coin. In the first game, you win a prize if you can throw between 45% and 55% heads. In the second game, you win if you can throw more than 60% heads. For each game, would you rather flip 20 times or 200 times?

Solution

The probability of throwing heads is 0.5. By the law of large numbers, the more times you throw the coin, the more the relative frequency tends to become closer to this probability. With fewer tosses, there is greater chance of wide swings in the relative frequency. Thus, in the first game, you would rather have 200 flips, whereas in the second game, you would rather have only 20 flips.

Example 2.7

Suppose you have a coin and believe it may have been tampered with so that heads is a less likely outcome than tails. To test this, you flip the coin over and over, noting the proportion of heads after each toss. The following graph shows the results of 500 coin tosses.



[Follow for extended description](#)

What does this graph tell you about the fairness of the coin?

✓ Solution

In the long run, the proportion of heads settled down to around 0.4, indicating that this is probably not a fair coin.

Basic Probability Rules

➤ Example 2.8

A standard literacy test consists of 100 multiple-choice questions, each with five possible answers. There is no penalty for guessing. A score of 60 is considered passing, and a score of 80 is considered superior. When an answer is completely unknown, test takers employ one of three strategies: guess, choose answer (c), or choose the longest answer. The table below summarizes the results of 1,000 test takers.

		Score			Total
		0–59	60–79	80–100	
Strategy	Guess	40	160	100	300
	Choose answer (c)	35	170	115	320
	Choose longest answer	45	200	135	380
Total		120	530	350	1,000

[Follow for extended description](#)

Note that in analyzing tables such as the one above, it is usually helpful to sum rows and columns.

- What is the probability that someone in this group uses the “guess” strategy?
- What is the probability that someone in this group scores 60–79?
- What is the probability that someone in this group does not score 60–79?
- What is the probability that someone in this group chooses strategy “answer (c)” *and* scores 80–100 (the intersection of the two events)?
- What is the probability that someone in this group chooses strategy “longest answer” *or* scores 0–59 (the union of the two events: “or” here means one event or the other event or both events)?
- What is the probability that someone in this group chooses the strategy “guess” *given that* his or her score was 0–59?
- What is the probability that someone in this group scored 80–100 *given that* the person chose strategy “longest answer”?
- Are the strategy “guess” and scoring 0–59 *independent events*? That is, is whether a test taker used the strategy “guess” unaffected by whether the test taker scored 0–59?
- Are the strategy “longest answer” and scoring 80–100 *mutually exclusive events*? That is, are these two events disjoint and cannot simultaneously occur?

TIP

Conditional probabilities shrink the population of interest.

✓ Solutions

- (a) $P(\text{guess}) = \frac{300}{1,000} = 0.3$
- (b) $P(\text{score } 60\text{--}79) = \frac{530}{1,000} = 0.53$
- (c) $P(\text{does not score } 60\text{--}79) = 1 - P(\text{score } 60\text{--}79) = 1 - 0.53 = 0.47$
- (d) $P(\text{answer (c)} \cap \text{score } 80\text{--}100) = \frac{115}{1,000} = 0.115$
- (e) $P(\text{longest answer} \cup \text{score } 0\text{--}59) = \frac{45 + 200 + 135 + 40 + 35}{1,000} = \frac{455}{1,000} = 0.455$

NOTE

The subtraction of the intersection probability is so that it is not counted twice.

Note that

$$\begin{aligned} P(\text{longest answer} \cup \text{score } 0\text{--}59) &= \\ P(\text{longest answer}) + P(\text{score } 0\text{--}59) - P(\text{longest answer} \cap \text{score } 0\text{--}59) &= \\ 0.380 + 0.120 - 0.045 &= 0.455 \end{aligned}$$

(f) $P(\text{guess} \mid \text{score } 0\text{--}59) = \frac{40}{120} \approx 0.333$

Note that we narrowed our attention to the 120 test takers who scored 0–59 to calculate this *conditional probability*.

(g) $P(\text{score } 80\text{--}100 \mid \text{longest answer}) = \frac{135}{380} \approx 0.355$

Note that we narrowed our attention to the 380 test takers who chose “longest answer” to calculate this *conditional probability*.

- (h) We must check if $P(\text{guess} \mid \text{score } 0\text{--}59) = P(\text{guess})$. From (f) and (a), we see that these probabilities are not equal ($0.333 \neq 0.3$), so the strategy “guess” and scoring 0–59 are *not* independent events.
- (i) $\text{Longest answer} \cap \text{score } 80\text{--}100 \neq \emptyset$ and $P(\text{longest answer} \cap \text{score } 80\text{--}100) = \frac{135}{1,000} \neq 0$, so the strategy “longest answer” and scoring 80–100 are *not* mutually exclusive events.

Recap

- The probability that an event will not occur, that is, the probability of its complement, is equal to 1 minus the probability that the event will occur. Three notations for the complement of A are A^C , $\bar{A}$, and A' . Then $P(\text{not } A) = P(A^C) = P(\bar{A}) = P(A') = 1 - P(A)$.
- **The general addition rule:** $P(A \text{ or } B \text{ or both}) = P(A \text{ union } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- A and B are mutually exclusive if $A \cap B = \emptyset$ or, equivalently, $P(A \cap B) = 0$. In this case, $P(A \cup B) = P(A) + P(B)$. Don't add probabilities unless the events are mutually exclusive!
- **The general multiplication rule:** $P(A \text{ and } B) = P(A \text{ intersect } B) = P(A \cap B) = P(A)P(B|A)$. This comes immediately from the conditional probability formula $P(B|A) = \frac{P(A \cap B)}{P(A)}$.
- A and B are independent if $P(A|B) = P(A)$ or, equivalently, $P(B|A) = P(B)$. It is also true that A and B are independent if and only if $P(A \cap B) = P(A)P(B)$, that is, if and only if the probability of both events happening is the product of their probabilities. Don't multiply probabilities unless the events are independent!
- Don't confuse *mutually exclusive* with *independent*!
 - If A and B are mutually exclusive, $P(A \cap B) = 0$.
 - If A and B are independent, $P(A \cap B) = P(A)P(B)$.

► Example 2.9

What probability rule or formula is appropriate for solving each of the following verbal scenarios?

- (a) If 44% of Chicago residents are Cubs fans, what is the probability that a Chicago resident is not a Cubs fan?
- (b) A survey of all students at a large high school revealed that 58% pick "Fifteen" as their favorite single, 34% pick Taylor Swift as their favorite artist, and 15% pick both of these choices. If a student is selected at random, what is the probability that the student picks "Fifteen" or Taylor Swift or both?

- (c) Both blue eyes and gray eyes have a dark epithelium at the back of the iris and a relatively clear stroma at the front. In the population, 10% have blue eyes and only 1% have gray eyes. If a person is selected at random, what is the probability the person has either blue or gray eyes?
- (d) Among the students at a large state college, 72% use Pandora, 43% use Spotify, and 29% use both. If a randomly selected student uses Spotify, what is the probability the student also uses Pandora?
- (e) According to one survey, 55% of Americans “love pizza,” and 68% of those who love pizza choose pepperoni as their favorite topping. What is the probability that a randomly selected American loves pizza and picks pepperoni as their favorite topping?
- (f) In 1948, Kellogg’s introduced 18 prizes in their cereal boxes. Suppose 5% of boxes had a Popeye metal button, while 4% had a Superman metal button. What is the probability that for two randomly picked boxes, the first had a Popeye button and the second had a Superman button?

✓ Solutions

- (a) Complement rule: $P(A^C) = 1 - P(A)$ so $P(\text{not Cubs fan}) = 1 - P(\text{Cubs fan}) = 1 - 0.44 = 0.56$.
- (b) General addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ so $P(\text{“Fifteen” or Swift}) = P(\text{“Fifteen”}) + P(\text{Swift}) - P(\text{“Fifteen” and Swift}) = 0.58 + 0.34 - 0.15 = 0.77$.
- (c) Addition rule for mutually exclusive events: $P(A \cup B) = P(A) + P(B)$ so $P(\text{blue or gray}) = P(\text{blue}) + P(\text{gray}) = 0.10 + 0.01 = 0.11$.
- (d) Conditional probability formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ so
$$P(\text{Pandora}|\text{Spotify}) = \frac{P(\text{Pandora} \cap \text{Spotify})}{P(\text{Spotify})} = \frac{0.29}{0.43} = 0.674$$
- (e) General multiplication rule: $P(A \cap B) = P(A) \cdot P(B|A)$ so $P(\text{love pizza} \cap \text{pepperoni}) = P(\text{love pizza}) \cdot P(\text{pepperoni} | \text{love pizza}) = (0.55)(0.68) = 0.374$.
- (f) Multiplication rule for independent events: $P(A \cap B) = P(A) \cdot P(B)$ so $P(\text{Popeye} \cap \text{Superman}) = P(\text{Popeye}) \cdot P(\text{Superman}) = (0.05)(0.04) = 0.002$.

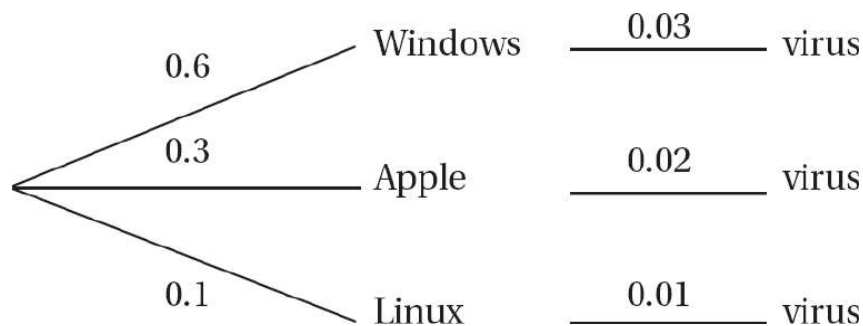
➤ Example 2.10

On a university campus, 60%, 30%, and 10% of the computers use Windows, Apple, and Linux operating systems, respectively. A new virus affects 3% of the Windows, 2% of the Apple, and 1% of the Linux operating systems.

- (a) What is the probability a computer on this campus has the virus?
- (b) If a randomly chosen computer on this campus has the virus, what is the probability it is a Windows machine? An Apple machine? A Linux machine?

✓ Solutions

- (a) Tree diagrams can be very useful in working with conditional probabilities.



[Follow for extended description](#)

We then have:

$$P(\text{Windows} \cap \text{virus}) = (0.6)(0.03) = 0.018$$

$$P(\text{Apple} \cap \text{virus}) = (0.3)(0.02) = 0.006$$

$$P(\text{Linux} \cap \text{virus}) = (0.1)(0.01) = 0.001$$

$$\begin{aligned} P(\text{virus}) &= P(\text{Windows} \cap \text{virus}) + P(\text{Apple} \cap \text{virus}) + P(\text{Linux} \cap \text{virus}) \\ &= 0.018 + 0.006 + 0.001 = 0.025 \end{aligned}$$

Important

Naked or bald answers will receive little or *no* credit. You must show how you arrived at your answer.

$$P(\text{Windows} \mid \text{virus}) = \frac{P(\text{Windows} \cap \text{virus})}{P(\text{virus})} = \frac{0.018}{0.025} = 0.72$$

$$(b) \quad P(\text{Apple} \mid \text{virus}) = \frac{P(\text{Apple} \cap \text{virus})}{P(\text{virus})} = \frac{0.006}{0.025} = 0.24$$

$$P(\text{Linux} \mid \text{virus}) = \frac{P(\text{Linux} \cap \text{virus})}{P(\text{virus})} = \frac{0.001}{0.025} = 0.04$$

► Example 2.11

A second way of analyzing [Example 2.10](#) is to construct a table of counts for a hypothetical population. In this case, let's assume that the original given percentages hold exactly for a population of size 1,000.

	Virus	No Virus	Total
<i>Windows</i>			
<i>Apple</i>			
<i>Linux</i>			
Total			1,000

Numbers in parentheses indicate the order in which the cells can be calculated.

	Virus	No Virus	Total
<i>Windows</i>	(4) 18	(5) 582	(1) 600
<i>Apple</i>	(6) 6	(7) 294	(2) 300
<i>Linux</i>	(8) 1	(9) 99	(3) 100
Total	(10) 25	(11) 975	1,000

Using the above table, calculate the conditional probabilities

✓ Solution

These conditional probabilities can be calculated straight from the first column (virus) of the table:

$$P(\text{Windows} \mid \text{virus}) = \frac{18}{25} = 0.72$$

$$P(\text{Apple} \mid \text{virus}) = \frac{6}{25} = 0.24$$

$$P(\text{Linux} \mid \text{virus}) = \frac{1}{25} = 0.04$$

Quiz 10

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

Questions 1–4 refer to the following study: Five hundred people used a home test for strep throat, and then all underwent more conclusive testing in a medical office. The accuracy of the home test was evidenced in the following table.

	Strep Throat	Healthy	
Positive test	35	25	60
Negative test	5	435	440
	40	460	

- What is the *predictive value* of the test? That is, what is the probability that a person has strep throat and tests positive?
 - $\frac{35}{500}$
 - $\frac{35}{60}$

- (C) $\frac{35}{40}$
(D) $\frac{35}{35 + 25 + 5}$

2. What is the *false-positive* rate? That is, what is the probability of testing positive given that the person does not have strep throat?

- (A) $\frac{25}{460}$
(B) $\frac{25}{60}$
(C) $\frac{25}{500}$
(D) $\frac{35 + 25 + 5}{500}$

3. What is the *sensitivity* of the test? That is, what is the probability of testing positive given that the person has strep throat?

- (A) $\frac{35}{500}$
(B) $\frac{35}{60}$
(C) $\frac{35}{40}$
(D) $\frac{35}{40}$

4. What is the *specificity* of the test? That is, what is the probability of testing negative given that the person does not have strep throat?

- (A) $\frac{35}{35 + 25 + 5}$
(B) $\frac{5}{40}$
(C) $\frac{435}{460}$
(D) $\frac{435}{500}$

5. In a 1974 “Dear Abby” letter, a woman lamented that she had just given birth to her eighth child and all were girls! Her doctor had assured her that the chance of the eighth child being a girl was less than 1 in 100. What was the real probability that the eighth child would be a girl?

- (A) 0.5

- (B) $(0.5)^7$
- (C) $(0.5)^8$
- (D) $\frac{(0.5)^7 + (0.5)^8}{2}$

6. Hospital A has about 20 births per day, and hospital B has about 40 births per day. About 50% of all babies are boys, but the percentage who are boys varies at each hospital from day to day. Over the course of a year, which hospital will have more days on which 60% or more of the births are boys?

- (A) Hospital A
- (B) Hospital B
- (C) The difference will be negligible.
- (D) By the law of large numbers, this cannot be predicted.

7. The probability that a college's mathematics department will fill an opening with a female professor is 0.55, while the probability that the college's physics department will fill an opening with a female professor is 0.3. Assuming these decisions are independent, what is the probability that exactly one of these two departments will fill their position with a female professor?

- (A) $(0.55)(0.3)$
- (B) $0.55 + 0.3 - (0.55)(0.3)$
- (C) $(0.55)(0.7) + (0.3)(0.45)$
- (D) $(0.55)(0.7) + (0.3)(0.45) - (0.55)(0.3)$

8. Suppose that 2% of a clinic's patients are known to have cancer. A blood test is developed that is positive in 98% of patients with cancer but is also positive in 3% of patients who do not have cancer. If a person who is chosen at random from the clinic's patients is given the test and it comes out positive, what is the probability that the person actually has cancer?

- (A) 0.02
- (B) $(0.02)(0.98)$

(C) $(0.02)(0.98) + (0.98)(0.03)$

(C) $\frac{(0.02)(0.98)}{(0.02)(0.98) + (0.98)(0.03)}$

9. A teacher has 90 students. Of those students, 21 have at least two analog clocks in their homes, and 62 have at least three television sets in their homes. There are 15 students who do not have at least two analog clocks and do not have at least three television sets in their homes. Let event C represent the event of selecting a student with at least two analog clocks at home, and let event T represent the event of selecting a student with at least three television sets at home.

Are C and T mutually exclusive events?

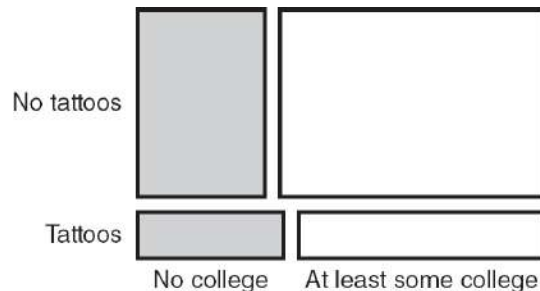
(A) No, because $P(C \cap T) = \frac{8}{90}$

(B) No, because $P(C \cap T) = \frac{83}{90}$

(C) Yes, because $P(C \cap T) = \frac{8}{90}$

(D) Yes, because $P(C \cap T) = \frac{83}{90}$

10. A random sample of adults aged 25 and up were surveyed as to whether or not they had tattoos and whether or not they had ever taken a college course. The mosaic plot below displays the data.



Which is greater, the probability that someone in this sample has taken a college course given that they have a tattoo or the probability that someone in this sample has a tattoo, given that they have taken a college course?

- (A) The probability that someone in this sample has taken a college course given that the person has a tattoo is the greater of the two probabilities.
- (B) The probability that someone in this sample has a tattoo given that the person has taken a college course is the greater of the two probabilities.
- (C) The two probabilities are equal.
- (D) This cannot be answered without knowing (1) the probability that someone in this sample has taken a college course, (2) the probability that someone in this sample has a tattoo, and (3) the probability that someone in this sample has taken a college course and has a tattoo.
11. For which of the following probability assignments are events E and F independent?
- (A) $P(E \cap F^c) = 0.2, P(E \cap F) = 0.2, P(E^c \cap F) = 0.2$
- (B) $P(E \cap F^c) = 0.3, P(E \cap F) = 0.0, P(E^c \cap F) = 0.3$
- (C) $P(E \cap F^c) = 0.3, P(E \cap F) = 0.2, P(E^c \cap F) = 0.1$
- (D) $P(E \cap F^c) = 0.4, P(E \cap F) = 0.1, P(E^c \cap F) = 0.1$
12. What is the proper order of size for the probabilities of the following three events?
- (I) A randomly selected high school senior eats breakfast.
- (II) A randomly selected teenager is a high school senior who eats breakfast.
- (III) A randomly selected teenager who eats breakfast is a high school senior.
- (A) $P(I) < P(II) < P(III)$
- (B) $P(I) > P(II) > P(III)$
- (C) $P(I) < P(III) < P(II)$
- (D) $P(I) > P(III) > P(II)$

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. A sample of applicants for a management position yields the following numbers with regard to age and experience:

	Years of experience		
	0–5	6–10	>10
Less than 50 years old	80	125	20
More than 50 years old	10	75	50

[Follow for extended description](#)

- (a) What is the probability that a randomly picked applicant from this sample is less than 50 years old? Has more than 10 years of experience? Is more than 50 years old and has five or fewer years of experience?
- (b) What is the probability that a randomly picked applicant from this sample is less than 50 years old, given that she has between 6 and 10 years of experience?
- (c) Are the two events “less than 50 years old” and “more than 10 years of experience” independent events? How about the two events “more than 50 years old and between 6 and 10 years of experience”? Explain.
2. Last year, the acceptance rates at Stanford and MIT were 5% and 7%, respectively.

- (a) A guidance counselor calculates the probability of a student getting into both of these colleges to be $(0.05)(0.07) = 0.0035$. What assumption is she making, and does her assumption seem reasonable? Explain.
 - (b) She also calculates the probability of a student getting into at least one of these colleges to be $0.05 + 0.07 = 0.12$. What assumption is she making here, and does it seem reasonable?
3. In one community, 26% of adults identify as Generation X, 38% as Generation Y (Millennials), and 36% as Generation Z. Nine percent of Generation X, 12% of Generation Y, and 15% of Generation Z keep their smartphones at least 3 years. An adult from this community is picked at random.
- (a) What is the probability the person identifies as Generation X and keeps their smartphone at least 3 years?
 - (b) What is the probability the person keeps their smartphone at least 3 years?
 - (c) If the person keeps their smartphone at least 3 years, what is the probability the person identifies as Generation X?

The answers for this quiz can be found beginning on [page 482](#).

Random Variables, Means (Expected Values), and Standard Deviations

Discrete random variables are discussed here, while *continuous* random variables will be looked at later in this unit.

Often each outcome of an experiment has not only an associated probability but also an associated *real number*. For example, the probability may be 0.5 that a student is taking 0 AP classes, 0.3 that she is taking 1 AP class, and 0.2 that she is taking 2 AP classes. If X represents the different numbers associated with the potential outcomes of some chance situation, we call X a random variable.

While the mean of the set $\{2, 7, 12, 15\}$ is $\frac{2+7+12+15}{4} = 9$, or $2(\frac{1}{4}) + 7(\frac{1}{4}) + 12(\frac{1}{4}) + 15(\frac{1}{4}) = 9$, the *expected value* or mean of a discrete random variable takes into account that the various outcomes may not be equally likely. The expected value or mean of a discrete random variable X is given by $\mu_X = E(X) = \sum xP(x)$, where $P(x)$ is the probability of outcome x . [We also write $\mu_X = \sum x_i p_i$, where p_i is the probability of outcome x_i .]

NOTE

The expected value is sometimes referred to as a *weighted average*.

Example 2.12

A charity holds a lottery in which 10,000 tickets are sold at \$1 each and with a prize of \$7,500 for one winner.

- Letting X be the payoff random variable for each ticket holder, make a table showing the values and the probability of each value for X .
- What is the average payoff for each ticket holder?
- Interpret the expected value of X in context.
- What is the most likely outcome for a ticket holder?

NOTE

The distribution of a variable describes the possible values the variable could assume and how often each of those values occurs.

Solutions

- The actual winning payoff is \$7,499 because the winner paid \$1 for a ticket.

Letting X be the payoff random variable, we have:

Outcome	Random Variable X	Probability $P(x)$
---------	---------------------	--------------------

Win	7,499	$\frac{1}{10,000}$
Lose	-1	$\frac{9,999}{10,000}$

(b) Expected value =

$$\text{Expected value} = \mu_x = \sum xP(x) = 7,499\left(\frac{1}{10,000}\right) + (-1)\left(\frac{9,999}{10,000}\right) = -0.25$$

Important

The expected value of a random variable does not, *in any way*, imply that the value is expected to occur.

Thus, the *average* result for each ticket holder is a \$0.25 loss.

(Alternatively, we can say that the expected payoff to the charity is \$0.25 for each ticket sold.)

- (c) The expected value is a long-term average. If this lottery is held many times and someone buys a ticket each time, in the long run the purchaser should expect to lose an average of \$0.25 per lottery.
- (d) The most likely outcome for a ticket purchase is the loss of \$1. (Note that the expected value of -\$0.25 is not the most likely outcome and, in fact, will never occur.)

We have seen that the mean of a discrete random variable is $\sum x_i p_i$. In addition to calculating the mean, we would like to measure the variability for the values taken on by a discrete random variable. Since we are dealing with chance events, the proper tool is variance (along with standard deviation). In Unit 1, *variance* was defined to be the mean of the squared deviations $(x - \mu)^2$. If we regard the $(x - \mu)^2$ terms as the values of some discrete random variable (whose probability is the same as the probability of x), the mean of this new discrete random variable is $\sum (x_i - \mu_x)^2 p_i$, which is precisely how we define the variance, σ^2 , of a discrete random variable:

NOTE

It is also true that $\text{var}(x) = \sum x_i^2 p_i - \mu_x^2$

$$\text{var}(X) = V(X) = \sigma_x^2 = \sum (x_i - \mu_x)^2 p_i$$

As before, the standard deviation σ is the square root of the variance.

$$SD(X) = \sigma_X = \sqrt{\sum (x_i - \mu_x)^2 p_i}$$

The standard deviation can be interpreted as the typical deviation of individual values of the random variable from the mean of the random variable.

Example 2.13

TIP

In a problem like this, there is no shortcut. You must show your work. However, the calculator shortcut of using two lists and running 1-Variable statistics can be useful for checking your answers.

A highway engineer knows that the workers can lay 5 miles of highway on a clear day, 2 miles on a rainy day, and only 1 mile on a snowy day. Suppose the probabilities are as follows:

Outcome	Random Variable X (Miles of highway)	Probability P(x)
Clear	5	0.6
Rain	2	0.3
Snow	1	0.1

- What are the mean (expected value), variance, and standard deviation of the random variable?
- Interpret the mean and standard deviation in context.

- (c) Would it be surprising if the workers laid 10 miles of highway one day?

NOTE

Alternatively, you can calculate:

$$\begin{aligned}\sigma_x^2 &= 5^2(0.6) \\ &\quad + 2^2(0.3) \\ &\quad + 1^2(0.1) \\ &\quad - 3.7^2 = 2.61\end{aligned}$$

✓ Solutions

$$\mu_x = \sum x_i p_i = 5(0.6) + 2(0.3) + 1(0.1) = 3.7$$

(a) $\sigma_x^2 = \sum (x_i - \mu_x)^2 p_i = (5 - 3.7)^2(0.6) + (2 - 3.7)^2(0.3) + (1 - 3.7)^2(0.1) = 2.61$

$$\sigma_x = \sqrt{2.61} = 1.62$$

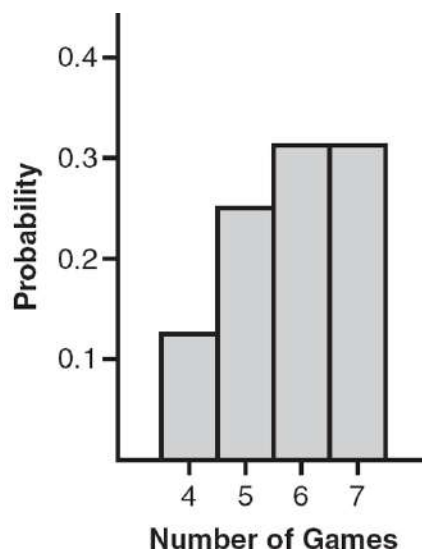
- (b) In the long run, the workers will lay an average of 3.7 miles of highway per day. The number of miles the workers lay on a randomly selected day will generally vary about 1.62 from the mean of 3.7 miles.
- (c) 10 is $\frac{10 - 3.7}{1.62} = 3.9$ standard deviations above the mean, so this would be very surprising!

➤ Example 2.14

The World Series for Major League Baseball is decided by the first team to win four games. Letting the random variable X be the number of games needed to determine a winner in a series where the two teams are evenly matched, we have the following probabilities with accompanying histogram:

Number of Games x	Probability $P(x)$
4	0.125
5	0.25
6	0.3125

Number of Games x	Probability $P(x)$
7	0.3125



- Describe the above distribution.
- Interpret the mean and standard deviation in context.

✓ Solutions

- Shape:* The distribution for the number of games needed to determine a World Series winner is skewed left.

Center: The median is 6, and the mean is $\sum xP(x) = 4(0.125) + 5(0.25) + 6(0.3125) + 7(0.3125) = 5.8125$.

Spread: The range is $7 - 4 = 3$, and the standard deviation is $\sqrt{\sum (x - \mu)^2 P(x)} = \sqrt{(4 - 5.8125)^2(0.125) + \dots + (7 - 5.8125)^2(0.3125)} = 1.0136$.
- In the long run, the number of games needed to determine a winner in a World Series where the two teams are evenly matched will average 5.8125 games, with the length of a randomly selected individual series generally varying about 1.0136 games from the average of 5.8125 games.

Although not explicitly part of the AP curriculum, we note the following with regard to combining and transforming random variables:

■ $\mu_{X \pm Y} = \mu_X \pm \mu_Y$

- If X and Y are independent, $\sigma_{X+Y} = \sigma_{X-Y} = \sqrt{(\sigma_X)^2 + (\sigma_Y)^2}$
- $E(X \pm a) = E(X) \pm a$ and $SD(X \pm a) = SD(X)$
- $E(bX) = bE(X)$ and $SD(bX) = |b|SD(X)$

Important

Variances of independent random variables can be added together, but standard deviations cannot!

Quiz 11

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

- Over several years, data were collected on the number of cloudy days per week in a certain locale. The values varied from 0 to 7, and a distribution of relative frequencies was found. Let the random variable X represent the number of cloudy days in a week. The mean of X is 2.73. Which of the following statements is the best interpretation of the mean?
 - During half the weeks, the number of cloudy days is less than 2.73, and during half the weeks, the number of cloudy days is greater than 2.73.
 - The majority of weeks have either 2 or 3 cloudy days.
 - In any given sample of weeks, the average number of cloudy days per week will be 2.73.
 - The long-run average for number of cloudy days in a week is 2.73.

2. A high school principal records the number of teacher absences per day. Let the random variable Y represent the number of teacher absences in a given day. The standard deviation is 4.31. Which of the following statements is the best interpretation of the standard deviation?
- (A) The average number of teacher absences per day is 4.31.
 - (B) The long-term average number of teacher absences per day is 4.31.
 - (C) On average, the number of teacher absences per day varies from the mean by 4.31.
 - (D) In a random sample of days, the number of teacher absences per day is expected to be 4.31 away from the mean.

3. Let X denote the number of accidents in a 1-month period at a certain intersection. Suppose the probability distribution of X is shown in this table.

X	0	1	2	3	4
$P(x)$	0.16	0.24	0.37	0.15	0.08

Which of the following would be evaluated to calculate the expected value of X ?

- (A) $0(0.16) + 1(0.24) + 2(0.37) + 3(0.15) + 4(0.08)$
 - (B) $\frac{0(0.16) + 1(0.24) + 2(0.37) + 3(0.15) + 4(0.08)}{5}$
 - (C) $\frac{0 + 1 + 2 + 3 + 4}{4}$
 - (D) $\frac{0 + 1 + 2 + 3 + 4}{5}$
4. An insurance company charges \$800 annually for car insurance. The policy specifies that the company will pay \$1,000 for a minor accident and \$5,000 for a major accident. If the probability of a motorist having a minor accident during the year is 0.2 and of having a major accident is 0.05, how much can the insurance company expect to make on a policy?

- (A) \$250
- (B) \$300
- (C) \$350
- (D) \$450

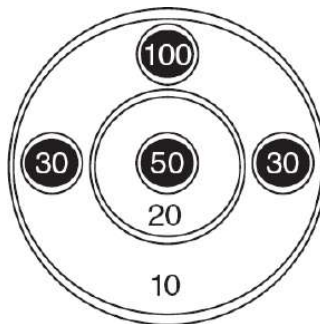
5. Suppose you are one of 7.5 million people who send in their name for a drawing with 1 top prize of \$1 million, 5 second-place prizes of \$10,000, and 20 third-place prizes of \$100. Is it worth the \$0.78 postage it costs you to send in your name?

- (A) Yes, because $\frac{1,000,000}{0.78} = 1,282,051$, which is less than 7,500,000.
- (B) No, because your expected winnings are only \$0.14.
- (C) Yes, because $\frac{7,500,000}{1 + 5 + 20} = 288,642$.
- (D) No, because $1,052,000 < 7,500,000$.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your results and explanations.

1. In Skee-Ball, a player rolls a ball up a ramp and tries to land the ball in the holes with the greatest point values.



[Follow for extended description](#)

One particular player has the following point distribution for any given roll.

X (points)	10	20	30	50	100
Probability	0.50	0.25	0.15	0.05	0.05

Determine the expected value and interpret it in context.

2. Suppose you are asked to play a game where you roll a single fair six-sided die. If it lands on 1 or 6, you win \$5 and the game is over. If it doesn't land on 1 or 6, you flip a fair coin, winning \$1 for heads and \$2 for tails, and the game is then over. Let X be the amount of money won.
- (a) Create a probability distribution for X .
 - (b) What are the expected value and standard deviation for X ? Interpret them in context.
 - (c) How much money should you pay to play this game for it to be a fair game?
3. A roulette wheel has numbers 1 through 36, half black and half red, plus two green numbers, 0 and 00. Suppose you plan to always bet \$5 on "lucky" 13. If the ball lands on 13, you receive \$175 back, otherwise nothing. Let X be the amount of money you are ahead after one spin of the wheel.
- (a) Create a probability distribution for X .
 - (b) What are the expected value and standard deviation for X ? Interpret them in context.
 - (c) One day you quit when you're over \$100 ahead. What is the most likely number of games you played? Explain.

The answers for this quiz can be found beginning on [page 485](#).

Binomial Distribution

A probability distribution is a list or formula that gives the probability of each outcome.

In many applications, such as coin tossing, there are only *two* possible outcomes. We have a *binomial* setting when the following conditions are met:

- There are two possible outcomes, often referred to as *success* and *failure*.
- There are a fixed number of trials, n .
- The trials are independent; that is, the outcome of one trial does not affect the outcome of any other trial.
- There is the same probability of success p on each trial.

The binomial probability function refers to a binomial random variable X with n independent trials and probability of success p . Then the probability of exactly x successes is shown as the following:

NOTE

A question on conditions for a binomial setting could appear either as a multiple-choice question or as part of a free-response question.

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} = \frac{n!}{x!(n-x)!} p^x (1 - p)^{n-x} \text{ for } x = 0, 1, 2, 3, \dots, n.$$

Software can be used to calculate binomial probabilities:

binompdf (n, p, x) gives the probability of exactly x successes in n trials, where p is the probability of success on a single trial.

binomcdf (n, p, x) gives the cumulative probability of x or fewer successes in n trials, where p is the probability of success on a single trial.

NOTE

The binomial coefficient $\binom{n}{x}$ is sometimes written as ${}_nC_x$.

NOTE

The *pdf* (such as in `binompdf`) stands for “probability density function.” That is, it’s a function that tells you the probability of certain events occurring.

Example 2.15

In cereal boxes that advertised an enclosed miniature toy, one-third of the toys were purple dragons. If six boxes of cereal are purchased, what is the probability of exactly two purple dragons?

Solution

If the probability of a purple dragon is $\frac{1}{3}$, the probability of no purple dragon is $1 - \frac{1}{3} = \frac{2}{3}$. If two out of six boxes have a purple dragon, $6 - 2 = 4$ do not.

The probability of getting exactly two purple dragons when six boxes of cereal are purchased is `binompdf(6, 1/3, 2) = 0.329`. [Or we could calculate $\binom{6}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^4 = 15 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^4 = \frac{80}{243} = 0.329$.]

For full credit on the exam, write either of the following two lines:

$$\text{Binomial, } n = 6, p = \frac{1}{3}, P(X = 2) = 0.329$$

$$B(6, 1/3), P(X = 2) = 0.329$$

In some situations it is easier to calculate the probability of the complementary event and subtract this value from 1.

Example 2.16

The baseball player Joe DiMaggio had a career batting average of 0.325. What was the probability that he would get at least one hit in five official times at bat?

Solution

This is a binomial distribution with $n = 5$ and $p = 0.325$.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{5}{0}(0.325)^0(0.675)^5 = 1 - 0.140 = 0.860$$

[Or we could calculate: $1 - \text{binompdf}(n = 5, p = 0.325, X = 0) = 0.860$.]

To receive full credit for probability calculations using the probability distributions from the AP Statistics course, you need to show:

1. Name of the distribution (“binomial” in the previous example)
2. Parameters (“ $n = 5$ and $p = 0.325$ ” in the previous example)
3. Values of interest (“ $X \geq 1$ ” in the previous example)
4. Correct probability (“0.860” in the previous example)

Many applications of probability involve such phrases as *at least*, *at most*, *less than*, and *more than*. In these cases, solutions involve summing two or more cases. For such calculations, `binomcdf` is very useful, as it gives the probability of x or fewer successes.

Example 2.17

One day a grocery store manager notes seven customers buying a particular product and is interested in the number who will make use of a store coupon to receive a discount. Suppose 35% of customers in general make use of a store coupon to receive a discount when buying that product.

- (a) Define the random variable of interest for the manager and state how the random variable is distributed.
- (b) Determine the probability that fewer than four of the customers will use a store coupon.

Solutions

- (a) The random variable of interest is the number of customers using a store coupon to receive a discount when buying this product among

seven customers. The distribution is binomial with parameters $n = 7$ and $p = 0.35$.

- (b) In this situation, “fewer than four” means zero, one, two, or three. We have a binomial with $n = 7$ and $p = 0.35$. Then $P(X \leq 3) = \text{binomcdf}(7, 0.35, 3) = 0.800$. [Or we could calculate:

$$\begin{aligned} & \binom{7}{0}(0.35)^0(0.65)^7 + \binom{7}{1}(0.35)^1(0.65)^6 + \binom{7}{2}(0.35)^2(0.65)^5 + \binom{7}{3}(0.35)^3(0.65)^4 \\ &= (0.65)^7 + 7(0.35)(0.65)^6 + 21(0.35)^2(0.65)^5 + 35(0.35)^3(0.65)^4 \\ &= 0.800.] \end{aligned}$$

TIP

While using `binomcdf` is the quick and recommended solution if part of a free-response question, the answer choices in a multiple-choice question here could be the expanded four-term binomial formula form.

Example 2.18

A manufacturer has the following quality control check at the end of a production line. If at least 8 of 10 randomly picked articles meet all specifications, the whole shipment is approved. Suppose 85% of an entire shipment meets all specifications.

- (a) Define the random variable of interest for the manufacturer and state how the random variable is distributed.
- (b) Determine the probability that this shipment will pass the quality control check.

Solutions

- (a) The random variable of interest is the number of articles meeting all specifications among 10 randomly selected articles. The distribution is binomial with parameters $n = 10$ and $p = 0.85$.
- (b) We have a binomial distribution with $n = 10$ and $p = 0.85$.

$$P(X \geq 8) = 1 - P(X \leq 7) = 1 - \text{binomcdf}(10, 0.85, 7) = 0.820$$

[Or we could calculate: $P(X = 8) + P(X = 9) + P(X = 10) =$

$$\binom{10}{8} (0.85)^8 (0.15)^2 + \binom{10}{9} (0.85)^9 (0.15)^1 + \binom{10}{10} (0.85)^{10} (0.15)^0$$

$$= \text{binompdf}(10, 0.85, 8) + \text{binompdf}(10, 0.85, 9) + \text{binompdf}(10, 0.85, 10) = 0.820.]$$

For a binomial probability distribution with the probability of a success equal to p and the number of trials equal to n , we have:

- The expected value or mean number of successes is np .
- The standard deviation is $\sqrt{np(1-p)}$.

Example 2.19

Of the automobiles produced at a particular plant, 40% had a certain defect. Suppose a company purchases five of these cars and is concerned about the number with that defect.

- (a) Define the random variable of interest for the company and state how the random variable is distributed.
- (b) What is the mean (expected value) for the number of cars with defects? Interpret in context.
- (c) What is the standard deviation for the number of cars with defects? Interpret in context.

Solutions

- (a) The random variable of interest is the number of cars with a defect among five cars. The distribution is binomial with parameters $n = 5$ and $p = 0.40$.
- (b) The mean or expected value for the number of cars with defects is $np = 5(0.4) = 2$. If five cars were purchased many times, the average number of cars with defects would be about 2.
- (c) The standard deviation for the number of cars with defects is $\sqrt{np(1-p)} = \sqrt{5(0.4)(0.6)} = \sqrt{1.2} = 1.095$. In any random sample of five cars, the number of cars with defects will typically vary about 1.095 from the mean of 2 cars.

Thus, for a random variable X :

Mean or expected value:

$$\mu_X = \sum x_i p_i$$

Variance:

$$\sigma_X^2 = \sum (x_i - \mu_X)^2 p_i$$

Standard deviation:

$$\sigma_X = \sqrt{\sum (x_i - \mu_X)^2 p_i}$$

In the case of a binomial probability distribution with probability of a success equal to p and number of trials equal to n , if we let X be the number of successes in the n trials, the above equations simplify to:

Mean or expected value:

$$\mu_X = np$$

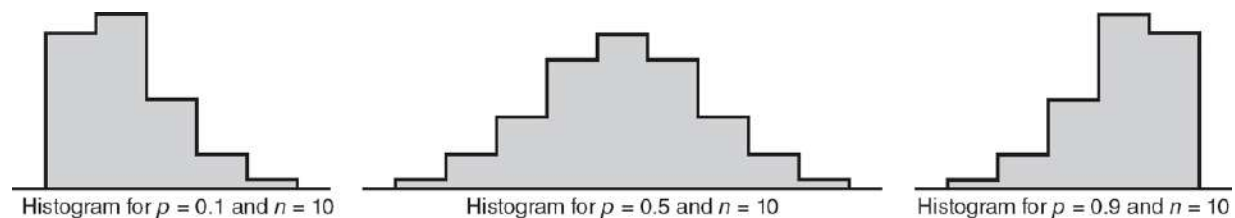
Variance:

$$\sigma_X^2 = npq = np(1 - p)$$

Standard deviation:

$$\sigma_X = \sqrt{npq} = \sqrt{np(1 - p)}$$

The shape of a binomial distribution is skewed right when $p < 0.5$, symmetric when $p = 0.5$, and skewed left when $p > 0.5$.



However, for all p , as n becomes larger without bound, the resulting histograms approach a smooth bell-shaped curve.

Example 2.20

- Ninety-five percent of all buyers of new cars choose automatic transmissions. For a group of 50 buyers of new cars, calculate and interpret the mean and standard deviation for the number of buyers choosing automatic transmissions.
- Would it be surprising if only 40 buyers chose automatic transmissions in a random sample of 50 new car buyers?

✓ Solutions

$$\mu_X = np = 50(0.95) = 47.5$$

(a) $\sigma_X = \sqrt{np(1-p)} = \sqrt{50(0.95)(0.05)} = 1.54$

In many random samples of size 50 buyers of new cars, we can expect an average of 47.5 buyers who will choose automatic transmissions, and for any particular random sample of 50 buyers, the number of buyers choosing automatic transmissions will typically vary by about 1.54 from the mean of 47.5 buyers.

- (b) Yes, that would be surprising because 40 is $\frac{7.5}{1.54} = 4.87$ standard deviations below the mean of 47.5.

Quiz 12

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. An inspection procedure at a manufacturing plant involves picking three items at random and then accepting the whole lot if at least two of the three items are in perfect condition. If in reality 90% of the whole lot are perfect, what is the probability that the lot will be accepted?

(A) $(0.9)^3 + 3(0.9)^2(0.1)$
(B) $(0.1)^3 + 3(0.1)^2(0.9)$
(C) $3(0.1)^2(0.9)$
(D) $3(0.9)^2(0.1)$

2. Suppose we have a random variable X where, for the values $k = 0, \dots, 10$, the associated probabilities are $\binom{10}{k}(0.37)^k(0.63)^{10-k}$. What is the

mean of X ?

- (A) 0.37
- (B) 0.63
- (C) 3.7
- (D) 6.3

3. One in six adults in the workplace has experienced cyberbullying. In a random sample of five adults in the workplace, what is the distribution for the number who have experienced cyberbullying?

- (A) Binomial with $n = 6$ and $p = \frac{1}{6}$
- (B) Binomial with $n = 5$ and $p = \frac{1}{6}$
- (C) Binomial with $n = 5$ and $p = \frac{1}{5}$
- (D) Binomial with $n = 6$ and $p = \frac{1}{5}$

Questions 4–5 refer to the following.

It is estimated that two out of five high school students would fall victim to a phishing email (an online scam asking for sensitive information) if it appears to originate from their high school main office.

4. In a random sample of five high school students, what is the probability that exactly two fall victim to a phishing email that appears to originate from their high school main office?

- (A) 0.4
- (B) $(0.4)^2 (0.6)^3$
- (C) $\binom{5}{2}(0.4)^2(0.6)^3$
- (D) $\binom{5}{2}(0.4)^3(0.6)^2$

5. What is the probability that the first student to fall victim will be the third student who is sent a phishing email that appears to originate from their high school main office?

- (A) $(0.4)^3$
- (B) $(0.6)^2(0.4)$
- (C) $(0.6)(0.4)^2$
- (D) $10(0.6)^2(0.4)$

6. A fair coin is flipped 5 times. Consider the following 3 outcomes where H refers to a toss resulting in “heads” and T refers to a toss resulting in “tails”:

- (I) The result in order is HHHTT.
- (II) The result in order is HTTHH.
- (III) The result includes 3 H’s and 2 T’s.

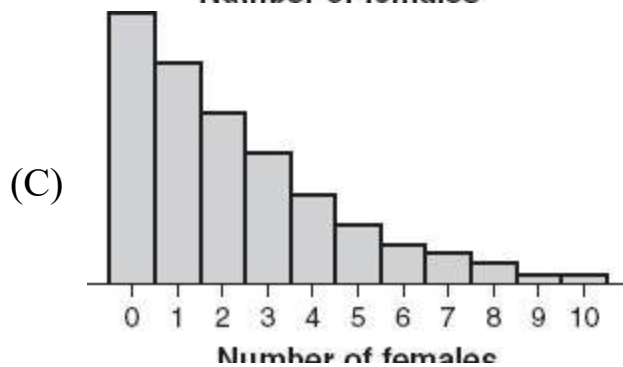
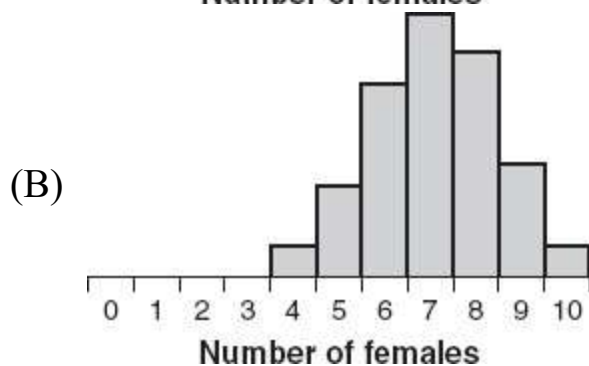
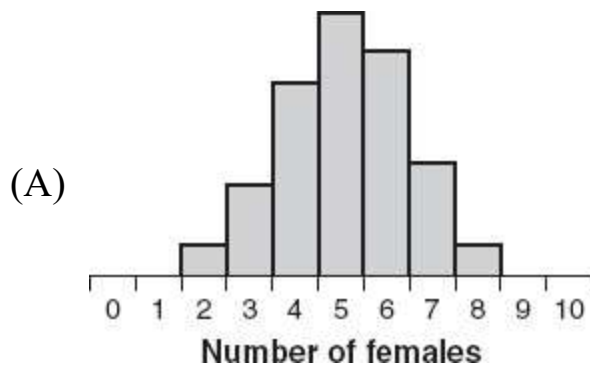
What is a comparison of the three probabilities?

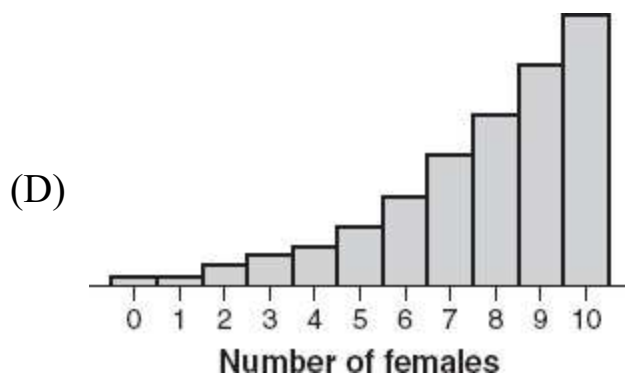
- (A) $P(I) < P(II) < P(III)$
- (B) $P(II) < P(I) < P(III)$
- (C) $P(III) < P(I) = P(II)$
- (D) $P(I) = P(II) < P(III)$

7. A statistician working for an online dating service determines that the binomial random variable D , representing the number of people in a sample who are happy with the service, has a mean of 5 with a standard deviation of 2. What is the probability that someone is happy with the service?

- (A) 0.2
- (B) 0.4
- (C) 0.6
- (D) 0.8

8. A college lecture class has 500 students, of which 350 are female. Let the random variable F represent the number of females in a random sample of 10 students from the class. Random variable F follows a binomial distribution. A simulation of distribution F is performed by randomly picking 10 students from this class a large number of times. Which of the following best represents this simulation?





9. If it were possible, would you want to live forever? When asked, 34% of people answer yes. Let the random variable L represent the number of people who would answer yes in a random sample of 50 people. Random variable L follows a binomial distribution with a mean of 17 and a standard deviation of 3.35. Which of the following is the best interpretation of the mean?
- (A) Every random sample of 50 people will have 34% answering yes.
 - (B) On average, 17 people will have to be surveyed before finding one who answers yes.
 - (C) In repeated random sampling of 50 people, the average number of people answering yes is 17.
 - (D) In repeated random sampling of 50 people, the average number of people answering yes will vary from 17 by an average of 3.35.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Could hands-free, automatic faucets actually be housing more bacteria than the manual kind? The concern is that decreased water flow may increase the chance that bacteria grow because the automatic faucets are not being thoroughly flushed through. It is known that 15% of

water cultures from manual faucets in hospital patient care areas test positive for *Legionella* bacteria. A recent study at Johns Hopkins Hospital found *Legionella* bacteria growing in 10 of cultured water samples from 20 automatic faucets.

- (a) If the probability of *Legionella* bacteria growing in a faucet is 0.15, what is the probability that in a sample of 20 faucets, 10 or more have the bacteria growing?
- (b) Does the Johns Hopkins study provide sufficient evidence that the probability of *Legionella* bacteria growing in automatic faucets is greater than 15%? Explain.

2. Consider the following probability distribution for the number of tattoos college students have:

Tattoos	0	1	2	3	4
Probability	0.85	0.11	0.02	0.014	0.006

- (a) What is the probability a student has two tattoos given that he has at least one tattoo?
- (b) In a random sample of three students, what is the probability that one of the students has at least two tattoos while the other two students have no tattoos?
- (c) What are the mean and standard deviation for number of tattoos? Interpret in context.

3. Suppose 35% of teenagers say that *The Hunger Games* is the best book series they ever read.

- (a) What is the probability that exactly two in a random sample of four teenagers say that *The Hunger Games* is the best series they ever read?
- (b) What is the probability that at least two in a random sample of four teens say that *The Hunger Games* is the best series they ever read?

- (c) What is the probability that the first teen who says that *The Hunger Games* is the best series he or she ever read is the second teen interviewed?

The answers for this quiz can be found beginning on [page 486](#).

Cumulative Probability Distribution

A probability distribution is a function, table, or graph that links outcomes of a statistical experiment with its probability of occurrence. A *cumulative* probability distribution is a function, table, or graph that links outcomes with the probability of less than or equal to that outcome occurring.

Example 2.21

The scores received on a recent AP Statistics exam are illustrated in the following table:

Score	Probability	Cumulative Probability
1	0.231	0.231
2	0.165	0.396
3	0.234	0.630
4	0.222	0.852
5	0.148	1.000

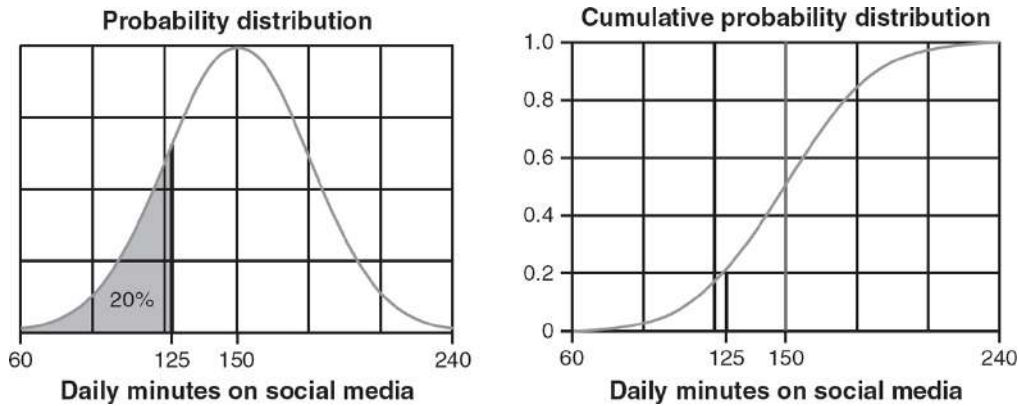
What is the probability that a student does not receive college credit (assuming you need a 3 or higher to receive college credit)?

Solution

The probability is $0.231 + 0.165 = 0.396$ that a student receives a 1 or 2 and thus does not receive college credit.

Example 2.22

The number of minutes that teenagers spend on social media per day is illustrated in the following graphical displays: a probability distribution and the corresponding cumulative probability distribution.



Note how, for example, 20% of the area under the probability distribution graph is to the left of 125 minutes and the point (125, 0.2) is on the cumulative probability distribution graph. Both indicate that the probability is 0.2 that a teenager spends 125 minutes or less per day on social media.

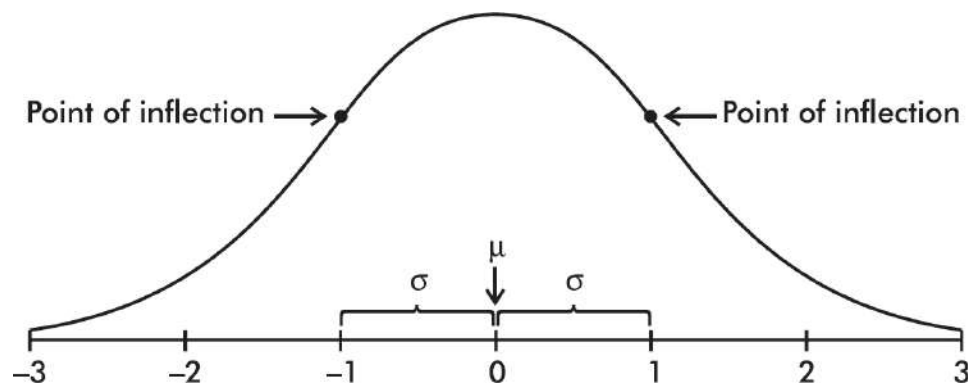
The Normal Distribution

One bell-shaped distribution, called the normal distribution, is valuable for providing a useful model in describing various natural phenomena. However, we will see that the real importance of the normal distribution in statistics is that it can be used to describe the results of many sampling procedures.

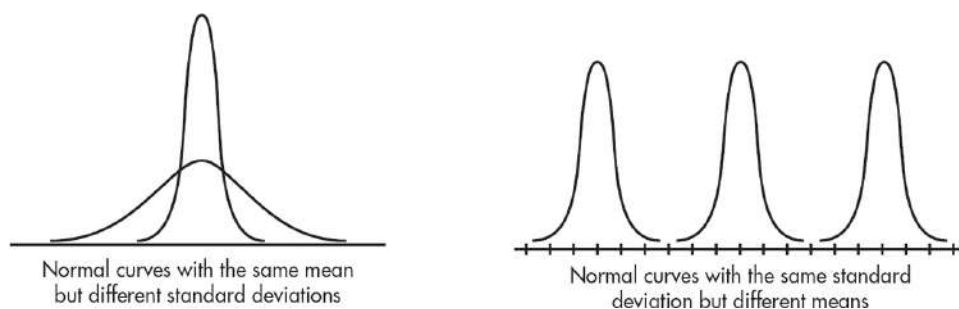
The normal distribution curve is bell-shaped and symmetric and has an infinite base. Long, flat-looking tails cover many values but only a small proportion of the area.

The mean of a normal distribution is equal to the median and is located at the center. We want a unit of measurement that applies equally well to any normal distribution, and we choose a unit that arises naturally out of the curve's shape. There is a point on each side where the slope is steepest. These two points are called *points of inflection*, and the distance from the mean to either point is precisely equal to one standard deviation. Thus, it is convenient to measure distances under the normal curve in terms of z -

scores. (Recall from earlier that z -scores represent the number of standard deviations a particular value is above or below the mean.)

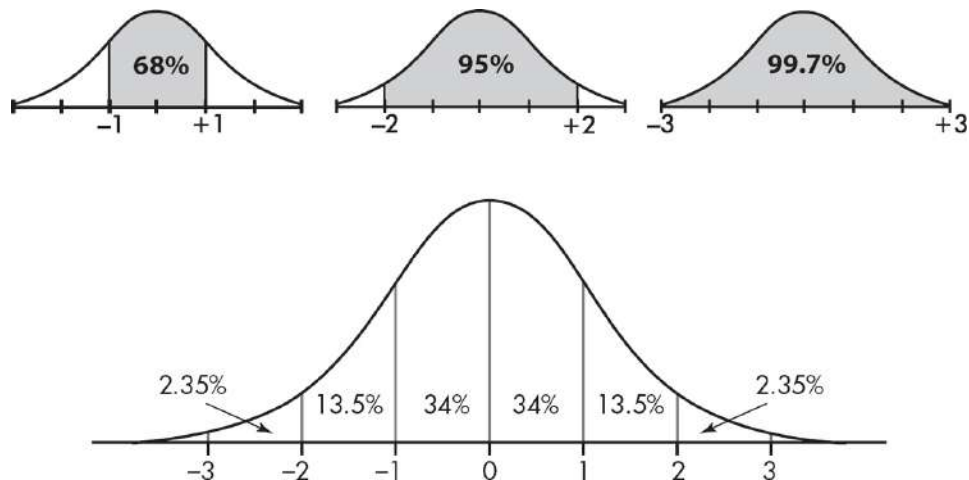


Every normal distribution is completely determined by two parameters: its mean μ and its standard deviation s .



The *empirical rule* (also called the *68-95-99.7 rule*) applies specifically to normal distributions. About 68% of the values lie within 1 standard deviation of the mean, about 95% of the values lie within 2 standard deviations of the mean, and about 99.7% of the values lie within 3 standard deviations of the mean.

In the following figures, the horizontal axes show z -scores:



However, don't be confused into thinking that z-scores mean normality. A z-score can be calculated whenever the mean and standard deviation are known, no matter what the distribution. Only in the very special case of nearly normal distributions can one move easily from z-scores to percentiles and probabilities.

► Example 2.23

Suppose that taxicabs in New York City are driven an average of 75,000 miles per year with a standard deviation of 12,000 miles. Assuming a roughly normal distribution, what information does the empirical rule give us?

✓ Solution

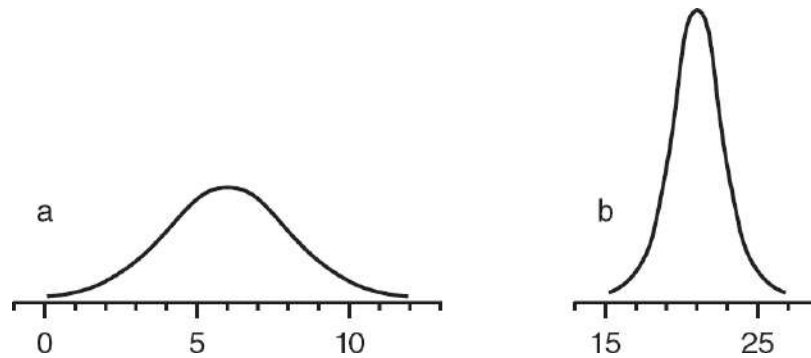
Given that the distribution is roughly normal, the empirical rule applies. Thus approximately 68% of the taxis are driven between 63,000 and 87,000 miles per year, approximately 95% are driven between 51,000 and 99,000 miles, and virtually all are driven between 39,000 and 111,000 miles.

Quiz 13

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

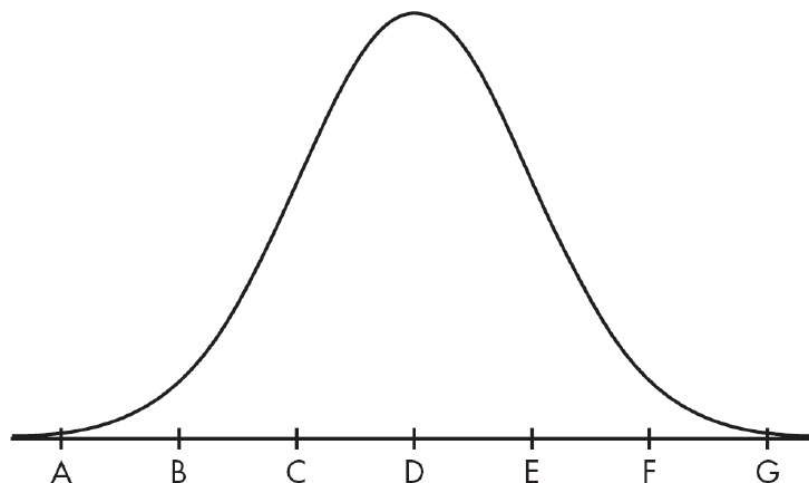
1. Consider the following two normal curves.



Which has the smaller mean, and which has the smaller standard deviation?

- (A) Smaller mean, a ; smaller standard deviation, a
- (B) Smaller mean, a ; same standard deviation
- (C) Smaller mean, a ; smaller standard deviation, b
- (D) Smaller mean, b ; smaller standard deviation, b

Questions 2–4 refer to the following. Suppose an average life expectancy in a particular country is 74.3 years with a standard deviation of 9.7 years. Assuming a roughly normal distribution, consider the following graph of the distribution.



[Follow for extended description](#)

2. Point E on this normal curve corresponds to
- (A) 54.9 years.
 - (B) 64.6 years.
 - (C) 84.0 years.
 - (D) 93.7 years.
3. About what percent of the years are between points B and F?
- (A) 50%
 - (B) 68%
 - (C) 95%
 - (D) 99.7%
4. About what percent of the years are less than point C?
- (A) 16%
 - (B) 34%
 - (C) 68%
 - (D) 95%
-
5. The distribution of body fat percentages of adult males is approximately normal with a standard deviation of 2.5%. A male whose body fat percentage is over 25% is considered obese. If 16% of males are obese, what is the average body fat percentage?
- (A) 16%
 - (B) 17.5%
 - (C) 20%
 - (D) 22.5%

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. It is estimated that 23% of American teen spending goes toward food. Assume that yearly amounts spent on food can be approximated by a normal model with a mean of \$600 and a standard deviation of \$120.
 - (a) A teen spending \$720 would be at what percentile?
 - (b) What percent of American teens spend between \$360 and \$840 per year on food?
 - (c) 99.7% of American teens spend between what two dollar amounts on food per year?
2. A high school guidance counselor tallied the number of colleges that seniors applied to in Fall 2023 and created the following descriptive statistics for the data set.

Variable	n	Mean	Median	StDev	Min	Max	Q ₁	Q ₃
Colleges	50	6.4	5	4.80	0	15	2	10

Do the numbers of colleges applied to by this school's seniors in Fall 2023 seem to roughly follow a normal model?

Normal Distribution Calculations

The normal distribution is valuable in describing various natural phenomena. However, the real importance of the normal distribution in statistics is that it can be used to describe sampling distributions, an important topic of this and later units. The normal distribution provides a valuable model for how many sample *statistics* vary, under repeated random

sampling from a population. Calculations involving normal distributions are often made through z-scores, which measure standard deviations from the mean.

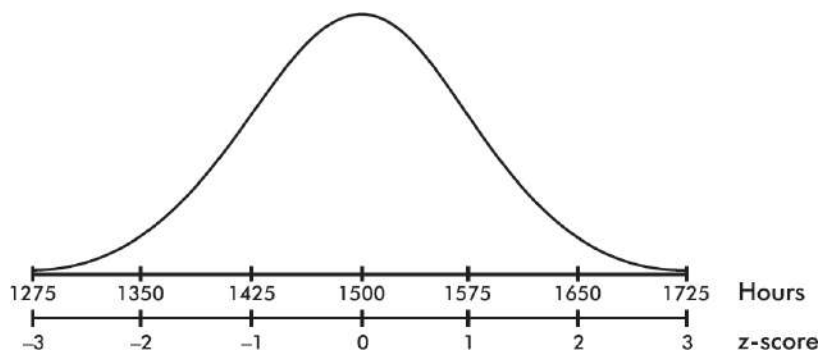
Both `normalcdf` and `invNorm` can work with z-scores or raw scores:

Because it is quicker and gives more accurate calculations than the normal table, a calculator is preferred when working with the normal distribution. Many experienced AP Statistics teachers do not teach or encourage use of the normal table for this reason.

- `normalcdf(lowerbd, upperbd)` = probability between two z-scores for the standard normal distribution (mean = 0 and SD = 1)
- `normalcdf(lowerbd, upperbd, mean, SD)` = probability between two raw scores for a normal distribution with the given mean and SD
- `invNorm(probability)` = z-score with the given probability to the left for the standard normal distribution (mean = 0 and SD = 1)
- `invNorm(probability, mean, SD)` = raw score with the given probability to the left for a normal distribution with the given mean and SD

► **Example 2.24**

The life expectancy of a particular brand of light bulb is roughly normally distributed with a mean of 1500 hours and a standard deviation of 75 hours.



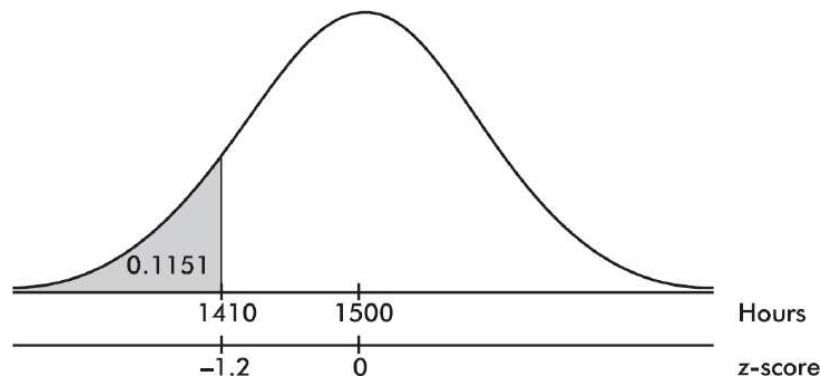
- (a) What is the probability that a light bulb will last less than 1410 hours?

- (b) What is the probability that a light bulb will last between 1563 and 1648 hours?

✓ Solutions

Draw a picture! Errors are much less likely if normal probability calculations are supported with a sketch.

- a. Calculate $\text{normalcdf}(0, 1410, 1500, 75) = 0.1151$. [Or the z-score of 1410 is $\frac{1410 - 1500}{75} = -1.2$, and $\text{normalcdf}(-1.2, 0) = 0.1151$.]



[Follow for extended description](#)

The answers for this quiz can be found beginning on [page 487](#).

The probability that a light bulb will last less than 1410 hours is 0.1151. We can also say that we expect that 11.51% of the light bulbs will last less than 1410 hours.

Either of the following two lines will receive full credit:

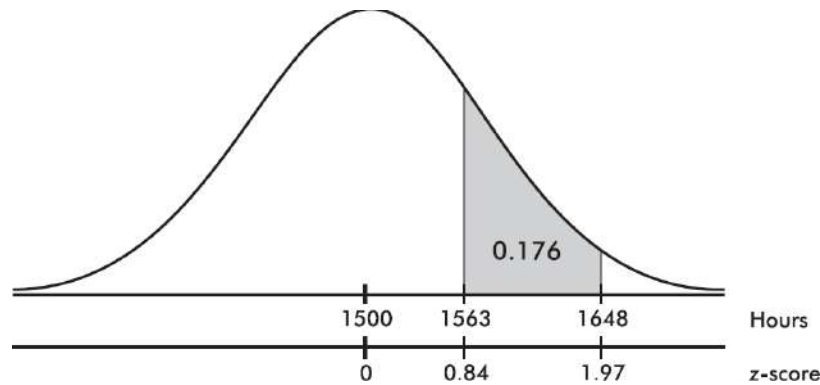
Normal distribution, $\mu = 1500$, $\sigma = 75$, $P(X < 1410) = 0.1151$

$N(1500, 75)$, $P(X < 1410) = 0.1151$

- b. For a normal distribution with $\mu = 1500$ and $\sigma = 75$,

$$P(1563 < X < 1648) = P\left(\frac{1563 - 1500}{75} < z < \frac{1648 - 1500}{75}\right) = P(0.84 < z < 1.97) = 0.176$$

$[\text{normalcdf}(1563, 1648, 1500, 75) = 0.176$ and $\text{normalcdf}(0.84, 1.97) = 0.176.]$



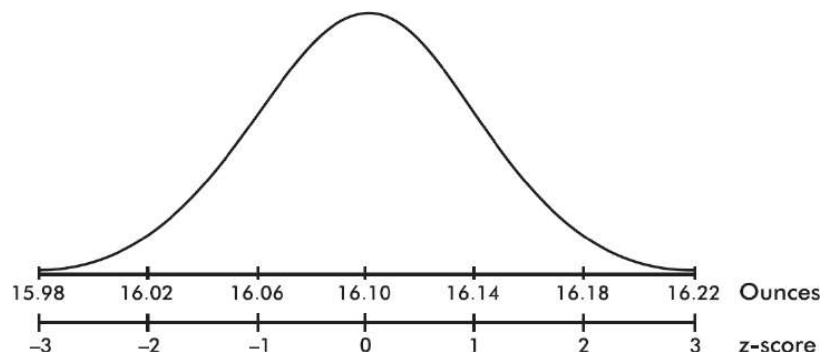
The probability that a light bulb will last between 1563 and 1648 hours is 0.176. We can also say that we expect that 17.6% of the light bulbs will last between 1563 and 1648 hours.

To receive full credit for probability calculations using the probability distributions, you need to show:

1. Name of the distribution (“normal” in the example above)
2. Parameters (“ $m = 1500, \sigma = 75$ ” in the example above)
3. Values of interest (“ $1563 < X < 1648$ ” in (b) of the example above)
4. Correct probability (“0.176” in (b) of the example above)

➤ Example 2.25

A packing machine is set to fill a cardboard box with a mean of 16.1 ounces of cereal. Suppose the amounts per box form an approximately normal distribution with a standard deviation equal to 0.04 ounce.

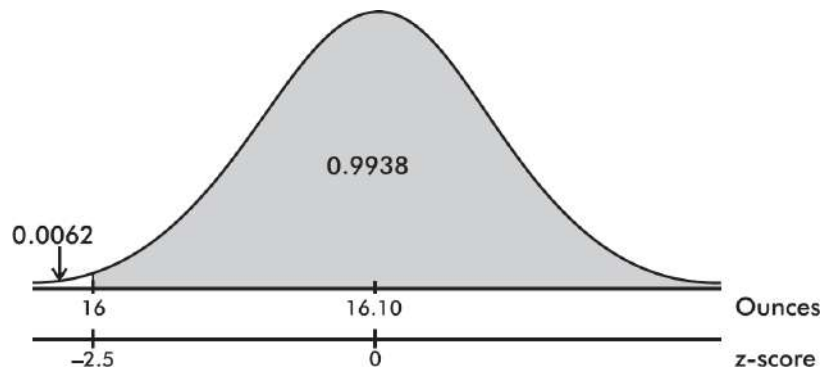


[Follow for extended description](#)

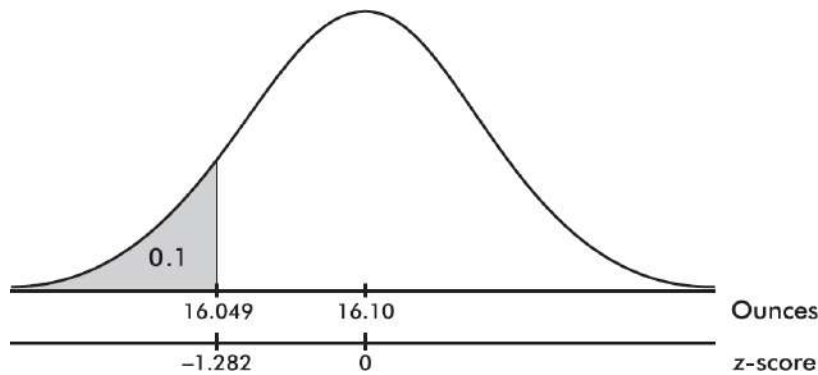
- (a) What percentage of the boxes will end up with at least 1 pound of cereal? (Note: 1 pound equals 16 ounces.)
- (b) Ten percent of the boxes will contain less than what number of ounces?
- (c) Eighty percent of the boxes will contain more than what number of ounces?
- (d) The middle 90% of the boxes will be between what two weights?

✓ Solutions

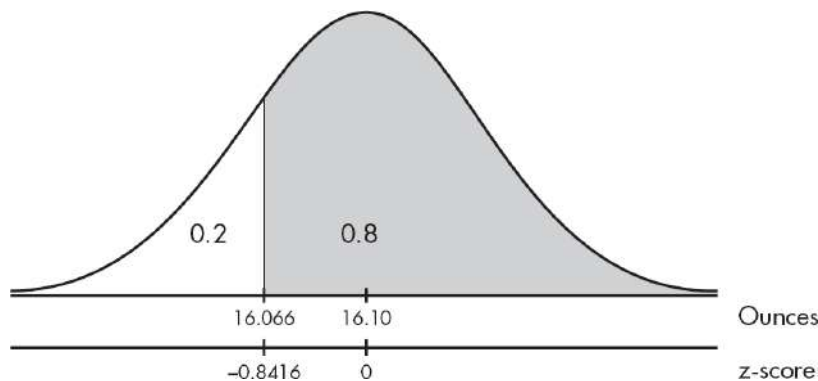
- (a) For a normal distribution with $\mu = 16.10$ and $\sigma = 0.04$, $P(X > 16) = P\left(z > \frac{16 - 16.10}{0.04}\right) = 0.9938$. [normalcdf(16, 1000, 16.10, 0.04) = 0.9938 and normalcdf(-2.5, 10) = 0.9938.] Conclude that 99.38% of the boxes will end up with at least 1 pound of cereal.



- (b) For a normal distribution with $\mu = 16.10$ and $\sigma = 0.04$, $\text{invNorm}(0.1) = -1.282$, and then $16.1 - 1.282(0.04) = 16.049$. [invNorm(0.1, 16.10, 0.04) = 16.049.] Conclude that 10% of the boxes will contain less than 16.049 ounces.



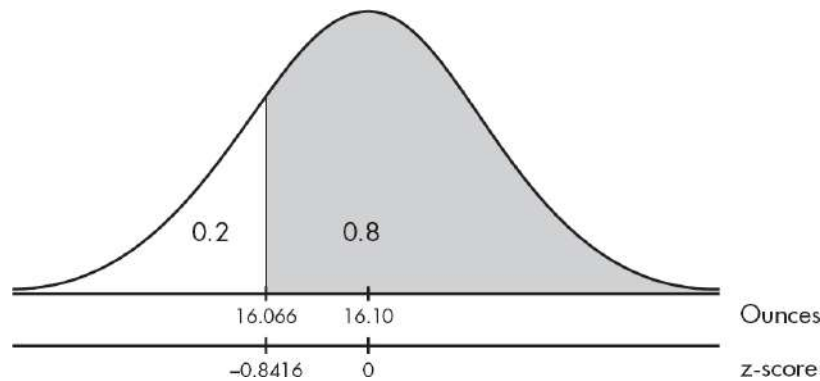
- (c) Eighty percent to the right corresponds to a probability of 0.2 to the left. For a normal distribution with $\mu = 16.10$ and $\sigma = 0.04$, $\text{invNorm}(0.2) = -0.8416$, and then $16.1 - 0.8416(0.04) = 16.066$. [$\text{invNorm}(0.2, 16.10, 0.04) = 16.066$.] Conclude that 80% of the boxes will contain more than 16.066 ounces.



NOTE

We chose 1000 as the upper bound (and -1000 as the lower bound) in [Example 2.24](#) as it is many standard deviations from the mean. On the TI-84, we could also have used $1E99$ for the upper bound (and $-1E99$ for the lower bound).

- (d) Ninety percent in the middle leaves 5% in each tail. For a normal distribution with $\mu = 16.10$ and $\sigma = 0.04$, $\text{invNorm}(0.05) = -1.645$, $\text{invNorm}(0.95) = 1.645$, $16.1 - 1.645(0.04) = 16.034$, and $16.1 + 1.645(0.04) = 16.166$. [$\text{invNorm}(0.05, 16.10, 0.04) = 16.034$ and $\text{invNorm}(0.95, 16.10, 0.04) = 16.166$.] Conclude that the middle 90% of the boxes are between 16.034 and 16.166 ounces.



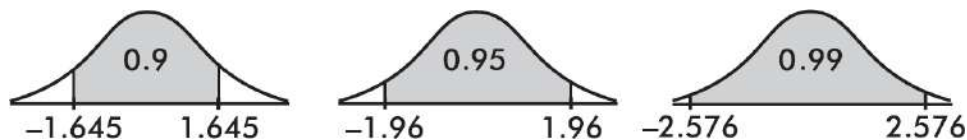
As can be seen above, there is often an interest in the limits enclosing some specified middle percentage of the data. For future reference, the limits most frequently asked for are noted below in terms of z-scores.

NOTE

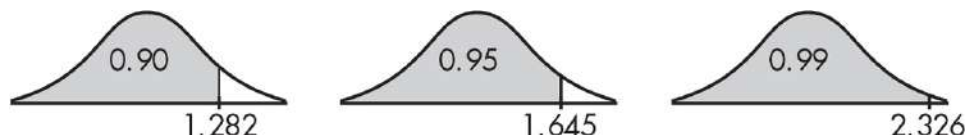
These critical z-scores can be found at the bottom of Table B in the Appendix.

NOTE

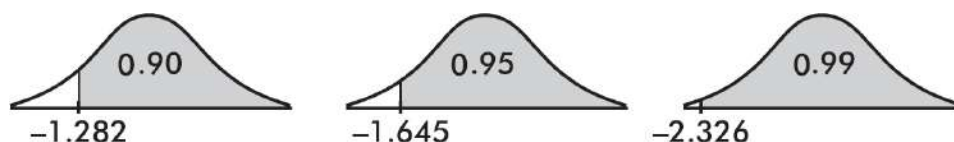
The 2σ corresponding to 95% in the 68–95–99.7 rule is an approximation to the more exact 1.96σ .



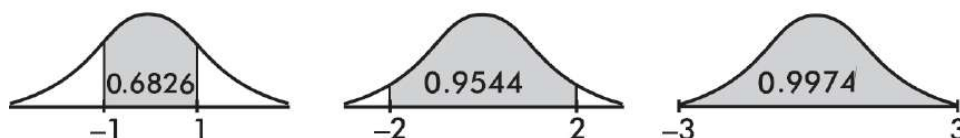
Sometimes the interest is in values with particular percentile rankings. For example,



There are corresponding conclusions for negative z-scores:



It is also useful to note the percentages corresponding to values falling between integer z -scores. For example,



[Follow for extended description](#)

NOTE

The empirical rule's 68%, 95%, and 99.7% are approximations of 68.26%, 95.44%, and 99.74%.

Example 2.26

Suppose that the average height of adult males in a particular locality is 70 inches with a standard deviation of 2.5 inches. Assume the distribution is approximately normal.

- The middle 95% of males are between what two heights?
- Ninety percent of the heights are below what value?
- Ninety-nine percent of the heights are above what value?
- Approximately what percentage of the heights are between z -scores of ± 1 ? Of ± 2 ? Of ± 3 ?

Solutions

- As noted above, the critical z -scores in this case are ± 1.96 , and so the two limiting heights are 1.96 standard deviations from the mean. Therefore, $70 \pm 1.96(2.5) = 70 \pm 4.9$, or from 65.1 to 74.9 inches.

- (b) The critical z -score is 1.282, and so the value in question is $70 + 1.282(2.5) = 73.205$ inches.
- (c) The critical z -score is -2.326 , and so the value in question is $70 - 2.326(2.5) = 64.185$ inches.
- (s) 68.26%, 95.44%, and 99.74%, respectively

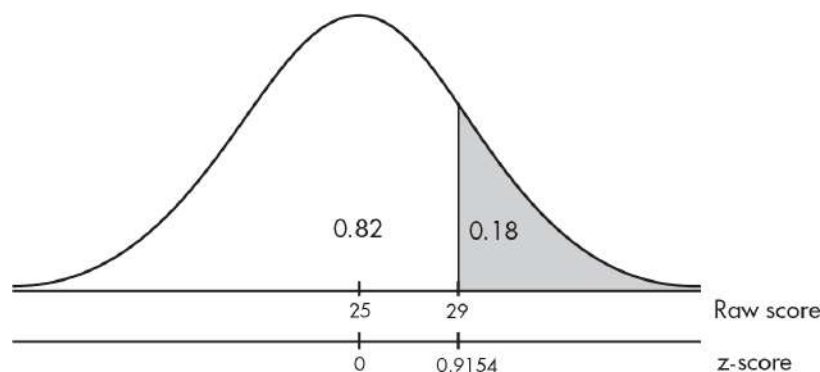
If we know that a distribution is normal, we can calculate the mean m and the standard deviation σ using percentage information from the population.

➤ Example 2.27

Given a normal distribution with a mean of 25, what is the standard deviation if 18% of the values are above 29?

✓ Solution

The 18% of values above corresponds to a probability of 0.82 below. Then $\text{invNorm}(0.82)$ gives the critical z -score of 0.9154. Thus, $29 - 25 = 4$ is equal to 0.9154 standard deviations; that is, $0.9154\sigma = 4$, and $\sigma = \frac{4}{0.9154} = 4.37$.

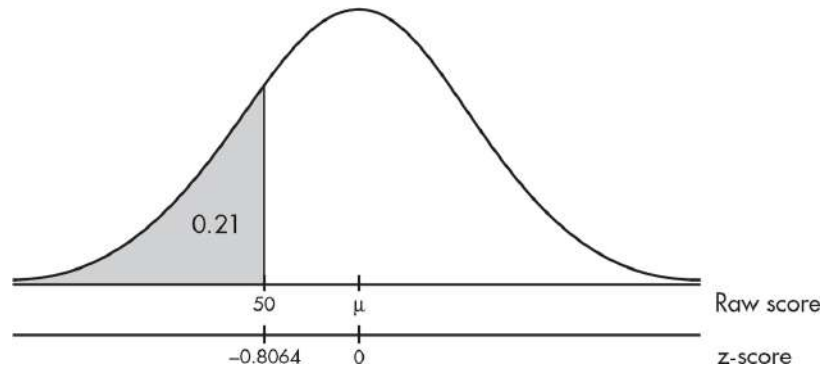


➤ Example 2.28

Given a normal distribution with a standard deviation of 10, what is the mean if 21% of the values are below 50?

✓ Solution

Using $\text{invNorm}(0.21)$ gives the critical z -score of -0.8064 . Thus, 50 is -0.8064 standard deviations from the mean, and so $m = 50 + 0.8064(10) = 58.06$.



In the above examples, we have assumed the population has a normal or approximately normal distribution. When you collect your own data, you must decide whether it is reasonable to assume the data come from a normal population before you can apply these procedures. In later units, we will have to check for normality before applying certain inference procedures.

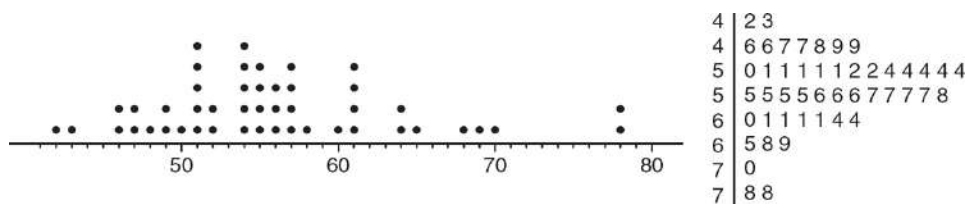
The initial check should be to draw a picture. Dotplots, stemplots, and histograms are all useful graphical displays to show that data are unimodal and roughly symmetric.

➤ Example 2.29

The ages at inauguration of U.S. presidents through 2025 were {57, 61, 57, 57, 58, 57, 61, 54, 68, 51, 49, 64, 50, 48, 65, 52, 56, 46, 54, 49, 51, 47, 55, 55, 54, 42, 51, 56, 55, 51, 54, 51, 60, 61, 43, 55, 56, 61, 52, 69, 64, 46, 54, 47, 70, 78, 78}. Can we conclude that the distribution is roughly normal?

✓ Solution

A dotplot and a stemplot of these data are as follows:



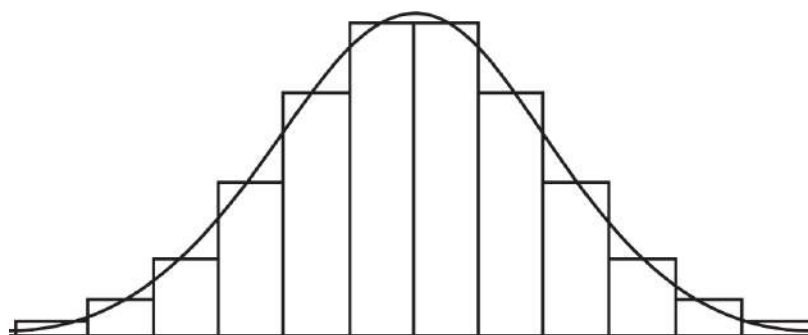
These two plots indicate a distribution that is unimodal and very roughly symmetric, with two outliers at 78 (Biden and Trump).

We can check rough normality by calculating summary statistics and counting values to check against the 68-95-99.7 rule. In this example, 1-VarStat gives $\bar{x} = 55.48$ and $s = 7.33$. Counting values (ages) gives the following:

Mean $\pm$ 1 SD	48.00 to 63.92	33 out of 47 = 70.2%
Mean $\pm$ 2 SD	40.04 to 71.88	45 out of 47 = 95.7%
Mean $\pm$ 3 SD	32.08 to 79.84	47 out of 47 = 100%

These results are close to the 68%, 95%, 99.7% expected from a normal distribution and so do suggest a roughly normal distribution.

The binomial takes values only at integers, while the normal is continuous with probabilities corresponding to areas over intervals. However, for large enough n , binomial distributions can be approximated by normal distributions.



Is the normal a good approximation? The answer, of course, depends on the error tolerances in particular situations. A general rule of thumb is that

the normal is a good approximation to the binomial whenever both np and $n(1 - p)$ are at least 10.

Proportions involve counting the presence or absence of some attribute, which suggests a binomial distribution. Inference procedures for proportions will rely on the normal distribution, and this makes sense only because the binomial can be approximated by the normal.

Quiz 14

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. Which of the following is a true statement?
 - (A) The area under a normal curve is always equal to 1, no matter what the mean and standard deviation are.
 - (B) All bell-shaped curves are normal distributions for some choice of m and σ .
 - (C) The smaller the standard deviation of a normal curve, the lower and more spread out the graph.
 - (D) Depending upon the value of the standard deviation, the mean and median of a particular normal distribution may be different.

2. The continuous random variable X has a normal distribution with a mean of 20 and a standard deviation of 4. For which of the following is the probability equal to 0?
 - (A) $P(X = 22)$
 - (B) $P(X < 22)$
 - (C) $P(X > 22)$
 - (D) $P(21 < X < 22)$

3. Populations P1 and P2 are normally distributed and have identical means. However, the standard deviation of P1 is twice the standard deviation of P2. What can be said about the percentage of observations falling within 2 standard deviations of the mean for each population?
- (A) The percentage for P1 is twice the percentage for P2.
 - (B) The percentage for P1 is greater than but not twice as great as the percentage for P2.
 - (C) The percentage for P2 is twice the percentage for P1.
 - (D) The percentages are identical.

In Questions 4–9, assume the given distributions are approximately normal.

4. It is estimated that a single cigarette will take an average of 20 minutes off of your life expectancy. Assuming a standard deviation of 5 minutes, what is the probability that a single cigarette will take over 30 minutes off of your life expectancy?
- (A) $P(z > 2)$
 - (A) $P(z > 5)$
 - (A) $P(z > 10)$
 - (A) $P(z > 30)$
5. An electronic product takes an average of 3.4 hours to move through an assembly line. If the standard deviation is 0.5 hours, what is the probability that an item will take between 3 and 4 hours to move through the assembly line?
- (A) $P(3 < z < 4)$
 - (B) $P\left(\frac{3}{0.5} < z < \frac{4}{0.5}\right)$
 - (C) $P((3 - 3.4)(0.5) < z < (4 - 3.4)(0.5))$
 - (D) $P\left(\frac{3 - 3.4}{0.5} < z < \frac{4 - 3.4}{0.5}\right)$

6. The mean noise level in a restaurant is 70 decibels with a standard deviation of 4 decibels. Ninety-nine percent of the time the noise level is below what value?
- (A) $70 + 1.96(4)$
(B) $70 - 2.326(4)$
(C) $70 + 2.326(4)$
(D) $70 + 2.576(4)$
7. The mean income per household in a certain state is \$9,500 with a standard deviation of \$1,750. The middle 95% of incomes are between what two values?
- (A) $9,500 \pm 1.645(1,750)$
(B) $9,500 \pm 1.96(1,750)$
(C) $9,500 \pm 1.645\left(\frac{1,750}{2}\right)$
(A) $9,500 \pm \frac{1,750}{1.96}$
8. Cucumbers grown on a certain farm have weights with a standard deviation of 2 ounces. What is the mean weight if 85% of the cucumbers weigh less than 16 ounces?
- (A) $16 - 0.85(2)$
(B) $16 - 1.036(2)$
(C) $16 + 0.85(2)$
(D) $16 + 1.036(2)$
9. The mean and standard deviation of the sample data collected on a continuous variable x are 10 and 5, respectively. The following table shows the relative frequencies of the data in the given intervals.

Interval	Relative Frequency
$-5 \leq x < 0$	0.03
$0 \leq x < 5$	0.12

Interval	Relative Frequency
$5 \leq x < 10$	0.35
$10 \leq x < 15$	0.34
$15 \leq x < 20$	0.14
$20 \leq x < 25$	0.02

Based on the table, do the data support using a normal model to approximate population characteristics?

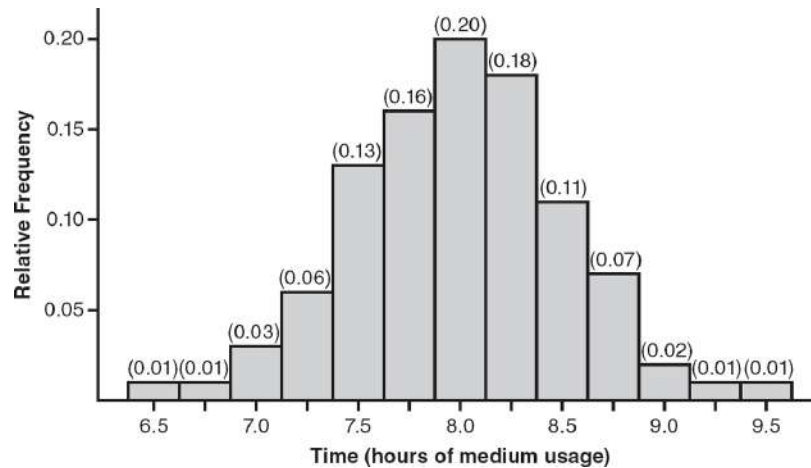
- (A) No, because some values are negative.
- (B) Yes, because the sum of the relative frequencies is 1.00.
- (C) Yes, because the relative frequency distribution roughly follows the empirical rule.
- (D) This cannot be answered without knowing the sample size.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Scores on a university entrance exam are approximately normally distributed with a mean of 150 and a standard deviation of 25.
 - (a) The university will accept 90% of applicants (based on entrance exam score). What is the cutoff score for acceptance (rounded to the nearest integer)?
 - (b) Given that a student scores over 100, what is the probability of acceptance?
 - (c) Given that three random students all score over 100, what is the probability that at least one is not accepted?

2. The following histogram shows the relative frequencies of the times of medium usage between necessary charging, recorded to the nearest 0.25 hour, of a sample of smartphones of a particular model. The mean is 8.03 hours, and the standard deviation is 0.49 hours.



One smartphone is selected at random.

- Based on the histogram, what is the probability that the selected smartphone will last at least 8.5 hours of medium usage before charging is necessary?
- The histogram displays a discrete probability model for time. However, time is often considered a continuous variable. Assuming a roughly normal model using the given mean and standard deviation, what is the probability that the selected smartphone will last at least 8.5 hours of medium usage before charging is necessary?
- Explain why your answers to (a) and (b) are or are not the same.
- To estimate the probability that this model smartphone will last exactly 7.25 hours before charging is necessary (assuming medium usage), should the above discrete or above continuous model for time be used? Explain.
- To estimate the probability that this model smartphone will last at least 7.1 hours before charging is necessary (assuming medium usage), should the above discrete or above continuous model for time be used? Explain.

The answers for this quiz can be found beginning on [page 488](#).

Sampling Distributions and the Central Limit Theorem

Important

A *sampling distribution* is not the same thing as the distribution of a sample.

The population mean μ , the population standard deviation σ , the population proportion p , and the population slope β are all examples of *population parameters*. A sample mean $\bar{x}$, a sample standard deviation s , a sample proportion $\hat{p}$, and a sample slope b are all examples of *sample statistics*. While a population parameter is a fixed quantity, statistics vary depending on the particular sample chosen. The probability distribution showing how a statistic varies is called a *sampling distribution*. Although different samples from the same population produce different statistics, the distribution of statistics from many different random samples of the same size is often predictable. For large enough samples, the normal distribution can be used to approximate many of these sampling distributions. These concepts will be critical for understanding statistical inference, the key topic that runs through the rest of the curriculum.

Real-life distributions are seldom, if ever, exactly normal. However, it can be shown mathematically that no matter how the original population is distributed, if n is large enough, the set of sample means is approximately normally distributed. For example, there is no reason to suppose that the amounts of money that different people spend in grocery stores are normally distributed. However, if each day we survey 50 people leaving a store and determine the average grocery bill, these daily averages will have an approximately normal distribution.

The following principle, called the *central limit theorem* of statistics (abbreviated CLT), forms the basis of much of what we discuss in this and the following units.

Start with a population with any shape distribution whatsoever. Pick n sufficiently large (at least 30), and take all samples of size n . Compute the mean of each of these samples. Then the set of all sample means is approximately normally distributed.

We say that for sufficiently large n , the *sampling distribution* of the sample mean $\bar{x}$ is approximately normal. Note that the CLT is only about the shape of the sampling distribution of the sample mean, not about its center or variability. However, we will see that

- $\mu_{\bar{x}} = \mu$; that is, the mean $\mu_{\bar{x}}$ of the sampling distribution of $\bar{x}$ equals the population mean μ .
- $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$; that is, the standard deviation $\sigma_{\bar{x}}$ of the sampling distribution of $\bar{x}$ equals the population standard deviation σ divided by the square root of the sample size.

While we mention $n \geq 30$ as a rough rule of thumb (and is sufficient for the AP exam), n should be chosen larger if more accuracy is required or if the original population is far from normal. However, **if the population itself has a normal distribution, then the sampling distribution of the sample mean is normal no matter what the sample size.** When working with a specific sample, we will also have the assumptions of a random sample and of sample size n no larger than 10% of the population (if sampling without replacement).

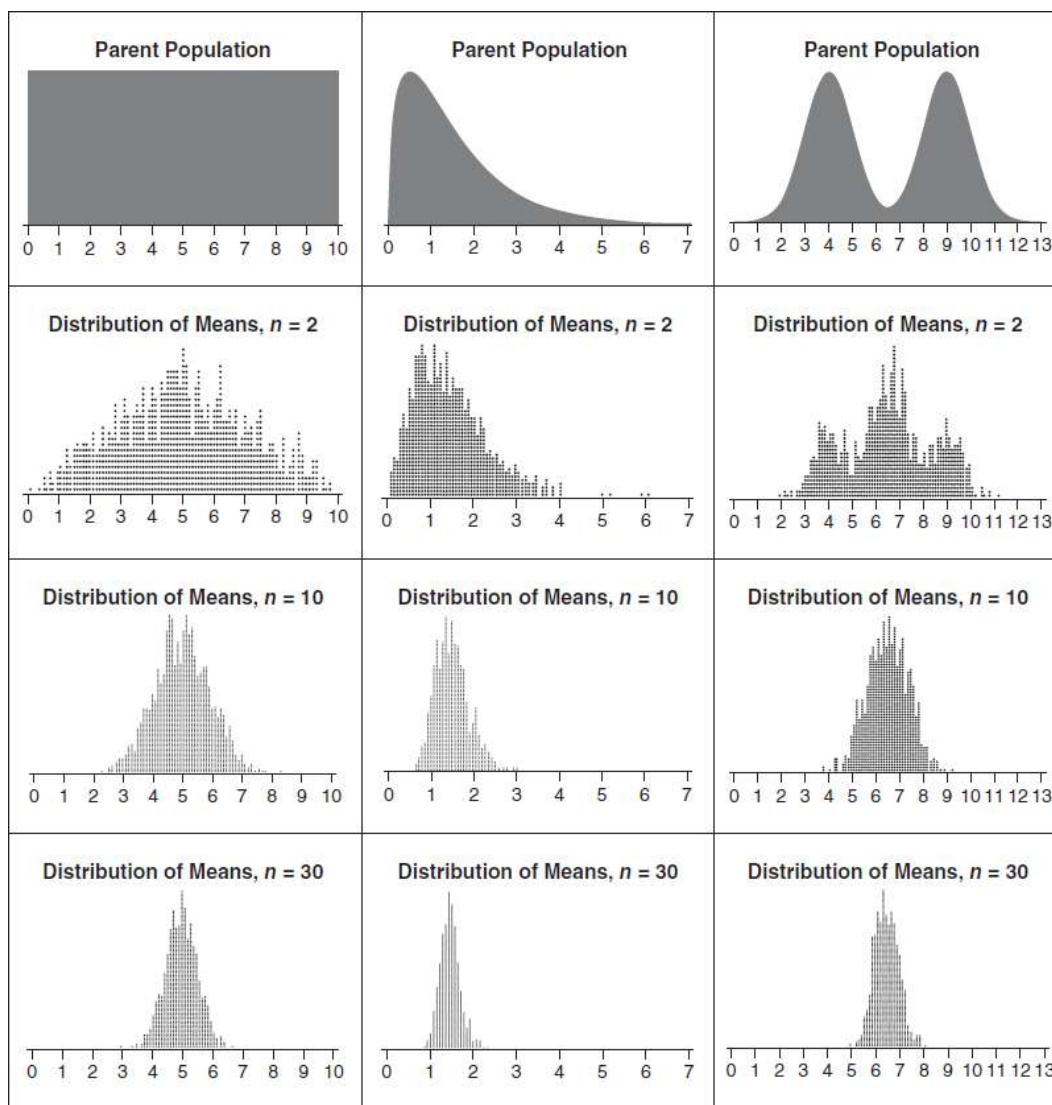
There are six key ideas to keep in mind:

- Averages vary less than individual values.
- Averages based on larger samples vary less than averages based on smaller samples.
- The central limit theorem (CLT) states that when the sample size is sufficiently large ($n \geq 30$ is used in AP Statistics), the sampling distribution of the mean will be approximately normal.
- The larger the sample size n , the closer the *sample distribution* is to the population distribution.
- The larger the sample size n , the closer the *sampling distribution* of $\bar{x}$ is to a normal distribution.

- If the original population has a normal distribution, then the sampling distribution of $\bar{x}$ has a normal distribution, no matter what the sample size n .

➤ Example 2.30

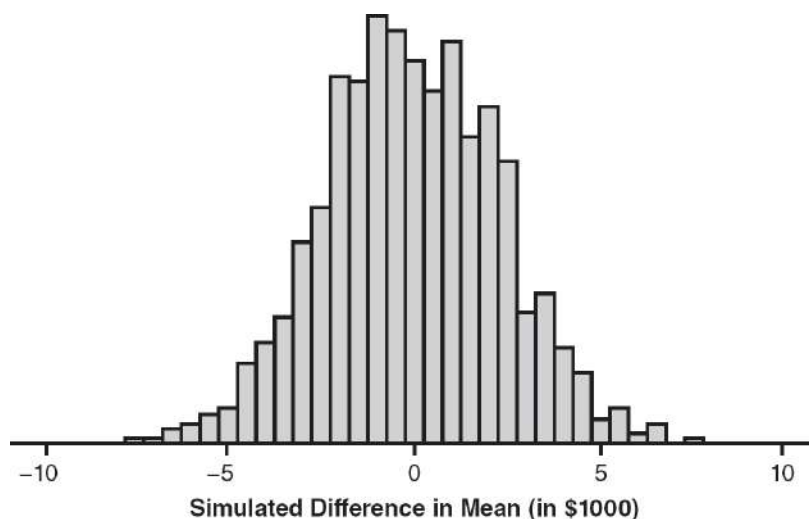
Following are simulations for sampling distributions for samples of sizes $n = 2$, $n = 10$, and $n = 30$ from three different populations. Note that as the sample size increases, the sampling distribution of $\bar{x}$ looks less like the population distribution and more like a normal distribution.



One example of a sampling distribution is a *randomization distribution*, a collection of statistics generated by simulation. For a sample survey, we can repeatedly randomly reassign individual responses to the different survey groups to see if our observed difference is significant. For an experiment, we can repeatedly randomly reassign the individual response values to treatment groups to see if our observed difference is significant.

➤ Example 2.31

In a random sample of 15 male Information Technology professionals, the average salary was \$59,325, while in an independent random sample of 15 female Information Technology professionals the average salary was \$51,850. The question of interest is whether or not the observed difference of $\$59,325 - \$51,850 = \$7,475$ is significant. The randomization distribution method proceeds as follows. All 30 salaries are put together, and 15 are randomly reassigned to the “Male” subgroup and the remaining 15 to the “Female” subgroup. The mean difference in salary between the two groups is calculated, and the procedure is repeated over and over. Suppose the resulting histogram of differences, a simulated sampling distribution of the differences, is below.



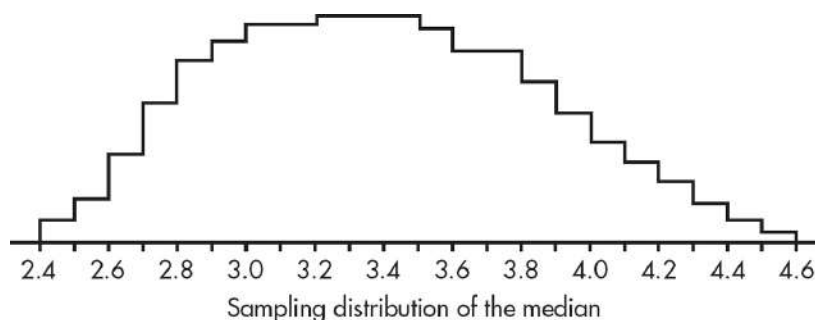
As can be seen, the observed difference of \$7,475, while possible, is very rare among the simulated differences. We conclude that the salary gap between male and female IT professionals is statistically significant.

Don't be confused by the several different distributions being discussed! First, there's the distribution of the original population, whose shape may be uniform, bell-shaped, strongly skewed, or any other possible shape. Second, there's the distribution of the data in the sample, and the larger the sample size, the more this will look like the population distribution. Third, there's the distribution of the means of many samples of a given size. The amazing fact is that this *sampling distribution* can be described by a normal model, regardless of the shape of the original population.

Does the central limit theorem say anything about proportions? If the population values are comprised of just 0s and 1s (numerical codes for failures and successes), we have a distribution that is bimodal in the extreme. Means of samples drawn from that population are sample proportions, so the sampling distribution of sample proportions is a special case of the CLT. The CLT doesn't kick in until the sample size gets fairly large, especially if the population is far from symmetric (p far from 0.5). Requiring $np \geq 10$ and $nq \geq 10$ is a rule of thumb suggesting how large n must be.

► Example 2.32

A study is made of the number of dreams high school students remember having every night. The median number is 3.41 with a variance of 1.46 and a minimum of 0. Now taking a large number of random samples of 15 students, we calculate the median for each sample and graph the resulting simulated sampling distribution.



The simulated sampling distribution for the sample median is roughly bell-shaped and centered around 3.4 with a range of approximately $4.6 - 2.4 = 2.2$.

Assuming the range of a roughly normal finite set is approximately six standard deviations, we can estimate the standard deviation to be about $\frac{\sigma}{\sqrt{50}}$

Quiz 15

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the questions.

1. Which of the following is the best interpretation of the sampling distribution of a sample mean?
 - (A) The distribution of all the values in a sample of size n
 - (B) The distribution of the population
 - (C) The distribution of all possible values of the parameter μ
 - (E) The distribution of the statistic $\bar{x}$ for all possible samples of a given size n from the population

2. A population is strongly skewed. For which of the following sample sizes would the sampling distribution of the sample mean be closest to normal in shape?
 - (A) $n = 35$
 - (B) $n = 115$
 - (C) The sampling distribution of the sample mean cannot be close to normal if the population is strongly skewed.
 - (D) The sampling distribution of the sample mean is always approximately normal.

3. In a recent survey of new mothers, the mean number of diapers per day over the first month of a baby's life was 10.5. Two simulations will be conducted for the sampling distribution of a sample mean from a population with a true mean of 10.5. Simulation A will consist of

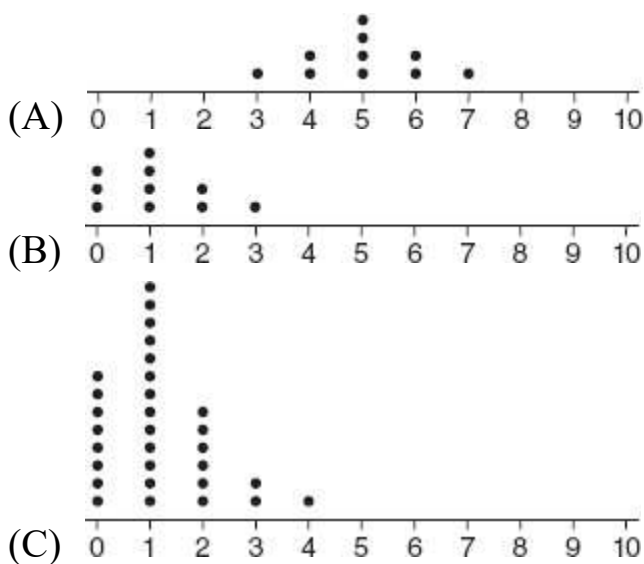
1,000 trials with a sample size of 100. Simulation B will consist of 2,000 trials with a sample size of 50.

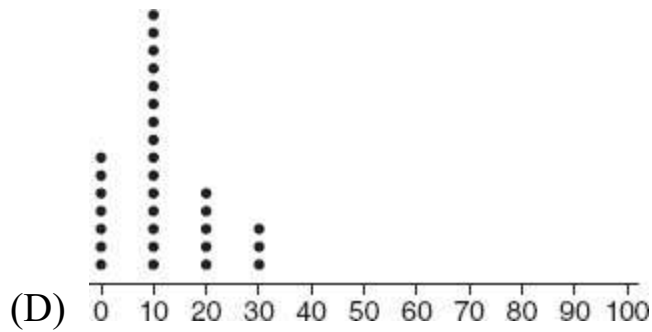
Which of the following describes the center and variability of the two simulations?

- (A) Their centers will be roughly equal, and their variabilities will be roughly equal.
- (B) Their centers will be roughly equal; however, the variability of Simulation A will be less than the variability of Simulation B.
- (C) Their centers will be roughly equal; however, the variability of Simulation A will be greater than the variability of Simulation B.
- (D) Their variabilities will be roughly equal; however, the mean of Simulation A will be less than the mean of Simulation B.

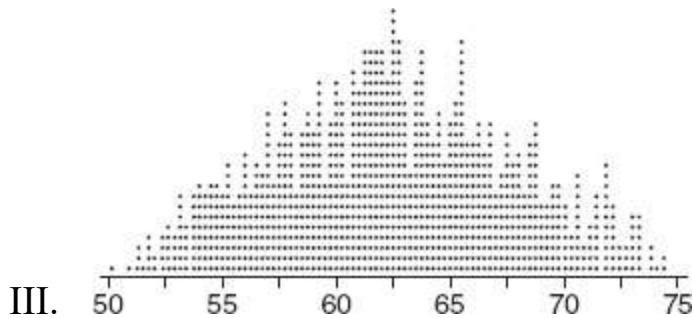
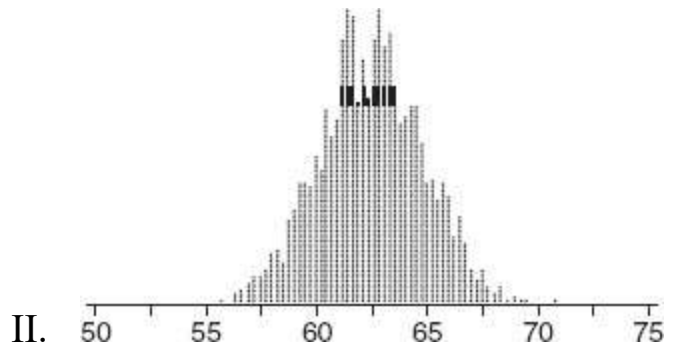
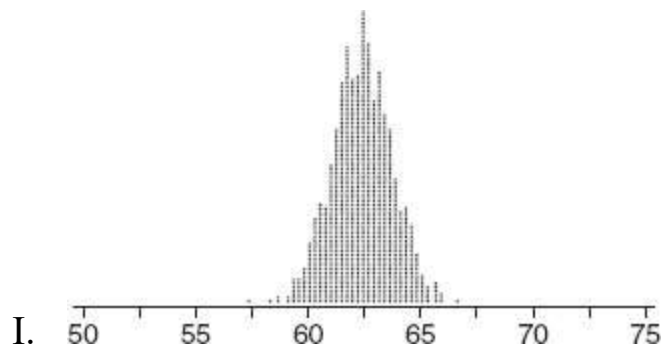
4. Approximately 10% of Americans identify as being multiracial. A simulation was conducted that consisted of 30 trials with a population parameter of 0.1. Each trial consisted of a sample size of 10. The number of successes out of 10 was recorded where success represented a person identifying as multiracial.

Which of the following dotplots best represents a possible simulation?





5. Three sampling distributions of the sample mean are constructed from the same population but use different sample sizes.

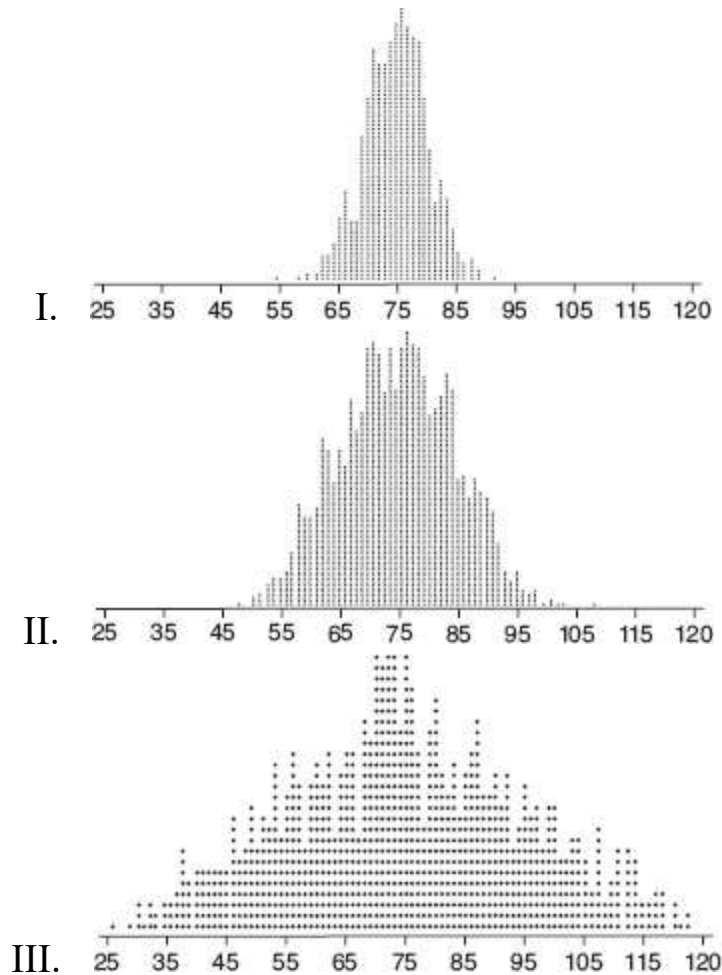


Which histogram is based on samples with the largest sample size?

- (A) I
(B) II

- (C) III
 (D) It is not possible to tell by looking at the histograms.

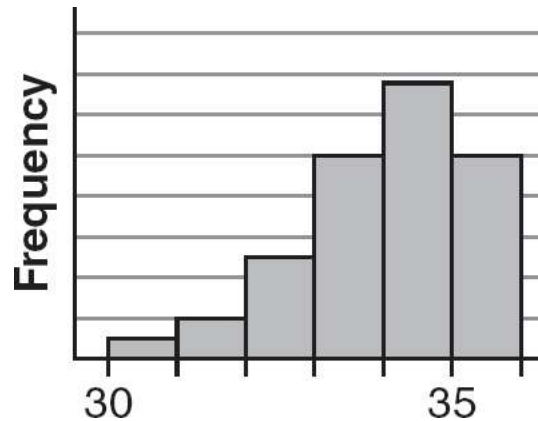
6. Three sampling distributions of the sample mean are constructed from different populations; however, all use the same sample size.



Which histogram is based on samples from the population with the largest standard deviation?

- (A) I
 (B) II
 (C) III
 (D) It is not possible to tell by looking at the histograms.

7. A recent survey indicated that the average hours worked per week in Switzerland is 34.1. A simulation was conducted to create a sampling distribution of the sample mean for a population with a mean of 34.1. The following is the resulting histogram of sample means.



[Follow for extended description](#)

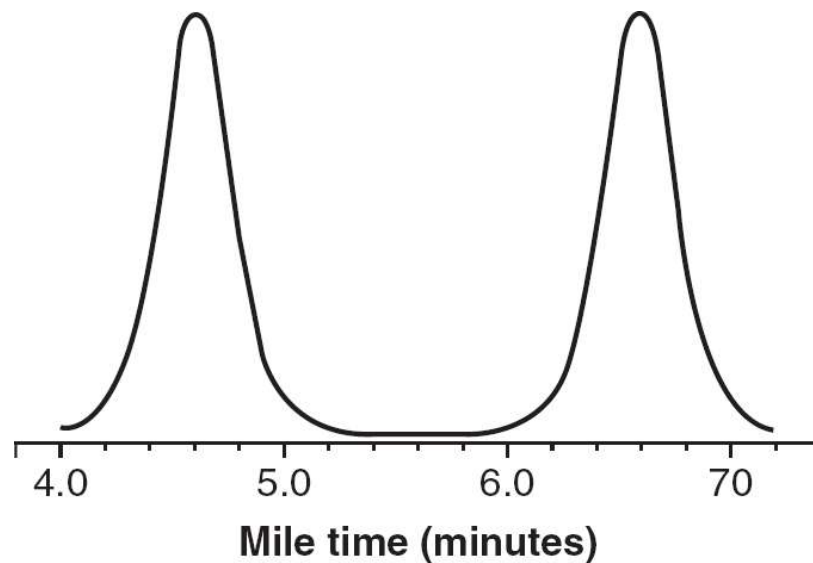
Which of the following is the most likely reason why the distribution is not approximately normal?

- (A) The original population is not approximately normal.
 - (B) The samples were not randomly selected.
 - (C) The sample distributions were mostly skewed left.
 - (D) The sample size was too small.
8. A manager at a big-box warehouse store claims that customers spend an average of \$120 per visit. A supervisor picks a random sample of 50 customers and calculates the mean spent among these customers during a single visit. He then repeats this process on a random sample of days.

Which of the following statements follows from the central limit theorem?

- (A) The distribution of amounts spent by the customers on a given day is approximately normal because $50 \geq 30$.

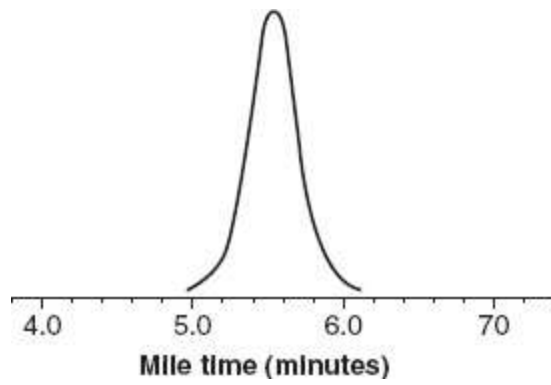
- (B) The distribution of amounts spent by the customers on a given day is approximately normal because $120 \geq 30$.
- (C) The distribution of sample means of amounts spent by customers from different days is approximately normal because $50 \geq 30$.
- (D) The distribution of sample means of amounts spent by customers from different days is approximately normal because $120 \geq 30$.
9. The histogram below represents the data obtained from timing mile runs (in minutes) from a large population of middle school and high school track team members.



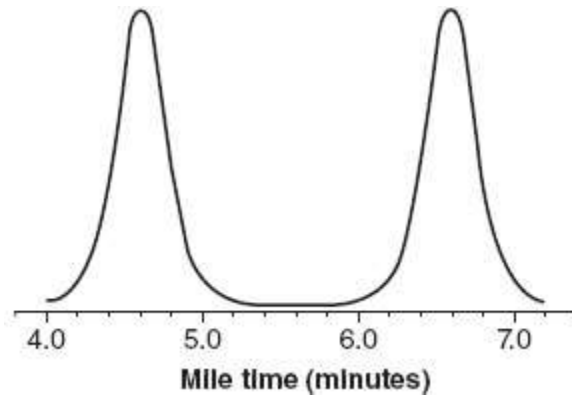
[Follow for extended description](#)

The sampling distribution of sample means based on samples of size 2 from this population is simulated and graphed as a histogram. Which of the following histograms is most likely that of the sampling distribution?

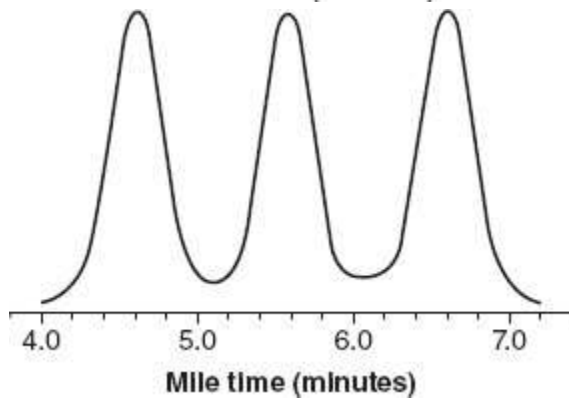
(A)



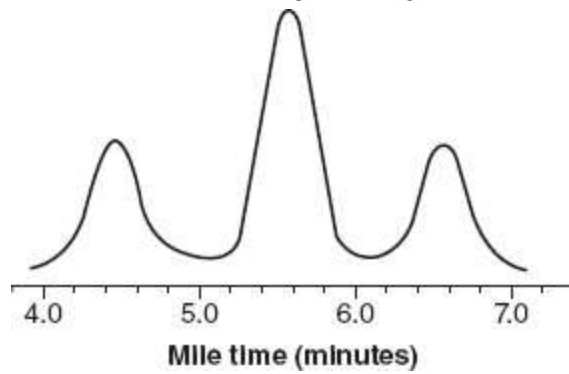
(B)



(C)



(D)



Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. The distribution of ages of death from heart disease is strongly skewed left because most such deaths occur at older ages with a few at younger ages.
 - (a) Describe the shape of the sampling distribution for the sample mean with a sample size of $n = 5$. Explain.
 - (b) Describe the shape of the sampling distribution for the sample mean with a sample size of $n = 50$. Explain.
2. Major League Baseball (MLB) salaries have a mean of approximately \$5,000,000 and a median of approximately \$2,000,000.
 - (a) What is the probable shape of the distribution of MLB salaries? Explain.
 - (b) What is the probable shape of the distribution of salaries from a random sample of 50 players? Explain.
 - (c) What is the probable shape of the distribution of mean salaries from random samples, each of sample size 50? Explain.
 - (d) How would the distribution of sample means in (c) change if the sample size increased? Explain.
 - (e) How would the distribution of sample means in (c) change if the number of samples increased? Explain.

The answers for this quiz can be found beginning on [page 489](#).

Summary

- Two-way tables are useful in showing relationships between two categorical variables.
- The row and column totals lead to calculations of the marginal distributions.

- Focusing on single rows or columns leads to calculations of conditional distributions.
- Segmented bar charts and mosaic plots are useful visual tools to show conditional distributions.
- The law of large numbers states that in the long run, a cumulative relative frequency tends closer and closer to what is called the probability of an event.
- The probability of the complement of an event is equal to 1 minus the probability of the event.
- Two events are mutually exclusive if they cannot occur simultaneously, and so $E \cap F = \emptyset$ and $P(E \cap F) = 0$.
- If two events are mutually exclusive, the probability that at least one will occur is the sum of their respective probabilities.
- More generally, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- The conditional probability of A given B is given by: $P(A|B) = \frac{P(A \cap B)}{P(B)}$; that is, the population of interest is shrunk.
- Two events A and B are independent if $P(A|B) = P(A)$; that is, the probability of one event occurring does not influence whether or not the other occurs.
- If two events A and B are independent, $P(A \cap B) = P(A) \times P(B)$.
- A random variable takes various numeric values, each with a given probability.
- The mean or expected value of a discrete random variable is calculated by $\mu = \sum xP(x)$, while the variance is $\sigma^2 = \sum (x - \mu)^2 P(x)$, and the standard deviation is $\sqrt{\sum (x - \mu)^2 P(x)}$.
- The binomial distribution arises when there are only two possible outcomes, the probability of success is constant, and there are a fixed number of trials.
- For a binomial distribution, the probability of exactly x successes in n trials is $\binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$.
- For a binomial distribution, the mean number of successes is np and the standard deviation is $\sqrt{np(1-p)}$.
- Drawing pictures of normal curves with horizontal lines showing raw scores and z-scores is usually helpful.

- Areas (probabilities) under a normal curve and critical values corresponding to given areas (probabilities) can be found using calculator software, such as `normalcdf` and `invNorm`.
- A sampling distribution of a statistic is the distribution of values for the statistic for all possible samples of the same size from a given population.
- The central limit theorem (CLT) states that when the sample size is sufficiently large, the sampling distribution of the mean will be approximately normal.

3

Inference for Categorical Data: Proportions

(6–10 multiple-choice questions on the AP exam)

Learning Objectives

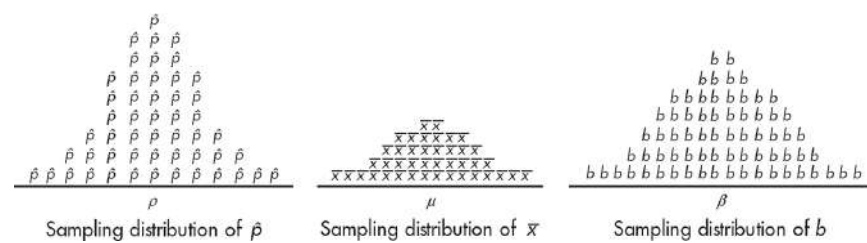
In this unit, you will learn how

- To be able to justify why an estimator is or is not unbiased.
- To be able to describe the shape and parameters of a sampling distribution for a sample proportion.
- To be able to check conditions to perform inference involving population proportions.
- To be able to construct and interpret a confidence interval for a population proportion and for the difference of population proportions.
- To understand the relationships among confidence level, sample size, margin of error, and confidence interval width.
- To be able to interpret confidence levels.
- To be able to perform a hypothesis test to evaluate a claim about a population proportion and about a difference between proportions.
- To be able to interpret a p -value in context.
- To be able to conduct a chi-square test for independence for associations between two categorical variables within a single population.
- To be able to conduct a chi-square test for homogeneity for comparing distributions of a categorical variable across two or more populations.

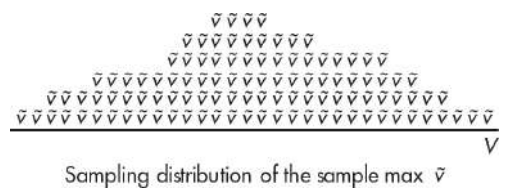
Biased and Unbiased Estimators

In Unit 1, we discussed *bias* in the context of surveys as a tendency to favor the selection of certain members of a population. In the context of sampling distributions, *bias* means that the sampling distribution of an estimator is not *centered* on the population parameter. An estimator is *unbiased* if, on average, it does not underestimate or overestimate the population parameter. The sampling

distributions of proportions, means, and slopes are *unbiased*. That is, for a given sample size, the set of all sample proportions, $\hat{p}$, is centered on the population proportion, p ; the set of all sample means, $\bar{x}$, is centered on the population mean, μ ; and the set of all sample slopes, b , is centered on the population slope, β . Here are some illustrative simulations:

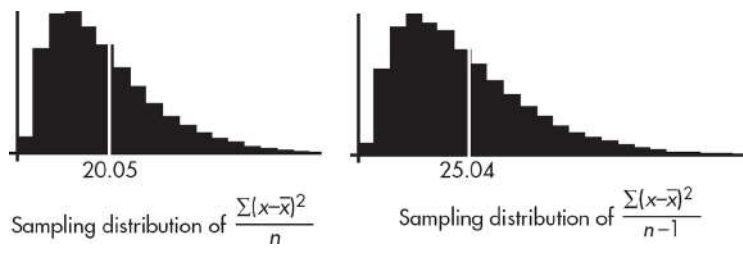


However, the sampling distribution for the maximum is clearly *biased*. That is, for a given sample size, the set of all sample maxima, $\tilde{v}$, is not centered on the population maximum, V . The estimator $\tilde{v}$ consistently underestimates the parameter V . For example, here is one simulation of sample maxima. Note that V falls far right of the center of the distribution of $\tilde{v}$.



Example 3.1

Note the following simulations for sampling distributions of $\frac{\Sigma(x - \bar{x})^2}{n}$ and $\frac{\Sigma(x - \bar{x})^2}{n - 1}$, both from a population with variance $\sigma^2 = 25$.



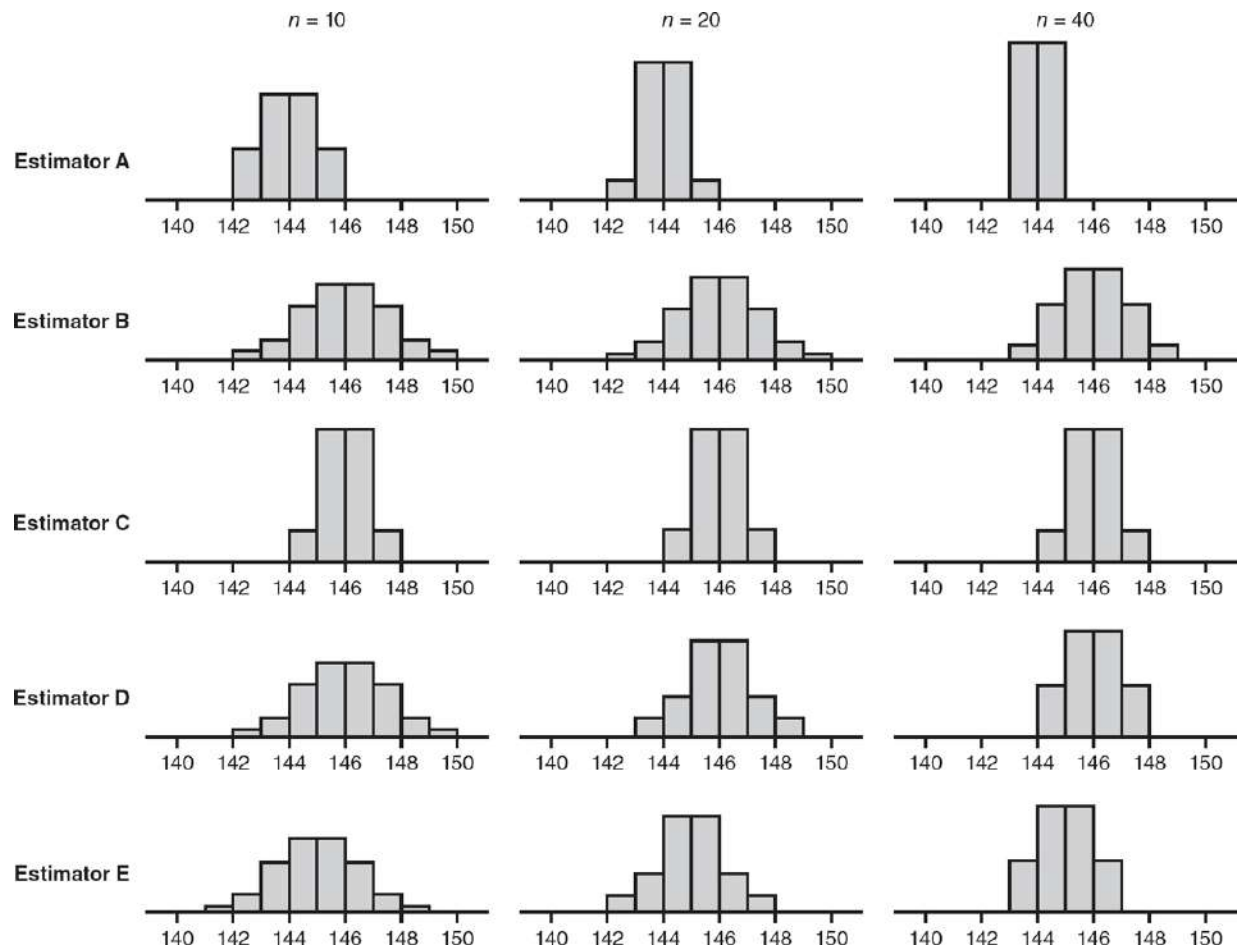
NOTE

This shows why we divide by $n - 1$ when calculating a sample variance and standard deviation.

We see that $\frac{\sum(x - \bar{x})^2}{n}$ is a *biased* estimator for σ^2 (that is, the distribution, with mean 20.05, is *not* approximately centered at σ^2), while $\frac{\sum(x - \bar{x})^2}{n-1}$ is an *unbiased* estimator for σ^2 (that is, the distribution, with mean 25.04, is approximately centered at σ^2).

Example 3.2

Five new estimators are being evaluated with regard to quality control in manufacturing professional baseballs of a given weight. Each estimator is tested every day for a month on samples of sizes $n = 10$, $n = 20$, and $n = 40$. The baseballs actually produced that month had a consistent mean weight of 146 grams. The distributions given by each estimator are as follows:



- Which of the above appear to be unbiased estimators of the population parameter?
- Which of the above exhibits the lowest variability for $n = 40$?

- (c) Which of the above is the best estimator if the selected estimator will eventually be used with a sample of size $n = 100$?

✓ Solutions

- (a) Estimators B, C, and D are unbiased estimators because they appear to have means approximately equal to the population mean of 146. A statistic used to estimate whether a population parameter is unbiased is if the mean of the sampling distribution of the statistic is equal to the true value of the parameter being estimated.
- (b) For $n = 40$, Estimator A exhibits the lowest variability, with a range of only 2 grams compared with the other ranges of 6 grams, 4 grams, 4 grams, and 4 grams, respectively.
- (c) Estimator D is best because we should choose an unbiased estimator with low variability. From part (a), we have Estimators B, C, and D as unbiased estimators. Now we look at the variability of these three statistics. As n increases, D shows tighter clustering around 146 than does B. Finally, while C looks better than D for $n = 40$, the estimator will be used with $n = 100$, and the D distribution is clearly converging as the sample size increases while the C distribution remains the same. Choose Estimator D.

There are three key ideas to keep in mind:

- Sample statistics are point estimators for population parameters.
- Estimators have variability, which can be modeled using probability.
- An estimator is unbiased if, on average, the value of the estimator equals the population parameter. Stated another way, an estimator is unbiased if, on average, it does not underestimate or overestimate the population parameter.

Sampling Distribution for Sample Proportions

We are interested in estimating a population proportion p by considering a single sample proportion $\hat{p}$. This sample proportion is just one of a whole universe of sample proportions. To judge the significance of a sample proportion, we must know how sample proportions vary; that is, we need to know the shape and parameters of the sampling distribution of $\hat{p}$.

Start with a population with a given proportion p . Take all samples of size n . Compute the proportion $\hat{p}$ in each of these samples. Once this is done:

1. The set of all sample proportions is approximately normally distributed.
2. The mean $\mu_{\hat{p}}$ of the set of sample proportions equals p , the population proportion.
3. The standard deviation $\sigma_{\hat{p}}$ of the set of sample proportions is approximately equal to $\sqrt{\frac{p(1-p)}{n}}$.

We say that the sampling distribution of $\hat{p}$ is approximately normal with mean p and standard deviation $\sqrt{\frac{p(1-p)}{n}}$.

When making calculations and drawing conclusions from a specific sample, the following three conditions must be checked:

- **The randomization condition:** The data should be collected using a random sample.
- **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
- **The normality condition:** The number of successes, np , and the number of failures, $n(1-p)$, should both be at least 10.

NOTE

The third condition is also written as $np \geq 10$ and $nq \geq 10$ where $q = 1 - p$.

Example 3.3

If we pick a simple random sample of size 80 from a large population, which of the following values of the population proportion p would allow use of the normal model for the sampling distribution of $\hat{p}$?

- (A) 0.10
- (B) 0.15
- (C) 0.90

(D) 0.95

(E) 0.99

 Solution

(B) The relevant condition is that both np and $nq \geq 10$. In (A), $np = (80)(0.10) = 8$; in (C), $nq = (80)(0.10) = 8$; in (D), $nq = (80)(0.05) = 4$; and in (E), $nq = (80)(0.01) = 0.8$. However, in (B), $np = (80)(0.15) = 12$ and $nq = (0.85)(80) = 68$ are both ≥ 10 .

 Example 3.4

It is hypothesized that the ratio $\frac{\text{outer edges of lips}}{\text{upper ridges of lips}}$ is most pleasing when equal to the golden ratio, $\varphi \approx 1.618$. A student would like to calculate the mean ratio for students in her high school and plans to randomly sample 50 out of the 780 students at the school. Does this satisfy the 10% condition?

 Solution

10% of 780 = 78, and $50 < 78$, so yes, the 10% condition is satisfied.

 Example 3.5

A human resources director is interested in the proportion of employees in an 85-member department who believe they are saving enough for retirement. She randomly selects 12 employees to interview. Explain whether conditions for inference are satisfied.

 Solution

It is stated that a random sample was taken, satisfying the randomization condition. The 10% condition is violated because 10% of 85 is 8.5, and 12 is greater than 8.5. The normality condition is also violated because np and nq total 12, so both cannot be ≥ 10 .

 Example 3.6

It is estimated that 80% of people with high math anxiety experience brain activity similar to that experienced under physical pain when anticipating doing a math problem.

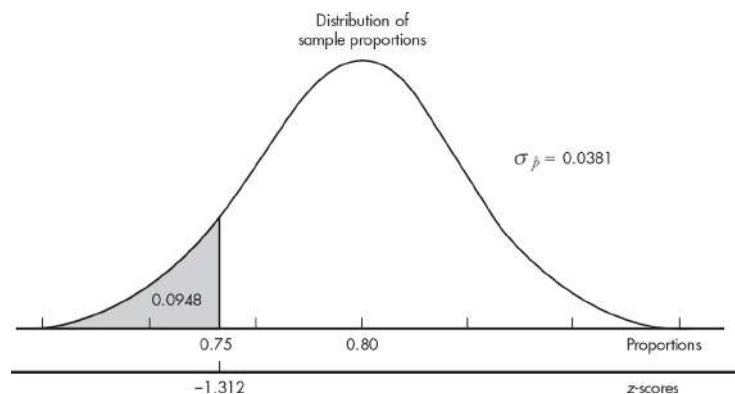
- If a simple random sample of 110 people with high math anxiety is taken, are the conditions for inference satisfied?
- In a simple random sample of 110 people with high math anxiety, what is the probability that less than 75% experience the physical pain brain activity?
- For this problem, interpret the mean and standard deviation of the sampling distribution of $\hat{p}$ in context.

✓ Solutions

- The sample is given to be random, both $np = (110)(0.80) = 88 \geq 10$ and $n(1 - p) = (110)(0.20) = 22 \geq 10$, and our sample is clearly less than 10% of all people with math anxiety.
- The sampling distribution of $\hat{p}$ is approximately normal with mean 0.80 and standard deviation

$$\sigma_{\hat{p}} = \sqrt{\frac{(0.80)(0.20)}{110}} = 0.0381. P(\hat{p} < 0.75) = P\left(z < \frac{0.75 - 0.80}{0.0381}\right) = P(z < -1.312) = 0.0948$$

[normalcdf(-1000, -1.312) = 0.0948 and normalcdf(-1000, 0.75, 0.80, 0.0381) = 0.0947]



Getting a sample proportion of 0.75 or less happens in about 9.48% of all possible samples of size 110 from this population.

- We say: For all random samples of size $n = 110$ from the population of people with high math anxiety, the sample proportions of people who experience physical pain brain activity when anticipating doing a math problem will have a mean of 0.80 and will typically vary by about 0.0381 from the population proportion of 0.80.

Quiz 16

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. A sample of size n is to be selected from a population with population proportion p . For the sampling distribution of the sample proportion to be approximately normal, which of the following should be checked?
 - (A) The sample size n should be at least 30.
 - (B) Both np and $n(1 - p)$ should be at least 10.
 - (C) The mean of the population should be np .
 - (D) The standard deviation of the population should be $\sqrt{np(1 - p)}$

2. A statistic $\hat{w}$ is an estimator for a population parameter W . The sampling distribution of $\hat{w}$ is not centered at W . What property does $\hat{w}$ exhibit?
 - (A) The sampling distribution of $\hat{w}$ is normal.
 - (B) The sampling distribution of $\hat{w}$ is not normal.
 - (C) $\hat{w}$ is a biased estimator.
 - (D) $\hat{w}$ is an unbiased estimator.

3. During a recent season, there were 560 players in the NBA. A sports statistician plans to sample 75 players randomly as to their views on whether they hope to stay with their current team for the rest of their career. Are all conditions for the sampling distribution of the sample proportion satisfied?
 - (A) Yes, all conditions are satisfied.
 - (B) No, the randomization condition is not satisfied.
 - (C) No, the 10% condition is not satisfied.
 - (D) No, the normality condition is not satisfied.

4. Seven percent of American adults believe that chocolate milk comes from brown cows. In a random sample of 400 American adults, what is the probability that between 5% and 10% will say that they believe that chocolate milk comes from brown cows?
- (A) $P(0.05 - 0.07 < z < 0.10 - 0.07)$
- (B) $P\left(\frac{0.05 - 0.07}{\sqrt{400(0.07)(0.93)}} < z < \frac{0.10 - 0.07}{\sqrt{400(0.07)(0.93)}}\right)$
- (C) $P\left(\frac{0.05 - 0.07}{\sqrt{\frac{(0.07)(0.93)}{400}}} < z < \frac{0.10 - 0.07}{\sqrt{\frac{(0.07)(0.93)}{400}}}\right)$
- (D) $P\left(\frac{0.05 - 0.07}{\sqrt{\frac{(0.05)(0.95)}{400}}} < z < \frac{0.10 - 0.07}{\sqrt{\frac{(0.10)(0.90)}{400}}}\right)$
5. A study is to be performed to estimate the proportion of voters who believe the economy is “heading in the right direction.” Which of the following pairs of sample size and population proportion p will result in the smallest variance for the sampling distribution of $\hat{p}$?
- (A) $n = 100$ and $p = 0.1$
- (B) $n = 100$ and $p = 0.5$
- (C) $n = 1,000$ and $p = 0.1$
- (D) $n = 1,000$ and $p = 0.5$
6. In a study of a large number of patients arriving at a city hospital emergency room, the proportion of initial misdiagnoses was 0.14. The hospital typically averages between 100 and 200 emergency arrivals per day. Two simulations will be conducted for the sampling distribution of a sample proportion with a true proportion of 0.14. Simulation A will consist of 4,000 trials with a sample size of 100, while simulation B will consist of 2,000 trials with a sample size of 200. Which of the following statements correctly compares the center and variability of the two simulations?
- (A) The centers will be roughly equal; however, the variability of simulation A will be less than the variability of simulation B .
- (B) The centers will be roughly equal; however, the variability of simulation A will be greater than the variability of simulation B .

- (C) The center of simulation A will be less than the center of simulation B ; however, the variability of simulation A will be greater than the variability of simulation B .
- (D) The center of simulation A will be greater than the center of simulation B ; however, the variability of simulation A will be less than the variability of simulation B .

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Suppose that 16% of teenagers who play video games would enjoy playing a new Super Mario game. The manufacturer doesn't know this actual proportion and plans on sampling a random sample of 80 teenagers who play video games.
 - (a) Calculate $\sigma_{\hat{p}}$, the standard deviation of the sampling distribution of $\hat{p}$.
 - (b) Are the conditions for inference met? Explain.
 - (c) If a different sample size would result in $\sigma_{\hat{p}} = 0.037$, is the sample size smaller or larger than 80? Explain.
2. It is estimated that 58% of all Americans sleep on their sides.
 - (a) What is the probability that a randomly chosen American sleeps on his or her side?
 - (b) In a random sample of five Americans, what is the probability that at least three sleep on their sides?
 - (c) In a random sample of 350 Americans, what is the probability that at least 50% sleep on their sides?

The answers for this Quiz can be found beginning on [page 490](#).

Confidence Interval for a Proportion

By using what we know about sampling distributions, we can establish a certain confidence that a sample proportion lies within a specified interval around the population proportion. Equivalently, we can say that we are equally confident that the population proportion lies within a specified interval around the sample proportion.

Typically, we consider 90%, 95%, and 99% confidence interval estimates, but any percent can be used. The percent is the percent of samples that would pinpoint the unknown p within plus or minus respective margins of error. We cannot say there is a 0.90, 0.95, or 0.99 *probability* that the proportion p is within a certain margin of error of a given sample proportion. The population proportion p either is or isn't within the specified interval, and so the probability it is within this interval is either 1 or 0.

As will be seen, there are two aspects to this concept:

- First, there is the *confidence interval*. This gives an interval of plausible values and is expressed in the following form:

$$\text{estimate} \pm \text{margin of error}$$

- Second, there is the success rate for the method, called the *confidence level*. This is the proportion of times repeated applications of this method would capture the true population parameter.

All of the above assume that certain conditions are met. For confidence interval estimates, the true population proportion p is unknown, so we use the sample proportion $\hat{p}$ in checking the normality condition. Similarly, since p is unknown, we estimate the standard deviation

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \text{ with the standard error defined by } SE(\hat{p}) = SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

The three conditions to be met and that we must check when finding a confidence interval for a proportion are as follows:

- **The randomization condition:** The data should be collected using a random sample.
- **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.

- **The normality condition:** The number of successes, $n\hat{p}$, and the number of failures, $n(1 - \hat{p})$, should be at least 10.

Our confidence interval will take the form:

$$\hat{p} \pm z^* SE(\hat{p}) = \hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where the critical value, z^* , depends on the chosen confidence level.

In the following examples, important points to keep in mind include:

- *When naming the parameter*, reference the proportion, the response variable, and the population in context.
- *Naming the procedure* is most easily done by simply stating “one-sample z-interval for a population proportion.”
- *Verifying conditions* means more than simply listing them with check marks. You must show work or give some reason to confirm verification.
- *Finding the interval* can be accomplished using calculator software (such as 1-PropZInt) instead of using the above formula, but answer choices using formulas may well appear on multiple-choice questions.
- *Interpreting the interval in context* must include mentioning the confidence level and parameter, and it must refer to the population, not the sample.
- *Interpreting the confidence level* never involves mentioning a specific interval.

► Example 3.7

- In a simple random sample of couples kissing in public places, such as airports, beaches, and parks, 80 out of 124 leaned their heads to the right. Establish a 95% confidence interval for the true proportion of public kissing couples who lean their heads to the right.
- Suppose a sociologist claimed that over 60% of all public kissing couples lean their heads to the right. Does this confidence interval give convincing evidence in support of the sociologist’s claim?
- Give an interpretation of the confidence level.

NOTE

$$n\hat{p} = (124)\left(\frac{80}{124}\right) = 80 \text{ is the number of “successes.”}$$

✓ Solutions

- (a) *Parameter:* Let p represent the proportion of the population of public kissing couples who lean their heads to the right.

Procedure: One sample z -interval for a population proportion.

Checks: We are given that the sample is an SRS, and we assume that 124 is less than 10% of all public kissing couples. For normality,

$$n\hat{p} = 80 \geq 10 \text{ and } n(1 - \hat{p}) = 124 - 80 = 44 \geq 10. .$$

Mechanics: Calculator software (such as 1-PropZInt) gives (0.56095, 0.72938).

[For instructional purposes, we note that $\hat{p} = \frac{80}{124} = 0.64516$,

$$SE(\hat{p}) = \sqrt{\frac{(0.64516)(0.35484)}{124}} = 0.04297, \text{ and } 0.64516 \pm 1.96(0.04297) = 0.64516 \pm 0.08422.]$$

Conclusion in context: We are 95% confident that the true proportion of all public kissing couples who lean their heads to the right is between 0.561 and 0.729. Alternatively, we can say the following: “We are 95% confident that the interval from 0.561 to 0.729 captures the true proportion of all public kissing couples who lean their heads to the right.”

- (b) No, the confidence interval (0.561 to 0.729) gives an interval of plausible values. Because all the values in the interval are not greater than 0.60, this confidence interval does not give convincing evidence in support of the sociologist’s claim that over 60% of all public kissing couples lean their heads to the right.
- (c) If all possible samples of 124 couples kissing in public places were observed, about 95% of the resulting confidence intervals would capture the true proportion of couples who lean their heads to the right.

In the above example, note the following:

- On the TI-84, for a confidence interval of a proportion, use 1-PropZInt, *not* ZInterval. Additionally, x must be an integer.
- The words “true” and “all” in the conclusion satisfy the requirement of referring to the population.
- In (b), the fact that $\hat{p} = 0.64516 > 0.60$ is not enough evidence to support the sociologist’s claim.

➤ Example 3.8

- (a) If 42% of a simple random sample of 550 young adults say that whoever asks for the date should pay for the first date, determine a 99% confidence interval estimate for the true proportion of all young adults who would say that whoever asks for the date should pay for the first date.
- (b) Does this confidence interval give convincing evidence in support of the claim that fewer than 50% of young adults would say that whoever asks for the date should pay for the first date?
- (c) Determine a 90% confidence interval. Note what effect going from 99% to 90% confidence has on the interval size.
- (d) What would the 99% confidence interval be if the sample proportion of 42% had come from a sample of size 2,200, which is four times the sample size of 550? Note what increasing the sample size fourfold does to the interval size.

Solutions

- (a) *Parameter:* Let p represent the proportion of the population of young adults who would say that whoever asks for the date should pay for the first date.

Procedure: One-sample z -interval for a population proportion

Checks: We are given that the sample is an SRS, and 550 is clearly less than 10% of the number of all young adults. For normality, n

$$n\hat{p} = 550(0.42) = 231 \geq 10 \text{ and } n(1 - \hat{p}) = 550(0.58) = 319 \geq 10.$$

Mechanics: Calculator software gives (0.36579, 0.47421). [For instructional purposes, we note that

$$\text{invNorm}(0.995) = 2.576, SE(\hat{p}) = \sqrt{\frac{(0.42)(0.58)}{550}} = 0.021, \text{ and } 0.42 \pm 2.576(0.021) = 0.42 \pm 0.054 \text{ or } (0.366, 0.474).$$

Conclusion in context: We are 99% confident that the true proportion of all young adults who would say that whoever asks for the date should pay for the first date is between 0.366 and 0.474. [Alternatively, we can say: “We are 99% confident that the interval from 0.366 to 0.474 captures the true proportion of all young adults who would say that whoever asks for the date should pay for the first date.”]

- (b) Yes, the confidence interval (0.366 to 0.474) gives an interval of plausible values. Because all these values are less than 0.50, this confidence interval gives convincing evidence in support of the claim that fewer than 50% of young adults would say that whoever asks for the date should pay for the first date.

- (c) Here, 1-PropZInt gives (0.38538, 0.45462). Note that the interval width decreased from $0.47421 - 0.36579 = 0.10842$ to $0.45462 - 0.38538 = 0.06924$. If we're willing to settle for a lower confidence (90% instead of 99%), the result is a smaller interval width.
- (d) Here, 1-PropZInt gives (0.3929, 0.4471). Note that the interval width decreased from $0.47421 - 0.36579 = 0.10842$ to $0.4471 - 0.3929 = 0.0542$. If the sample size had been fourfold larger, the interval width would have been cut in half.

As seen in the above example, for a given sample, *decreasing the confidence level* will result in the following:

- The critical value z^* will decrease.
- The margin of error $z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ will decrease.
- The width of the confidence interval will decrease.

As seen in the above example, when all other things remain the same, *increasing the sample size* will result in the following:

- The margin of error $z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ will decrease.
- The width of the confidence interval will decrease by a factor of $\frac{1}{d}$ if the sample size is increased by a factor of d^2 .

▶ Example 3.9

In a telephone survey of a random sample of 1,000 high school seniors, 832 answered yes to the question: “Do you want to get married someday?”

- (a) What is the parameter of interest?
- (b) Are the conditions for inference satisfied?
- (c) With what confidence can it be asserted that $83.2\% \pm 3\%$ of the high school population want to get married someday?

✓ Solutions

- (a) The parameter is p , which represents the proportion of the population of high school seniors who would answer yes to the question: “Do you want to get married someday?”

(b) We are given that there was a random sample, and clearly 1,000 is less than 10% of all high school seniors. For normality, $n\hat{p} = 832 \geq 10$ and $n(1 - \hat{p}) = 1,000 - 832 = 168 \geq 10$.

(c) $\hat{p} = \frac{832}{1,000} = 0.832$ and so $SE(\hat{p}) = \sqrt{\frac{(0.832)(0.168)}{1,000}} = 0.0118$

The relevant z-scores are $\pm \frac{0.03}{0.0118} = \pm 2.54$. $\text{Normalcdf}(-2.54, 2.54) = 0.9889$.

In other words, $83.2\% \pm 3\%$ is a 98.89% confidence interval estimate for the proportion of all high school seniors who would answer yes to the question: “Do you want to get married someday?”

As we have seen, there are two types of statements that come out of confidence intervals. First, we can interpret the **confidence interval** and say we are 90% confident that, for example, between 60% and 66% of all voters favor a bond issue. Second, we can interpret the **confidence level** and say that if this survey were conducted many times, about 90% of the resulting confidence intervals would contain the true proportion of voters who favor the bond issue.

If (0.60, 0.66) is a 90% confidence interval for the proportion of all voters who favor a bond issue, following are three common misconceptions:

MISCONCEPTION #1: The true proportion of all voters who support the bond issue is between 0.60 and 0.66.

Fact: The true proportion may or may not be in the specified interval; we are only 90% confident that it is in the interval.

MISCONCEPTION #2: There is a 0.90 probability that the true proportion of all voters who favor the bond issue is between 0.60 and 0.66.

Fact: For a given confidence interval, the true proportion either is or is not within the specified interval, and so the probability is either 1 or 0.

MISCONCEPTION #3: If this survey were conducted many times, about 90% of the sample proportions would be in the interval (0.60, 0.66).

Fact: (0.60, 0.66) is one specific interval derived from one sample proportion and has nothing to do with other samples and other sample proportions.

One important consideration in setting up a survey is the choice of sample size. To obtain a smaller, more precise interval estimate of the population proportion, we must either decrease the degree of confidence or increase the sample size. Similarly, if we want to increase the degree of confidence, we can either accept a wider interval estimate or increase the sample size. Again, while choosing a larger sample size may seem desirable, in the real world this decision involves time and cost considerations.

In setting up a survey to obtain a confidence interval estimate of the population proportion, what should we use for $\sigma_{\hat{p}}$? Algebraically it can be

shown that $\sqrt{p(1-p)}$ is greatest when $p = 0.5$, and in this case $\sqrt{p(1-p)} = \sqrt{0.5(1-0.5)} = 0.5$. Thus, $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ is at most $\frac{0.5}{\sqrt{n}}$. We make use of this fact to determine sample sizes in questions such as those in [Examples 3.10](#) and [3.11](#).

➤ Example 3.10

A CDC investigator wants to know the proportion of all adults who are fully vaccinated and boosted against Covid-19. If the answer must be within ± 0.03 at the 96% confidence level, how many adults should be in the selected sample?

For 96% confidence, there is 0.02 in each tail and $\text{invNorm}(0.02) = -2.05$ or $\text{invNorm}(0.98) = 2.05$.

TIP

Noting the inequality, you must round a result “up” if it is a decimal.

✓ Solution

The parameter is p , which represents the proportion of all adults who are fully vaccinated and boosted against Covid-19. We want $2.05\sigma_{\hat{p}} \leq 0.03$. From the above $\sigma_{\hat{p}}$ is at most $\frac{0.5}{\sqrt{n}}$, and so it is sufficient to consider $2.05\left(\frac{0.5}{\sqrt{n}}\right) \leq 0.03$.

Algebraically, we have $\sqrt{n} \geq \frac{2.05(0.5)}{0.03} = 34.17$ and $n \geq 1,167.4$. Therefore, choosing a sample of 1,168 adults gives the true vaccinated and boosted proportion to within ± 0.03 at the 96% level.

Note that the accuracy of the estimate does *not* depend on what fraction of the whole population we have sampled. What is critical is the *absolute size* of the sample. Is some minimal value of n necessary for the procedures we are using to be meaningful? Since we are using the normal approximation to the binomial, both np and $n(1-p)$ should be at least 10.

➤ Example 3.11

A study is undertaken to determine the proportion of industry executives who believe that workers’ pay should be based on individual performance. How

many executives should be interviewed if an estimate is desired at the 99% confidence level to within ± 0.06 ? To within ± 0.03 ? To within ± 0.02 ?

Solution

The parameter is p , which represents the proportion of the population of industry executives who believe that workers' pay should be based on individual performance. Algebraically, $2.576\left(\frac{0.5}{\sqrt{n}}\right) \leq 0.06$

$\sqrt{n} \geq \frac{2.576(0.5)}{0.06} = 21.5$, so $n \geq 462.25$. Similarly, $2.576\left(\frac{0.5}{\sqrt{n}}\right) \leq 0.03$ gives $\sqrt{n} \geq \frac{2.576(0.5)}{0.03} = 42.9$, so $n \geq 1,840.4$.

Finally, $2.576\left(\frac{0.5}{\sqrt{n}}\right) \leq 0.02$ gives $\sqrt{n} \geq \frac{2.576(0.5)}{0.02} = 64.4$, so $n \geq 4,147.4$. Thus, at the 99% confidence level, at least 463 executives should be interviewed for a margin of error of ± 0.06 , at least 1,841 should be interviewed for a margin of error of ± 0.03 , and at least 4,148 should be interviewed for a margin of error of ± 0.02 .

Note that to cut the interval estimate in half (from ± 0.06 to ± 0.03), we would have to increase the sample size fourfold. To cut the interval estimate to a third (from ± 0.06 to ± 0.02), a ninefold increase in the sample size would be required (answers are not exact because of round-off error).

More generally, to divide the interval estimate by d without affecting the confidence level, we must increase the sample size by a multiple of d^2 .

A formula for choosing the sample size n for a specific margin of error, MOE, can be found by algebraically solving for n in the equation

$\text{MOE} = z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$ to obtain:

$$n = \frac{(z^*)^2 \hat{p}(1 - \hat{p})}{\text{MOE}^2}$$

When $\hat{p}$ cannot be roughly calculated, use $\hat{p} = 0.5$ to obtain:

$$n = \frac{(z^*)^2 (0.25)}{\text{MOE}^2}$$

So in the previous example, with 99% confidence and

$\text{MOE} = 0.02$, $n = \frac{2.576^2 (0.25)}{0.02^2} = 4,147.36$, which can be rounded up to 4,148.

Quiz 17

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. For a class project, an AP Statistics student wishes to determine a 95% confidence interval for the proportion of students in the school building who have social media profile pages. The total number of students in the building is 3,520. Which of the following combinations of n and $\hat{p}$ will satisfy the conditions for inference?
 - (A) $n = 25$ and $\hat{p} = 0.35$
 - (B) $n = 50$ and $\hat{p} = 0.17$
 - (C) $n = 75$ and $\hat{p} = 0.15$
 - (D) $n = 100$ and $\hat{p} = 0.91$

2. In a simple random sample of 60 teenagers, two-thirds said they would rather text a friend than call. What is the 98% confidence interval for the proportion of all teenagers who would rather text than call a friend?
 - (A) $\frac{2}{3} \pm 1.96\sqrt{\frac{(2/3)(1/3)}{60}}$
 - (B) $\frac{2}{3} \pm 2.054\sqrt{\frac{(2/3)(1/3)}{60}}$
 - (C) $\frac{2}{3} \pm 2.326\sqrt{\frac{(2/3)(1/3)}{60}}$
 - (D) $\frac{2}{3} \pm 2.391\sqrt{\frac{(2/3)(1/3)}{60}}$

3. A school district has 250 teachers. A member of the Board of Education wanted to estimate the percentage of teachers who would be willing to accept a lower pay raise in return for a substantial increase in health benefits. Thirty randomly selected teachers were surveyed. Of the 30 selected teachers, 14 expressed a willingness to accept a lower pay raise in return for a substantial increase in health benefits. Have the conditions been met for inference with a one-sample z -interval?
 - (A) Yes, the conditions for inference have been met.
 - (B) No, because the sample likely has undercoverage bias.

- (C) No, because the sample size is not large enough to ensure that the sampling distribution is approximately normal.
- (D) No, because the sample size violates the 10% condition.

4. In a random sample of 150 adults, 60 admitted to feeling lonely at least once during the previous week. A 95 percent confidence interval for the proportion of all adults who would admit to feeling lonely at least once during the previous week is given by (0.3216, 0.4784). Which of the following is a correct interpretation of the confidence level of 95 percent?

- (A) We are 95 percent confident that the true proportion of all adults who would admit to feeling lonely at least once during the previous week is between 0.3216 and 0.4784.
- (B) If the proportion of adults who would admit to feeling lonely at least once during the previous week is calculated for repeated samples of 150 adults, 95 percent of these sample proportions will fall between 0.3216 and 0.4784.
- (C) In repeated samplings of 150 adults, approximately 95 percent of the calculated confidence intervals for the proportion of adults who would admit to feeling lonely at least once during the previous week will capture the associated sample proportion.
- (D) If a confidence interval for the proportion of adults who would admit to feeling lonely at least once during the previous week is calculated for repeated samples of 150 adults, approximately 95 percent of these intervals will contain the true population proportion.

5. A 2020 random survey of 1,470 adult Americans found that 29% of the American adult population would be willing to pay higher taxes if the government offered “Medicare for All.” With what degree of confidence can it be concluded that $29\% \pm 3\%$ of the American adult population would be willing to pay higher taxes if the government offered “Medicare for All”?

- (A) 49.5%
- (B) 90%
- (C) 95%
- (D) 99%

6. The population of the United States is about 330,000,000. During a time that it is strongly suggested to regularly wear face masks, suppose that a public health official wants to estimate, with 95 percent confidence, the proportion of Americans who do so within a margin of error of 2.5 percentage points. Approximately how many people would she need to randomly sample?
- (A) 150
 - (B) 1,500
 - (C) 15,000
 - (D) 150,000
7. From a random sample of dog owners, 43% indicated that they allow their dogs to sleep in bed with them. A 95 percent confidence interval was constructed from the sample, and the margin of error for the estimate was 3%. Which of the following is the best interpretation of the interval?
- (A) We are 95 percent confident that between 40% and 46% of the sample dog owners allow their dogs to sleep in bed with them.
 - (B) We are 95 percent confident that between 40% and 46% of the population of dog owners allow their dogs to sleep in bed with them.
 - (C) There is a 0.95 probability that between 40% and 46% of the population of dog owners allow their dogs to sleep in bed with them.
 - (D) If 95 percent confidence intervals are calculated from all possible samples of the given size, 95 percent of these intervals will capture the true percentage of the population of dog owners who allow their dogs to sleep in bed with them.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Net neutrality is the idea that all Internet traffic should be treated equally—that is, no Internet service provider (ISP) should have the power to favor

one source over another by blocking or paid prioritization. An online survey, responded to by 155,612 Internet users, reported that 93% favor net neutrality with a margin of error of 0.1%.

- (a) Is “93%” a statistic or a parameter? Explain.
 - (b) Without doing an actual calculation, explain why the margin of error is so small.
 - (c) How confident should you be that between 92.9% and 93.1% of all Internet users favor net neutrality?
2. In a random sample of 500 American women between the ages of 45 and 64, 305 said that breast cancer was the medical condition they feared most.
- (a) Calculate a 99% confidence interval for the proportion of all American women between the ages of 45 and 64 who would say that breast cancer was the medical condition they fear most.
 - (b) Explain the meaning of “99% confidence level” in this example.
 - (c) Currently, about 3% of female deaths are due to breast cancer. Is the value 3% consistent with the confidence interval obtained in (a)? Explain.

The answers to this Quiz can be found beginning on [page 491](#).

Quiz 18

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. An environmental activist, interested in advocating for the implementation of sustainable energy utilization, interviews a random sample of 30 white-collar workers. Only eight of them said they would be willing to accept cuts in their standard of living to protect the environment. Letting p be the proportion of all white-collar workers who would be willing to accept cuts in their standard of living to protect the environment, is it appropriate to

assume that the corresponding sampling distribution of $\hat{p}$ is approximately normal?

- (A) Yes, because the sample was chosen randomly.
- (B) Yes, because sampling distributions of the sample proportion are modeled with normal distributions.
- (C) Yes, because the sample size is ≥ 30 .
- (D) No, because the sample size is not large enough for the normality condition.

2. In a random sample of 500 adults, 480 said that they found it unacceptable that the federal government would hack into their emails. A 90% confidence interval for the proportion of all adults who would find it unacceptable that the federal government would hack into their emails is (0.946, 0.974). Which of the following is a correct interpretation of the confidence level 90 percent?

- (A) We are 90% confident that the true proportion of all adults who would find it unacceptable that the federal government would hack into their emails is between 0.946 and 0.974.
- (B) If a confidence interval for the proportion of adults who would find it unacceptable that the federal government would hack into their emails is calculated for repeated samples of 500 adults, 90% of these intervals will contain the true population proportion.
- (C) There is a 0.90 probability that the true proportion of all adults who would find it unacceptable that the federal government would hack into their emails is between 0.946 and 0.974.
- (D) If the proportion of adults who would find it unacceptable that the federal government would hack into their emails is calculated for repeated samples of 500 adults, 90% of these sample proportions will fall between 0.946 and 0.974.

3. Consider a sample with sample proportion $\hat{p}$. Which of the following best describes the purpose of a one-sample z-interval?

- (A) To estimate $\hat{p}$
- (B) To establish whether or not the sampling distribution of $\hat{p}$ is approximately normal

- (C) To estimate a margin of error for $\hat{p}$
- (D) To ascertain whether or not there is convincing evidence that $\hat{p}$ is the true population proportion

4. One month, the actual unemployment rate in Spain was 13.4%. If during that month you took a simple random sample of 100 Spaniards of working age and constructed a confidence interval estimate of the unemployment rate, which of the following would be true?

- (A) The center of the interval is 13.4.
- (B) The interval contains 13.4.
- (C) A 99% confidence interval estimate contains 13.4.
- (D) None of the above are true statements.

5. In a confidence interval using z-scores, the margin of error covers which of the following?

- (A) Sampling variability (sampling error)
- (B) Errors due to undercoverage and nonresponse in obtaining sample surveys
- (C) Errors due to using sample standard deviations as estimates for population standard deviations
- (D) Use of the normal curve as an estimation to the binomial

6. In a random sample of voters, 87% indicated that they do not have confidence in what the White House physician says about the president's health. A 95% confidence interval was constructed from the sample, and the margin of error for the estimate was 2%. Which of the following is the best interpretation of the interval?

- (A) There is a 0.95 probability that between 85% and 89% of the population of voters do not have confidence in what the White House physician says about the president's health.
- (B) We are 95% confident that between 85% and 89% of the sample voters do not have confidence in what the White House physician says about the president's health.

- (C) We are 95% confident that between 85% and 89% of the population of voters do not have confidence in what the White House physician says about the president's health.
- (D) If 95% confidence intervals are calculated from all possible samples of the given size, 95% of these intervals will capture the true percentage of the population of voters who do not have confidence in what the White House physician says about the president's health.

7. A politician wants to know what percentage of all voters support her position on the issue of forced busing for the racial integration of public schools. What size voter sample should be obtained to determine with 90% confidence the support level to within 4%?

- (A) 21
- (B) 25
- (C) 423
- (D) 600

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. During the H1N1 pandemic, one published study concluded that if someone in your family had H1N1, you had a 1 in 8 chance of also contracting the disease. A state health officer tracked a random sample of new H1N1 cases in her state and noted that 129 out of 876 family members later contracted the disease.
 - (a) Calculate a 90% confidence interval for the proportion of family members who contract H1N1 after an initial family member does in this state.
 - (b) Based on this confidence interval, is there evidence that the proportion of family members who contract H1N1 after an initial

family member does in this state is different from the 1 in 8 chance concluded in the published study? Explain.

- (c) Would the conclusion in (b) be any different with a 99% confidence interval? Explain.

2. In a July 2020 study of 1,050 randomly selected smokers, 74% said they would like to give up smoking.

- (a) At the 95% confidence level, what is the margin of error?
- (b) Calculate and explain the meaning of the “95% confidence interval” in this example.
- (c) Explain the meaning of “95% confidence level” in this example.
- (d) Give an example of possible response bias in this example.
- (e) If we want instead to be 99% confident, would our confidence interval need to be wider or narrower?
- (f) If the sample size were greater, would the margin of error be smaller or greater?

The answers to this Quiz can be found beginning on [page 492](#).

Logic of Hypothesis Testing

Closely related to the problem of estimating a population proportion or mean is the problem of testing a hypothesis about a population proportion or mean. For example, a travel agency might determine an interval estimate for the proportion of sunny days in the Virgin Islands or, alternatively, might test a tourist bureau’s claim about the proportion of sunny days. A consumer protection agency might determine an interval estimate for the mean nicotine content of a particular brand of cigarettes or, alternatively, might test a manufacturer’s claim about the mean nicotine content of its cigarettes. In each of these cases, the experimenter must decide whether the interest lies in estimating a population proportion or mean or in testing a claimed proportion or mean.

The general testing procedure is to choose a specific hypothesis to be tested, called the *null hypothesis*, pick an appropriate random sample, and then use measurements from the sample to determine the likelihood of the null hypothesis. If the sample statistic is far enough away from the claimed population parameter, we say that there is sufficient evidence to reject the null

hypothesis. We attempt to show that the null hypothesis is unacceptable by showing that it is improbable.

Consider the context of the population proportion. The null hypothesis H_0 is stated in the form of an equality statement about the *population* proportion (for example, $H_0: p = 0.37$). There is an *alternative hypothesis*, stated in the form of a strict inequality (for example, $H_a: p < 0.37$ or $H_a: p > 0.37$ or $H_a: p \neq 0.37$). The strength of the sample statistic $\hat{p}$ can be gauged through its associated *p-value*, which is the probability of obtaining a sample statistic as extreme (or more extreme) as the one obtained if the null hypothesis is assumed to be true. The smaller the *p-value*, the more significant the difference between the null hypothesis and the sample results.

Several points worth noting:

- Understand that the sample statistic almost never equals the claimed population parameter. Either the sample statistic differs by random chance or the claimed parameter is wrong. If you can eliminate random chance as a plausible explanation for the sample statistic, then you can say that there is convincing evidence that the claimed parameter is wrong.
- The equality reference in the null hypothesis can be stated using $\geq$ or $\leq$ when the alternative is one-sided in the other direction.
- Never base your hypotheses on what you find in the sample data.
- Note that only population parameter symbols like μ , p , and β appear in H_0 and H_a . Sample statistics like $\bar{x}$, $\hat{p}$, and b do NOT appear in H_0 or H_a .
- The *p-value* is a *conditional probability*; it assumes the null hypothesis is true. Assuming that H_0 is true, the *p-value* measures how likely it is to get evidence for H_a as strong as or stronger than the observed evidence by chance alone.
- For a one-sided test (the alternative is $<$ or $>$), the *p-value* is the proportion of values at or more extreme than the test statistic (in the direction of the alternative). For a two-sided test (the alternative is $\neq$), the *p-value* is the proportion of values at or more extreme than the test statistic in either direction; that is, it is twice the calculated tail probability.
- The *p-value* is NOT the probability of the null hypothesis being true.
- The significance level, α , is fixed at the beginning of the hypothesis test. The conclusion to the test will be based upon whether or not the calculated *p-value* is less than or greater than α . If the *p-value* $\leq \alpha$, reject

H_0 ; there is convincing evidence for H_a . If the p -value $> \alpha$, fail to reject H_0 ; there is not convincing evidence for H_a .

There are two types of possible errors: the error of mistakenly rejecting a true null hypothesis and the error of mistakenly failing to reject a false null hypothesis. The α -risk, also called the *significance level* of the test, is the probability of committing a *Type I error* and mistakenly rejecting a true null hypothesis. A *Type II error*, a mistaken failure to reject a false null hypothesis, has associated probability β . There is a different value of β for each possible correct value for the population parameter p . For each β , $1 - \beta$ is called the “power” of the test against the associated correct value. The *power* of a hypothesis test is the probability that a Type II error is not committed. That is, given a true alternative, the power is the probability of rejecting the false null hypothesis. Increasing the sample size and increasing the significance level are both ways of increasing the power. Also note that a true null that is further away from the hypothesized null is more likely to be detected, thus offering a more powerful test.

		Population truth	
		H_0 true	H_0 false
Decision based on sample	Reject H_0	Type I error	Correct decision
	Fail to reject H_0	Correct decision	Type II error

TIP

Be able to identify Type I and Type II errors in context and give possible consequences of each in context.

A simple illustration of the difference between a Type I and a Type II error is as follows. Suppose the null hypothesis is that all systems are operating satisfactorily with regard to a NASA launch. A Type I error would be to delay the launch mistakenly thinking that something was malfunctioning when everything was actually OK. A Type II error would be to fail to delay the launch mistakenly thinking everything was OK when something was actually malfunctioning. The power is the probability of recognizing a particular

malfunction. (Note the complementary aspect of power, a “good” thing, with Type II error, a “bad” thing.)

The U.S. justice system provides another often-quoted illustration. If the null hypothesis is that a person is innocent, a Type I error results when an innocent person is found guilty while a Type II error results when a guilty person is not convicted. We try to minimize Type I errors in criminal trials by demanding unanimous jury guilty verdicts. In civil suits, however, many states try to minimize Type II errors by accepting simple majority verdicts.

It should be emphasized that with regard to calculations, questions like “What is the *power* of this test?” and “What is the probability of a *Type II error* in this test?” cannot be answered without reference to a specific alternative hypothesis. Furthermore, AP students are not required to know how to calculate these probabilities. However, you are required to understand the concepts and interactions among the concepts.

Example 3.12

Research is conducted on a new weight loss medication with the following null hypothesis:

H_0 : The medication is effective in promoting weight loss

- (a) What would be the results of a Type I error?
- (b) What would be the results of a Type II error?

Solutions

- (a) A Type I error is falsely rejecting a true null hypothesis. In this case, if H_0 were true, the new medication would be effective in promoting weight loss. If you reject H_0 , the result would be a failure to recognize, manufacture, and make available an effective weight loss medication.
- (b) A Type II error is failing to reject a false null hypothesis. In this case, if H_0 were false, the new medication would not be effective in promoting weight loss. If you fail to reject H_0 , an ineffective drug would be needlessly produced and marketed to the public.

For a hypothesis test, we check if np_0 and $n(1 - p_0)$ are ≥ 10 . For a confidence interval, we check if $n\hat{p}$ and $n(1 - \hat{p})$ are ≥ 10 .

Hypothesis Test for a Proportion

Three conditions must be met and checked for:

- **The randomization condition:** The data should be collected using a random sample.
- **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
- **The normality condition:** The number of successes, np_0 , and the number of failures, $n(1 - p_0)$, should both be at least 10, where p_0 is the claimed null hypothesis proportion.

Example 3.13

- (a) A union spokesperson claims that 75% of union members will support a strike if their basic demands are not met. A company negotiator believes the true percentage is lower and runs a hypothesis test. What is the conclusion if 87 out of a simple random sample of 125 union members say they will strike? (Give conclusions both at the 0.05 and the 0.10 significance levels.)
- (b) For each of the two possible answers to part (a), what error might have been committed, Type I or Type II, and what would be a possible consequence?
- (c) Interpret the p -value in context.

Solutions

TIP

Hypotheses are always about the population, never about a sample; note that H_0 is $p = 0.75$, not $\hat{p} = 0.75$.

- (a) *Parameter:* Let p represent the proportion of all union members who will support a strike if their basic demands are not met.

Hypotheses: $H_0: p = 0.75$ and $H_a: p < 0.75$.

Procedure: One-sample z -test for a population proportion.

Checks: It is given that we have an SRS, and we must assume that 125 is less than 10% of the total union membership. For normality, $np_0 = 125(0.75) = 93.75 \geq 10$ and $n(1 - p_0) = 125(0.25) = 31.25 \geq 10$.

Mechanics: Calculator software (1-PropZTest) gives $z = -1.394$ and a p -value of 0.0816.

[For instructional purposes, we note that the observed sample proportion is $\hat{p} = \frac{151}{500} = 0.302$, and using the claimed proportion, 0.75, we have $\sigma_{\hat{p}} = \sqrt{\frac{(0.75)(0.25)}{125}} = 0.03873$ and $z = \frac{0.302 - 0.75}{0.03873} = -1.394$ with a resulting p -value of 0.0817.]

Important

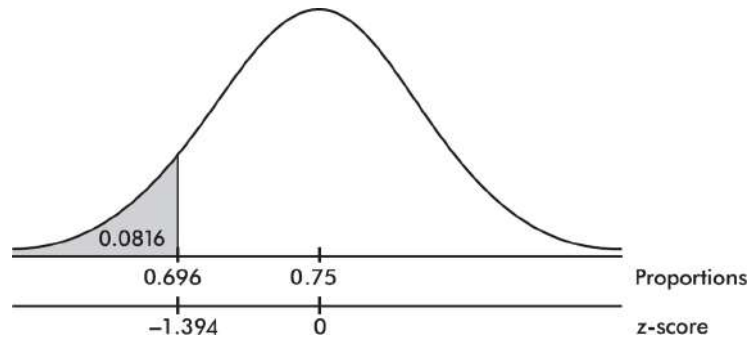
We never “accept” a null hypothesis; we either do or do not have sufficient evidence to reject it.

Conclusion in context with linkage to the p -value: There are different answers depending on the choice of significance level. Answers using $\alpha = 0.05$ and $\alpha = 0.10$ are as follows:

1. With this large of a p -value, $0.0816 > 0.05$, there is not sufficient evidence to reject H_0 ; that is, there is not convincing evidence at the 5% significance level that the true percentage of union members who support a strike is less than 75%.
2. With this small of a p -value, $0.0816 < 0.10$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence at the 10% significance level that the true percentage of union members who support a strike is less than 75%.

TIP

Always give the conclusion in context, and, in general, the conclusion should be about the alternative hypothesis, H_a .



- (b) If the p -value is considered large, $0.0816 > 0.05$, so that there is not sufficient evidence to reject the null hypothesis, there is the possibility that a false null hypothesis would mistakenly not be rejected and thus a Type II error would be committed. In this case, the union might call a strike thinking they have greater support than they actually do. If the p -value is considered small, $0.0816 < 0.10$, so that there is sufficient evidence to reject the null hypothesis, there is the possibility that a true null hypothesis would mistakenly be rejected, and thus a Type I error would be committed. In this case, the union might not call for a strike thinking they don't have sufficient support when they actually do have support.
- (c) Assuming that 75% of union members will support a strike, there is a probability of 0.0817 of getting a sample proportion of 0.696 or lower by chance alone in a random sample of 125 union members.

NOTE

Absence of evidence does not constitute evidence of absence. Just because there is not convincing evidence to reject H_0 does not mean that there is evidence that H_0 is true or that H_a is false.

TIP

If the question does not suggest a direction for the alternative, use a two-sided test.

Example 3.14

- (a) A cancer research group surveys a random sample of 500 women more than 40 years old to test the hypothesis that 28% of women in this age group have regularly scheduled mammograms. Should the hypothesis

be rejected at the 5% significance level if 151 of the women respond affirmatively?

- (b) In the answer to (a) above, what error might have been committed, Type I or Type II, and what would be a possible consequence?
- (c) Interpret the p -value in context.

✓ Solutions

- (a) *Parameter:* Let p represent the proportion of the population of women over 40 years old who have regularly scheduled mammograms.

Hypotheses: $H_0: p = 0.28$ and $H_a: p \neq 0.28$. (Since no suspicion is voiced that the 28% claim is low or high, we run a two-sided z -test for proportions.)

Procedure: One-sample z -test for a population proportion.

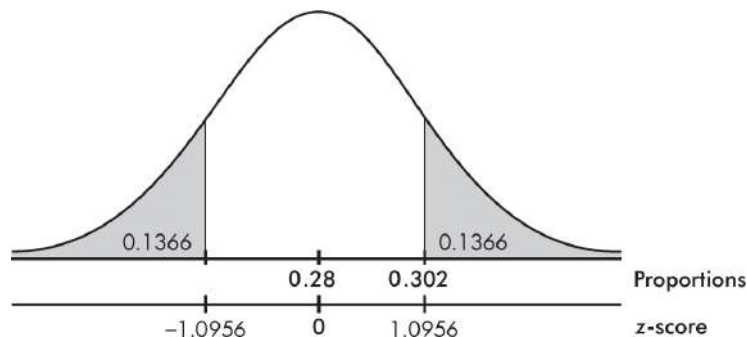
Checks: We are given a random sample, and clearly $500 < 10\%$ of all women over 40 years old. For normality, $np_0 = 500(0.28) = 140 \geq 10$ and $n(1 - p_0) = 500(0.72) = 360 \geq 10$.

Mechanics: Calculator software gives $z = 1.0956$ and $p = 0.2732$.

[For instructional purposes, we note that the observed $\hat{p} = \frac{151}{500} = 0.302$,

$\sigma_{\hat{p}} = \sqrt{\frac{(0.28)(0.72)}{500}} = 0.02008$ and $z = \frac{0.302 - 0.28}{0.02008} = 1.0956$ which corresponds to a tail probability of 0.1366. Doubling this value (because the test is two-sided) gives a p -value of 0.2732.]

Conclusion in context with linkage to the p -value: With this large of a p -value, $0.2732 > 0.05$, there is not sufficient evidence to reject H_0 ; that is, the cancer research group does not have convincing evidence to dispute the claim that 28% of all women in this age group have regularly scheduled mammograms.



- (b) There was not sufficient evidence to reject the null hypothesis. So a Type II error might be committed, that is, failing to reject a false null hypothesis. If so, the cancer research group would be doing their research assuming that 28% of women over 40 years old have regularly scheduled mammograms whereas the true figure is different.
- (c) Interpretation of p -value: Assuming that 28% of women over 40 years old have regularly scheduled mammograms, there is a 0.2732 probability of getting a sample proportion as extreme or more extreme than 0.302 in either direction by chance alone in a random sample of 500 women over 40 years old.

TIP

When we fail to reject H_0 , we are *not* accepting it. It simply means that there is not sufficient evidence against H_0 ; that is, we cannot rule out random chance as the explanation.

If $H_0: p = 0.28$ and $H_a: p \neq 0.28$, where p = the proportion of women over 40 years old who have regularly scheduled mammograms, with $\hat{p} = 0.302$ and a p -value of 0.2732, and with a significance level of $\alpha = 0.05$, following are three common misconceptions:

MISCONCEPTION #1: H_0 and H_a are alternative *conclusions*.

Fact: A hypothesis test is a decision-making process in order to reject or fail to reject a null hypothesis, H_0 . You are not deciding that one of H_0 or H_a is true. Your conclusion should be either that there is sufficient evidence to reject H_0 (that is, there is convincing evidence in support of H_a) or that there is not sufficient evidence to reject H_0 (that is, there is not convincing evidence in support of H_a).

MISCONCEPTION #2: The significance level $\alpha = 0.05$ is the probability that the alternative hypothesis is true. That is, the probability is 0.05 that the proportion of women over 40 years old who have regularly scheduled mammograms is different from 0.28.

Fact: The significance level is a fixed value we use to decide how small a p -value is required to be in order to reject the null hypothesis.

MISCONCEPTION #3: The p -value of 0.2732 is the probability of obtaining by chance alone a statistic as extreme or more extreme than the observed event of $\hat{p} = 0.302$.

Fact: This misses that the p -value is a conditional probability. The p -value is the probability of obtaining a statistic as extreme or more extreme than the one actually observed, by chance alone, when assuming that the null hypothesis is true.

Quiz 19

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. Consider the null and alternative hypotheses in a hypothesis test. Are these statements about a parameter or a statistic?
 - (A) Null hypothesis: parameter Alternative hypothesis: parameter
 - (B) Null hypothesis: parameter Alternative hypothesis: statistic
 - (C) Null hypothesis: statistic Alternative hypothesis: parameter
 - (D) Null hypothesis: statistic Alternative hypothesis: statistic

2. Past studies indicate that 70 percent of teens in a large city pass their driving test on the first attempt. A city administrator believes that because of an economic slowdown resulting in fewer cars on the road, the proportion might be greater than 0.70. A random sample of teens taking their driving test for the first time was surveyed. With $H_0: p = 0.70$ and $H_a: p > 0.70$, the p -value of the test is 0.035. Which of the following is a correct interpretation of the p -value?
 - (A) If it is true that 70 percent of teens in the city pass their driving test on the first attempt, 0.035 is the probability of obtaining a population proportion greater than 0.70.
 - (B) If it is true that 70 percent of teens in the city pass their driving test on the first attempt, 0.035 is the probability of obtaining a sample proportion as small as or smaller than the one obtained by the administrator.
 - (C) If it is true that 70 percent of teens in the city pass their driving test on the first attempt, 0.035 is the probability of obtaining a sample proportion as large as or larger than the one obtained by the administrator.
 - (D) If it is *not* true that 70 percent of teens in the city pass their driving test on the first attempt, 0.035 is the probability of obtaining a sample proportion as large as or larger than the one obtained by the administrator.

3. One extrasensory perception (ESP) test asks the subject to view the backs of cards and identify whether a circle, square, star, or cross is on the front of each card. If p is the proportion of correct answers, this may be viewed as a hypothesis test with $H_0: p = 0.25$ and $H_a: p > 0.25$. The subject is recognized to have ESP when the null hypothesis is rejected. What would a Type II error result in?

(A) Correctly recognizing someone has ESP
(B) Mistakenly thinking someone has ESP
(C) Not recognizing that someone really has ESP
(D) Correctly realizing that someone doesn't have ESP

4. A research dermatologist believes that skin cancer presenting on the head or neck will occur most often on the left side, the side next to a window when a person is driving. In a review of 565 cases of skin cancer on the head or neck, 305 occurred on the left side. What is the resulting p -value?

(A) $P\left(z > \frac{0.54 - 0.50}{\sqrt{(0.5)(0.5)/565}}\right)$
(B) $2P\left(z > \frac{0.54 - 0.50}{\sqrt{(0.5)(0.5)/565}}\right)$
(C) $P\left(z > \frac{0.54 - 0.50}{\sqrt{(0.54)(0.46)/565}}\right)$
(D) $2P\left(z > \frac{0.54 - 0.50}{\sqrt{(0.54)(0.46)/565}}\right)$

5. A sociologist claims that over two-thirds of White adults have entirely White social networks without any minority presence. In a random sample of 200 White adults, 150 of them have entirely White social networks. Is this convincing evidence in support of the sociologist's claim?

(A) Yes, because the p -value is 0.0062.
(B) Yes, because the p -value is 0.0124.
(C) No, because the p -value is 0.0062.
(D) No, because the p -value is 0.0124.

6. The greater the difference between the null hypothesis claim and the true value of the population parameter,

- (A) the smaller the risk of a Type II error and the smaller the power.
- (B) the smaller the risk of a Type II error and the greater the power.
- (C) the greater the risk of a Type II error and the smaller the power.
- (D) the greater the risk of a Type II error and the greater the power.

7. Consider a claimed population proportion p_0 , and a sample with sample proportion $\hat{p}$. Which of the following best describes the purpose of a one-sample z -test?

- (A) To prove that p_0 either is or is not true
- (B) To estimate the population proportion
- (C) To estimate a margin of error for $\hat{p}$
- (D) To estimate the probability of observing a value as extreme or more extreme than $\hat{p}$ given that p_0 is true

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Microplastics are tiny polymer fragments. They are found in many popular foods and, in large quantities, can pose dangers to body organs and functions. An environmental health scientist is concerned that the percentage of animal and vegetable protein samples that contain measurable amounts of microplastics is now over 90%. A random sample of 500 animal and vegetable products is taken, and 462 are found to contain measurable amounts of microplastics.

- (a) What is the parameter of interest?
- (b) What are the hypotheses?
- (c) One condition for a hypothesis test of a population proportion is that the appropriate sampling distribution is approximately normal. Is that condition met? Explain.

- (d) The standard error of the sampling distribution is 0.0134. Show how this is calculated.
- (e) What is the test statistic z ? Show how it can be calculated.

2. No vaccination is 100% risk free, and the theoretical risk of rare complications always has to be balanced against the severity of the disease. Suppose the CDC (Centers for Disease Control) decides that a risk of one in a million is the maximum acceptable risk of Guillain-Barré syndrome (GBS) complications for a new vaccine for a particularly serious strain of influenza. A large sample study of the new vaccine is conducted with the following hypotheses:

H_0 : The proportion of GBS complications is 0.000001 (one in a million).

H_a : The proportion of GBS complications is greater than 0.000001 (one in a million).

The p -value of the test is 0.138.

- (a) Interpret the p -value in the context of this study.
- (b) What conclusion should be drawn at the $\alpha = 0.10$ significance level?
- (c) Using your conclusion from part (b), what possible error, Type I or Type II, might be committed? Give a possible consequence of committing this error.

3. It is estimated that 17.4% of all U.S. households own a Roth IRA, a retirement savings account. The American Association of University Professors (AAUP) believes this figure is higher among their members and commissions a study. In the study, 150 out of a random sample of 750 AAUP members own Roth IRAs.

- (a) Is this convincing evidence to support the AAUP belief?
- (b) What error might have been committed, Type I or Type II, and what would be a possible consequence?
- (c) Interpret the p -value in context.

The answers for this Quiz can be found beginning on [page 493](#).

[Sampling Distribution for the Difference Between Sample Proportions](#)

We are interested in comparing two population proportions, p_1 and p_2 , by considering two sample proportions, $\hat{p}_1$ and $\hat{p}_2$. These sample proportions are from a whole universe of sample proportions. To judge their significance, we must know how differences of sample proportions vary; that is, we need to know the shape and parameters of the sampling distribution of $\hat{p}_1 - \hat{p}_2$.

Start with two populations with given proportions p_1 and p_2 . Take all samples of sizes n_1 and n_2 , respectively. Compute the difference $\hat{p}_1 - \hat{p}_2$ of the two proportions in each pair of samples. The following are now true:

1. The set of all differences of sample proportions is approximately normally distributed.
2. The mean $\mu_{\hat{p}_1 - \hat{p}_2}$ of the set of differences of sample proportions equals $p_1 - p_2$, the difference of population proportions.
3. The standard deviation $\sigma_{\hat{p}_1 - \hat{p}_2}$ of the set of differences of sample proportions is approximately equal to $\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$.

We say that the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal with mean $p_1 - p_2$ and standard deviation $\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$.

When making calculations and drawing conclusions from two specific samples, the following three conditions must be checked:

- **The randomization condition:** The data should be collected using random samples, and the samples must be taken independently of each other.
- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of the respective populations.
- **The normality condition:** The numbers of successes, $n_1 p_1$ and $n_2 p_2$, and the numbers of failures, $n_1(1-p_1)$ and $n_2(1-p_2)$, should all be at least 10.

► Example 3.15

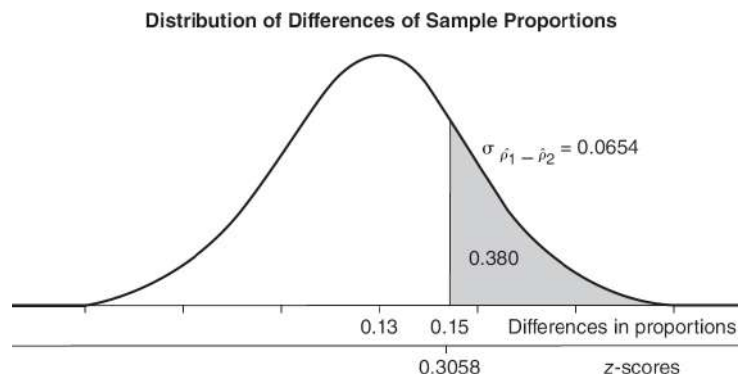
In an ideal world, everyone washes their hands after using the restroom. Unfortunately, this is not the reality. Previous epidemiological studies of college dormitory restrooms indicate that 54% of women and 41% of men wash their hands after using a restroom. In independent random samples of 110 women and 120 men living in college dormitories, what is the probability that the difference in percentages washing their hands after using a restroom (women minus men) is more than 15%?

✓ Solution

We have independent random samples, each less than 10% of their respective populations. So both the randomization condition and the 10% condition are met. For the normality condition, we note that $n_1p_1 = 110(0.54) = 59.4$, $n_1(1 - p_1) = 110(0.46) = 50.6$, $n_2p_2 = 120(0.41) = 49.2$, and $n_2(1 - p_2) = 120(0.59) = 70.8$ are all ≥ 10 . Thus, the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal with mean $\mu_{\hat{p}_1 - \hat{p}_2} = 0.54 - 0.41 = 0.13$ and standard deviation $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{(0.54)(0.46)}{110} + \frac{(0.41)(0.59)}{120}} = 0.0654$. Then

$$P(\hat{p}_1 - \hat{p}_2 > 0.15) = P\left(z > \frac{0.15 - 0.13}{0.0654}\right) = P(z > 0.3058) = 0.380$$

[normalcdf(0.3058, 1000) = 0.380 and normalcdf(0.15, 1000, 0.13, 0.0654) = 0.380]



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Confidence Interval for the Difference of Two Proportions

For confidence interval estimates, the true population proportions p_1 and p_2 are unknown, so we use the sample proportions $\hat{p}_1$ and $\hat{p}_2$ in checking the normality condition. Similarly, since p_1 and p_2 are unknown, we estimate the standard deviation $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$ with the *standard error* defined by:

$$SE(\hat{p}_1 - \hat{p}_2) = SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Thus, to find a confidence interval for the difference of two proportions, the three conditions to be met and that must be checked for are as follows:

- **The randomization condition:** The data should be collected using random samples, and the samples must be taken independently of each other. (Alternatively, the data should be from a randomized experiment.)
- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of the respective populations. (This condition is unnecessary when the data are from a randomized experiment.)
- **The normality condition:** $n_1 \hat{p}_1$, $n_2 \hat{p}_2$, $n_1(1 - \hat{p}_1)$, and $n_2(1 - \hat{p}_2)$ are all ≥ 10 .

A confidence interval for the difference of two proportions will take the following form:

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

► Example 3.16

Is high-speed Internet access critical to educational opportunities? A researcher believes that a greater proportion of adults living in urban areas than rural areas are in favor of spending more government funds to expand high-speed Internet access. To investigate this belief, independent simple random samples of 1,017 adults living in urban settings and 801 adults living in rural areas were selected. Of those surveyed, 612 of the urban residents and 137 of the rural residents said they were in favor of spending more government funds to expand high-speed Internet access.

- (a) Use a 99% confidence interval to estimate the difference in the proportions of urban residents and rural residents who are in favor of spending more

- government funds to expand high-speed Internet access.
- (b) Based on only this confidence interval, is there convincing evidence that a greater proportion of urban residents than rural residents are in favor of spending more government funds to expand high-speed Internet access?

Interpretation of the confidence level: If all possible random samples of 1,017 urban residents and 801 rural residents are surveyed and if a confidence interval for the difference in proportions of those who are in favor of spending more government funds to expand high-speed Internet access (urban residents minus rural residents) is constructed from each pair of samples, then 99% of these intervals would capture the true difference in proportions between urban residents and rural residents who are in favor of spending more government funds to expand high-speed Internet access.

Solutions

- (a) *Parameters:* Let p_U represent the proportion of the population of urban residents who are in favor of spending more government funds to expand high-speed Internet access. Let p_R represent the proportion of the population of rural residents who are in favor of spending more government funds to expand high-speed Internet access.

Procedure: Two-sample z -interval for the difference between population proportions, $p_U - p_R$.

Checks: We are given independent SRSs. The sample sizes are clearly less than 10% of the population of all urban residents and of all rural residents, respectively. For normality, $n_U \hat{p}_U =$

$$612 \geq 10, n_U(1 - \hat{p}_U) = 1,017 - 612 = 405 \geq 10, n_R \hat{p}_R = 137 \geq 10, \text{ and } n_R(1 - \hat{p}_R) = 801 - 137 = 664 \geq 10.$$

Mechanics: Computer software (2-PropZInt) gives (0.378, 0.483).

[For instructional purposes, we have $\hat{p}_U = \frac{612}{1,017} = 0.602$,

$$\hat{p}_R = \frac{137}{801} = 0.171, \text{ and } SE(\hat{p}_U - \hat{p}_R) =$$

$$\sqrt{\frac{(0.602)(0.398)}{1,017} + \frac{(0.171)(0.829)}{801}} = 0.0203.$$

The observed difference is $0.602 - 0.171 = 0.431$, and the critical z -scores are ± 2.576 . The confidence interval is $0.431 \pm 2.576(0.0203) = 0.431 \pm 0.052$ or (0.379, 0.483).]

Conclusion in context: We are 99% confident that the true proportion of urban residents who are in favor of spending more government funds to

expand high-speed Internet access is between 0.378 and 0.483 higher than the true proportion of rural residents who are in favor of spending more government funds to expand high-speed Internet access. [Or we are 99% confident that the interval from 0.378 and 0.483 captures the true difference (urban residents minus rural residents) in the proportions of urban residents and rural residents who are in favor of spending more government funds to expand high-speed Internet access.]

- (b) Yes, because the entire confidence interval from 0.378 to 0.483 is positive, there is convincing evidence that a greater proportion of urban residents than rural residents are in favor of spending more government funds to expand high-speed Internet access.

NOTE

If the value in question was ANYWHERE in the interval, even at the very edge, then there is not convincing evidence of a significant difference from that value.

Example 3.17

A dermatologist asked a random sample of 950 teenagers suffering from moderate cases of acne to choose from among the following three statements:

- **Glycemic diet statement:** A low-glycemic diet is the most important action one can take to control acne outbreaks.
- **Dairy statement:** Avoiding dairy products is the most important action one can take to control acne outbreaks.
- **No right diet:** Diets are of minor importance in controlling acne outbreaks.

The number of teenagers choosing each statement is given in the following table.

	Glycemic Diet	Dairy	No Right Diet
Number of teenagers	285	323	342

- (a) Assuming all conditions for inference are met, find a 95% confidence interval for the proportion of teenagers suffering from moderate cases of acne who would choose the glycemic diet statement.

- (b) One of the conditions that was met is that the number of teenagers who chose the glycemic diet statement and the number who did not choose the glycemic diet statement are both at least 10. Explain why it is necessary to satisfy this condition.
- (c) The dermatologist would like to use a two-sample z -interval to find a confidence interval for the difference in proportions of teenagers who would choose the glycemic diet statement and the proportion who would choose the dairy statement. Is this an appropriate procedure using these data? Explain.

Solutions

- (a) *Parameter:* Let p represent the proportion of the population of teenagers suffering from moderate cases of acne who would choose the glycemic diet statement.
Procedure: One-sample z -interval for a population proportion.
Checks: It is given that all conditions for inference are met.
Mechanics: Calculator software gives (0.271, 0.329).
Conclusion in context: We are 95% confident that the true proportion of teenagers suffering from moderate cases of acne who would choose the glycemic diet statement is between 0.271 and 0.329.
- (b) The formula for the confidence interval relies on the fact that the binomial distribution can be approximated by a normal distribution so that the sampling distribution of $\hat{p}$ is approximately normal. This works if both $n\hat{p}$ and $n(1 - \hat{p})$ are at least 10.
- (c) No, it would not be appropriate, because to use a two-sample z -interval to find a confidence interval for the difference in proportions, there has to be two independent random samples. The data here are from only a single sample.

Hypothesis Test for the Difference of Two Proportions

The null hypothesis for a difference of two proportions is $H_0 : p_1 = p_2$. The alternative hypothesis is $H_a : p_1 > p_2$, $H_a : p_1 < p_2$, or $H_a : p_1 \neq p_2$. Equivalently, we can instead use $H_0 : p_1 - p_2 = 0$ and $H_a : p_1 - p_2 > 0$, $H_a : p_1 - p_2 < 0$, or $H_a : p_1 - p_2 \neq 0$.

With a null hypothesis of $H_0 : p_1 = p_2$, we use the combined (or pooled) proportion $\hat{p}$ in the normality condition and in calculating the standard error. Thus, the three conditions to be met and must be checked for are as follows:

- **The randomization condition:** The data should be collected using random samples, and the samples must be taken independently of each other. (Alternatively, the data are from a randomized experiment.)
- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of their respective populations. (This condition is unnecessary when the data are from a randomized experiment.)
- **The normality condition:**

$$n_1\hat{p}_c, n_2\hat{p}_c, n_1(1 - \hat{p}_c), \text{ and } n_2(1 - \hat{p}_c) \text{ are all } \geq 10, \text{ where } \hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}.$$

The standard error becomes:

$$SE(\hat{p}_1 - \hat{p}_2) = SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

The relevant test statistic z takes the form:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

► Example 3.18

FEMA aid denial was a big problem after Hurricane Harvey hit the Gulf Coast in 2018. In a random sample of 775 District X homeowners, 403 said they were able to get the help they needed. In an independent random sample of 825 District Y homeowners, 264 said they were able to get the help they needed.

- (a) Do the above data give convincing evidence that a greater proportion of District X homeowners than District Y homeowners were able to get the help they needed?
- (b) The 95% confidence interval for the difference in proportions is (0.153, 0.257). Is this result consistent with the above conclusion?
- (c) Based on your conclusion in part (a), which type of error, Type I or Type II, could have been made? What is one potential consequence of this error?

✓ Solutions

Interpretation of p -value: Assuming that there is no difference in the proportions of District X and District Y homeowners who were able to get the help they needed, there is a probability of 0.000 of getting an observed proportion difference of 0.20 or greater by chance alone in random samples of 775 District X homeowners and 825 District Y homeowners.

- (a) *Parameters:* Let p_X represent the proportion of the population of District X homeowners who were able to get the help they needed. Let p_Y represent the proportion of the population of District Y homeowners who were able to get the help they needed.

Hypotheses: $H_0 : p_X - p_Y = 0$ or $H_0 : p_X = p_Y$ and $H_a : p_X - p_Y > 0$ or $H_a : p_X > p_Y$

Procedure: Two-sample z -test for a difference of two population proportions.

Checks: We are given independent random samples and must assume that the sample sizes are less than 10% of the respective populations. For normality and with $\hat{p}_c = \frac{403 + 264}{775 + 825} = 0.417$, we have $n_X \hat{p}_c = 775(0.417) = 323.2$, $n_X(1 - \hat{p}_c) = 775(0.583) = 451.8$, $n_Y \hat{p}_c = 825(0.417) = 344.0$, and $n_Y(1 - \hat{p}_c) = 825(0.583) = 481.0$ are all ≥ 10 .

Mechanics: Calculator software (2-PropZTest) gives $z = 8.11$ and $p \approx 0.000$.

[For instructional purposes, we note that $\hat{p}_X = \frac{403}{775} = 0.52$ and $\hat{p}_Y = \frac{264}{825} = 0.32$. The observed difference is $0.52 - 0.32 = 0.20$, $SE(\hat{p}_X - \hat{p}_Y) = \sqrt{(0.417)(0.583)\left(\frac{1}{775} + \frac{1}{825}\right)} = 0.02467$, $z = \frac{0.20 - 0}{0.02467} = 8.11$, and the tail probability gives a p -value of $\text{normalcdf}(8.11, 1000) \gg 0.000$.]

Conclusion in context with linkage to the p -value: With this small a p -value, $0.000 < 0.05$, there is sufficient evidence to reject H_0 . In other words, there is convincing evidence that after Hurricane Harvey hit the Gulf Coast, a greater proportion of all District X homeowners than all District Y homeowners were able to get the help they needed.

- (b) We are 95% confident that the true difference in proportions (true proportion of all District X homeowners who received the help they needed minus the true proportion of District Y homeowners who received

the help they needed) is between 0.153 and 0.247. Since this interval is entirely positive, it is consistent with the conclusion from the hypothesis test.

- (c) Since we rejected H_0 , there is the possibility that a true null hypothesis was mistakenly rejected, which would have been a Type I error. A possible consequence would have been claims of district inequities in FEMA aid distribution when none actually existed.

Quiz 20

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. In a simple random sample of 300 elderly men, 65% were married, while in an independent simple random sample of 400 elderly women, 48% were married. Determine a 99% confidence interval estimate for the difference between the proportions of all elderly men and women who are married.
 - (A) $(0.65 - 0.48) \pm 2.326 \sqrt{\frac{(0.65)(0.35)}{300} + \frac{(0.48)(0.52)}{400}}$
 - (B) $(0.65 - 0.48) \pm 2.576 \sqrt{\frac{(0.65)(0.35)}{300} + \frac{(0.48)(0.52)}{400}}$
 - (C) $(0.65 - 0.48) \pm 2.576 \left(\frac{(0.65)(0.35)}{\sqrt{300}} + \frac{(0.48)(0.52)}{\sqrt{400}} \right)$
 - (D) $\left(\frac{0.65 + 0.48}{2} \right) \pm 2.576 \sqrt{\frac{(0.65)(0.35)}{300} + \frac{(0.48)(0.52)}{400}}$

2. Is Internet usage different in the Middle East from Internet usage in Latin America? In a random sample of 500 adults in the Middle East, 151 claimed to be regular Internet users, while in a random sample of 1,000 adults in Latin America, 345 claimed to be regular users. What is the p -value for the appropriate hypothesis test?

$$(A) \quad P \left(z < \frac{0.302 - 0.345}{\sqrt{\frac{(0.302)(0.698)}{500} + \frac{(0.345)(0.655)}{1,000}}} \right)$$

$$(B) \quad 2P \left(z < \frac{0.302 - 0.345}{\sqrt{\frac{(0.302)(0.698)}{500} + \frac{(0.345)(0.655)}{1,000}}} \right)$$

$$(C) \quad P \left(z < \frac{0.302 - 0.345}{\sqrt{(0.331)(0.669) \left(\frac{1}{500} + \frac{1}{1,000} \right)}} \right)$$

$$(D) \quad 2P \left(z < \frac{0.302 - 0.345}{\sqrt{(0.331)(0.669) \left(\frac{1}{500} + \frac{1}{1,000} \right)}} \right)$$

3. Laura has a penny and a nickel and wonders if they have the same likelihood of showing heads when they are spun. She spins each coin 200 times to test if there is a significant difference in the proportion of spins that they each land showing heads. Suppose the test with $H_0: p_{\text{penny}} = p_{\text{nickel}}$ and $H_a: p_{\text{penny}} \neq p_{\text{nickel}}$ results in a p -value of 0.032. Which of the following is a proper conclusion?

- (A) The probability that the null hypothesis is true is 0.032.
- (B) The probability that the alternative hypothesis is true is 0.032.
- (C) The difference in population proportions is 0.032.
- (D) None of the above are proper conclusions.

4. A college guidance counselor is studying the incidence of reported cheating at her college. She surveys independent random samples of freshmen and seniors as to whether they were willing to report cheating by other students. A 95% confidence interval for the difference (proportion of seniors minus proportion of freshmen) was calculated as -0.06 ± 0.04 . Which of the following is the best interpretation of the confidence interval?

- (A) We are 95% confident that the difference in the sample proportions of seniors and freshmen who are willing to report cheating by other students is between -0.10 and -0.02.

- (B) We are 95% confident that the difference in proportions of all seniors and all freshmen who are willing to report cheating by other students is between -0.10 and -0.02.
- (C) In repeated samplings of freshmen and seniors, approximately 95% of the calculated confidence interval proportion differences will capture the true proportion difference between seniors and freshmen who are willing to report cheating by other students.
- (D) There is a 0.95 probability that the true difference between the proportions of seniors and freshmen who are willing to report cheating by other students is between -0.10 and -0.02.

5. Cyberbullying is a growing problem, and for those who suffer cyberbullying, the effects can be devastating. A company marketing a free antibullying app believes that different age groups of teens experience different proportions of cyberbullying. The company does a study to compare the experiences of 15-to-17-year-olds versus 13-to-14-year-olds. Let p_{15-17} represent the proportion of 15-to-17-year-olds who have experienced some form of cyberbullying, and let p_{13-14} represent the proportion of 13-to-14-year-olds who have experienced some form of cyberbullying. The study reports that the 90% confidence interval for the difference in proportion of teens experiencing some form of cyberbullying (15–17 minus 13–14) is (0.04, 0.16) while a 99% confidence interval for the difference is (0.02, 0.18). Which of the following is the most reasonable conclusion for the company's hypothesis at the $\alpha = 0.05$ significance level?

- (A) The p -value is less than 0.01, so there is convincing evidence at the 5% significance level that the proportion of teens experiencing cyberbullying is different for the 15–17 and 13–14 age groups.
- (B) The p -value is between 0.01 and 0.10, so there is convincing evidence at the 5% significance level that the proportion of teens experiencing cyberbullying is different for the 15–17 and 13–14 age groups.
- (C) The p -value is between 0.01 and 0.10, so at the 5% significance level it cannot be determined whether the proportion of teens experiencing cyberbullying is different for the 15–17 and 13–14 age groups.
- (D) The p -value is greater than 0.10, so there is not convincing evidence at the 5% significance level that the proportion of teens experiencing cyberbullying is different for the 15–17 and 13–14 age groups.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. In a random sample of 500 new births in the United States, 41.2% were to unmarried women, while in a random sample of 400 new births in the United Kingdom, 46.5% were to unmarried women.
 - (a) Calculate a 95% confidence interval for the difference in the proportions of new births to unmarried women in the United States and the United Kingdom.
 - (b) Does the confidence interval support the belief by a UN health care statistician that the proportions of new births to unmarried women is different in the United States and the United Kingdom? Explain.
2. A national survey before the death of Ruth Bader Ginsburg studied whether or not a random sample of American voters could name a single Supreme Court justice. The study compared survey responses between voters under the age of 35 and voters over the age of 65. The following table displays some of the results.

	All Voters	Voters under 35	Voters over 65	Difference (percentage points)
Number surveyed	2,050	1,500	550	
% who could not name a justice	54%	57%	46%	11%

- (a) Is random sampling, random assignment, both, or neither involved in this study? Explain.
- (b) What are the appropriate hypotheses for testing whether the two populations of voters differ with regard to the proportion who cannot name a single Supreme Court justice?
- (c) The test statistic is $z = 4.43$. What does this value mean in context?

- (d) Is there convincing evidence to reject the null hypothesis at the 0.01 significance level? Explain.
- (e) A 99% confidence interval based on these data is (0.04612, 0.17388). Interpret this interval in context.
- (f) Is this confidence interval consistent with the test decision in part (d)? Explain.

3. A randomized, double-blind, placebo-controlled trial across the United States and parts of Canada tested hydroxychloroquine as a postexposure prophylaxis for asymptomatic participants reporting a high-risk exposure to a confirmed Covid-19 contact. The incidence of Covid-19 among participants receiving hydroxychloroquine was 49 out of 414, while the incidence among participants receiving a placebo was 58 out of 407.

- (a) Explain the purpose of random assignment in this experiment.
- (b) State the hypotheses the researchers tested about hydroxychloroquine use as a postexposure prophylaxis for Covid-19.
- (c) Identify the hypothesis test the researchers used to analyze the results of this experiment and show that the conditions for this test were met.
- (d) The p -value of this test is 0.152. What conclusion should be made at the $\alpha = 0.05$ significance level?
- (e) What error might have been committed, Type I or Type II, and what would be a possible consequence?

The answers for this Quiz can be found beginning on [page 495](#).

Quiz 21

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

- 1. In a random sample of people ages 70 to 79, 84% said their overall health is good, very good, or excellent, while in an independent random sample of people ages 50 to 59, just 74% said this. A 90% confidence interval of the difference in proportions (ages 70–79 minus ages 50–59) was

calculated as (0.07, 0.13). Which of the following is the best interpretation of the confidence level?

- (A) We are 90% confident that (0.07, 0.13) captures the true difference in proportions (ages 70–79 minus ages 50–59) of people who would say their overall health is good, very good, or excellent
- (B) If all possible independent random samples of people ages 70–79 and of people ages 50–59 of the same size as in this study are sampled and if a confidence interval for the difference in proportions of health rating is constructed from each pair of samples, then approximately 90% of all these intervals would capture the difference in sample proportions.
- (C) If all possible independent random samples of people ages 70–79 and of people ages 50–59 of the same size as in this study are sampled and if a confidence interval for the difference in proportions of health rating is constructed from each pair of samples, then approximately 90% of all these intervals would capture the true difference in proportions.
- (D) There is a 0.95 probability that the true difference in proportions (ages 70–79 minus ages 50–59) of people who said their overall health is good, very good, or excellent is between 0.07 and 0.13.

2. In an effort to reduce tobacco use and prevent its detrimental health effects, New York T21 policies aim to limit youth access to tobacco products, including cigarettes and e-cigarettes. Studies were performed to determine whether or not e-cigarette use among high school students changed between pre-T21 and post-T21. In a T21 study, the 99% confidence interval for the difference in proportion of students using e-cigarettes (post-T21 minus pre-T21) was (0.12, 0.22). Which of the following is the most reasonable conclusion?

- (A) Because all the values in the confidence interval are greater than 0, there is convincing evidence that the proportion of high school students using e-cigarettes increased between pre-T21 and post-T21.
- (B) There is a 0.99 probability that the proportion of high school students using e-cigarettes increased between pre-T21 and post-T21.
- (C) We are 99% confident that the proportion of high school students using e-cigarettes increased between pre-T21 and post-T21.

- (D) If 99% confidence intervals for the difference in proportions are calculated from all possible samples of the same size samples as the original studies, 99% will show convincing evidence that the proportion of high school students using e-cigarettes increased between pre-T21 and post-T21.

3. Suppose that you want to compare the percentage of male high school students who hope someday to run for political office to the percentage of female high school students who hope someday to run for political office. A random sample of male high school students and a random sample of female high school students are selected. Which of the following results provides the strongest evidence that the percentages differ between male and female high school students who hope someday to run for political office?
- (A) 10 of 100 males hope someday to run for political office compared with 30 out of 100 females.
- (B) 15 of 100 males hope someday to run for political office compared with 25 out of 100 females.
- (C) 20 of 200 males hope someday to run for political office compared with 60 out of 200 females.
- (D) 25 of 200 males hope someday to run for political office compared with 55 out of 200 females.
4. Suppose that a two-sided hypothesis test of whether two population proportions differ results in a p -value of 0.069. What can be said about a 95% confidence interval for the difference in population proportions, calculated from the same sample data?
- (A) It includes only negative values.
- (B) It includes both positive and negative values.
- (C) It includes only positive values.
- (D) This cannot be answered without knowing which proportion is subtracted from which.
5. Alcohol consumption is one of many concerns of today's teens. In one study, 650 of 5,000 teens surveyed said they felt "a lot" or "some" pressure to drink alcohol. A study a year later reported that 800 of 5,000 teens

surveyed said they felt “a lot” or “some” pressure to drink alcohol. Assume all conditions for inference have been met. Which of the following gives a 95% confidence interval estimate of the difference in the numbers of teens (later year minus earlier year) from samples of 5,000 who say they feel “a lot” or “some” pressure to drink alcohol?

- (A) (0.0162, 0.0438)
- (B) (0.081, 0.219)
- (C) (16, 44)
- (D) (81, 219)

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. A behavior study of high school students looked at whether a higher proportion of boys than girls meets a recommended level of physical activity (increased heart rate for 60 minutes per day for at least 5 days during the 7 days before the survey). What is the proper conclusion if 370 out of a random sample of 850 boys and 218 out of an independent random sample of 580 girls met the recommended level of activity?
2. A national survey studied whether or not a random sample of American voters believed a particular false political claim. The study compared survey responses between voters 18 to 25 years old and voters over the age of 65. The following table displays some of the results.

	All Voters	Voters 18 to 25	Voters over 65	Difference (percentage points)
Number surveyed	1,500	900	600	
% who believed the false claim	10.8%	12%	9%	3%

- (a) Is random sampling, random assignment, both, or neither involved in this study? Explain.
- (b) What are the appropriate hypotheses for testing whether the two populations of voters differ with regard to the proportion who believe the false political claim?
- (c) The test statistic is $z = 1.834$. What does this value mean in context?
- (d) Is there convincing evidence to reject the null hypothesis at the 0.05 significance level? Explain.
- (e) A 95% confidence interval based on this data is $(-0.0012, 0.06123)$. Interpret this interval in context.
- (f) Is this confidence interval consistent with the test decision in part (d)? Explain.

3. Exercise addiction, an uncontrollable desire to exercise, is one example of behavior addiction. In one study of 1,000 randomly selected adults from Boulder, Colorado, psychologists investigated the relationship between exercise addiction and negative body image. The following two-way table summarizes the data:

		Exercise addiction diagnosis			Total
		None	Mild	Severe	
Negative body image	Yes	301	68	26	395
	No	500	83	22	605
Total		801	151	48	1000

- (a) In the sample, 4.8% of the adults were diagnosed with severe exercise addiction. To what population can this result be generalized? Explain.
- (b) If one person from the sample is selected at random, what is the probability the person is diagnosed to have some form of exercise addiction?
- (c) If one person from the sample is selected at random, what is the probability the person is diagnosed to have some form of exercise addiction given that the person has negative body image?

Let p_{neg} = the proportion of people with severe exercise addiction among those in the population who have negative body image, and let p_{pos} = the proportion of people with severe exercise addiction among those in the population who do not have negative body image. A 95% confidence interval for $p_{neg} - p_{pos}$ is (0.000815, 0.0581).

- (d) One condition for constructing this confidence interval is that the number of people with severe exercise addiction and the number of people who do not have severe exercise addiction in each of the two groups (with and without negative body image) is at least 10. Explain why it is necessary for this condition to be satisfied.
- (e) Interpret the confidence interval.
- (f) Even though all of the values in the confidence interval are positive, a cause-and-effect relationship between negative body image and severe exercise addiction cannot be established from this study due to possible confounding. Identify a potential confounding variable in this context and explain your answer.

The answers for this Quiz can be found beginning on [page 497](#).

Chi-Square Test for Independence or Homogeneity

There is a class of real-world problems in which we want to determine if there is a significant association between **two categorical variables**. We cross-classify a single sample of observations from a **single population** in two ways and then ask whether the two variables are independent of each other. This leads to *chi-square tests for independence* with hypotheses:

- H_0 : There is no association between two categorical variables in a given population, or equivalently, two categorical variables in a given population are independent.
- H_a : There is an association between two categorical variables in a given population, or equivalently, two categorical variables in a given population are not independent.

There is a related class of problems in which we have a **single categorical variable** and obtain independent samples from **two or more populations** to

determine whether the distributions of that variable are different for the populations. This leads to *chi-square tests for homogeneity* with hypotheses:

- H_0 : There is no difference in the distributions of the categorical variable across populations.
- H_a : There is a difference in the distributions of the categorical variable across populations.

Tests for independence and tests for homogeneity must meet the following set of three conditions:

- **The randomization condition:** For tests of independence, the data should be collected using a random sample. For tests of homogeneity, the data should be collected using independent random samples.
- **The 10% condition:** When sampling without replacement, sample sizes should be no larger than 10% of respective population sizes.
- **The expected values condition:** All expected values should be greater than 5.

Expected values are calculated assuming the null hypothesis is true. We then compare these expected values with the observed values and ask if the differences are reasonable if H_0 is true. The significance of the differences is gauged by a χ^2 -value of weighted squared differences defined by:

$$\chi^2 = \sum \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

where the sum is taken over all values in a two-way table. The chi-square statistics have a chi-square distribution with degrees of freedom equal to:

$$df = (\text{rows} - 1)(\text{columns} - 1)$$

The smaller the resulting χ^2 -value, the more reasonable is the null hypothesis. If the χ^2 -value is large enough—that is, if the p -value is small enough—we can say that the evidence is sufficient to reject the null hypothesis and to claim there is convincing evidence of the alternative hypothesis. To calculate p -values, we use software such as χ^2 -Test or χ^2 cdf.

A point worth noting is that even if there is sufficient evidence to reject the null hypothesis of independence, we cannot necessarily claim any direct *causal* relationship. In other words, although we can make a statement about some link

or relationship between two variables, we are *not* justified in claiming that one causes the other. For example, we may demonstrate a relationship between salary level and job satisfaction, but our methods would not show that higher salaries cause higher job satisfaction. Perhaps an employee's higher job satisfaction impresses his superiors and thus leads to larger increases in pay. Or perhaps there is a third variable, such as training, education, or personality, that has a direct causal relationship to both salary level and job satisfaction.

► Example 3.19

A growing number of states have legalized marijuana for medical or recreational purposes. In a nationwide telephone poll of 1,000 randomly selected adults representing Democrats, Republicans, and Independents, respondents were asked two questions: their party affiliation and if they supported the legalization of marijuana. The answers, cross-classified by party affiliation, are given in the following two-way table (also called a *contingency table*).

		Support Legalization			Totals
		Yes	No	No Opinion	
Party Affiliation	Democrats	280	110	15	405
	Republicans	155	190	10	355
	Independents	180	45	15	240
Totals		615	345	40	1000

Test the null hypothesis that support for legalizing marijuana is independent of party affiliation. Use a 5% significance level.

NOTE

Another acceptable statement of the hypotheses:

H_0 : There is no association between party affiliation and support for legalizing marijuana.

H_a : There is an association between party affiliation and support for legalizing marijuana.

 Solution

Hypotheses: H_0 : Party affiliation and support for legalizing marijuana are independent.

H_a : Party affiliation and support for legalizing marijuana are not independent.

Procedure: Chi-square test for independence.

Checks: We are given a random sample, and $n = 1,000$ is less than 10% of all adults. For the expected value condition, by putting the observed data into a calculator matrix, calculator software such as χ^2 -Test gives the expected cells in a second matrix and all are >5 :

249.1	139.7	16.2
218.3	122.5	14.2
147.6	82.8	9.6

NOTE

Put the data, but not the totals, into a matrix on your calculator.

Mechanics: The χ^2 -Test software gives $\chi^2 = 94.5$ and $p \approx 0.000$.

Conclusion with linkage to the p-value: With this small of a p -value, $0.000 < 0.05$, there is sufficient evidence to reject H_0 ; that is, among all adults there is convincing evidence of a relationship between party affiliation and support for legalizing marijuana.

[For instructional purposes, we note that the expected values, χ^2 , and p can be calculated as follows:

The formula for an expected cell count is the following: *expected count* = $\frac{(\text{row total})(\text{column total})}{\text{table total}}$. Thus, for example, the expected value for the middle cell, which corresponds to Republican *no* responses, is $\frac{(355)(345)}{1,000} = 122.5$.

Continuing in this manner, we fill in the table of expected values as follows:

Support Legalization				Totals
	Yes	No	No Opinion	
Democrats	249.1	139.7	16.2	405
Republicans	218.3	122.5	14.2	355
Independents	147.6	82.8	9.6	240
Totals	615	345	40	1000

(An appropriate check at this point is that each expected cell count is greater than 5.)

Next we calculate the value of the chi-square statistic:

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(280 - 249.1)^2}{249.1} + \frac{(110 - 139.7)^2}{139.7} + \frac{(15 - 16.2)^2}{16.2} + \frac{(155 - 218.3)^2}{218.3} \\ + \frac{(190 - 122.5)^2}{122.5} + \frac{(10 - 14.2)^2}{14.2} + \frac{(180 - 147.6)^2}{147.6} + \frac{(45 - 82.8)^2}{82.8} + \frac{(15 - 9.6)^2}{9.6} = 94.5$$

$$df = (r - 1)(c - 1) = (3 - 1)(3 - 1) = 4$$

The p -value is then calculated using calculator software such as $\chi^2_{\text{cdf}}(94.5, 1000, 4) \gg \mathbf{0.000.}$

If asked, the largest contributions to the chi-square statistics were two of the Republican responses, with fewer yes responses and more no responses than would be expected with independence.

While it is important to understand these calculations, on the exam the calculations should be made quickly by putting the observed data into a matrix and using the χ^2 -Test on your calculator.

➤ Example 3.20

Young adults from different generations have had entirely different religious and social experiences that have led to profound religious changes over the generations. Has this affected spirituality? In a nationwide poll of 500 randomly selected adults, respondents were asked two questions: their age and whether

spirituality was important in their lives. The cross-classified answers are given in the following two-way table:

		Age (Generation)			
		Baby Boomers	Generation X	Millennials	Generation Z
Is spirituality in one's life important?	Yes	73	71	79	69
	No	32	54	61	61

- Test the null hypothesis that importance of spirituality is independent of generational age. Use a 5% significance level.
- Interpret the p -value in context.

✓ Solutions

- Hypotheses:*

H_0 : Importance of spirituality and generational age are independent.

H_a : Importance of spirituality and generational age are not independent.

Procedure: Chi-square test of independence.

Checks: We are given a random sample, and $n = 500$ is less than 10% of all adults. All expected cells (calculated below) are greater than 5.

Mechanics: Using a calculator matrix and χ^2 -Test software gives $\chi^2 = 7.220$, $p = 0.0652$, $df = 3$, and the following expected cell counts:

61.3	73	81.8	75.9
43.7	52	58.2	54.1

Conclusion in context with linkage to the p -value: With this large a p -value, $0.0652 > 0.05$, there is *not* sufficient evidence to reject the null hypothesis. There is *not* convincing evidence of a relationship between importance of spirituality and generational age.

- Assuming that importance of spirituality and generational age are independent, there is a 0.0652 probability of getting a χ^2 -value of 7.220 or

greater by chance alone in a random sample of 500 adults.

Example 3.21

In a large city, a group of AP Statistics students work together on a project to determine which group of school employees has the greatest proportion who are satisfied with their jobs. In independent simple random samples of 100 teachers, 60 administrators, 45 custodians, and 55 secretaries, the numbers satisfied with their jobs were found to be 82, 38, 34, and 36, respectively. Is there convincing evidence that the proportion of employees satisfied with their jobs is different in different school system job categories?

Solution

Hypotheses:

H_0 : The proportion of employees satisfied with their jobs is the same across the various school system job categories.

H_a : At least two of the job categories differ in the proportion of employees satisfied with their jobs.

Procedure: A chi-square test for homogeneity with observed data:

	Teachers	Administrators	Custodians	Secretaries
Satisfied	82	38	34	36
Not satisfied	18	22	11	19

Checks: We are given independent random samples, and we assume that the four samples are each less than 10% of their respective populations in the large city. For the expected value condition, using a calculator matrix and χ^2 -Test gives the expected cells in a second matrix, and these are all >5 :

NOTE

Interpretation of the p -value: Assuming there is no difference in the proportion of employees satisfied with their jobs across the various school system job categories, there is a 0.0335 probability of getting a χ^2 of 8.707 or greater by chance alone in the random samples of the designated sizes.

73.1	43.8	32.9	40.2
26.9	16.2	12.1	14.8

Mechanics: The χ^2 -Test software gives $\chi^2 = 8.707$, $p = 0.0335$, and $df = 3$.

Conclusion in context with linkage to the p-value:

With this small of a p -value, $0.0335 < 0.05$, there is convincing evidence to reject H_0 ; that is, there is convincing evidence that the true proportion of employees satisfied with their jobs is not the same across all the school system job categories.

The difference between the test for independence and the test for homogeneity can be confusing. When we're simply given a two-way table, it's not obvious which test should be performed. The crucial difference is in the *design* of the study. Did we pick samples from each of two or more populations to compare the distribution of some variable (one question) among the different populations? If so, we are doing a test for homogeneity. Did we pick one sample from a single population and cross-categorize on two variables (two questions) to see if there is an association between the variables? If so, we are doing a test for independence.

For example, if we separately sample Democrats, Republicans, and Independents to determine whether they are for, against, or have no opinion with regard to stem cell research (several samples, one variable), we do a test for homogeneity with a null hypothesis that the distribution of opinions on stem cell research is the same among Democrats, Republicans, and Independents. However, if we sample the general population, noting the political preference and opinions on stem cell research of the respondents (one sample, two variables), we do a test for independence with a null hypothesis that political preference is independent of opinion on stem cell research.

Finally, it should be remembered that while we used χ^2 for categorical data, the χ^2 -distribution is a *continuous* distribution. Applying it to counting data is just an approximation.

Quiz 22

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. A random sample of 100 former student-athletes are picked from each of the eleven colleges that are members of the Big East conference. Students are surveyed about whether or not they feel they received a quality education while participating in varsity athletics. Which of the following is the most appropriate test to determine whether there is a difference among these schools as to the student-athlete perception of having received a quality education?
 - (A) A chi-square test of independence
 - (B) A chi-square test of homogeneity
 - (C) A z-test of eleven sample proportions
 - (D) A z-test of eleven population proportions
2. Given a two-way table, it's not obvious whether to perform a test for independence or a test for homogeneity. What is the main difference between the tests?
 - (A) How expected counts are calculated from the table
 - (B) How df , the degrees of freedom, are calculated from the table
 - (C) The number of rows in the table
 - (D) The number of samples from which the data in the table are obtained
3. Is there a relationship between education level and sports interest? A study cross-classified 1,500 randomly selected adults in three categories of education level (not a high school graduate, high school graduate, and college graduate) and five categories of major sports interest (baseball, basketball, football, hockey, and tennis). The χ^2 -value is 13.95. Is there evidence of a relationship between education level and sports interest?

(A) The evidence points to a cause-and-effect relationship between education and sports interest.

(B) There is sufficient evidence at the 5% significance level of a relationship between education level and sports interest.

(C) There is sufficient evidence at the 10% significance level, but not at the 5% significance level, of a relationship between education level and sports interest.

(D) The p -value is greater than 0.10, so there is not sufficient evidence of a relationship between education level and sports interest.

4. Random samples of 25 students are chosen from each high school class level. Students are asked whether or not they are satisfied with the school cafeteria food, and the results are summarized in the following table:

	Freshmen	Sophomores	Juniors	Seniors
Satisfied	15	12	9	7
Dissatisfied	10	13	16	18

A chi-square test for homogeneity gives $\chi^2 = 5.998$. Is there sufficient evidence of a difference in cafeteria food satisfaction among the class levels?

(A) The data prove that there is a difference in cafeteria food satisfaction among the class levels.

(B) There is sufficient evidence at the 1% significance level of a difference in cafeteria food satisfaction among the class levels.

(C) There is sufficient evidence at the 5% significance level, but not at the 1% significance level, of a difference in cafeteria food satisfaction among the class levels.

(D) With $P = 0.1117$, there is not sufficient evidence of a difference in cafeteria food satisfaction among the class levels.

5. With a chi-square test of independence and a 3 by 4 table, we have $\chi^2 = 12.7$. Assuming a significance level of 0.05, which of the following is

true?

- (A) While a Type II error was not committed, a Type I error may have been committed.
- (B) While a Type I error was not committed, a Type II error may have been committed.
- (C) Either a Type I or a Type II error may have been committed.
- (D) Neither a Type I nor a Type II error may have been committed.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Optometrists are interested in various causes of eye abnormalities. In a Philadelphia eye institute study, a random sample of parents were asked whether their children slept primarily in room light, darkness, or with a night-light before the age of two. They were also asked about each child's eyesight diagnosis (nearsighted, farsighted, or normal vision) from the child's most recent examination. The resulting counts are as follows:

	Room Light	Darkness	Night-light	Total
Nearsighted	41	18	78	137
Farsighted	12	40	39	91
Normal	22	114	115	251
Total	75	172	232	479

- (a) What were the explanatory and response variables?
- (b) Was this an observational study or an experiment? Explain.

- (c) What was the probability that a child who slept in darkness was farsighted? What was the probability that a farsighted child slept in darkness?
- (d) Is there convincing evidence to say that there is an association between the bedroom lighting condition and a child's eyesight? Perform an appropriate test.
- (e) Some researchers proposed that parents' eyesight might be a confounding variable in this study. How would that explain the observed association between the bedroom lighting condition and the child's eyesight?

2. For a study of Americans' core social networks, independent samples of 1,500 White, 250 Black, and 300 Hispanic Americans were asked, "Over the past six months, give the initials of people with whom you discussed matters important to you." Answers were used to estimate core social network sizes. The data are tabulated in the following table:

	Number of people in core social network			
	0 or 1	2	3	4 or more
White Americans	397	210	309	584
Black Americans	74	36	36	104
Hispanic Americans	90	35	40	135

- (a) Was this a retrospective observational study, a prospective observational study, or an experiment?
- (b) What is the probability that a person in this study had 4 or more people in his or her core social network?
- (c) What is the probability that a person in this study had exactly 3 people in his or her core social network given that he or she is a White American?
- (d) What is the probability that a person in this study who had exactly 2 people in his or her core social network is a Hispanic American?
- (e) Is there convincing evidence that there is a difference in the distributions of the number of people in the core social network

among White, Black, and Hispanic Americans? Perform an appropriate test.

The answers for this Quiz can be found beginning on [page 499](#).

Summary

- An estimator is unbiased if, on average, the value of the estimator is equal to the population parameter.
- Provided that conditions are met, the sampling distribution of $\hat{p}$ is approximately normal with mean $\mu_{\hat{p}} = p$ and standard deviation $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.
- **Statistical inference** draws a conclusion (i.e., infers something) about a population parameter based on a sample statistic.
 1. A **confidence interval** is used to estimate the value of a parameter with a range of plausible values. We estimate a population proportion p with a z -interval.
 2. A **hypothesis test** is used to assess the plausibility of a particular claim about a parameter. We test a claim about a population proportion p with a z -test.
- Conditions must be checked for inference on a population proportion:
 - **The randomization condition:** The data should be collected using a random sample.
 - **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
 - **The normality condition:** For confidence intervals, both $n\hat{p}$ and $n(1 - \hat{p})$ are ≥ 10 . For hypotheses tests, both np_0 and $n(1 - p_0)$ are ≥ 10 .
- A confidence interval for a population proportion takes the following form:

$$\text{estimate} \pm \text{margin of error} = \hat{p} \pm z^* SE(\hat{p}) = \hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- We are never able to say exactly what a population parameter is equal to; rather, we say that we have a certain confidence that it lies within a certain interval.

- When scoring a confidence interval free-response question on the exam, readers will look for whether or not you complete the following:
 1. Identify the procedure (such as one-sample z -interval for a population proportion) and check the conditions.
 2. Compute a proper interval. It is sufficient simply to copy the interval from your calculator calculation.
 3. Interpret the interval in context. The conclusion must mention the confidence level and the parameter, and it must refer to the *population*, not to the sample.
- The width of a confidence interval is proportional to $\frac{1}{\sqrt{n}}$.
- The margin of error is exactly one-half the width of the confidence interval.
- If we want a narrower interval, we must either decrease the confidence level or increase the sample size.
- If we want a higher level of confidence, we must either accept a wider interval or increase the sample size.
- The confidence level refers to the percentage of samples that produce intervals that capture the true population parameter.
- We can justify a claim based on a confidence interval only if all the values in the confidence interval are consistent with the claim.
- In a hypotheses test, the null hypothesis is stated in the form of an equality statement about a population parameter, while the alternative hypothesis is in the form of a strict inequality statement about the same population parameter.
- We attempt to show that a null hypothesis is unacceptable by showing it is improbable.
- The significance level α is the predetermined conditional probability of rejecting the null hypothesis given that it is true.
- Lack of sufficient evidence for the alternative hypothesis is not the same as evidence for the null hypothesis. That is, *absence of evidence does not constitute evidence of absence*.
- The p -value is the conditional probability of obtaining a sample statistic as extreme as (or more extreme than) the one obtained if the null hypothesis is assumed to be true.
- When the p -value is small (typically, less than 0.05 or 0.10), we say we have sufficient evidence to reject the null hypothesis.

- A recommended way of formatting conclusions to hypothesis tests is as follows:
 1. Because the p -value is small (less than alpha), there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that (alternative in context).
 2. Because the p -value is large (greater than alpha), there is not sufficient evidence to reject H_0 ; that is, there is not convincing evidence that (alternative in context).
- When scoring a hypothesis test free-response question on the exam, readers will look for whether or not you complete the following:
 1. State the hypotheses (both H_0 and H_a) and define the parameter of interest. (The hypotheses must refer to the *population*, not to the sample.)
 2. Identify the procedure (such as one-sample z -test for a population proportion) and check the conditions.
 3. Compute the test statistic and the p -value. It is sufficient simply to copy these from your calculator calculation.
 4. Give a conclusion in context with linkage to the p -value. The conclusion must refer to the parameter and to the population, not to the sample.
- A Type I error is the probability of mistakenly rejecting a true null hypothesis.
- A Type II error is the probability of mistakenly failing to reject a false null hypothesis.
- The power of a hypothesis test is the probability that a Type II error is not committed.
- $P(\text{Type I error}) = \text{significance level} = \alpha$.
- $P(\text{Type II error}) = 1 - \text{power}$.
- Provided that conditions are met, the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal with mean $\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$ and standard deviation $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$.
- Conditions must be checked for inference about a difference in proportions:

- **The randomization condition:** The data should be collected using two random samples, and the two samples must be taken independently of each other.
- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of the respective populations.
- **The normality condition:** For confidence intervals, $n_1\hat{p}_1$, $n_2\hat{p}_2$, $n_1(1 - \hat{p}_1)$, and $n_2(1 - \hat{p}_2)$ are all ≥ 10 . For hypothesis tests, $n_1\hat{p}_c$, $n_2\hat{p}_c$, $n_1(1 - \hat{p}_c)$, and $n_2(1 - \hat{p}_c)$ are all ≥ 10 , where $\hat{p}_c$ is the combined (or pooled) proportion, $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$.
- Chi-square analysis is an important tool for inference on distributions of counts.
- Chi-square distributions take only positive values and are skewed right.
- The chi-square statistic of weighted differences is defined by:

$$\chi^2 = \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

- A chi-square test of independence tests for evidence of an association between two categorical variables from a single random sample from one population.
- A chi-square test of homogeneity compares independent random samples from two or more populations with regard to distributions of a single categorical variable.
- For these chi-square tests, the observed counts must be integers; however, this is not true for the calculated expected counts.
- Three conditions must be met and checked for a chi-square test of independence and for a chi-square test of homogeneity:
 - **The randomization condition:** For tests of independence, the data should be collected using a random sample. For tests of homogeneity, the data should be collected from independent random samples.
 - **The 10% condition:** When sampling without replacement, the sample size (or sample sizes for homogeneity) should be no larger than 10% of the respective population(s).
 - **The expected values condition:** All expected values should be greater than 5.

- When scoring a chi-square free-response question on the exam, readers will look for whether or not you complete the following:
 1. State the hypotheses (both H_0 and H_a) and define the parameter of interest. The hypotheses must refer to the *population*, not to the sample.
 2. Name the test: independence or homogeneity. (How the sample(s) are chosen and how the hypotheses are stated determine which test you choose.)
 3. Check the conditions. This requires showing the expected values, which can be copied from the calculator.
 4. Report the resulting chi-square value, the p -value, and the df (the number of degrees of freedom).
 5. Give a conclusion in context with linkage to the p -value.

- One suggested acronym for confidence intervals is PANIC:
 - **P**arameter (state it)
 - **A**ssumptions (state and justify)
 - **N**ame the interval
 - **I**nterval (find it)
 - **C**onclusion (in context)

- One suggested acronym for hypothesis tests is PHANTOMS:
 - **P**arameter (state it)
 - **H**ypotheses (about the population)
 - **A**ssumptions (state and justify)
 - **N**ame the test
 - **T**est statistic (calculate z or t or χ^2)
 - **O**btain a p -value
 - **M**ake a decision (reject or fail to reject the null hypothesis)
 - **S**tate a conclusion (in context with linkage to the p -value)

4

Inference for Quantitative Data: Means

(4–8 multiple-choice questions on the AP exam)

Learning Objectives

In this unit, you will learn how

- To be able to describe the shape and parameters of a sampling distribution for a sample mean and for a sampling distribution for the difference of sample means.
- To be able to check conditions to perform inference involving population means.
- To understand that when the appropriate conditions are met, the statistic $\frac{\bar{x} - \mu}{s/\sqrt{n}}$ follows a t -distribution with $n - 1$ degrees of freedom.
- To be able to construct and interpret a confidence interval for a population mean and for the difference of population means.
- To understand the relationships among confidence level, sample size, sample standard deviation, and confidence interval width.
- To be able to interpret confidence levels.
- To be able to perform a hypothesis test to evaluate a claim about a population mean and about a difference between population means.
- To understand how simulation can be used to estimate a p -value.

Sampling Distribution for Sample Means

We are interested in estimating a population mean μ by considering a single sample mean $\bar{x}$. This sample mean is just one of a whole universe of sample means. To judge its significance, we must know how sample means

vary; that is, we need to know the shape and parameters of the sampling distribution of $\bar{x}$:

Start with a population with a given mean μ and a standard deviation σ . Take all samples of size n . Compute the mean in each of these samples. Then the following is true:

1. The mean $\mu_{\bar{x}}$ of the set of sample means equals μ , the population mean.
2. The standard deviation $\sigma_{\bar{x}}$ of the set of sample means is approximately equal to $\frac{\sigma}{\sqrt{n}}$.

We say that the sampling distribution of $\bar{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$. If the population is approximately normal, or if the sample size is large enough (at least 30) for the central limit theorem to apply, we also have that the set of sample means is approximately normally distributed.

When making calculations and drawing conclusions from a specific sample, the following three conditions must be checked:

- **The randomization condition:** The data should be collected using a random sample.
- **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
- **The normality condition:** Either the population distribution should be approximately normal or the sample size n should be ≥ 30 , or if $n < 30$, the sample data should be free from strong skewness and outliers.

➤ Example 4.1

It is estimated that domesticated cats who mainly live outdoors kill an average of 10 birds a year with a standard deviation of 3 birds. What is the probability an SRS of 35 such cats will kill an average of between 9 and 12 birds in a year?

We say, “For all random samples of size $n = 35$ from the population of domesticated cats who mainly live outdoors, the sample mean numbers of birds killed per year will

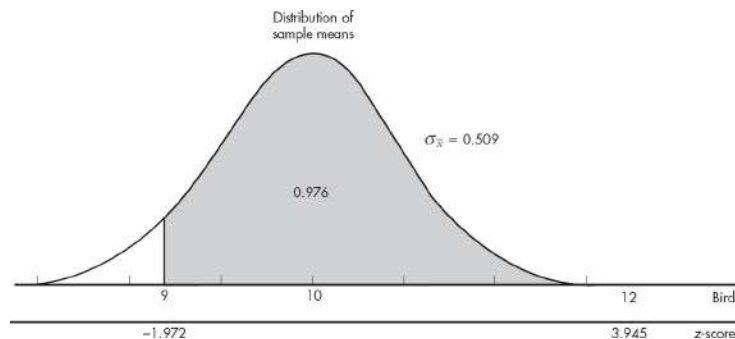
have a mean of 10 birds and will typically vary by about 0.6 birds from the population mean of 10 birds.”

✓ Solution

The sampling distribution of the sample mean is approximately normal (because the sample size $35 > 30$ so the CLT applies) with mean $\mu_{\bar{x}} = 10$ and standard deviation $\sigma_{\bar{x}} = \frac{3}{\sqrt{35}} = 0.507$.

$$P(9 < \bar{x} < 12) = P\left(\frac{9 - 10}{0.507} < z < \frac{12 - 10}{0.507}\right) = P(-1.972 < z < 3.945) = 0.976$$

[normalcdf(-1.972, 3.945) = 0.976 and normalcdf(9, 12, 10, 0.507) = 0.976.]



The probability an SRS of 35 domesticated cats who mainly live outdoors will kill an average of between 9 and 12 birds in a year is 0.976.

Quiz 23

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. The distribution for the number of lies college students tell to their mothers during phone conversations is approximately normal with mean 0.81 and standard deviation 0.24. Consider two different random

samples taken from the population, one of size 10 and one of size 100. Which of the following is true about the sampling distributions of the sample mean for the two sample sizes?

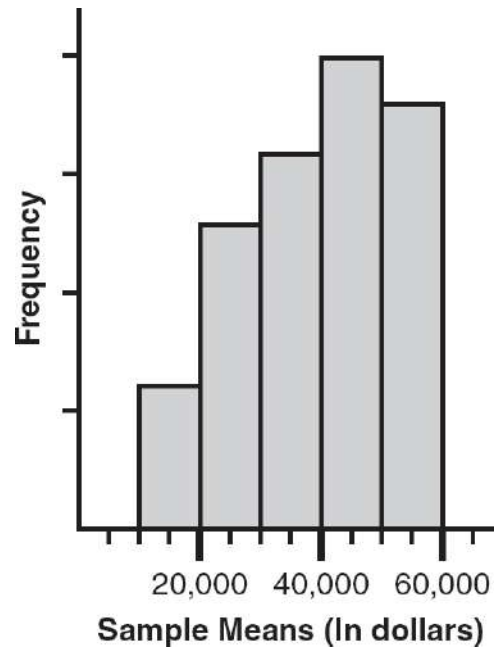
- (A) Both distributions are approximately normal with mean 0.81 and standard deviation 0.24.
- (B) Both distributions are approximately normal. Both the mean and the standard deviation for the $n = 10$ sampling distribution are greater than for the $n = 100$ distribution.
- (C) Both distributions are approximately normal with the same mean. The standard deviation for the $n = 10$ sampling distribution is greater than for the $n = 100$ distribution.
- (D) Only the $n = 100$ sampling distribution is approximately normal. Both distributions have the same mean. The standard deviation for the $n = 10$ sampling distribution is greater than for the $n = 100$ distribution.

2. Suppose that, using accelerometers in helmets, researchers determine that boys playing high school football absorb an average of 355 hits to the head with a standard deviation of 80 hits during a season (including both practices and games). What is the probability on a randomly selected team of 48 players that the average number of head hits per player is between 340 and 360?

- (A) $P(340 - 355 < z < 360 - 355)$
- (B) $P\left(\frac{340 - 355}{80} < z < \frac{360 - 355}{80}\right)$
- (C) $P\left(\frac{340 - 355}{\sqrt{48}} < z < \frac{360 - 355}{\sqrt{48}}\right)$
- (D) $P\left(\frac{340 - 355}{\frac{80}{\sqrt{48}}} < z < \frac{360 - 355}{\frac{80}{\sqrt{48}}}\right)$

3. A 2021 survey of private, not-for-profit, four-year colleges indicated the mean spent on tuition and fees per year is \$39,070. A simulation was conducted to create a sampling distribution of the sample mean

for a population with a mean of \$39,070. The following histogram shows the result of the simulation.



Among the following, which would be the best reason why the simulation of the sampling distribution does not appear roughly normal?

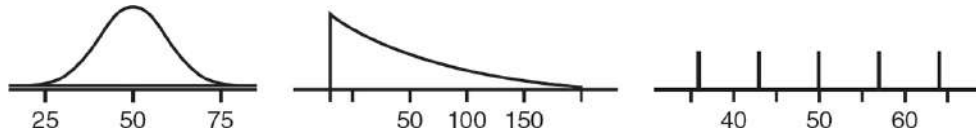
- (A) The sample size was small.
- (B) The population distribution has large variability.
- (C) The samples were not randomly selected.
- (D) The samples were selected without replacement.

4. A human resources specialist for a large international company wanted to estimate the mean amount of wasted time during 8-hour workdays for all employees of the company. The specialist took repeated random samples of size 25 employees and estimated the mean and standard deviation of the sampling distribution to be 95 minutes and 4 minutes, respectively. Based on the estimate for the sampling distribution, which of the following provides the best estimates of the population standard deviation?

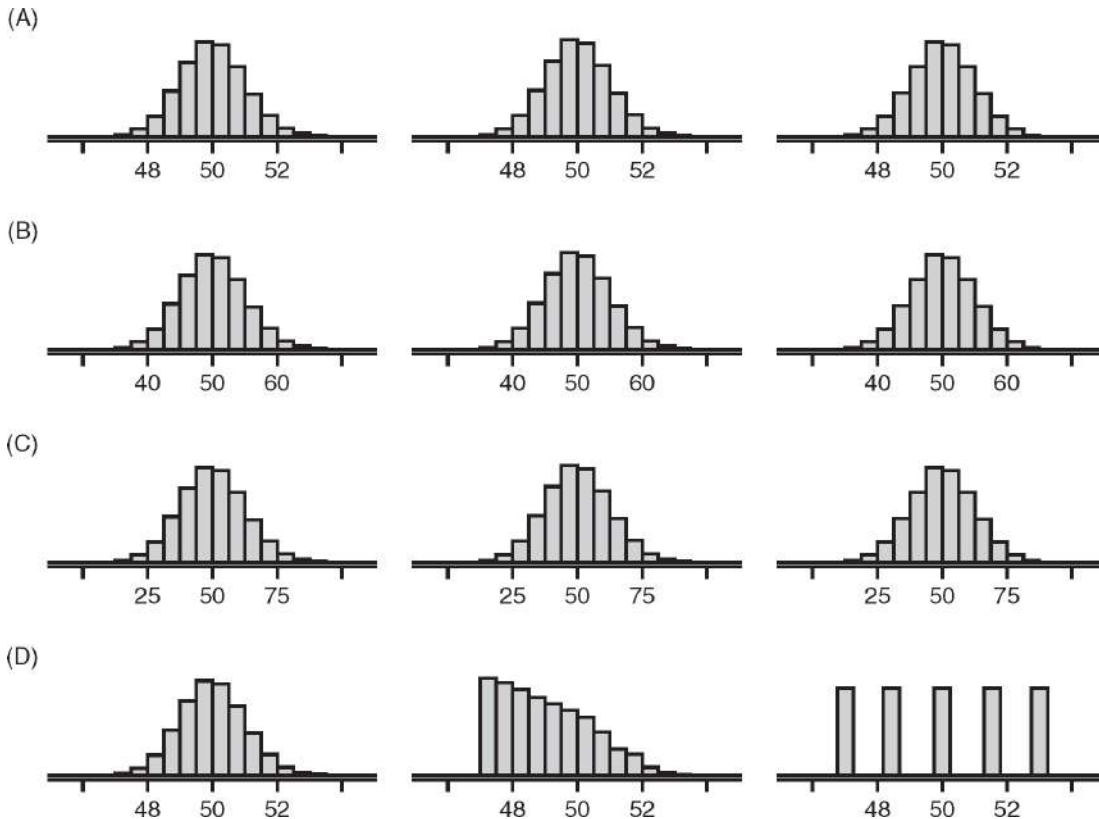
- (A) 0.16 minutes

- (B) 0.8 minutes
- (C) 4 minutes
- (D) 20 minutes

5. Three populations all have the same mean of 50 and same standard deviation of 10. They are represented by the following very different probability distributions:



Using software, a large number of samples, each of size 100, are selected from each population. Which of the following gives the resulting three most likely distributions of sample means?



Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. The amount of fuel used by jumbo jets to take off is approximately normally distributed with a mean of 4,000 gallons and a standard deviation of 125 gallons.
 - (a) What is the probability that a randomly selected jumbo jet will need more than 4,250 gallons of fuel to take off?
 - (b) What are the shape, mean, and standard deviation of the sampling distribution of the mean gallons of fuel to take off of 40 randomly selected jumbo jets?
 - (c) In a random sample of 40 jumbo jets, what is the probability that the mean number of gallons of fuel needed to take off is less than 3,950?
 - (d) Would the answer to (a), (b), or (c) be affected if the original population of gallons of fuel used by jumbo jets to take off were skewed instead of normal? Explain.
2. One estimate is that the average number of teenagers giving birth per day in the United States is 386.3 with a standard deviation of 21.5 and that the numbers per day are approximately normally distributed.
 - (a) What is the 90th percentile of the distribution of number of teenagers giving birth per day in the United States?
 - (b) What is the probability that at least 3 out of 5 randomly selected days will be such that the number of teenagers giving birth is at least the number calculated in (a)?
 - (c) What is the probability that a randomly selected sample of 5 days will have a mean number of teenagers giving birth at least the number calculated in (a)?

The answers for this Quiz can be found beginning on [page 501](#).

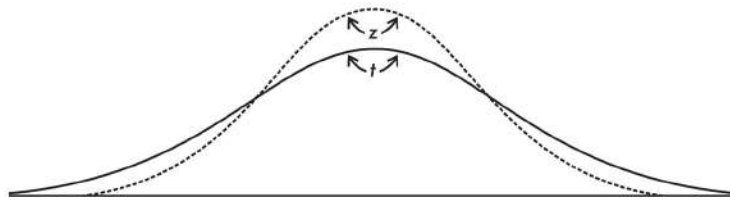
The t -Distribution

When the population standard deviation σ is unknown, we use the sample standard deviation s as an estimate for σ . But then $\frac{\bar{x} - \mu}{s/\sqrt{n}}$ does not follow a normal distribution. There is a distribution that can be used when working with the $s/\sqrt{n}$ ratios. This *Student's t -distribution* was introduced in 1908 by W. S. Gosset, a British mathematician employed by the Guinness Breweries.

For a sample from a *normally distributed population*, we work with the variable

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

with a resulting t -distribution that is bell-shaped and symmetric but lower at the mean, higher at the tails, and so more spread out than the normal distribution.



Like the binomial distribution, t -distributions are different for different values of n . In the tables, these distinct t -distributions are associated with the values for degrees of freedom (df). For this discussion, the df value is equal to the sample size minus 1. The smaller the df value, the larger the dispersion in the distribution. The larger the df value—that is, the larger the sample size—the closer the distribution to the normal distribution.

Since there is a separate t -distribution for each df value, fairly complete tables would involve many pages; therefore, in Table B of the Appendix, we list areas and t -values for only the more commonly used percentages or probabilities. The last row of Table B is the normal distribution, which is a special case of the t -distribution taken when n is infinite. Technology (such as `tcdf` and `invT`) easily handle calculations involving the t -distribution.

Thus, the t -distribution is the proper choice whenever the population standard deviation σ is unknown. In the real world, σ is almost always unknown, and so we should almost always use the t -distribution. The check

that must be satisfied before using the t -distribution is that either the original population is roughly normal or the sample size is large enough ($n \geq 30$) for the central limit theorem to apply. What happens if the sample size is less than 30 and we are not told that the parent population is roughly normal? Then we must check that the sample shows no outliers and no strong skewness, so that we can say it is not unreasonable to assume the parent population is roughly normal. (This check can quickly be made with a hand-drawn dotplot or stemplot of the sample data.)

Confidence Interval for a Mean

Three conditions must be met and must be checked for:

- **The randomization condition:** The data should be collected using a random sample.
- **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
- **The normality condition:** The population distribution should be roughly normal, or the sample size n should be ≥ 30 , or if $n < 30$, the sample data should be free from strong skewness and outliers.

Typically, we do not know σ , the population standard deviation. In such cases, we must use s , the *standard deviation of the sample*, as an estimate of σ . In this case $\frac{s}{\sqrt{n}}$ is called the *standard error*, $SE(\bar{x})$, and is used as an estimate for $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. (We will use t -distributions instead of the standard normal curve whenever σ is unknown, no matter what the sample size.)

Our confidence interval will take the form:

$$\bar{x} \pm t^* SE(\bar{x}) = \bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

where the critical value, t^* , depends on the chosen confidence level and on the degrees of freedom, $df = n - 1$.

In the following examples, important points to keep in mind include:

- *Naming the parameter* should reference “mean,” the variable, and the population in context.

- *Naming the procedure* is easiest done by simply stating “one-sample *t*-interval for a population mean.”
- *Verifying conditions* means more than simply listing them with check marks. You must show your work or give some reason to confirm verification.
- *Finding the interval* can be accomplished using calculator software (such as TInterval) instead of using the above formula. Note that answer choices using the formula may well appear on multiple-choice questions.
- *Interpreting the interval in context* must include mentioning the confidence level and parameter, and this must refer to the population, not the sample.
- *Interpreting the confidence level* never involves mentioning a specific interval.

► Example 4.2

When a random sample of 10 cars of a new model was tested for gas mileage, the results showed a mean of 27.2 miles per gallon with a standard deviation of 1.8 miles per gallon. (Assume that the population of mpg results for all the new model cars is approximately normally distributed.)

- (a) What is a 95% confidence interval estimate for the mean gas mileage achieved by this model?
- (b) Based on this confidence interval, do you think that the true mean mileage is significantly different from 25 mpg?
- (c) Determine a 99% confidence interval. Note what effect going from 95% to 99% confidence has on the interval size.
- (d) What would the 95% confidence interval be if the sample mean of 27.2 and standard deviation of 1.8 had come from a sample of 20 cars? Note what effect going from a sample size of 10 to one of 20 has on the interval size.
- (e) With the original data ($\bar{x} = 27.2$, $s = 1.8$, $n = 10$), with what confidence can we assert that the true mean gas mileage is 27.2 ± 1.04 ?

Interpretation of the confidence level: If all possible random samples of 10 cars of this model are tested for gas mileage, and a confidence interval for the mean is constructed from each sample, then 95% of all these intervals would capture the true mean gas mileage.

✓ Solutions

- (a) *Parameter:* Let μ represent the mean gas mileage (in miles per gallon) in the population of cars of a new model.

Procedure: A one-sample t -interval for a population mean (population SD is unknown).

Checks: The sample is given as random, and 10 cars are assumed to be less than 10% of all cars of the new model. For normality, the population is stated to be approximately normal. Calculation of a t -interval is appropriate.

Mechanics: Calculator software (TInterval) gives (25.912, 28.488).

[For instructional purposes, we note that the standard error of the sample means is $SE(\bar{x}) = \frac{1.8}{\sqrt{10}} = 0.569$. With $n - 1 = 10 - 1 = 9$ degrees of freedom and 2.5% in each tail, the appropriate t -scores are $\pm \text{invT}(0.975, 9) = \pm 2.262$. Then $27.2 \pm 2.262(0.569) = 27.2 \pm 1.29$.]

Conclusion in context: We are 95% confident that the true mean gas mileage of all cars of the new model is between 25.91 and 28.49 miles per gallon.

[Alternatively we can say: “We are 95% confident that the interval from 25.91 to 28.49 mpg captures the true mean gas mileage of all cars of the new model.”]

- (b) Yes; because 25 is not in the interval of plausible values from 25.9 to 28.5, there is convincing evidence that the true mean mileage is significantly different from 25 mpg.
- (c) Here, TInterval gives (25.35, 29.05). We are 99% confident that the true mean gas mileage of all cars of the new model is between 25.35 and 29.05 miles per gallon.

Note that when we want a higher confidence (99% instead of 95%), we have to settle for a larger, less specific interval (± 1.85)

instead of (± 1.29) .

- (d) Here, TInterval gives (26.36, 28.04). We are 95% confident that the true mean gas mileage of all cars of the new model is between 26.36 and 28.04 miles per gallon.

Note that when the sample size increased (from $n = 10$ to $n = 20$), the same sample mean and standard deviation resulted in a narrower, more specific interval (± 0.84) instead of (± 1.29) .

- (e) Converting ± 1.04 to t -scores yields

$$\frac{\pm 1.04}{1.8/\sqrt{10}} = \pm 1.827 \text{ and } \text{tcdf}(-1.827, 1.827, 9) = 0.899 = 89.9\%, = \pm 1.827 \text{ and } \text{tcdf}(-1.827, 1.827, 9) = 0.899 = 89.9\%.$$

We are 89.9% confident that the true mean gas mileage of all cars of the new model is between 26.16 and 28.24 miles per gallon.

As seen in the above example, for a given sample, *increasing the confidence level* will result in the following:

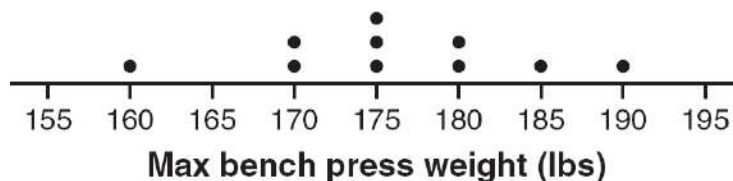
- The critical t -value will increase.
- The margin of error, $t^* \frac{s}{\sqrt{n}}$, will increase.
- The width of the confidence interval will increase.

Also as seen in the above example, when all other things remain the same, *increasing the sample size* will result in the following:

- The width of the confidence interval will decrease.
- If the sample size is increased by a factor of d , the width of the confidence interval will decrease by a factor of $\frac{1}{\sqrt{d}}$.

► Example 4.3

Maximum bench press weights were recorded for a random sample of ten 17-year-old males. The resulting data are displayed in the following dotplot.



- (a) Find a 90% confidence interval for the mean maximum bench press weight for all 17-year-old males.
- (b) Based on this interval, is there convincing evidence that the mean maximum bench press weight for all 17-year-old males is less than 180 lbs?
- (c) Interpret the confidence level in context.

NOTE

Understand that you are not trying to prove that the sample data are normal but, rather, trying to infer something about the parent population.

✓ Solutions

- (a) *Parameter:* Let μ represent the mean maximum bench press weight for all 17-year-old males.

Procedure: One-sample t -interval for a population mean (population SD is unknown)

Checks: The sample is given to be random, and the ten 17-year-old males are less than 10% of all 17-year-old males. For normality, a dotplot of the sample values is unimodal, reasonably symmetric, and with no outliers and no skewness, indicating it is not unreasonable to assume the data come from a roughly normal distribution. Calculating a t -interval is appropriate.

Mechanics: Calculator software (such as `TInterval` with the data put in a `List`) gives (171.11, 180.89).

[For instructional purposes, we note the following:

$$\bar{x} = \frac{\sum x}{n} = \frac{1,760}{10} = 176,$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{\sum (x - 176)^2}{9}} = 8.433, \text{ and}$$

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{8.433}{\sqrt{10}} = 2.667.$$

With $df = 10 - 1 = 9$ and 5% in each tail, the critical t -scores are $\pm \text{invT}(0.95, 9) = \pm 1.833$. Then $176 \pm 1.833(2.667) = 176 \pm 4.89$.]

Conclusion in context: We are 90% confident that the true mean maximum bench press weight for all 17-year-old males is between 171.11 and 180.89 lbs.

- (b) No; because the entire interval from 171.11 to 180.89 is not less than 180, there is not convincing evidence that the mean maximum bench press weight for all 17-year-old males is less than 180 lbs. [It is not sufficient that $176 < 180$, and it is not sufficient that almost the entire interval is < 180 . All values in the interval are considered to be “plausible” values, and 180 is in the interval.]
- (c) If all possible random samples of ten 17-year-old males are taken and if a confidence interval for the mean maximum bench press weight is constructed from each sample, then 90% of all these intervals would capture the true mean maximum bench press weight for all 17-year-old males.

Many tests involve comparing pairs of data. In this case, the proper procedure is to run a one-sample test on a single variable consisting of the differences from the paired data. For this *matched pairs design*, the procedure is called a “one-sample t -interval for a population mean difference.” Using the mean $\bar{x}_{\text{diff}}$ and the standard deviation s_{diff} from the set of differences, the confidence interval for a mean difference becomes:

$$\bar{x}_{\text{diff}} \pm t^* \frac{s_{\text{diff}}}{\sqrt{n_{\text{diff}}}}$$

This is sometimes simply written as:

$$\bar{x}_d \pm t^* \frac{s_d}{\sqrt{n_d}}$$

Example 4.4

An SAT preparation class of 30 randomly selected students produces the following total score summary:

	First Score	Second Score	Improvement (2nd Score–1st Score)
Mean	1093.33	1135.58	42.25
Standard Deviation	87.76	85.73	27.92

Find a 90% confidence interval of the mean improvement in test scores.

Solution

The proper procedure is a one-sample t -interval on the set of differences (improvement) between the scores for each of the 30 students.

Parameter: Let μ_{diff} represent the mean improvement (2nd score minus 1st score) in the SAT scores of the population of students who take this SAT preparation class.

Procedure: A one-sample t -interval for a population mean difference

Checks: We are given a random sample and $n = 30$ is less than 10% of all students. For normality, $n = 30$ is large enough for the CLT to apply.

Calculation of a t -interval is appropriate.

Mechanics: With an unknown population SD, it's necessary to find a t -interval. Using $\bar{x}_{\text{diff}} = 42.25$ and $s_{\text{diff}} = 27.92$, calculator software (TInterval) gives (33.59, 50.91).

Conclusion in context: We are 90% confident that the true mean improvement in test scores is between 33.59 and 50.91.

NOTE

Alternatively we can say: "We are 90% confident that the interval from 33.59 to 50.91 captures the true mean improvement in test scores."

Quiz 24

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. What happens to a t -distribution as df , the degrees of freedom, increase?
 - (A) The center remains the same, and the area in the tails decreases.
 - (B) The center remains the same, and the area in the tails remains the same.
 - (C) The center remains the same, and the area in the tails increases.
 - (D) The center increases, and the area in the tails increases.

2. The mayor of a small city would like to know how many weeks women think should be a standard for paid maternity leave. He asked the 38 women working in his office to anonymously answer the query. He plans to construct a one-sample t -interval for a population mean. Have the conditions for inference been met?
 - (A) Yes, all conditions have been met.
 - (B) No, the normality condition is violated.
 - (C) No, the randomization condition is violated.
 - (D) No, the 10% condition cannot be verified.

3. To estimate the average number of books high school students read during their four years in high school, a random sample of 95 high school graduates are interviewed, and a one-sample t -interval is found. However, it is noted that the sample data are skewed right. Have all conditions for inference been met?
 - (A) Yes, all conditions have been met.
 - (B) No, the normality condition is violated.
 - (C) No, the randomization condition is violated.
 - (D) No, it is not clear whether the 10% condition is satisfied.

4. Most recent tests and calculations estimate at the 95% confidence level that mitochondrial Eve, the maternal ancestor to all living humans, lived $138,000 \pm 18,000$ years ago. What is meant by “95% confidence” in this context?
- (A) A confidence interval of the true age of mitochondrial Eve has been calculated using t -scores consistent with $df = n - 1$ and tail probabilities of ± 0.025 .
 - (B) There is a 0.95 probability that mitochondrial Eve lived between 120,000 and 156,000 years ago.
 - (C) If 20 random samples of data are obtained by this method and a 95% confidence interval is calculated from each, the true age of mitochondrial Eve will be in 19 of these intervals.
 - (D) Of all random samples of data obtained by this method, 95% will yield intervals that capture the true age of mitochondrial Eve.
5. A confidence interval estimate is determined from the GPAs of a simple random sample of n students. All other things being equal, which of the following will result in a smaller margin of error?
- (A) A smaller confidence level
 - (B) A larger sample standard deviation
 - (C) A smaller sample size
 - (D) A smaller sample mean
6. One gallon of gasoline is put into each of an SRS of 30 test autos, and the resulting mileage figures are tabulated with $\bar{x} = 28.5$ and $s = 1.2$. Determine a 95% confidence interval estimate of the mean mileage of all comparable autos. (Assume that all conditions for inference are met.)
- (A) $28.5 \pm 1.96 \left(\frac{1.2}{\sqrt{29}} \right)$
 - (B) $28.5 \pm 1.96 \left(\frac{1.2}{\sqrt{30}} \right)$

(C) $28.5 \pm 2.045\left(\frac{1.2}{\sqrt{29}}\right)$

(D) $28.5 \pm 2.045\left(\frac{1.2}{\sqrt{30}}\right)$

7. Studies have found an association between drinking sugary beverages such as soda and the likelihood of developing liver cancer. A high school has 3,000 students, of whom 300 drink sugary beverages. In order to estimate the total quantity of sugary beverages consumed daily by students at this high school, two methods are proposed.

Method I: Sample 30 students at random. Calculate a confidence interval for the mean quantity of sugary beverages consumed daily by those 30 students. Multiply both ends of the interval by 3,000 to get a confidence interval estimate of the total quantity of sugary beverages consumed daily by students at this high school.

Method II: Identify the 300 students who drink sugary beverages. Sample 30 of these 300 students at random. Calculate a confidence interval for the mean quantity of sugary beverages consumed daily by those 30 students. Multiply both ends of the interval by 300 to get a confidence interval estimate of the total quantity of sugary beverages consumed daily by students at this high school.

Which is the better method for estimating the total quantity of sugary beverages consumed daily by students at this high school?

- (A) Method I is better.
(B) Method II is better.
(C) Both methods are good and will produce equivalent results.
(D) There is insufficient information given to compare the two methods.
8. To assess the effectiveness of a new over-the-counter sleep aid, 30 adults from a random sample were each assessed as to the time in minutes it took to fall asleep before beginning the medication regimen and also as to the time in minutes it took to fall asleep after being on the regimen for one week. For each adult, the difference in time was

calculated (before minus after) and used to create the 95 percent confidence interval (4.6, 7.3). Assuming all conditions for inference are met, which of the following is a correct interpretation of the interval?

- (A) There is a 0.95 probability that the mean difference in times for all adults taking this sleep aid is between 4.6 and 7.3 minutes.
- (B) For all adults taking this sleep aid, 95 percent will have a mean difference in times of between 4.6 and 7.3 minutes.
- (C) We are 95 percent confident that the mean difference in times for all adults in this sample is between 4.6 and 7.3 minutes.
- (D) We are 95 percent confident that the mean difference in times for all adults taking this sleep aid is between 4.6 and 7.3 minutes.

9. Hospital administrators wish to learn the average length of stay of all surgical patients. A statistician determines that, for a 95% confidence level estimate of the average length of stay to within ± 0.5 days, 50 surgical patients' records will have to be examined. How many records should be looked at to obtain a 95% confidence level estimate to within ± 0.25 days?

- (A) 25
- (B) 50
- (C) 100
- (D) 200

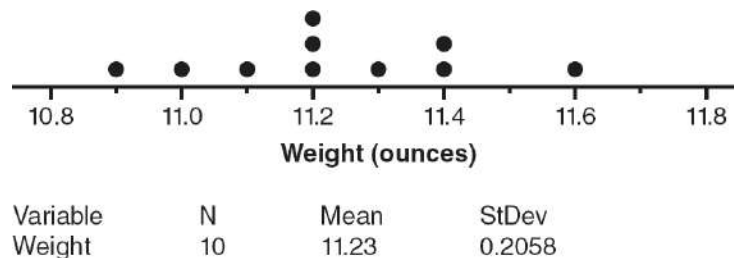
10. Researchers collected data from two different random samples, X and Y , on weight gain (in pounds) by teenagers with anorexia taking part in family therapy. The confidence interval 8.3 ± 0.9 was constructed from sample X , and the confidence interval 8.3 ± 1.4 was constructed from sample Y . Both samples had roughly the same standard deviation. Which of the following statements could explain why the width of the confidence interval from X is less than the width of the confidence interval from Y ?

- (A) The sample sizes of X and Y are the same, the confidence level is the same for both intervals, and the calculated difference in widths is due to sampling variability.
- (B) The sample size of X is less than the sample size of Y , and the confidence level is the same for both intervals.
- (C) The sample sizes of X and Y are the same, and the confidence level used for X is less than that used for Y .
- (D) The sample sizes of X and Y are the same, and the confidence level used for X is greater than that used for Y .

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Students gather a simple random sample of “11 oz” bags of tortilla chips and weigh the contents. Software output of the sample data follows:



- (a) Calculate a 95% confidence interval for the population mean.
- (b) What percentage of the bags in the sample have weights that fall within this interval?
- (c) Is the percentage in (b) close to 95%? Explain what went wrong or explain that nothing went wrong.

2. In a random sample of 30 white-collar crimes (such as embezzlement, securities fraud, and money laundering), the average monetary amount involved was \$2,900,000 with a standard deviation of \$850,000.

- (a) What is the point estimate of the population mean?
- (b) Find the standard error of the sample mean.
- (c) Are conditions for inference met? Explain.
- (d) The 95% confidence interval is (2,580,000, 3,222,000). Interpret this in context.
- (e) What does this confidence interval say, if anything, about the plausibility of a white-collar crime with a monetary amount of \$2,000,000?
- (f) What does this confidence interval say, if anything, about the plausibility that the mean of all white-collar crime monetary amounts is \$2,000,000?

3. An economics professor specializing in labor studies obtains a random sample of 60 recent college graduates. The sample is arranged in 30 pairs. Each pair consists of two people with the same GPA; however, one graduated in a STEM field and the other didn't. Weekly wages are noted for each subject. The data are summarized in the following table:

	STEM graduate	Non-STEM graduate	Paired differences
Mean	\$2,276	\$1,300	\$976
Standard deviation	\$458	\$215	\$41

- (a) Find a 90% confidence interval of the true mean difference (STEM minus non-STEM) between weekly wages of recent graduates.
- (b) Based on this confidence interval, is there significant evidence that the true mean difference is different from \$985?
- (c) Interpret the confidence level in context.

The answers for this Quiz can be found beginning on [page 502](#).

Hypothesis Test for a Mean

Three conditions must be met and must be checked for:

- **The randomization condition:** The data should be collected using a random sample.
- **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
- **The normality condition:** The population distribution should be roughly normal, or the sample size n should be ≥ 30 , or if $n < 30$, the sample data should be free from strong skewness and outliers.

Example 4.5

A manufacturer claims that a new brand of air-conditioning units uses only 6.5 kilowatts of electricity per day. A consumer agency believes the true figure is higher and runs a test on a random sample of size 50. The sample mean is 7.0 kilowatts with a standard deviation of 1.4. The consumer agency performs a test at the 5% significance level.

- (a) What is the parameter of interest?
- (b) What are the hypotheses?
- (c) What is the proper hypothesis test to be performed?
- (d) Check if the conditions are met for performing the test.
- (e) Calculate the standardized test statistic and the p -value.
- (f) What conclusion would you make?
- (g) Given the above conclusion, what type of error, Type I or Type II, might have been committed, and what would be a possible consequence?

NOTE

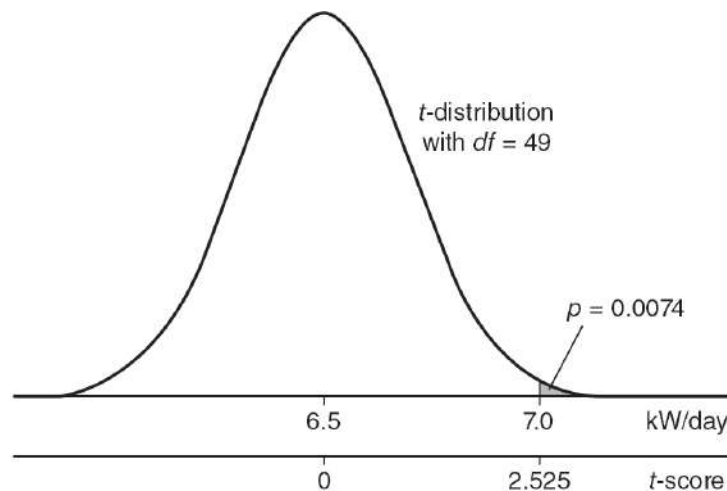
There are two possible explanations for the sample mean of 7.0 being greater than the claimed mean of 6.5. Either the manufacturer's claim is wrong, or by random chance (sample variability), the sample picked by the agency happened to have a mean greater than the claimed mean.

✓ Solutions

- (a) *Parameter:* Let μ represent the mean electricity usage (in kilowatts per day) of the population of this new brand of air-conditioning unit.
- (b) *Hypotheses:* $H_0: \mu = 6.5$ and $H_a: \mu > 6.5$.
- (c) *Procedure:* A one-sample t -test for a population mean (population SD is unknown)
- (d) *Checks:* We are given a random sample, and we assume that 50 is less than 10% of all the new AC units. For normality, $n = 50$, so the CLT applies. Conducting a t -test is appropriate.
- (e) *Mechanics:* Calculator software (such as T-Test on the TI-84) gives $t = 2.525$ and $p = 0.0074$.

[For instructional purposes, we note that $SE(\bar{x}) = \frac{1.4}{\sqrt{50}} = 0.198$ and $t = \frac{7.0 - 6.5}{0.198} = 2.525$ with a resulting p -value of $\text{tcdf}(2.525, 1000, 49) = 0.0074$.]

- (f) *Conclusion in context with linkage to the p -value:* With this small of a p -value, $0.0074 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence for the consumer agency to reject the manufacturer's claim that the new unit uses a mean of only 6.5 kilowatts of electricity per day.



[Follow for extended description](#)

- (g) There was sufficient evidence to reject the null hypothesis. If the null hypothesis were true, we would be committing a Type I error—that is,

mistakenly rejecting a true null hypothesis. A possible consequence here is that the consumer agency would discourage customers from purchasing a new brand of air-conditioning unit that really was saving on electricity consumption as advertised.

Interpretation of the p -value: Assuming that the true mean electricity usage of the new brand of this AC unit is 6.5 kilowatts per day, there is a probability of 0.0074 of getting a sample mean of 7.0 or greater by chance alone in a random sample of 50 units.

➤ Example 4.6

An ice cream parlor manager believes that the mean annual ice cream consumption for high school students is 44 pints. For a class project to test that claim, a random sample of 25 high school students kept careful track of the quantity of ice cream each consumed for a full year. Following are the data and summary statistics.

0			
1		5	
2		5 6	$\bar{x} = 50.04$
3		1 7 9	$s = 15.78$
4		2 2 5 7 8	
5		0 0 1 3 6 6 9	
6		2 4 4 7	
7		1 5 6	

(1 | 5 means 15 pints)

[Follow for extended description](#)

- At a significance level of $\alpha = 0.05$, is there convincing evidence to reject the manager's belief?
- In context of this study, what is the meaning of the p -value calculated in part (a)?
- Given the conclusion in part (a), which kind of error, Type I or Type II, might have been committed, and what would be a possible consequence?

NOTE

The question does not suggest a direction for the alternative so a two-sided test is called for.

✓ Solutions

- (a) *Parameter:* Let μ represent the true mean annual ice cream consumption for the population of all high school students.

Hypotheses: $H_0: \mu = 44$ and $H_a: \mu \neq 44$.

Procedure: A one-sample t -test for a population mean (population SD is unknown)

Checks: The sample is given to be random, and the 25 high school students are less than 10% of all high school students. For normality, a stemplot of the sample values is unimodal, reasonably symmetric, and with no strong skewness or outliers, indicating it is not unreasonable to assume the data come from a roughly normal distribution.

Conducting a t -test is appropriate.

Mechanics: Calculator software (T-Test) gives $t = 1.914$ and a p -value of 0.0675.

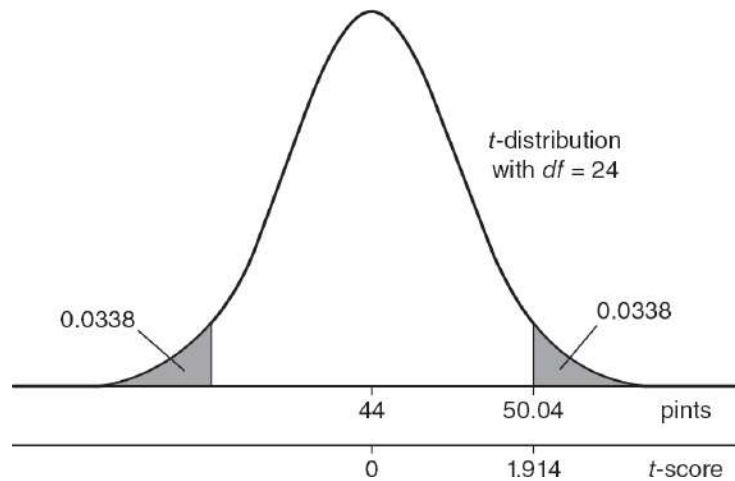
[For instructional purposes, we note that

$t SE(\bar{x}) = \frac{15.78}{\sqrt{25}} = 3.156$ and $t = \frac{50.04 - 44}{3.156} = 1.914$ with a resulting p -value of $2 \times \text{tcdf}(1.914, 1000, 24) = 0.0676$.]

Conclusion in context with linkage to the p -value: With this large a p -value, $0.0676 > 0.05$, there is not sufficient evidence to reject H_0 . In other words, there is not convincing evidence to reject the manager's belief that the true mean annual ice cream consumption for high school students is 44 pints.

- (b) Assuming that the true mean annual ice cream consumption for high school students is 44 pints, there is a probability of 0.0675 of getting a sample mean as extreme or more extreme than 50.04 pints in either direction by chance alone in a random sample of 25 high school students.
- (c) There was not sufficient evidence to reject H_0 . If the null hypothesis were false, we would be committing a Type II error—that is, mistakenly failing to reject a false null hypothesis. A possible

consequence here is that the manager will incorrectly estimate future sales, either underordering or overordering for future demand.



[Follow for extended description](#)

Like with confidence intervals, if the data come in pairs, the proper procedure is a one-sample t -test for a population mean difference.

► Example 4.7

Does a particular drug slow reaction times? If so, the government might require a warning label concerning driving a car while taking the medication. An SRS of 30 people who might benefit from the drug are tested before and after taking the drug, and their reaction times (in seconds) to a standard testing procedure are noted. The resulting histograms and summary statistics are as follows:



	n	Mean	SD
Before	30	1.46433	0.26267
After	30	1.50200	0.20942
Differences	30	-0.037667	0.11755

- (a) Perform the appropriate inference test.
- (b) Given your conclusion, what type of error, Type I or Type II, might have been committed, and what would be a possible consequence?

✓ Solutions

- (a) The proper procedure is to form the set of 30 differences and to perform a one-sample hypothesis test as follows.

Parameter: Let μ represent the mean difference in reaction time (before taking drug minus after taking drug) in the population of people taking a particular drug.

Hypotheses:

H_0 : The reaction times of individuals to a standard testing procedure are the same before and after they take a particular drug; the mean difference is zero: $\mu_d = 0$.

H_a : The reaction time is greater after they take the drug; the mean difference (before minus after) is less than zero: $\mu_d < 0$.

Procedure: A one-sample t -test for a population mean difference (the mean difference of reaction times, before minus after)

TIP

Use proper terminology! In a two-sample t -test, the hypotheses are about the difference of means. In a paired t -test, the hypotheses are about the mean difference.

Checks:

1. The *data are paired* because they are measurements on the same individuals before and after taking the drug.
2. The reaction times of any individual are independent of the reaction times of the others, so the *differences are independent*.
3. A *random sample* of people is tested.
4. The histogram of the differences looks *nearly normal* (roughly unimodal and symmetric), or we could use that the sample size, $n =$

30, is large enough for the CLT to apply.

5. The sample size, $n = 30$, is assumed to be less than 10% of the population of all people who may take the drug.

Mechanics: T-Test gives $t = -1.755$ with a p -value of 0.0449.

[Or we can calculate

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{0.11755}{\sqrt{30}} = 0.021462 \text{ and } t = \frac{\bar{x} - 0}{\sigma_{\bar{x}}} = \frac{-0.037667}{0.021462} = -1.755.$$

With $df = n - 1 = 29$, the p -value is $P(t < -1.755) = \text{tcdf}(-1000, -1.755, 29) = 0.0449.$]

Conclusion in context with linkage to the p -value: With this small of a p -value, $0.0449 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that the true mean of the observed rise in reaction times after taking the drug is significant.

- (b) There was sufficient evidence to reject the null hypothesis. If the null hypothesis were true, we would be committing a Type I error—that is, mistakenly rejecting a true null hypothesis. A possible consequence here is that the drug really doesn't slow reaction times, but the government requires the company to put an unnecessary, and possibly costly, label on the drug. Furthermore, people might choose not to take necessary medication.

Quiz 25

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. In a government consumer publication, it is stated that middle-income families average \$13,000 on child-related costs per year. A sociologist is investigating whether the true mean is now greater than what was published. A random sample of middle-income families had a sample

mean $\bar{x}$ of \$13,800 per year. Assuming all conditions for inference are met, what are the hypotheses for a one-sample t -test ?

(A) $H_0: \mu = 13,000, H_a: \mu > 13,000$

(B) $H_0: \mu = 13,000, H_a: \mu < 13,800$

(C) $H_0: \bar{x} = 13,800, H_a: \bar{x} > 13,000$

(D) $H_0: \bar{x} = 13,800, H_a: \bar{x} < 13,800$

2. A consumer group believes that a wine-dispensing vending machine is pouring less than the advertised 5 ounces. They randomly sample 22 glasses of wine from the machine. In the sample, most of the wine weights are between 4.95 and 5.03 ounces with a few of the weights being substantially larger. Is it appropriate for the consumer group to perform a one-sample t -test for the mean wine weight poured by the machine?

(A) Yes, all conditions have been met.

(B) No, np is not given to be ≥ 10 .

(C) No, the normality condition is violated.

(D) No, this is a convenience sample so any conclusion would be suspect.

3. A baseball manufacturer believes that an average of 98 baseballs are used in MLB games and tests this hypothesis on a random sample of 100 games. The sample is found to have a mean of $\bar{x} = 103.2$ with a standard deviation of 4.8. Which of the following is the correct test statistic?

(A) $t = \frac{103.2 - 98}{4.8/10}$

(B) $t = \frac{103.2 - 98}{4.8/100}$

(C) $t = \frac{98 - 103.2}{4.8/10}$

(D) $t = \frac{98 - 103.2}{4.8/100}$

4. A pharmaceutical company claims that a medicine will produce a desired effect for a mean time of 58.4 minutes. A government

researcher runs a hypothesis test of 40 patients and calculates a mean of $\bar{x}$ 59.5 with a standard deviation of $s = 8.3$. What is the p -value? (Assume that all conditions for inference are met.)

- (A) $P\left(t > \frac{59.5 - 58.4}{8.3/\sqrt{40}}\right)$ with $df = 39$
- (B) $P\left(t > \frac{59.5 - 58.4}{8.3/\sqrt{40}}\right)$ with $df = 40$
- (C) $2P\left(t > \frac{59.5 - 58.4}{8.3/\sqrt{40}}\right)$ with $df = 39$
- (D) $2P\left(t > \frac{59.5 - 58.4}{8.3/\sqrt{40}}\right)$ with $df = 40$

5. A Fish and Game Department inspector is testing mercury content of fish in an area and will close the area to fishing if fish have a mean mercury content over 5 ppm.

Suppose that a hypothesis test results in a t -statistic of 0.035. Which of the following is the best interpretation of this value?

- (A) The probability is 0.035 that the null hypothesis is true.
- (B) The probability is 0.035 that the null hypothesis is false.
- (C) If the null hypothesis were true, the probability would be 0.035 of obtaining a sample mean as great as or greater than the observed sample mean.
- (D) The sample mean is 0.035 standard errors greater than the hypothesized mean of 5.

6. A published study states that U.S. beer consumption per adult age 21 and over is 28.2 gallons per person, per year. A brewer believes that because of difficult economic times during the past several years, this number has gone up. He surveys 50 randomly selected adults over age 21 as to yearly beer consumption and obtains a sample mean of 29.4 gallons. A one-sample t -test results in a p -value of 0.051. Which of the following is a correct interpretation of the p -value?

- (A) The probability is 0.051 that the mean beer consumption per adult age 21 and over is greater than 28.2 gallons.
- (B) The probability is 0.051 that the mean beer consumption per adult age 21 and over is greater than 29.4 gallons.

- (C) If the mean beer consumption per adult age 21 and over is 28.2 gallons, the probability is 0.051 that a randomly selected group of adults age 21 and over consumes a mean of 29.4 gallons or more.
- (D) If the mean beer consumption per adult age 21 and over is greater than 28.2 gallons, the probability is 0.051 that a randomly selected group of adults age 21 and over consumes a mean of 29.4 gallons or more.

7. A company manufactures a synthetic rubber bungee cord with a braided covering of natural rubber and a minimum breaking strength of 450 kg. If the mean breaking strength of a sample drops below a specified level, the production process is halted and the machinery inspected. Which of the following would result from a Type I error?

- (A) Halting the production process when too many cords break
- (B) Halting the production process when the breaking strength is below the specified level
- (C) Halting the production process when the breaking strength is within specifications
- (D) Allowing the production process to continue when the breaking strength is within specifications

8. Suppose you do five independent tests of the form $H_0: \mu = 38$ versus $H_a: \mu > 38$, each at the $\alpha = 0.01$ significance level. What is the probability of committing a Type I error and incorrectly rejecting a true null hypothesis with at least one of the five tests?

- (A) 0.01
- (B) 0.049
- (C) 0.05
- (D) 0.951

9. Given an experiment with $H_0: \mu = 35$, $H_a: \mu < 35$, and a possible correct value of 32, you obtain a sample statistic of $\bar{x} = 33$. After doing analysis, you realize that the sample size n is actually larger than you first thought. Which of the following results from reworking with the increase in sample size?

- (A) Decrease in probability of a Type I error; decrease in probability of a Type II error; decrease in power
- (B) Increase in probability of a Type I error; increase in probability of a Type II error; decrease in power
- (C) Decrease in probability of a Type I error; decrease in probability of a Type II error; increase in power
- (D) Decrease in probability of a Type I error; increase in probability of a Type II error; increase in power

10. A recording industry producer believes that teenagers will rate the sound quality of LPs (vinyl records) greater than that of CDs (compact discs). He randomly selects 100 teenagers to participate in an experiment. For each teen, the producer plays the same music selection, either first on an LP and then on a CD or in the other order depending on a flip of a coin. The teens listen through earphones and are blinded as to the source of the music. Each teen rates each of the two recordings on a 1 to 10 quality scale. Using a one-sample t -test for a population mean difference (with a matched pairs design), the producer obtains a p -value of 0.295. Assuming the conditions for inference have been met, which of the following is the best conclusion?

- (A) Because the p -value is less than any reasonable level of significance, there is convincing evidence to support the belief that teenagers will rate the sound quality of LPs greater than that of CDs.
- (B) Because the p -value is less than any reasonable level of significance, there is not convincing evidence to support the belief that teenagers will rate the sound quality of LPs greater than that of CDs.
- (C) Because the p -value is greater than any reasonable level of significance, there is convincing evidence to support the belief that teenagers will rate the sound quality of LPs greater than that of CDs.
- (D) Because the p -value is greater than any reasonable level of significance, there is not convincing evidence to support the

belief that teenagers will rate the sound quality of LPs greater than that of CDs.

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. A particular wastewater treatment system aims at reducing the most probable number per mL (MPN per mL) of *E. coli* to 1,000 MPN per 100 mL. A random study of 40 of these systems in current use is conducted with the data showing a mean of 1,002.4 MPN per 100 mL and a standard deviation of 7.12 MPN per 100 mL. A test of significance is conducted with:

H_0 : The mean concentration of *E. coli* after treatment under this system is 1,000 MPN per 100 mL.

H_a : The mean concentration of *E. coli* after treatment under this system is greater than 1,000 MPN per 100 mL.

The resulting p -value is 0.0197 with $df = 39$ and $t = 2.132$.

- (a) Interpret the p -value in the context of this study.
 - (b) What conclusion should be drawn at a 5% significance level?
 - (c) Given this conclusion, what possible error, Type I or Type II, might be committed? Give a possible consequence of committing this error.
2. A recent study reported that college students work an average of 25 hours per week over the summer break. A guidance counselor believes that students at her large college work more than this. To collect data, she will randomly sample 50 out of the 6,000 students at the college.

- (a) What is the population of interest?
- (b) Explain how the counselor could obtain an SRS of 50 students.
- (c) Based on the sample, the counselor calculates $\bar{x} = 26.4$ with $s = 7.1$. At a significance level of $\alpha = 0.05$, do the data provide convincing statistical evidence to support the counselor's claim? Complete an appropriate inference procedure to support your answer.

3. A random sample of 20 nonmanagerial employees in the restaurant industry are matched into 10 pairs, where each pair has employees of similar ages, similar job descriptions, and similar years of experience. However, one of each pair belongs to a union and the other does not. Statistics from their weekly salaries excluding tips are calculated for union employees, for nonunion employees, and for differences (union minus nonunion) from each pair.

Union	■ ■ ■ ■ ■ ■ ■ ■ ■
Nonunion	■ ■ ■ ■ ■ ■ ■ ■ ■ ■
Differences	■ ■ ■ ■ ■ ■ ■ ■ ■ ■

[Follow for extended description](#)

	Weekly salary excluding tips (in dollars)		
	Number	Mean	Standard deviation
Union	10	791.4	94.3
Nonunion	10	665.0	92.1
Differences	10	126.4	66.4

Do the data provide convincing statistical evidence that the mean salary of the nonmanagerial union employees is greater than the mean salary of the nonmanagerial nonunion employees? Complete an appropriate inference procedure at the 1% significance level to support your answer.

The answers for this Quiz can be found beginning on [page 504](#).

Sampling Distribution for the Difference Between Sample Means

We have the following about the sampling distribution of $\bar{x}_1 - \bar{x}_2$

Start with two populations with means μ_1 and μ_2 and standard deviations σ_1 and σ_2 . Take all samples of sizes n_1 and n_2 , respectively. Compute the difference $\bar{x}_1 - \bar{x}_2$ of the two means in each pair of these samples:

1. The mean $\mu_{\bar{x}_1 - \bar{x}_2}$ of the set of differences of sample means equals $\mu_1 - \mu_2$, the difference of population means.
2. The standard deviation $\sigma_{\bar{x}_1 - \bar{x}_2}$ of the set of differences of sample means is approximately equal to $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.
3. If the populations are normal or if the sample sizes are large enough (at least 30) for the central limit theorem to apply, we also have that the set of all differences of sample means is approximately normally distributed.

When analyzing two samples, one from each population, we have the assumption of independent simple random samples and, if sampling without replacement, the assumption of sample sizes no larger than 10% of their respective populations.

Example 4.8

It is estimated that 40-year-old men contribute an average of 65 genetic mutations to their new children, whereas 20-year-old men contribute an average of only 25. Assuming standard deviations of 15 and 5 mutations, respectively, for the 40- and 20-year-olds, what is the probability that the mean number of mutations in a random sample of thirty-five 40-year-old new fathers is between 35 and 45 more than the mean number in a random sample of forty 20-year-old new fathers?

We say, “For all random samples of thirty-five 40-year-old new fathers and forty 20-year-old new fathers, the differences (40-year-olds – 20-year-olds) in sample mean genetic mutations passed to their new children will have a mean of 40 mutations and will typically vary by about 2.656 mutations from the true difference of 40 mutations.”

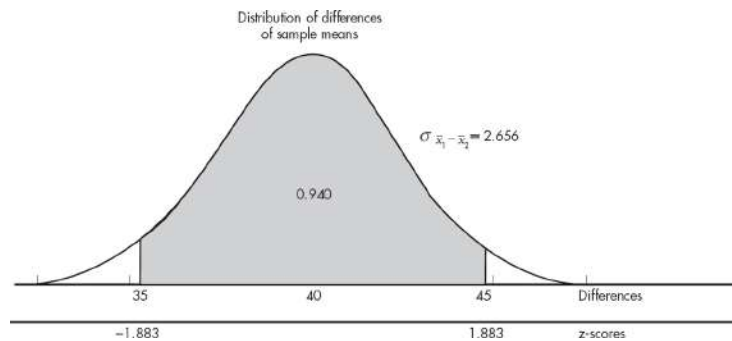
✓ Solution

We have independent random samples, each less than 10% of their age groups, and both sample sizes are over 30. So the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is roughly normal with mean $\mu_{\bar{x}_1 - \bar{x}_2} = 65 - 25 = 40$ and standard deviation

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{15^2}{35} + \frac{5^2}{40}} = 2.656.$$

$$\begin{aligned} P(35 < \bar{x}_1 - \bar{x}_2 < 45) &= P\left(\frac{35 - 40}{2.656} < z < \frac{45 - 40}{2.656}\right) \\ &= P(-1.883 < z < 1.883) = 0.940 \end{aligned}$$

[normalcdf(−1.883, 1.883) = 0.940 and normalcdf(35, 45, 40, 2.656) = 0.940.]



A difference (40-year-olds – 20-year-olds) in sample mean genetic mutations passed to their new children of between 35 and 45 mutations happens in about 94.0% of all possible samples of thirty-five 40-year-old new fathers and forty 20-year-old new fathers.

NOTE

If the data come from an experiment, the 10% condition is unnecessary, and randomization is satisfied if the treatments are randomly assigned.

Confidence Interval for the Difference of Two Means

Three conditions must be met and must be checked for:

- **The randomization condition:** The data should be collected using random samples, and the samples must be taken independently of each other.
- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of their respective populations.
- **The normality condition:** The population distributions should both be approximately normal, or the sample sizes should both be ≥ 30 , or the sample distributions should both be free of strong skewness and outliers.

When the population standard deviations are unknown, we use a t -distribution with $SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$. The confidence interval takes the following form:

$$(\bar{x}_1 - \bar{x}_2) \pm t^* SE(\bar{x}_1 - \bar{x}_2) = (\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where the degrees of freedom df for the t -distribution can be found using technology.

NOTE

Interpretation of the confidence level: If all possible random samples of 30 college professors and 30 police officers are analyzed as to planned retirement ages and if a confidence interval for the difference in means is constructed for each pair of samples, then 95% of these intervals would capture the true difference (professors minus police) in mean planned retirement ages.

➤ Example 4.9

College professors' lives are often so immersed in their careers that they find it difficult to accept retirement. Although police officers identify with their careers, their stressful discipline takes its toll. A sociologist surveys 30 randomly chosen college professors and 30 randomly chosen police officers as to their planned retirement ages. In the sample of college professors, the mean planned retirement age was 66 with a standard deviation of 3.5, while in the sample of police officers, the mean planned retirement age was 55 with a standard deviation of 5.1.

- (a) Determine a 95% confidence interval for the difference in mean planned retirement ages (professors minus police).
- (b) Based on this confidence interval, is there convincing evidence that the mean retirement age for college professors is more than 5 years greater than for police officers?

Solutions

- (a) *Parameters:* Let μ_c represent the true mean planned retirement age for college professors. Let μ_p represent the true mean planned retirement age for police officers.

Procedure: A two-sample t -interval for the difference of two population means, $\mu_c - \mu_p$

Checks: We are given independent random samples, and the sample sizes, $n_c = n_p = 30$, are less than 10% of the respective populations of professors and police. For normality, the sample sizes, $n_c = n_p = 30 \geq 30$, are large enough for the CLT to apply. Calculating a t -interval is appropriate.

Mechanics: Calculator software (2-SampTInt) gives (8.73, 13.27) with $df = 51.36$.

Conclusion in context: We are 95% confident that the true difference in mean planned retirement ages (professors minus police) between all college professors and police officers is between 8.73 and 13.27.

- (b) Yes, there is convincing evidence that the mean retirement age for college professors is at least 5 years greater than for police officers because the entire interval (8.73, 13.27) is greater than 5.

As in the previous unit about proportions, the null hypothesis is usually a claim about “no difference” or “no change” or “no treatment effect” and is stated as an equality. The alternative hypothesis is a claim that we hope to support with evidence from the collected data and is stated as a strict inequality. Both hypotheses are about population parameters.

Hypothesis Test for the Difference of Two Means

The null hypothesis is usually that the means of the populations are the same or, equivalently, that their difference is 0:

$$H_0: \mu_1 - \mu_2 = 0$$

The alternative hypothesis is then:

$$H_a: \mu_1 - \mu_2 < 0, \quad H_a: \mu_1 - \mu_2 > 0, \quad \text{or} \quad H_a: \mu_1 - \mu_2 \neq 0$$

Alternatively, we can use $H_0: \mu_1 = \mu_2$ and $H_a: \mu_1 < \mu_2$ or $H_a: \mu_1 > \mu_2$ or $H_a: \mu_1 \neq \mu_2$.

The first two possibilities lead to one-sided tests, and the third possibility leads to two-sided tests.

The three conditions to be met and must be checked for are as follows:

- **The randomization condition:** The data should be collected using two independent random samples or a randomized experiment.
- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of the respective populations. (This condition is unnecessary for randomized experiments.)
- **The normality condition:** The population distributions should both be approximately normal, or the sample sizes should both be ≥ 30 , or the sample distributions should both be free of strong skewness and outliers.

Example 4.10

A city council member claims that male and female officers wait equal times for promotion in the police department. A women's spokesperson, however, argues that women wait longer than men. She obtains a random sample of five promoted men officers with wait times of 8, 7, 10, 5, and 7 years for promotion and an independent random sample of promoted four women officers with wait times of 9, 5, 12, and 8 years.

- What are the parameters and hypotheses of interest for the women's spokesperson?
- Are the conditions for a two-sample t -test for the difference of population means satisfied? Explain.

The t -test yields $t = -0.6641$ and $p = 0.2685$.

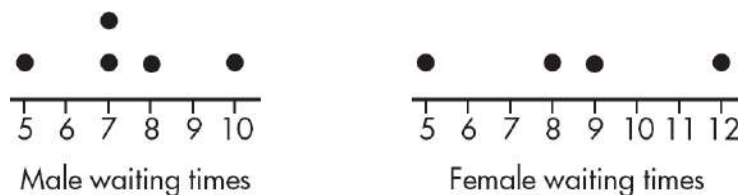
- Interpret the p -value in context.
- Give a conclusion in context.
- Given the above conclusion, what type of error, Type I or Type II, might have been committed, and what would be a possible consequence?

✓ Solutions

- Parameters:* Let μ_M represent the mean wait time for promotion in the population of male officers. Let μ_F represent the mean wait time for promotion in the population of female officers.

Hypotheses: $H_0: \mu_M = \mu_F$ and $H_a: \mu_M < \mu_F$

- We are given random samples and must assume that they are independent and less than 10% of total numbers of promoted male and female officers. For normality, dotplots of the sample data show no strong skewness or outliers, so it is not unreasonable to assume the data come from roughly normal population distributions.

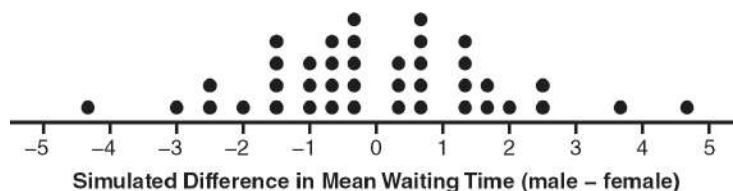


- Assuming that the mean wait times for promotion for male and female officers are the same, there is a 0.2685 probability in getting a difference (male minus female) in sample means of $7.4 - 8.5 = -1.1$ or

less (more negative) by chance alone in random samples of five promoted male officers and four promoted female officers.

- (d) With this large a p -value, $0.2685 > 0.05$, there is not sufficient evidence to reject H_0 . In other words, there is not convincing evidence of the women's spokesperson's claim that the mean wait time for promotion is longer for women than for men.
- (e) There was not sufficient evidence to reject the null hypothesis. If the null hypothesis were false, we would be committing a Type II error—that is, mistakenly failing to reject a false null hypothesis. A possible consequence would be that female officers really do have to wait longer times for promotions but nothing is done to remedy the situation.

The above example could have been approached using a simulation. Could the above observed difference of -1.1 have occurred by chance? We can randomly assign the observed nine waiting times, (5, 7, 7, 8, 10, 5, 8, 9, 12), into two groups, five to the male group and the remaining four to the female group. For each such regrouping, the difference in mean waiting times (male – female) is calculated. One such simulation is illustrated in the dotplot below.



As can be seen from this simulation, a difference of -1.1 or less is not unusual. We conclude that the observed difference could reasonably have occurred by chance, and is thus not significant. (Note that there are 9 out of 40 dots that are -1.1 or less. This gives $\frac{9}{40} = 0.225$ as a simulated estimate for the p -value.)

▶ Example 4.11

A study is conducted to determine whether or not there is convincing evidence that professional tennis players and professional basketball players put on different mean numbers of miles during a game. Thirty-five players are randomly selected from each sport and are outfitted with pedometers to

measure distances run during their next game. Summary statistics are shown in the table.

Sport	Sample Size	Mean (miles)	Standard Deviation (miles)
Tennis	35	2.8	0.41
Basketball	35	2.5	0.32

- (a) Do the data provide convincing evidence at the $\alpha = 0.01$ significance level of a difference in mean miles put on during a game between professional tennis players and professional basketball players?
- (b) Interpret in context the p -value from part (a).

NOTE

Always remember to define each parameter so as to identify which parameter refers to which subscript.

 Solutions

- (a) *Parameters:* Let μ_T represent the true mean miles put on during games by the population of professional tennis players. Let μ_B represent the true mean miles put on during games by the population of professional basketball players.

Hypotheses: $H_0: \mu_T = \mu_B$ and $H_a: \mu_T \neq \mu_B$

Procedure: A two-sample t -test for the difference of two population means

Checks: We are given independent random samples and assume that the sample sizes are less than 10% of the respective populations of professional players. For normality, the sample sizes, $n_T = 35 \geq 30$ and $n_B = 35 \geq 30$, are large enough for the CLT to apply. Conducting a t -test is appropriate.

Mechanics: Calculator software gives $t = 3.41$ and $p = 0.0011$ with $df = 64.21$.

Conclusion in context with linkage to the p-value: With this small a p -value, $0.0011 < 0.01$, there is sufficient evidence to reject H_0 . In other words, there is convincing evidence that there is a difference in mean miles put on during a game between professional tennis players and professional basketball players.

- (b) Assuming that there is no difference in the mean miles put on during a game between professional tennis players and professional basketball players, there is a probability of 0.0011 of getting an observed mean difference of 0.3 or greater in either direction by chance alone in random samples of 35 professional tennis players and 35 professional basketball players.

NOTE

You should *not* pool in a two-sample t -test like we did in a two- proportion z -test.

Quiz 26

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. It is hypothesized that grandmothers spend more hours per week on FaceTime with their grandchildren than do grandfathers. A random sample of 100 couples who have at least one grandchild are interviewed, and a two-sample t -test is performed. Have all conditions for inference been met?
 - (A) Yes, all conditions have been met.
 - (B) No, while a random sample was selected, there is no mention of random assignment.
 - (C) No, there is no indication of whether the sample distribution is approximately normal.

(D) No, independence is violated.

2. Head injuries are a major concern in the sport of football. Five hundred D-1 football players were randomly picked. Half were randomly assigned to wear helmets based on foam padding. The other half were randomly assigned to wear helmets using padding that was custom 3D-printed to the shape of their heads. The helmets were used for an entire season including practices, preseason games, and regular season games. The mean number of head injuries was recorded for each group. Which of the following is an appropriate inference procedure for analyzing the results of the investigation?

- (A) A one-sample t -test for a population mean
- (B) A matched pairs t -test for a mean difference
- (C) A two-sample t -test for a difference between sample means
- (D) A two-sample t -test for a difference between population means

3. In a study aimed at reducing developmental problems in low birth weight babies (under 2,500 grams), 347 infants were exposed to a special educational curriculum while 561 did not receive any special help. After 3 years, the children exposed to the special curriculum showed a mean IQ of 93.5 with a standard deviation of 19.1; the other children had a mean IQ of 84.5 with a standard deviation of 19.9. Find a 95% confidence interval estimate for the difference in mean IQs of all low birth weight babies who receive special intervention and those who do not. (Assume that all conditions for inference are met.)

- (A) $(93.5 - 84.5) \pm 1.97 \sqrt{\frac{(19.1)^2}{347} + \frac{(19.9)^2}{561}}$
- (B) $(93.5 - 84.5) \pm 1.97 \left(\frac{19.1}{\sqrt{347}} + \frac{19.9}{\sqrt{561}} \right)$
- (C) $(93.5 - 84.5) \pm 1.65 \sqrt{\frac{(19.1)^2}{347} + \frac{(19.9)^2}{561}}$
- (D) $(93.5 - 84.5) \pm 1.65 \left(\frac{19.1}{\sqrt{347}} + \frac{19.9}{\sqrt{561}} \right)$

4. A researcher believes a new diet should improve weight gain in laboratory mice. If ten control mice on the old diet gain an average of 4 ounces with a standard deviation of 0.3 ounces, while the average gain for ten mice on the new diet is 4.8 ounces with a standard deviation of 0.2 ounces, what is the test statistic? (Assume that all conditions for inference are met.)

(A) $t = \frac{4 - 4.8}{\sqrt{\frac{(0.3)^2}{10} + \frac{(0.2)^2}{10}}}$

(B) $z = \frac{4 - 4.8}{\sqrt{\frac{(0.3)^2}{10} + \frac{(0.2)^2}{10}}}$

(C) $t = \frac{4 - 4.8}{\frac{0.3}{\sqrt{10}} + \frac{0.2}{\sqrt{10}}}$

(D) $z = \frac{4 - 4.8}{\frac{0.3}{\sqrt{10}} + \frac{0.2}{\sqrt{10}}}$

5. Do high school girls apply to more colleges than high school boys? A two-sample t -test of the hypotheses $H_0: \mu_{\text{girls}} = \mu_{\text{boys}}$ versus $H_a: \mu_{\text{girls}} > \mu_{\text{boys}}$ results in a p -value of 0.02. Which of the following statements must be true?

- I. A 90% confidence interval for the difference in means contains 0.
- II. A 95% confidence interval for the difference in means contains 0.
- III. A 99% confidence interval for the difference in means contains 0.

- (A) I and II only
- (B) III only
- (C) II and III only
- (D) I, II, and III

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. Bisphenol A (BPA), a synthetic estrogen found in packaging materials, has been shown to leach into infant formula and beverages. More recently, BPA has been detected in high concentrations in cash register receipts. Random sample biomonitoring data to see if retail workers carry higher amounts of BPA concentration in their bodies than nonretail workers are as follows:

		BPA concentration ($\mu\text{g/L}$)	
		Mean	Standard deviation
Nonretail workers	528	2.43	0.45
Retail workers	197	3.28	0.48

- (a) Calculate a 99% confidence interval for the difference in mean BPA body concentrations of nonretail and retail workers.
 - (b) Does the confidence interval support the belief that retail workers carry higher amounts of BPA in their bodies than nonretail workers? Explain.
2. A random sample of 50 married couples were surveyed as to age at wedding. Following are data with regard to husbands, wives, and couple differences (husband's age minus wife's age).

Ages at date of wedding			
	Sample size n	Mean	Standard deviation
Husbands	50	30.6	8.792

Ages at date of wedding			
Wives	50	28.6	7.977
Differences	50	2.0	3.742

- A sociologist believes that the above table is evidence that the mean age at wedding for all husbands is greater than the mean age at wedding for all wives. He calculates $(-1.332, 5.332)$ to be a two-sample, 95% confidence interval for the true difference in mean ages of husbands and wives. He then concludes that there is not convincing evidence that the true mean age at wedding for husbands is greater than the true mean age at wedding for wives because the interval contains both positive and negative values. Is this a statistically valid argument? Explain.
- Perform a hypothesis test at the 0.05 significance level with regard to the sociologist's claim.
- Given your conclusion, what type of error, Type I or Type II, might have been committed, and what would be a possible consequence?

The answers for this Quiz can be found beginning on [page 505](#).

Simulations and p -Values

When our learned test procedures do not apply, either because conditions are not met or because we are working with a parameter for which we have not learned a test procedure, we may be able to proceed with a simulation. We can use a simulation to determine what values of a test statistic are likely to occur by random chance alone, assuming the null hypothesis is true. Then looking at where our test statistic falls, we can estimate a p -value.

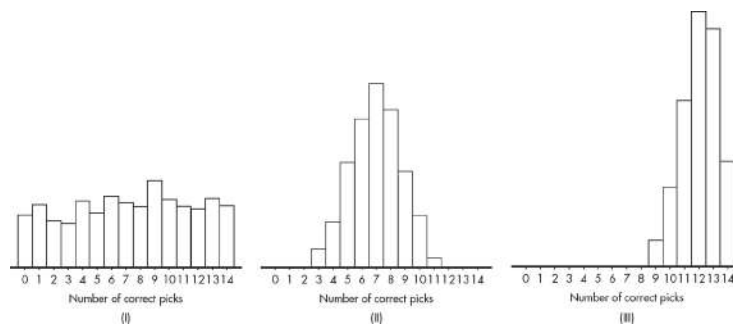
NOTE

These numbers are too small to use in our standard proportion hypothesis tests, so we use a simulation.

▶ Example 4.12

Researchers want to study whether Paul the octopus, who lives in a tank at Sea Life Centre in Oberhausen, Germany, could correctly predict the winner of soccer matches. In a random sample of 14 matches, Paul correctly predicted the winner in 12 out of 14 soccer matches (by choosing to eat mussels from the boxes labeled with national flags of the eventual winning teams).

- (a) What is the parameter of interest?
- (b) What are the null and alternative hypotheses?
- (c) If you conduct a simulation to investigate whether the observed result provides strong evidence that Paul can correctly predict winners, what would you use for the probability of success, for the sample size, and for the number of samples?
- (d) Which of the following graphs could reasonably have come from the simulation?



- (e) Estimate the p -value, and interpret it in context.
- (f) Make a conclusion based on the simulation and p -value.

✓ Solutions

- (a) Let p represent the proportion of correct predictions in the population of all soccer match winner predictions by Paul the octopus.
- (b) H_0 : Paul can do no better than guesswork in predicting winners, $H_0: p = 0.5$.
 H_a : Paul can correctly predict winners better than guesswork, $H_a: p > 0.5$.
- (c) $p = 0.5$, $n = 14$, and a large number for number of samples.

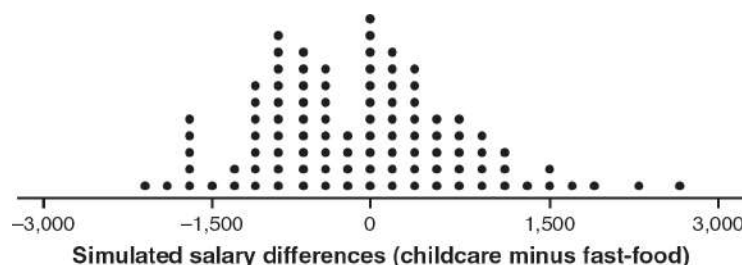
- (d) Graph (II). The simulation should be centered at the null hypothesis expected value, not the observed value.
- (e) Looking at the correct graph, (II), 12 or greater correct picks seems to have a probability of 0. If the null hypothesis were true—that is, if Paul can do no better than guesswork—the probability of Paul correctly predicting 12 or more out of 14 soccer matches is close to 0.00.
- (f) With this small of a p -value, $0.00 < 0.05$, there is strong evidence to reject H_0 ; that is, there is convincing evidence that Paul can pick the winner of soccer matches better than guesswork.

NOTE

We have not learned a test procedure for medians, so a simulation is needed.

Example 4.13

Childcare work is critically important, but it isn't well paid. Do childcare workers have greater salaries than fast-food workers? A random sample of 30 childcare workers' salaries has a median of \$23,240, while a random sample of 35 fast-food workers' salaries has a median of \$21,740. Could the observed difference have occurred by chance? We can randomly assign the 65 observed salaries into two groups, 30 to the childcare workers group and the remaining 35 to the fast-food workers group. For each regrouping, the difference in median salaries (childcare minus fast food) is calculated. One such simulation of 100 such regroupings is illustrated in the dotplot below.



- (a) What are the parameters of interest?
- (b) What are the hypotheses?
- (c) What is the observed difference?

- (d) What is the estimated p -value?
- (e) Based on the estimated p -value, what is the appropriate conclusion in context at a significance level of 0.05?

✓ Solutions

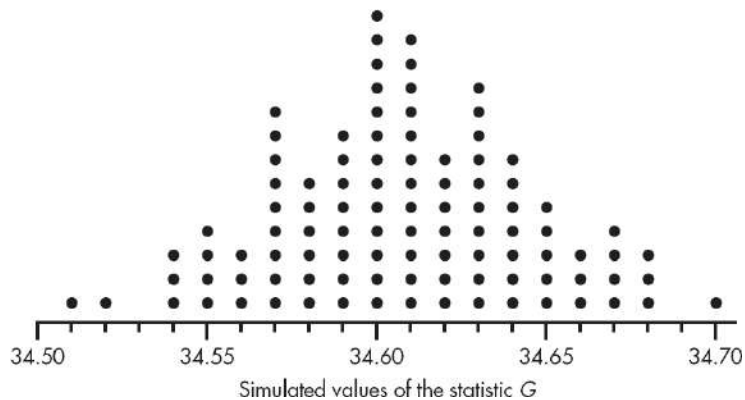
- (a) Let M_C represent the median salary in the population of childcare workers. Let M_F represent the median salary in the population of fast-food workers.
- (b) $H_0: M_C = M_F$ and $H_a: M_C > M_F$.
- (c) The observed difference is $\$23,240 - \$21,740 = \$1,500$.
- (d) There are 6 differences 1,500 or greater. So the estimated p -value is $\frac{6}{100} = 0.06$.
- (e) With this large of a p -value, $0.06 > 0.05$, there is not sufficient evidence to reject the null hypothesis. In other words, there is not convincing evidence that the median salary of all childcare workers is greater than the median salary of all fast-food workers.

Quiz 27

Multiple-Choice Questions

DIRECTIONS: The questions that follow are followed by four suggested answers. Choose the response that best answers the question.

1. A company engineer creates a diagnostic measurement, $G = \frac{\text{Max} + \text{Min}}{2}$, which should be at least 34.60 in a sample of size 10 if certain machinery is operating correctly. To explore this diagnostic measurement, the machine is perfectly calibrated, and 100 random samples of size 10 of the product are taken from the assembly line. For each of these 100 samples, the diagnostic measurement G is calculated and shown plotted below.



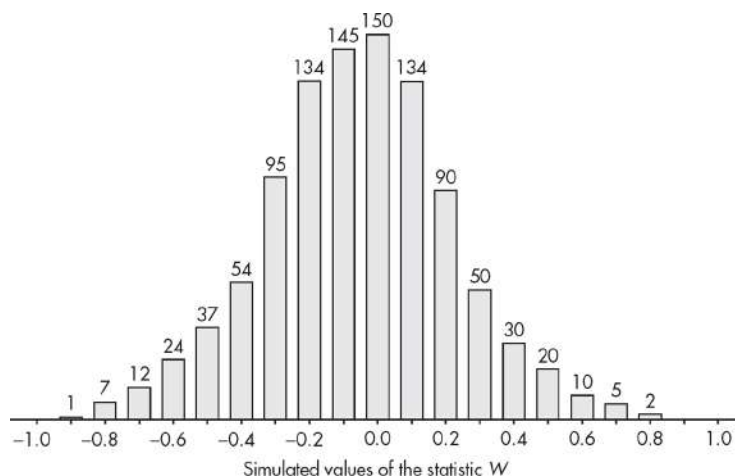
Each day, one sample of size 10 is taken from the assembly line and the diagnostic measurement G is calculated. If G drops too low, a decision to recalibrate the machinery is made.

One day the random sample is $\{34.89, 35.55, 33.93, 34.4, 35.51, 33.51, 35.0, 34.73, 35.19, 33.71\}$.

For the hypothesis test $H_0: G = 34.60$, $H_a: G < 34.60$, what is the estimated p -value?

- (A) 0.01
- (B) 0.02
- (C) 0.03
- (D) 0.04

2. A dietitian compares the effects of two diet pills. Using 25 participants taking each pill, she calculates a statistic W by computing a mean difference in weight loss expressed in pooled standard deviation units. The dietitian calculates $W = -0.7$ in her experiment. One thousand simulations assuming no real difference between the two pills gives the plot below for W .



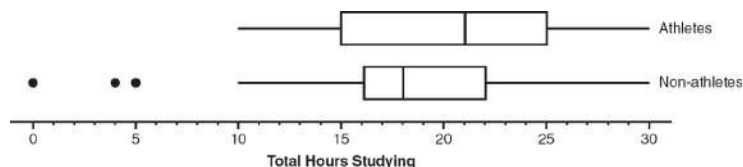
What is the estimated p -value for her observed value of $W = -0.7$ for the null hypothesis of no difference in effect versus the alternative hypothesis that there is a difference in effect?

- (A) 0.007
- (B) 0.01
- (C) 0.02
- (D) 0.027

Free-Response Questions

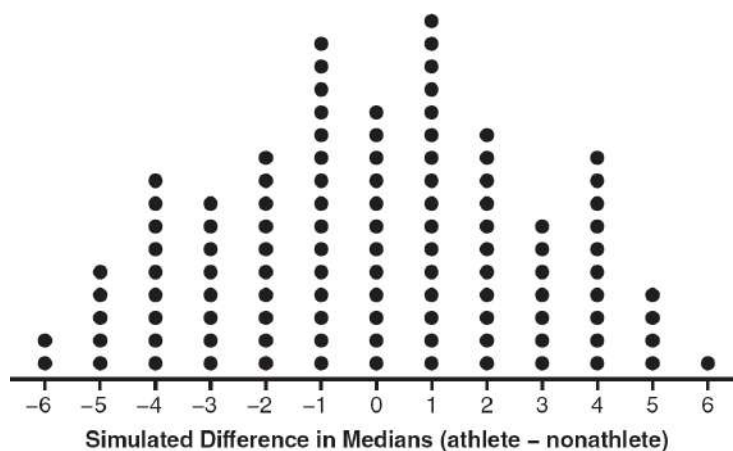
DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. College coaches often strongly stress study hours out of concern that their players remain academically eligible to play. In one study at a Division III college, 25 students participating and 25 students not participating in a varsity sport were surveyed as to the number of hours per week spent with academic studies outside the classroom. The survey results are summarized in the following parallel boxplots.



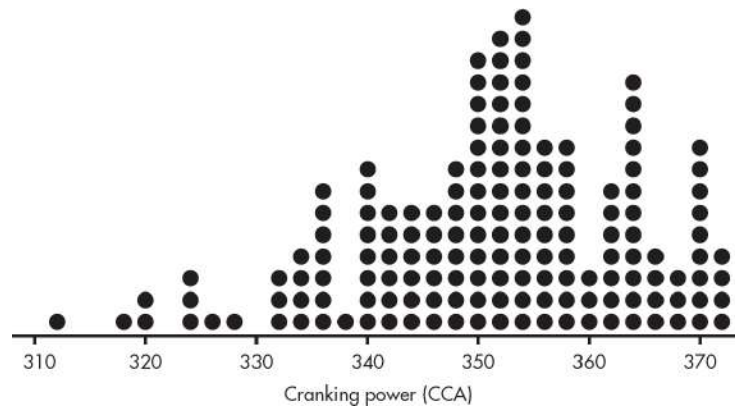
(a) Compare the two distributions.

To test whether the observed difference in medians is significant, a simulation is conducted. For each trial of the simulation, the 25 values from each survey were combined into one group of size 50 and then randomly split into two groups of 25. The difference in medians (athletes minus nonathletes) of the groups were calculated for each trial. The following dotplot shows the difference in the medians for 110 trials of the simulation.



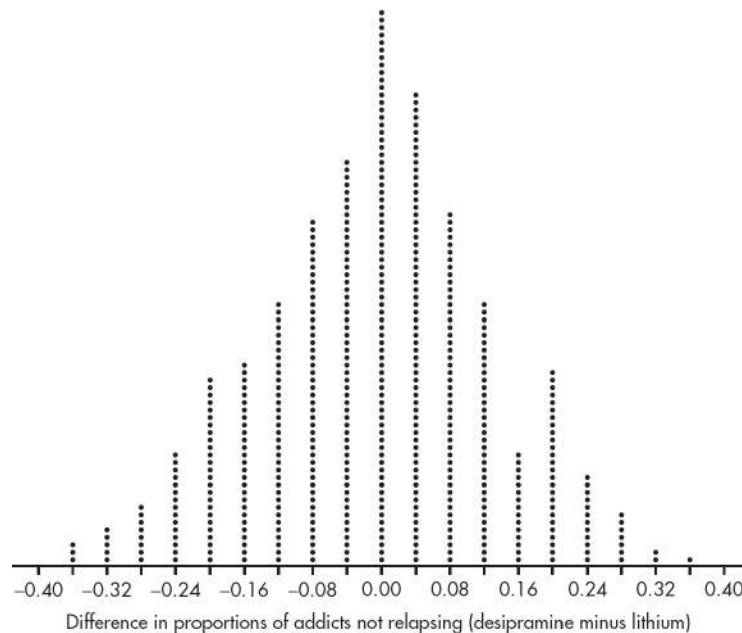
- In how many of the trials was the athletes' median 3 or more than the nonathletes' median?
- Using the observed difference (athletes – nonathletes) in median study hours and using the results of the simulation, what is the estimated probability of the observed difference or more occurring if there were really no difference in study hours (the assumption of the simulation methodology)?
- Based on the answer to (b), is there convincing statistical evidence that athletes participating in varsity sports at this college spend more hours per week with academic studies outside the classroom than do nonathletes?

2. A battery manufacturer advertises that a 12-volt lawn and garden battery it sells has an average cranking power of 350 CCA (cold cranking amps). A consumer group believes that the true figure is lower. The consumer group obtains a random sample of 5 of the company's batteries and determines an average of 320 CCA. Is this significant? The company claims that perfectly good batteries do vary in CCA. To illustrate its point, the company runs a simulation by randomly picking 5 batteries 160 times from a group of new batteries manufactured to have 350 CCA strength and calculating the resulting averages to show that the consumer group's sample average of 320 CCA was possible. The company makes the following dotplot.



- (a) What are the hypotheses for an appropriate test?
 - (b) Explain why or why not the conditions for a t -test of a mean are met.
 - (c) What is the estimated p -value?
 - (d) Based on the estimated p -value, what is the appropriate conclusion in context?
3. A medical researcher believes that the antidepressant desipramine is more effective than the mood stabilizer lithium in treating cocaine addiction. In a randomized experiment, 50 cocaine addicts were randomly assigned to take either desipramine or lithium. Fourteen out of the 25 addicts taking desipramine had not relapsed after one year, while only 6 of the 25 addicts taking lithium had not relapsed. These numbers were too small to use in our standard proportion hypothesis tests. In a simulation of 500 sets of 50 addicts undergoing this

experiment, assuming the null hypothesis of no difference in the treatments, the difference in proportions of addicts not relapsing (desipramine minus lithium) yields:



- (a) In terms of appropriate proportions, what are the hypotheses?
- (b) What is the observed difference in proportions?
- (c) What is the estimated p -value?
- (d) What is the appropriate conclusion in context?

The answers for this Quiz can be found beginning on [page 507](#).

[More on Power, Type I Errors, and Type II Errors](#)

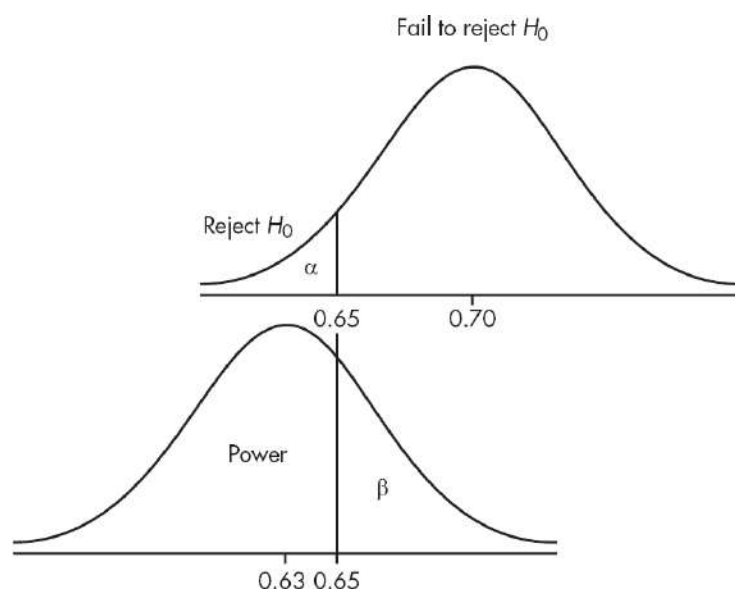
Given a specific alternative hypothesis, a Type II error is a mistaken failure to reject the false null hypothesis, while the *power* is the probability of rejecting that false null hypothesis.

TIP

While calculations are not called for, you must understand the interactions among the errors, confidence level, power, *effect*, and sample size.

► **Example 4.14**

A candidate claims to have the support of 70% of the people, but you believe that the true figure is lower. You plan to gather an SRS and will reject the 70% claim if your sample shows 65% or less support. What if, in reality, only 63% of the people support the candidate?



The upper graph shows the null hypothesis model with the claim that $p_0 = 0.70$ and the plan to reject H_0 if $\hat{p} < 0.65$. The lower graph shows the true model with $p = 0.63$. When will we fail to recognize that the null hypothesis is incorrect? Answer: precisely when the sample proportion is greater than 0.65. This is a Type II error with probability β . When will we rightly conclude that the null hypothesis is incorrect? Answer: when the sample proportion is less than 0.65. This is the “power” of the test and has probability $1 - \beta$.

The following points should be emphasized:

TIP

Both blocking in experiments and stratification in sampling reduce variability and so can improve power.

- Power gives the probability of avoiding a Type II error.
- Power has a different value for different possible correct values of the population parameter; thus, it is actually a *function* where the independent variable ranges over specific alternative hypotheses.
- Choosing a smaller α (that is, a tougher standard to reject H_0) results in a higher risk of a Type II error and a lower power—observe in the above graphs how making α smaller (in this case moving the critical cutoff value to the left) makes the power less and β more!
- The greater the difference between the null hypothesis p_0 and the true value p , the smaller the risk of a Type II error and the greater the power—observe in the above picture how moving the lower graph to the left makes the power greater and β less. (The difference between p_0 and p is sometimes called the *effect*—thus the greater the effect, the greater is the power.)
- A larger sample size n will reduce the standard deviations, making both graphs narrower, and—if the rejection value is unchanged—the result is smaller α , smaller β , and larger power!

A general conclusion is that if the other three values below are unchanged, the power will increase (and the probability of a Type II error will decrease) if any of the following occur:

1. Significance level, *alpha*, increases (but this also increases the chance of a Type I error).
2. Standard error decreases (for example, blocking on something strongly related to the response variable can do this).
3. Sample size, n , increases (of course, more data also means greater expense in time and money).
4. True parameter is farther from the null (this is called the *effect size* and is something you really have no control over).

Quiz 28

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. An air quality monitor gives a warning when the air quality is unhealthy for sensitive groups. The situation can be stated in terms of hypotheses as follows.

H_0 : The air quality is good.

H_a : The air quality is poor.

Which of the following best describes the power of the test?

- (A) The probability of a warning when the air quality is poor
- (B) The probability of a warning when the air quality is good
- (C) The probability of not giving a warning when the air quality is poor
- (D) The probability of not giving a warning when the air quality is good

2. Which of the following gives the probability of committing a Type I error?

- (A) The power
- (B) The standard error
- (C) The significance level
- (D) The p -value

3. If the significance level is 0.1, and the probability of a Type II error is 0.4, what is the resulting power?

- (A) 0.1
- (B) 0.5

- (C) 0.6
- (D) 0.9

4. A nutritionist conducts a study as to whether children prefer candy or toys for Halloween treats. The hypotheses are as follows.

H_0 : The probability a child chooses candy is 0.50.

H_a : The probability a child chooses candy is greater than 0.50.

Suppose the null hypothesis is false. What is the effect on power and the probability of a Type II error if the significance level is changed from 0.05 to 0.01?

- (A) The power of the test would increase, and the probability of a Type II error would decrease.
- (B) The power of the test would decrease, and the probability of a Type II error would increase.
- (C) Both the power of the test and the probability of a Type II error would increase.
- (D) Both the power of the test and the probability of a Type II error would decrease.

5. Which of the following is the best explanation of the power of a test?

- (A) The probability of correctly detecting an effect that exists
- (B) The probability of incorrectly detecting an effect when no effect exists
- (C) The probability of correctly not detecting an effect when no effect exists
- (D) The probability of detecting an effect whether or not one exists

6. A guidance counselor believes that students studying for an AP exam get less than 8 hours of sleep the night before the exam. The hypotheses are as follows:

$H_0: \mu = 8$

$$H_a: \mu < 8$$

Assume that the guidance counselor is correct that $\mu < 8$. For a fixed sample size and significance level, the power of the test will be greatest if the actual mean hours of sleep before the AP exam is which of the following?

- (A) 6
- (B) 7
- (C) 9
- (D) 11

7. A better business agency claims that over 20 percent of merchants favor eliminating the penny from circulation. To test this claim at a 5 percent significance level, a random sample of n merchants will be interviewed. The hypotheses are as follows.

H_0 : Twenty percent of merchants favor eliminating the penny from circulation.

H_a : More than twenty percent of merchants favor eliminating the penny from circulation.

Which of the following would result in the greatest power for this test?

- (A) A sample size of 300 and a true proportion of 40 percent
- (B) A sample size of 500 and a true proportion of 20 percent
- (C) A sample size of 500 and a true proportion of 40 percent
- (D) A sample size of 700 and a true proportion of 40 percent

8. Suppose that your friend claims that he can shoot free throws at a 70 percent rate. You don't believe he is that good and decide to test him. The hypotheses are as follows.

$$H_0: p = 0.70$$

$$H_a: p < 0.70$$

Suppose his real rate is 65 percent. Which of the following would result in the highest power for this test?

- (A) A significance level of 0.01 and 100 shots
- (B) A significance level of 0.01 and 200 shots
- (C) A significance level of 0.05 and 200 shots
- (D) A significance level of 0.05 and 300 shots

Free-Response Question

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. The amount of active ingredient per tablet of an advertised medication has an approximately normal distribution with mean 81 mg and standard deviation 0.3 mg. To pass quality inspection, tablets should have between 80.5 and 81.5 mg of active ingredient.

- (a) What is the probability that a single selected tablet passes inspection?

As each shipment arrives, an inspector randomly chooses five tablets from that shipment. She finds the mean active ingredient in the five tablets and rejects the entire shipment if the sample mean is less than 80.5 mg or greater than 81.5 mg.

- (b) Describe the sampling distribution of the sample mean for samples of size five.
 - (c) What is the probability that a Type I error is committed?

Suppose the true mean active ingredient in a particular shipment is 80.4 mg with a standard deviation of 0.3 mg.

- (d) What is the probability a Type II error is committed for this shipment?

The answers to this Quiz are found beginning on [page 508](#).

Confidence Intervals Versus Hypothesis Tests

A question sometimes arises as to whether a problem calls for calculating a confidence interval or performing a hypothesis test. Generally:

- An estimate of an unknown population parameter indicates a confidence interval.
- A claim about a population parameter indicates a hypothesis test.
- Asking how much or how effective requires a confidence interval calculation.
- Asking whether or not there is evidence requires a hypothesis test.

Sometimes it is possible to conduct a hypothesis test by constructing a confidence interval of the parameter and then checking whether or not the null hypothesis value is in the interval. For example:

- If a 95% confidence interval for μ captures μ_0 , there is not sufficient evidence to reject $H_0 : \mu = \mu_0$ in a two-sided test with $\alpha = 0.05$.
- If a 95% confidence interval for μ does not capture μ_0 , there is sufficient evidence to reject $H_0 : \mu = \mu_0$ in a two-sided test with $\alpha = 0.05$.

There have also been AP free-response questions where part (a) calls for a confidence interval calculation and part (b) asks if this calculation provides evidence of a claim about a population parameter. We can justify a claim based on a confidence interval only if all the values in the confidence interval are consistent with the claim.

Although alternative approaches, such as confidence intervals versus hypothesis tests, are accepted on the AP exam, these do require very special care in properly stating hypotheses, in dealing with one-sided versus two-sided questions, and in how standard deviations are calculated in problems involving proportions. So the recommendation is to avoid the alternative approach and instead conduct the more clearly indicated procedure.

Example 4.15

Suppose it is reported that random samples of 3-point shots by basketball players Stephen Curry and Michael Jordan show a 43% rate for Curry and a 33% rate for Jordan.

- (a) Are these numbers parameters or statistics?
- (b) State appropriate hypotheses for testing whether the difference is statistically significant.
- (c) Suppose the sample sizes were both 100. Find the z -statistic and the p -value, and give an appropriate conclusion at the 5% significance level.
- (d) Calculate and interpret a 95% confidence interval for the difference in population proportions.
- (e) Are the test decision and confidence interval consistent with each other?
- (f) Repeat (c), (d), and (e) as if the sample size had been 200 for each.

Solutions

- (a) These two numbers are statistics because they describe samples, not all 3-point shots ever taken by these two players.
- (b) $H_0: p_{\text{Curry}} = p_{\text{Jordan}}$ and $H_a: p_{\text{Curry}} \neq p_{\text{Jordan}}$
- (c) Calculator software gives $z = 1.46$ and $P = 0.145$. With this large of a p -value, $0.145 > 0.05$, there is not sufficient evidence to reject H_0 ; that is, there is not convincing evidence of a difference in the true 3-point percentage rates of Curry and Jordan.
- (d) Calculator software gives $(-0.034, 0.234)$. We are 95% confident that the true difference in 3-point percentage rates (Curry minus Jordan) is between -3.4% and 23.4% .
- (e) Yes, they are consistent. We did not conclude that the two percentages differ, and the confidence interval (for the difference in population proportions) includes the value zero.
- (f) With a sample size of 200, calculator software gives $z = 2.06$ and $P = 0.039$. With this small of a p -value, $0.039 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence of a difference in the true 3-point percentage rates of Curry and Jordan. With a sample size of 200, calculator software gives a confidence interval of $(0.005, 0.195)$. We are 95% confident that the true

difference in 3-point percentage rates (Curry minus Jordan) is between 0.5% and 19.5%. This interval does not include zero, so it is convincing evidence of a difference in the true 3-point percentage rates of Curry and Jordan. This is again consistent with the hypothesis test.

TIP

Sample size plays a substantial role in inferential statistics!

Summary

- Provided that conditions are met, the sampling distribution of sample means is approximately normal with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.
- Provided that conditions are met, the sampling distribution of the difference of sample means is approximately normal with mean $\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2$ and standard deviation $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.
- Three conditions must be met and must be checked for inference on a population mean:
 - **The randomization condition:** The data should be collected using a random sample.
 - **The 10% condition:** When sampling without replacement, the sample size n should be no larger than 10% of the population.
 - **The normality condition:** The population distribution should be approximately normal, or the sample size n should be ≥ 30 , or the sample distribution should be free of both strong skewness and outliers.
- For inference on the difference of population means, these conditions are the following:
 - **The randomization condition:** The data should be collected using random samples, and the samples must be taken independently of

each other.

- **The 10% condition:** When sampling without replacement, the sample sizes n_1 and n_2 should be no larger than 10% of the respective populations.
 - **The normality condition:** The population distributions should both be approximately normal, or the sample sizes should both be ≥ 30 , or the sample distributions should both be free of strong skewness and outliers.
- When the population standard deviation is unknown (which is almost always the case):
- Use $SE(\bar{x}) = \frac{s}{\sqrt{n}}$ as an estimate of $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.
 - Use $SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ as an estimate of $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.
 - Use a t -distribution with $df = n - 1$ for inference on a mean.
 - Use a t -distribution where df can be found using technology for inference on the difference of two means.
- When scoring a confidence interval free-response question on the exam, readers will look for whether you:
1. Identify the procedure (such as one-sample t -interval for a population mean) and check the conditions.
 2. Compute a proper interval (it is sufficient to simply copy the interval from your calculator calculation).
 3. Interpret the interval in context (the conclusion must mention the confidence level and the parameter and refer to the population, not to the sample).
- When asked to justify a claim based on a confidence interval:
1. If all the values in the confidence interval are consistent with the claim, then the confidence interval gives convincing evidence in support of the claim.
 2. If even a single value in the confidence interval is not consistent with the claim, then the confidence interval does not give convincing evidence in support of the claim.

- When scoring a hypothesis test free-response question on the exam, readers will look for whether you:
 1. State the hypotheses (both H_0 and H_a) and define the parameter of interest. (The hypotheses must refer to the population, not to the sample. The alternative hypothesis must be stated in terms of a strict inequality.)
 2. Identify the procedure (such as one-sample t -test for a population mean) and check the conditions.
 3. Compute the test statistic, p -value, and degrees of freedom if appropriate. (It is sufficient to simply copy all of these from your calculator calculation.)
 4. Give a conclusion in context with linkage to the p -value. (The conclusion must refer to the parameter and to the population, not to the sample.)

NOTE

You do not need to interpret the p -value unless specifically asked to do so.

- The conclusion to a hypothesis test should start with a comparison between the p -value and a significance level.
 1. With small p -values—that is, when the p -value $\leq \alpha$ —the test statistic is unlikely to have occurred by random chance alone. We have sufficient evidence to reject H_0 , and there is convincing evidence in support of H_a .
 2. With large p -values—that is, when the p -value $> \alpha$ —the test statistic is likely to have occurred by random chance alone. We do not have sufficient evidence to reject H_0 , and there is not convincing evidence in support of H_a .

NOTE

For paired data, be sure to indicate the direction of the difference.

- To distinguish between matched pair versus two-sample inference procedures:
 1. If the data come from separate (independent) random samples or are obtained from randomly assigned treatment groups in an experiment, then a two-sample inference procedure is appropriate.
 2. If the data come from two measurements taken from a pairing or from a single individual completing two tasks or if they are obtained from a matched pairs experiment, then a matched pair (one-sample) inference procedure is appropriate.
- A general conclusion about power is that the power will increase (and the probability of a Type II error will decrease) if any of the following occur:
 1. Significance level, *alpha*, increases.
 2. Standard error decreases.
 3. Sample size, *n*, increases.
 4. True parameter is farther from the null.
- When our learned test procedures do not apply, we may be able to proceed with a simulation.

5

Regression Analysis

(4–8 multiple-choice questions on the AP exam)

Learning Objectives

In this unit, you will learn how

- To be able to describe bivariate data from a scatterplot using terminology, such as direction (positive or negative), form (linear or nonlinear), strength (such as weak, moderate, or strong), and unusual features, and always do so in context.
- To be able to find the equation of a least squares regression line from generic computer regression output.
- To be able to interpret the slope and y-intercept of a linear regression line in context, using nondeterministic language.
- To be able to calculate and interpret a residual.
- To be able to interpret the coefficient of determination in context.

Two Quantitative Variables

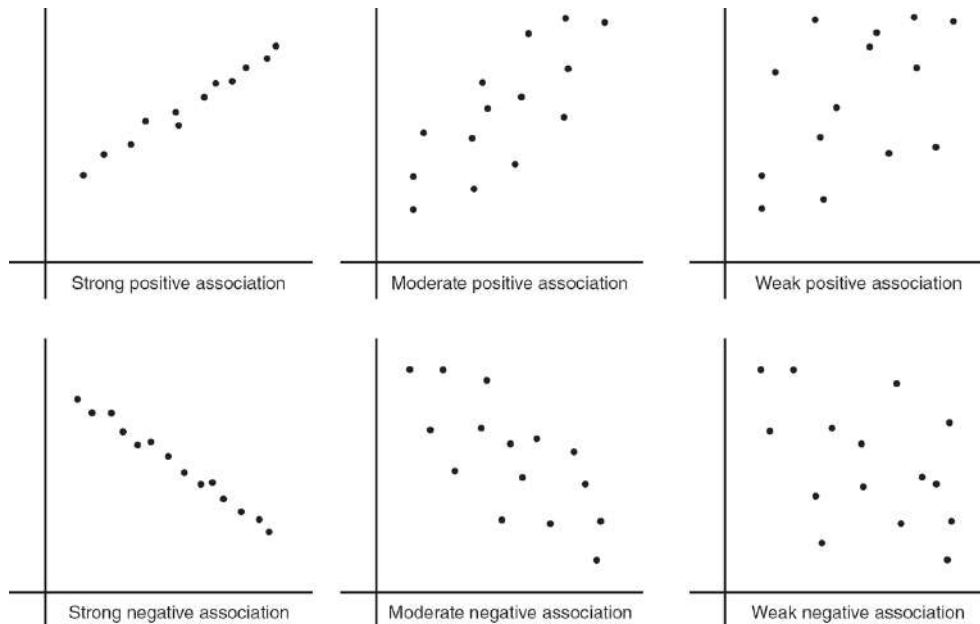
Many important applications of statistics involve examining whether two quantitative (numerical) variables are related to one another. For example, we may want to know if there is a relationship between the smoking histories of pregnant women and the birth weights of their children? Between SAT scores and success in college? Between amount of fertilizer used and amount of crop harvested?

Several questions immediately arise:

1. How can the strength of an apparent relationship be measured?
 - The strength of an apparent linear relationship is typically described as weak, moderate, or strong.
 - A *scatterplot* gives an immediate visual impression of the strength of a linear relationship. The strength of a linear association can be gauged by how close the plotted points are to a straight line.
 - A numerical measurement, r , which is called the *correlation coefficient* (or simply the *correlation*), gives a quantitative value of the strength of a linear relationship. The strength of a linear association can be gauged by how close $|r|$ is to 1.
 - However, strength of a relationship is not evidence of *causation*.
2. How can an observed relationship be put into functional terms?
 - The equation of a *best-fitting straight line* will usually be derived from generic computer regression output.
 - The interpretation of the slope and y-intercept of a linear regression line in context, using *nondeterministic* language (such as “predicted,” “estimated,” “expected,” or “average”), is imperative.
3. How well does the linear model explain the observed outcomes?
 - The *coefficient of determination*, r^2 , is the proportion of variation in the observed values that is explained by the linear model.
4. When is an association positive or negative?
 - When larger values of one variable are associated with larger values of a second variable, the variables are called *positively associated*. In this case, both the slope and the correlation will be

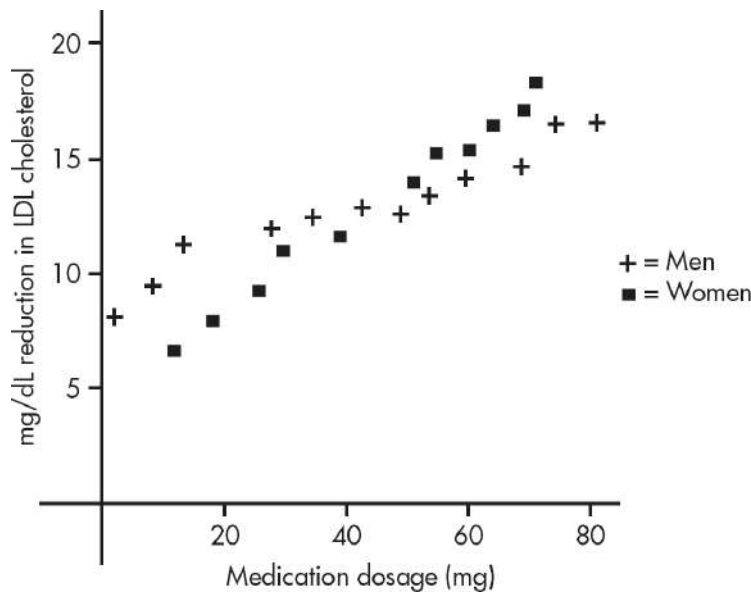
positive.

- When larger values of one variable are associated with smaller values of a second variable, the variables are called *negatively associated*. In this case, both the slope and the correlation will be negative.
- However, no matter whether the variables are positively or negatively associated, for any two points, anything is possible and either point could be higher or lower than the other.



Sometimes different dots in a scatterplot are labeled with different symbols or different colors to show a categorical variable. The resulting labeled scatterplot might distinguish, for example, between men and women, or between stocks and bonds, and so on.

➤ Example 5.1



NOTE

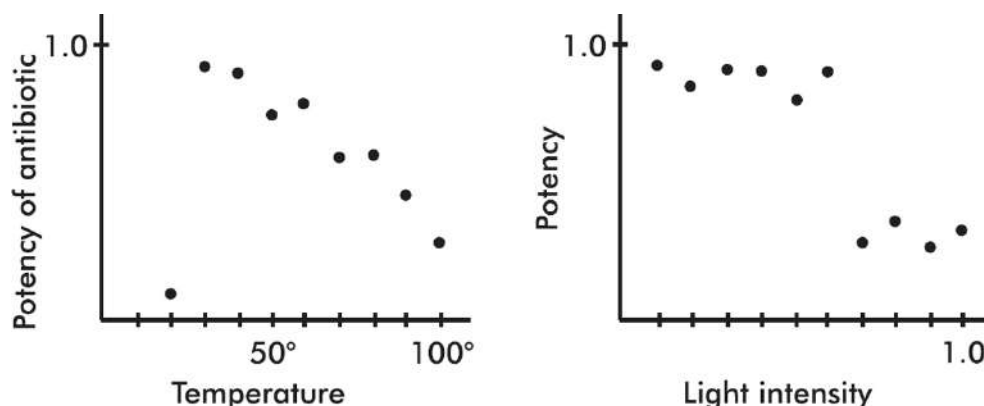
Be careful about *extrapolation* beyond the observed x -values. Although one should not ask what this data set predicts for a medication dosage of 200 mg, it is appropriate to ask the prediction of a 70 mg dosage.

This diagram is a labeled scatterplot distinguishing men with plus signs and women with square dots to show a categorical variable, gender. The plot indicates that higher doses of the medication lead to greater reductions in LDL (bad cholesterol) for both men and women. However, lower doses of the medication help men more than women in lowering LDL, while higher doses help women more than men.

When analyzing the overall pattern in a scatterplot, it is also important to note *clusters* and *outliers*.

Example 5.2

An experiment was conducted to determine the effect of temperature and light on the potency of a particular antibiotic. One set of vials of the antibiotic was stored under different temperatures but under the same lighting, while a second set of vials was stored under different lightings but under the same temperature.



TIP

Notice when the data fall in distinct groups.

In the first scatterplot, notice the roughly linear pattern with one outlier far outside this pattern. A possible explanation is that the antibiotic is more potent at lower temperatures but only down to a certain temperature, at which it drastically loses potency.

In the second scatterplot, notice the two clusters. It appears that below a certain light intensity the potency is one value, while above that intensity it is another value. In each cluster, there seems to be no association between intensity and potency.

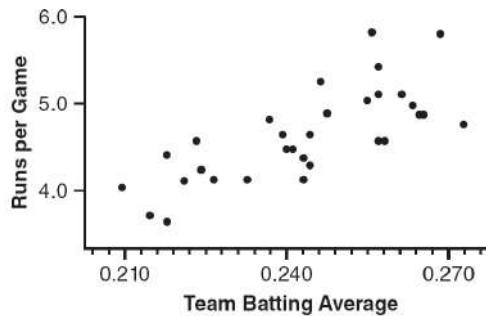
As noted in the above examples, when asked to describe a scatterplot, you must consider:

- *form* (linear or nonlinear)
- *direction* (positive or negative)
- *strength* (weak, moderate, or strong)
- *unusual features* (such as outliers and clusters)
- *context* (context must be mentioned somewhere in the scatterplot description)

TIP

Example 5.3

The following is a scatterplot of average runs per game versus team batting average for all Major League teams during a recent season.



Describe the scatterplot.

Solution

The scatterplot of average runs per game versus team batting average for all Major League teams during this season has form: linear; direction: positive; and strength: moderate.

Correlation

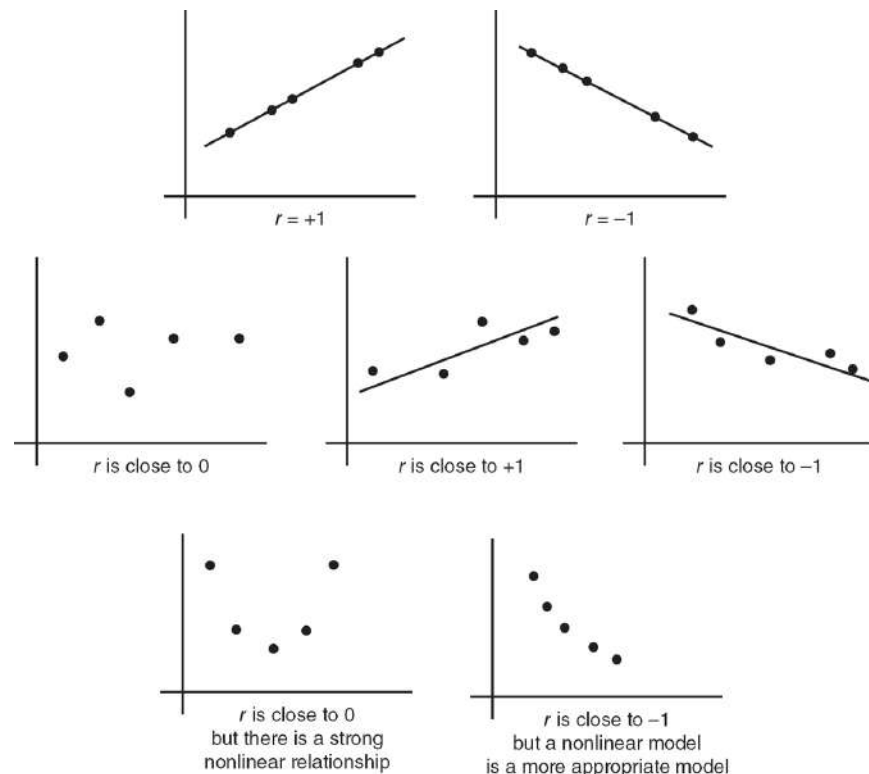
Although a scatterplot usually gives an intuitive visual indication when a linear relationship is strong, in most cases it is quite difficult to visually judge the specific strength of a relationship. For this reason, there is a mathematical measure called *correlation coefficient* (or simply the *correlation*). Correlation measures strength and gives the direction of a linear relationship. However, you should establish a reasonable case for linearity (eyeball the scatterplot) before calculating the correlation r . Linearity is a prerequisite, not a conclusion.

As important as correlation is, we always need to keep in mind that significant correlation does not necessarily indicate causation. For example, there is a very strong correlation between ice cream sales and frequency of forest fires! (As ice cream sales rise, in general so does temperature, and with higher temperatures, there are more forest fires.) Furthermore, correlation measures the strength only of a linear relationship.

Remember these important points about correlation:

- The value of r always falls between -1 and $+1$.
- $r < 0$ indicates a negative association, and $r > 0$ indicates a positive association.
- A value of $r = 0$ indicates that there is no *linear* association, while a value of $r = 1$ or $r = -1$ indicates that there is a perfect linear association.
- A correlation at or near 0 does not mean there is not a relationship between the variables; there may still be a strong *nonlinear* relationship.
- A correlation close to -1 or $+1$ does not necessarily mean that a linear model is the most appropriate model.
- The correlation r itself is unit-free.
- Changing the units in either the x - or y -variable will not change r .
- Correlation does not distinguish between which variable is called x and which is called y , so interchanging the variables (on the scatterplot) will not change the value of r .

- Correlation does not imply causation!



The *coefficient of determination*, r^2 , is the proportion of variation in the response variable y that is explained by the linear relationship with the explanatory variable x . That is,

$$r^2 = \frac{\text{variability explained by the model}}{\text{total variability in the } y\text{-values}}. \text{ Important points are as follows:}$$

- The coefficient of determination takes on values between 0.0 and 1.0, or equivalently between 0% and 100%.
- If $r^2 = 100\%$, then all the data points fall on a straight line, and thus all the variation in the response variable y is explained by the linear relationship with the explanatory variable x . In other words, all the variation in the response variable y is predictable by variation in the explanatory variable x .
- As a formula, $r^2 = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$, but you will not have to use this formula as r^2 will almost always be given in generic computer output, usually as “R-Sq =.”
- When asked to interpret r^2 , fill in “___% of the variation in (the y -variable in context) is explained by the linear relationship with (the x -variable in context).”
- When calculating r from r^2 , the square root function on your calculator always gives the positive root; however, r may be positive or negative. To decide whether r is positive or negative, remember that r will always take the same sign as the slope.
- In generic computer output, do not confuse “R-Sq =,” which gives our desired r^2 , with “R-Sq(adj) =,” which refers to a multiple regression topic that is not covered in this course. You will be marked wrong if you use the value of “R-Sq(adj).”
- Note that although a correlation, r , of 0.6 is twice a correlation, r , of 0.3, the corresponding r^2 of 36% is *four* times the corresponding r^2 of 9%.
- Note that a high value for r^2 does NOT indicate causation between variables.

Example 5.4

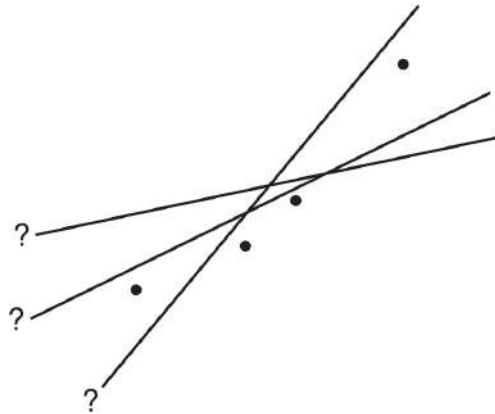
The correlation between Total Points Scored and Total Yards Gained for the 2025 season among a set of college football teams is $r = 0.84$. What information is given by the coefficient of determination?

Solution

$r^2 = (0.84)^2 = 0.7056$. Thus, 70.56% of the variation in Total Points Scored is explained by the linear relationship with Total Yards Gained.

Least Squares Regression

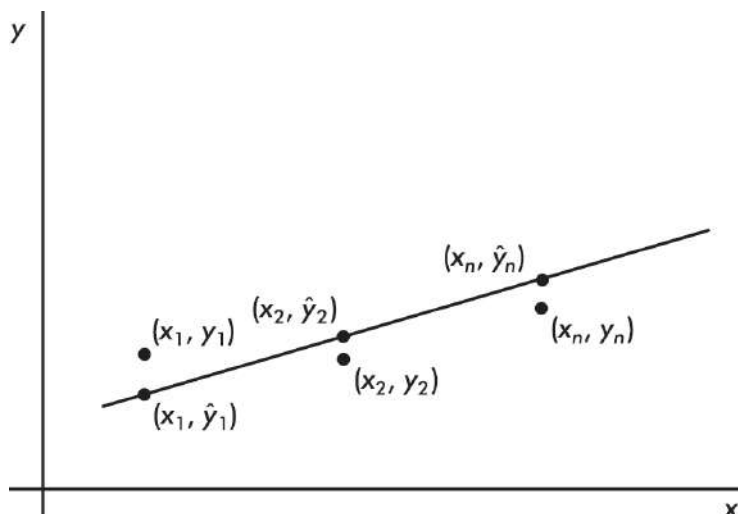
What is the best-fitting straight line that can be drawn through a set of points?



TIP

If a scatterplot indicates a nonlinear relationship, don't try to force a straight-line fit.

On the basis of our experience with measuring variances, by *best-fitting straight line* we mean the straight line that minimizes the sum of the squares of the vertical differences between the observed values and the values predicted by the line.



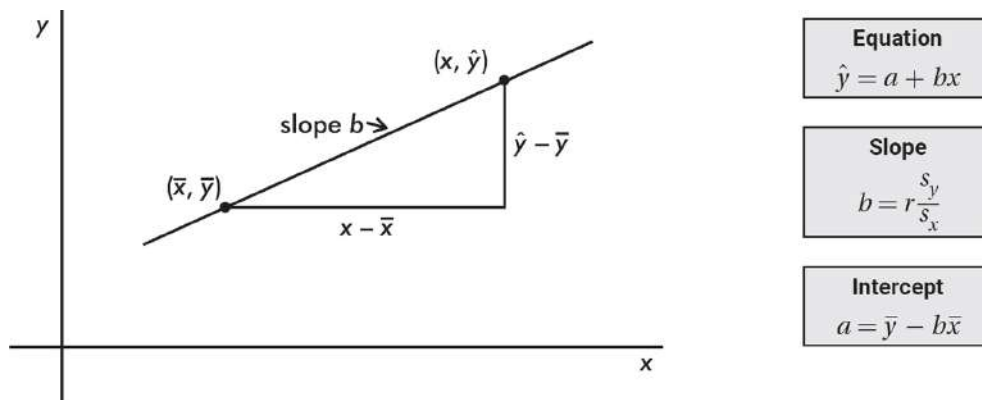
IMPORTANT

A variable with a hat ($\hat{y}$) is a predicted value of the variable.

That is, in this figure, we wish to minimize

$$(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \cdots + (y_n - \hat{y}_n)^2$$

It is reasonable, intuitive, and correct that the best-fitting line will pass through $(\bar{x}, \bar{y})$, where $\bar{x}$ and $\bar{y}$ are the means of the variables X and Y . The expression for a line with a given slope through a given point is as follows:



The slope is the amount that the predicted y -value changes for every unit increase in x . Similarly, the y -intercept is the predicted y -value for $x = 0$.

The slope b can be determined from the formula

$$b = r \frac{s_y}{s_x} = \frac{rs_y}{s_x}$$

where r is the correlation and s_x and s_y are the standard deviations of the two sets. That is, each standard deviation change in x results in a change of r standard deviations in $\hat{y}$. If you graph z -scores for the y -variable against z -scores for the x -variable, the slope of the regression line is precisely r , and in fact, the linear equation becomes $z_y = rz_x$.

This best-fitting straight line—that is, the line that minimizes the sum of the squares of the differences between the observed values and the values predicted by the line—is called the *least squares regression line*, or simply the *regression line*. It can be calculated directly by entering the two data sets and using the statistics package on your calculator. However, on the exam you will almost always be provided generic computer output giving the intercept and slope.

Example 5.5

Consider the three points $(2, 11)$, $(3, 17)$, and $(4, 29)$. Given any straight line, we can calculate the sum of the squares of the three vertical distances from these points to the line. What is the smallest possible value this sum can be?

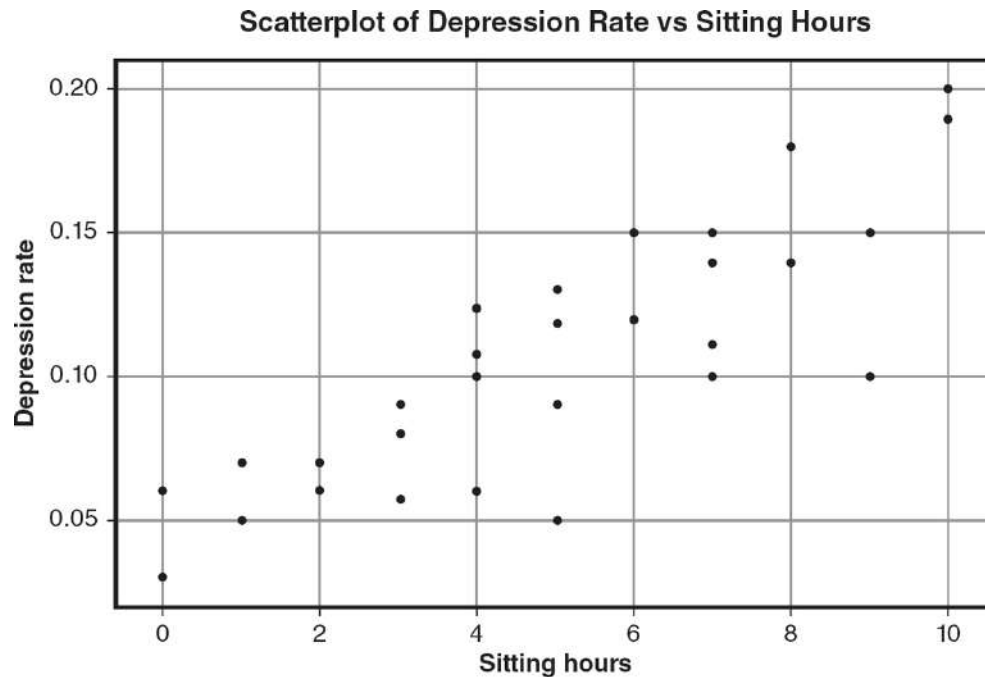
Solution

Calculator software gives the regression line to be $\hat{y} = -8 + 9x$. This regression line minimizes the sum of the squares of the vertical distances between the points and the line. In this case $(2, 10)$, $(3, 19)$, and $(4, 28)$ are on the line, so the minimum sum is $(10 - 11)^2 + (19 - 17)^2 + (28 - 29)^2 = 6$.

It is very important to be able to interpret generic computer output as this is the format in which most linear regression problems will appear.

➤ Example 5.6

An epidemiologist studied a random sample of participants, ages 35 to 70, as to the number of hours they sat per day versus whether or not they suffered from depression. The following is the resulting computer output.



[Follow for extended description](#)

Term	Coef	SE Coef	T-Value	P-Value
Constant	0.04174	0.00935	4.47	0.000
Sitting hours	0.01293	0.00163	7.95	0.000
	S	R-sq	R-sq(adj)	
	0.0248623	70.09%	68.98%	

- Identify the explanatory and response variables.
- Describe the scatterplot.
- What is the equation of the regression line?
- Interpret the slope in context.
- Interpret the y -intercept in context.
- Interpret the coefficient of determination in context.
- Predict the depression rate for someone who sits 4.6 hours per day.

✓ Solutions

- (a) The explanatory variable, X , is the number of hours someone sits per day. The response variable, Y , is the depression rate.
- (b) The relationship between the number of hours someone sits per day and depression rate appears to be linear, positive, and moderate to strong.
- (c) $\hat{y} = 0.04174 + 0.01293x$ where y = depression rate and x = sitting hours
Or: Predicted depression rate = $0.04174 + 0.01293$ (Sitting hours)
- (d) Each additional sitting hour leads to a *predicted* rise of 0.01293 in the depression rate.
- (e) For a person who sits 0 hours per day, the *predicted* depression rate is 0.04174.
- (f) 70.09% of the variation in depression rate is explained by the linear relationship with the numbers of sitting hours per day.
- (g) $0.04174 + 0.01293(4.6) = 0.101$, so conclude that people who sit for 4.6 hours per day are *predicted* to have a depression rate of 0.101.

► Example 5.7

A sociologist conducts a survey of 15 teens. The number of “close friends” and the number of times Facebook is checked every evening are noted for each student. Letting X and Y represent the number of close friends and the number of Facebook checks, respectively, gives:

Close friends: X	25	23	30	25	20	33	18	21	22	30	26	26	27	29	20
Facebook checks: Y	10	11	14	12	8	18	9	10	10	15	11	15	12	14	11

TIP

Be sure to label axes and show number scales whenever possible.

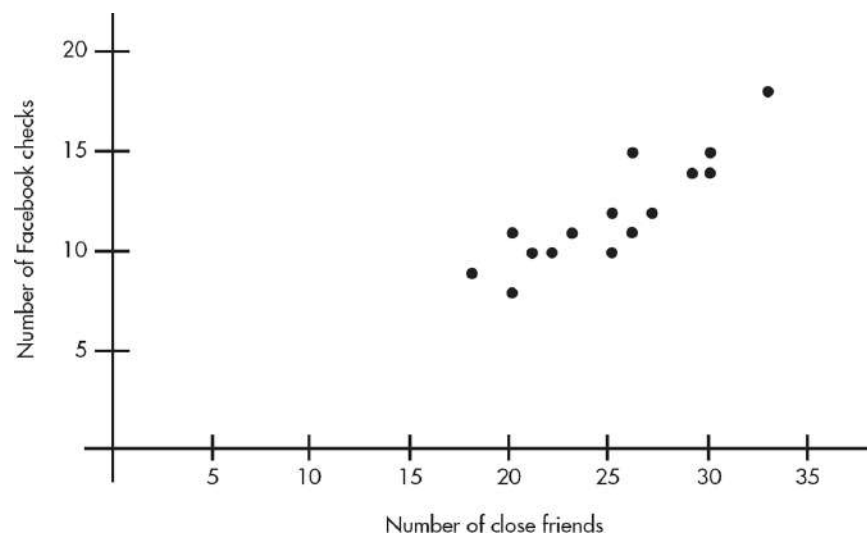
- (a) Identify the explanatory and response variables.
- (b) Draw a scatterplot.
- (c) Describe the scatterplot.
- (D) What is the equation of the regression line?
- (e) Interpret the slope in context.
- (f) Interpret the coefficient of determination in context.
- (g) Predict the number of evening Facebook checks for a student with 24 close friends.

NOTE

Although the overall pattern is *positive*, given any two points, it is still possible for the first point to have a larger x -value and a smaller y -value than the second point.

✓ Solutions

- (a) The explanatory variable, X , is the number of close friends, and the response variable, Y , is the number of evening Facebook checks.
- (b) Plotting the 15 points (25, 10), (23, 11), . . . , (20, 11) gives an intuitive visual impression of the relationship:



- (c) The relationship between the number of close friends and the number of evening Facebook checks appears to be linear, positive, and strong.

NOTE

The “hat” above a variable refers to the *predicted* value of that variable.

- (d) Calculator software gives $\hat{y} = -1.73 + 0.5492x$, where x is the number of close friends and y is the number of evening Facebook checks. We can instead write the following:

$$\text{Predicted Facebook checks} = -1.73 + 0.5492(\text{Close friends}) \text{ or}$$

$$\widehat{\text{Facebook checks}} = -1.73 + 0.5492(\text{Close friends})$$

- (e) The slope is 0.5492. Each additional close friend leads to a predicted increase of 0.5492 more evening Facebook checks.
- (f) Calculator software gives $r = 0.8836$, so $r^2 = 0.78$. Thus, 78% of the variation in the number of evening Facebook checks is explained by the linear relationship with the number of close friends.
- (g) $-1.73 + 0.5492(24) = 11.45$ evening Facebook checks. Students with 24 close friends are predicted to have 11.45 evening Facebook checks.

NOTE

We can also say that 22% of the variation in number of evening Facebook checks is *not* accounted for by the linear regression model.

Example 5.8

By using a respirometer while subjects exercise, scientists can calculate calories burned. The following are walking times (at 4 mph on a treadmill) and calories burned for a sample of six people weighing between 120 and 140 pounds.

Time (minutes): x	30	90	90	75	60	50
Calories burned: y	185	630	585	500	430	400

The following are the resulting computer outputs first using walking time as the independent variable and then using calories burned as the independent variable.

Term	Coef	SE Coef	T-Value	P-Value
Constant	23.8	46.6	0.51	0.636
Walking time	6.549	0.672	9.74	0.001
	S	R-sq	R-sq(adj)	
	35.7012	95.96%	94.95%	

Term	Coef	SE Coef	T-Value	P-Value
Constant	−0.83	7.18	−0.12	0.914
Calories burned	0.1465	0.0150	9.74	0.001
	S	R-sq	R-sq(adj)	
	5.33966	95.96%	94.95%	

NOTE

While the slope of the regression line will always have some meaning in context, the y -intercept may or may not be meaningful.

For those under 30, RNA viral diseases are no fun, but are not typically deadly. However, above age 70, RNA viral diseases are very dangerous. One study produced the following statistics:

Ages of RNA viral disease victims: $\bar{x} = 52.5$, $s_x = 29.96$

Probability of dying from the RNA viral disease: $\bar{y} = 0.0465$, $s_y = 0.0580$

Correlation $r = 0.817$

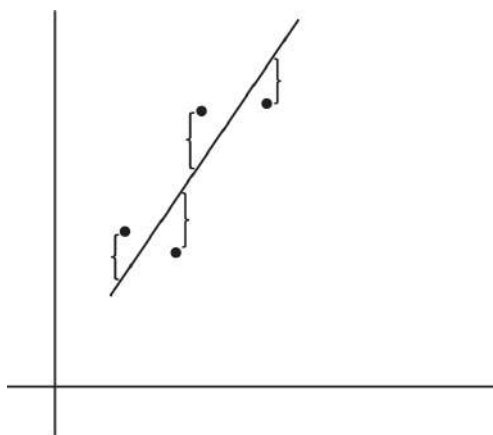
- (a) Find the slope of the least squares regression line and interpret it in context.
- (b) Find the y -intercept of the least squares regression line and interpret it in context.
- (c) What is the equation of the least squares regression line with age as the explanatory variable?
- (d) Based on this study, what is the predicted probability of dying for a 70-year-old with an RNA viral disease?

✓ Solutions

- (a) The slope is $b = r \frac{s_y}{s_x} = 0.817 \left(\frac{0.0580}{29.96} \right) = 0.001582$. As age increases by one year, the model predicts that the probability of dying from an RNA viral disease will increase by an average of 0.001582.
- (b) The y -intercept $= \bar{y} - b\bar{x} = 0.0465 - 0.001582(52.5) = -0.03656$. This refers to a predicted probability when $x = 0$ and is meaningless in this context. Probabilities cannot be negative, and it is unclear what an age of 0 years might refer to.
- (c) Predicted probability of dying $= -0.03656 + 0.001582(\text{Age})$
- (d) The predicted probability of dying for a 70-year old with an RNA viral disease is $-0.03656 + 0.001582(70) = 0.074$.

Residuals

The difference between an observed and a predicted value is called the *residual*. When the regression line is graphed on the scatterplot, the residual of a point is the vertical distance the point is from the regression line.

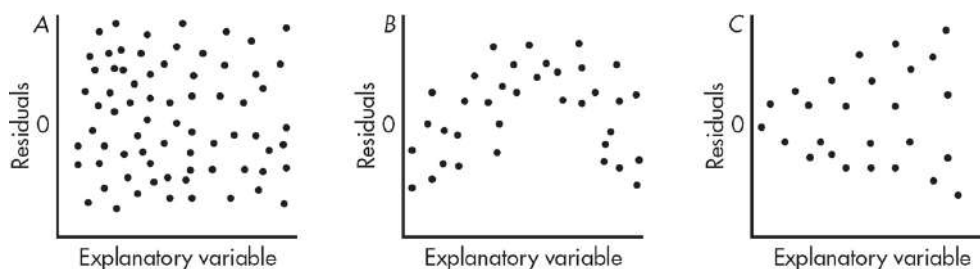


Important points about *residuals* are as follows:

- Order is important: $\text{residual} = y - \hat{y} = \text{observed } y \text{ minus predicted } y$.
- When the data point is above the regression line, the residual is positive; a data point below the regression line gives a negative residual.
- A positive residual means the linear model *underestimated* the actual response value, while a negative residual means the linear model *overestimated* the actual response value.
- The sum, and thus the mean, of the residuals is always zero.
- The regression line is the line that minimizes the sum of the *squares* of the residuals.
- The standard deviation of the residuals gives a measure of how the points are spread around the regression line. In computer output, the standard deviation of the residuals is typically given as “S =.” (The standard deviation of the residuals is helpful for your understanding, but you will not be tested on this topic.)
- To obtain a residual plot, the residuals can be plotted against either the x -values or the $\hat{y}$ -values.
- A residual plot with a definite pattern is an indication that a nonlinear model will show a better fit to the data than the straight regression line.
- Understand that although a linear model may be appropriate, it may be weak with a low r^2 -value. Alternatively, a linear model may not be the best model (as evidenced by the residual plot), but it still might be a very good model with high r^2 -value.

► Example 5.10

What do each of the following residual plots indicate about the linear model?



✓ Solution

In *A*, there is no apparent pattern in the residual plot, indicating that a linear model is appropriate.

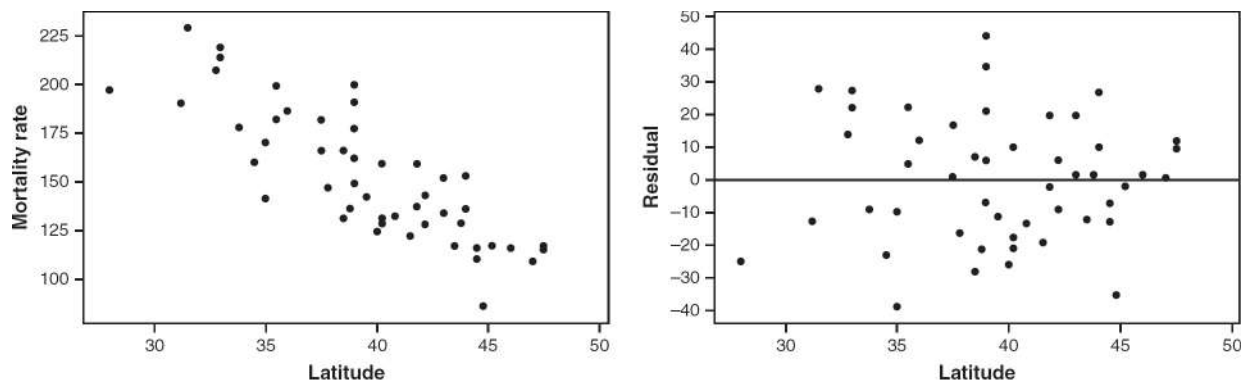
In *B*, the residual plot shows a strong pattern, indicating that a nonlinear model will be a better fit than a straight-line model.

In *C*, the residual plot shows “fanning,” indicating that the linear model gives a stronger fit for smaller x -values than for larger x -values.

The ability to interpret computer output is important not only to do well on the AP Statistics exam but also to understand statistical reports in the business and scientific worlds.

► Example 5.11

Does strength of the sun, as indicated by latitude, have an effect on skin cancer mortality? A study looked at 49 randomly selected regions in North America and noted the skin cancer mortality rate (deaths per 100,000 population) and latitude of the region. Calculating a least squares regression line gives the following computer output:



Predictor	Coef	SE Coef	T	P
Constant	389.19	23.81	16.34	0.000
Latitude	-5.9776	0.5984	-9.99	0.000
S = 19.115		R-Sq = 68.0% R-Sq(adj) = 67.3%		

[Follow for extended description](#)

- Identify the explanatory and response variables.
- Describe the relationship given by the scatterplot of Mortality rate and Latitude.
- Does it appear that a line is an appropriate model for the data? Explain.
- What is the slope of the regression line, and what does it signify?
- Is it reasonable to conclude a cause-and-effect relationship between skin cancer mortality and latitude (probably due to strength of the sun at different latitudes)?
- One of the regions was a location in Maryland with a latitude of 39 and a skin cancer mortality rate of 162 per 100,000. Calculate the residual for that point, and interpret it in context.
- Interpret the coefficient of determination in context.
- What is the correlation r ?
- Interpret the value “S = 19.115” in context.

Solutions

NOTE

Look in the box and notice in the first column the words “Constant” and “Latitude.” In the second column, to the right of “Constant,” is the number 389.19, which is the y -intercept of the regression line; to the right of “Latitude” (the explanatory variable) is the number -5.9776, which is the slope of the regression line.

- Latitude is the explanatory variable, and skin cancer mortality rate per 100,000 population is the response variable.
- There is a moderate, negative, linear relationship between latitude and skin cancer mortality rate.
- Yes, a line appears to be an appropriate model between latitude and skin cancer mortality rate because the scatterplot looks linear and there is no apparent pattern in the residual plot.
- The slope of the regression line is -5.9776 , signifying that each unit increase in latitude gives a predicted decrease of 5.9776 in the skin cancer mortality rate per 100,000 population.
- No, correlation does not imply causation.
- For a latitude of 39, the predicted skin cancer mortality rate is $389.19 + (-5.9776)(39) = 156.06$. The residual is actual minus predicted $= 162 - 156.06 = +5.94$. The observed skin cancer mortality rate is 5.94 more than what is predicted by the regression model.

- (g) 68.0% of the variation in skin cancer mortality rate is explained by the linear relationship with the latitude.
- (h) $r = \pm\sqrt{r^2} = -\sqrt{0.680} = -0.825$ where the negative square root is used because the slope is negative (the correlation and the slope always have the same sign).
- (i) The standard deviation of the residuals, $S = 19.115$, is a “typical value” of the residuals and gives a measure of how the data points are spread around the regression line. It is an estimate of the “typical” prediction error when using the least squares regression line. (Note: you will not be tested on this topic, the standard deviation of the residuals.)

Quiz 29

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. A rural college is considering constructing a windmill to generate electricity but is concerned over noise levels. A study is performed measuring noise levels (in decibels) at various distances (in feet) from the campus library, and a least squares regression line is calculated with a correlation of 0.74. Which of the following is a proper and most informative conclusion for an observation with a negative residual?
 - (A) The predicted noise level is greater than the measured noise level.
 - (B) The measured noise level is greater than the predicted noise level.
 - (C) The predicted noise level is 0.74 times the measured noise level.
 - (D) The slope of the regression line at that point must also be negative.
2. A college baseball coach tabulates the number of practice hours in a batting cage and the number of hits over the course of a season for a random sample of players. When the least squares regression analysis is performed, the correlation is 0.67. Which of the following is the correct way to label the correlation?
 - (A) 0.67
 - (B) 0.67 hits per hour
 - (C) 0.67 hours per hit
 - (D) 0.67 hit hours

Questions 3–5 refer to the following:

The relationship between winning game proportions when facing the sun and when the sun is on one’s back is analyzed for a random sample of 10 professional tennis players. The computer printout for regression is below:

Predictor	Coef	SE Coef	T	P
Constant	0.05590	0.02368	2.36	0.046
Facing	0.92003	0.03902	23.58	0.000
S = 0.0242922 R-Sq = 98.6% R-Sq(adj) = 98.4%				

[Follow for extended description](#)

3. What is the equation of the regression line, where “facing” and “back” are the winning game proportions when facing the sun and with back to the sun, respectively?

(A) $\widehat{\text{facing}} = 0.056 + 0.920(\text{back})$
(B) $\widehat{\text{back}} = 0.056 + 0.920(\text{facing})$
(C) $\widehat{\text{facing}} = 0.920 + 0.056(\text{back})$
(D) $\widehat{\text{back}} = 0.920 + 0.056(\text{facing})$

4. What is the correlation?

(A) -0.986
(B) 0.984
(C) 0.986
(D) 0.993

5. For one player, the winning game proportions were 0.55 and 0.59 for “facing” and “back,” respectively. What was the associated residual?

(A) -0.0488
(B) -0.028
(C) 0.028
(D) 0.0488

-
6. Data are obtained for a group of college freshmen examining their total SAT scores from their senior year of high school and their GPAs during their first year of college. The resulting regression equation is

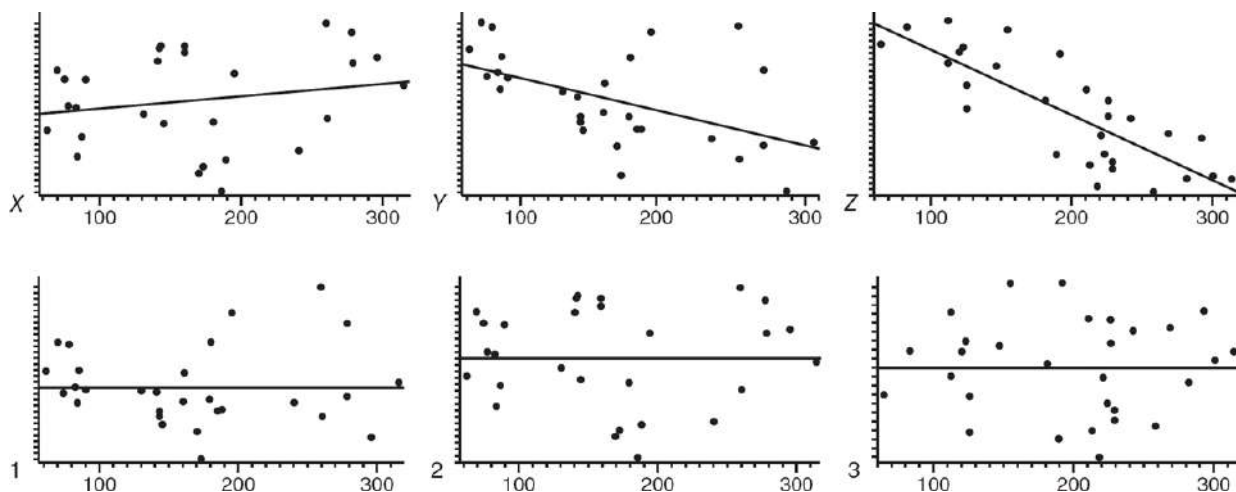
$$\widehat{\text{GPA}} = 0.55 + 0.00161(\text{SAT score}) \text{ with } r = 0.632$$

What percentage of the variation in GPAs can be accounted for by looking at the linear relationship between GPAs and SAT scores?

- (A) 0.161%
(B) 16.1%
(C) 39.9%
(D) 63.2%
7. Suppose the correlation between two variables is $r = 0.23$. What will the new correlation be if 0.14 is added to all values of the x -variable, every value of the y -variable is doubled, and the two variables are interchanged?
- (A) 0.23
(B) 0.74
(C) -0.23
(D) -0.74

8. A study of the number of highway deaths due to failure to wear seat belts in a year versus percentage seat belt usage in that year in 300 national regions shows a strong negative linear association. The least squares regression equation is: Predicted number of deaths = $11,100 - 305.1(\text{Belt usage})$. What does “negative” mean in this context?
- (A) If a given region has a lower percentage of seat belt usage than a second region, the given region will have a higher number of highway deaths due to failure to wear seat belts than the second region.
- (B) Regions with a higher percentage of seat belt usage tend to have lower numbers of highway deaths due to failure to wear seat belts.
- (C) If a region has a 1 percent gain in seat belt usage, then it will have a reduction of 305 highway deaths due to failure to wear seat belts, on average.
- (D) The correlation, r , between the number of highway deaths due to failure to wear seat belts and the percentage of seat belt usage is negative.
9. Consider the set of points $\{(2, 5), (3, 7), (4, 9), (5, 12), (10, n)\}$. What should n be so that the correlation between the x - and y -values is 1?
- (A) 21
- (B) 24
- (C) 25
- (D) No value for n can make $r = 1$.

10. The following are the graphs of three scatterplots, X , Y , Z , and of three residual plots, 1, 2, 3.



[Follow for extended description](#)

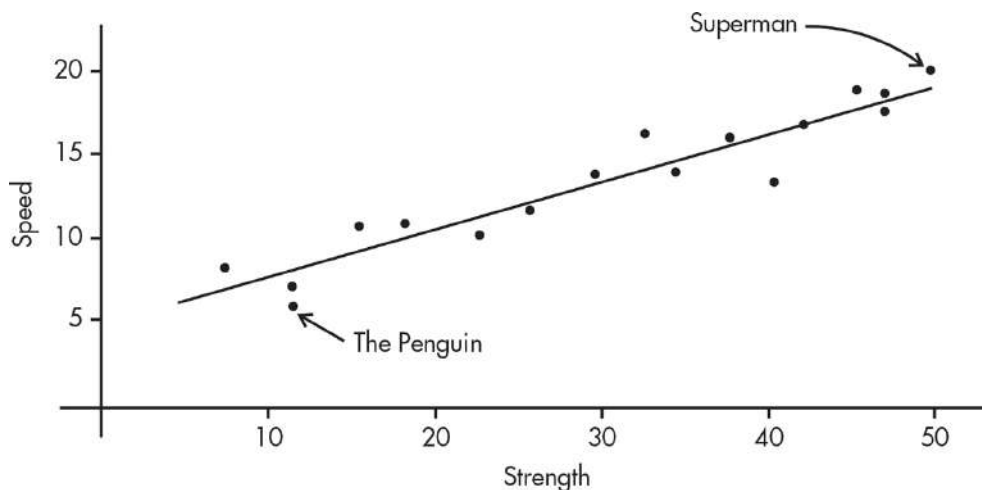
Which scatterplot corresponds to which residual plot?

- (A) X and 1, Y and 3, Z and 2
- (B) X and 2, Y and 1, Z and 3
- (C) X and 2, Y and 3, Z and 1
- (D) X and 3, Y and 1, Z and 2

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

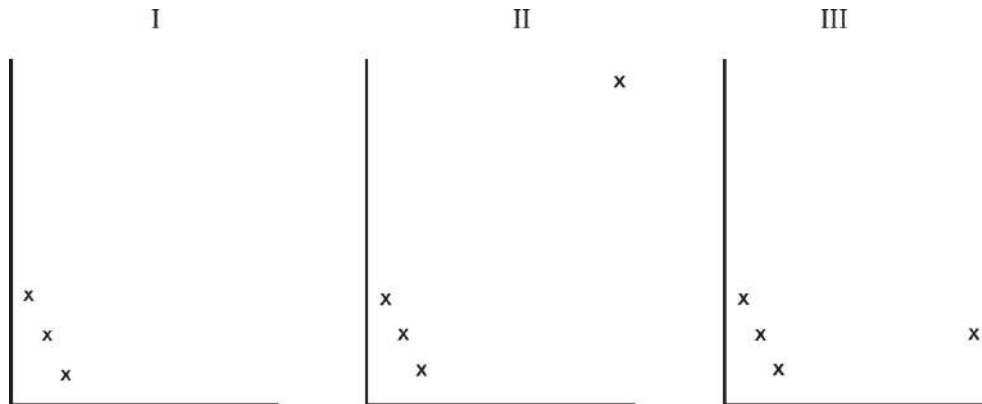
1. Comic book heroes and villains can be compared with regard to various attributes. The scatterplot below looks at speed (measured on a 20-point scale) versus strength (measured on a 50-point scale) for 17 such characters.



Computer software showing the results of fitting a straight line to the data by the method of least squares gives:

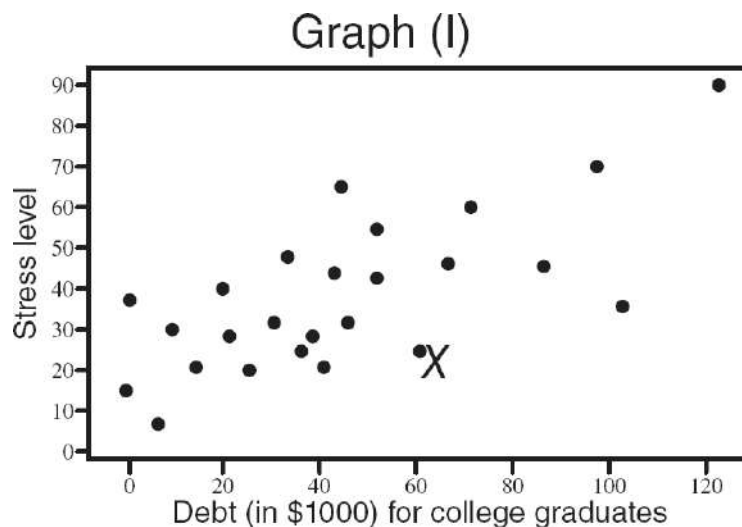
$$\text{Predicted speed} = 4.626 + 0.283(\text{Strength}) \quad R\text{-sq} = 84.28\%$$

- Find the correlation for the linear relationship between speed and strength.
 - What is the slope of the regression line, and what does it signify?
 - Given that the mean of the strength values is 30.47, what is the mean of the speed values?
 - Interpret the coefficient of determination in context.
 - Thanos has a strength score of 45 and a speed score of 18. Does the linear model underestimate or overestimate his speed and by how much?
 - How will the correlation change if the point (42, 16.5), which corresponds to Supergirl, is removed? Explain.
 - If the points corresponding to both Superman and the Penguin are removed, will the slope increase, decrease, or remain unchanged? Explain.
2. An outlier can have a striking effect on the correlation r . Consider the following three scatterplots:



- (a) How do the outliers in II and III affect the correlation from I?
 (b) Insert a fourth point in I such that the correlation doesn't change.

3. A continuing political debate concerns how to address the large student debt carried by college graduates. Graph (I) is a scatterplot showing the amount of student debt of 25 randomly selected college graduates and their stress level as self-reported on a 1–100 scale.

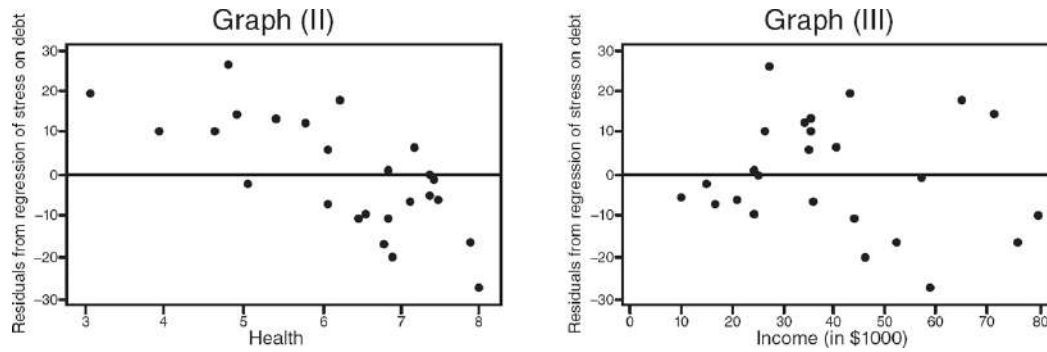


Computer output from linear regression gives:

Predictor	Coef	SE Coef
Constant	19.376	4.782
Debt	0.42640	0.08906

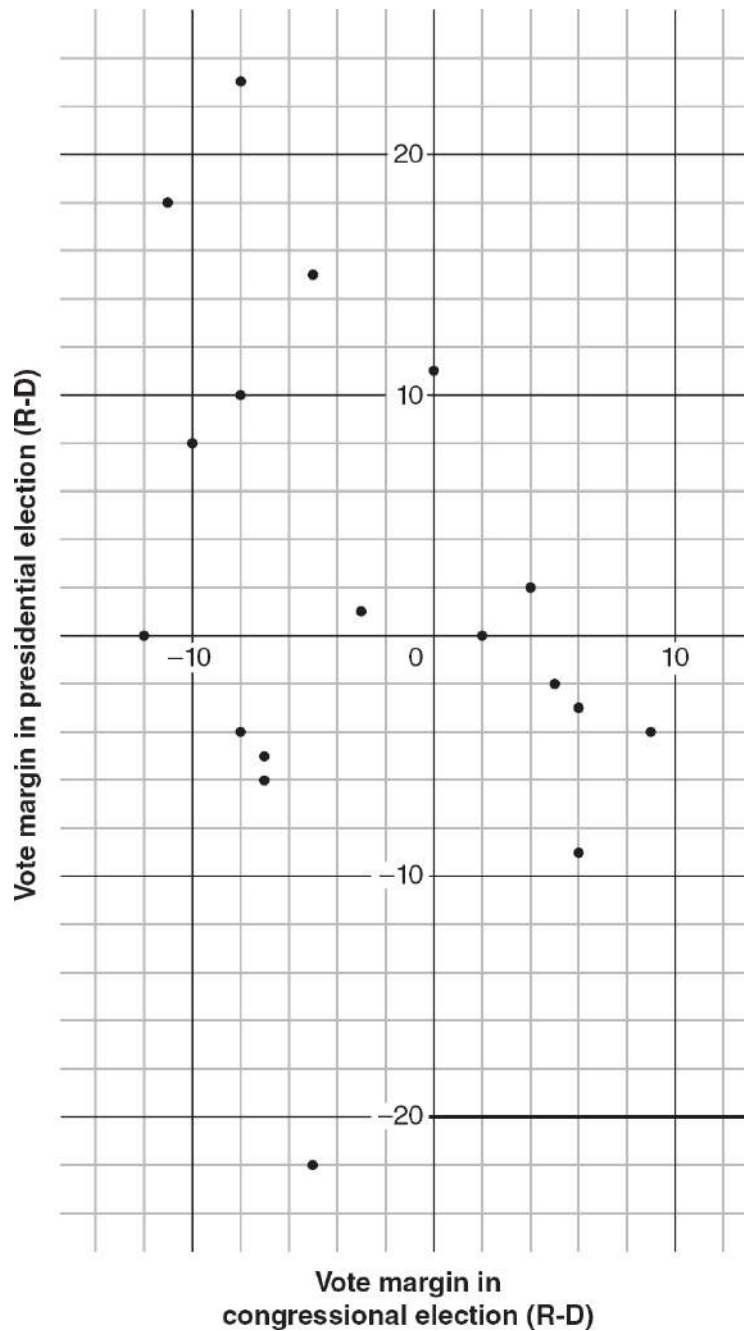
- (a) Describe the association between stress level and debt for this sample of 25 college graduates.
 (b) The point labeled *X* represents a student with \$60,000 debt and a stress level of 25. Calculate and interpret the residual of that point.

Two other possible explanatory variables with regard to stress level are income and health as measured on a 3 to 8 HRQoL (health-related quality of life) index. Graph (II) is a scatterplot of health plotted with the corresponding residuals from the regression of stress on debt. Graph (III) is a scatterplot of income plotted with the corresponding residuals from the regression of stress on debt.



(c) Which variable, health or income, in addition to debt should be used to improve the prediction of stress level? Explain.

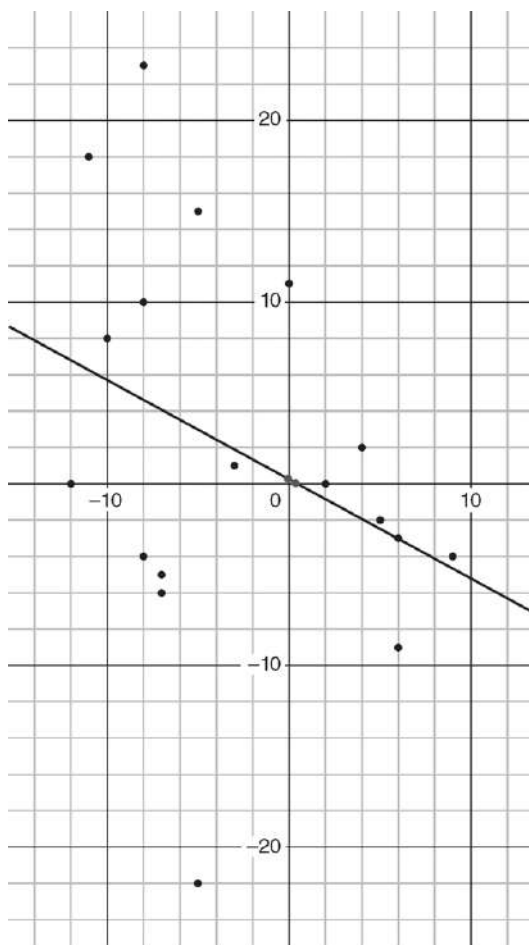
4. A researcher was interested in whether or not there is an association between House membership in midterm elections and House membership in the following presidential elections. Data were collected on the difference in the number of House members (Republicans minus Democrats) for midterm elections and the difference in number of House members (Republicans minus Democrats) for the following presidential elections for the years 1946 through 2020. The data are summarized in the following scatterplot.



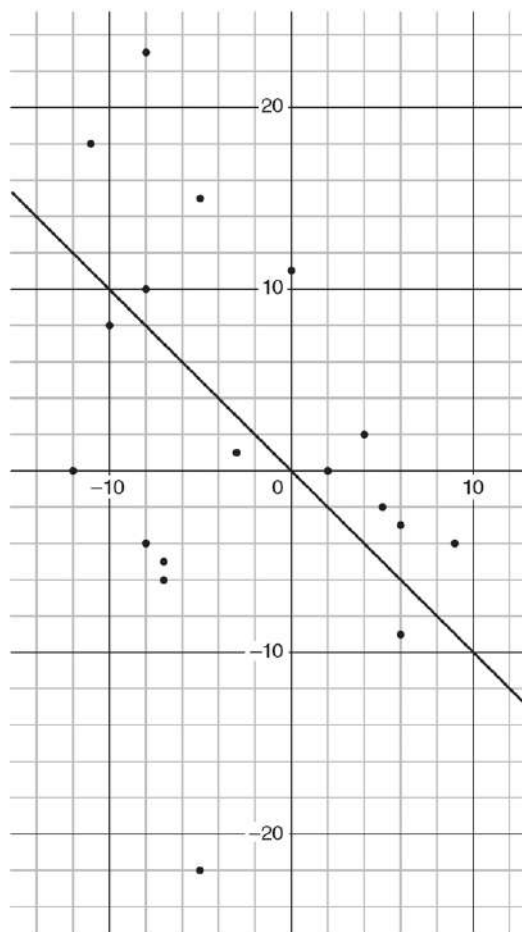
[Follow for extended description](#)

- (a) Based on the scatterplot, describe the relationship between the difference in House membership (Republicans minus Democrats) in midterm elections and the difference in House membership (Republicans minus Democrats) in the following presidential elections.

Let x represent the difference in number of House members (Republicans minus Democrats) for midterm elections, and let y represent the difference in number of House members (Republicans minus Democrats) for the following presidential elections. Graph 1 shows the data with the least squares regression line, $y = 0.257 - 0.546x$. Graph 2 shows the data with the line $y = -x$.



Graph 1



Graph 2

[Follow for extended description](#)

Consider the following criteria for classifying overall vote margins.

Republican majority	Balanced	Democrat majority
$x + y > 0$ ($y > -x$)	$x + y = 0$ ($y = -x$)	$x + y < 0$ ($y < -x$)

- For which graph, 1 or 2, is the line more helpful classifying overall vote margins? Explain.
- Using the answer from part (b), how many of the points in the scatterplot should be classified in each of the three categories?
- Using the best model for prediction, predict the difference in number of House members (Republicans minus Democrats) for the following presidential election if the difference in number of House members (Republicans minus Democrats) for a congressional election is $R - D = 7$ where R is Republicans and D is Democrats.

The answers for this quiz can be found beginning on [page 509](#).

[Quiz 30](#)

Multiple-Choice Questions

DIRECTIONS: The questions that follow are each followed by four suggested answers. Choose the response that best answers the question.

1. A study collects data on average total SAT scores and percentage of students who took the exam at 100 randomly selected high schools. The following is part of the computer printout for regression:

Variable	Coefficient	s.e. of coeff	t-ratio	prob
Constant	1176.32	12.65	92.99	≤ 0.0001
SAT	-2.84276	0.2461	-11.55	≤ 0.0001
R-squared = 76.5%		R-squared (adj) = 76.1%		

Which of the following is a correct conclusion?

- (A) The correlation is 0.875.
- (B) The y -intercept indicates the mean total SAT score if percent of students taking the exam has no effect on total SAT scores.
- (C) The r^2 -value indicates that the residual plot does not show a strong pattern.
- (D) Schools with lower percentages of students taking the exam tend to have higher average total SAT scores.
2. Suppose the correlation is negative. Given two points from the scatterplot, which of the following is possible?
- I. The first point has a larger x -value and a smaller y -value than the second point.
- II. The first point has a larger x -value and a larger y -value than the second point.
- III. The first point has a smaller x -value and a larger y -value than the second point.
- (A) I only
- (B) III only
- (C) I and III only
- (D) I, II, and III
3. Suppose the regression line for a set of data, $a + 3x$, passes through the point (2, 5). If $\bar{x}$ and $\bar{y}$ are the sample means of the x - and y -values, respectively, then $\bar{y} =$
- (A) $-1 + 3\bar{x}$
- (B) $-2 + \bar{x}$
- (C) $5 + \bar{x}$
- (D) $3\bar{x}$
4. Which of the following statements about correlation r is true?

- (A) A correlation of 0.2 means that 20% of the points are highly correlated.
- (B) Perfect correlation, that is, when the points lie exactly on a straight line, results in $r = 0$.
- (C) Correlation is not affected by which variable is called x and which is called y .
- (D) A correlation of 0.75 indicates a relationship that is 3 times as linear as one for which the correlation is only 0.25.

5. Which of the following statements about residuals from the least squares line are true?

- I. Residuals may be positive, negative, or zero.
- II. The linear regression line maximizes the sum of the squares of the residuals.
- III. A definite pattern in the residual plot is an indication that a nonlinear model will show a better fit to the data than the straight regression line.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II, and III

6. In a study of winning percentage in home games versus average home attendance for professional baseball teams, the resulting regression line is

$$\text{Predicted winning percentage} = 44 + 0.0003(\text{Attendance})$$

What is the residual if a team has a winning percentage of 55% with an average attendance of 34,000?

- (A) -11.0
- (B) -0.8
- (C) 0.8
- (D) 11.0

7. Suppose the correlation between two variables is -0.57 . If each of the y -scores is multiplied by -1 , which of the following is true about the new scatterplot?

- (A) It slopes up to the right, and the correlation is -0.57 .
- (B) It slopes up to the right, and the correlation is $+0.57$.
- (C) It slopes down to the right, and the correlation is -0.57 .
- (D) It slopes down to the right, and the correlation is $+0.57$.

8. A study of selling prices of homes in a southern California community (in \$1,000) versus size of the homes (in 1,000s of square feet) shows a moderate positive linear association. The least squares regression equation is Predicted selling price = $35.3 + 214.1(\text{Size})$. What does “linear” mean in this context?

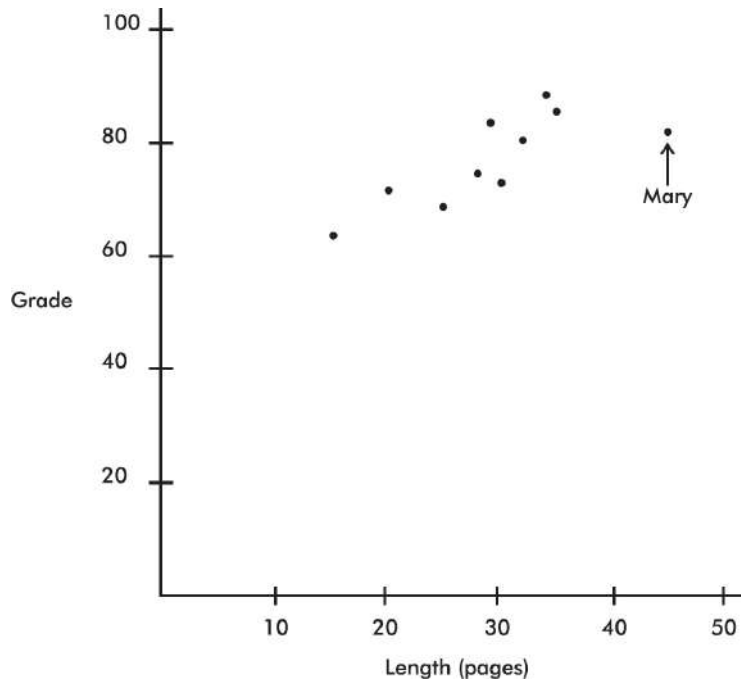
- (A) The points in the scatterplot line up in a straight line.
- (B) There is no distinct pattern in the residual plot.
- (C) As home size increases by 1,000 square feet, the selling price tends to change by a constant amount, on average.

- (D) Each increase of 1,000 square feet in home size gives an increase of 214.1(\$1,000) in selling price.
9. Does lower sun exposure, as measured by greater distance from the equator, result in lower incidence of skin cancer? A study of skin cancer mortality versus latitude of 100 Northern Hemisphere locales shows a strong negative linear association. What does “strong” mean in this context?
- (A) The actual incidence of skin cancer at a given latitude will be very close to what is predicted by a least squares model.
- (B) More sun exposure is associated with greater numbers of skin cancer deaths.
- (C) A locale’s incidence of skin cancer deaths has a linear association with its latitude.
- (D) A least squares model predicts that the greater the latitude, the lower the average incidence of skin cancer.
10. With regard to a least squares regression line, the sum of the residuals is always zero. Does it follow that the mean or the median of the residuals also must be zero?
- (A) Mean of the residuals: Yes, must be zero
Median of the residuals: Yes, must be zero
- (B) Mean of the residuals: Yes, must be zero
Median of the residuals: No, may or may not be zero
- (C) Mean of the residuals: No, may or may not be zero
Median of the residuals: Yes, must be zero
- (D) Mean of the residuals: No, may or may not be zero
Median of the residuals: No, may or may not be zero

Free-Response Questions

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your final answers.

1. The following scatterplot shows the grades for research papers for a sociology professor’s class plotted against the lengths of the papers (in pages).

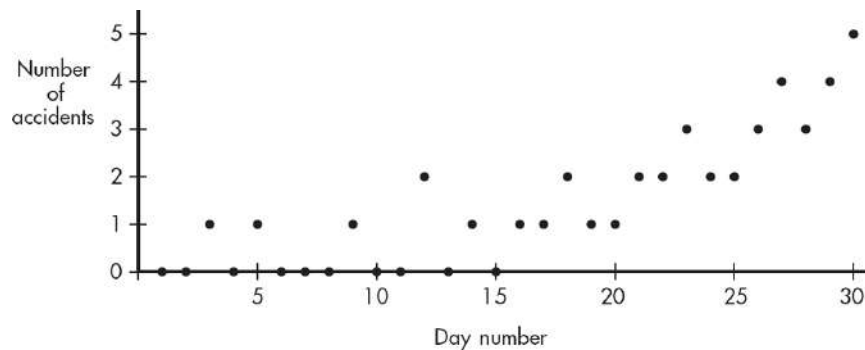


Mary turned in her paper late and was told by the professor that her grade would have been higher if she had turned it in on time. A computer printout fitting a straight line to the data (not including Mary's score) by the method of least squares gives:

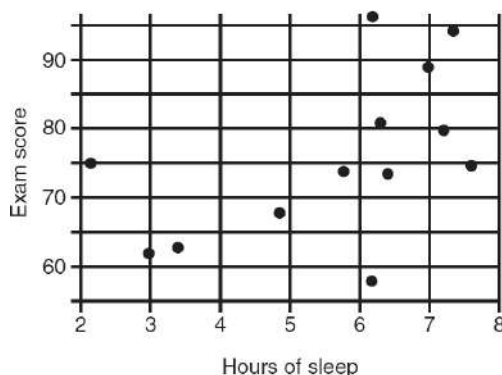
$$\text{Grade} = 46.51 + 1.106 \text{ Length}$$

$$R\text{-sq} = 74.6\%$$

- Find the correlation coefficient for the relationship between grade and length of paper based on these data (excluding Mary's paper).
 - What is the slope of the regression line and what does it signify?
 - How will the correlation coefficient change if Mary's paper is included? Explain your answer.
 - How will the slope of the regression line change if Mary's paper is included? Explain your answer.
 - Interpret the coefficient of determination in context.
2. A scatterplot of the number of accidents per day on a particular interstate highway during a 30-day month is as follows:



- (a) Draw a histogram of the frequencies of the number of accidents.
 - (b) Name a feature apparent in the scatterplot but not in the histogram.
 - (c) Name a feature clearly shown by the histogram but not as obvious in the scatterplot.
3. In a study of how exam scores are influenced by hours of sleep the night before the exam, a random sample of 13 students was outfitted with wristbands that tracked sleep time. The scatterplot of the resulting data follows.



- (a) Describe the association between exam score and hours of sleep for these 13 students.
 - (b) Describe the association between exam score and hours of sleep for the 8 students who received at least 6 hours of sleep the night before the exam.
 - (c) Explain how the appropriateness of using linear regression is critically influenced by one student's data.
4. Researchers studying the association between developing dementia and consumption of ultra-processed foods collected data on people 55 years of age or older who showed no signs of dementia at the beginning of the study. Participants filled out questionnaires about daily diet from which the researchers determined how much ultra-processed food was consumed daily on average. The participants were followed for 10 years at which time they were again evaluated for signs of dementia (on a 10-point scale). A scatterplot dementia score on the 10-point scale versus daily ultra-processed food consumed in grams revealed an association described as strong, linear, and positive.
- (a) Was this an experiment or an observational study? Explain.
 - (b) For the situation described above, explain what is meant by each of the following:
 - i. Strong
 - ii. Linear
 - iii. Positive
 - (c) Is it reasonable to conclude that consumption of ultra-processed foods is one cause of dementia? Explain.
 - (d) What would a positive residual mean in this context?
- The answers for this quiz can be found beginning on [page 510](#).

Summary

- A description of the association shown in a scatterplot should include:

- Form (linear or nonlinear)
 - Direction (positive or negative)
 - Strength (weak, moderate, or strong)
 - Unusual features (outliers, clusters)
 - Context!
- The correlation coefficient, r , summarizes the strength and direction of the linear association between two quantitative variables. Remember these important points about correlation:
- The value of r always falls between -1 and $+1$.
 - $r < 0$ indicates a negative association, and $r > 0$ indicates a positive association.
 - A value of $r = 0$ indicates that there is no *linear* association, while a value of $r = 1$ or $r = -1$ indicates that there is a perfect linear association.
 - A correlation at or near 0 does not mean there is not a relationship between the variables; there may still be a strong *nonlinear* relationship.
 - A correlation close to -1 or $+1$ does not necessarily mean that a linear model is the most appropriate model.
 - The correlation r itself is unit-free.
 - Changing the units in either the x - or y -variable will not change r .
 - Correlation does not distinguish between which variable is called x and which is called y , so interchanging the variables (on the scatterplot) will not change the value of r .
 - Correlation does not imply causation!
- From the linear regression formula $\hat{y} = a + bx$:
- The y -intercept a describes the average or predicted outcome of y if $x = 0$.
 - The slope b describes the average or predicted increase or decrease in the y -variable for each 1 unit increase in the x -variable.
- Remember these important points about residuals:
- Order is important: residual equals observed minus predicted.
 - When the data point is above the regression line, the residual is positive; a data point below the regression line gives a negative residual.
 - A positive residual means the linear model underestimated the actual response value, while a negative residual means the linear model overestimated the actual response value.
 - A residual plot with a definite pattern suggests that a nonlinear model will show a better fit to the data than will the linear regression line.
- It is important to be able to read generic computer output (where a is the y -intercept, b is the slope, and r^2 is the coefficient of determination):

Dependent variable: Math					y-variable
					r ²
S = 16.69	R-Sq = 93.5%	R-Sq (adj) = 93.0%			
Variable	Coef	SE Coef	T	Prob	
Constant	92.5724	31.75	2.92	0.012	
Verbal	0.763604	0.05597	13.6	0.000	
Intercept					
x-variable	b	a			

[Follow for extended description](#)

Part 3

Final Review

Final Review

After completing Units 1–5, their associated quizzes, and the following review sections, you'll be ready to try the practice tests. Wishing you good luck will not be appropriate, because you'll be so well prepared that you won't need luck!

Selecting an Appropriate Inference Procedure

Of the six free-response questions on the AP Statistics exam, at least one—and often two—involves inference. For these problems, the first thought should be to identify the appropriate inference method. While knowing how to perform the different procedures is relatively easy when the question appears on a page with the unit heading at the top, the exam provides no such help! Thus, time devoted to recognition and decision-making should be an important part of any review. This unit provides that valuable practice.

To increase success, stop and think before proceeding. Hasty decisions often lead to choosing the wrong statistical procedure needed to solve the problem. Take a few moments to answer several questions that should be at the forefront of your mind. Use the test booklet to jot down notes. The following are important questions and tips:

Are there key terms or phrases? (Underline or circle them!)

- Words such as *proportion*, *mean*, or *association*?
- Expressions such as *find a confidence interval* or *do a hypothesis test*?

What kind of data are presented?

- Categorical (indicating proportions or chi-square)?
- Quantitative (indicating means)?

What summary statistics are given?

- A sample proportion?
- A sample mean and standard deviation?
- Sample counts (leading to chi-square inference)?
- A sample slope?

What specifically are you asked to do?

- Estimate a quantity (indicating a confidence interval)?
- Look for evidence to test a claim (indicating a hypothesis test)?

How was the data collected?

- From one sample or two samples?
- Is there pairing of data (is information lost if the data in either set are shuffled around)?

If a hypothesis test is indicated, should you use a one-tailed or two-tailed test?

- Are we looking for a difference from some particular value or to see whether nothing has changed (indicating a two-tailed test)?
- Are we looking for a change in a specific direction (indicating a one-tailed test)?

Which distribution should be used for inference problems?

- Proportions? Use z .
- Means? Use t .
- Counts? Use χ^2 .

What is the context of the question?

- Underline or circle the context!
- A response should always refer to the context in the conclusion.

What conditions need to be addressed?

- Independence assumption
 1. Random sampling or random assignment
 2. $n \leq 0.10N$ if sampling without replacement

3. Independence across samples if comparing more than one population
- Normality assumption
 1. For proportions, check that np and nq are both ≥ 10 .
 2. For means, check that either the population is roughly normal or the sample size is large ($n \geq 30$).
 3. For means, if we are not given that the population is roughly normal, for small samples ($n < 30$) the sample data should be free from strong skewness and outliers.
 4. For counts, check that all expected cells are > 5 (for increasing df , the χ^2 distribution approaches a normal distribution).

► Example

What procedure is called for with the data below?

29	32	50	41	32
34	50	48	44	40

Of course this is impossible to answer without knowing context!

- (a) A random sample of five students signing up for a private tutoring service take a pretest and a posttest with scores on the 50-point tests given below.

Student	A	B	C	D	E
Pretest score	29	32	50	41	32
Posttest score	34	50	48	44	40

What procedure should be used to find a 95% confidence interval for the mean difference in score results (posttest score minus pretest score) for students who use the tutoring service?

- (b) In a random sample of 400 attendees at a Cubs–White Sox game, favorite team is cross-classified against favorite reading genre. The following table shows the data.

	Science Fiction	Romance	Sports	Mystery	History
Cubs	29	32	50	41	32
White Sox	34	50	48	44	40

Which procedure should be used to test whether or not there is an association between favorite team and favorite reading genre?

- (c) A miniature golf course offers two nine-hole courses. A random sample of five players who play both courses shows the following scores.

	A	B	C	D	E
Course 1	29	32	50	41	32
Course 2	34	50	48	44	40

The franchise owner is interested in using a player's score on one of the courses to predict his or her score on the other course. What procedure should be used?

- (d) A school's AP coordinator picks a random sample of five students who claim to have stayed up all night studying for a next day's AP exam and an independent random sample of five students who claim to have had a good night's sleep the night before the AP exam. Their scores on the exam are in the table below.

Student	A	B	C	D	E
No sleep	29	32	50	41	32
Sleep	34	50	48	44	40

What procedure should the AP coordinator use to determine if there is a statistically significant difference in the mean scores obtained by students who stay up all night studying and those who have a good night's sleep before the exam?

- (a) The pretest and posttest data are not independent. This is a paired data test, so we cannot use two-sample inference. No population standard deviation is given, so we will use a t -procedure rather than a z -procedure. Use a one-sample t -interval on the set of differences.
- (b) There is a single sample cross-classified on two variables (rather than samples from two or more populations and a single categorical variable), so use a chi-square test of independence (rather than a chi-square test of homogeneity).
- (c) See if a scatterplot looks roughly linear. If so, find the regression line equation. and use for predicting the value of one variable from the value of the other.
- (d) We have two independent samples, are asked to determine if there is a significant difference (rather than a confidence interval) between two means, and are not given any population standard deviation (so we must use a t -procedure rather than a z -procedure). Use a two-sample t -test for the difference of two population means.

The following quizzes, [Quiz 31](#) and [Quiz 32](#), evaluate your ability to name procedures, define parameters, list conditions to be checked, and state hypotheses if appropriate.

Quiz 31

DIRECTIONS: For each of the following problems, name the procedure you would use, define any parameter you use, list the conditions to be checked, and state hypotheses if appropriate.

1. At the high school level, do men and women play varsity sports for the same reasons? Of all students who tried out for varsity sports, a random sample of men and a random sample of women are picked. The selected students are given questionnaires asking which of the following is their primary reason for playing sports: social, health, or status.

2. Estimate the proportion of AP Statistics students who will graduate in the top ten percent of their senior class. There were 150 AP Statistics students tracked.
3. Estimate the average number of books read by high school students during their four years in secondary school. A random sample of 125 students are followed during their high school years.
4. To test the claim that over 70 percent of racially motivated hate crimes are motivated by anti-Black bias, a criminologist obtains data from a random sample of police commissioners.
5. Do baseball pitchers throw harder if they participate in ten-minute yoga sessions before games? A pitcher who normally averages 95 mph with his fastballs participated in ten-minute yoga sessions for a random sample of his games. His fastballs for those games were clocked.
6. Does family income have an impact on school performance? In a random sample of 100 students, family income and student GPA were recorded. What percentage of the variation in student GPA can be explained by variation in family income?
7. A study is planned to find the difference in the proportion of patients with warts who are cured with a treatment of cryotherapy and the proportion cured with a treatment of duct tape occlusion.
8. A study is planned to determine if there is significant evidence that the mean cholesterol level (measured in milligrams per deciliter) for people living in “Western” countries is higher than that of people living in “non-Western” countries.
9. What is the average increase in CO₂ emissions (in tons per person) for each unit increase in income level (in GDP per capita) among countries of the world?

10. An economist is interested in whether there is an association between race/ethnicity and whether an adult is unbanked (does not have a checking or savings account). A random sample of 500 adults gives the following table:

	White	Black	Hispanic (non-White)	Native American
Banked	203	98	77	49
Unbanked	18	25	20	10

11. For the prevention of strokes, at-risk patients are often given anticlotting drugs. A study is being designed to compare the proportion of patients receiving rivaroxaban who still have strokes to the proportion receiving warfarin who still have strokes.
12. Malaria is endemic in Liberia. Many people contract and are cured of malaria repeatedly during their lifetimes. A new prophylactic medication was developed as a pill taken weekly to prevent the contraction of malaria. A random sample of adults took the pill for one year and took a placebo pill for another year. The year each adult took the prophylactic was decided randomly for each participant. Summary statistics were as follows.

	<i>n</i>	Mean	Standard Deviation
Episodes on the prophylactic	1,500	0.22	0.11
Episodes on the placebo	1,500	0.84	1.04
Difference (on – off)	1,500	-0.62	0.35

13. In a random sample of 5,000 live births in the U.S., the infant mortality rate was 5.8 (infant deaths per 1,000 live births). In a random sample of 2,000 live births in Japan, the infant mortality rate was 2.0 (infant deaths per 1,000 live births). What is the 99 percent confidence interval for the difference in mortality rates in the United States and Japan?

14. How much better do students do on national standardized exams if they take them a second time? The scores of 500 students who took a national standardized exam twice are charted.
15. What is the mean waist size of male high school math teachers? The waist sizes of a random sample of 50 male high school math teachers are measured. From past studies, it is known that the standard deviation of such waist sizes is 2.15 inches.

The answers for this quiz can be found beginning on [page 512](#).

Quiz 32

DIRECTIONS: For each of the following problems, name the procedure you would use, define any parameter you use, list the conditions to be checked, and state hypotheses if appropriate.

1. College advisers suspect that first-year students study more hours per week than sophomores.
2. Are students willing to report cheating by other students? Students from a random sample are asked to fill out an anonymous questionnaire asking whether they would report cheating by other students.
3. A cardiologist is interested in whether there is a relationship between blood pressure level (low, average, or high) and personality type (high-strung or easygoing).
4. During a previous season, Division 1 college basketball players successfully made 34% of their 3-point shot attempts. A sports magazine is interested in determining if the percentage is lower this year.

5. The director of a health clinic is interested in finding out if participating in a particular month-long workout program will increase the average number of push-ups that can be completed by clinic members. Enrollees will be asked to do as many push-ups as possible in a 90-second period before the program begins and then again a month later when the program ends.
6. It is hypothesized that the average body temperature is less than 98.6°F.
7. By how much does exercise lower resting heart rates for teenagers? Ninety teenagers took part in a survey where their resting heart rates were tested before and after an extensive exercise program.
8. A study about racism in the business world involved sending resumes with stereotypically “White” names and identical resumes except with stereotypically “Black” names. Twelve out of 110 job applications with stereotypically “White” names received callbacks, while only 10 out of 155 job applications with stereotypically “Black” names received callbacks. Establish a 90 percent confidence interval for the difference in proportions of callbacks between applications with stereotypically “White” and stereotypically “Black” names.
9. Is there a significant difference in the distribution of highest school level attained between Whites, Blacks, Asians, and Hispanics (non-White)? Random samples of 200 people from each population interviewed resulted in the following summary table.

	High School or Less	Undergraduate Degree	Graduate Degree
White	130	58	12
Black	168	27	5
Asian	98	82	20
Hispanic (non-White)	172	24	4

- [10.](#) A reporter for a national car magazine plans to sample used cars of a certain model in hopes of being able to determine what percent of the variation in selling price can be explained by variation in miles noted on the odometer.
- [11.](#) After answering your phone, if there is an initial pause on the other end, it often means that the call is from a telemarketer. A study is conducted to estimate the average pause length for telemarketing calls.
- [12.](#) In a random sample of 550 White students, 4.1 percent reported that, at times, they felt unsafe to go to school, while in an independent random sample of 425 non-White students, 7.8 percent reported that, at times, they felt unsafe to go to school. A sociologist believes that the difference is significant, meaning non-White students are more likely than White students to feel unsafe to go to school, at times.
- [13.](#) Conservationists plan to determine the mean difference in weights between elephants living in captivity and those in the wild.
- [14.](#) What is the predicted increase in sitting pulse rate for each additional ounce of coffee consumed?
- [15.](#) It is hypothesized that the mean hourly wage of summer work available for high school students is \$10.50. A city planner believes the true mean is lower. She randomly surveys 75 high school students with summer jobs. From past studies, it is known that the standard deviation of such jobs is \$1.83.

The answers for this quiz can be found beginning on [page 514.](#)

Statistical Insights into Social Issues

Quiz 33 Topics

- 1. DISTRIBUTION OF WEALTH**
- 2. INCARCERATION BY AGE AND race/ethnicity**
- 3. FAMILY INCOME AND SAT SCORES**
- 4. DEVELOPMENT AID**

- 5. REPORTING CONFLICT VIOLENCE**
- 6. JURY SELECTION**
- 7. ESSENTIAL HISTORY**
- 8. ENVIRONMENTAL RACISM**
- 9. DEATH PENALTY CONSIDERATIONS**
- 10. FACIAL RECOGNITION**
- 11. FLAWED FORENSIC TESTIMONY**
- 12. MODERN-DAY SLAVERY**
- 13. HATE CRIMES**

Quiz 34 Topics

- 1. BULLYING**
- 2. FIREARM SUICIDES**
- 3. DANGEROUS LEAD LEVELS**
- 4. RACIALLY IDENTIFIED NAMES**
- 5. CHILDREN AND GUN SAFETY**
- 6. LAW ENFORCEMENT–ASSISTED DIVERSION**
- 7. POLICE PROFILING**
- 8. EDUCATION AND EARNINGS**
- 9. AWARENESS OF RELIGIOUS DISCRIMINATION**
- 10. VACCINES AND AUTISM**
- 11. GLOBAL WARMING**

In the spirit of AP Statistics, the following two quizzes encompass comprehensive review questions that aim to give an appreciation of the power of statistics to better understand some of society's most pressing issues. As you review and apply the statistical tools you have learned, you should also reflect on the content of the examples and the empowerment that statistics gives you to be aware of, to discuss, and maybe someday to help overcome some of the ills present in the broad social world in which we live. More specifically, statistics can help you understand the relationships among power, resource inequities, and uneven opportunities among different social groups and to understand explicit discrimination based on race/ethnicity, class, and gender. In many ways, the AP Statistics curriculum is just as much a civics course as it is a math course. Of course, in addition to reflecting on these significant issues, do not forget to read

carefully what the questions ask, to check assumptions and conditions when appropriate, to show your work, and to communicate your answers clearly.

There is great interest in how statistics can help better society. Statistics can be an incredibly powerful tool both in helping you understand social issues in terms of our current information age and in helping you effect social change. From disparities in income levels to disparities in the distribution of wealth, from unequal access to education to unequal access to health care, from unfair criminal justice enforcement to unfair housing market discrimination, statistics can be the key for you to become an engaged participant in our democracy. Based on your background, your interests, and your available time, you can decide how far to take the opportunities for deeper discussion raised in the following exercises.

Following are two quizzes. [Quiz 33](#) has 13 multisection free-response questions reviewing the concepts and skills from [Units 1](#), [2](#), and [5](#) on exploratory analysis, collecting data, and probability. [Quiz 34](#) has 11 multisection free-response questions reviewing the concepts and skills from Units 3 and 4 on statistical inference. All the quiz questions are set in the context of contemporary social justice issues, and they are specifically designed to evaluate your understanding of the entire curriculum and reinforce your communication skills. The answers can be found beginning on [page 517](#).

[Quiz 33](#)

1. A 2023 Oxfam press release stated that the richest 1% take in twice as much new wealth as the rest of the world put together.¹ One way in which the distribution of wealth throughout the world can be analyzed is illustrated below (where GDP refers to Gross Domestic Product and Oceania is a region centered on the islands of the tropical Pacific Ocean).

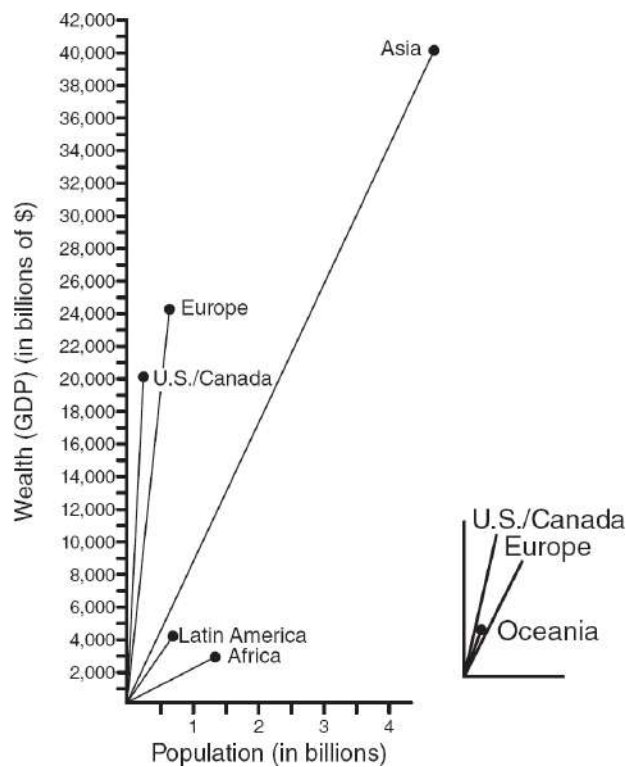
(a) Fill in the blank cells in the table below.

Region	Population (in millions)	% of World Population	Wealth (GDP) (in billions)	% of World GDP
Africa	1,341		2,995	

Region	Population (in millions)	% of World Population	Wealth (GDP) (in billions)	% of World GDP
Asia	4,641		40,248	
Oceania	43		2,002	
Europe	748		24,881	
U.S./Canada	369		31,319	
Latin America	654		4,068	
World Total	7,796	100.0	105,513	100.0

(b) Construct a segmented bar chart with two bars from the above table, and comment on its significance.

(c) Consider the following graph:

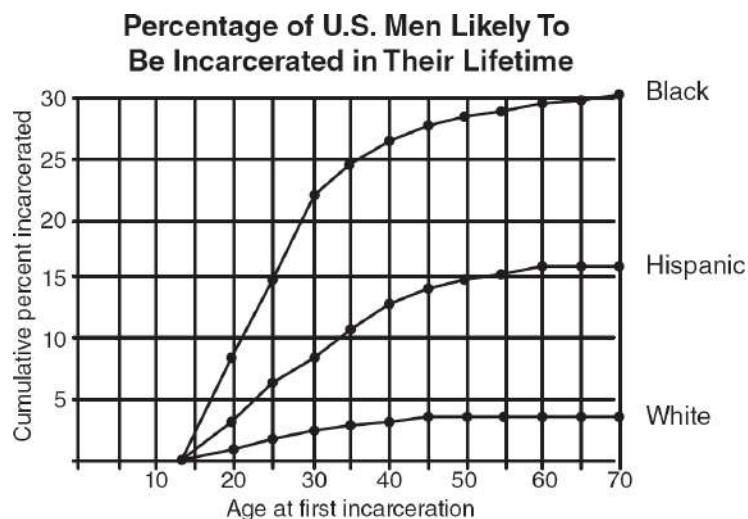


What do the slopes of the six slanted lines represent? Use these slopes to give an appropriate ranking of the six regions.

(d) The particular quantitative index used in a statistical study can make a marked difference. For example, an alternative to the

GDP for measuring the wealth of a nation is the GPI (Genuine Progress Indicator).² The GPI takes into account negative factors, such as industrial pollution, depletion of natural resources, and crime rates, as well as positive factors, such as socially productive time uses like volunteerism. Give an example of how changing the quantitative index from GDP to GPI might affect the relative position of U.S./Canada and Oceania in part (c).

2. The United States has the second highest incarceration rate in the world (second only to the small African country of Seychelles).³ One study⁴ of incarceration rates and ages summarized their findings in the following graph:



- What is the meaning of the point (25, 15) on the Black graph?
- What is the meaning of the horizontal segment on the White graph from ages 60 to 70?
- What is the median age at first incarceration of Hispanics who ever went to prison?
- Some states are considering legislation to address the disproportionate school suspensions among students of color. How might this affect the above graphs?

3. SAT used to stand for “Scholastic Aptitude Test,” but some researchers suggest it also stands for “Student Affluence Test.”⁵ There appears to be an association between mean total SAT scores and family income level. One study is summarized in the following table:

2015 College-Bound High School Students

Income Bracket (\$)	Total SAT Score
0–20,000	876
20,000–40,000	933
40,000–60,000	972
60,000–80,000	1000
80,000–100,000	1029
100,000–120,000	1054
120,000–140,000	1061
140,000–160,000	1077
160,000–200,000	1091
Over 200,000	1147

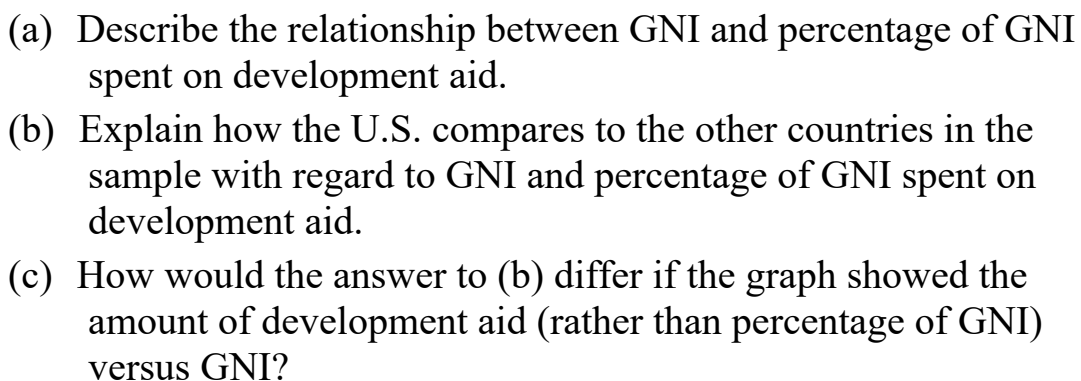
Using the median income within each of the first 9 income brackets and a \$210,000 income from the last bracket, computer output yields the following:

Predictor	Coef	SE Coef	T	P
Constant	903.36	112.57	71.85	0.000
Income	0.00116	0.000104	11.17	0.000

S = 20.936 R-Sq = 94.0%

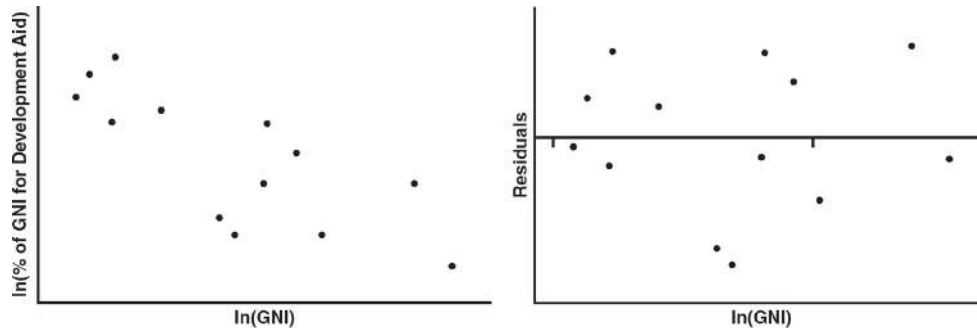
- What is the equation of the regression line?
- Interpret the slope of the regression line in context.
- What is the coefficient of determination? Interpret it in context.

4. It is estimated that 700 million people worldwide live in extreme poverty. With regard to foreign policy, governments invest in hard power (military) and soft power (development aid). The cost of deploying one soldier overseas is the same as producing 100 village wells. Countries spend widely different percentages of their Gross National Income (GNI) on development aid. A random sample of 12 countries plus the United States yields the following:



- (d) The five lowest values for GNI in this sample come from Belgium, Denmark, Netherlands, Norway, and Sweden. What is apparent about the use of soft power by these countries?

Following is a scatterplot and some computer linear regression output showing the relationship between the natural logarithm of percentage of GNI spent on development aid and the natural logarithm of GNI, where GNI is in billions of dollars.



Predictor	Coef	SE Coef	T	P
Constant	2.4135	0.7914	3.05	0.011
ln(GNI)	-0.4179	0.1028	-4.06	0.002

S = 0.467076 R-Sq = 60.0% R-Sq(adj) = 56.4%

[Follow for extended description](#)

- (e) Describe the relationship between $\ln(\text{GNI})$ and $\ln(\text{percentage of GNI spent on development aid})$.
- (f) Based on this output, explain why it would be reasonable to use this regression to describe the relationship between $\ln(\text{GNI})$ and $\ln(\text{percentage of GNI spent on development aid})$.
- (g) What is the correlation coefficient r for this regression?
- (h) Interpret the slope for this regression line in context.
- (i) Using the computer output, give the equation of the least-squares regression line.
- (j) Using the model from (i) above, predict the percentage of GNI spent on development aid for the U.S. with a GNI value of \$23,400 (billion). Is this prediction an overestimate or an underestimate for the true value of 0.15 percent?

5. In analyzing conflict violence, the data are often incomplete, as obtaining data in such situations is very difficult if not impossible. However, accurate data are crucial for systematic studies of the origins of conflict, conflict dynamics, and conflict resolution.

- (a) Suppose during one conflict month early during the Syrian civil war, the Syrian Observation for Human Rights (SOHR) reported 3,060 deaths, the United Nations (UN) reported 3,627 deaths, and 2,635 of the deaths were found on *both* lists. Letting D be the actual number of deaths, fill in the following table:

SOHR				
		Reported Deaths	Unreported Deaths	Totals
UN	Reported Deaths			3,627
	Unreported Deaths			$D - 3,627$
	Totals	3,060	$D - 3,060$	D

- (b) Assuming the two reports above represented independent random samples of all deaths that month, solve for D .
- (c) In reality, only *convenience samples* were available—perhaps posted videos of public executions, perhaps text messages sent out by protesters, or perhaps victims’ testimonies recorded by local reporters. In general, why do convenience samples often lead to misleading conclusions? In our context, how are assumptions made in part (b) above violated?
- (d) Another source of *undercoverage bias* found in conflict situations can be called *size bias*. Events involving large numbers of casualties are much more likely to be documented than events involving only one victim. For example, a cluster bombing of a marketplace with many victims will attract media attention while targeted assassinations may be unreported because the victim’s family is afraid to make a report or because the body is never found. Explain how this affects the independence assumption made in part (b) above.
- (e) Would you expect the answer found for D above to be an underestimate or an overestimate? Explain.

6. In 1986, the Supreme Court ruled that it is unconstitutional to exclude potential jurors because of race/ethnicity.⁶ However, many studies since then have shown that racial discrimination in jury selection remains widespread, particularly in murder trials. A university law department plans a study using mock trials to determine if outcomes are different with all-White versus mixed-race/ethnicity juries. The law department will first randomly select two counties from among its state's 67 counties.

- (a) What is this sampling method called, and what is an advantage of using this method over a simple random sample?
- (b) Give a procedure for randomly selecting the two counties.
- (c) Forty-five people will then be randomly selected from among the pool of all eligible White jurors in each of the two counties to obtain a sample of size 90. What is the purpose of randomization in this two-step process?
- (d) The proposed study will involve 10 mock cases involving a young Black man charged with murdering an elderly White woman. The sample of 90 White jurors will be randomly assigned to two groups—60 to make up five all-White juries (each of size 12) and 30 to be part of five mixed-race/ethnicity juries (each with 6 White jurors and 6 jurors of color). What is the purpose of random assignment here? How might allowing jurors to self-select whether or not to be part of mixed-race/ethnicity juries lead to bias?

7. Book banning is a controversial topic in school districts across the country. Suppose a questionnaire is to be designed as to parents' opinions on whether books with depictions of slavery should be banned from school libraries. *Wording bias* occurs when the wording of a question influences the responses.

- (a) Give an example of a neutrally designed question on this topic.
- (b) Give an example of a question on this topic illustrating wording bias that will lead to an underestimate of the true proportion of parents in favor of banning books with depictions of slavery.

- (c) Give an example of a question on this topic illustrating wording bias that will lead to an overestimate of the true proportion of parents in favor of banning books with depictions of slavery.

In a recent survey, 26% of American adults described their political views as liberal, 38% as moderate, and 36% as conservative. Given a neutrally designed question, 9% percent of liberals, 12% of moderates, and 15% of conservatives answered that books with depictions of slavery should be banned from school libraries. An American adult is picked at random.

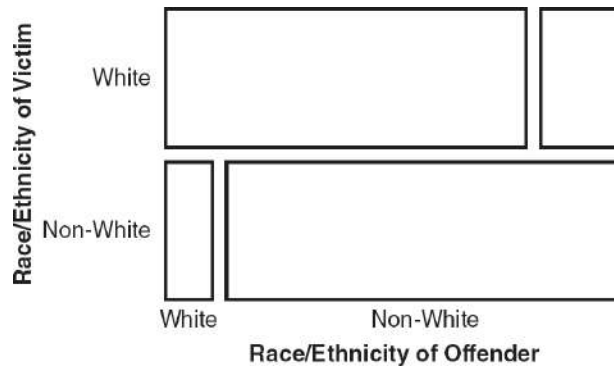
- (d) What is the probability the person is liberal and believes that books depicting slavery should be banned from school libraries?
- (e) What is the probability the person believes that books depicting slavery should be banned from school libraries?
- (f) If the person believes that books depicting slavery should be banned from school libraries, what is the probability the person is liberal?

8. “Environmental racism” is said to occur when race/ethnicity is associated with the location of hazardous sites. A 2015 California study was designed to assess who was at the greatest risk from oil trains, which can derail and explode. Of the 5.5 million Californians who live within a blast zone (defined as within one mile of the railroad tracks), 75% are Latino and only 10% are White. California’s population of 38.8 million is 39.0% Latino and 38.9% White.

- (a) What is the probability a randomly selected Californian is a Latino living within a blast zone?
- (b) What is the probability that a randomly selected Californian Latino lives within a blast zone?
- (c) In California, are being Latino and living within a blast zone independent events? Justify your answer mathematically.
- (d) In a random sample of 5 Californians, what is the probability that at least 2 of them live in a blast zone?

- (e) What is another explanatory variable that could account for the relatively high probability that someone living in a blast zone is a person of color?

9. Consider the following mosaic plot of murder cases.



- (a) What can be concluded from the above mosaic plot?

Studies seem to indicate that Black lives matter less than White lives with regard to the death penalty.⁷ One long-term study found that while the race/ethnicity of the offender did not seem to predict the likelihood of a death sentence, the race/ethnicity of the victim did. In this study, 50.5 percent of murder victims were White, 49.5 percent were non-White. In the cases that resulted in executions, 76% of the murder victims were White and 24% were non-White. Finally, 1.2% of the murder cases resulted in the execution of the defendant. Use these findings to answer the following questions.

- (b) What is the probability that there is a White murder victim and the defendant is executed? What is the probability that there is a non-White murder victim and the defendant is executed?
- (c) Given that the murder victim is White, what is the probability that the defendant is executed? Given that the murder victim is non-White, what is the probability that the defendant is executed?
- (d) In (c), what can be concluded about the likelihood of a victim being White or non-White if the defendant is executed?

10. Law enforcement agencies across the United States are rapidly adopting facial-recognition surveillance.⁸ A false positive, which incorrectly identifies an innocent person as being a criminal, generally occurs less than 10% of the time. While this percent may sound low, the question is whether it is a small enough risk when the end result may be the arrest of, or even worse, the use of force (including deadly force) against, an innocent person.⁹

Suppose facial recognition software used by a police department to identify possible suspects is 99 percent accurate. That is, the software will correctly identify a criminal as being a criminal 99 percent of the time, and it will correctly identify an innocent person as being innocent 99 percent of the time. Assume 0.88 percent of the adult population are criminals (in 2020, about 0.88% of the United States adult population were behind bars).

- (a) What is the probability that someone is identified to be a criminal by this software?
 - (b) What is the probability that someone identified as being a criminal by this software really is a criminal?
 - (c) What concern regarding the use of this software is raised by the answer to (b)? Explain.
 - (d) Statisticians should always consider ethical issues when gathering data. Numerous studies have found that facial recognition is significantly less accurate when identifying people of color. Does this raise ethical issues? Explain.
11. Although there is no accepted research on how often hair from different people may appear the same, for over two decades the FBI used visual hair comparison matches to help obtain convictions in major criminal cases. Acknowledging possible flawed forensic testimony based on unsound science, the FBI agreed to review over 2,500 cases. Flawed forensic testimony was found in 257 of the first 268 cases reviewed.¹⁰ In the following, assume that the sample proportion of cases involving flawed testimony is an accurate measure of the population proportion.

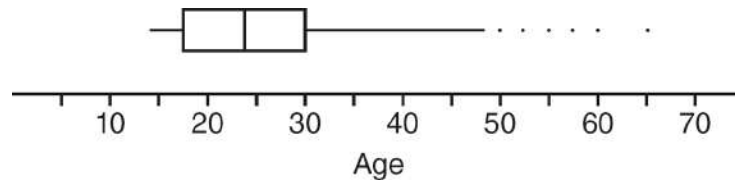
- (a) Was this a retrospective observational study, a prospective observational study, or an experiment? Explain.
- (b) The FBI agreed to review 2,500 cases involving possible flawed forensic testimony. Assuming this is a random sample of relevant cases, what is the expected number of these 2,500 cases with actual flawed forensic testimony? What is the standard deviation for the expected number of cases with actual flawed forensic testimony?
- (c) Among the 257 cases with flawed forensic testimony, 32 resulted in death penalty convictions. Assume this is illustrative of the overall proportion. In examining a random sequence of cases involving flawed testimony, what is the probability that the first case with a resulting death penalty doesn't occur until at least the third case examined?
- (d) According to the Innocence Project, the misapplication of forensic science is the second-leading contributor to wrongful conviction, playing a role in 50% of known wrongful convictions, while eyewitness misidentification has played a role in nearly 75% of known wrongful convictions.¹¹ In wrongful conviction cases, can “the misapplication of forensic science” and “eyewitness misidentification” be mutually exclusive? Can they be independent? Explain.

¹². We often think that slavery ended in the 19th century, but the sad fact is that there are more people enslaved today than at any other time in human history. According to the not-for-profit Force4Compassion, an average of 3,287 people are sold or kidnapped and forced into slavery every day.¹² Assume this daily total has a roughly normal distribution with a standard deviation of 250 people.

- (a) What is the probability that on a randomly selected day, over 3,500 people are forced into slavery?
- (b) In a random sample of 5 days, what is the probability that on a majority of the days over 3,500 people are forced into slavery?
- (c) In a random sample of 5 days, what is the probability that the average number of people forced into slavery is over 3,500?

(d) Are there any statistics that, if more widely disseminated, might mobilize the world to fight this social blight?

[13.](#) When bias-motivated crimes (triggered by prejudice and hate) are committed, whole communities can feel victimized. A study^{[13](#)} of profiles on America's most popular online hate site revealed that the site members are very young. Consider the following boxplot of self-reported ages of the site members:



- (a) Describe the distribution of ages.
- (b) Is the quotient $\frac{\text{Mean age}}{\text{Median age}}$ greater than 1, less than 1, or approximately 1? Explain.
- (c) Members of online social networks tend to be younger than the general population. What does this possibly indicate about the median age of all hate group members? Explain.
- (d) According to recent FBI statistics,^{[14](#)} 18.6% of bias incidents are religious based. In a random sample of 3 bias incidents, what is the probability that a majority are religious based?

The answers for this quiz can be found beginning on [page 517](#).

[Quiz 34](#)

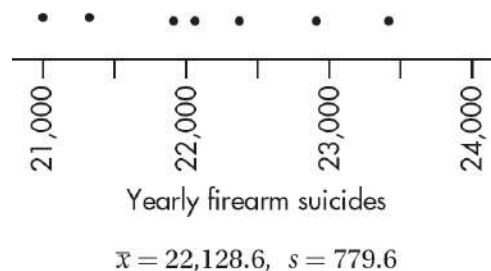
[1.](#) Schoolyard and online bullying are leading to increased levels of violence and depression among students. In a national survey^{[15](#)} of 15,686 students in grades six through ten, 19% were involved in moderate to frequent bullying as perpetrators, 17% as victims, and 6% as both victims and perpetrators.

- (a) What proportion of the students were involved in moderate to frequent bullying?

- (b) Find a 95% confidence interval for the proportion of students in grades six through ten who are victims of moderate to frequent bullying.
- (c) Based on this confidence interval, is there convincing evidence that the proportion of students who are victims of moderate to frequent bullying is different from the 22% claimed in one study?¹⁶ Explain.
- (d) School administrators tend to underestimate the rates of bullying because students rarely report it and it often happens when adults are not present. However, knowing the numbers and types of bullying at all grade levels in your school is critical before a plan for bullying prevention and intervention can be developed. How would you plan a survey at your school to assess the frequency and types of bullying present?

2. Using the Second Amendment to the U.S. Constitution, gun rights proponents have argued for nearly or completely unchecked rights of personal gun ownership. However, because the U.S. has one of the highest rates of gun violence in the world, others have sought to regulate gun ownership. In the news, deaths by firearms are most often associated with murders; however, most firearm-related deaths in the United States are suicides, and most suicides are by guns. According to the CDC,¹⁷ in 2021 there were 26,328 suicides in the U.S. using firearms.

- (a) Assume we have a random sample of 7 years that are considered representative with regard to numbers of firearm suicides:

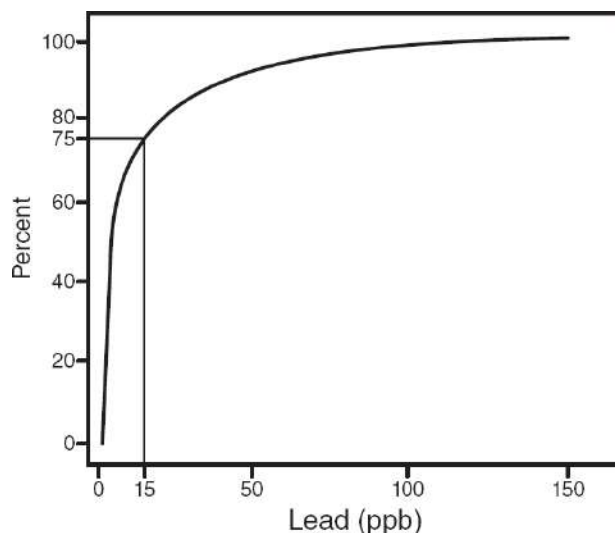


Calculate a 95% confidence interval estimate for the mean number of firearm suicides in the U.S. each year.

- (b) According to a major recognized survey,¹⁸ the U.S. has the highest rate of gun ownership in the world with 120.5 guns per 100 people. The same survey reports Israel with 6.7 guns per 100 people. Given that these figures report only privately owned guns and exclude weapons that are technically owned by the government, how might the apparent wide discrepancy between the U.S. and Israel not be as wide as indicated?
- (c) When presented with any data set, we should always try to understand what and whose values are implicitly represented. Israel rejects 40% of gun applications and requires yearly renewals for continuing gun ownership. How would this affect the conversation from the statistics quoted in part (b)?
- (d) The average U.S. gun owner owns 3 guns.¹⁹ Comment on measuring rates of suicide by firearms in terms of guns per 100 people (120.5 in the U.S.) versus measuring rates of suicide by firearms in terms of gun owners per 100 people.

3. As a cost-cutting measure in April 2014, Flint, Michigan, began temporarily drawing its drinking water from the Flint River. The result was dangerous lead contamination in the water supply. (Lead poisoning especially impacts a child's brain and nervous system.)

- (a) An independent research team²⁰ tested the Flint water supply for lead. The resulting cumulative frequency plot of the lead levels in ppb (parts per billion) in the water supply of sampled homes is as follows:

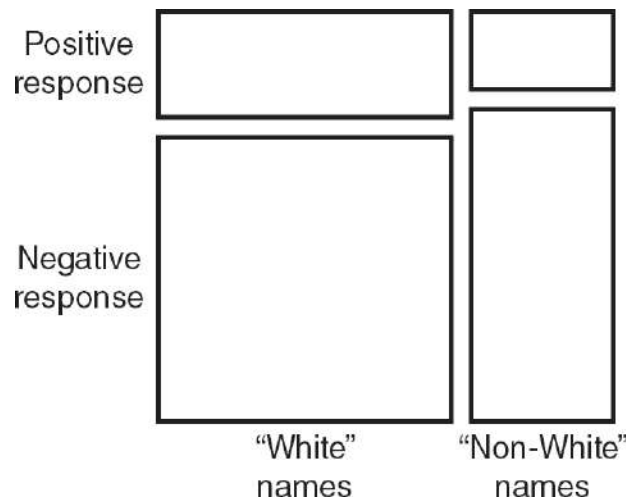


What is the shape of the histogram of the lead levels in the water supply of sampled homes? How would you expect the mean and median lead levels to compare with each other? Explain.

- (b) It is recognized that if over 10% of a city's homes have lead levels in their water supply of over 5 ppb, the city has "a very serious problem with lead."²¹ The research team determined that 15 ppb corresponded to the 75th percentile of Flint home water supply lead levels. Does this indicate a very serious problem?
- (c) Suppose the concern is whether or not the mean lead level in the water supply of the homes is greater than 10 ppb. (The EPA regards 15 ppb in the water supply as a very dangerous level.²²) Describe Type I and Type II errors in context. Which is of greater concern, a Type I or Type II error, and what would be a possible consequence? Explain.

4. Even in the highest-paying occupations, non-White workers face racism in the workplace.²³ In one study, racially identified names were randomly assigned to resumes, and the number of interviews offered (positive responses) for each group was tabulated.

- (a) The positive response rate for applicants with "White" names had a 95% confidence interval of (0.21882, 0.28258). What was the sample size for applicants with "White" names?
- (b) A mosaic plot of responses is shown below. What does it show?



- (c) A hypothesis test, with $H_a: p_W > p_{NW}$ where p_W = the true proportion of resumes with “White” names that result in positive responses, and p_{NW} = the true proportion of resumes with “non-White” names that result in positive responses, yields $z = 2.585$ and $P = 0.00486$. Is there convincing evidence that resumes with “White” names result in a greater proportion of positive responses than resumes with “non-White” names?
- (d) A measure of discrimination, the *discrimination ratio*, for African Americans contrasted to Whites (with positive values indicating discrimination against African Americans and negative values indicating discrimination against Whites) was calculated over a 25-year period, resulting in a confidence interval for the slope of $(-0.007, 0.015)$. Is there convincing evidence that discrimination against African Americans is trending higher?

5. Gun violence is now the leading cause of death for children in the United States, and all sides of the political spectrum are concerned. In a recent study, 226 children, ages 8 to 12, were tested in pairs of kids who knew each other, including siblings, cousins, and friends. One of each pair was randomly picked to watch a short video on car safety, while the other of each pair was shown a short video on gun safety. A week later, all 226 children came to a large playroom filled with toys and with two disabled 9 mm handguns. Of those who watched the gun safety video, 34% told an adult that there were guns in the room,

while of those who watched the car safety video, 11% told an adult there were guns in the room.²⁴

- (a) Is random sampling, random assignment, both, or neither involved in this study? Explain.
 - (b) What are the appropriate hypotheses for testing whether there is convincing evidence that the proportion of all youths 8 to 12 years old who will tell an adult about a gun in the room is greater for those children shown a gun safety video than for those children shown a car safety video?
 - (c) The test statistic is $z = 4.167$. What does this value mean in context?
 - (d) Is there convincing evidence to reject the null hypothesis at the 0.01 significance level? Explain.
 - (e) A 99% confidence interval for the difference in proportions (gun safety video viewers minus car safety video viewers) based on these data is (0.09342, 0.36676). Interpret this interval in context.
 - (f) Is this confidence interval consistent with the test decision in part (d)? Explain.
 - (g) The above two videos both featured a police chief in full uniform. A second experiment where both videos featured a cartoon figure called Eddie Eagle showed no statistical difference between the proportions of the two groups telling an adult about seeing a gun in the playroom. Give a possible explanation.
 - (h) Another study involved children in homes with a gun in the home and children in homes without a gun in the home. Those children living in homes with a gun in the home were more likely to tell an adult about seeing a gun in the playroom. Give a possible explanation.
6. Law Enforcement Assisted Diversion (LEAD) is a prebooking diversion program whereby low-level drug and prostitution offenders are redirected to community-based services instead of jail and

prosecution.²⁵ There were 318 offenders who participated in a pilot program in which they were assigned to either the LEAD program ($n = 203$) or control (i.e., booking as usual; $n = 115$). All 318 offenders were deemed eligible for participation in the LEAD program.

- (a) The average yearly criminal and legal system costs of those participating in the LEAD program were \$4,763. For the control group, the average was \$11,695. (Assume standard deviations of \$875 and \$2,125, respectively.) Is this statistically significant evidence that the LEAD program has lower costs than the usual booking program?
- (b) After participation in the program, the LEAD group showed significantly fewer average number of days per year in jail than the control group ($p < 0.05$). Which error, Type I or Type II, might have been committed and what would be a possible consequence?
- (c) Even though it was felt that both the LEAD and the control groups were representative of the targeted offender population, a nonrandomized design was used. This may be of concern. What is the purpose of *random assignment* in such studies?
- (d) Using your answer from part (a), make an argument in favor of the LEAD program to a city council whose members are unfamiliar with statistical arguments.

7. Local activists and community leaders have long maintained that local law enforcement “overpolice” people of color in Berkeley, California. One claim is that people of color are being stopped at a higher rate than their population would indicate and that the stops are often for no legitimate reasons. A study²⁶ of the Police Department found that during the period from January 26 through August 12, 2015, 4,658 civilians were stopped. Of that total, 1,710 were White, 1,423 were African American, and 543 were Hispanic. Of those stopped, 652 of the Whites, 942 of the African Americans, and 306 of the Hispanics were released without arrest or citation.

According to 2013 census data, the population of Berkeley was 116,774, of which 64,412 were White, 10,076 were African American,

and 11,600 were Hispanic.

- (a) Sketch side-by-side bar charts to illustrate the population percentage distribution and the percentage distribution of who was stopped by race/ethnicity.
- (b) Is there statistically significant evidence that African Americans are being stopped at a higher rate than their population would indicate? (Assume that we have a representative sample of common practices for years.)
- (c) Is there statistically significant evidence of a relationship between race/ethnicity (White, African American, Hispanic) and disposition (arrest/citation or not)? (Assume that we have a representative sample of common practices for years.)

8. Many studies have shown that, in general, higher educational attainment correlates with higher earnings (Carnevale, Anthony, *The College Payoff*, Georgetown University, 2021). Consider the following data from independent random samples of workers with high school degrees and workers with some college experience.

Lifetime earnings	High school degree	College experience
Sample size n	250	200
Mean	\$1,600,000	\$1,900,000
Standard deviation	\$350,000	\$420,000

- (a) Determine a 95% confidence interval for the difference in mean lifetime earnings (college experience minus high school degree).
- (b) Based on this confidence interval, is there convincing evidence that the mean lifetime earnings of workers with some college experience is more than \$250,000 greater than the mean lifetime earnings of workers with a high school degree?
- (c) Name two variables that could impact any conclusion about using the above data to encourage people to attend college.

9. In 2021 and 2024, the Pew Research Center conducted independent random surveys of U.S. adults asking if they see discrimination against Jews. The following table summarizes the data.

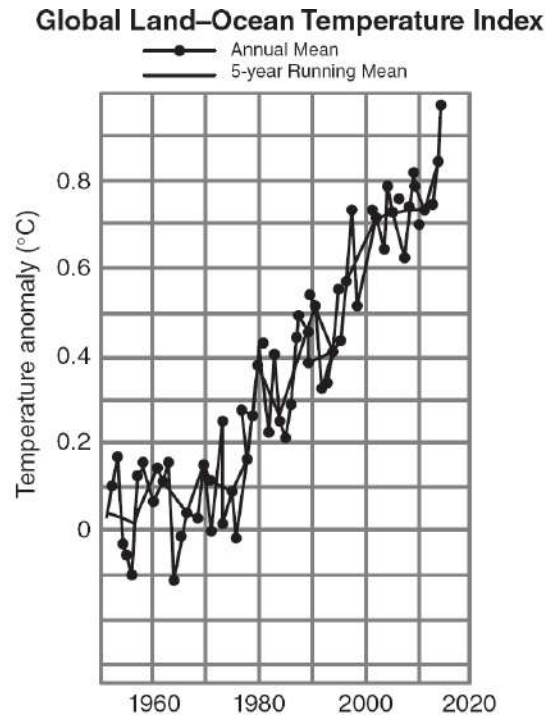
Year	2021	2024
Sample size	12,055	12,693
# who see discrimination against Jews	2,411	5,077

- (a) Find a 99% confidence interval for the difference in the proportion of U.S. adults between 2024 and 2021 who would say they see discrimination against Jews.
- (b) Based only on this confidence interval, is there convincing evidence that the difference (2024 minus 2021) in the proportions of U.S. adults who would say they see discrimination against Jews is greater than 15%?
- (c) How would your answer to (a) probably have differed if the surveys had just been from among the population of Jewish U.S. adults?
10. Immunizations are a cornerstone of the world's efforts to protect people from infectious diseases. After a 1998 research paper falsely claimed a link between MMR (measles, mumps, rubella) vaccines and ASD (autism spectrum disorder), vaccination rates dropped sharply.²⁷ The World Health Organization estimated that in 2018, over 140,000 people worldwide died of measles. In 2019, the U.S. experienced a large, multistate measles outbreak, involving mostly unvaccinated children.
- (a) Is there a link between vaccines and autism? A major study²⁸ concluded that between 30% and 36% of all parents have some such concerns. Assuming this is a 95% confidence interval, what was the sample size?
- (b) The study performed a large-scale test with the following:
 H_0 : There is no association between autism and vaccines.

H_a : There is an association between autism and vaccines.

Given a very large p -value, what was the proper conclusion?

- (c) In a 2020 Pew Institute Study, 10% of American parents expressed belief that risks outweighed benefits with regard to the MMR (measles, mumps, and rubella) vaccine. Assuming the 10% figure is correct, in a random sample of 500 parents, what is the probability that over 12% express belief that risks outweigh benefits with regard to the MMR vaccine?
 - (d) It has been theorized that the MMR vaccine may protect against or reduce the severity of coronavirus disease (Covid-19) infection. In an NIH (National Institute of Health) study, in a random sample of 50 recovered Covid-19 patients, a linear regression analysis matching a y -variable, Covid-19 severity (on a 30-point scale), versus an x -variable, mumps IgG titer (measuring mumps antibodies), resulted in a p -value of 0.000 with a correlation of -0.58 . Interpret the coefficient of determination in context.
11. There is a strong consensus among scientists that global surface temperatures have increased markedly in recent years and that the trend is caused by emissions of greenhouse gases. However, in the popular media, especially in the United States, there is much debate over whether there is any problem—and even if there is a problem, the debate continues over whether humans are responsible. (China and the U.S. are the two largest producers of greenhouse gases.)
- (a) Given the following graph, describe the relationship between Temperature Anomaly (deviation) and Time.



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- (b) In a Pew Global Attitudes Project survey in a random sample of 2,029 Indian people, 1,319 expressed a great deal of worry about global warming. In a random sample of 2,180 Chinese people, 436 expressed a great deal of worry about global warming. Is the difference statistically significant?
- (c) Suppose we wish to conduct a survey among the student body of your high school and calculate a confidence interval for the proportion who worry a great deal about global warming. Comment about bias and direction of bias, from the wording of the following survey:

King Charles III has consistently taken a strong stance criticizing both climate change deniers and corporate lobbyists by likening the Earth to a dying patient. Do you have a great deal of worry about global warming?

The answers for this quiz can be found beginning on [page 522](#).

50 Misconceptions

Misconceptions About Graphs

Misconception #1: In a boxplot, the median is in the interquartile range (IRQ) box.

Fact: The IQR is a number that represents the length of the box. The box itself is not the IQR.

Misconception #2: A symmetric boxplot indicates an approximately normal distribution.

Fact: Many differently shaped distributions can have identical boxplots; in general, it is very difficult to say anything about the shape of a distribution from its boxplot.

Misconception #3: When comparing two distributions, it is sufficient to list the center, spread, shape, and unusual features of each distribution.

Fact: *Comparative* language must be used to state which distribution has the greater center and which has the greater spread.

Misconceptions About Linear Regression

Misconception #4: When $r = 0$, there is no relationship between the variables.

Fact: Correlation measures only linearity. When $r = 0$, there is no LINEAR relationship between the variables, but there might be a strong NONLINEAR relationship between these variables.

Misconception #5: When $r = 0.3$, then 30 percent of the variables are closely related.

Fact: When $r^2 = 0.3$, then 30 percent of the variation in the dependent variable can be explained by the linear relationship with the independent variable.

Misconception #6: When $r = 1$, there is a perfect cause-and-effect relationship between the variables.

Fact: Correlation shows association, not cause and effect.

Misconception #7: A correlation close to 1 means that a linear model will give the best fit to the data.

Fact: Curved data can also have a correlation near 1.

Misconception #8: The slope of a linear regression line gives the change in the dependent variable, y , for each unit change in the independent variable, x .

Fact: The slope gives the “predicted,” “estimated,” “expected,” or “average” change in the y -variable for each one unit increase in the x -variable.

Misconception #9: If $\hat{y} = a + bx$, then we can algebraically solve to get $\hat{x} = -\frac{a}{b} + \frac{y}{b}$.

Fact: The variables x and $\hat{y}$ are not the same as $\hat{x}$ and y . The equation that predicts $\hat{y}$ from x does not correctly predict $\hat{x}$ from y .

Misconception #10: If $\hat{y} = a + bx$ and $\hat{x} = c + dy$ are derived from the same set of data, then b and d are reciprocals.

Fact: $b = r \frac{s_y}{s_x}$ and $d = r \frac{s_x}{s_y}$ are not reciprocals.

Misconceptions About Collecting Data

Misconception #11: An experiment cannot be double-blind if anyone knows who gets which treatment.

Fact: Someone always knows, but as long as this person doesn’t interact with the subjects or measure the response, the experiment can still be double-blind.

Misconception #12: There is no difference between a stratified random sample and a randomized block design experiment.

Fact: Although the ideas are similar, as both involve forming groups of similar objects, stratified sampling occurs when taking a sample from a population, while blocking occurs before assigning units to treatments in an experiment.

Misconception #13: Blocks and treatment groups are basically the same.

Fact: Blocks are not formed at random but are formed by grouping together similar units. Each block contains units with a certain characteristic. Treatment groups are formed at random with the goal of making the different groups as similar as possible, other than the treatment received.

Misconceptions About Probability

Misconception #14: The complement rule gives that $0.3^C = 0.7$ or $P(E)^C = 0.7$.

Fact: Probabilities do not have complements; events do. The “C” should appear in the superscript of the event E ; that is, $P(E^C) = 1 - 0.3 = 0.7$.

Misconception #15: If E and F are mutually exclusive, they are independent.

Fact: If E and F are mutually exclusive, $P(E \cap F) = 0$ and $P(E | F) = 0$. If E and F are independent, $P(E \cap F) = P(E)P(F)$ and $P(E | F) = P(E)$. “Mutually exclusive” and “independent” are two very different concepts.

Misconception #16: The probability a student taking AP Statistics owns a graphing calculator is the same as the probability a student who owns a graphing calculator is taking AP Statistics.

Fact: $P(E | F) \neq P(F | E)$. This can be seen as $P(E | F) = \frac{P(E \cap F)}{P(F)}$ while $P(F | E) = \frac{P(E \cap F)}{P(E)}$.

Misconception #17: If a fair coin comes up “heads” five times in a row, by the “law of averages,” the probability that the next toss is “tails” is greater than 0.5.

Fact: There is no “law of averages” or “law of small numbers.” By the “law of large numbers,” in the long run, the proportion of heads will tend toward 0.5. However, coins have no memories, and the probability the next toss comes up “tails” is still 0.5.

Misconception #18: Writing `normalcdf(20, 30, 35, 15) = 0.211` will receive full credit on a Normal calculation question, and writing `binompdf(10, 0.2,`

3) = 0.201 will receive full credit on a Binomial calculation question.

Fact: Do not use “calculator-speak.” You may use standard notation such as $N(35, 15)$ for a normal distribution with mean 35 and standard deviation 15, and you may use $B(10, 0.2)$ for a binomial distribution with 10 trials and probability 0.2.

Misconception #19: If a random process is repeated long enough, eventually the expected value will occur.

Fact: The expected value of a random process may never occur. It is simply a long-run average.

Misconception #20: The expected value of a random variable will always equal one of the possible values of that variable.

Fact: There is no reason why $\sum xP(x)$ should equal one of the x -values. In fact, $\sum xP(x)$ often equals some nonwhole number. Do NOT round the expected value to a whole number after calculation.

Misconceptions About Sampling Distributions

Misconception #21: The larger the sample size, the closer the sample distribution is to a normal distribution.

Fact: The larger the sample size, the closer the sample distribution is to the *population* distribution; the larger the sample size, the closer the *sampling* distribution of the sample mean is to a normal distribution.

Misconception #22: The larger the sample size, the closer the sampling distribution is to the population distribution.

Fact: The larger the sample size, the closer the sampling distribution of the sample mean is to a normal distribution.

Misconception #23: If the sample size is large, the sampling distribution has the same mean as the population and has a standard deviation that is close to the standard deviation of the population.

Fact: No matter what the sample size, the mean of the sampling distribution of $\bar{x}$ equals the population mean, $\mu_{\bar{x}} = \mu$, but the standard deviation of the sampling distribution of $\bar{x}$ equals the population

standard deviation divided by the square root of the sample size,

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

Misconception #24: The larger the sample size, the closer the distribution of any statistic will approximate a normal distribution.

Fact: The *central limit theorem* states that for sufficiently large sample sizes, the sampling distribution of the sample mean can be approximated by a normal distribution. This is not true, for example, for the sampling distribution of statistics such as the sample *maximum* or the sample *range*.

Misconception #25: Accuracy and precision are the same thing.

Fact: Accuracy means that a statistic has low or no bias, while precision means that a statistic has low variability.

Misconception #26: The *central limit theorem* has universal validity; that is, it can always be applied to give normal approximations to appropriate sampling distributions.

Fact: There is a critical condition regarding a large enough sample size that must be satisfied (for the sampling distribution of $\bar{x}$, we use $n \geq 30$).

Misconceptions About Confidence Intervals and Levels

Misconception #27: If (35, 40) is a 95% confidence interval of a population mean, there is a 95% probability that the true mean is between 35 and 40.

Fact: The probability that the population mean is between 35 and 40 is either 0 or 1 depending upon whether or not it is in the interval (35, 40).

Misconception #28: If (35, 40) is a 95% confidence interval of a population mean, 95% of the subjects are between 35 and 40.

Fact: The confidence interval is about a population mean, not about individual values.

Misconception #29: If (35, 40) is a 95% confidence interval of a population mean, 95% of sample means from all such samples will be

between 35 and 40.

Fact: (35, 40) is one specific interval derived from one sample mean and has nothing to do with other samples and other sample means.

Misconception #30: Interpreting a confidence interval is the same as interpreting a confidence level.

Fact: When constructing a confidence interval, the confidence interval must be interpreted in context. Only interpret the confidence level when specifically asked, for example, “What does 95% confidence mean?”

Misconception #31: In general, increasing the sample size makes us more confident.

Fact: In general, the confidence level (for example, 95%) doesn’t change. Increasing the sample size while keeping the same confidence level gives a narrower interval—that is, a *more precise* estimate.

Misconception #32: For the same data, a 90% confidence interval is wider than a 95% confidence interval.

Fact: The 95% confidence interval is wider. If we want higher confidence, we must accept a wider, less precise interval.

Misconception #33: The *Normal/Large Sample condition* can be simplified to saying that the sample must have a normal distribution.

Fact: A finite sample cannot have a normal distribution. When samples are too small for the *central limit theorem* to apply, we look to see if the sample has no outliers and no extreme skewness in order to conclude that it is not unreasonable to say “the sample came from a roughly normal population.”

Misconceptions About Hypothesis Testing

Misconception #34: H_0 and H_a are alternative conclusions.

Fact: Hypotheses testing is a decision-making process in order to reject or fail to reject a null hypothesis, H_0 . You are not deciding that one of H_0 or H_a is true. Your conclusion should be either that there *is* sufficient evidence to reject H_0 (that is, there *is* convincing evidence in

support of H_a) or that there is *not* sufficient evidence to reject H_0 (that is, there is *not* convincing evidence in support of H_a).

Misconception #35: The hypotheses can refer to either the population or the sample.

Fact: The hypotheses **always** refer to the population, never to the sample.

Misconception #36: The null hypothesis and the failure-to-reject region are the same thing.

Fact: The null hypothesis is a claim about a population parameter. The failure-to-reject region is an interval of values for the sample statistic that would indicate insufficient evidence to reject the given null hypothesis claim.

Misconception #37: The choice of alternative hypothesis depends on the sample statistic—that is, whether the sample statistic is positive or negative.

Fact: Both the null and alternative hypotheses are decided before the data are collected. One should never decide if a test is one-sided (and in which direction) or two-sided based upon the data.

Misconception #38: The significance level α is the probability that the null hypothesis is true given that it has been rejected.

Fact: The significance level is a fixed value we use to decide how small a p -value is required to be in order to reject the null hypothesis.

Misconception #39: The significance level α is the probability that the alternative hypothesis is true.

Fact: The significance level is a fixed value we use to decide how small a p -value is required to be in order to reject the null hypothesis.

Misconception #40: A hypotheses test, if correctly performed, establishes the truth of one of the two hypotheses, either the null or the alternative.

Fact: Truth is never established. The conclusion is simply whether or not there is sufficient evidence to reject the null hypothesis.

Misconception #41: Statistical significance and practical significance are the same.

Fact: A statistically significant result might have no practical significance, and a practically significant result might turn out to *not* be statistically significant.

Misconception #42: The probability of a Type I error plus the probability of a Type II error should equal 1.

Fact: The probability of a Type I error, α , is a fixed value giving the threshold value of the hypothesis test, while the probability of a Type II error, β , is different for each possible alternative parameter value.

Misconception #43: A well-planned test of significance should result in a statement either that the null hypothesis is true or that it is false.

Fact: Tests of significance are designed only to measure the strength of evidence against the null hypothesis. They do not establish truth.

Misconceptions About the p -Value

Misconception #44: If the p -value is 0.16, the probability that the null hypothesis is correct is 0.16.

Fact: The p -value of a test is the conditional probability of obtaining a result as extreme as or more extreme than the one obtained assuming the null hypothesis is true.

Misconception #45: When the null hypothesis is rejected, it is because it is not true.

Fact: The null hypothesis is rejected when the p -value is small. The small p -value indicates there is sufficient evidence to doubt the null hypothesis, but it is always possible that we mistakenly reject a true null hypothesis.

Misconception #46: A very large p -value provides convincing evidence that the null hypothesis is true.

Fact: A very large p -value simply says that there is not sufficient evidence to reject the null hypothesis; it does not say that the null hypothesis is true and does not say that we can accept the null hypothesis.

Misconception #47: The p -value is the probability that the observed event has happened by chance.

Fact: This misses that the p -value is a conditional probability. The p -value is the probability of obtaining a statistic as extreme as or more extreme than the one actually observed, by chance alone, when assuming that the null hypothesis is true.

Misconceptions About χ^2 —Tests

Misconception #48: The chi-square statistic, $\sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$, can be calculated using either counts or proportions.

Fact: The chi-square statistic is **always** calculated using integers for the observed counts, and never using proportions for either the observed or expected cells.

Misconception #49: For chi-square homogeneity tests, the null hypothesis is that some observed distributions are identical.

Fact: The hypotheses **never** refer to the sample. The null hypothesis is that the *population* distributions are the same.

Misconception #50: Expected counts should always be rounded to integer values.

Fact: While observed counts are always integers, the expected counts need not be, and students may lose points with this rounding.

50 Common Errors on the AP Exam

Common Errors: Exploratory Analysis

1. When asked to draw a graph, many students forget *labeling* and *keys*, which are critical for stemplots, and *labeling* and *scaling*, which are critical for histograms, boxplots, and scatterplots. For example, if asked to draw parallel boxplots, be sure to label which is which. Proper labeling is more important than getting every point in exactly the right place.

2. When asked to describe a distribution from a graph such as a histogram or dotplot, students address “center, spread, shape, and unusual features” and forget to mention context.
3. When describing a distribution given only a boxplot, students may try to conclude shape like approximate normality or even bell-shaped. In general, shape cannot be concluded from a boxplot; however, a boxplot may suggest symmetry or skewness.
4. When describing center or spread given only a histogram, definitive language like “the median is 120” will be penalized. One needs to use language that conveys uncertainty, such as “the median is approximately 120” or “the median is in the 115–130 interval.”
5. When claiming a value is an outlier or claiming that there are no outliers, some calculation is necessary to communicate how the decision was made, otherwise credit is lost.
6. When asked to *compare* distributions, it is not enough to address center, spread, shape, and unusual features, in context, separately for each distribution. In addition, the characteristics of center and spread must be *compared* using comparative expressions like “greater than” or “less than” or “equal to.”
7. When mentioning skew, students sometimes get the direction wrong. A distribution is skewed in the direction the tail goes, not in the direction where the peak is.
8. When using words with a statistical meaning, incorrect use will be penalized. For example, in statistics, “range” refers to a single number, the maximum minus the minimum. Writing “the data range from 50 to 100” will be marked wrong. You can say that all the data are in the interval from 50 to 100.
9. When combining independent events, variances add, but it is a mistake to add standard deviations.
10. When writing the equation of a regression line, students forget to include the “hat” on the y-variable (or use the word “expected”) and forget to define the variables.
11. When asked if a linear model is appropriate, students need to refer to the residual plot, not to whether or not the correlation is close to ± 1 .

12. When a correlation is close to ± 1 , students incorrectly think correlation implies causation.
13. When explaining the slope of a regression line, students need to use a word like “predicted,” “estimated,” “expected,” or “average” for the change in the y -variable for each one unit increase in the x -variable. Lack of such a nondeterministic word will be penalized.
14. When asked to describe a scatterplot, students address “form, direction, and strength” but forget context.
15. When interpreting a residual, students address the distance between observed and predicted but forget the *direction* of the error—that is, whether the predicted value is an underestimate or overestimate. Remember that the residual is calculated as the observed minus the predicted, not the other way around.

Common Errors: Collecting Data

16. When describing how to select a sample using a random integer generator or a random integer table, students forget to explicitly state that repeated integers are to be ignored and forget to state that the selected integers then correspond to the subjects to be chosen for the sample.
17. When asked about a possible source of bias, students name the bias but are penalized for not clearly describing the problem caused by the bias and giving consequences in context.
18. When looking at data collection, students confuse *stratification* and *blocking*. While both involve dividing into homogeneous groups, stratification is a sampling method while blocking is an experimental design.
19. When looking at data collection, students confuse *stratified* and *cluster* random sampling. Stratified random sampling involves homogeneous groups and random sampling from each strata, while cluster random sampling involves heterogeneous groups and a random sample of clusters.
20. When analyzing experimental design, students confuse *treatment groups* and *blocks*. A treatment group is a randomly formed collection of individuals to which a treatment is applied. A block is a group of

similar experimental units that is then randomly allocated into each of the treatment groups. The goal is for treatment groups to be as similar as possible to each other (other than the treatment), while blocks are chosen to be very different from each other.

21. When looking at data collection, students fail to understand that *random selection* and *random assignment* have different purposes. Random selection in sampling is used to generalize to a population, while random assignment in experiments is used to minimize the effect of confounding variables.
22. When looking at data collection, students may correctly state what variable is a confounding variable but are penalized for not explaining *why* the variable is a confounding variable in the context of the problem.

Common Errors: Probability

23. When calculating probabilities, minor arithmetic errors might not be penalized; however, reporting that a probability is a negative number or a number greater than 1 will always be penalized.
24. When studying two events, students confuse mutually exclusive and independent events. E and F are mutually exclusive if $P(E \cap F) = 0$. E and F are independent if $P(E \cap F) = P(E)P(F)$.
25. When using the conditional probability formula, students should show the numerator and denominator of the probability, not just a reduced fraction and not just a decimal answer.
26. When interpreting probabilities, there is a “law of large numbers” (in the long run, proportions will tend toward a specific probability); however, students incorrectly think there is a “law of averages” or a “law of small numbers.”
27. When calculating probabilities, students confuse “intersection” and “given that.”
28. When calculating probabilities, students confuse $P(E | F)$ with $P(F | E)$.
29. When given a straightforward probability problem, students sometimes incorrectly try to turn it into a full-blown inference problem.

30. When using the normal distribution for probability calculations, students forget to state that they are using a normal distribution and forget to give the values of the two parameters, the mean μ and the standard deviation σ .
31. When using the binomial distribution for probability calculations, students forget to state that they are using a binomial distribution and forget to give the values of the two parameters, the number of trials n and the probability of success p .
32. When using the binomial distribution for cumulative probability calculations, students confuse what should be used for the x -value in $\text{binomcdf}(n, p, x)$ especially with complementary events.
33. When calculating probabilities, students forget to clearly identify the *boundary* and *direction* for the outcome of interest.
34. Do not use “calculator-speak”, but you may use standard notation such as $N(50, 15)$ to identify a normal distribution with mean 50 and standard deviation 15, and $B(15, 0.4)$ to identify a binomial distribution with 15 trials and probability 0.4.
35. When calculating expected values, rounding to integers will often be penalized. Expected values do not have to be integers!
36. When describing a distribution, students are unclear as to whether they are describing a sample distribution, a population distribution, or a sampling distribution.

Common Errors: Inference

37. When beginning an inference procedure, students sometimes mistakenly refer to the *sample* instead of the *population* in stating hypotheses. There is nothing to hypothesize about the sample since the sample results are known exactly.
38. When communicating which procedure is being used, it is generally easier to use *words* rather than writing out a *formula*. In free-response inference questions, more mistakes are made when trying to describe using formulas than when using words.
39. When defining a parameter, do so in context and refer to the population (not mistakenly to the sample!).

40. When a formula is used, it is often safer to start with numbers plugged in and not to include the symbolic formula, because including an incorrect symbol will be penalized.
41. When concluding an inference procedure, students confuse interpreting a confidence *interval* and interpreting a confidence *level*.
42. When concluding an inference procedure, students sometimes mistakenly refer to the sample instead of the population in giving the conclusion. For example, students mistakenly refer to a confidence interval of people who gave a particular answer to a survey question rather than to the people who would have answered in a certain way (only those in the sample actually “answered”).
43. When analyzing an inference procedure, students confuse *parameters* and *statistics*. A parameter is a fixed, usually unknowable, truth about a population such as p , μ , σ , or β , while statistics are measurable values such as $\hat{p}$, $\bar{x}$, s , or b coming from samples. We use sample statistics to make inferences about population parameters.
44. When conducting an inference procedure, students incorrectly use the expression to “prove” or “accept” the null or alternative hypothesis. Conclusions should always be that there is or is not “convincing evidence” of some claim. (Often, the alternative hypothesis is the claim.)
45. When concluding a confidence interval or hypothesis test question about the population mean, students forget to include the word “mean” in the conclusion.
46. When performing an inference procedure, there is no reason to conduct both a hypothesis test and a confidence interval unless specifically asked to. Students will be graded on whichever is the weaker response.
47. When concluding a hypothesis test, students are often incomplete in their conclusions, forgetting to involve linkage to a p -value (a direct comparison of the p -value to α), or forgetting context, or forgetting to refer to the parameter (such as “true proportion” or “true mean”).
48. When asked about interpretations of errors on multiple-choice questions, students confuse the definitions of Type I and Type II

errors.

49. When asked which type of error might have been committed based on the result of a hypothesis test, students often are unable to pick between Type I and Type II.
50. When asked to explain or analyze possible errors, students often cannot provide or explain possible consequences of Type I and Type II errors.

50 AP Exam Hints, Advice, and Reminders

1. There is no penalty for wrong answers on the multiple-choice section, so **answer every question**.
2. For free-response questions, let the graders understand *what* you are doing, *why* you are doing it, and *how* you are doing it.
Communication is just as important as statistical knowledge! Graders want to give you credit—help them! Don't make the readers guess at what you are doing.
3. Read each of the multiple-choice and free-response questions *carefully* while underlining key words and phrases. Be sure you understand **exactly what you are being asked to do, find, or explain**.
4. *Check assumptions*. **Don't just state them!** Be sure the assumptions to be checked are stated correctly. Verifying assumptions and conditions means more than simply listing them with check marks—you must show work or give some reason to confirm verification.
5. Learn and practice how to read *generic computer output*. Linear regression questions are often very straightforward from such output. Also note that when answering questions using computer output, you usually will **not** need to use all the information given.
6. Show where answers come from. *Naked or bald answers* will receive little or **no** credit! If you are using a formula, start with showing the numbers substituted rather than writing the formula with symbols, which is easy to mess up. Also, don't give more than one solution to the same problem—you will receive credit only for the weaker one.

7. Sketch, with labeled axes and numbered scales when possible, any graph to which you refer. This includes histograms, boxplots, stemplots, scatterplots, residuals plots, normal probability plots, or any other kind of graph. It is not enough to simply say, “I did a normal probability plot of the residuals on my calculator and it looked linear.”
8. **Use proper terminology!** For example, the language of experiments is different from the language of observational studies—you shouldn’t mix up *blocking* and *stratification*. Know what *confounding* means and when it is proper to use this term. Words like *bias*, *correlation*, *normal*, *power*, *range*, *skew*, and even *statistic* have specific statistical meanings and should not be used colloquially. Don’t use statistical vocabulary unless you use it correctly.
9. **Avoid “calculator-speak”!** For example, do not write “2-SampTTest...” for “2-sample t-test for the difference between two population means.”
10. When part of a question says to use results from previous parts of the question, you must explicitly refer to the previous parts in your answer.
11. Write null and alternative hypotheses in terms of the population parameter. If the question refers to sample data, don’t automatically parrot the stem of the problem in your hypotheses.
12. When describing a one-variable distribution, you need to mention context, but you need only to mention it once, either with regard to shape, center, or spread.
13. Approach each problem systematically. Some problems look scary on first reading but are not overly difficult and are surprisingly straightforward. Other questions might take you beyond the scope of the AP curriculum; however, remember that they will be phrased in ways that you should be able to answer them based on what you have learned in your AP Statistics class.
14. Read through all four free-response questions, underlining key points. Go back and answer the questions you think are easiest (this will usually, but not always, include Question 1). Finally, try the remaining problems.

- 15.** Be sure that your methods, reasoning, and calculations are clear to the reader and that the explanations and conclusions are given *in the context of the problem*. Answers do not have to be in paragraph form. Symbols and algebra are fine.
- 16.** Show calculations carefully. A wrong answer due to a computational error might still result in full credit.
- 17.** For any part of a multistep problem that you can't answer but a solution is needed in a later step, make up a reasonable answer. Later steps might be easy, so don't give up.
- 18.** State the values that you are substituting into any formula that you use. You do not have to write down the actual formula.
- 19.** Don't waste time erasing. Cross out wrong answers and draw arrows to help the reader follow your work. You are given lots of space to answer, but be succinct—there's no reason to fill in all the space you are given. When you have answered the question, stop writing!
- 20.** Read carefully and recognize that sometimes very different hypothesis tests or very different approaches are required in different parts of the same problem.
- 21.** Realize that there may be several reasonable approaches to a given problem. In such a case, pick the one with which you are most comfortable or the one you feel will require the least amount of time to complete.
- 22.** Realize that there may not be one clear, correct answer. Any of several different answers may receive full credit if they are backed up with well-reasoned arguments and explanations.
- 23.** Punching a long list of data into the calculator does not show statistical knowledge. Be sure it is necessary!
- 24.** Remember that uniform and symmetric are not the same, and that not all symmetric unimodal distributions are normal.
- 25.** Boxplots show center (median), variability (IQR and range), and outliers but not shape. Differently shaped distributions can have identical boxplots.
- 26.** Use comparative words like “more than,” “less than,” “equal to,” “same as,” or “different from” when comparing distributions. A

“laundry list” of *each* distribution’s shape, center, spread, and unusual features is not enough.

27. Do not refer to the “law of averages” or “law of small numbers” since there are no such things. There *is* a “law of large numbers,” which refers to the relative frequency of an event becoming closer to the true probability of the event as the experiment is performed more and more times.
28. Comment on whether or not there are nonlinear patterns, increasing or decreasing spread, and whether the residuals are small or large compared with the different x -values when describing residual plots. “Randomly scattered” is not the same as “half below and half above.”
29. Know how to *interpret* the slope, the y -intercept, and the coefficient of determination in context when using a calculator or computer printout to find a regression line.
30. Think about the smallest collection of things to which a single treatment is applied when identifying experimental units.
31. Understand that there are two types of control groups—experimental units that receive a placebo and experimental units that receive no treatment at all.
32. Explain how you will “randomly assign” subjects to treatment groups when asked to design an experiment. For example, you must state that you will use and how you will use a random number table or a random number generator on a calculator.
33. Follow these three steps when using a random number generator to pick a sample: (1) number the subjects; (2) define the interval of values from which integers are being selected, indicate that repeats are to be ignored, and note the stopping point; and (3) state that the sample is the group of subjects associated with the selected integers.
34. Understand that the expected value of a random variable does *not* mean the value that is expected to occur. In fact, it may have probability 0 of ever occurring. It is simply a long-run average that is somewhere between the lowest and highest values of the random variable.
35. Know the difference between a *sample distribution* and a *sampling distribution*. Do not state that the sample distribution will be close to

normal because it is large! The larger the sample, the closer the *sample distribution* is to the population distribution. On the other hand, the larger the sample size, the closer the *sampling distribution* of the sample mean is to a normal distribution (due to the central limit theorem).

36. Look carefully at the original data scatterplot together with the residual plot as they often give critical insights in understanding a regression question.
37. Determine for inference problems whether a variable is *categorical* (leading to proportions or chi-square) or *quantitative* (leading to means or linear regression), whether you are given *raw data* or *summary statistics*, whether there is a *single population* of interest or *two populations* being compared, and, in the case of comparison, whether there are *independent samples* or a *paired comparison*.
38. Always define parameters! For example: Let p be the proportion of the population of all AP students who plan to major in statistics when they go to college.
39. Use clear subscripts if there are two populations. For example, μ new drug and μ old drug.
40. Follow four steps when answering a confidence interval problem: (1) identify the procedure by name or formula (usually name is easier); (2) check conditions (do not just state them); (3) calculate the confidence interval (calculator software does this quickly); and (4) state a conclusion with reference to the confidence level, the parameter, and the population, and use context.

NOTE

In general with inference, do not show work with formulas as it is easy to mess this up.

41. Follow five steps when answering a hypothesis test problem: (1) state the hypotheses in terms of population parameters (don't forget to define the parameters); (2) state the test by name or formula (usually by name is easier); (3) check conditions (do not just state them); (4)

calculate the test statistic and associated p -value (calculator software does this quickly); and (5) state a decision based upon the p -value and write a conclusion *in context* about the alternative hypothesis with reference to the parameter and the population.

42. Follow five steps when answering a chi-square problem: (1) state the hypotheses (both H_0 and H_a in context); (2) name the test (independence or homogeneity), which can often be inferred from the way the hypotheses are stated; (3) check the conditions, which requires showing the expected values (they can be copied from the calculator); (4) report the resulting chi-square value, the p -value, and df , the number of degrees of freedom; and (5) give a conclusion in context with linkage to the p -value.
43. Communicate that there is convincing evidence to support the *alternative hypothesis* if the decision is to reject the null hypothesis.
44. Know which type of error, Type I or Type II, might have been committed based on the conclusion to a hypothesis test, and be able to give a possible reasonable consequence of committing that error.
45. Use nondeterministic language like “on average,” “estimated,” “expected,” and “predicted.” Use words like “approximately” often, especially in front of “normal.”
46. When asked to calculate a confidence interval, you need a conclusion in context about the *interval*; *however*, you do not need to interpret the confidence *level* unless specifically asked.
47. When performing a hypothesis test, you need to *calculate* a p -value, but you do not need to *interpret* the p -value unless specifically asked.
48. Describe both an advantage of one thing and a disadvantage of the other if asked why one option is better than another.
49. In a yes or no question, be sure to answer yes or no, in addition to your explanation.
50. Follow directions on free-response questions. They often have a flow leading to solutions.

carefully vetted by AP Statistics teachers who have given their feedback with regard to fairness, balance, and rigor.

- ¹Oxfam International, January 16, 2023.
- ²Talberth, Cobb, and Slattery, “Gross Domestic Product, A Tool for Sustainable Development,” February 2007.
- ³“U.S. Prison Boom: Impact on Age of State Prisoners,” *Journalist’s Resource*, February 2016.
- ⁴“The Use of Incarceration in the United States,” *American Society of Criminology*, November 2000.
- ⁵“SAT Scores and Income Inequality: How Wealthier Kids Rank Higher,” *Wall Street Journal*, October 2014.
- ⁶*Batson v. Kentucky*, 476 U.S. 79 (1986).
- ⁷“A Vast Racial Gap in Death Penalty Cases,” Adam Liptak, *New York Times*, Aug. 3, 2020.
- ⁸Laperruque, Jake, “Unmasking the Realities of Facial Recognition,” in POGO (Project on Government Oversight), December 5, 2018.
- ⁹Wagner, Peter, and Wanda Bertram, “What percent of the U.S. is incarcerated?”, in Prison Policy Initiative, January 16, 2020.
- ¹⁰National Association of Criminal Defense Lawyers and Innocence Project Analysis of FBI and Justice Department Data as of March 2015.
- ¹¹<http://www.innocenceproject.org>
- ¹²<http://www.f-4-c.org/statistics/>
- ¹³“The Data of Hate,” *New York Times*, July 13, 2014.
- ¹⁴“Bias Breakdown,” www.fbi.gov, November 2015.
- ¹⁵Nansel, T. R., Overpeck, M., Pilla, R. S., Ruan, W. J., Simons-Morton, B., & Scheidt, P. “Bullying Behaviors Among U.S. Youth: Prevalence and Association with Psychosocial Adjustment,” *Journal of the American Medical Association*, 285(16), 2094–2100.
- ¹⁶National Center for Educational Statistics, 2015.
- ¹⁷<https://www.cdc.gov/nchs/fastats/suicide.htm>
- ¹⁸<https://www.cnn.com/2021/11/26/world/us-gun-culture-world-comparison-intl-cmd/index.html>
- ¹⁹<https://qz.com/1095899/gun-ownership-in-america-in-three-charts/>
- ²⁰2015 Virginia Tech Study.
- ²¹<http://flintwaterstudy.org>
- ²²CDC 24/7: “Saving Lives, Protecting People.”
- ²³“Minorities who ‘Whiten’ Job Resumes Get More Interviews,” *Working Knowledge*, Harvard Business School, May 17, 2017.
- ²⁴<https://jamanetwork.com/journals/jamapediatrics/article-abstract/2807325>
- ²⁵“LEAD: Reducing the Role of Criminalization in Local Drug Control,” <http://www.drugpolicy.org>
- ²⁶<https://www.nbcbayarea.com/news/local/racial-profiling-rampant-in-berkeley-police-department-report/103331/>
- ²⁷<https://www.sciencedaily.com/releases/2012/06/120604142726.htm>
- ²⁸DeStefano, Price, and Weintraub, “Increasing Exposure to Antibody-Stimulating Proteins and Polysaccharides in Vaccines Is Not Associated with Risk of Autism,” 2013.

Part 4

Practice Tests

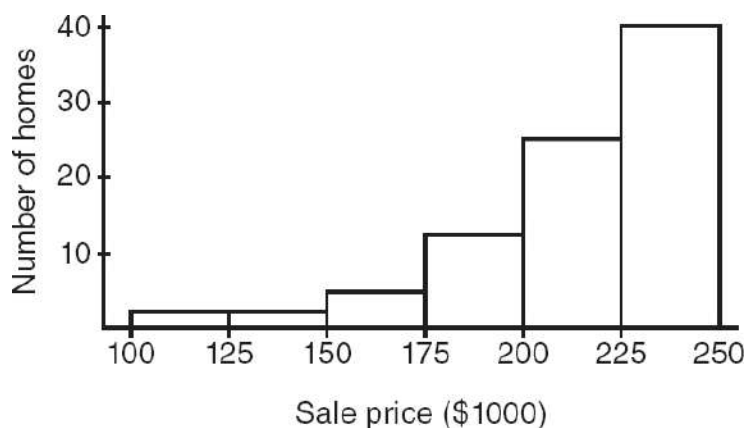
Practice Test 1

Section I: Questions 1–42

SPEND 90 MINUTES ON THIS PART OF THE EXAM.

DIRECTIONS: The questions or incomplete statements that follow are each followed by four suggested answers or completions. Choose the response that best answers the question or completes the statement.

1. The following is a histogram of 87 home sale prices (in thousands of dollars) in one community:



Which of the following could be the median of the sale prices?

- (A) \$175,000
- (B) \$187,500
- (C) \$212,500

(D) \$237,500

2. Which of the following is most useful in establishing cause-and-effect relationships?

(A) A complete census

(B) A least squares regression line showing high correlation

(C) A simple random sample

(D) An experiment

3. The average height of ancient Egyptian men was 5'6". Assuming a roughly normal distribution of heights, with a standard deviation of 2.5", would Pharaoh Sa-Nakht's height of 6'1" be considered an outlier using our standard definition of outliers? ($\text{invNorm}(0.75) = 0.6745$)

(A) Yes, because $6'1" > 5'6" + 2(0.6745)(2.5")$

(B) Yes, because $6'1" > 5'6" + 3(0.6745)(2.5")$

(C) Yes, because $6'1" > 5'6" + 4(0.6745)(2.5")$

(D) No, because $6'1" < 5'6" + 3(2.5")$

4. To determine the average cost of running for a congressional seat, a simple random sample of 50 politicians is chosen and the politicians' records examined. The cost figures show a mean of \$1,825,000 with a standard deviation of \$132,000. Which of the following is the best interpretation of a 90% confidence interval estimate for the average cost of running for office?

(A) Of all politicians running for a congressional seat, 90% spend between \$1,794,000 and \$1,856,000.

(B) Of all politicians running for a congressional seat, 90% spend a mean dollar amount that is between \$1,794,000 and \$1,856,000.

(C) We are 90% confident that politicians running for a congressional seat spend between \$1,794,000 and \$1,856,000.

(D) We are 90% confident that politicians running for a congressional seat spend a mean dollar amount between

\$1,794,000 and \$1,856,000.

5. In one study on the effect that eating meat products has on weight level, a simple random sample (SRS) of 500 subjects who admitted to eating meat at least once a day had their weights compared with those of an independent SRS of 500 people who claimed to be vegetarians. In a second study, an SRS of 500 subjects was served at least one meat meal per day for 6 months, while an independent SRS of 500 others were chosen to receive a strictly vegetarian diet for 6 months, with weights compared after 6 months. Which of the following is a true statement?
- (A) The first study is a controlled experiment, while the second is an observational study.
 - (B) The first study is an observational study, while the second is a controlled experiment.
 - (C) Both studies are controlled experiments.
 - (D) Both studies are observational studies.
6. A plumbing contractor obtains 60% of her boiler circulators from a company whose defect rate is 0.005 and the rest from a company whose defect rate is 0.010. If a circulator is defective, what is the probability that it came from the first company?
- (A) $\frac{(0.6)(0.005)}{(0.6)(0.005) + (0.4)(0.010)}$
 - (B) $\frac{0.6}{(0.6)(0.005) + (0.4)(0.010)}$
 - (C) $\frac{0.005}{(0.6)(0.005) + (0.4)(0.010)}$
 - (D) $(0.6)(0.005)$
7. A kidney dialysis center periodically checks a sample of its equipment and performs a major recalibration if readings are sufficiently off target. Similarly, a fabric factory periodically checks the sizes of towels coming off an assembly line and halts production if measurements are sufficiently off target. In both situations, we have the null hypothesis that the equipment is performing satisfactorily. For

each situation, which is the *more* serious concern, a Type I or Type II error?

- (A) Dialysis center: Type I error; towel manufacturer: Type I error
- (B) Dialysis center: Type I error; towel manufacturer: Type II error
- (C) Dialysis center: Type II error; towel manufacturer: Type I error
- (D) Dialysis center: Type II error; towel manufacturer: Type II error

8. A coin is weighted so that the probability of heads is 0.75. The coin is tossed 10 times, and the number of heads is noted. This procedure is repeated a total of 50 times, and the number of heads is recorded each time. What kind of distribution has been simulated?

- (A) The sampling distribution of the sample proportion with $n = 10$ and $p = 0.75$
- (B) The sampling distribution of the sample mean with $\bar{x} = 10(0.75)$ and $\sigma = \sqrt{10(0.75)(0.25)}$
- (C) The binomial distribution with $n = 10$ and $p = 0.75$
- (D) The binomial distribution with $n = 50$ and $p = 0.75$

Questions 9–11 refer to the following:

Past studies indicate that the probability of a tropical cyclone becoming a hurricane is 0.61. Assume that the probability of any cyclone becoming a hurricane is independent of what happens to any other cyclone.

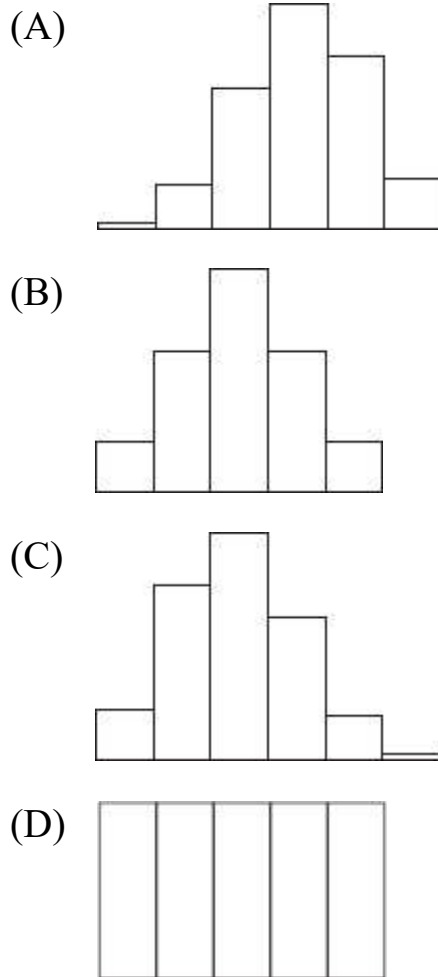
9. If there are five cyclones in one year, what is the probability that at least three become hurricanes?

- (A) $\binom{5}{3}(0.61)^3(0.39)^2$
- (B) $(0.61)^3(0.39)^2 + (0.61)^4(0.39) + (0.61)^5$
- (C) $1 - [(0.39)^5 + 5(0.61)(0.39)^4 + 10(0.61)^2(0.39)^3 + 10(0.61)^3(0.39)^2]$
- (D) $\binom{5}{3}(0.61)^3(0.39)^2 + \binom{5}{4}(0.61)^4(0.39) + (0.61)^5$

10. For years with five cyclones, what is the expected number of those that become hurricanes?

- (A) $5(0.39)$
- (B) $5(0.61)$
- (C) $5(0.39)(0.61)$
- (D) $\sqrt{5(0.39)(0.61)}$

11. Which of the following histograms best illustrates a binomial probability distribution with $n = 5$ and $p = 0.61$?



12. Given the probabilities $P(A) = 0.3$ and $P(B) = 0.2$, what is the probability of the union $P(A \cup B)$ if A and B are mutually exclusive? If A and B are independent? If B is a subset of A ?

- (A) 0.5, 0.44, 0.3
- (B) 0.5, 0.44, 0.2
- (C) 0.44, 0.5, 0.3
- (D) 0, 0.5, 0.3

13. A social science researcher is testing the claim that 69 percent of Americans use social media every day. In a survey involving a simple random sample (SRS) of 1,100 Americans, 770 say that they use social media every day. With $H_0: p = 0.69$ and $H_a: p \neq 0.69$, what is the value of the test statistic?

- (A) $z = \frac{0.7 - 0.69}{\sqrt{1,100(0.69)(1 - 0.69)}}$
- (B) $z = 2 \frac{0.7 - 0.69}{\sqrt{1,100(0.69)(1 - 0.69)}}$
- (C) $z = \frac{0.7 - 0.69}{\sqrt{\frac{(0.69)(1 - 0.69)}{1,100}}}$
- (D) $z = 2 \frac{0.7 - 0.69}{\sqrt{\frac{(0.69)(1 - 0.69)}{1,100}}}$

14. A medical research team tests for tumor reduction in a sample of patients using three different dosages of an experimental cancer drug. Which of the following is true?

- (A) Each explanatory level has an associated level of response.
- (B) Tumor reduction is the only explanatory variable, but there are three response variables corresponding to the different dosages.
- (C) There are three explanatory variables and one response variable.
- (D) There are three levels of a single explanatory variable.

15. Suppose that a study was carried out in which each CEO in a random sample of U.S. electric utility companies was asked whether he or she thought the federal government is doing enough to safeguard against a cyberattack on the electric grid. Which of the following is a correct

understanding of the type of investigative question this study might address?

- (A) The data could be used to estimate a population mean since the data collected are numerical.
- (B) The data could be used to estimate a population proportion since the data collected are numerical.
- (C) The data could be used to estimate a population mean since the data collected are categorical.
- (D) The data could be used to estimate a population proportion since the data collected are categorical.

16. Mr. Bee's statistics class had a standard deviation of 11.2 on a standardized test, while Mr. Em's class had a standard deviation of 5.6 on the same test. Which of the following is the most reasonable conclusion concerning the two classes' performance on the test?

- (A) Mr. Bee's class is less heterogeneous than Ms. Em's.
- (B) Ms. Em's class is more homogeneous than Mr. Bee's.
- (C) Mr. Bee's class performed twice as well as Ms. Em's.
- (D) Ms. Em's class did not do as well as Mr. Bee's.

17. A college admissions officer is interested in comparing the SAT Math scores of high school applicants who have and have not taken AP Statistics. She randomly pulls the files of five applicants who took AP Statistics and five applicants who did not, and proceeds to run a t -test to compare the mean SAT Math scores of the two groups. Which of the following is a necessary assumption?

- (A) The population variances from each group are known.
- (B) The population variances from the two groups are equal.
- (C) The population of all SAT scores from each group is roughly normally distributed.
- (D) The samples must be independent simple random samples, and for each sample, np and $n(1 - p)$ must both be at least 10.

18. The appraised values of houses in a city have a mean of \$195,000 with a standard deviation of \$23,000. Because of a new teachers' contract, the school district needs an extra 10% in funds compared with the previous year. To raise this additional money, the city instructs the assessment office to raise all appraised house values by \$5,000. What will be the new standard deviation of the appraised values of houses in the city?

- (A) \$23,000
- (B) \$25,300
- (C) \$28,000
- (D) \$30,300

19. Past studies indicate that 20% of tennis players say they would give up their life savings to be able to play in a grand slam. A tournament director believes that the proportion might be greater than 0.20. A random sample of players was surveyed. With $H_0: p = 0.20$ and $H_a: p > 0.20$, the p -value of the test is 0.041. Which of the following is a correct interpretation of the p -value?

- (A) 0.041 is the probability that the null hypothesis is false.
- (B) 0.041 is the probability of obtaining a sample proportion as large as or larger than the one obtained by the tournament director.
- (C) If it is true that 20% of tennis players would give up their life savings to be able to play in a grand slam, 0.041 is the probability of obtaining a sample proportion equal to or greater than 0.20.
- (D) If it is true that 20% of tennis players would give up their life savings to be able to play in a grand slam, 0.041 is the probability of obtaining a sample proportion as large as or larger than the one obtained by the tournament director.

20. A talk show host recently reported that in response to his on-air question, 82% of the more than 2,500 email messages received through his publicized address supported the death penalty for anyone convicted of selling drugs to children. What does this show?

- (A) The survey is meaningless because of voluntary response bias.
- (B) No meaningful conclusion is possible without knowing something more about the characteristics of his listeners.
- (C) The survey would have been more meaningful if he had picked a random sample of the 2,500 listeners who responded.
- (D) This was a legitimate sample, randomly drawn from his listeners, and of sufficient size to be able to conclude that most of his listeners support the death penalty for such a crime.

21. At a certain bakery, the number of bagels purchased by customers is a random variable with a standard deviation of 1.5. Which of the following is the best interpretation of the standard deviation?

- (A) The number of bagels purchased by a random sample of customers is expected to be 1.5 from the mean.
- (B) The number of bagels purchased by a randomly selected customer will be 1.5 from the mean.
- (C) On average, the number of bagels purchased by customers varies from the mean by 1.5.
- (D) On average, the number of bagels purchased by customers will be 1.5 standard deviations from the mean.

22. Which of the following can affect the value of the correlation r ?

- (A) A change in measurement units
- (B) A change in which variable is called x and which is called y
- (C) Adding the same constant to all values of the x -variable
- (D) None of the above can affect the r -value.

23. The American Medical Association (AMA) wishes to determine the percentage of obstetricians who are considering leaving the profession because of the rapidly increasing number of lawsuits against obstetricians. The AMA would like an answer to within $\pm 3\%$ at the 95% confidence level. Which of the following should be used to find the sample size (n) needed?

- (A) $1.645\sqrt{\frac{0.5}{n}} \leq 0.03$
- (B) $1.96\sqrt{\frac{(0.5)(0.5)}{n}} \leq 0.03$
- (C) $1.645\sqrt{\frac{(0.5)(0.5)}{n}} \leq 0.06$
- (D) $1.96\sqrt{\frac{0.5}{n}} \leq 0.03$

24. Suppose you did 10 independent tests of the form $H_0: \mu = 25$ versus $H_a: \mu < 25$, each at the $\alpha = 0.05$ significance level. What is the probability of committing a Type I error and incorrectly rejecting a true H_0 with at least one of the 10 tests?

- (A) 0.05
- (B) 0.40
- (C) 0.60
- (D) 0.95

25. Which of the following sets has the smallest standard deviation? Which has the largest?

- I. 1, 2, 3, 4, 5, 6, 7
- II. 1, 1, 1, 4, 7, 7, 7
- III. 1, 4, 4, 4, 4, 4, 7

- (A) Smallest SD: I; largest SD: II
- (B) Smallest SD: II; largest SD: III
- (C) Smallest SD: III; largest SD: I
- (D) Smallest SD: III; largest SD: II

26. A researcher plans a study to examine long-term confidence in the U.S. economy among the adult population. She obtains a simple random sample of 30 adults as they leave a Wall Street office building one weekday afternoon. All but two of the adults agree to participate in the survey. Which of the following conclusions is correct?

- (A) Proper use of chance as evidenced by the simple random sample makes this a well-designed survey.
- (B) The high response rate makes this a well-designed survey.
- (C) Selection bias makes this a poorly designed survey.
- (D) Lack of anonymity makes this a poorly designed survey.

27. Which among the following would result in the narrowest confidence interval?

- (A) Small sample size and 95% confidence
- (B) Small sample size and 99% confidence
- (C) Large sample size and 95% confidence
- (D) Large sample size and 99% confidence

28. In a random sample of 500 working adults, heights x (in inches) and annual salaries y (in dollars) are tabulated. When the least-squares regression analysis is performed, the correlation is 0.38. Which of the following is the correct way to label the correlation?

- (A) 0.38
- (B) 0.38 dollars per inch
- (C) 0.38 inches per dollar
- (D) 0.38 dollar inches

29. There are two games involving flipping a fair coin. In the first game, you win a prize if you can throw between 45% and 55% heads. In the second game, you win if you can throw more than 80% heads. For each game, would you rather flip the coin 30 times or 300 times?

- (A) The outcomes of the games do not depend on the number of flips.
- (B) 300 times for each game
- (C) 30 times for the first game and 300 times for the second
- (D) 300 times for the first game and 30 times for the second

30. City planners are trying to decide among various parking plan options ranging from more on-street spaces to multilevel facilities to spread-out small lots. Before making a decision, they wish to test the downtown merchants' claim that shoppers park for an average of only 47 minutes in the downtown area. The planners have decided to tabulate parking durations for 225 shoppers and to reject the merchants' claim if the sample mean exceeds 50 minutes. If the merchants' claim is wrong and the true mean is 51 minutes, what is the probability that the random sample will lead to a mistaken failure to reject the merchants' claim? Assume that the standard deviation in parking durations is 27 minutes.

(A) $P\left(z < \frac{50 - 47}{27/\sqrt{225}}\right)$

(B) $P\left(z > \frac{50 - 47}{27/\sqrt{225}}\right)$

(C) $P\left(z < \frac{50 - 51}{27/\sqrt{225}}\right)$

(D) $P\left(z > \frac{50 - 51}{27/\sqrt{225}}\right)$

31. Consider the following parallel boxplots indicating the starting salaries (in thousands of dollars) for blue-collar and white-collar workers at a particular production plant:



Which of the following is a correct conclusion?

- (A) The ranges of the distributions are the same.
- (B) In each distribution, the mean is equal to the median.
- (C) Each distribution is symmetric.

(D) The distributions are outliers of each other.

32. The mean Law School Aptitude Test (LSAT) score for applicants to a particular law school is 650 with a standard deviation of 45. Suppose that only applicants with scores above 700 are considered. What percentage of the applicants considered have scores below 740? (Assume the scores are approximately normally distributed.)

(A) 17.1%

(B) 82.9%

(C) 86.7%

(D) 97.7%

33. It has been reported that 1 in 3 older adults struggle to meet basic needs. A health statistician believes that this problem is even worse today and would like to create a study to test her belief. Which of the following would be an appropriate component for the health statistician's investigative question?

(A) The parameter is the mean number of older adults who struggle to meet basic needs, and the alternative hypothesis is greater than $\frac{1}{3}$.

(B) The parameter is the mean number of older adults who struggle to meet basic needs, and the alternative hypothesis is not equal to $\frac{1}{3}$.

(C) The parameter is the proportion of all sampled older adults who struggle to meet basic needs, and the alternative hypothesis is greater than $\frac{1}{3}$.

(D) The parameter is the proportion of all older adults who struggle to meet basic needs, and the alternative hypothesis is greater than $\frac{1}{3}$.

34. A researcher planning a survey of school principals in a particular state has lists of the school principals employed in each of the 125 school districts. The procedure is to obtain a random sample of principals from each of the districts rather than grouping all the lists

together and obtaining a sample from the entire group. Which of the following is a correct conclusion?

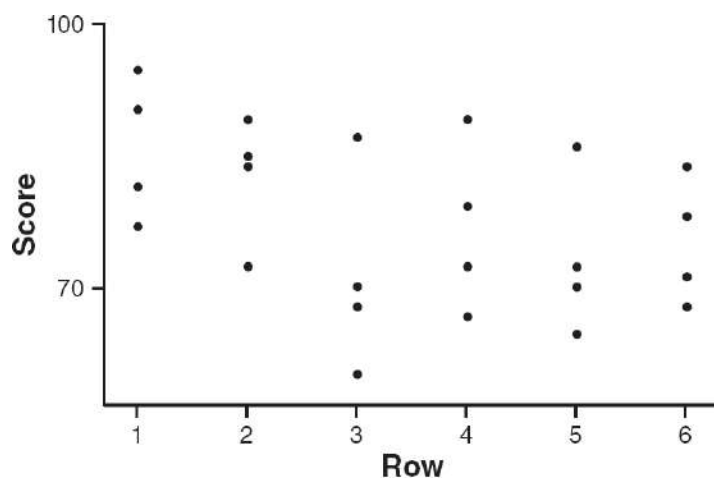
- (A) This is a simple random sample obtained in an easier and less costly manner than procedures involving sampling from the entire population of principals.
- (B) This is a cluster random sample in which the population was divided into heterogeneous groups called clusters.
- (C) This is an example of systematic random sampling, which gives a reasonable sample as long as the original order of the list is not related to the variables under consideration.
- (D) This is a stratified random sample, which may give comparative information that a simple random sample wouldn't give.

35. A simple random sample is defined by

- (A) the method of selection.
- (B) examination of the outcome.
- (C) how representative the sample is of the population.
- (D) the size of the sample versus the size of the population.

Questions 36–38 refer to the following situation:

A teacher randomly assigns students to seat locations and then notes final test scores versus assigned row. Some computer output of the regression analysis follows.



Predictor	Coef	SE Coef
Constant	86.267	3.665
Row	-1.9214	0.9410
S = 7.87264 R-Sq = 15.9%		

36. The association is weak, negative, and linear. What does “negative” mean in this context?
- (A) If a student is seated in a lower-numbered row than a second student, the first student will have a higher test score than the second student.
 - (B) Students sitting in lower-numbered rows tend to have higher test scores.
 - (C) A student moving to the next lower-numbered row will have an increased test score.
 - (D) The points in the scatterplot fall on a straight line sloping downward.
37. If a student moves a row closer to the front, what does the linear model predict as a change in that student’s final exam score?
- (A) A decrease of 0.9410
 - (B) A decrease of 1.9214
 - (C) An increase of 1.9214

(D) An increase of 7.87264

38. One student in row 4 scored 80 on the final exam. Which of the following is a correct interpretation of the residual?

- (A) The linear model underestimated the student's score by 1.4186.
 - (B) The linear model overestimated the student's score by 1.4186.
 - (C) The linear model underestimated the student's score by 10.031.
 - (D) The linear model overestimated the student's score by 10.031.
-

39. Which of the following is a true statement about hypothesis testing?

- (A) If there is sufficient evidence to reject a null hypothesis at the 10% level, there is sufficient evidence to reject it at the 5% level.
- (B) Whether to use a one- or a two-sided test is typically decided after the data are gathered.
- (C) If a hypothesis test is conducted at the 1% level, there is a 1% chance of rejecting the null hypothesis.
- (D) The power of a test concerns its ability to detect an alternative hypothesis.

40. A population is normally distributed with mean 25. Consider all samples of size 10. The variable $\frac{\bar{x} - 25}{\frac{s}{\sqrt{10}}}$

- (A) has a normal distribution.
- (B) has a t -distribution with $df = 10$.
- (C) has a t -distribution with $df = 9$.
- (D) has either a normal or a t -distribution depending on the characteristics of the population standard deviation.

41. A correlation of 0.6 indicates that the percentage of variation in y that is explained by the variation in x is how many times the percentage indicated by a correlation of 0.3?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

42. In a high school of 1,650 students, 132 have personal investments in the stock market. To estimate the total stock investment by students in this school, two plans are proposed. Plan I would sample 30 students at random, find a confidence interval estimate of their average investment, and then multiply both ends of this interval by 1,650 to get an interval estimate of the total investment. Plan II would sample 30 students at random from among the 132 who have investments in the market, find a confidence interval estimate of their average investment, and then multiply both ends of this interval by 132 to get an interval estimate of the total investment. Which is the better plan for estimating the total stock market investment by students in this school?

- (A) Plan I
- (B) Plan II
- (C) Both plans use random samples and so will produce equivalent results.
- (D) The resulting data must be seen to evaluate which is the better plan.

STOP

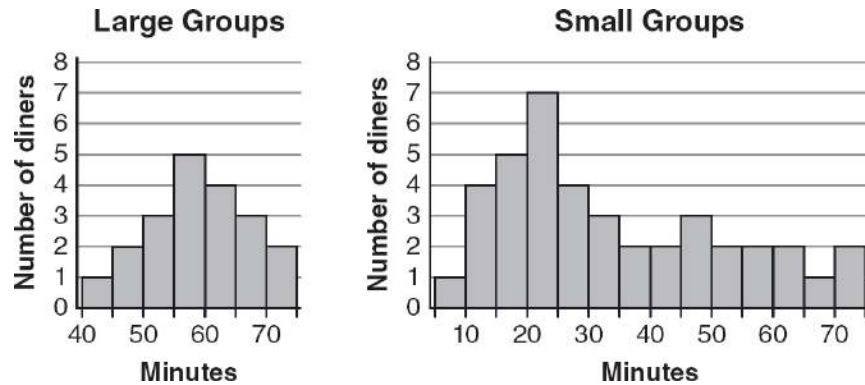
If there is still time remaining, you may check your work on this section.

Section II

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your results and explanations.

1. When students do not understand an author's vocabulary, they often miss subtleties of meaning in the text; thus, learning vocabulary is recognized as one of the most important skills in all subject areas. A school system is planning a study to see which of two programs is more effective in ninth grade. Program A involves teaching students to apply morphemic analysis, which is the process of deriving a word's meaning by analyzing its meaningful parts, such as roots, prefixes, and suffixes. Program B involves teaching students to apply contextual analysis, which is the process of inferring the meaning of an unfamiliar word by examining the surrounding text. To comply with informed consent laws, parents will be asked to give consent for their children to be randomly assigned to be taught in either one of the new programs or using the local standard method. Ninety students will be randomly selected from ninth-grade students whose parents returned the form giving their consent.

- A. **Identify** and **classify** the variable(s) in the study.
 - B.
 - i. **Explain** the purpose of incorporating a control group in this experiment.
 - ii. **Explain** why students without parental consent should not be included in this experiment, not even in the control group.
 - C.
 - i. **Explain** how teachers could be a confounding variable.
 - ii. Suppose the students were allowed to select which instructional program they preferred to take. **Explain** how that self-selection process might jeopardize a statistically valid comparison between the programs of study. Provide a specific example to support your answer.
 - D. **Determine** whether it is or is not reasonable to generalize the results of this study beyond the 90 participants.
 - E. The school would like to compare the performance of students taking Programs A and B using national standardized tests. **Determine** a valid investigative question. Ensure all three components of a valid investigative question are included in your response.
2. A diner manager is interested in the length of time customers spend at their table for dinner. He randomly samples 20 large groups and 40 small groups. The following histograms summarize the times spent.



[Follow for extended description](#)

- A.
 - i. **Identify** the type of study the manager conducted.
 - ii. **Identify** the categorical variable in the manager's study.
 - B.
 - i. **Justify** whether the proportion of diners in large groups spending over 70 minutes is less than, equal to, or greater than the proportion of diners in small groups spending over 70 minutes.
 - ii. **Calculate** how many of the diners spent over 60 minutes at their tables.
 - C.
 - i. **Compare** the shapes of the two distributions.
 - ii. **Compare** the centers and spreads of the two distributions.
 - D. Suppose the data from the large and small groups are combined.
 - i. **Describe** the shape of the histogram of the combined data.
 - ii. **Estimate** the median time that the combined 60 groups spend at their table for dinner.
 - iii. **Estimate** the range of the combined data.
 - E. **Determine** the population for a valid investigative question for the manager's study.
3. A new anti-spam software program is field-tested on 250 randomly selected spam emails and on 750 randomly selected legitimate emails. The results are summarized in the following table (positive test = program labels email as spam):

	Spam	Legitimate	Total
Positive test	205	90	295
Negative test	45	660	705
Total	250	750	1,000

The software developer claims that the proportion of all spam emails for which the tests are positive equals the proportion of all legitimate emails for which the tests are negative.

- A. **Identify** and **classify** the variable(s) in this study.
 - B. **Identify** the appropriate inference procedure.
 - C. **Complete** the inference procedure, including **calculating** the appropriate statistics.
 - D. **Justify** a conclusion in context.
4. Cumulative exposure to nitrogen dioxide (NO_2) is a major risk factor for lung disease in tunnel construction workers. In one study, researchers compared cumulative exposure to NO_2 (in parts per million per year, ppm/yr) for a random sample of drill and blast workers and for an independent random sample of concrete workers. Summary statistics are shown in the following table:

Tunnel Worker Activity	Sample Size	Mean Cumulative NO_2 Exposure (ppm/yr)	Standard Deviation in Cumulative NO_2 Exposure (ppm/yr)
Drill and blast	115	4.1	1.8
Concrete	69	4.8	2.4

- A. i. **Determine** whether random sampling, random assignment, both, or neither is involved in this study.
ii. **Calculate** what levels of NO₂ exposure would be considered upper outliers for each of these professions based on the above data.
- B. The researchers believe that the average cumulative NO₂ exposure for the concrete workers is greater than for the drill and blast workers. **Determine** whether this conclusion is supported by the data and justify your response without conducting an inference procedure.
- C. The researchers want to conduct an inference procedure to **estimate** the difference in the true mean cumulative NO₂ exposure between drill and blast workers and concrete workers. **Identify** the appropriate inference procedure.
- D. Assume the conditions for the inference procedure identified in part C are satisfied and that a 95% confidence level is used.
i. **Calculate** the results from the appropriate inference procedure.
ii. **Interpret** a conclusion in context.
iii. Based on the results of this inference procedure, **determine** whether there is sufficient statistical evidence of a difference in the true mean cumulative NO₂ exposure between concrete workers and drill and blast workers. Justify your response.
- E. If it turns out that both data sets were somewhat skewed, **determine** whether this invalidates your analysis. **Explain**.

STOP

If there is still time remaining, you may check your work on this section.

[Answer Explanations](#)

Section I

1. (C) With 87 home sale prices, the median is between the 43rd and 44th values when placed in either ascending or descending order. There appear to be around 40 values between \$225,000 and \$250,000 and 25

more values between \$200,000 and \$225,000. So, both the 43rd and 44th values are in the 200 to 225 interval in the histogram.

2. **(D)** Regression lines show association, not causation. Surveys suggest relationships, which experiments can help to show to be cause and effect.
3. **(C)** $\text{invNorm}(0.75) = 0.6745$, so $Q_3 = 5'6'' + 0.6745(2.5'')$, $Q_1 = 5'6'' - 0.6745(2.5'')$, and $\text{IQR} = Q_3 - Q_1 = 2(0.6745)(2.5'')$. On the higher side, outliers are values greater than $Q_3 + 1.5(\text{IQR}) = [5'6'' + 0.6745(2.5'')] + 1.5[2(0.6745)(2.5'')] = 5'6'' + 4(0.6745)(2.5'') = 6'0.745''$ and $6'1'' > 6'0.745''$.
4. **(D)** Calculator software gives (1,794,000, 1,856,000). We are 90% confident that the population mean is within the interval calculated using the data from the sample.
5. **(B)** The first study was observational because the subjects were not chosen for treatment; their current eating habits were simply noted. In the second study, two treatments were applied (meat meals and vegetarian meals).
6. **(A)**

$$\begin{aligned} P(\text{def}) &= P(1\text{st} \cap \text{def}) + P(2\text{nd} \cap \text{def}) \\ &= (0.6)(0.005) + (0.4)(0.010) \end{aligned}$$

$$\begin{aligned} P(1\text{st} \mid \text{def}) &= \frac{P(1\text{st} \cap \text{def})}{P(\text{def})} \\ &= \frac{(0.6)(0.005)}{(0.6)(0.005) + (0.4)(0.010)} \end{aligned}$$

7. **(C)** At the dialysis center the more serious concern would be a Type II error, which is that the equipment is not performing correctly yet the check does not pick this up; however, at the towel manufacturing plant the more serious concern would be a Type I error, which is that the

equipment is performing correctly yet the check causes a production halt.

8. **(C)** There are two possible outcomes (heads and tails), with the probability of heads always 0.75 (independent of what happened on the previous toss), and we are interested in the number of heads in 10 tosses. Thus, this is a binomial model with $n = 10$ and $p = 0.75$. Repeating this over and over (in this case 50 times) simulates the resulting binomial distribution.
9. **(D)** In a binomial distribution with $n = 5$ and $p = 0.61$, the probability of at least 3 successes is $P(3 \text{ successes}) + P(4 \text{ successes}) + P(5 \text{ successes}) = \binom{5}{3} (0.61)^3 (0.39)^2 + \binom{5}{4} (0.61)^4 (0.39) + (0.61)^5$
10. **(B)** The mean or expected value of a binomial = np . Here we have $np = 5(0.61)$.
11. **(A)** With $n = 5$, we expect 6 bars corresponding to 0, 1, 2, 3, 4, and 5 “successes.” With $p = 0.61$, there are higher probabilities associated with the larger number of successes. We expect the histogram to be slightly skewed left.
12. **(A)** If A and B are mutually exclusive, then $P(A \cap B) = 0$, and so $P(A \cup B) = 0.3 + 0.2 - 0 = 0.5$. If A and B are independent, then $P(A \cap B) = P(A)P(B)$, and so $P(A \cup B) = 0.3 + 0.2 - (0.3)(0.2) = 0.44$. If B is a subset of A , then $A \cup B = A$, and so $P(A \cup B) = P(A) = 0.3$.
13. **(C)** The standard deviation of the test statistic is $\sigma_{\hat{p}} = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{(0.69)(1-0.69)}{1,100}}$ and $\hat{p} = \frac{770}{1,100} = 0.7$. Then $z = \frac{\hat{p} - p_0}{\sigma_{\hat{p}}} = \frac{0.7 - 0.69}{\sqrt{\frac{(0.69)(1-0.69)}{1,100}}}$. Since this is a two-sided test, the p -value will be twice the tail probability of the test statistic; however, the test statistic itself is not doubled.
14. **(D)** Dosage is the only explanatory variable, and it is being tested at three levels. Tumor reduction is the single response variable.
15. **(D)** The data are categorical as each CEO responds either “yes” or “no.” This leads to estimating a population proportion—that is, the

proportion of all CEOs who feel that the federal government is doing enough to safeguard against a cyberattack on the electric grid.

16. **(B)** Standard deviation is a measure of variability. The less variability, the more homogeneity there is.
17. **(C)** Since the sample sizes are small, the samples must come from roughly normally distributed populations. While the samples should be independent simple random samples, np and $n(1 - p)$ refer to conditions for tests involving sample proportions, not means.
18. **(A)** Adding the same constant to all values in a set will increase the mean by that constant but will leave the standard deviation unchanged.
19. **(D)** The null hypothesis is that 20% of tennis players would give up their life savings to be able to play in a grand slam. The p -value, 0.041, is the probability of obtaining a sample statistic, $\tilde{p}$, as extreme as or more extreme than the one obtained if the null hypothesis is assumed to be true.
20. **(A)** This is a good example of voluntary response bias, which often overrepresents strong or negative opinions. The people who chose to respond were very possibly the parents of children facing drug problems or people who had had bad experiences with drugs being sold in their neighborhoods. There is very little chance that the 2,500 respondents were representative of the population. Knowing more about his listeners or taking a sample of the sample would not have helped.
21. **(C)** The standard deviation of a random variable is roughly the average or typical amount of variability from the mean.
22. **(D)** The correlation coefficient r is not affected by changes in units, by which variable is called x or y , by adding the same number to all values of a variable, or by multiplying all values of a variable by the same positive number.
23. **(B)** The critical z -score for a 95% confidence interval is $\text{invNorm}(0.975) = 1.96$. With unknown p , we use $p = 0.5$ in our calculation of n .

24. **(B)** With $\alpha = 0.05$, the probability of committing a Type I error is 0.05, and the probability of not committing a Type I error is 0.95. $P(\text{at least one Type I error in 10 tests}) = 1 - P(\text{no Type I errors in 10 tests}) = 1 - (0.95)^{10} = 0.40$.
25. **(D)** Note that all three sets have the same mean and the same range. The third set has most of its values concentrated right at the mean, while the second set has most of its values concentrated far from the mean.
26. **(C)** People coming out of a Wall Street office building are a very unrepresentative sample of the adult population, especially given the question under consideration. Using chance and obtaining a high response rate will not change the selection bias and make this into a well-designed survey. This is a convenience sample, not a voluntary response sample.
27. **(C)** Larger samples (so $\frac{\sigma}{\sqrt{n}}$ is smaller) and less confidence (so the critical z or t is smaller) both result in smaller intervals.
28. **(A)** The correlation r is unit-free.
29. **(D)** The probability of throwing heads is 0.5. By the law of large numbers, the more times you flip the coin, the more the relative frequency tends to become closer to this probability. With fewer tosses there is more chance for wide swings in the relative frequency.
30. **(C)** $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{27}{\sqrt{225}}$. If the true mean parking duration is 51 minutes, the normal curve should be centered at 51. The critical value of 50 has a z -score of $\frac{50 - 51}{27/\sqrt{225}}$. The city planners will reject the merchants' claim if the sample mean is greater than 50 and so will fail to reject if less than 50.
31. **(A)** The overall lengths (between tips of whiskers) are the same, and so the ranges are the same. Just because the min and max are equidistant from the median, and Q_1 and Q_3 are equidistant from the median, does not imply that a distribution is symmetric or that the mean and median are equal. Even if a distribution is symmetric, this does not imply that

it is roughly normal. Particular values, not distributions, may be outliers.

32. (B) Critical z-scores are $\frac{700 - 650}{45} = 1.11$ and $\frac{740 - 650}{45} = 2$ with right tail probabilities of 0.1335 and 0.0228, respectively. The percentage below 740 given that the scores are above 700 is $\frac{0.1335 - 0.0228}{0.1335} = 82.9\%$ Or $\frac{\text{normalcdf}(700, 740, 650, 45)}{\text{normalcdf}(700, 1000000, 650, 45)} = 0.82928$.
33. (D) The parameter involves a proportion of older adults who struggle to meet basic needs, and the parameter refers to a population, not a sample.
34. (D) Stratified random samples are often easier and less costly to obtain and also make comparative data available. In this case, responses can be compared among various districts. This is not a simple random sample because all possible sets of the required size do not have the same chance of being picked. For example, a set of principals all from just half the school districts has no chance of being picked to be the sample. This is not a cluster random sample in that there is no reason to believe that each school district resembles the population as a whole, and furthermore, there was no random sample taken of the school districts. This is not systematic random sampling as the districts were not put in some order with every n th district chosen.
35. (A) A simple random sample can be any size and may or may not be representative of the population. It is a method of selection in which every possible sample of the desired size has an equal chance of being selected.
36. (B) The word “negative” means that generally as one variable increases, the other variable decreases. Equivalently, as one decreases, the other increases. Although the overall pattern is negative, anything can happen with regard to two specific points.
37. (C) The slope, -1.9214 , gives the predicted change in the dependent variable for a 1 unit increase in the independent variable. In this case, *decreasing* a row by moving toward the front will give a predicted *increase* of 1.9214 in the final exam score.

38. (A) The predicted final exam score for a student in row 4 is $86.267 - 1.9214(4) = 78.5814$, giving a residual of $80 - 78.5814 = 1.4186$. The linear model underestimated the student's actual score by 1.4186.
39. (D) If the p -value is less than 0.10, it does not follow that it is less than 0.05. Decisions such as whether a test should be one- or two-sided are made before the data are gathered. If $\alpha = 0.01$, there is a 1% chance of rejecting the null hypothesis *if* the null hypothesis is true. There is one probability of a Type I error, the significance level, while there is a different probability of a Type II error associated with each possible correct alternative, so the sum does not equal 1. When given a true alternative, power is the probability of rejecting the false null hypothesis.
40. (C) If the population is normally distributed, then $\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ is normally distributed, but $\frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ has a t -distribution with $df = n - 1$.
41. (C) A correlation of 0.6 explains $(0.6)^2$, or 36%, of the variation in y , while a correlation of 0.3 explains only $(0.3)^2$, or 9%, of the variation in y .
42. (B) With Plan I, the expected number of students with stock investments is only $30\left(\frac{132}{1,650}\right) = 2.4$ out of 30. Plan II allows an estimate to be made using a full 30 investors.

Section II

1. Model Solution:

- A. The explanatory variable is the version of ninth-grade program used to learn vocabulary. The variable is categorical, with categories Program A, Program B, and a local standard method of instruction.
- B. i. Incorporating a control group would allow the school system to compare the effectiveness of each of the new programs to the local standard method currently being used.

- ii. Parents who fail to return the consent form are a group who may well place less priority on education. The effect of using their children may distort results. Also, their children could be placed only in the control group.
- C.
 - i. If an exceptional teacher is teaching one of the groups and that group does particularly well, it would not be clear if the reason is due to the program used or due to the instructional technique of the exceptional teacher.
 - ii. For example, if all the students with the highest grade point averages selected the same program, this would be a confounding variable. The success of that group could be because of the program itself or could be because of the academically better students in that class.
- D. Results of the study can be generalized to all ninth-grade students whose parents are willing to give parental consent to their participation in such a study.
- E. Is there convincing statistical evidence to suggest that the mean scores on national standardized tests are different for students learning ninth-grade vocabulary using Program A and Program B?

A.		Identify the variable(s).	Ninth-grade program used to learn vocabulary.	1 point
		Classify the variable(s).	Categorical.	1 point
B.	i.	Explain the use of a control group.	To compare the effectiveness of each of the new programs to the local standard method.	1 point
	ii.	Explain why students without parental consent	Parents who fail to return the consent form are a group who may well place less priority on education.	1 point

		should not be part of this study.		
C.	i.	Explain how teachers could be a confounding variable.	If an exceptional teacher is teaching one of the groups and that group does particularly well, it would not be clear if the reason is due to the program used or due to the instructional technique of the exceptional teacher.	1 point
	ii.	Explain how allowing students to self-select might jeopardize the study.	If all the students with the highest GPA selected the same program, success of that group could be because of the program itself or could be because of the academically better students in that class.	1 point
D.		Determine whether the results can be generalized.	The study results can be generalized to all ninth-grade students whose parents are willing to give parental consent to their participation	1 point
E.		First component: identify the variables of interest.	The variable is the ninth-grade program used to learn vocabulary, either Program A or Program B.	1 point
		Second component: identify the parameter and direction of the	The mean scores on national standardized tests are different for students	1 point

	alternative hypothesis.	taking Program A and Program B.	
	Third component: identify the population to which the conclusion is applicable.	The conclusion can be applied to all ninth-grade students at the school who have parental consent to participate in such a study.	1 point

2. Model Solution:

- A.
 - i. Observational study
 - ii. Group size, either large or small
- B.
 - i. The proportion of diners in large groups spending over 70 minutes, $\frac{2}{20} = 0.10$, is greater than the proportion of diners in small groups spending over 70 minutes, $\frac{2}{40} = 0.05$.
 - ii. $4 + 3 + 2 = 9$ of diners in large groups and $2 + 1 + 2 = 5$ of diners in small groups spent over 60 minutes at their tables, for a total of $9 + 5 = 14$.
- C.
 - i. The distribution of times for the large groups is roughly symmetric, while the distribution of times for the small groups is skewed right. Both are unimodal.
 - ii. The median time of the large group distribution is greater than the median time of the small group distribution. The spread of the large group distribution is less than the spread of the small group distribution.
- D.
 - i. The shape of a histogram of the combined data is bimodal with one cluster centered between 20 and 25 and the other cluster centered between 55 and 60. The middle bar of each cluster shows 7 diners.
 - ii. There are a total of 60 times, so the median will be between the 30th and 31st times. The time of 45 has $1 + 28 = 29$ times below it, and the time of 50 has $3 + 31 = 34$ times below it. So the median time is between 45 and 50 minutes.
 - iii. The range of the combined data is between $70 - 10 = 60$ minutes and $75 - 5 = 70$ minutes.

E. The population is all small and large groups of customers coming to the diner for dinner.

A.	i.	Identify the type of study.	Observational.	1 point
	ii.	Identify the variable.	Group size.	1 point
B.	i.	Compare the proportions of diners in each group who spent over 70 minutes.	Proportion of diners in large groups is greater than proportion of diners in small groups. $\frac{2}{20} = 0.10 > \frac{2}{40} = 0.05$	1 point
	ii.	Calculate the number of diners in both groups who spent over 60 minutes.	$(4 + 3 + 2) + (2 + 1 + 2) = 14$	1 point
C.	i.	Compare the shapes of the distributions.	The distribution of times for the large groups is roughly symmetric while the distribution of times for the small group is skewed right.	1 point
	ii.	Compare the centers and ranges.	The distribution for large groups has a greater center and smaller spread than the distribution for small groups.	1 point
D.	i.	Describe the shape of a histogram showing the combined data.	The combined histogram is bimodal with one cluster centered between 20 and 25 and the other	1 point

		cluster centered between 55 and 60.	
ii.	Estimate the median of the combined data.	The median is between the 30th and 31st times. 45 has $1 + 28 = 29$ times below it, and 50 has $3 + 31 = 34$ times below it. So the median time is between 45 and 50 minutes.	1 point
iii.	Estimate the range of the combined data.	The range is between $70 - 10 = 60$ minutes and $75 - 5 = 70$ minutes.	1 point
E.	Determine the population for an investigative question.	The population consists of all small and large groups of customers coming to the diner for dinner.	1 point

3. Model Solution:

- A. The explanatory variables are whether an email is spam or legitimate and whether the test is positive or negative. Both variables are categorical.
- B. An appropriate inference procedure is a two-sample z-test for the difference between two population proportions.
- C. The null hypothesis is $H_0: p_1 = p_2$, and the alternative hypothesis is $H_a: p_1 \neq p_2$, where p_1 is the proportion of all positive tests from among the emails that are spam and p_2 is the proportion of all negative tests from among the emails that are legitimate. The randomization condition is satisfied as the spam and legitimate emails were randomly and independently chosen. The sample sizes can be assumed to be less than 10% of the respective populations. The combined proportion is $\hat{p}_c = \frac{205 + 660}{250 + 750} = 0.865$. The normality condition is met because $250(0.865) = 216.25$,

$250(0.135) = 33.75$, $750(0.865) = 648.75$, and $750(0.135) = 101.25$ are all at least 10. The value of the test statistic is $z = -2.404$, and the p -value is 0.0162.

- D. With this small a p -value, $0.016 < 0.05$, there is sufficient evidence to reject the null hypothesis. There is convincing evidence that the proportion of all spam emails for which the tests are positive does not equal the proportion of all legitimate emails for which the tests are negative. This goes against the developer's claim.

A.	Identify the variable(s).	Spam or legitimate email and positive or negative test.	1 point
	Classify the variable(s).	Both are categorical variables.	1 point
B.	Identify the inference procedure.	Use a two-sample z -test for the difference between two population proportions.	1 point
C.	Identify H_0 and H_a .	$H_0: p_1 = p_2$ and $H_a: p_1 \neq p_2$	1 point
	Identify the parameters.	p_1 is the proportion of all positive tests for which the emails are spam, and p_2 is the proportion of all negative tests for which the emails are legitimate.	1 point
	Justify whether the randomization condition is met.	The spam and legitimate emails were randomly and independently chosen. The sample sizes can be assumed to be less than 10% of the respective populations.	1 point

	Justify whether the normality condition is met.	$\hat{p}_c = \frac{205 + 660}{250 + 750} = 0.865$. The normality condition is met because $250(0.865) = 216.25$, $250(0.135) = 33.75$, $750(0.865) = 648.75$, and $750(0.135) = 101.25$ are all ≥ 10 .	1 point
	Calculate the test statistic and the p -value.	$z = -2.404$, and the p -value is 0.0162.	1 point
D.	Compare the p -value to the significance level.	$0.016 < 0.05$ so the null hypothesis should be rejected.	1 point
	Justify whether the claim is or is not supported.	There is convincing evidence that the proportion of all spam emails for which the tests are positive does not equal the proportion of all legitimate emails for which the tests are negative. This goes against the developer's claim.	1 point

4. Model Solution:

- A.
- Random sampling is involved as two independent random samples of workers are chosen, a random sample of drill and blast workers and an independent random sample of concrete workers. There was no random assignment of workers to positions.
 - A standard definition of outliers is all values more than two standard deviations below or above the mean. Upper outliers in this example are values greater than $4.1 + 2(1.8) = 7.7$

ppm/yr for drill and blast workers and values greater than $4.8 + 2(2.40) = 9.6$ ppm/yr for concrete workers.

- B. Yes, the data support the researchers' belief that the average cumulative NO_2 exposure for the concrete workers is greater than for drill and blast workers because $4.8 > 4.1$.
- C. The appropriate inference procedure would be a two-sample t -interval for the difference between population means.
- D. i. The 95% confidence interval is $(-1.362, -0.0381)$ ppm/yr for drill and blast workers minus concrete workers mean cumulative NO_2 exposure.
- ii. The researchers are 95% confident that the true mean difference in cumulative NO_2 exposure between drill and blast workers and concrete workers falls between -1.362 and -0.0381 .
- iii. There is sufficient statistical evidence of a difference in the true mean cumulative NO_2 exposure between drill and blast workers and concrete workers because 0 is not in the interval $(-1.362, -0.0381)$.
- E. With sample sizes this large, both 115 and 69 are greater than 30, so the central limit theorem applies and our analysis is valid.

A.	i.	Determine whether random sampling, random assignment, both, or neither is used.	Random sampling is involved as two independent random samples are chosen. There is no random assignment of workers to positions.	1 point
	ii.	Calculate the upper outliers.	The upper outlier values are greater than $4.1 + 2(1.8) = 7.7$ ppm/yr for drill and blast workers and are greater than $4.8 + 2(2.40) = 9.6$ ppm/yr for concrete workers.	1 point

B.	Determine and justify whether a claim is supported.	Yes, the researchers' belief is supported by the data because $4.8 > 4.1$.	1 point
C.	Identify the appropriate inference procedure.	Two-sample t -interval for the difference between population means.	1 point
D.	i. Calculate a confidence interval.	$(-1.362, -0.0381)$ ppm/yr for drill and blast workers minus concrete workers mean cumulative NO_2 exposure.	1 point
	ii. Interpret a conclusion.	The researchers are 95% confident that the true mean difference in cumulative NO_2 exposure between drill and blast workers and concrete workers falls between -1.362 and -0.0381 .	1 point
	iii. Determine whether a claim is supported.	There is sufficient statistical evidence of a difference in the true mean cumulative NO_2 exposure between drill and blast workers and concrete workers	1 point
	Justify a claim.	0 is not in the interval $(-1.362, -0.0381)$.	1 point

E.	Determine whether the analysis is valid.	With sample sizes this large, our analysis is valid.	1 point
	Determine if a condition is violated.	Both 115 and 69 are greater than 30, so the central limit theorem applies.	1 point

Practice Test 2

Section I: Questions 1–42

SPEND 90 MINUTES ON THIS PART OF THE EXAM.

DIRECTIONS: The questions or incomplete statements that follow are each followed by four suggested answers or completions. Choose the response that best answers the question or completes the statement.

1. Suppose that the regression line for a set of data, $\hat{y} = 3 + bx$, passes through the point $(2, 7)$. If $\bar{x}$ and $\bar{y}$ are the sample means of the x - and y -values, respectively, which of the following is equal to $\bar{y}$?
 - (A) $\bar{x}$
 - (B) $3 + \bar{x}$
 - (C) $3 + 2\bar{x}$
 - (D) $3 + 3.5\bar{x}$
2. A study is made to determine whether more hours of academic studying leads to a greater number of points scored by basketball players. In surveying 50 basketball players, it is noted that the 25 who claim to study the most hours have a higher point average than the 25 who study less. Based on this study, the coach begins requiring the players to spend more time studying. Which of the following is a correct statement?
 - (A) While this study may have its faults, it still does prove causation.
 - (B) There could well be a confounding variable responsible for the seeming relationship.
 - (C) While this is a controlled experiment, the conclusion of the coach is not justified.
 - (D) To get the athletes to study more, it would be more meaningful to have them put in more practice time on the court to boost their point averages,

as higher point averages seem to be associated with more study time.

3. In a social experiment, students were asked to take a selfie and post it to social media. Unknown to the students, what the study was really about was counting how many selfies a student would take before feeling satisfied to post one. The standard deviation for number of selfies taken before posting was 4.3, and 30% of numbers of selfies taken were over 15. Assuming an approximately normal distribution for numbers of selfies taken before posting, what was the mean number?
- (A) $15 + 0.30(4.3)$
(B) $15 - 0.4756(4.3)$
(C) $15 + 0.5244(4.3)$
(D) $15 - 0.5244(4.3)$
4. Which of the following is a correct statement about correlation?
- (A) If the slope of the regression line is exactly 1, the correlation is exactly 1.
(B) If the correlation is 0, the slope of the regression line is undefined.
(C) Switching which variable is called x and which is called y changes the sign of the correlation.
(D) Changing units does not change the correlation r .
5. Which of the following are affected by outliers?
- I. Mean
II. Median
III. Standard deviation
IV. Range
V. Interquartile range
- (A) I and II only
(B) III and IV only
(C) III, IV, and V only
(D) I, III, and IV only
6. A company that produces facial tissues continually monitors tissue strength. If the mean strength from sample data drops below a specified level, the production process is halted and the machinery inspected. Which of the following would result from a Type I error?

- (A) Halting the production process when sufficient customer complaints are received.
- (B) Halting the production process when the tissue strength is below specifications.
- (C) Halting the production process when the tissue strength is within specifications.
- (D) Allowing the production process to continue when the tissue strength is within specifications.

7. Two possible wordings for a questionnaire on a proposed school budget increase are as follows:

- I. This school district has one of the highest per student expenditure rates in the state. This has resulted in low failure rates, high standardized test scores, and most students going on to good colleges and universities. Do you support the proposed school budget increase?
- II. This school district has one of the highest per student expenditure rates in the state. This has resulted in high property taxes, with many people on fixed incomes having to give up their homes because they cannot pay the school tax. Do you support the proposed school budget increase?

One of these questions showed that 58% of the population favor the proposed school budget increase, while the other question showed that only 13% of the population support the proposed increase. Which produced which result and why?

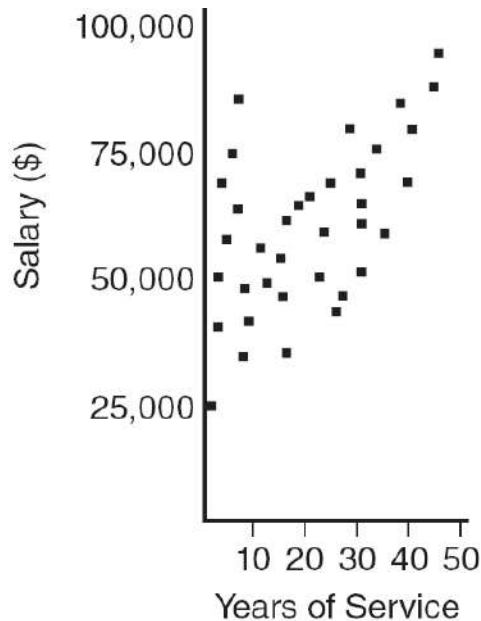
- (A) The first showed 58% and the second 13% because of the lack of randomization as evidenced by the wording of the questions.
- (B) The first showed 13% and the second 58% because of a placebo effect due to the wording of the questions.
- (C) The first showed 58% and the second 13% because of response bias due to the wording of the questions.
- (D) The first showed 13% and the second 58% because of response bias due to the wording of the questions.

8. Suppose that for a certain Caribbean island, in any 3-year period the probability of a major hurricane is 0.25, the probability of water damage is 0.44, and the probability of both a hurricane and water damage is 0.22. What is the probability of water damage given that there is a hurricane?

- (A) $0.25 + 0.44 - 0.22$

- (B) $\frac{0.22}{0.44}$
- (C) $0.25 + 0.44$
- (D) $\frac{0.22}{0.25}$

9. A union spokesperson is trying to encourage a college faculty to join the union. She would like to argue that faculty salaries are not truly based on years of service as most faculty believe. She gathers data and notes the following scatterplot of salary versus years of service.



Which of the following *most* correctly interprets the overall scatterplot?

- (A) The faculty member with the fewest years of service makes the lowest salary, and the faculty member with the most years of service makes the highest salary.
 - (B) A faculty member with more years of service than another has a higher salary than the other.
 - (C) There is a strong positive correlation with little deviation.
 - (D) While there is a strong positive correlation, there is a distinct deviation from the overall pattern for faculty with fewer than ten years of service.
10. Two random samples of students are chosen, the first set from those taking an AP Statistics class and the second set from those not taking an AP Statistics class. The following back-to-back stemplots compare the GPAs.

AP Statistics		No AP Statistics
	1	89
97653	2	015688
98775332110	3	133344777888
1100	4	

Key: 1 | 8 means 1.8

Which of the following is true about the ranges and standard deviations?

- (A) The first set has both a greater range and a greater standard deviation.
- (B) The first set has a greater range, while the second has a greater standard deviation.
- (C) The first set has a greater standard deviation, while the second has a greater range.
- (D) The second set has both a greater range and a greater standard deviation.

11. In a group of 10 scores, the largest score is increased by 40 points. What will happen to the mean?

- (A) It will remain the same.
- (B) It will increase by 4 points.
- (C) It will increase by 10 points.
- (D) There is not sufficient information to answer this question.

12. Past studies indicate that 85% of forest fires in the U.S. every year are caused by humans. A forest ranger believes that the proportion is greater than 0.85. A random sample of forest fires is analyzed. With $H_0: p = 0.85$ and $H_a: p > 0.85$, the p -value of the test is 0.093. Which of the following is a correct interpretation of the p -value?

- (A) 0.093 is the probability that the null hypothesis is false.
- (B) 0.093 is the probability of obtaining a sample proportion as large as or larger than the one obtained by the ranger.
- (C) If it is true that 85% of forest fires in the U.S. every year are caused by humans, 0.093 is the probability of obtaining a sample proportion equal to

or greater than 0.85.

- (D) If it is true that 85% of forest fires in the U.S. every year are caused by humans, 0.093 is the probability of obtaining a sample proportion as large as or larger than the one obtained by the ranger.

13. Suppose you toss a fair die three times and it comes up an even number each time. Which of the following is a true statement?

- (A) The probability that the next toss will again be even is 0.5.
(B) Based on the properties of conditional probability, the next toss is more likely to be an even number given that three in a row have been even.
(C) By the law of large numbers, the next toss is more likely to be an odd number than another even number.
(D) The law of large numbers tells how many tosses will be necessary before the percentages of evens and odds are again in balance.

Questions 14–16 refer to the following situation:

A pharmaceutical company is interested in the association between advertising expenditures and sales for various over-the-counter products. A sales associate collects data on nine products, looking at sales (in \$1,000) versus advertising expenditures (in \$1,000). The results of the regression analysis are shown below.

Dependent variable: Sales

Variable	Coefficient	SE Coef	t-ratio	P
Constant	123.800	1.798	68.84	0.000
Advertising	12.633	0.378	33.44	0.000
R-Sq = 99.4% R-Sq (adj) = 99.3%				
s = 2.926 with 9-2 = 7 degrees of freedom				

[Follow for extended description](#)

14. Which of the following statements is the best interpretation of the value 123.800 in the output?

- (A) Every dollar increase in advertising is predicted to give an increase of \$123.80 in sales.
(B) If advertising has no effect on sales, sales are predicted to be 123.800 (in \$1,000).
(C) If \$0 are spent on advertising, sales are predicted to be 123.800 (in \$1,000).

(D) In this context, 123.800 has no logical meaning.

15. Which of the following statements is the best interpretation of $R\text{-Sq} = 99.4\%$ in the output?

- (A) 99.4% of the data points are highly correlated.
- (B) 99.4% of the variation in sales is accounted for by the linear model relating sales to advertising expenditures.
- (C) 99.4% of the variation in predicted sales is accounted for by the linear model relating sales to advertising expenditures.
- (D) The very high value of $R\text{-Sq}$ indicates a cause-and-effect association between sales and advertising expenditures.

16. What is the value of the correlation r ?

- (A) 0.000
 - (B) 0.378
 - (C) $\sqrt{0.993}$
 - (D) $\sqrt{0.994}$
-

17. A school principal wanted to estimate the hours per month that parents spend volunteering at their children's school. The principal took repeated random samples of size 25 parents and estimated the mean and standard deviation of the sampling distribution to be 5.5 hours and 0.4 hours, respectively. Based on the estimates for the sampling distribution, which of the following provides the best estimates of the population parameters?

- (A) Mean of 5.5 hours and standard deviation of 0.016 hours
- (B) Mean of 5.5 hours and standard deviation of 0.08 hours
- (C) Mean of 5.5 hours and standard deviation of 1 hour
- (D) Mean of 5.5 hours and standard deviation of 2 hours

18. Jonathan obtained a score of 80 on a statistics exam, placing him at the 90th percentile. Suppose 5 points are added to everyone's score. Jonathan's new score will be at the

- (A) 80th percentile.
- (B) 85th percentile.
- (C) 90th percentile.

(D) 95th percentile.

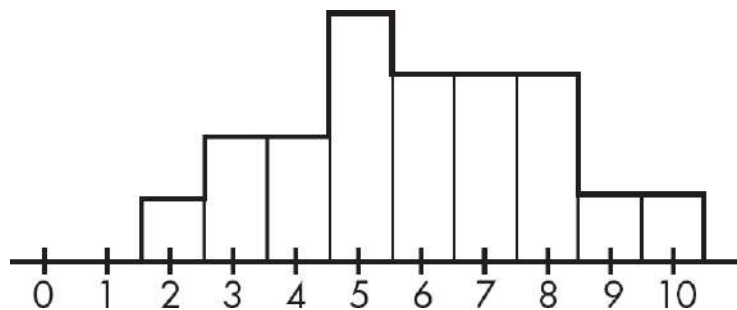
19. To study the effect of music on piecework output at a clothing manufacturer, two experimental treatments are planned: day-long classical music for one group versus day-long light rock music for another. Which one of the following groups would serve best as a control for this study?

- (A) A third group for which no music is played
- (B) A third group that hears day-long R & B music
- (C) A third group that hears classical music every morning and light rock every afternoon
- (D) A third group in which each worker has earphones and chooses his or her own favorite music

20. Suppose $H_0: p = 0.6$ and the power of the test for the alternative $p = 0.7$ is 0.8. Which of the following is a valid conclusion?

- (A) The probability of committing a Type I error is 0.1.
- (B) If the alternative is true, the probability of failing to reject H_0 is 0.2.
- (C) The probability of committing a Type II error is 0.3.
- (D) All of the above are valid conclusions.

21. The following is a histogram of the numbers of ties owned by bank executives.



Which of the following is a correct statement?

- (A) The median number of ties is five.
- (B) More than four executives own over eight ties each.
- (C) An executive is equally likely to own fewer than five ties or more than seven ties.

- (D) Removing all the executives with three, nine, and ten ties may change the median.

22. A policewoman counts the number of motorists not wearing seat belts in a random sample of five drivers. Define the random variable of interest for the policewoman, and state how the random variable is distributed.

- (A) The random variable of interest is the proportion of motorists wearing seat belts in a random sample of five drivers. The distribution is binomial with parameters $n = 5$ and $p =$ probability a motorist is wearing a seat belt.
- (B) The random variable of interest is the number of motorists wearing seat belts in a random sample of five drivers. The distribution is binomial with parameters $n = 5$ and $p =$ probability a motorist is not wearing a seat belt.
- (C) The random variable of interest is the number of motorists not wearing seat belts in a random sample of five drivers. The distribution is binomial with parameters $n = 5$ and $p =$ probability a motorist is not wearing a seat belt.
- (D) The random variable of interest is the percent of motorists not wearing seat belts in a random sample of five drivers. The distribution is binomial with parameters $n = 5$ and $p =$ probability a motorist is wearing a seat belt.

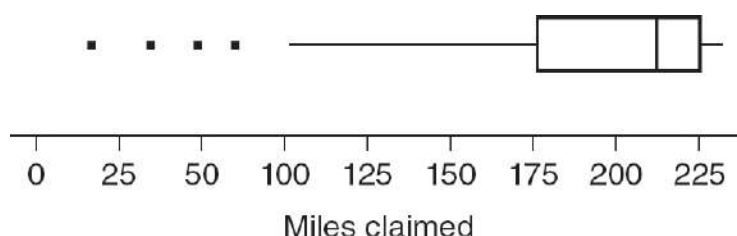
23. Company I manufactures demolition fuses that burn an average of 50 minutes with a standard deviation of 10 minutes, while Company II advertises fuses that burn an average of 55 minutes with a standard deviation of 5 minutes. Which company's fuse is more likely to last at least 1 hour? Assume roughly normal distributions of fuse times.

- (A) Company I's, because of its greater standard deviation
- (B) Company II's, because of its greater mean
- (C) For both companies, the probability that a fuse will last at least 1 hour is 0.159.
- (D) For both companies, the probability that a fuse will last at least 1 hour is 0.841.

24. Which of the following is *not* important in the design of experiments?

- (A) Control of confounding variables
- (B) Randomization in assigning subjects to different treatments
- (C) Use of a confounding variable to control the placebo effect
- (D) Replication of the experiment using sufficient numbers of subjects

25. The travel miles claimed in weekly expense reports of the sales personnel at a corporation are summarized in the following boxplot.



Which of the following is the most reasonable conclusion?

- (A) The mean and median numbers of travel miles are roughly equal.
 - (B) The mean number of travel miles is probably greater than the median number.
 - (C) Most of the claimed numbers of travel miles are in the $[0, 200]$ interval.
 - (D) Most of the claimed numbers of travel miles are in the $[200, 240]$ interval.
26. For the least squares regression line, which of the following statements about residuals is true?
- (A) A negative residual means the regression model underestimates the observed value.
 - (B) If the linear model is good, the number of positive residuals will be the same as the number of negative residuals.
 - (C) The mean of the residuals is always zero.
 - (D) If the correlation is 0, there will be a distinct pattern in the residual plot.

Questions 27–29 refer to the following situation:

Assume that 12% of the public favor legalizing all drugs.

27. In a simple random sample of 6 people, what is the probability that at least 1 person favors legalization?
- (A) $(0.88)^6$
 - (B) $1 - (0.88)^6$
 - (C) $1 - 6(0.12)(0.88)^5$
 - (D) $6(0.12)(0.88)^5 + (0.88)^6$

28. For a random sample of 6 people, what is the standard deviation for the number of people who favor legalization?

- (A) $(6)(0.12)$
- (B) $(6)(0.12)(0.88)$
- (C) $\sqrt{(6)(0.12)(0.88)}$
- (D) $\sqrt{\frac{(0.12)(0.88)}{6}}$

29. For a random sample of 6 people, is the normal distribution a good approximation for the binomial distribution of the number of people who favor legalization?

- (A) No, because $(6)(0.12) < 10$.
- (B) No, because $6 < 30$.
- (C) Yes, because 6 people is less than 10% of the population of all people.
- (D) Yes, because the normal is always a good approximation of the binomial.

30. An administrator wishes to determine a 95% confidence interval for the proportion of college students who would be willing to participate in a demonstration that closed down the campus if they strongly believed in a cause. Which of the following combinations of n and $\hat{p}$ will satisfy the conditions for inference?

- (A) $n = 20$ and $\hat{p} = 0.66$
- (B) $n = 30$ and $\hat{p} = 0.26$
- (C) $n = 50$ and $\hat{p} = 0.16$
- (D) $n = 80$ and $\hat{p} = 0.13$

31. A telecommunications executive instructs an associate to contact 104 customers using their service to obtain their opinions in regard to an idea for a new pricing package. The associate notes the number of customers whose names begin with A and uses a random number generator to pick four of these names. She then proceeds to use the same procedure for each letter of the alphabet and combines the $4 \times 26 = 104$ results into a group to be contacted. Which of the following is a correct conclusion?

- (A) Her procedure makes use of chance.
- (B) Her procedure results in a simple random sample.
- (C) Each customer has an equal probability of being included in the survey.

- (D) With this small a sample size, it is better to let the surveyor use the company's data banks to pick representative customers to be surveyed based on features such as gender, political affiliation, income level, race/ethnicity, age, and so on.

32. A random sample of 300 high school students are to be selected, and each will be asked if they have ever been anonymously bullied over the Internet. Which of the following is a reason for increasing the size of this study from 300 to 500 students?

- (A) To reduce response bias
(B) To reduce nonresponse bias
(C) To reduce undercoverage bias
(D) To more likely obtain an estimate that is closer to the population proportion

33. PCB (polychlorinated biphenyl) contamination of a river by a manufacturer is being measured by amounts of the pollutant found in fish. A company scientist claims that the fish contain only 5 parts per million, but an investigator believes the figure is higher. The investigator catches six fish that show the following amounts of PCB (in parts per million): 6.8, 5.6, 5.2, 4.7, 6.3, and 5.4 with a mean of 5.67 and a standard deviation of 0.763. In performing a hypothesis test with $H_0: \mu = 5$ and $H_a: \mu > 5$, what is the test statistic?

(A) $t = \frac{5.67 - 5}{\left(\frac{0.763}{\sqrt{5}}\right)}$

(B) $t = \frac{5.67 - 5}{\left(\frac{0.763}{\sqrt{6}}\right)}$

(C) $t = \frac{5.67 - 5}{\sqrt{\frac{0.763}{5}}}$

(D) $t = \frac{5.67 - 5}{\sqrt{\frac{0.763}{6}}}$

34. Consider two sets, A and B , both roughly normally distributed with $n = 1,000$. Set A has mean 10 and standard deviation 2, while set B has mean 20 and standard deviation 1. Define two new sets, X and Y , where X is formed by combining all the values from A and B into a larger set with $n = 2,000$, while Y is formed by adding each value in A to each value in B to form a set with $n = 1,000 \times 1,000$. Which of the following best describes the distributions of X and Y ?

- (A) Both are roughly normal with identical means.
- (B) Both are roughly normal, but one has a mean twice the other.
- (C) One is roughly normal, the other is bimodal, and they have identical means.
- (D) One is roughly normal, the other is bimodal, and one has a mean twice the other.

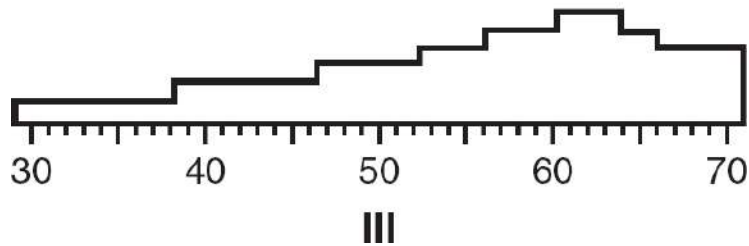
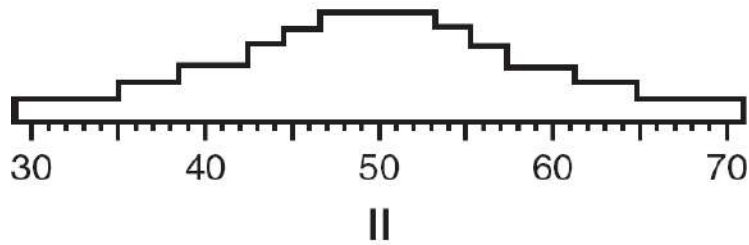
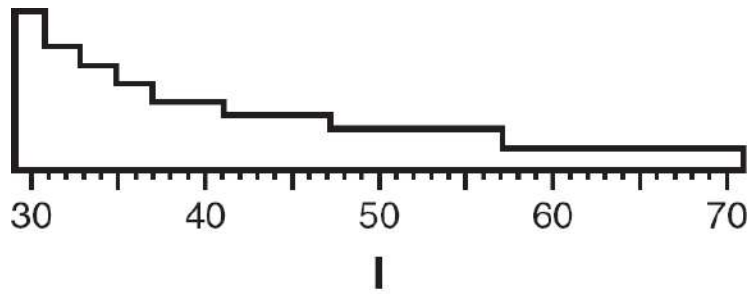
35. In general, how does tripling the sample size change the confidence interval width?

- (A) It multiplies the interval width by 1.732.
- (B) It divides the interval width by 1.732.
- (C) It triples the interval width.
- (D) It divides the interval width by 3.

36. Which of the following statements is *false*?

- (A) Like the normal distribution, the t -distributions are symmetric.
- (B) The t -distributions are lower at the mean and higher at the tails and so are more spread out than the normal distribution.
- (C) The greater the df , the closer the t -distributions are to the normal distribution.
- (D) The smaller the df , the better the 68-95-99.7 rule works for t -models.

37. Consider the following three histograms:



Which of the following correctly orders the histograms from the one with the smallest proportion of data above its mean to the one with the largest proportion of data above its mean?

- (A) I, II, III
- (B) I, III, II
- (C) III, II, I
- (D) II, III, I

38. The sampling distribution of the sample mean is close to the normal distribution

- (A) only if both the original population has a normal distribution and n is large.
- (B) if the standard deviation of the original population is known.
- (C) if n is large, no matter what the distribution of the original population.
- (D) only if the original population is not badly skewed and does not have outliers.

39. In a random sample of 500 American adults, 210 expressed the belief that journalists have “low” ethical standards. A 95% confidence interval for the proportion of all American adults who believe that journalists have “low” ethical standards is given by (0.3767, 0.4633). Which of the following is a correct interpretation of the confidence *level* of 95%?

- (A) There is a 0.95 probability that the true proportion of all American adults who believe that journalists have “low” ethical standards is between 0.3767 and 0.4633.
- (B) We are 95% confident that the true proportion of all American adults who believe that journalists have “low” ethical standards is between 0.3767 and 0.4633.
- (C) If the proportion of American adults who believe that journalists have “low” ethical standards is calculated for repeated random samples of 500 adults, 95% of these sample proportions will fall between 0.3767 and 0.4633.
- (D) If a confidence interval for the proportion of American adults who believe that journalists have “low” ethical standards is calculated for repeated random samples of 500 American adults, approximately 95% of these intervals will contain the true population proportion.

40.



Above is the dotplot for a set of numbers. One element is labeled X . Which of the following is a correct statement?

- (A) X has the largest z -score, in absolute value, of any element in the set.
 - (B) A boxplot will plot an outlier like X as an isolated point.
 - (C) A stemplot will show X isolated from two clusters.
 - (D) The interquartile range (IRQ) is exactly half the range.
41. A 2019 survey of 500 adults concluded that 82% of adults support increasing the minimum wage by \$5. Which of the following best describes what is meant by the poll having a margin of error of $\pm 3\%$?
- (A) It would not be unexpected for 3% of adults to begin supporting an increase of the minimum wage by \$5 or to stop supporting an increase of

the minimum wage by \$5.

- (B) Between 395 and 425 of the adults surveyed responded that they support increasing the minimum wage by \$5.
- (C) If a similar survey of 500 adults were taken weekly, a 3% change in each week's results would not be unexpected.
- (D) It is likely that between 79% and 85% of all adults support increasing the minimum wage by \$5.

42.

		College Plans	
		Public	Private
Taking AP Statistics:		18	27
Not taking AP Statistics:		26	40

[Follow for extended description](#)

The above two-way table summarizes the results of a survey of a random sample of high school seniors conducted to determine if there is a relationship between whether or not a student is taking AP Statistics and whether the student plans to attend a public or a private college after graduation. Which of the following is the most reasonable conclusion about the relationship between taking AP Statistics and the type of college a student plans to attend?

- (A) There appears to be no association since the proportion of AP Statistics students planning to attend public schools is almost identical to the proportion of students not taking AP Statistics who plan to attend public schools.
- (B) There appears to be an association since the proportion of AP Statistics students planning to attend public schools is almost identical to the proportion of students not taking AP Statistics who plan to attend public schools.
- (C) There appears to be an association since more students plan to attend private than public schools.
- (D) There appears to be an association since fewer students are taking AP Statistics than are not taking AP Statistics.

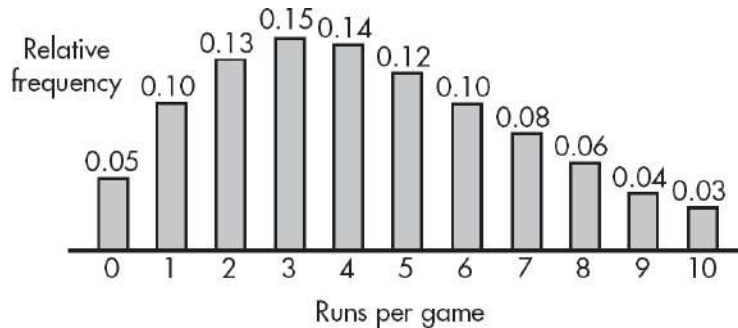
STOP

If there is still time remaining, you may check your work on this section.

Section II

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your results and explanations.

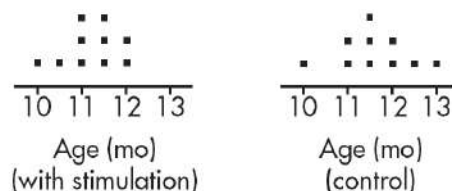
1. Ten volunteer male subjects are to be used for an experiment to study 4 substances (aloe, camphor, eucalyptus oil, and benzocaine) and a placebo with regard to itch relief. Itching is induced on a forearm with an itch stimulus (cowage), a drug is topically administered, and the duration of itching is recorded.
 - A. **Identify** the treatments, the experimental units, and the response variable.
 - B. Two experimental designs are under consideration:
 - Design X will randomly pick 2 volunteers to receive each of the 5 treatments.
 - Design Y gives each volunteer all 5 treatments, 1 each day, with the time order randomized.
 - i. **Identify** the experimental design illustrated by Design X.
 - ii. **Identify** the experimental design illustrated by Design Y.
 - C. **Describe** a statistical advantage of using Design Y rather than Design X.
 - D.
 - i. **Describe** any limits there are on generalizations of any results of this study.
 - ii. Suppose 100 male volunteers were used and each was allowed to pick among the 4 drugs and placebo where all 5 substances are unlabeled. **Explain** why such a method of assignment might lead to an incorrect conclusion.
 - E. A study is to be performed about the average duration of itching for camphor versus for the placebo. The researchers believe that camphor will provide greater relief than the placebo. **Determine** a valid investigative question for this study. Ensure all three components of a valid investigative question are included in your response.
2. A comprehensive study of more than 3,000 randomly picked major league baseball games results in the graph below showing the relative frequencies of runs scored by home teams.



- A.
 - i. **Identify** the type of study.
 - ii. **Identify** the numerical variable in the study.
 - B.
 - i. **Calculate** the mean runs per game in this sample.
 - ii. **Calculate** the median runs per game in this sample.
 - iii. Of the mean and the median, is the one that is actually greater the same as the one that was expected to be greater? **Explain**.
 - C. **Calculate** the probability that in 4 randomly selected games, the home team is shut out (scores no runs) in at least one of the games.
 - D. If X is the random variable for the runs scored per game by home teams, its standard deviation is 2.578. Suppose that samples of size $n = 200$ games are selected at random and that $\bar{x}$, the mean number of runs scored by the home teams, is calculated for each sample.
 - i. **Describe** the shape of the sampling distribution of $\bar{x}$.
 - ii. **Calculate** the standard deviation of the sampling distribution of $\bar{x}$.
 - E. A sports statistician believes the mean runs per game scored by home teams is greater than 4. **Determine** the population for a valid investigative study.
3. An experiment is run to test whether daily stimulation of specific reflexes in young infants will lead to earlier walking. Twenty infants were recruited through a pediatrician's service and were randomly split into 2 groups of 10. One group received the daily stimulation, while the other was considered a control group. The ages (in months) at which the infants first walked alone were recorded.

Ages in Months at Which First Steps Alone Were Taken

With stimulation:	10, 12, 11, 10.5, 11, 11.5, 11.5, 11, 12, 11.5	Mean = 11.2, SD = 0.63246
Control:	10, 13, 12, 11, 11.5, 11.5, 12.5, 12, 11.5, 11	Mean = 11.6, SD = 0.84327



[Follow for extended description](#)

- A. **Identify** and **classify** the variable(s) in this study.
 - B. **Identify** the appropriate inference procedure.
 - C. **Complete** the inference procedure, including calculating the appropriate statistics.
 - D. **Justify** a conclusion in context.
4. A production manager measured the volume (fluid ounces) of liquid dispensed into soft drink bottles models and calculated the following descriptive statistics for the data set:

Variable	<i>n</i>	Mean	Median	StDev	Min	Max	Q_1	Q_3
Volume	25	16.121	16.152	0.040	15.991	16.283	16.091	16.155

- A. Do the sample volumes appear to follow a roughly normal model? **Explain**.
- B. Assume the population distribution is roughly as indicated in part A with a mean of 16.121 and a standard deviation of 0.040.
 - i. **Calculate** the probability that a bottle will have less than 16 fluid ounces.
 - ii. **Calculate** the probability that a random sample of 25 bottles will have a mean less than 16 fluid ounces.
- C. The advertised mean is 16 fluid ounces. Suppose the manufacturer wants to make a change that will result in fewer bottles weighing less than the advertised mean.
 - i. If the manufacturer changes only the advertised volume, should it make the advertised volume smaller or larger? **Explain**.
 - ii. If the manufacturer changes only the mean volume of liquid dispensed, should it make the mean volume dispensed smaller or larger? **Explain**.
- D. The production manager wants to conduct an inference procedure to estimate the true mean volume of liquid dispensed into the bottles.
 - i. **Identify** the appropriate inference procedure the manager should use.
 - ii. Assuming the conditions for inference are satisfied, **calculate** the results for the appropriate inference procedure using a 99% confidence level.
 - iii. **Interpret** a conclusion in context.
- E. Based on the results from the manager's inference procedure, **determine** whether there is sufficient statistical evidence that the true mean is greater than 16 fluid ounces. **Explain** your reasoning.

STOP

If there is still time remaining, you may check your work on this section.

Answer Explanations

Section I

1. **(C)** Since $(2, 7)$ is on the line, $\hat{y} = 3 + bx$, we have $7 = 3 + 2b$ and $b = 2$. Thus, the regression line is $\hat{y} = 3 + 2x$. The point $(\bar{x}, \bar{y})$ is always on the regression line, so we have $\bar{y} = 3 + 2\bar{x}$.
2. **(B)** It could well be that conscientious students are the same ones who both study and do well on the basketball court. If students could be randomly assigned to study or not study, the results would be more meaningful. Of course, ethical considerations might make it impossible to isolate the confounding variable in this way.
3. **(D)** $\text{invNorm}(0.7) = 0.5244$ so $15 - \mu = 0.5244(4.3)$, which gives $\mu = 15 - 0.5244(4.3)$.
4. **(D)** The slope of the regression line and the correlation are related by $b = r \frac{s_y}{s_x}$. When using z -scores, the standard deviations s_x and s_y are 1. If $r = 0$, then $b = 0$. Switching which variable is x and which is y , or changing units, will not change the correlation.
5. **(D)** The median and interquartile range are specifically used when outliers are suspected of unduly influencing the mean, range, or standard deviation.
6. **(C)** This is a hypothesis test with H_0 : tissue strength is within specifications, and H_a : tissue strength is below specifications. A Type I error is committed when a true null hypothesis is mistakenly rejected.
7. **(C)** The wording of questions can lead to response bias. The neutral way of asking this question would simply have been, "Do you support the proposed school budget increase?"
8. **(D)** $P(\text{water} \mid \text{hurricane}) = \frac{P(\text{water} \cap \text{hurricane})}{P(\text{hurricane})} = \frac{0.22}{0.25}$
9. **(D)** Although it is important to look for basic patterns, it is also important to look for deviations from these patterns. In this case, there is an overall positive correlation; however, those faculty with under ten years of service show little relationship between years of service and salary. Although (A) is a true statement, it does not give an overall interpretation of the scatterplot.
10. **(D)** The second set has a greater range, $3.8 - 1.8 = 2.0$ as compared with $4.1 - 2.3 = 1.8$, and with its skewness it also has a greater standard deviation.

11. **(B)** With $n = 10$, increasing $\sum x$ by 40 increases $\frac{\sum x}{n}$ by $\frac{40}{10} = 4$.
12. **(D)** The null hypothesis is that 85% of forest fires in the U.S. every year are caused by humans. The p-value, 0.093, is the probability of obtaining a sample statistic, $\hat{p}$, as extreme as or more extreme than the one obtained if the null hypothesis is assumed to be true.
13. **(A)** Dice have no memory, so the probability that the next toss will be an even number is 0.5 and the probability that it will be an odd number is 0.5. The law of large numbers says that as the number of tosses becomes larger, the proportion of even numbers tends to become closer to 0.5.
14. **(C)** 123.800 (in \$1,000) is the y-intercept, which gives the predicted outcome if $x = 0$.
15. **(B)** R-Sq, the coefficient of determination, gives the variation in the dependent variable, y , that is accounted for by the linear model relating the y -variable to the independent variable, x . Note that 100% of the variation in the *predicted* y -values is accounted for by the linear model.
16. **(D)** The correlation r is the square root of R-Sq, where the sign, plus or minus, is the same as the slope.
17. **(D)** The mean of the sampling distribution is equal to the population mean. The standard deviation of the sampling distribution is the population standard deviation divided by the square root of the sample size. In this case, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ gives $\frac{\sigma}{\sqrt{25}} = 0.4$, and thus the population standard deviation equals $\sigma = (0.4)\sqrt{25} = 2$.
18. **(C)** Percentile ranking is a measure of relative position. Adding 5 points to everyone's score will not change the relative positions.
19. **(A)** The control group should have experiences identical to those of the experimental groups except for the treatment under examination. They should not be given a new treatment.
20. **(B)** If the alternative $p = 0.7$ is true, the probability of failing to reject H_0 and thus committing a Type II error is 1 minus the power, that is, $1 - 0.8 = 0.2$.
21. **(C)** In histograms, relative area corresponds to relative frequency. The area from 1.5 to 4.5 (fewer than 5) appears to be the same as the area between 7.5 and 10.5 (more than 7). Five does not split the area in half, so 5 is not the median. Histograms such as these show relative frequencies, not actual frequencies. Given the spread, 1 is too small an estimate of the standard deviation. The area

above 3 looks to be the same as the area above 9 and 10, so the median won't change.

22. **(C)** The policewoman is *counting*, not calculating a *proportion*. More specifically, she is counting the number of motorists *not* wearing seat belts rather than the number wearing seat belts. It seems reasonable to assume that for any given driver, the probability of not wearing a seat belt is the same and is independent from driver to driver. There is a fixed number, $n = 5$, of repetitions, and p = probability a motorist is not wearing a seat belt.
23. **(C)** In both cases, 1 hour is one standard deviation from the mean with a right tail probability of 0.159.
24. **(C)** Control, randomization, and replication are all important aspects of well-designed experiments. We try to control confounding variables, not to use them to control something else.
25. **(D)** The median appears to be roughly 215, indicating that the interval [200, 240] probably has more than 50% of the values. While the shape of a distribution is difficult to discern from a boxplot, the data do appear to be skewed to the left, indicating that the mean is less than the median. While in a boxplot without outliers each whisker contains 25% of the values, this boxplot shows four outliers on the left, and so the left whisker has four fewer values than the right whisker.
26. **(C)** For the least squares regression line, the sum and thus the mean of the residuals are always zero. A residual equals an observed value minus the predicted value, so a negative residual means the regression model overestimates the observed value.
27. **(B)** $P(\text{at least } 1) = 1 - P(\text{none}) = 1 - (0.88)^6$.
28. **(C)** For a binomial probability distribution with the probability of success equal to p and the number of trials equal to n , the expected value or mean number of successes is np and the standard deviation for the number of successes is $\sqrt{np(1 - p)}$.
29. **(A)** The normal is a good approximation to the binomial when both np and $n(1 - p)$ are ≥ 10 .
30. **(D)** The relevant condition is that both $n\hat{p} \geq 10$ and $n\hat{q} \geq 10$, and only (D) with $n\hat{p} = (80)(0.13) = 10.4$ and $n\hat{q} = (80)(0.87) = 69.6$ satisfies this.
31. **(A)** While the associate does use chance, each customer would have the same chance of being selected only if the same number of customers had names starting with each letter of the alphabet. This selection does not result in a simple random sample because each possible set of 104 customers does not have the

same chance of being picked as part of the sample. For example, a group of customers whose names all start with A will not be chosen. Sampling error (sampling variability), the natural variation inherent in a survey, is always present and is not a source of bias. Letting the surveyor have free choice in selecting the sample, rather than incorporating chance in the selection process, is a recipe for disaster!

- 32. (D)** Increasing the sample size increases the accuracy but has no effect on any bias that may be present.
- 33. (B)** $SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{0.763}{\sqrt{6}}$ and the test statistic is $t = \frac{\bar{x} - x_0}{SE(\bar{x})} = \frac{5.67 - 5}{\left(\frac{0.763}{\sqrt{6}}\right)}$.
- 34. (D)** The set $X = A \cup B$ will be bimodal with modes at the values 10 and 20. Since both A and B are of the same size, the mean of X will be $\frac{(10 + 20)}{2} = 15$. $Y = A + B$ will be roughly normal because both A and B are roughly normal. The mean of Y will be $10 + 20 = 30$, the sum of the means of A and B .
- 35. (B)** Increasing the sample size by a multiple of d divides the interval width by $\sqrt{d}$.
- 36. (D)** The t -distributions are symmetric; however, they are lower at the mean and higher at the tails and so are more spread out than the normal distribution. The greater the df , the closer the t -distributions are to the normal distribution. The 68-95-99.7 rule applies to the z -distribution and will work for t -models with very large df . All probability density curves have an area of 1 below them.
- 37. (A)** Right-skewed distributions, as in I, generally have means greater than their medians, so less than 50% of the data are greater than the mean. Symmetric distributions, as in II, have means equal to their medians, so 50% of the data are greater than the mean. Left-skewed distributions, as in III, generally have means less than their medians, so more than 50% of the data are greater than the mean.
- 38. (C)** This follows from the central limit theorem.
- 39. (D)** Confidence level (95% in this example) is the proportion of times repeated applications of this method (random samples of 500 American adults in this example) would capture the true population parameter (proportion in this example).
- 40. (C)** X and the two clusters will be clearly visible in a stemplot of these data. X is close to the mean and so will have a z -score close to 0. While boxplots do show outliers as isolated points, outliers are from the mean, so X is not an outlier. In symmetric distributions, the mean and median are equal. The IQR here is close to the range.

41. (D) When using a measurement from a sample, we are never able to say *exactly* what a population proportion is; rather, we always say we have a certain *confidence* that the population proportion lies in a particular *interval*. In this case, that interval is $82\% \pm 3\%$ or between 79% and 85%.
42. (A) Whether or not students are taking AP Statistics seems to have no relationship to which type of school they are planning to go to. Chi-square is close to 0.

Section II

1. Model Solution:

- A. There are 5 treatments: aloe, camphor, eucalyptus oil, benzocaine, and the placebo. The subjects are the 10 male volunteers. The response variable is duration of itching.
- B. i. Design X illustrates a completely randomized experimental design.
ii. Design Y illustrates a randomized block design where the volunteers are the blocks.
- C. The randomized block design (RB) of Design Y assures that all treatments are given to each individual with their own outcome-related characteristics. In the completely randomized design (CR) of Design X, for example, the two individuals receiving a particular treatment might be particularly sensitive to itching or might have resistance to topical treatments.
- D. i. For example, results of this study cannot be generalized to women as all of the participants were men.
ii. For example, men especially sensitive to certain treatments might not pick those treatments, rendering results of certain treatments incorrect as certain populations were excluded.
- E. Is there convincing statistical evidence to suggest that topically administered camphor will provide greater mean duration of itch relief than a topically administered placebo for males similar to those volunteering for this study?

A.	Identify the treatments.	Aloe, camphor, eucalyptus oil, benzocaine, and the placebo.	1 point
	Identify the subjects.	The 10 male volunteers.	
	Identify the response variable.	Duration of itching.	

B.	i.	Identify the experimental design of Design X.	Completely randomized design.	1 point
	ii.	Identify the experimental design of Design Y.	Randomized block design.	1 point
C.		Describe a statistical advantage of RB.	RB assures that all treatments are given to each individual with their own outcome-related characteristics.	1 point
		Describe a statistical disadvantage of CR.	The two individuals receiving one of the treatments might be particularly sensitive to itching or might have resistance to topical treatments.	1 point
D.	i.	Describe a limit to making generalizations.	This cannot be generalized to women.	1 point
	ii.	Describe a drawback.	For example, men especially sensitive to certain treatments might intuitively not pick those treatments, rendering results of those treatments incorrect as certain populations were excluded.	1 point
E.		First component: identify the variables of interest.	Topically administered camphor or placebo.	1 point
		Second component: identify the parameter and direction of the alternative hypothesis.	The mean duration of itch relief from topically administered camphor is greater than the mean duration of itch relief from a topically administered placebo.	1 point
		Third component: identify the population	The conclusion can be applied to all males similar to those	1 point

	to which the conclusion is applicable.	volunteering for this study.	
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2. Model Solution:

- A. i. This is an observational study.
 ii. The numerical variable is runs per game.
- B. i. The mean is 4.27 because $\bar{x} = \sum xP(x) = 0(0.05) + 1(0.10) + 2(0.13) + 3(0.15) + 4(0.14) + 5(0.12) + 6(0.10) + 7(0.08) + 8(0.06) + 9(0.04) + 10(0.03) = 4.27$.
 ii. With $0.05 + 0.10 + 0.13 + 0.15 = 0.43$ below 4 runs and with $0.12 + 0.10 + 0.08 + 0.06 + 0.04 + 0.03 = 0.43$ above 4 runs, the median must be 4.
 iii. The mean is greater than the median as was to be expected because the distribution is slightly skewed to the right.
- C. This is a binomial with $n = 4$ and $p = 0.05$. $P(x \geq 1) = 1 - P(x = 0) = 1 - (0.95)^4 = 0.1855$.
- D. i. By the CLT, the shape of the sampling distribution of $\bar{x}$ is approximately normal.
 ii. The standard deviation of the sampling distribution is $\sigma_{\bar{x}} = \frac{2.578}{\sqrt{200}} = 0.1823$.
- E. The population for a valid investigative study is all home games of major league teams.

A.	i.	Identify the type of study.	Observational.	1 point
	ii.	Identify the variable in the study.	Runs per game.	1 point
B.	i.	Calculate a mean.	$\bar{x} = 0(0.05) + 1(0.10) + 2(0.13) + 3(0.15) + 4(0.14) + 5(0.12) + 6(0.10) + 7(0.08) + 8(0.06) + 9(0.04) + 10(0.03) = 4.27$	1 point
	ii.	Calculate a median.	With $0.05 + 0.10 + 0.13 + 0.15 = 0.43$ below 4 runs and with $0.12 + 0.10 + 0.08 + 0.06 + 0.04 + 0.03 = 0.43$ above 4 runs, the median must be 4.	1 point

iii.	Explain the direction of a difference.	The mean is greater than the median as expected because the distribution is skewed right.	1 point
C.	Identify the distribution and parameters.	This is binomial with $n = 4$ and $p = 0.05$.	1 point
	Calculate a probability.	$P(x \geq 1) = 1 - P(x = 0) = 1 - 0.05 = 0.95$	1 point
D.	i. Describe a shape.	The shape is approximately normal.	1 point
	ii. Calculate a standard deviation.	$\sigma_{\bar{x}} = \frac{2.578}{\sqrt{200}} = 0.1823$	1 point
E.	Identify a population.	All home games of major league teams.	1 point

3. Model Solution:

- The explanatory variable is categorical; the infants received either daily stimulation or no stimulation. Months before walking is a numerical variable.
- An appropriate inference procedure is a two-sample t -test for the difference between two population means.
- The null hypothesis is $H_0: \mu_1 = \mu_2$, and the alternative hypothesis is $H_a: \mu_1 < \mu_2$. Let μ_1 be the true mean number of months at which first steps are taken by infants receiving daily stimulation. Let μ_2 be the true mean number of months at which first steps are taken by infants not receiving daily stimulation. The randomization condition is satisfied as we are given that the infants are randomly assigned to the two groups. The normality condition is satisfied because dotplots of the two groups show no outliers and no strong skew. So it is not unreasonable to assume the data come from roughly normal distributions. The value of the test statistic is $t = -1.20$ with a p -value of 0.1234 and $df = 16.69$.
- With this large a p -value, $0.1234 > 0.05$, there is not sufficient evidence to reject H_0 . There is not convincing evidence that the true mean number of months at which first steps are taken is less for infants receiving daily stimulation than for infants not receiving daily stimulation.

A.	Identify the variable.	With or without stimulation; months before walking.	1 point
	Classify the variable.	With or without stimulation is a categorical variable; months before walking is a numerical variable.	1 point
B.	Identify the procedure.	Two-sample t -test for the difference between two population means.	1 point
C.	Identify H_0 and H_a .	$H_0: \mu_1 = \mu_2$ and $H_a: \mu_1 < \mu_2$	1 point
	Identify the parameters.	μ_1 is the true mean number of months at which first steps are taken by infants receiving daily stimulation. μ_2 is the true mean number of months at which first steps are taken by infants not receiving daily stimulation.	1 point
	Justify whether the randomization condition is met.	We are given that the infants are randomly assigned to the two groups.	1 point
	Justify whether the normality condition is met.	Dotplots of the two groups show no outliers and no strong skew, so it is not unreasonable to assume the data come from roughly normal distributions.	1 point
	Calculate the test statistic and p -value.	$t = -1.20$ with a p -value of 0.1234 and $df = 16.69$	1 point
D.	Compare the p -value to the significance level.	With this large a p -value, $0.1234 > 0.05$, there is not sufficient evidence to reject H_0 .	1 point
	Justify whether the claim is or is not supported.	There is not convincing evidence that the true mean number of months at which first steps are taken is less for infants	1 point

		receiving daily stimulation than for infants not receiving daily stimulation.	
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4. Model Solution:

- A. In a normal distribution, the mean and median should be nearly equal, and here $16.121 \approx 16.152$. For a discrete set, the min and max should be roughly 3 standard deviations from the mean, and here $16.121 - 3(0.040) = 16.001 \approx 15.991$ and $16.121 + 3(0.040) = 16.241 \approx 16.283$. Under a normal curve, Q_1 and Q_3 are roughly $\text{invNorm}(0.75) \approx 0.674$ standard deviations from the mean, and here $16.121 - (0.674)(0.040) = 16.094 \approx 16.091$ and $16.121 + (0.674)(0.040) = 16.148 \approx 16.155$. Thus, yes, the sample volumes appear to roughly follow a roughly normal distribution.
- B. i. Given a normal distribution with $\mu = 16.121$ and $\sigma = 0.040$, $P(x < 16) = 0.0012$.
 ii. Given a normal distribution with $\mu \bar{x} = 16.121$ and $\sigma_{\bar{x}} = \frac{0.040}{\sqrt{25}}$, $P(\bar{x} < 16) \approx 0.000$.
- C. i. The manufacturer should make the advertised volume smaller. To reduce the percentage of bottles weighing less than advertised, without changing the mean or standard deviation, the manufacturer would have to decrease the advertised volume. In other words, the manufacturer would have to make the advertised volume further into the left tail of the roughly normal distribution of bottle volumes.
 ii. The manufacturer should make the mean volume of liquid dispensed larger. Decreasing the percentage of underweight bottles would require putting more volume into each bottle. So the mean of the distribution of bottle volumes would increase, and the advertised volume would then fall further into the left tail of the roughly normal distribution of bottle volumes.
- D. i. Use a one-sample t -interval for a population mean.
 ii. (16.0986, 16.1434)
 iii. We are 99% confident that the true mean volume dispensed into the soft drink bottles models is between 16.0986 and 16.1434 fluid ounces.
- E. There is sufficient statistical evidence that the true mean volume dispensed is greater than 16 fluid ounces because the entire confidence interval (16.0986, 16.1434) is greater than 16.

A.		Explain why a distribution appears roughly normal.	Any 2 of the following 3 points should be met: The mean and median should be nearly equal; $16.121 \approx 16.152$. Min and max should be roughly 3 standard deviations from the mean; $16.121 - 3(0.040) = 16.001 \approx 15.991$ and $16.121 + 3(0.040) = 16.241 \approx 16.283$. Q_1 and Q_3 should be roughly $\text{invNorm}(0.75) \approx 0.674$ standard deviations from the mean; $16.121 - (0.674)(0.040) = 16.094 \approx 16.091$ and $16.121 + (0.674)(0.040) = 16.148 \approx 16.149$. Thus, yes, the sample volumes appear to follow a roughly normal distribution.	1 point
B.	i.	Calculate a probability.	$N(16.121, 0.040)$, $P(x < 16) = 0.0012$	1 point
	ii.	Calculate a probability.	$N(16.121, \frac{0.040}{\sqrt{25}})$, $P(\bar{x} < 16) \approx 0.000$	1 point
C.	i.	Explain how to make a change.	The advertised volume should be smaller. To reduce the percentage of bottles weighing less than advertised, make the advertised volume further into the left tail of the roughly normal distribution of bottle volumes.	1 point
	ii.	Explain how to make a change.	The mean volume of liquid dispensed should be larger. Put more volume into each bottle. The mean will increase, and the advertised volume will fall further into the left tail of the	1 point

		roughly normal distribution of bottle volumes.		
D.	i.	Identify the procedure.	One-sample t -interval for a population mean.	1 point
	ii.	Calculate the interval.	(16.0986, 16.1434)	1 point
	iii.	Interpret the result.	We are 99% confident that the true mean volume dispensed into the bottles models is between 16.0986 and 16.1434 fluid ounces.	1 point
E.		Determine if there is sufficient evidence.	There is sufficient statistical evidence that the true mean volume dispensed is greater than 16 fluid ounces because the entire confidence interval (16.0986, 16.1434) is greater than 16.	1 point

Practice Test 3

Section I: Questions 1–42

SPEND 90 MINUTES ON THIS PART OF THE EXAM.

DIRECTIONS: The questions or incomplete statements that follow are each followed by four suggested answers or completions. Choose the response that best answers the question or completes the statement.

1. Which of the following is a true statement?
 - (A) While properly designed experiments can strongly suggest cause-and-effect relationships, a complete census is the only way of establishing such a relationship.
 - (B) If properly designed, observational studies can establish cause-and-effect relationships just as strongly as can properly designed experiments.
 - (C) Controlled experiments are often undertaken later to establish cause-and-effect relationships first suggested by observational studies.
 - (D) A useful approach to overcome bias in observational studies is to increase the sample size.
2. Two classes take the same exam. Suppose a certain score is at the 40th percentile for the first class and at the 80th percentile for the second class. Which of the following is the most reasonable conclusion?

- (A) Students in the first class generally scored higher than students in the second class.
- (B) Students in the second class generally scored higher than students in the first class.
- (C) A score at the 20th percentile for the first class is at the 40th percentile for the second class.
- (D) One of the classes has twice the number of students as the other.

3. In an experiment, the control group should receive

- (A) treatment opposite that given to the experimental group.
- (B) the same treatment given to the experimental group without knowing they are receiving the treatment.
- (C) a procedure identical to that given to the experimental group except for receiving the treatment under examination.
- (D) none of the procedures given to the experimental group.

4. In a simple random sample (SRS) of 625 families who do not live near any chemical plant, 10 had children with leukemia. In an SRS of 412 families living near chemical plants, 15 had children with leukemia. A 90% confidence interval of the difference in proportions is reported to be -0.0204 ± 0.0173 . What is a proper conclusion?

- (A) The interval is invalid because probabilities cannot be negative.
- (B) Families living near chemical plants are approximately 2.04% more likely to have children with leukemia than families who do not live near any chemical plant.
- (C) Ninety percent of families living near chemical plants are approximately 2.04% more likely to have children with leukemia than families who do not live near any chemical plant.
- (D) We are 90% confident that the difference in proportions between families who do not live near any chemical plant having children with leukemia and families who live near chemical plants having children with leukemia is between -0.0377 and -0.0031 .

5. From a random sample of registered voters, 41% thought the age of our political leaders is a major problem. A 95% confidence interval was constructed from the sample, and the margin of error for the estimate was 3%. Which of the following is the best interpretation of the confidence interval?
- (A) There is a 0.95 probability that between 38% and 44% of the population of registered voters think the age of our political leaders is a major problem.
 - (B) We are 95% confident that between 38% and 44% of the sample of registered voters think the age of our political leaders is a major problem.
 - (C) We are 95% confident that between 38% and 44% of the population of registered voters think the age of our political leaders is a major problem.
 - (D) If 95% confidence intervals are calculated from all possible samples of the given size, 95% of these intervals will capture the true percentage of the population of registered voters who think the age of our political leaders is a major problem.

Questions 6–8 refer to the following situation:

A computer manufacturer sets up three locations to provide technical support for its customers. Logs are kept, noting whether or not calls about problems are solved successfully. Data from a sample of 1,000 calls are summarized in the following table:

	Location			
	1	2	3	Total
Problem solved	325	225	150	700
Problem not solved	125	100	75	300
Total	450	325	225	1,000

6. What is the probability that a call from location 2 is solved?

(A) $\frac{225}{325}$

(B) $\frac{225}{700}$

(C) $\frac{225}{1,000}$

(D) $\frac{325 + 700 - 225}{1,000}$

7. Are “problem solved” and “location 2” independent events?

(A) Yes, because $\frac{700}{1,000} \times \frac{325}{1,000} \neq \frac{225}{1,000}$.

(B) No, because $\frac{700}{1,000} \times \frac{325}{1,000} \neq \frac{225}{1,000}$.

(C) Yes, because $P(\text{problem solved} \mid \text{location 2}) \neq P(\text{location 2} \mid \text{problem solved})$.

(D) No, because $P(\text{problem solved} \mid \text{location 2}) \neq P(\text{location 2} \mid \text{problem solved})$.

8. Assuming there is no association between location and whether or not a problem is resolved successfully, what is the expected number of successful calls (problem solved) from location 1?

(A) $\frac{(325)(450)}{700}$

(B) $\frac{(450)(700)}{1,000}$

(C) $\frac{(325)(700)}{450}$

(D) $\frac{(325)(700)}{1,000}$

9. Which of the following statements about the correlation coefficient is true?

- (A) The correlation coefficient and the slope of the regression line may have opposite signs.
- (B) A correlation of 1 indicates a perfect cause-and-effect relationship between the variables.
- (C) Correlations of $+0.87$ and -0.87 indicate the same degree of clustering around the regression line.
- (D) A correlation of 0 shows little or no association between two variables.

10. A school district has 450 staff members, including secretaries, registrars, and counselors. A member of the Board of Education wanted to estimate the percentage of staff who would join the picket line if the teachers went out on strike. Fifty randomly selected staff were surveyed. Of the 50 selected staff, 34 expressed a willingness to join the picket line if the teachers went out on strike. Have the conditions been met for inference with a one-sample z -interval?

- (A) No, because the sample size violates the 10% condition.
- (B) No, because the sample size is not large enough to ensure that the sampling distribution is approximately normal.
- (C) No, because both $n\hat{p}$ and $n(1 - \hat{p})$ have to be ≥ 10 .
- (D) Yes, the conditions for inference have been met.

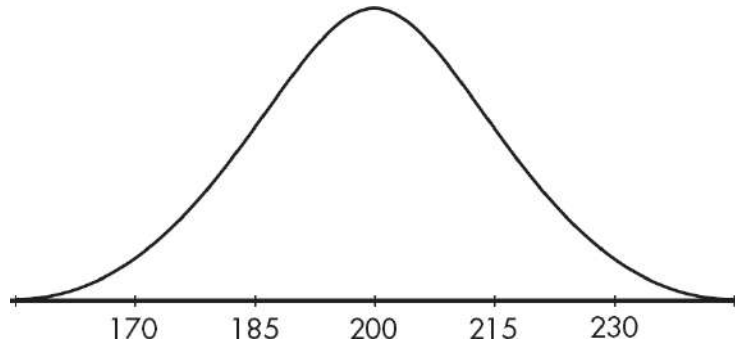
11. What is the proper order of size for the probabilities of the following three events?

- I. A randomly selected person over 80 years old is registered to vote.
- II. A randomly selected person is over 80 years old and registered to vote.
- III. A randomly selected person who is registered to vote is over 80 years old.

- (A) $P(I) < P(II) < P(III)$
- (B) $P(I) > P(II) > P(III)$
- (C) $P(I) < P(III) < P(II)$

(D) $P(I) > P(III) > P(II)$

12. A histogram of the cholesterol levels of all employees at a large law firm is as follows:



Which of the following is the best estimate of the standard deviation of this distribution?

(A) $\frac{230 - 170}{6} = 10$

(B) 15

(C) $\frac{200}{6} = 33.3$

(D) $230 - 170 = 60$

13. A soft drink dispenser can be adjusted to deliver any fixed number of ounces. If the machine is operating with a standard deviation in delivery equal to 0.3 ounce, what should be the mean setting so that a 12-ounce cup will overflow less than 1% of the time? Assume an approximately normal distribution for ounces delivered.

(A) $12 + 2.326(0.3)$ ounces

(B) $12 - 2.326(0.3)$ ounces

(C) $12 + 2.576(0.3)$ ounces

(D) $12 - 2.576(0.3)$ ounces

14. An insurance company wishes to study the number of years drivers in a large city go between automobile accidents. They plan to obtain and analyze the data from a sample of drivers. Which of the following is a true statement?

- (A) A reasonable time-and-cost-saving procedure would be to use systematic random sampling on an available list of all AAA (Automobile Association of America) members in the city.
- (B) A reasonable time-and-cost-saving procedure would be to randomly choose families and include all drivers in each of these families in the sample.
- (C) To determine the mean number of years between accidents, randomness in choosing a sample of drivers is not important as long as the sample size is very large.
- (D) None of the above are true statements.

15. The probability that a person will show a certain gene-transmitted trait is 0.8 if the father shows the trait and 0.06 if the father doesn't show the trait. Suppose that the children in a certain community come from families in 25% of which the father shows the trait. Given that a child shows the trait, what is the probability that her father shows the trait?

(A) $(0.25)(0.8) + (0.75)(0.06)$

(B) $\frac{0.25}{(0.25)(0.8) + (0.75)(0.06)}$

(C) $\frac{0.8}{(0.25)(0.8) + (0.75)(0.06)}$

(D) $\frac{(0.25)(0.8)}{(0.25)(0.8) + (0.75)(0.06)}$

16. Given an experiment with $H_0: \mu = 10$, $H_a: \mu > 10$, and a sample mean of 11, which of the following increases if n is discovered to have been greater than originally thought?

- I. The p -value
- II. The probability of a Type II error
- III. The power of the test

- (A) I only
- (B) II only
- (C) III only

(D) II and III

17. If all the values of a data set are the same, all of the following must equal zero except for which one?

- (A) Mean
- (B) Standard deviation
- (C) Range
- (D) Interquartile range

18. Peer relationships are an important part of the socialization process of children and require the use of many complex social skills. Substantial evidence suggests that children with poor peer relationships are more likely to be truant, repeat grade levels, and drop out of school. In a random sample of 15 fifth-grade students, peer acceptance as measured on a sociometric scale showed a mean of 2.72 with a standard deviation of 0.15. Which of the following gives a 90% confidence interval for the mean peer acceptance of all fifth-grade students?

- (A) $2.72 \pm 1.761 \frac{0.15}{\sqrt{15}}$
- (B) $2.72 \pm 1.753 \frac{0.15}{\sqrt{15}}$
- (C) $2.72 \pm 1.761 \frac{0.15}{\sqrt{14}}$
- (D) $2.72 \pm 1.753 \frac{0.15}{\sqrt{14}}$

19. A company has 1,000 employees evenly distributed throughout five assembly plants. A sample of 30 employees is to be chosen as follows. Each of the five managers will be asked to place the 200 time cards of their respective employees into a bag, shake them up, and randomly draw out six names. The six names from each plant will be put together to make up the sample. Will this method result in a simple random sample of the 1,000 employees?

- (A) No, because the plants are not chosen randomly.
- (B) No, because not every group of 30 employees has the same chance of being selected.
- (C) Yes, because every employee has the same chance of being selected.
- (D) Yes, because every plant is equally represented.

20. Given that $P(E) = 0.32$, $P(F) = 0.15$, and $P(E \cap F) = 0.048$, which of the following is a correct conclusion?

- (A) The events E and F are both independent and mutually exclusive.
- (B) The events E and F are neither independent nor mutually exclusive.
- (C) The events E and F are mutually exclusive but not independent.
- (D) The events E and F are independent but not mutually exclusive.

21. Researchers surveyed a random sample of 55 adults in neighborhood A and 64 adults in neighborhood B from populations of over 1,000 adults in each neighborhood. The sampled individuals were asked what their families anticipated spending on Halloween. Let $\bar{x}_A$ be the calculated mean spending of the 55 adults in neighborhood A , and let $\bar{x}_B$ be the calculated mean spending of the 64 adults in neighborhood B . Which of the following is the best explanation for why the sampling distribution of $\bar{x}_A - \bar{x}_B$ can be modeled with a normal distribution?

- (A) There are at least 30 people in each population.
- (B) The sample sizes are sufficiently large.
- (C) The population distributions are assumed to be roughly normal.
- (D) The sample distributions are assumed to be unimodal and roughly symmetric.

22. Consider the following back-to-back stemplot:

	0	348	Key: 2 5 6 represents a value of 52 on the left and 56 on the right
	1	01256	
843	2	29	
65210	3	2557	
92	4		
7552	5	6	
	6	1458	
6	7	09	
8541	8		
90	9		

Which of the following is a correct statement?

- (A) The distributions have the same mean.
- (B) The distributions have the same median.
- (C) The interquartile range of the distribution to the left is 20 greater than the interquartile range of the distribution to the right.
- (D) The distributions have the same variance.

23. Which of the following is a correct statement?

- (A) A study results in a 99% confidence interval estimate of (34.2, 67.3). This means that in about 99% of all samples selected by this method, the sample means will fall between 34.2 and 67.3.
- (B) A high confidence level may be obtained no matter what the sample size.
- (C) The sampling distribution for a mean has standard deviation $\frac{\sigma}{\sqrt{n}}$ only when n is sufficiently large (typically one uses $n \geq 30$).
- (D) The center of any confidence interval is the population parameter.

24. A fair coin is flipped 6 times. Consider the following 3 outcomes where H refers to a toss landing heads, and T refers to a toss landing

tails:

- I. The result in order is HHHTTT.
- II. The result in order is HTTHHT.
- III. The result includes 3 Hs and 3 Ts.

What is a comparison of the three probabilities?

- (A) $P(\text{I}) < P(\text{II}) < P(\text{III})$
- (B) $P(\text{III}) < P(\text{I}) < P(\text{II})$
- (C) $P(\text{I}) = P(\text{II}) < P(\text{III})$
- (D) $P(\text{III}) < P(\text{I}) = P(\text{II})$

25. Suppose two events, E and F , have nonzero probabilities p and q , respectively. Which of the following is impossible?

- (A) $p + q > 1$
- (B) $\frac{p}{q} > 1$
- (C) E and F are both independent and mutually exclusive.
- (D) E and F are neither independent nor mutually exclusive.

26. An inspection procedure at a manufacturing plant involves picking four items at random and accepting the whole lot if at least three of the four items are in perfect condition. If in reality 90% of the whole lot are perfect, what is the probability that the lot will be accepted?

- (A) $4(0.9)^3(0.1) + (0.9)^4$
- (B) $0.1 - 4(0.9)^3(0.1)$
- (C) $4(0.9)^3(0.1)$
- (D) $1 - (0.9)^4$

27. A town has one high school, which buses students from urban, suburban, and rural communities. Which of the following samples is recommended in studying attitudes toward tracking of students in honors, regular, and below-grade classes?

- (A) Convenience sample
- (B) Simple random sample
- (C) Stratified random sample
- (D) Systematic random sample

28. Suppose there is a correlation of $r = 0.9$ between number of hours per day students study and GPAs. Which of the following is a reasonable conclusion?

- (A) 81% of the variation in GPAs can be explained by variation in number of study hours per day.
- (B) 90% of students who receive high grades study a lot.
- (C) 90% of the variation in GPAs can be explained by variation in number of study hours per day.
- (D) 10% of the variation in GPAs cannot be explained by variation in number of study hours per day.

29. To determine the average number of children living in single-family homes, a researcher picks a simple random sample of 50 such homes. However, even after one follow-up visit, the interviewer is unable to make contact with anyone in 8 of these homes. Concerned about nonresponse bias, the researcher picks another simple random sample and instructs the interviewer to keep trying until contact is made with someone in 8 more homes for a total of 50 homes. The average number of children is determined to be 1.73. Is this estimate probably too low or too high?

- (A) Too low, because of undercoverage bias
- (B) Too low, because convenience samples overestimate average results
- (C) Too high, because of undercoverage bias
- (D) Too high, because voluntary response samples overestimate average results

30. Past studies indicate that 63% of high school students believe that mothers are more strict than fathers when it comes to doing homework. A family therapist believes that the proportion is greater than 0.63. A random sample of students was surveyed. With $H_0: p = 0.63$ and $H_a: p > 0.63$, the test statistic is $z = 1.623$ with a p -value of 0.0523. Which of the following is a correct interpretation of the test statistic?

- (A) 0.0523 is the probability of obtaining a sample proportion as extreme as or more extreme than the one obtained by the therapist.
- (B) If it is true that 63% of high school students believe that mothers are more strict than fathers when it comes to doing homework, the sample proportion and the claimed proportion of 0.63 are 1.623 standard errors apart.
- (C) If it is true that 63% of high school students believe that mothers are more strict than fathers when it comes to doing homework, 0.0523 is the probability of obtaining a sample proportion as extreme as or more extreme than the one obtained by the therapist.
- (D) If confidence intervals are calculated from all possible samples of the given size, in 95% of these intervals the sample proportion will be within 1.623 standard errors from the true population proportion.

31. An experiment is to be conducted to determine whether taking fish oil capsules or garlic capsules has more of an effect on cholesterol levels. In past studies, it was noted that daily exercise intensity (low, moderate, high) is associated with cholesterol level, but average sleep length (< 5 , $5-8$, > 8 hours) is not associated with cholesterol level. This experiment should be done

- (A) by blocking on exercise intensity.
- (B) by blocking on sleep length.
- (C) by blocking on cholesterol level.
- (D) by blocking on capsule type.

32. A nutritionist believes that taking elderberry extract will shorten the mean duration of flu symptoms to less than 5 days. The nutritionist would like to create a study to test her belief. Which of the following would be the appropriate components of the investigative question?

- (A) The parameter is the mean quantity of elderberry extract found in commonly available pills, and the direction of the alternative is less than 5.
- (B) The parameter is the mean duration of flu symptoms among all flu sufferers in the sample who take elderberry extract, and the direction of the alternative is less than 5.
- (C) The parameter is the proportion of all flu sufferers taking elderberry extract whose symptoms last less than 5 days, and the direction of the alternative is less than 5.
- (D) The parameter is the mean duration of flu symptoms among all flu sufferers taking elderberry extract, and the direction of the alternative is less than 5.

33. A sales researcher randomly chooses 100 female high school students to participate in a study to investigate whether Nike Air Force 1s or Claire's are more comfortable shoes. The researcher is considering two study designs.

- I. Randomly assign 50 participants to wear Nike Air Force 1 shoes for 3 months and then Claire's shoes for the next 3 months. The remaining participants will wear Claire's shoes for the first 3 months and then Nike Air Force 1 shoes for the next 3 months.
- II. Randomly assign 50 participants to wear Nike Air Force 1 shoes for 6 months and the remaining participants to wear Claire's shoes for 6 months.

Which of the following best describes the two designs?

- (A) Both designs are completely randomized experiments.
- (B) Both designs are matched pairs experiments.

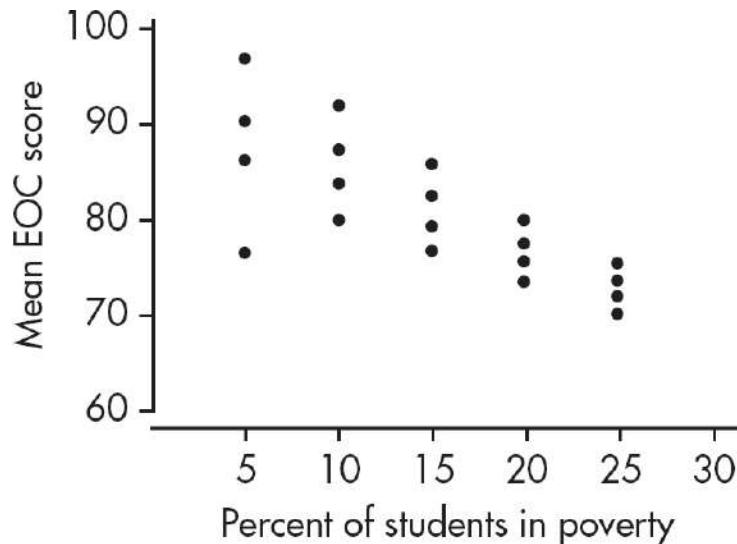
- (C) Design I is a completely randomized experiment, and design II is a matched pairs experiment.
- (D) Design I is a matched pairs experiment, and design II is a completely randomized experiment.
34. The weight of an aspirin tablet is 300 milligrams according to the bottle label. An FDA investigator weighs a simple random sample of seven tablets, obtains weights of 299, 300, 305, 302, 299, 301, and 303, and runs a hypothesis test of the manufacturer's claim. Which of the following gives the p -value of this test?
- (A) $P(t > 1.54)$ with $df = 6$
- (B) $P(t > 1.54)$ with $df = 7$
- (C) $2P(t > 1.54)$ with $df = 6$
- (D) $2P(t > 1.54)$ with $df = 7$
35. Past studies indicate that the average number of acres of wildland burned per wildfire each year in the United States is 170 acres. A forest ranger believes that the average might now be greater. A random sample of wildfires was surveyed. With $H_0: \mu = 170$ and $H_a: \mu > 170$, the p -value of the test is 0.022. Which of the following is a correct interpretation of the p -value?
- (A) 0.022 is the probability the population mean is greater than the one obtained by the ranger.
- (B) 0.022 is the probability of obtaining a sample mean as large as or larger than the one obtained by the ranger.
- (C) If it is true that the average number of acres of wildland burned per wildfire each year in the United States is 170 acres, 0.022 is the probability of obtaining a sample mean as large as or larger than 170.
- (D) If it is true that the average number of acres of wildland burned per wildfire each year in the United States is 170 acres, 0.022 is the probability of obtaining a sample mean as large as or larger than the one obtained by the ranger.

36. Which of the following is *not* true with regard to contingency tables for chi-square tests for independence?

- (A) Both variables are categorical.
- (B) Observed frequencies should be whole numbers.
- (C) Expected frequencies should be whole numbers.
- (D) The expected frequency for any cell can be found by multiplying the row total by the column total and dividing by the table total.

Questions 37–39 refer to the following situation:

A random sample of 20 school districts looked at average EOC (end-of-course) test scores versus percent of students living in poverty. A scatterplot of the data is below.



37. The scatterplot can be described as negative, linear, and moderate to strong. What does “linear” mean in this context?

- (A) There is no distinct pattern in the residual plot.
- (B) The coefficient of determination, r^2 , is large (close to 1).
- (C) As percent of students in poverty increases by 1%, the mean EOC score tends to change by a constant amount on average.

- (D) The dependent variable, y = mean EOC score, and the independent variable, x = percent of students in poverty, are related by a linear equation, $y = a + bx$.

38. How would the correlation, r , change if the four values corresponding to 5% were removed?

- (A) The correlation would not change.
(B) The correlation would decrease.
(C) The correlation would increase.
(D) This cannot be answered from viewing the scatterplot.

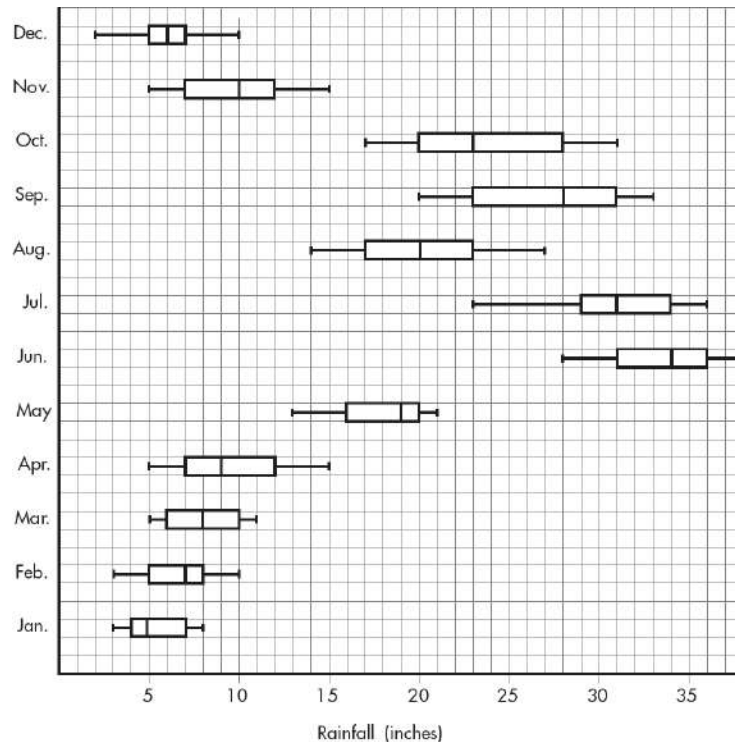
39. Which of the following would be revealed by a graph of the residuals versus percent of students in poverty?

- (A) There is a negative linear relationship between the residuals and the percent in poverty.
(B) The sum of the residuals is greater than 0.
(C) The mean of the residuals is less than 0.
(D) The variation in the mean EOC scores is not the same across the percents in poverty.

40. Studies have shown that only 36% of young adults feel that the death penalty is applied fairly by courts. In a random sample of 50 young adults, what is the expected number of young adults that do not feel that the death penalty is applied fairly by courts?

- (A) $50(0.36)$
(B) $50(0.64)$
(C) $\sqrt{50(0.36)(0.64)}$
(D) $\sqrt{\frac{(0.36)(0.64)}{50}}$

41. The parallel boxplots below show monthly rainfall summaries for Liberia, West Africa.



Which of the following months has the least variability as measured by *interquartile range*?

- (A) January
- (B) February
- (C) March
- (D) December

42. In a random sample of 750 teenagers, 36% said they picked losing weight as one of their New Year's resolutions. A 90% confidence interval for the proportion of all teenagers who picked losing weight as one of their New Year's resolutions is given by (0.3312, 0.3888). Which of the following is a correct interpretation of the confidence level of 90%?

- (A) There is a 0.90 probability that the true proportion of all teenagers who picked losing weight as one of their New Year's resolutions is between 0.3312 and 0.3888.
- (B) We are 90% confident that the true proportion of all teenagers who picked losing weight as one of their New Year's resolutions

is between 0.3312 and 0.3888.

- (C) If the proportion of teenagers who picked losing weight as one of their New Year's resolutions is calculated for repeated samples of 750 teenagers, 90% of these sample proportions will fall between 0.3312 and 0.3888.
- (D) If a confidence interval for the proportion of teenagers who picked losing weight as one of their New Year's resolutions is calculated for repeated samples of 750 teenagers, approximately 90% of these intervals will capture the true population proportion.

STOP

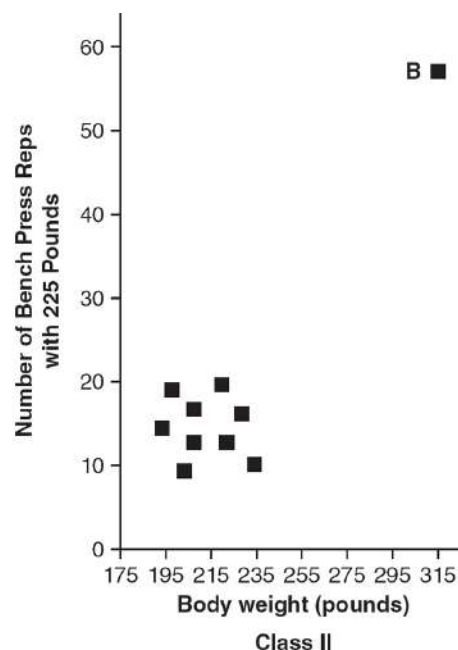
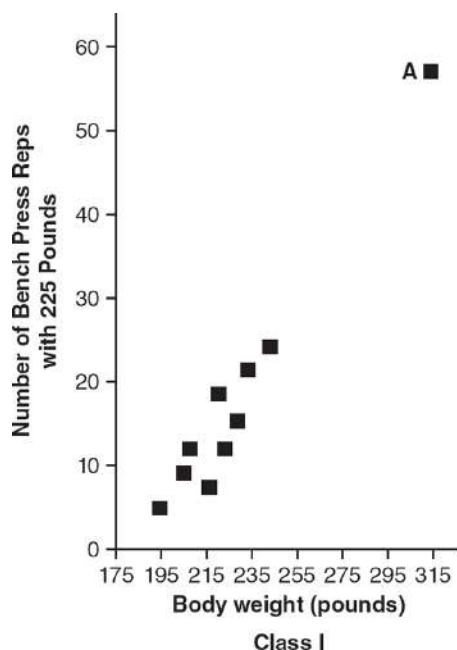
If there is still time remaining, you may check your work on this section.

Section II

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your results and explanations.

1. A study is proposed to compare two treatments for patients with significantly narrowed neck arteries (carotid stenosis). Some patients will be treated with surgery to remove built-up plaque, while others will be treated with stents to improve circulation. The response variable will be the proportion of patients who suffer a major complication such as a stroke or heart attack within 5 years after the treatment.
 - A.
 - i. **Identify** the type of study.
 - ii. **Identify** and **classify** the categorical explanatory variable in this study.
 - B.
 - i. Suppose the researchers allowed the patients' doctors to decide which of their patients should receive surgery and which should receive stents. **Explain** why such a method of assignment might lead to an incorrect conclusion.

- ii. Suppose half of the patients available for this study have had a ministroke in the previous year. **Explain** why using a randomized block design would be more appropriate than using a fully randomized design.
 - C.
 - i. **Explain** how choosing a different sample size can increase the power of detecting a difference between using surgery versus stents for patients with this medical condition.
 - ii. **Explain** how increasing the significance level α can increase the power of detecting a difference between using surgery versus stents for patients with this medical condition.
 - D. The experiment will last over the course of 5 years. **Identify** a possible ethical concern that could arise.
 - E. Medical researchers are interested in whether there is a difference in the proportions of patients suffering major complications between the surgery and stent recipients. **Determine** a valid investigative question.
2. A trainer has two weight lifting classes, Class I and Class II. Each class has 9 amateur adult males and 1 professional adult male. Their body weights and number of bench press reps with 225 pounds are shown in the following dotplots. The professionals are designated as A and B.



[Follow for extended description](#)

- A.
 - i. **Identify** the type of study the trainer conducted.
 - ii. **Identify** and **classify** the quantitative variable(s) in this study.
- B. **Compare** the two scatterplots.
- C. The equation of the least squares regression line for the Class I data is

$$\text{Predicted reps} = -73.38 + 0.4029(\text{Body weight})$$

Suppose a new male amateur joining Class I has body weight 255 pounds and can do 30 reps.

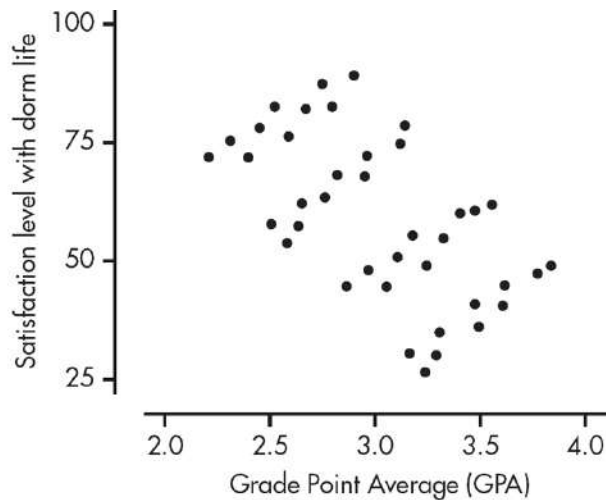
- i. Using the regression line formula, **calculate** his predicted number of reps. Show your work.
 - ii. **Calculate** and **interpret** the residual for this person. Show your work.
- D. Suppose the professionals are removed from the study.
 - i. **Describe** the change in the slope and correlation in a regression study of the Class I data.
 - ii. **Describe** the change in the slope and correlation in a regression study of the Class II data.
- E. The trainer would like to study the relationship between body weight and number of bench press reps with 225 pounds for amateur males signing up for his classes. **Determine** the population for a valid investigative question for this study.

3. Snails are featured heavily as part of the local cuisine in France, where they are called *escargot*. An escargot harvester, interested in estimating the quantity consumed in French restaurants, obtains statistics from a random sample of restaurants over a random sample of 50 days. The mean number of snails consumed was 101,000 with a standard deviation of 12,500. At the 95% confidence level, complete the appropriate inference procedure.

- A. **Identify** and **classify** the variable(s) in this study.
- B. **Identify** the type of study and the appropriate inference procedure.
- C. **Complete** the inference procedure, including calculating the appropriate statistics.

D. **Justify** a conclusion in context.

4. A college guidance counselor obtains a survey from a random sample of 40 students who live in the college dormitories. He concludes that students with higher GPAs (on a 4-point scale) appear to be less satisfied (on a 100-point scale) with dorm life. A scatterplot and linear regression analysis show the following computer output:



Predicted satisfaction level = $117.5 - 19.93(\text{GPA})$

$S = 14.80$

$df = 38$

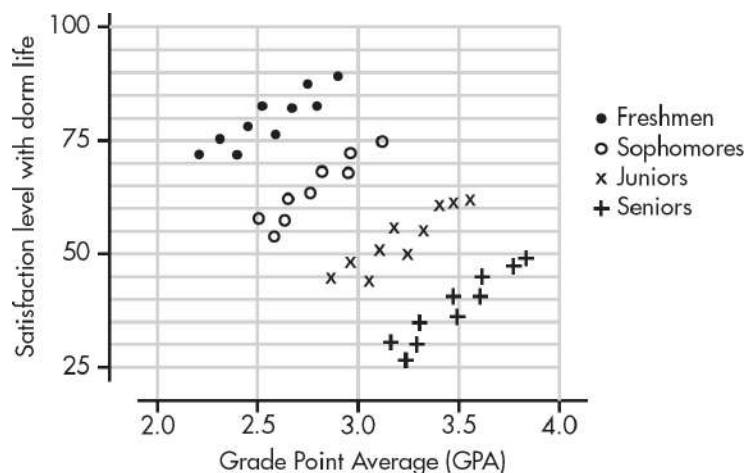
$t = -3.559$

$P = 0.001$

$R\text{-Sq} = 0.250$

[Follow for extended description](#)

- A. **Determine** if the scatterplot and regression analysis support the guidance counselor's conclusion. **Justify** your conclusion.
- B. **Calculate** the correlation. Show your work.
- C. If one student's GPA is 0.5 greater than a second student's GPA, **determine** how their predicted satisfaction levels compare.
- D. A student who had taken AP Statistics in high school reads that the survey was obtained from a stratified random sample. She analyzes the data and produces the following revised scatterplot:



[Follow for extended description](#)

- i. If one student's GPA is 0.5 greater than a second student's GPA, **estimate** how their predicted satisfaction levels would compare if both students are sophomores.
 - ii. Based on the student's analysis, **determine** how the guidance counselor's conclusion should be modified to give a better description of the relationship between GPA and satisfaction level for the students in the sample.
- E. **Determine** the population for a valid investigative question for these data.

STOP

If there is still time remaining, you may check your work on this section.

[Answer Explanations](#)

Section I

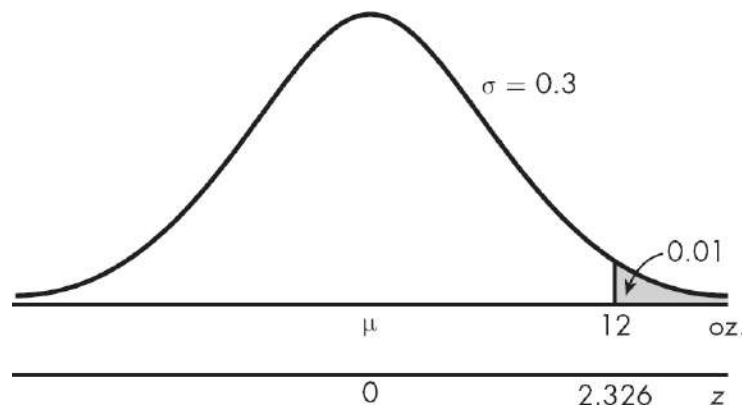
1. **(C)** A complete census can give much information about a population, but it doesn't necessarily establish a cause-and-effect relationship among seemingly related population parameters. While the results of well-designed observational studies might suggest relationships, it is difficult to conclude that there is cause and effect without running a well-designed experiment. If bias is present, increasing the sample size

simply magnifies the bias. The control group is selected by the researchers making use of chance procedures.

2. **(A)** In the first class, only 40% of the students scored below the given score, while in the second class, 80% scored below the same score.
3. **(C)** The control group should have experiences identical to those of the experimental groups except for the treatment under examination. The control group should not be given a new treatment.
4. **(D)** The negative sign comes about because we are dealing with the difference of proportions. The confidence interval estimate means that we have a certain *confidence* that the difference in population proportions lies in a particular *interval*. Note that $-0.0204 - 0.0173 = -0.0377$ and $-0.0204 + 0.0173 = -0.0031$.
5. **(C)** The interval is a statement about how confident we are that the interval has captured the population parameter. The confidence interval is about the population, not the sample. The probability that the true percentage is between 38% and 44% is either 1 or 0 depending upon whether it is or is not in the interval. Choice (D) is a correct interpretation of the confidence *level*, not the confidence *interval*.
6. **(A)** $P(\text{solved} \mid \text{location 2}) = \frac{P(\text{solved} \cap \text{location 2})}{P(\text{location 2})} = \frac{225/1,000}{325/1,000} = \frac{225}{325}$
7. **(B)** E and F are independent if $P\{E \mid F\} = P(E)$ or $P\{F \mid E\} = P(F)$ or $P(E \cap F) = P(E)P(F)$. We have $P(\text{solved} \cap \text{location 2}) \neq P(\text{solved}) \times P(\text{location 2})$; that is $\frac{225}{1,000} \neq \frac{700}{1,000} \times \frac{325}{1,000}$.
8. **(B)** The proportion of successful calls (problem solved) is $\frac{700}{1,000}$, so $\frac{700}{1,000}(450)$ is the expected number of calls from location 1 that are successful. Alternatively, the proportion of calls from location 1 is $\frac{450}{1,000}$, so $\frac{450}{1,000}(700)$ gives the expected number of successful calls from location 1.

$$\left[\text{Or, Expected} = \frac{(\text{column total})(\text{row total})}{(\text{grand total})} \right. \\ \left. = \frac{(450)(700)}{1,000} \right]$$

9. **(C)** The slope and the correlation always have the same sign. Correlation shows association, not causation. Correlation does not apply to categorical data. Correlation measures linear association, so even with a correlation of 0, there may be very strong nonlinear association.
10. **(A)** The sample size of 50 is greater than 10% of the population; that is, $50 > (0.10)(450) = 45$. This violates the 10% condition. Note that $n\hat{p} = 34 \geq 10$, and $n(1 - \hat{p}) = 16 \geq 10$, which ensure that the sampling distribution of $\hat{p}$ is approximately normal.
11. **(D)** $P(\text{I}) = P(\text{vote}|\text{over } 80) = \frac{P(\text{vote} \cap \text{over } 80)}{P(\text{over } 80)}$, $P(\text{II}) = P(\text{vote} \cap \text{over } 80)$, and $P(\text{III}) = P(\text{over } 80|\text{vote}) = \frac{P(\text{vote} \cap \text{over } 80)}{P(\text{vote})}$. $P(\text{I}) > P(\text{III})$ because the denominator in (I) is smaller than the denominator in (III). $P(\text{II})$ is the smallest because (I) and (III) have denominators that are less than one.
12. **(B)** The markings, spaced 15 apart, clearly look like the standard deviation spacings associated with a roughly normal curve. The curve seems to be the steepest above the points 185 and 215, and this too indicates that the standard deviation is 15.
13. **(B)** With a right tail having probability 0.01, the critical z-score is $\text{invNorm}(0.99) = 2.326$. Thus, $\mu + 2.326(0.3) = 12$, giving $\mu = 12 - 2.326(0.3)$.



14. **(D)** There is no reason to think that AAA members are representative of the city's drivers. Family members may have similar driving habits, and the independence condition would be violated. Random selection is important regardless of the sample size. The larger a random sample, the closer its standard deviation will be to the population standard deviation.

15. **(D)**

$$\begin{aligned}
 P\left(\begin{array}{c} \text{child} \\ \text{shows} \end{array}\right) &= P\left(\begin{array}{c} \text{father} \\ \text{shows} \end{array} \cap \begin{array}{c} \text{child} \\ \text{shows} \end{array}\right) \\
 &\quad + P\left(\begin{array}{c} \text{father} \\ \text{doesn't} \end{array} \cap \begin{array}{c} \text{child} \\ \text{shows} \end{array}\right) \\
 &= (0.25)(0.8) + (0.75)(0.06) \\
 P\left(\begin{array}{c} \text{father} \\ \text{shows} \end{array} \middle| \begin{array}{c} \text{child} \\ \text{shows} \end{array}\right) &= \frac{P\left(\begin{array}{c} \text{father} \\ \text{shows} \end{array} \cap \begin{array}{c} \text{child} \\ \text{shows} \end{array}\right)}{P\left(\begin{array}{c} \text{child} \\ \text{shows} \end{array}\right)} \\
 &= \frac{(0.25)(0.8)}{(0.25)(0.8) + (0.75)(0.06)}
 \end{aligned}$$

16. **(C)** With a greater n , the standard error is smaller, resulting in the p -value and the probability of a Type II error both being less. Power equals one minus the probability of a Type II error, so it will be greater.

17. **(A)** The mean equals the common value of all the data elements. The other terms all measure variability, which is zero when all the data elements are equal.

18. **(A)** $SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{0.15}{\sqrt{15}}$ and with $df = n - 1 = 15 - 1 = 14$, the critical t -scores are $\pm_{\text{invT}}(0.95, 14) = \pm 1.761$. The 90% confidence interval is

$$\bar{x} \pm t^*(SE(\bar{x})) = 2.72 \pm 1.761\left(\frac{0.15}{\sqrt{15}}\right)$$

19. **(B)** In a simple random sample, every possible group of the given size has to be equally likely to be selected, and this is not true here. For example, with this procedure it is impossible for the employees in the

final sample to all be from a single plant. This method is an example of stratified random sampling, but stratified random sampling does not result in simple random samples.

20. **(D)** $(0.32)(0.15) = 0.048$, so $P(E \cap F) = P(E)P(F)$ and, thus, E and F are independent. $P(E \cap F) \neq 0$, so E and F are not mutually exclusive.
21. **(B)** Both sample sizes are at least 30, so by the central limit theorem, the sampling distribution of $\bar{x}_A - \bar{x}_B$ can be modeled with a normal distribution.
22. **(D)** One set is a shift of 20 units from the other, so they have different means and medians, but they have identical shapes and thus the same variability, including IQR, standard deviation, and variance.
23. **(B)** The wider the confidence interval, the higher the confidence level, so one may obtain as high a confidence level as one wishes if a very wide confidence interval is accepted. A 99% confidence interval estimate means that in about 99% of all samples selected by this method, the population mean will be included in the confidence interval. The central limit theorem applies to any population, no matter if it is normally distributed or not. The sampling distribution for a mean always has standard deviation $\frac{\sigma}{\sqrt{n}}$; large enough sample size n refers to the closer the distribution will be to a normal distribution. The center of a confidence interval is the sample statistic, not the population parameter.
24. **(C)** $P(I) = P(II) = (0.5)^6$, while $P(III) = {}_6C_3(0.5)^3(0.5)^3 = 20(0.5)^6$
25. **(C)** Independent implies $P(E \cap F) = P(E)P(F)$, while mutually exclusive implies $P(E \cap F) = 0$. So, to be both independent and mutually exclusive in this example would mean $pq = 0$, which is impossible because p and q are given to be nonzero.
26. **(A)** In a binomial with $n = 4$ and $p = 0.9$, $P(\text{at least 3 successes}) = P(\text{exactly 3 successes}) + P(\text{exactly 4 successes}) = 4(0.9)^3(0.1) + (0.9)^4$.

27. **(C)** In stratified random sampling, the population is divided into representative groups, and random samples of persons from each group are chosen. In this case, it might well be important to have sufficient numbers from each group and to be able to consider separately the responses from each of the three groups—urban, suburban, and rural.
28. **(A)** The coefficient of determination, r^2 , indicates the percentage of variation in the response variable y that is explained by variation in the explanatory variable x .
29. **(C)** It is most likely that the homes at which the interviewer had difficulty finding someone home were homes with fewer children living in them. Replacing these homes with other randomly picked homes will most likely replace homes with fewer children with homes with more children.
30. **(B)** We have $z = \frac{\hat{p} - 0.63}{SE(\hat{p})}$. Choice (C) is an interpretation of the p -value.
31. **(A)** Blocking divides the subjects into groups of similar individuals, in this case individuals with similar exercise habits, and runs the experiment on each separate group. This controls the known effect of variation in exercise level on cholesterol level.
32. **(D)** The parameter involves a mean (days of flu symptom duration). The parameter refers to a population, not a sample.
33. **(D)** Design I is a matched pairs experiment because each person is considered a pair, receiving both kinds of shoes, with the type of shoe randomly assigned to either the first or the last 3 months. Design II is a completely randomized experiment because the type of shoe is randomly assigned to each participant.
34. **(C)** Since we are not told that the investigator suspects that the average weight is over 300 mg or is under 300 mg and since a tablet containing too little or too much of a drug clearly should be brought to the manufacturer's attention, this is a two-sided test. Thus, the p -value is twice the tail probability obtained (using the t -distribution with $df = n - 1 = 6$). Note that while t can be calculated by putting the data in a list

and using T-Test, this is unnecessary here because every answer choice involves “ $t > 1.54$.”

35. **(D)** If the null hypothesis is true, the p -value represents the probability of obtaining a sample mean as extreme as or more extreme than the one obtained for the hypothesis test, in the direction of the alternative hypothesis. In this case, the null hypothesis is that the average number of acres of wildland burned per wildfire each year in the United States is 170 acres. Given the direction of the alternative hypothesis, “as extreme as or more extreme than” means “as large as or larger than.”
36. **(C)** The expected frequencies, as calculated by
$$\frac{(\text{row total}) \times (\text{column total})}{\text{table total}}$$
, may not be whole numbers.
37. **(C)** A linear association means that as the independent variable (percent of students in poverty) changes by a constant amount, the dependent variable (EOC score) also changes by a constant amount on average. Unless there was a perfect linear correlation, the points will not line up on a straight line. No distinct pattern in the residual plot simply means that there is not an obviously better model to be found, but it does not necessarily mean that the data are linear. The coefficient of determination indicates something about the strength of the relationship but does not define linearity. The resulting linear equation is about x and $\hat{y}$, not about x and y .
38. **(B)** The four points corresponding to 5% appear to be spread about what would be the regression line, and they are more spread out than any of the other points. Removing these four values would result in a stronger association. In other words, removing them would increase the correlation in absolute value. Because the correlation is negative, this means a decrease (becoming closer to -1) in the correlation.
39. **(D)** The variation in the mean EOC scores is larger for lower percent poverty and smaller for higher percent poverty, and this will be shown by residuals with larger absolute values and then smaller absolute values. The sum and the mean of the residuals are always exactly 0, and the residual plot will never show either a positive or a negative linear relationship.

- [40.](#) **(B)** The probability of an adult feeling that the death penalty is not applied fairly by the courts is $1 - 0.36 = 0.64$. The expected value of a binomial with $n = 50$ and $p = 0.64$ is $np = 50(0.64)$.
- [41.](#) **(D)** A boxplot gives a five-number summary: smallest value, 25th percentile (Q_1), median, 75th percentile (Q_3), and largest value. The interquartile range is given by $Q_3 - Q_1$, or the total length of the two “boxes,” ignoring the “whiskers.”
- [42.](#) **(D)** A confidence *level* gives the percent of all possible confidence intervals of the given size that will capture the population parameter. Choice (B) is a correct interpretation of the confidence *interval*.

Section II

1. Model Solution:

- A.
 - i. This is an experiment.
 - ii. The explanatory variable is the type of treatment, either surgery or stents, and it is a categorical variable.
- B.
 - i. The severity of the disease could be a confounding variable. Doctors might recommend one of the treatments for their patients with more severe disease. If these patients don't do well (have major complications), it would be unclear if the reason was the particular treatment or because their disease was more severe.
 - ii. If there is an important difference among the subjects, similar subjects should be grouped into blocks and treatments should be randomized within the blocks. Patients who have had a ministroke in the previous year might respond differently to the treatments than patients who have not had this episode.
- C.
 - i. Increasing the sample size will result in a reduction in the standard error of the sampling distribution. This will increase the probability of rejecting the null hypothesis if it is false, and thus power will increase.

- ii. Increasing the significance level α makes the rejection region greater. So there is increased probability of detecting any difference in the proportions of patients suffering major complications between the surgery recipients and the stent recipients. Thus, power is increased.
- D. If, during the course of the study, the early results show a marked difference in outcomes between the two treatments, the experiment should be stopped and all patients should receive the better treatment.
- E. Is there convincing statistical evidence to suggest that the proportion of all carotid stenosis patients who suffer major complications (a stroke or heart attack) after a surgery treatment is different from the proportion of all carotid stenosis patients who suffer major complications after receiving a stent treatment?

A.	i.	Identify the study.	Experiment.	1 point
	ii.	Identify and classify the explanatory variable.	Type of treatment, either surgery or stents, is a categorical variable.	1 point
B.	i.	Explain why an assignment could lead to incorrect conclusions.	Doctors might recommend one of the treatments for their patients with more severe disease. If these patients don't do well (have major complications), it would be unclear if the reason was the particular treatment or because their disease was more severe.	1 point
	ii.	Explain why a block design is	Patients who have had a ministroke in the previous	1 point

		appropriate.	year might respond differently to the treatments than patients who have not had this episode.	
C.	i.	Explain sample size and power.	Increasing the sample size will result in a reduction in the standard error of the sampling distribution, increasing the probability of rejecting the null hypothesis if it is false. Thus, power will increase.	1 point
	ii.	Explain significance level α and power.	Increasing the significance level α makes the rejection region greater, increasing the probability of detecting any response differences between the surgery and stent recipients. Thus, power is increased.	1 point
D.		Explain an ethical concern.	If early results show a marked difference in outcomes between the two treatments, the experiment should be stopped and all patients should receive the better treatment.	1 point
E.	i.	First component: identify variables of interest.	Surgery treatment or stent treatment.	1 point

ii.	Second component: identify the parameter and direction of the alternative hypothesis.	The proportion of carotid stenosis patients who suffer major complications after a surgery treatment is different from the proportion of carotid stenosis patients who suffer major complications after receiving a stent treatment.	1 point
iii.	Third component: identify the population to which the conclusion is applicable.	All carotid stenosis patients.	1 point

2. Model Solution:

- A.
 - i. This is an observational study.
 - ii. The variables are body weight and the number of bench press reps with 225 pounds. Both are numerical variables.
- B. The relationship between body weight and lifting reps for both plots is linear and positive. The plot from Class I shows a stronger linear relationship than the plot from Class II.
- C.
 - i. $-73.38 + 0.4029(255) = 29.36$
 - ii. $30 - 29.36 = 0.64$. The linear model underestimates the observed number of bench presses by 0.64.
- D.
 - i. The slope would remain about the same, and the correlation would decrease slightly.
 - ii. The slope and correlation would both decrease to approximately 0.
- E. The population is all amateur males signing up for the trainer's classes.

A.	i.	Identify the type of study.	Observational study.	1 point
	ii.	Identify and classify the variables.	Body weight and number of bench press reps with 225 pounds. Both are numerical variables.	1 point
B.		Compare the forms.	The relationship between body weight and lifting reps for both plots is linear.	1 point
		Compare the direction.	Both are positive.	1 point
		Compare the strength.	The plot from Class I is stronger than the plot from Class II.	1 point
C.	i.	Calculate a predicted number from a regression line formula.	$-73.38 + 0.4029(255) = 29.36$	1 point
	ii.	Calculate and interpret a residual.	$30 - 29.36 = 0.64$, so the linear model underestimates the true number of bench presses by 0.64	1 point
D.	i.	Describe a change.	The slope would remain about the same, and the correlation would decrease slightly.	1 point

ii.	Describe a change.	The slope and correlation would both decrease to approximately 0.	1 point
E.	Determine a population.	All amateur males signing up for the trainer's classes.	1 point

3. Model Solution:

- A. The explanatory variable is numerical: the daily quantity of snails consumed in French restaurants.
- B. This was an observational study. An appropriate inference procedure is a one-sample t -interval for a population mean.
- C. We are given that the days were randomly chosen, and 50 is less than 10% of all possible days. The sample size $n = 50 \geq 30$ is sufficiently large for the central limit theorem to apply.
- D. We are 95% confident that the true mean number of snails consumed per day in French restaurants is between 97,448 and 104,552.

A.	Identify the variables.	Daily quantity of snails consumed in French restaurants.	1 point
	Classify the variables.	The quantity of snails is a numerical variable.	1 point
B.	Identify a study.	Observational.	1 point
	Identify a procedure.	One-sample t -interval for a population mean where $\mu =$ mean daily snail consumption in French restaurants	1 point

C.	Justify whether the randomization condition is met.	We are given that the days were randomly chosen, and 50 is less than 10% of all possible days.	1 point
	Justify whether the normality condition is met.	The sample size $n = 50 \geq 30$ is sufficiently large for the central limit theorem to apply.	1 point
	Calculate the confidence interval.	(97,448, 104,552) with $df = 49$	1 point
D.	Identify the confidence level and calculate the interval.	95% confidence; (97,448, 104,552)	1 point
	Identify the population parameter.	Mean daily snail consumption.	1 point
	Justify the context.	The context is the true mean for all days in French restaurants.	1 point

4. Model Solution:

- A. The scatterplot is generally negative. In fact, the slope is given to be a negative number. So, yes, this supports the guidance counselor's conclusion.
- B. The correlation $r = -\sqrt{0.25} = -0.5$.
- C. $(-19.93)(0.5) = -9.965$. So the student with the greater GPA is predicted to have a satisfaction level 9.965 lower than that of the other student.

- D. i. The sophomore points on the regression line are strongly linear. Two points on the sophomore regression line that are 0.5 apart on the x -axis are approximately (2.5, 55) and (3.0, 75). So the sophomore student with the GPA that is 0.5 higher is estimated to have a $75 - 55 = 20$ higher satisfaction level.
- ii. Overall, higher grade point averages are associated with lower satisfaction with dorm life. However, within each class level (freshmen, sophomores, juniors, and seniors), students with higher grade point averages tend to have higher satisfaction with dorm life.
- E. The population for a valid investigative question is all students living in the college dormitories.

A.	Determine if data support a conclusion.	Yes, this supports the guidance counselor's conclusion.	1 point
	Justify a claim.	The slope is negative.	1 point
B.	Calculate the correlation.	$r = -\sqrt{0.25} = -0.5$	1 point
C.	Determine how two predicted satisfaction levels compare.	$(-19.93)(0.5) = -9.965$	1 point
		The student with the greater GPA is predicted to have a satisfaction level 9.965 lower than that of the other student.	1 point
D. i.	Estimate how levels would compare.	For sophomores, the regression line is strongly linear. So pick two points that are 0.5 apart on the x -	1 point

ii.		axis. For example, pick approximately (2.5, 55) and (3.0, 75).	
		So the sophomore student with the GPA that is 0.5 higher is estimated to have a $75 - 55 = 20$ higher satisfaction level.	1 point
	Determine a conclusion.	Overall, higher grade point averages are associated with lower satisfaction with dorm life.	1 point
		However, within each class level, students with higher grade point averages tend to have higher satisfaction with dorm life.	1 point
E.	Determine the population.	The population is all students living in the college dormitories.	1 point

Practice Test 4

Section I: Questions 1–42

SPEND 90 MINUTES ON THIS PART OF THE EXAM.

DIRECTIONS: The questions or incomplete statements that follow are each followed by four suggested answers or completions. Choose the response that best answers the question or completes the statement.

Questions 1–3 refer to the following situation:

A company wishes to determine the relationship between the number of days spent training employees and their performances on a job aptitude test. Collected data result in a least squares regression line, $\hat{y} = 12.1 + 6.2x$, where x is the number of training days and $\hat{y}$ is the predicted score on the aptitude test. The correlation is 0.73, and the association is described as strong, positive, and linear.

1. What is a correct interpretation of the “6.2” in the regression equation?
 - (A) For every additional day of training, an employee’s score on the aptitude test will increase by 6.2.
 - (B) For each additional 6.2 days of training, an employee’s score on the aptitude test will increase by 1.
 - (C) An employee’s score on the aptitude test equals 6.2 times the number of days of training.

- (D) For every additional day of training, an employee's predicted score on the aptitude test will increase by 6.2.

2. What does “strong” mean in this context?

- (A) The least squares model predicts that the greater the number of training days, the higher the aptitude test score will be.
- (B) The actual aptitude test score for a given number of training days will be very close to what is predicted by the least squares model.
- (C) The least squares model proves that the greater the number of training days, the higher the aptitude test score will be.
- (D) The least squares model gives a cause-and-effect association between the number of training days and the aptitude test score.

3. What is the correct way to label the correlation?

- (A) 0.73
- (B) 0.73 test points per training days
- (C) 0.73 training days per test points
- (D) 0.73 test points training days

4. To survey the opinions of the students at your high school, a researcher plans to select every twenty-fifth student entering the school in the morning. Assuming there are no absences, will this result in a simple random sample of students attending your school?

- (A) Yes, because every student has the same chance of being selected.
- (B) Yes, but only if there is a single entrance to the school.
- (C) Yes, because this is an example of systematic random sampling, which is a special case of simple random sampling.
- (D) No, because not every sample of the intended size has an equal chance of being selected.

5. Consider a hypothesis test with $H_0: \mu = 70$ and $H_a: \mu < 70$. Which of the following choices of significance level and sample size results in the greatest power of the test when $\mu = 65$?
- (A) $\alpha = 0.05, n = 15$
 - (B) $\alpha = 0.01, n = 15$
 - (C) $\alpha = 0.05, n = 30$
 - (D) $\alpha = 0.01, n = 30$
6. In 2018, it was estimated that there were roughly 554,000 homeless people in the United States. The average shelter stay for homeless families with kids is 435 days. Assume a skewed left distribution with a standard deviation of 85 days. Consider random samples of size 100 taken from the distribution with the mean length of stay, $\bar{x}$, recorded for each sample. Which of the following is the best description of the sampling distribution of $\bar{x}$?
- (A) Skewed left with mean 435 days and standard deviation 0.85 days
 - (B) Skewed left with mean 435 days and standard deviation 8.5 days
 - (C) Approximately normal with mean 435 days and standard deviation 0.85 days
 - (D) Approximately normal with mean 435 days and standard deviation 8.5 days
7. The label on a package of cords claims that the breaking strength of a cord is 3.5 pounds, but a hardware store owner believes the real value is less. She plans to test 36 such cords; if their mean breaking strength is less than 3.25 pounds, she will reject the claim on the label. If the standard deviation for the breaking strengths of all such cords is 0.9 pounds, what is the probability of mistakenly rejecting a true claim?
- (A) $P\left(z > \frac{3.25 - 3.5}{0.9}\right)$
 - (B) $P\left(z < \frac{3.25 - 3.5}{\left(\frac{0.9}{\sqrt{36}}\right)}\right)$

$$(C) \quad 2P\left(z < \frac{3.25 - 3.5}{\left(\frac{0.9}{\sqrt{36}}\right)}\right)$$

$$(D) \quad P\left(z > \frac{3.25 - 3.5}{\left(\frac{0.9}{\sqrt{36}}\right)}\right)$$

8. Suppose that a study is carried out in which the number of tennis balls used in a random sample of tournaments is tabulated. Which of the following is a correct understanding of the type of investigative question this study might address?

- (A) The data could be used to estimate a population mean since the data collected are numerical.
- (B) The data could be used to estimate a population proportion since the data collected are numerical.
- (C) The data could be used to estimate a population mean since the data collected are categorical.
- (D) The data could be used to estimate a population proportion since the data collected are categorical.

9. For which of the following probability assignments are events E and F independent?

- (A) $P(E \cap F^c) = 0.2, P(E \cap F) = 0.0, P(E^c \cap F) = 0.2$
- (B) $P(E \cap F^c) = 0.3, P(E \cap F) = 0.3, P(E^c \cap F) = 0.3$
- (C) $P(E \cap F^c) = 0.4, P(E \cap F) = 0.1, P(E^c \cap F) = 0.1$
- (D) $P(E \cap F^c) = 0.4, P(E \cap F) = 0.2, P(E^c \cap F) = 0.1$

10. To study the effect of alcohol on reaction time, subjects were randomly selected and given three beers to consume. Their reaction time to a simple stimulus was measured before and after drinking the alcohol. Which of the following is a correct statement?

- (A) This study was an observational study.
- (B) The placebo effect is irrelevant in this type of study.

- (C) This study was an experiment in which the subjects were used as their own controls.
- (D) This study was an experiment with no controls.
11. Suppose that the regression line for a set of data, $\hat{y} = a + 7x$, passes through the point $(-2, 4)$. If $\bar{x}$ and $\bar{y}$ are the sample means of the x - and y -values, respectively, then which of the following is equal to $\bar{y}$?
- (A) $18 + 7\bar{x}$
- (B) $2 + \bar{x}$
- (C) $-4 + \bar{x}$
- (D) $7\bar{x}$
12. Which of the following is a *false* statement about simple random samples?
- (A) A sample must be reasonably large to be properly considered a simple random sample.
- (B) Inspection of a sample will give no indication of whether or not it is a simple random sample.
- (C) Every element of the population has an equal chance of being picked.
- (D) Every sample of the desired size has an equal chance of being picked.
13. Studies have shown that drinking sugary beverages can increase your likelihood of developing liver cancer. From a random sample of adults, 65% reported consuming at least some sweetened drinks on any given day. A 90% confidence interval was constructed from the sample, and the margin of error for the estimate was 4%. Which of the following is the best interpretation of the confidence interval?
- (A) There is a 0.90 probability that between 61% and 69% of all adults consume at least some sweetened drinks on any given day.
- (B) We are 90% confident that between 61% and 69% of the sample of adults consume at least some sweetened drinks on any given

day.

- (C) We are 90% confident that between 61% and 69% of all adults consume at least some sweetened drinks on any given day.
- (D) If 90% confidence intervals are calculated from all possible samples of the given size, 90% of these intervals will capture the true percentage of all adults who consume at least some sweetened drinks on any given day.

14. To conduct a survey of which long-distance carriers are used in a particular locality, a researcher opens a telephone book to a random page, closes his eyes, puts his finger down on the page, and then calls the next 75 names. Which of the following is a correct statement?

- (A) The procedure results in a simple random sample.
- (B) While the survey design does incorporate chance, the procedure could easily result in selection bias.
- (C) This is an example of stratified random sampling with 26 strata.
- (D) Given that the researcher truly keeps his eyes closed, this is a good example of blinding.

15. Which of the following is *not* true about t -distributions?

- (A) There are different t -distributions for different values of df (degrees of freedom).
- (B) t -distributions are bell-shaped and symmetric.
- (C) t -distributions always have mean 0 and standard deviation 1.
- (D) The larger the df value, the closer the distribution is to the normal distribution.

16. Suppose the probability that a person picked at random has lung cancer is 0.035 and the probability that the person both has lung cancer and is a heavy smoker is 0.014. Given that someone picked at random has lung cancer, what is the probability that the person is a heavy smoker?

- (A) $\frac{0.014}{0.035}$

- (B) $0.035 + 0.014$
- (C) $0.035 + 0.014 - (0.035)(0.014)$
- (D) $\frac{0.035}{1 - 0.014}$

17. Suppose computer science graduates earn an average starting salary of \$75,000 with a standard deviation of \$12,000. What is the probability that a randomly selected computer science graduate has a starting salary less than \$100,000 if it is known that his or her starting salary is over \$80,000? Assume a roughly normal distribution of starting salaries of computer science graduates.

- (A) $\frac{P(80,000 < z < 100,000)}{P(z > 80,000)}$
- (B) $\frac{P\left(z < \frac{100,000}{12,000}\right)}{P\left(z > \frac{80,000}{12,000}\right)}$
- (C) $\frac{P\left(\frac{80,000}{12,000} < z < \frac{100,000}{12,000}\right)}{P\left(z > \frac{80,000}{12,000}\right)}$
- (D) $\frac{P\left(\frac{80,000 - 75,000}{12,000} < z < \frac{100,000 - 75,000}{12,000}\right)}{P\left(z > \frac{80,000 - 75,000}{12,000}\right)}$

18. There are 4,900,000 people in a large city. A nutritionist wishes to determine the proportion of people in the city who like tofu. She wants to be 95% confident of the answer to within ± 0.014 . Which of the following is closest to the sample size she should survey?

- (A) 490
- (B) 4,900
- (C) 49,000
- (D) 490,000

19. Suppose the correlation between two variables is $r = 0.19$. What is the new correlation if 0.23 is added to all values of the x -variable, every

value of the y -variable is doubled, and the two variables are interchanged?

- (A) 0.84
- (B) -0.84
- (C) 0.19
- (D) -0.19

20. Should the United Nations take a more active role in combating poverty around the world? A human rights nongovernmental organization (NGO) conducts interviews of a random sample of 35 high-level government officials. Of these officials, 28 of them expressed the belief that the United Nations should take a more active role in combating poverty around the world. Letting p be the proportion of all high-level government officials who would express the belief that the United Nations should take a more active role in combating poverty around the world, is it appropriate to assume that the corresponding sampling distribution of $\hat{p}$ is approximately normal?

- (A) Yes, because the sample was chosen randomly.
- (B) Yes, because sampling distributions of the sample proportion are modeled with normal distributions.
- (C) Yes, because the sample size is ≥ 30 .
- (D) No, because the sample size is not large enough for the normality condition.

21. Texas has more lightning strikes than any other state. For a period of time, a researcher records the number of lightning strikes each hour in Texas. Let the random variable L represent the number of lightning strikes in Texas in 1 hour. Assume the standard deviation of L is 587. Which of the following is the best interpretation of the standard deviation?

- (A) The number of lightning strikes in Texas in a randomly selected hour is expected to be 587 away from the mean.

- (B) The average number of lightning strikes per hour in Texas for a random sample of hours is expected to be 587.
- (C) On average, the number of lightning strikes per hour in Texas varies from the mean by about 587.
- (D) The number of lightning strikes per hour in Texas for a random sample of hours is expected to be 587 away from the mean.

22. Suppose $P(X) = 0.35$ and $P(Y) = 0.40$. If $P(X|Y) = 0.28$, what is $P(Y|X)$?

- (A) $\frac{(0.28)(0.35)}{0.40}$
- (B) $\frac{(0.28)(0.40)}{0.35}$
- (C) $\frac{(0.35)(0.40)}{0.28}$
- (D) $\frac{0.28}{0.40}$

23. To test whether extensive exercise lowers the resting heart rate, a study is performed by randomly selecting half of a group of volunteers to exercise 1 hour each morning, while the rest are instructed to perform no exercise. Is this study an experiment or an observational study?

- (A) An experiment with a control group and blinding
- (B) An observational study with comparison and randomization
- (C) An observational study with little if any bias
- (D) None of the above statements are correct.

Questions 24–26 refer to the following situation:

At an amusement park, the waiting times for a new roller coaster are approximately normally distributed with a mean of 35 minutes and a standard deviation of 10 minutes.

24. What is the 75th percentile for waiting times?

- (A) 28.26 minutes
- (B) 35.75 minutes
- (C) 41.74 minutes
- (D) 55.00 minutes

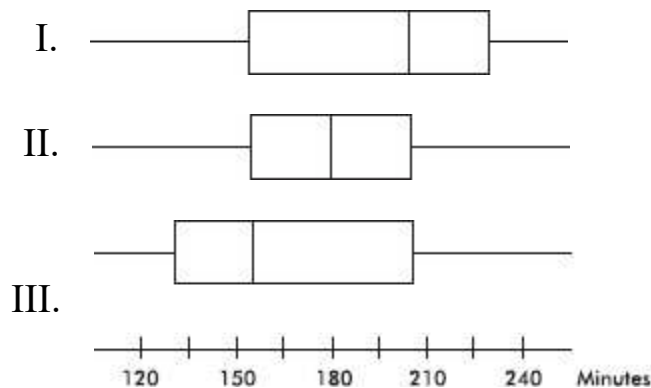
25. In a random sample of 5 people, what is the probability that at least 2 have to wait over 45 minutes?

- (A) 0.031
- (B) 0.150
- (C) 0.181
- (D) 0.819

26. In a random sample of 5 people, what is the probability that their mean waiting time is over 45 minutes?

- (A) 0.0127
- (B) 0.159
- (C) 0.841
- (D) 0.9873

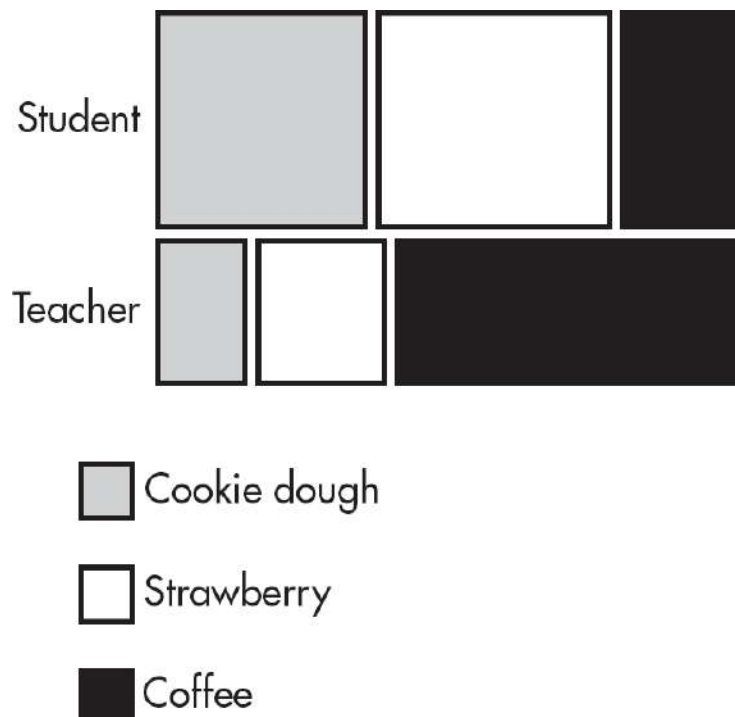
27. A medical researcher, studying the effective durations of three over-the-counter pain relievers, obtains the following boxplots from three equal-sized groups of patients, one group using each of the pain relievers.



Which of the following is a correct statement with regard to comparing the effective durations in minutes of the three pain relievers?

- (A) The durations of pain relief in the (II) group form a roughly normal distribution.
- (B) More patients had over 210 minutes of pain relief in the (I) group than in either of the other two groups.
- (C) More patients had over 240 minutes of pain relief in the (I) group than in either of the other two groups.
- (D) More patients had less than 120 minutes of pain relief in the (III) group than in either of the other two groups.

28. People at a high school were surveyed as to whether they were students or teachers and what their favorite ice cream flavor was, excluding vanilla and chocolate. The results are displayed in the following mosaic plot.



Based on this plot, which of the following is a true statement?

- (A) The number of students answering “coffee” is greater than the number of teachers answering “coffee.”
 - (B) More teachers answered either “cookie dough” or “strawberry” than students who answered “cookie dough.”
 - (C) Of those who answered “strawberry,” a greater proportion were teachers than students.
 - (D) There were more students than teachers in the survey.
29. Suppose that 54% of the graduates from your high school go on to 4-year colleges, 20% go on to 2-year colleges, 19% find employment, and the remaining 7% search for a job. If a randomly selected student is not going on to a 2-year college, what is the probability he or she will be going on to a 4-year college?
- (A) 0.460
 - (B) 0.540
 - (C) 0.630
 - (D) 0.675
30. A congresswoman is interested in the proportion p of registered voters in her district who favor legalizing medical marijuana. Forty-four percent of a simple random sample of 750 registered voters in her district favor legalizing medical marijuana. What is the midpoint for a 95% confidence interval estimate of p ?
- (A) 0.025
 - (B) 0.418
 - (C) 0.44
 - (D) p
31. Random samples of size n are drawn from a population. The mean of each sample is calculated, and the standard deviation of this set of sample means is found. Then the procedure is repeated, this time with samples of size $4n$. How does the standard deviation of the second group compare with the standard deviation of the first group?

- (A) It will be one-quarter as large.
- (B) It will be half as large.
- (C) It will be twice as large.
- (D) It will be four times as large.

32. Leech therapy is used in traditional medicine for treating localized pain. In a double-blind experiment on 50 patients with osteoarthritis of the knee, half are randomly selected to receive injections of leech saliva, while the rest receive a placebo. Pain levels 7 days later among those receiving the saliva show a mean of 19.5, while pain levels among those receiving the placebo show a mean of 25.6 (higher numbers indicate more pain). Partial calculator output is shown below.

2-SampTTest

$\mu 1 < \mu 2$

$t = -3.939503313$

$df = 43.43159286$

$\bar{x} 1 = 19.5$

$\bar{x} 2 = 25.6$

$Sx1 = 4.5$

$Sx2 = 6.3$

$n1 = 25$

$n2 = 25$

[Follow for extended description](#)

Which of the following is a correct conclusion?

- (A) After 7 days, the mean pain level with the leech treatment is significantly lower than the mean pain level with the placebo at the 0.01 significance level.
- (B) After 7 days, the mean pain level with the leech treatment is significantly lower than the mean pain level with the placebo at the 0.05 significance level but not at the 0.01 level.

- (C) After 7 days, the mean pain level with the leech treatment is significantly lower than the mean pain level with the placebo at the 0.10 significance level but not at the 0.05 level.
- (D) After 7 days, the mean pain level with the leech treatment is not significantly lower than the mean pain level with the placebo at the 0.10 significance level.

33. Two meteorologists use the same sample data to calculate the average temperature across the United States. The first meteorologist constructs a 90% confidence interval, while the second constructs a 95% confidence interval. Which of the following statements is true?

- (A) The midpoint of the first meteorologist's interval will be less than the midpoint of the second meteorologist's interval.
- (B) The midpoint of the first meteorologist's interval will be greater than the midpoint of the second meteorologist's interval.
- (C) The width of the first meteorologist's interval will be less than the width of the second meteorologist's interval.
- (D) The width of the first meteorologist's interval will be greater than the width of the second meteorologist's interval.

34. It is estimated that 30% of all cars parked in a metered lot outside city hall receive tickets for meter violations. In a random sample of 5 cars parked in this lot, what is the probability that at least one receives a parking ticket?

- (A) $1 - (0.3)^5$
- (B) $1 - (0.7)^5$
- (C) $5(0.3)^4(0.7)$
- (D) $5(0.3)^4(0.7) + 10(0.3)^3(0.7)^2 + 10(0.3)^2(0.7)^3 + 5(0.3)(0.7)^4 + (0.7)^5$

35. Past studies indicate that 65% of Americans have a positive view of capitalism. A political scientist believes that the proportion is now less than 0.65. In a random sample of Americans, 57% have a positive

view of capitalism. With $H_0: p = 0.65$ and $H_a: p < 0.65$, the p -value of the test is 0.047. Which of the following is a correct interpretation of the p -value?

- (A) 0.047 is the probability that the population proportion is less than 0.57.
- (B) 0.047 is the probability that the population proportion is less than 0.65.
- (C) 0.047 is the probability of obtaining a sample proportion less than or equal to 0.57.
- (D) If it is true that 65% of Americans have a positive view of capitalism, 0.047 is the probability of obtaining a sample proportion less than or equal to 0.57.

36. In one study, half of a class were instructed to watch exactly 1 hour of television per day, the other half were told to watch 5 hours per day, and then their class grades were compared. In a second study, students in a class responded to a questionnaire asking about their television usage and their class grades.

- (A) The first study was an experiment without a control group, and the second was an observational study.
- (B) The first study was an observational study, and the second was a controlled experiment.
- (C) Both studies were controlled experiments.
- (D) Both studies were observational studies.

37. The table below shows data that were collected from 125 middle school children. It indicates their gender and whether or not they believe that their father could overpower a sumo wrestler.

Believe father can overpower a sumo wrestler			
	Yes	No	Total
Boy	41	30	71

Believe father can overpower a sumo wrestler			
	Yes	No	Total
Girl	31	23	54
Total	72	53	125

One of these children is picked at random. Which of the following correctly interprets mutually exclusive events represented by the table?

- (A) Believing one's father can overpower a sumo wrestler and not believing one's father can overpower a sumo wrestler
- (B) Believing one's father can overpower a sumo wrestler and being a boy
- (C) Believing one's father can overpower a sumo wrestler and being a girl
- (D) Not believing one's father can overpower a sumo wrestler and being a boy

38. In leaving for school on an overcast April morning, you make a judgment on the null hypothesis: the weather will remain dry. What would the results be of Type I and Type II errors?

- (A) Type I error: get drenched
Type II error: needlessly carry around an umbrella
- (B) Type I error: needlessly carry around an umbrella
Type II error: get drenched
- (C) Type I error: get drenched
Type II error: carry no umbrella, and weather remains dry
- (D) Type I error: get drenched
Type II error: carry an umbrella, and it rains

39. The mean thrust of a certain model jet engine is 9,500 pounds. Concerned that a production process change might have lowered the thrust, an inspector tests a sample of units, calculating a mean of 9,350

pounds with a z -score of -2.46 and a p -value of 0.0069 . Which of the following is the most reasonable conclusion?

- (A) There is convincing evidence to conclude that the new process is producing engines with a mean thrust under 9,350 pounds.
- (B) There is convincing evidence to conclude that the new process is producing engines with a mean thrust under 9,500 pounds.
- (C) 99.31% of the engines produced under the new process will have a thrust under 9,350 pounds.
- (D) 99.31% of the engines produced under the new process will have a thrust under 9,500 pounds.

40. In a group of 10 third graders, the mean height is 50 inches with a median of 47 inches, while in a group of 12 fourth graders, the mean height is 54 inches with a median of 49 inches. What is the median height of the combined group?

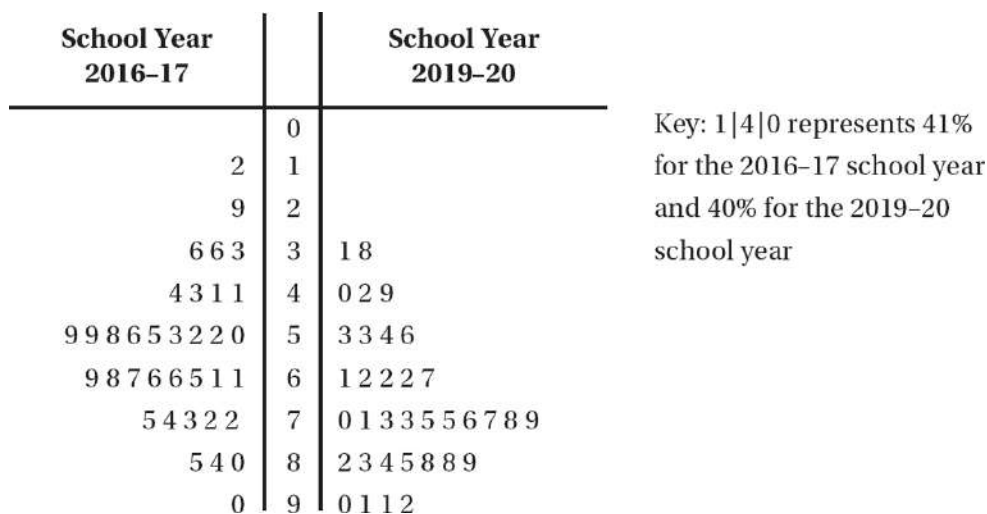
- (A) 48 inches
- (B) $\frac{10(47) + 12(49)}{22}$ inches
- (C) $\frac{10(50) + 12(54)}{22}$ inches
- (D) All that can be said is that the median is between 47 and 49 inches.

41. A reading specialist in a large public school system believes that the more time students spend reading, the better they will do in school. She plans a middle school experiment in which a simple random sample (SRS) of 30 eighth graders will be assigned four extra hours of reading per week, an SRS of 30 seventh graders will be assigned two extra hours of reading per week, and an SRS of 30 sixth graders with no extra assigned reading will be a control group. After one school year, the mean GPAs from each group will be compared. Is this a good experimental design?

- (A) Yes, because simple random samples were used.

- (B) No, because while this design may point out an association between reading and GPA, it cannot establish a cause-and-effect relationship.
- (C) No, because grade level is a confounding variable.
- (D) No, because any conclusion would be flawed because of blocking bias.

42. A study at 35 large city high schools gives the following back-to-back stemplot of the percentages of students who say they have tried alcohol.



Which of the following does *not* follow from the above data?

- (A) In general, the percentage of students trying alcohol seems to have increased from 2016–17 to 2019–20.
- (B) The median alcohol percentage among the 35 schools increased from 2016–17 to 2019–20.
- (C) The spread between the lowest and highest alcohol percentages decreased from 2016–17 to 2019–20.
- (D) The percentage of students trying alcohol increased in each of the schools between 2016–17 and 2019–20.

STOP

If there is still time remaining, you may check your work on this section.

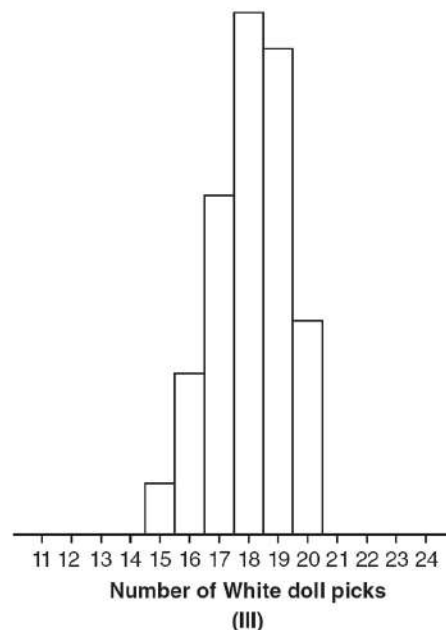
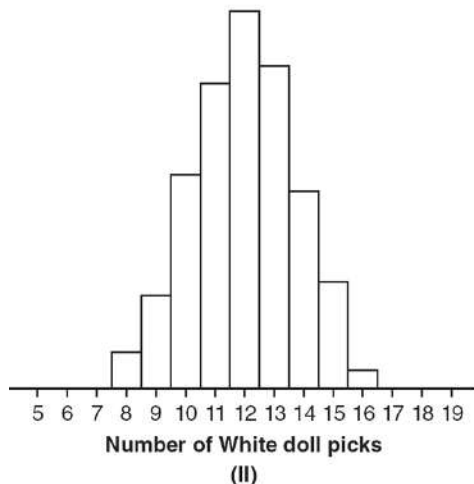
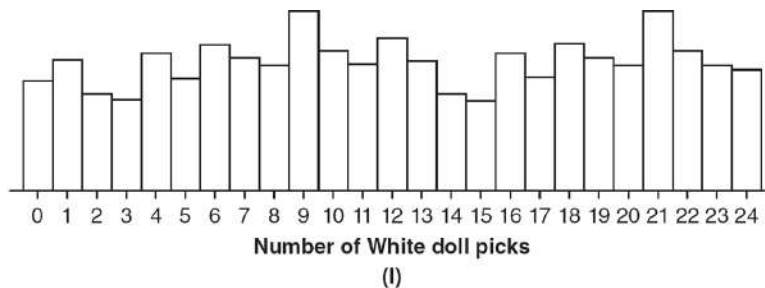
Section II

DIRECTIONS: You must show all work and indicate the methods you use. You will be graded on the correctness of your methods and on the accuracy of your results and explanations.

1. A guidance counselor at a large college is interested in interviewing students about their college experience. After talking with a local AP Statistics teacher, the guidance counselor plans the following three-stage sampling procedure.
 - (1) The counselor will obtain separate lists of the four groups of students on the college campus: freshmen, sophomores, juniors, and seniors.
 - (2) Each of the four groups takes a number of large classes together. The counselor will use a random number generator to pick three of these classes for each of the four groups.
 - (3) In each of the chosen classes, the counselor will pick every fifth student entering the classroom.
 - A.
 - i. **Identify** the sampling procedure in the first stage above.
 - ii. **Identify** an advantage in using this procedure in this context.
 - B.
 - i. **Identify** the sampling procedure in the second stage above.
 - ii. **Identify** an advantage in using it in this context.
 - C.
 - i. **Identify** the sampling procedure in the third stage above.
 - ii. **Identify** a disadvantage in using it in this context.
 - D. The guidance counselor is also considering first stratifying by gender and then proceeding with steps (1), (2), and (3) in each strata. **Identify** an advantage to this added step.
 - E. The guidance counselor wants to perform a study about the proportion of students at her college who feel positively about their college experience. **Determine** a valid investigative question for this study. Ensure all three components of a valid investigative question are included in your response.

2. In the famous doll study, researchers studied racial awareness and child development by investigating whether preschool African Americans would tend to select the White doll when offered a choice of playing with a White doll or a Black doll. In one study, 18 out of 24 preschool African Americans chose the White doll.

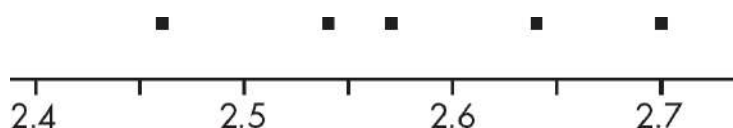
- A. i. **Identify** the parameter of interest for the above study.
 ii. **Identify** the hypotheses.
- B. A simulation analysis is performed to investigate whether the observed result provides strong evidence that preschool African Americans prefer to play with White, rather than Black, dolls. **Identify** which of the following graphs could most reasonably have come from the simulation? **Explain**.



[Follow for extended description](#)

- C.
 - i. Based on the correct graph and the results of the study, **identify** which is closest to the p -value of this test: 1.000, 0.500, 0.100, 0.050, or 0.001? **Explain**.
 - ii. Based on this test and the simulation, **justify** a conclusion.
- D.
 - i. **Determine** what type of error, Type I or Type II, may have been committed.
 - ii. **Identify** a possible consequence.
- E. **Determine** the population for a valid investigative question for this study.

3. A laboratory is testing to estimate the mean concentration level in milligrams per milliliter for the active ingredient found in a pharmaceutical product. In a random sample of five vials of the product, the concentrations were measured at 2.46, 2.57, 2.70, 2.64, and 2.54 mg/mL.



[Follow for extended description](#)

- A. **Identify** and **classify** the variable(s) in this study.
 - B. **Identify** the appropriate inference procedure.
 - C. **Complete** the inference procedure, including **calculating** the appropriate statistics.
 - D. **Justify** a conclusion in context.
4. It was recently reported that in the United States, 40.3% of all births are to unmarried mothers. A county health administrator is investigating whether births to unmarried women is higher in her county than in the national average. If so, she will propose additional funding to counsel unmarried mothers. A random sample of 100 births in the county will be looked at. Let p = the proportion of the population of women giving birth in the county who are unmarried. Consider the following hypotheses.

$$H_0: p = 0.403, H_a: p > 0.403$$

- A.
 - i. **Describe** a Type II error in context.
 - ii. **Describe** a possible consequence of a Type II error in context.
- B.
 - i. Assuming a true $p = 0.403$, the sampling distribution of $\hat{p}$ will be approximately normal with a mean of $\mu_{\hat{p}} = 0.403$. **Calculate** the standard deviation $\sigma_{\hat{p}}$ of the sampling distribution of $\hat{p}$.
 - ii. **Determine** what values of the sample proportion $\hat{p}$ would represent sufficient evidence to reject the null hypothesis at a significance level of $\alpha = 0.05$. Show your work.
- C. Suppose the actual proportion of all women giving birth in the county who are unmarried is 0.45.
 - i. Assuming a true $p = 0.45$, this sampling distribution of $\hat{p}$ will be approximately normal with a mean of $\mu_{\hat{p}} = 0.45$. **Calculate** the standard deviation of this sampling distribution of $\hat{p}$.
 - ii. Using the actual proportion of 0.45 and the answer from part B (ii), **calculate** the probability that the null hypothesis will be rejected. Show your work.
- D. **Identify** the statistical term describing the probability calculated in part C (ii).
- E. Suppose the size of the sample was greater than 100. **Explain** how that would affect the probability of rejecting the null hypothesis calculated in part C.

STOP

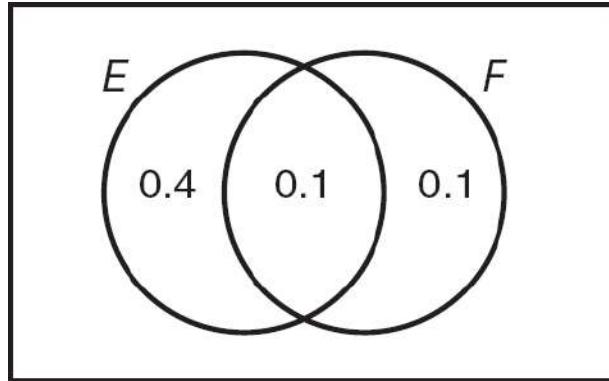
If there is still time remaining, you may check your work on this section.

Answer Explanations

Section I

1. **(D)** 6.2 is the slope of the regression line relating $\hat{y}$, the predicted value of the dependent variable, to x , the independent variable. In context, for every additional day of training, an employee's *predicted* score on the aptitude test will increase by 6.2.

2. **(B)** “Strong” means that the points in the related scatterplot fall close to the least squares regression line. In context, this means that the actual aptitude test score for a given number of training days will be very close to what is predicted by the least squares model. Association, even strong association, does not imply causation or “prove” anything.
3. **(A)** The correlation is unit-free.
4. **(D)** For a simple random sample, every possible group of the given size has to be equally likely to be selected, and this is not true here. For example, with this procedure it will be impossible for all the early arrivals to be together in the final sample. This procedure is an example of systematic random sampling (although the starting point should be randomly selected), but systematic random sampling does not result in simple random samples.
5. **(C)** $\text{Power} = 1 - \beta$, and β is smallest when both the significance level α and the sample size n are greater.
6. **(D)** By the central limit theorem, the sampling distribution of $\bar{x}$ is approximately normal with mean equal to the population mean and with standard deviation equal to the population standard deviation divided by the square root of the sample size. In this example, $\mu_{\bar{x}} = \mu = 435$ and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{85}{\sqrt{100}} = 8.5$.
7. **(B)** We have $H_0: \mu = 3.5$ and $H_a: \mu < 3.5$. Then $\sigma_{\bar{x}} = \frac{0.9}{\sqrt{36}}$ and the z -score of 3.25 is $\frac{3.25 - 3.5}{\left(\frac{0.9}{\sqrt{36}}\right)}$.
8. **(A)** The data are numerical because the number of tennis balls used takes on numerical values for a counted quantity. This leads to estimating a population mean—that is, the mean number of tennis balls used in tournaments.
9. **(C)** Events E and F are independent if $P(E \cap F) = P(E)P(F)$. Here, $P(E) = 0.4 + 0.1 = 0.5$ and $P(F) = 0.1 + 0.1 = 0.2$. Then $P(E)P(F) = (0.5)(0.2) = 0.1 = P(E \cap F)$.



10. (C) In experiments on people, subjects can be used as their own controls, with responses noted before and after the treatment. However, with such designs there is always the danger of a placebo effect. In this case, subjects might well have slower reaction times after drinking the alcohol because they think they should.
11. (A) Since $(-2, 4)$ is on the line $\hat{y} = a + 7x$, we have $4 = a + 7(-2)$ and $a = 18$. Thus, the regression line is $\hat{y} = 18 + 7x$. The point $(\bar{x}, \bar{y})$ is always on the regression line, and so we have $\hat{y} = 18 + 7\bar{x}$.
12. (A) A simple random sample can be any size.
13. (C) The interval is a statement about how confident we are that the interval has captured the population parameter. The confidence interval is about the population, not the sample. The probability that the true percentage is between 61% and 69% is either 1 or 0, depending upon whether it is or is not in the interval. Choice (D) is a correct interpretation of the confidence *level*, not the confidence *interval*.
14. (B) While the procedure does use some element of chance, all possible groups of size 75 do not have the same chance of being picked, so the result is not a simple random sample. There is a real chance of selection bias. For example, a number of relatives with the same last name and all using the same long-distance carrier might be selected.
15. (C) While t -distributions do have mean 0, their standard deviations are greater than 1.
16. (A)

$$P(\text{smoker}|\text{cancer}) = \frac{P(\text{smoker} \cap \text{cancer})}{P(\text{cancer})}$$

$$= \frac{0.014}{0.035}$$

17. **(D)** The critical z-scores are $\frac{80,000 - 75,000}{12,000}$ and $\frac{100,000 - 75,000}{12,000}$. The probability of being less than \$100,000 given that the starting salary is over \$80,000 is

$$\frac{P(80,000 < x < 100,000)}{P(x > 80,000)} =$$

$$\frac{P\left(\frac{80,000 - 75,000}{12,000} < z < \frac{100,000 - 75,000}{12,000}\right)}{P\left(z > \frac{80,000 - 75,000}{12,000}\right)}$$

18. **(B)**

$$1.96 \sqrt{\frac{(0.5)(0.5)}{n}} \leq 0.014 \text{ gives}$$

$$\sqrt{n} \geq \frac{1.96(0.5)}{0.014} = 70 \text{ and } n = 4,900.$$

19. **(C)** The correlation coefficient is not changed by adding the same number to every value of one of the variables, by multiplying every value of one of the variables by the same positive number, or by interchanging which are the x- and y-variables.
20. **(D)** Both $n\hat{p}$ and $n(1 - \hat{p})$ have to be ≥ 10 . Although $n\hat{p} = 28$, $n(1 - \hat{p})$ is only $35 - 28 = 7$ in this example. So the normality condition is violated.
21. **(C)** The standard deviation of a random variable roughly gives the average or typical amount of variability from the mean of the population.
22. **(B)** $P(X|Y) = \frac{P(X \cap Y)}{P(Y)}$ gives that $P(X \cap Y) = P(X|Y) P(Y) = (0.28)(0.40)$.

$$\text{Then } P(Y|X) = \frac{P(X \cap Y)}{P(X)} = \frac{(0.28)(0.40)}{0.35}.$$

23. **(D)** This study is an experiment because a treatment (extensive exercise) is imposed. There is no blinding because subjects clearly know whether or not they are exercising. There is no blocking because subjects are not divided into blocks before random assignment to treatments. For example, blocking would have been used if subjects had been separated by gender or age before random assignment to exercise or not.
24. **(C)** $\text{invNorm}(0.75, 35, 10) = 41.74$
25. **(C)** $\text{normalcdf}(45, 1000, 35, 10) = 0.1587$
Binomial, $n = 5, p = 0.1587; P(x \geq 2) = 0.181$
26. **(A)** $\text{normalcdf}(45, 1000, 35, \frac{10}{\sqrt{5}}) = 0.0127$
27. **(B)** More than 25% of the patients in the (I) group had over 210 minutes of pain relief, which is not the case for the other two groups. The interquartile range is the length of the box, so they are not all equal. There is no way to positively conclude a normal distribution from a boxplot.
28. **(D)** In mosaic plots, the area of a box is proportional to the count corresponding to that box. The vertical axis in this plot indicates the proportion of students versus teachers. The horizontal axis in this plot indicates the proportion of people choosing each of the three ice cream flavors for students and for teachers. Choice (D) is true because the three boxes corresponding to students have a total area greater than the three boxes corresponding to teachers, or noting that along the vertical axis, “Student” has a greater length than “Teacher.”
29. **(D)** $\frac{0.54}{0.54 + 0.19 + 0.07} = 0.675$
30. **(C)** The midpoint of the confidence interval is the sample proportion $\hat{p} = 0.44$.
31. **(B)** $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, so the standard deviation of sample means is inversely related to the square root of the sample size. Thus, increasing the sample size by a multiple of d^2 divides the standard deviation of the set of sample means by d .

32. **(A)** With $df = 43.43$ and $t = -3.94$, tcdf gives that the p -value is $0.000146 < 0.01$. With a p -value this small, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that the mean pain level with the leech treatment is lower than the mean pain level with the placebo.
33. **(C)** For the same sample, as the confidence level increases, the width of the interval increases.
34. **(B)** This is a binomial distribution with $n = 5$ and $p = 0.3$. The probability that a car does not receive a ticket is $1 - 0.3 = 0.7$, the probability that none of the five cars receives a ticket is $(0.7)^5$, and thus the probability that at least one receives a ticket is $1 - (0.7)^5$.
35. **(D)** The p -value is a conditional probability. It assumes the null hypothesis ($p = 0.65$ in this question) is true, and it gives the probability of obtaining a sample statistic $\hat{p}$ as extreme as or more extreme than the one obtained (0.57 in this question).
36. **(A)** The first study is an experiment with two treatment groups (1 hour and 5 hours of television per night) and no control group. The second study is observational; the researcher simply noted the students' self-reported responses.
37. **(A)** Based on the table, children can either believe their father can overpower a sumo wrestler or not believe their father can overpower a sumo wrestler. Because the events cannot occur at the same time, the events are mutually exclusive.
38. **(B)** A Type I error means that the null hypothesis is correct (the weather will remain dry) but you reject it (thus you needlessly carry around an umbrella). A Type II error means that the null hypothesis is wrong (it will rain) but you fail to reject it (thus you get drenched).
39. **(B)** If the sample statistic is far enough away from the claimed population parameter, we say that there is sufficient evidence to reject the null hypothesis. In this case, the null hypothesis is that $\mu = 9,500$. The p -value is the probability of obtaining a sample statistic as extreme as, or more extreme than, the one obtained if the null hypothesis is

assumed to be true. The smaller the p -value, the more significant the difference between the null hypothesis and the sample results. With a p -value of 0.0069, there is strong evidence to reject H_0 .

40. (D) There are $10 + 12 = 22$ students in the combined group. In ascending order, where are the two middle scores? At least 5 third graders and 6 fourth graders have heights less than or equal to 49 inches, so at most 11 students have heights greater than or equal to 49. Thus, the median is less than or equal to 49. At least 5 third graders and 6 fourth graders have heights greater than or equal to 47 inches, so at most 11 students have heights less than or equal to 47. Thus, the median is greater than or equal to 47. All that can be said about the median of the combined group is that it is between 47 and 49 inches.
41. (C) Good experimental design aims to give each group the same experiences except for the treatment under consideration. Thus, all three SRSs should be picked from the same grade level.
42. (D) The stemplot does not indicate what happened for any individual school.

Section II

1. Model Solution:

- A. i. This is an example of *stratified random sampling*, where the groups (freshmen, sophomores, juniors, and seniors) are strata.
- ii. The advantage is that the guidance counselor will ensure that the final sample will represent all four groups of students, each of which might have different views of their college experience.
- B. i. This is an example of *cluster random sampling*, where each class (cluster) resembles the overall group from which it is selected.
- ii. The advantage is that using these clusters is much more practical than trying to sample from among the entire groups

(freshmen, sophomores, juniors, and seniors).

- C. i. This is an example of *systematic random sampling*, which is quicker and easier than many other procedures.
- ii. A possible disadvantage occurs if ordering is related to the variable under consideration. For example, if the order students enter the classroom is related to their views of their college experience, the counselor could end up with a nonrepresentative sample.
- D. Stratifying by gender will guarantee that both males and females will be represented in the sample. Without this step, it is possible that the sample may have few or no males or few or no females. This matters because males and females may well have different views on their college experience.
- E. Is there convincing statistical evidence to suggest that a majority ($p > 0.5$) of the students at the guidance counselor's college feel positively about their college experience?

A.	i.	Identify the procedure.	Stratified random sampling.	1 point
	ii.	Identify an advantage.	This procedure will ensure that the final sample will represent all four groups of students.	1 point
B.	i.	Identify the procedure.	Cluster random sampling.	1 point
	ii.	Identify an advantage.	This procedure is more practical than trying to sample from among the entire groups.	1 point
C.	i.	Identify the procedure.	Systematic random sampling.	1 point

ii.	Identify a disadvantage.	If the order students enter the classroom is related to their views of their college experience, the guidance counselor could end up with a nonrepresentative sample.	1 point
D.	Identify an advantage.	Stratifying by gender guarantees that both males and females will be represented in the sample. This matters because males and females may have different views.	1 point
E.	First component: identify the variable(s) of interest.	The variable is whether or not students feel positively about their college experience.	1 point
	Second component: identify the parameter and direction of the alternative hypothesis.	A majority of the students at the guidance counselor's college feel positively about their college experience. (True proportion $p > 0.5$)	1 point
	Third component: identify the population to which the conclusion is applicable.	All students at the guidance counselor's college.	1 point

2. Model Solution:

- A. i. Let p represent the proportion of the population of all preschool African Americans who would select the White doll when offered a choice of playing with a White doll or a Black doll.
- ii. $H_0: p = 0.5$ and $H_a: p > 0.5$
- B. If there was no preference (the null hypothesis), we would expect the distribution to be centered around $(0.5)(24) = 12$, tapering off in each direction from there. Choice II is the only reasonable graph.
- C. i. For Graph II, very few, if any, of the simulations produced 18 or more White doll picks (as did our sample). Thus, the p -value closest to the given choices is the smallest, 0.001.
- ii. With this small a p -value, $0.001 < 0.05$, there is convincing evidence that the proportion of the population of preschool African Americans who would select to play with a White doll when offered a choice of playing with a White doll or a Black doll is greater than 0.5.
- D. i. We found sufficient evidence to reject the null hypothesis. It is possible that a Type I error may have been committed—that is, mistakenly rejecting a true null hypothesis.
- ii. A possible consequence is that funding would be committed to create more empowering learning environments for African American children and the funding would be unnecessary.
- E. The population would be all preschool African American children.

A.	i.	Identify the parameter.	p = the proportion of the population of all preschool African Americans who would select the White doll when offered a choice of	1 point
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		playing with a White or Black one.	
ii.	Identify the hypotheses.	$H_0: p = 0.5$ and $H_a: p > 0.5$	1 point
B.	Identify the graph.	Choice II.	1 point
	Explain your choice of graph.	If there was no preference (the null hypothesis), we would expect the distribution to be centered around $(0.5)(24) = 12$, tapering off in each direction from there.	1 point
C.	i. Identify the correct p -value, and explain your reasoning.	For Graph II, very few, if any, of the simulations produced 18 or more White doll picks (as did our sample). The p -value closest to the given choices is the smallest, 0.001.	1 point
	ii. Justify a conclusion.	This is a small p -value, $0.001 < 0.05$.	1 point
	Conclude in context.	Thus, there is convincing evidence that the proportion of the population of preschool African Americans who would select to play with a white doll when offered a choice of playing with a White doll or a Black doll is greater than 0.5.	1 point

D.	i.	Determine the type of error.	Type I error.	1 point
	ii.	Identify a possible consequence.	Funding would be committed to create more empowering learning environments for African American children, and the funding would be unnecessary.	1 point
E.		Determine the population.	Preschool African American children.	1 point

3. Model Solution:

- A. The explanatory variable is the concentration level in milligrams per milliliter for the active ingredient found in vials of a pharmaceutical product. The variable is numerical.
- B. This was an observational study. An appropriate inference procedure is a one-sample t -interval for a population mean where μ = the mean concentration level for the active ingredient found in the population of vials of this pharmaceutical product.
- C. We are given that this is a random sample, and $n = 5$ is less than 10% of all vials of this product. The normality condition is satisfied because the dotplot of the sample data shows no outliers or strong skewness. The interval is (2.4674, 2.6966).
- D. We are 95% confident that the true mean concentration level in milligrams per milliliter for the active ingredient found in all vials of the pharmaceutical product is between 2.4674 and 2.6966.

A.	Identify the variable(s).	The variable is the concentration level in milligrams per milliliter for the active ingredient found	1 point
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		in vials of a pharmaceutical product.	
	Classify the variable(s).	Numerical.	1 point
B.	Identify the study.	Observational.	1 point
	Identify the procedure.	Use a one-sample t -interval for a population mean where μ = mean concentration level for the active ingredient found in the population of vials of this pharmaceutical product.	1 point
C.	Justify whether the randomization condition and 10% condition are met.	We are given that this is a random sample, and $n = 5$ is less than 10% of all vials of this product.	1 point
	Justify whether the normality condition is met.	The dotplot of the sample data shows no outliers or strong skewness. So it is not unreasonable to assume the sample comes from a roughly normal population.	1 point
	Calculate the confidence interval.	(2.4674, 2.6966) and $df = 4$	1 point
D.	Identify the confidence level and the interval.	95% confidence; (2.4674, 2.6966)	1 point
	Identify the population parameter.	Mean concentration level in milligrams per milliliter for the active ingredient found	1 point

		in a pharmaceutical product.	
	Identify the context.	True mean concentration for all vials of the product.	1 point

4. Model Solution:

- A. i. A Type II error is mistakenly failing to reject a false null hypothesis. In this situation, it would happen if the proportion of all women giving birth in the county who are unmarried is greater than 0.403 but the sample proportion does not provide convincing evidence that it is.
- ii. A possible consequence would be that necessary additional funding to counsel unmarried mothers would not be provided.
- B. i. The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.403)(0.597)}{100}} = 0.04905.$$
- ii. The critical z-score is $\text{invNorm}(0.95) = 1.645$, and then

$$\frac{\hat{p} - 0.043}{0.04905} > 1.6454 \text{ gives } \hat{p} > 0.4837.$$
- C. i.
$$\sigma_{\hat{p}} = \sqrt{\frac{(0.45)(0.55)}{100}} = 0.04975$$
- ii.
$$P(\hat{p} > 0.4837) = P\left(z > \frac{0.4837 - 0.45}{0.04975}\right) = 0.249$$
- D. This is called the power of the test.
- E. If the sample size was increased, although the rejection region would still be $z > 1.645$, the sampling distribution of $\hat{p}$ would have a smaller standard deviation. Then the minimum value of $\hat{p}$ for which we would reject H_0 would be lower. So the probability of rejecting H_0 would be greater.

A.	i.	Describe a Type II error.	A Type II error is mistakenly failing to	1 point
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		reject a false null hypothesis. In this situation, it would happen if the proportion of all women giving birth in the county who are unmarried is greater than 0.403 but the sample proportion does not provide convincing evidence that it is.	
ii.	Describe a possible consequence.	Necessary additional funding to counsel unmarried mothers would not be provided.	1 point
B.	i.	Calculate $\sigma_{\hat{p}}$.	$\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.403)(0.597)}{100}} = 0.04905$ 1 point
	ii.	Calculate the rejection region.	$\frac{\hat{p} - 0.043}{0.04905} > 1.6454$ gives $\hat{p} > 0.4837$. 1 point
C.	i.	Calculate $\sigma_{\hat{p}}$.	$\sqrt{\frac{(0.45)(0.55)}{100}} = 0.04975$ 1 point
	ii.	Calculate the probability of rejecting the null hypothesis.	$P(\hat{p} > 0.4837) = P\left(z > \frac{0.4837 - 0.45}{0.04975}\right) = 0.249$ 1 point
D.		Identify a type of calculation.	This is called the power of the test. 1 point
E.		Identify the change in $\sigma_{\hat{p}}$.	The sampling distribution of $\hat{p}$ will have a smaller $\sigma_{\hat{p}}$. 1 point
		Determine the direction of change	The minimum value of $\hat{p}$ to reject H_0 would be 1 point

	in the rejection region.	lower.	
	Determine the direction of change in the probability of rejecting H_0 .	The probability of rejecting H_0 would be greater.	1 point

Part 5

Appendices

Appendices

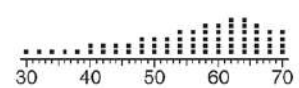
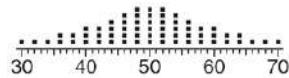
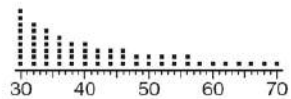
Graphical Displays

Right-skewed

Symmetric

Left-skewed

Dotplots



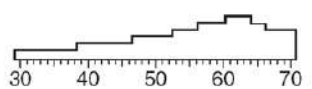
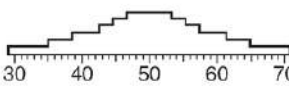
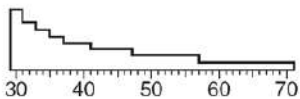
Stemplots

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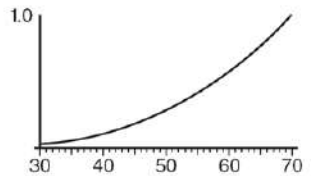
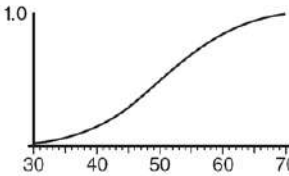
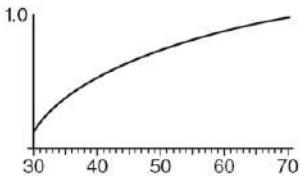
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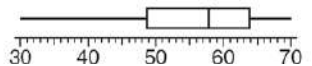
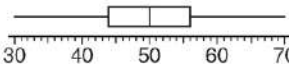
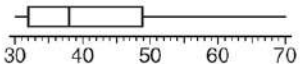
Histograms



Cumulative Frequency Plots

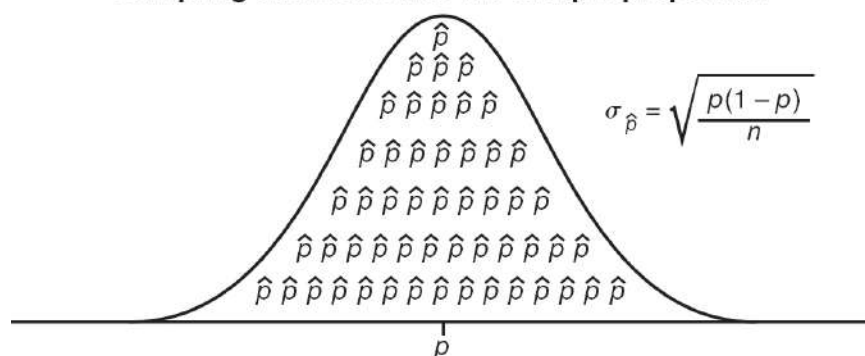


Boxplots



[Follow for extended description](#)

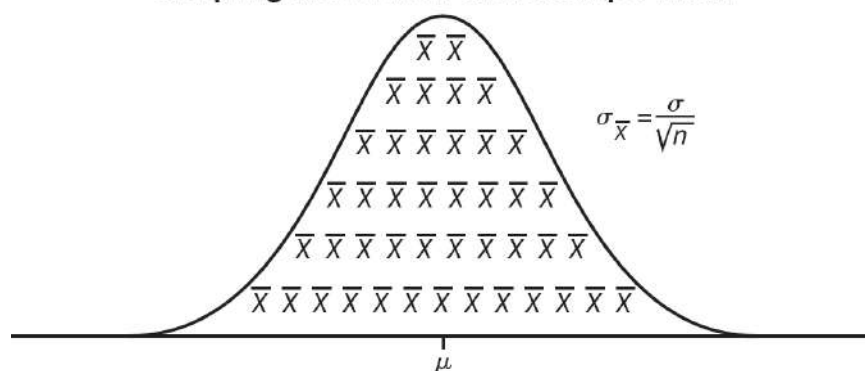
Sampling distribution of the sample proportion



[Follow for extended description](#)

If both $np \geq 10$ and $n(1-p) \geq 10$, then $\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$ has a roughly normal distribution.

Sampling distribution of the sample mean



[Follow for extended description](#)

If either the original population is normal or the sample size is large enough ($n \geq 30$), then $\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ has a roughly normal distribution, and $\frac{\bar{x} - \mu}{s/\sqrt{n}}$ has a roughly t -distribution with $df = n - 1$.

[Guide to Inference](#)

Confidence interval for proportions:

Procedure	Parameter	Statistic	Conditions	Formula	Calculators*
One-sample z-interval for a proportion	p	$\hat{p}$	- Random sample $n \leq 10\%N$ $n\hat{p} \geq 10, n(1-\hat{p}) \geq 10$	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	1-PropZInt 1-Props Z Interval Z-Int: 1π
Two-sample z-interval for a difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ $n_1\hat{p}_1 \geq 10, n_1(1-\hat{p}_1) \geq 10$ $n_2\hat{p}_2 \geq 10, n_2(1-\hat{p}_2) \geq 10$	$\hat{p}_1 - \hat{p}_2 \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	2-PropZInt 2-Prop Z Interval Z-Int: $\pi_1 - \pi_2$

[Follow for extended description](#)

Confidence interval for means:

Procedure	Parameter	Statistic	Conditions	Formula	Calculators*
One-sample t-interval for a mean	μ	$\bar{x}$	- Random sample $n \leq 10\%N$ - Population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$ with $df = n - 1$	TInterval 1-Sample t Interval T-Int: 1μ
Two-sample t-interval for a difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ - For each sample, population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}$ with df from calculator	2-SampTInt 2-Sample t Interval T-Int: $\mu_1 - \mu_2$
Paired data t-interval for a mean difference	μ_{diff}	$\bar{x}_{diff}$	- Random sample of paired data $n_{diff} \leq 10\%N_{diff}$ - Population of differences $\approx$ normal OR $n_{diff} \geq 30$ OR sample differences shows no strong skewness or outliers	$\bar{x}_{diff} \pm t^* \frac{s_{diff}}{\sqrt{n_{diff}}}$ with $df = n_{diff} - 1$	TInterval 1-Sample t Interval T-Int: 1μ

[Follow for extended description](#)

Hypothesis test for proportions:

Procedure	Null Hypothesis	Conditions	Formula	Calculators*
One-sample z-test for a proportion	$H_0: p = p_0$	- Random sample $n \leq 10\%N$ $np_0 \geq 10, n(1 - p_0) \geq 10$	$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$	1-PropZTest 1-Prop Z Test Z-Test: 1π
Two-sample z-test for a difference in proportions	$H_0: p_1 - p_2 = 0$ OR $H_0: p_1 = p_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ $n_1 \hat{p}_c \geq 10, n_1(1 - \hat{p}_c) \geq 10$ $n_2 \hat{p}_c \geq 10, n_2(1 - \hat{p}_c) \geq 10$ - where $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$	$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_c(1 - \hat{p}_c)}{n_1} + \frac{\hat{p}_c(1 - \hat{p}_c)}{n_2}}}$	2-PropZTest 2-Prop Z Test Z-Test: $\pi_1 - \pi_2$

[Follow for extended description](#)

Hypothesis test for means:

Procedure	Null Hypothesis	Conditions	Formula	Calculators*
One-sample t-test for a mean	$H_0: \mu = \mu_0$	- Random sample $n \leq 10\%N$ - Population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$ with $df = n - 1$	T-Test 1-Sample tTest T-Test: 1μ
Two-sample t-test for a difference in means	$H_0: \mu_1 - \mu_2 = 0$ OR $H_0: \mu_1 = \mu_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ - For each sample, population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}}$ with df from calculator	2-SampTTest 2-Sample tTest T-Test: $\mu_1 - \mu_2$
Paired data t-test for a mean difference	$H_0: \mu_{diff} = 0$	- Random sample of paired data $n_{diff} \leq 10\%N_{diff}$ - Population of differences $\approx$ normal OR $n_{diff} \geq 30$ OR sample differences shows no strong skewness or outliers	$t = \frac{\bar{x}_{diff} - 0}{\frac{s_{diff}}{\sqrt{n_{diff}}}}$ with $df = n_{diff} - 1$	T-Test 1-Sample tTest T-Test: 1μ

[Follow for extended description](#)

Hypothesis test for chi-square:

Procedure	Hypotheses	Conditions	Formula	Calculators*
χ^2 -test for independence	H_0 : Two variables are independent. H_a : Two variables are not independent.	- Random sample $n \leq 10\%N$ - All expected counts > 5	$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$ $df = (\# \text{ of rows} - 1)(\# \text{ of columns} - 1)$	χ^2 -Test χ^2 two-way Test χ^2 Test two-way test
χ^2 -test for homogeneity	H_0 : A distribution is the same across 2 or more populations. H_a : A distribution differs across 2 or more populations.	- Random samples from each population $n_i \leq 10\%N_i$ each sample - All expected counts > 5	$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$ $df = (\# \text{ of rows} - 1)(\# \text{ of columns} - 1)$	χ^2 -Test χ^2 two-way Test χ^2 Test two-way test

[Follow for extended description](#)

*Calculators refers to TI-84, Casio Prizm, and HP Prime.

TEMPLATES

Following are 25 templates with examples to help you word your free-response answers. Remember, it is imperative that you communicate in your responses what you want the reader to understand. Correct calculations and answers will not give you full credit if you do not explain what the numbers mean within the context of the problem.

Describing a one-variable distribution

The distribution of (*context*) has a shape that is roughly (*symmetric/unimodal/bimodal/bell-shaped/skewed left/skewed right/uniform*), (*center/mean/median*) of ____, (*range of ____/IQR of ____/spread from ____ to ____*), and unusual features of (*outliers, gaps, clusters*).

Example: In a competition among Gen X gamers, the distribution of Tetris scores was unimodal and roughly symmetric with a median of 160,000, a range of 175,000, a possible outlier at 260,000, and a gap between 240,000 and 260,000.

Comparing one-variable distributions

The (*first distribution in context*) has a ____ shape that is (*similar/not similar*) to the (*second distribution in context*) that has a ____ shape, the (*first distribution in context*) has a (*larger/smaller/equal*) center to the (*second distribution in context*), the (*first distribution in context*) has a (*larger/smaller/equal*) variability to the (*second distribution in context*), and the (*distribution in context*) has (*outliers/gaps/clusters*) although the (*distribution in context*) (*also does/doesn't*).

Example: At the end of the 2018–2019 season, the distribution of wins among the 15 teams in the Eastern NBA Conference (EC) was roughly bell-shaped, which is different from the distribution of wins among the 15 teams in the Western NBA Conference (WC) that was roughly uniform with a low outlier. The WC distribution had a greater center, while the EC distribution had a greater spread. The WC distribution had an apparent outlier at 19 and a gap between 19 and 33, which was different from the EC distribution that had no apparent outliers or gaps.

Interpreting the standard deviation

The mean distance between individual (*values in context*) and the mean (*context*) is approximately ____.

Example: Given that the number of Facebook friends of high school students has a mean of 13.5 and a standard deviation of 3.4, the mean distance between the number of Facebook friends of individual high school students and 13.5 is approximately 3.4.

Interpreting a z-score

The (*value in context*) is ____ standard deviations (*above/below*) the mean (*value in context*).

Example: A student has a GPA, which is 0.85 standard deviations below the mean GPA of all students.

Describing a scatterplot

There is a (*strong/moderate/weak*), (*positive/negative*), (*linear/nonlinear*) association between (*the explanatory variable x in context*) and (*the response variable y in context*).

Example: There is a moderate, negative, roughly linear association between the average number of minutes people sit during a day and their life expectancy in years.

Interpreting the slope

The predicted increase in (*response variable y in context*) is ____ for each unit increase in (*explanatory variable x in context*).

Example: The predicted increase in number of calories burned is 6.55 for each additional minute walked at 4 mph.

Interpreting the y-intercept

____ is the predicted value of the (*response variable y in context*) when the (*explanatory variable x in context*) is equal to 0.

Example: 0.142 is the predicted batting average for a high school player who spends 0 time in the batting cage.

Interpreting R-squared, the coefficient of determination

____% of the variation in (*the y-variable in context*) is accounted for by the linear model relating (*the y-variable in context*) to (*the x-variable in context*).

Example: 38% of the variation in incidence of skin cancer is accounted for by the linear model relating incidence of skin cancer to the SPF (sun protection factor) value of sunscreen used.

Picking a reasonable value for a correlation

Given a (*weak/moderate/strong*), (*positive/negative*), linear association between ____ and ____, a reasonable value for the correlation is (*a number between -1 and +1*).

Example: Given a strong, negative, linear association between resting heart rate and weekly hours of exercise, a reasonable value for the correlation is -0.83.

Making a conclusion when the scatterplot shows a linear association and the residual plot has no definite pattern

Because the scatterplot shows a linear association and the residual plot shows a random scatter, it appears that a linear model relating (*the y-variable in context*) to (*the x-variable in context*) (*is*) reasonable.

Example: Because the scatterplot shows a linear association and the residual plot shows a random scatter, it appears that a linear model relating grades for research papers to length of the papers is reasonable.

Making a conclusion when the residual plot has a definite pattern

Because the residual plot shows a definite pattern, a linear model relating (*the y-variable in context*) to (*the x-variable in context*) (is not) not appropriate.

Example: Because the residual plot shows a definite pattern, $\cap$ -shaped, a linear model relating mpg to car speed is not appropriate.

Describing how to use a random number generator to pick a simple random sample

Number the subjects 1, 2, 3, ..., n . Use a random number generator to generate integers between 1 and n , throwing out repeats, until x unique integers have been selected. The sample consists of the subjects with numbers corresponding to the x generated integers.

Example: To pick an SRS of 10 students from a classroom of 25 students, number the students from 1 through 25. Use a random number generator to generate integers between 1 and 25, throwing out repeats, until 10 unique integers have been selected. The sample consists of the 10 students with numbers corresponding to the 10 generated integers.

Performing a normal distribution probability calculation

We have a normal distribution with $\mu = \underline{\hspace{1cm}}$ and $\sigma = \underline{\hspace{1cm}}$. $P(X > / < \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$.
[You do not have to show whether you used a formula or calculator software.]

Example: We have a roughly normal distribution with mean $\mu = 25$ and standard deviation $\sigma = 10$. Then the probability a student's backpack weighs over 30 pounds is $P(X > 30) = 0.309$.

Performing a binomial distribution probability calculation

We have a binomial distribution with $n = \underline{\hspace{1cm}}$ and $p = \underline{\hspace{1cm}}$. $P(X = / < / > / \leq / \geq \underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$.
[You do not have to show whether you used a formula or calculator software.]

Example: We have a binomial distribution with $n = 25$ and $p = 0.63$. Then $P(X \geq 15) = 0.702$.

Describing a sampling distribution

The sampling distribution for $(\hat{p} / \bar{x} / b)$ is approximately normal with mean ($p = \underline{\hspace{1cm}}$ OR $\mu = \underline{\hspace{1cm}}$ OR $\beta = \underline{\hspace{1cm}}$) and standard deviation ($\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \underline{\hspace{1cm}}$ OR $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \underline{\hspace{1cm}}$ OR $\sigma_b = \underline{\hspace{1cm}}$).

Example: The sampling distribution of $\bar{x}$ is approximately normal with mean $\mu = 25$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{45}} = 1.19$.

Checking conditions for inference

We check (1) the *randomization condition* (random sample), (2) the *10% condition* (if sampling without replacement, the sample is less than the 10% of the population), and (3) the *normality condition* (for proportions: $np \geq 10$ and $n(1-p) \geq 10$; and for means: either the population is roughly normal OR the sample size $n \geq 30$ OR the sample distribution has no strong skewness or outliers).

Example (proportions): We check the randomization condition (we are given an SRS), the 10% condition ($n = 125 < 10\%$ of all students), and the normality condition ($n\hat{p} = (125)(0.61) = 76.25 \geq 10$ and $n(1-\hat{p}) = (125)(0.39) = 48.75 \geq 10$).

Example (means): We check the randomization condition (it is stated that the sample was randomly chosen), the 10% condition ($n = 35 < 10\%$ of all teachers), and the normality condition ($n = 35 \geq 30$).

Interpreting a confidence interval

We are ____ % confident that the true (proportion/mean in context) is between ____ and ____.

Example: We are 95% confident that the true proportion of people who leave a tip at fast-food restaurants like McDonald's and Starbucks is between 0.13 and 0.19.

Interpreting a confidence level

If we take all possible samples of (size in context) and if a confidence interval for the (proportion/mean in context) is constructed from each sample, then ____ % of these intervals will capture the true (proportion/mean in context).

Example: If all possible samples of 40 smartphones are analyzed as to battery length and if a confidence interval for the mean battery length is constructed from each sample, then 90% of these intervals would capture the true mean battery length.

Interpreting a p-value

Assuming that (the null hypothesis in context) is true, there is a probability of ____ of getting a sample (proportion/mean) as extreme as or more extreme than the one observed by chance alone in a random sample of (size in context).

Example: Assuming that 12% of American adults believe that Sodom and Gomorrah were husband and wife, there is a probability of 0.083 of getting a sample proportion of 0.135 or higher by chance alone in a random sample of 900 American adults.

Concluding a hypothesis test when rejecting the null hypothesis

Because the p -value is so small, $p = ___ < (\text{significance level})$, there is sufficient evidence to reject the null hypothesis. There is convincing evidence that (alternative hypothesis in context).

Example: With this small a p -value, $p = 0.031 < 0.05$, there is sufficient evidence to reject the null hypothesis. There is convincing evidence that less than 30% of all teenagers can name two current Supreme Court justices.

Concluding a hypothesis test when failing to reject the null hypothesis

Because the p -value is so large, $p = ___ > (\text{significance level})$, there is not sufficient evidence to reject the null hypothesis. There is not convincing evidence that (alternative hypothesis in context).

Example: With this large a p -value, $p = 0.18 > 0.05$, there is not sufficient evidence to reject the null hypothesis. There is not convincing evidence that the mean age of all high school teachers is over 45.

Interpreting a Type I error

Because the p -value was small, there was sufficient evidence to reject the (null hypothesis in context). However, the null hypothesis might have been true, in which case we committed a Type I error. A possible consequence would be ____.

Example: Because of a small p -value, there was sufficient evidence to reject the null hypothesis that a medication is not effective. However, if the medication really is not effective, a Type I error would have

been committed. A possible consequence is that patients will not recover because they will continue to be treated with an ineffective medication.

Interpreting a Type II error

Because the p -value was large, there was not sufficient evidence to reject the (*null hypothesis in context*). However, the null hypothesis might have been false, in which case we committed a Type II error. A possible consequence would be ____.

Example: Because of a large p -value, there is not sufficient evidence to reject the null hypothesis that a new teaching method fails to improve student performance. However, if the new teaching method really does improve student performance, a Type II error would have been committed. A possible consequence is that a school district will not adopt a new teaching method that would have helped their students.

Interpreting power

There is a (*power*) probability of correctly rejecting the null hypothesis.

Example: With H_0 : a medical device is operating correctly, there is a 0.92 probability of correctly recognizing that the medical device is not operating correctly.

Names of Procedures

When asked to identify the appropriate inference procedure, the answer should be a complete proper name as in the following expressions.

- One-sample z -interval for a population proportion
- One-sample z -test for a population proportion
- Two-sample z -interval for a difference between two population proportions
- Two-sample z -test for a difference between two population proportions
- Chi-square test for homogeneity
- Chi-square test for independence
- One-sample t -interval for a population mean
- One-sample t -interval for a population mean difference (with a matched pairs design)
- One-sample t -test for a population mean
- One-sample t -test for a population mean difference (with a matched pairs design)
- Two-sample t -interval for a difference between two population means
- Two-sample t -test for a difference between two population means

Answer Explanations for Quizzes 1–34

Quiz 1 (Pages 51–56)

Multiple-Choice

1. (C) Primary color is categorical (it takes values that are names), and width is quantitative (it takes values that can be measured).
2. (B) Number of passengers is an integer, a countable variable, thus is discrete. Weight can take on an infinite number of values between any two values, thus is continuous.
3. (D) The data are categorical because each student responds yes or no. This leads to estimating a population proportion, that is, the proportion of students who feel that too much homework is assigned in

most classes.

4. (D) The minimum time is somewhere between 5 and 10 minutes but might not be exactly 5 minutes. Similarly, the maximum time is somewhere between 30 and 35 minutes. With 155 times, the middle time will be the 78th time if the times are arranged in order. There are 70 times less than 15 minutes and 40 times between 15 and 20 minutes, so the 78th time must be between 15 and 20 minutes.
5. (C) Stemplots are not used for categorical data sets, are too unwieldy to be used for very large data sets, and show every individual value. Stems should never be skipped over—gaps are important to see.
6. (B) Histograms give information about relative frequencies (relative areas correspond to relative frequencies) and may or may not have an axis with actual frequencies. Symmetric histograms can have any number of peaks. Choice of width and number of classes changes the appearance of a histogram. Stemplots clearly show outliers; however, in histograms outliers may be hidden in large class widths.
7. (B) The percentage of Generation X who preferred Company B is $\frac{31}{50} = 0.62 = 62\%$, while the percentage of Millennials who preferred Company B is $\frac{32}{75} = 0.427 = 42.7\%$.
8. (B) The number of students not choosing Comedy or Romance was $20 - (4 + 6)$ or $5 + 1 + 4$. Of these, $4 + 5$ chose Sci-Fi or Action for a proportion of $\frac{4 + 5}{20 - (4 + 6)}$.

Free-Response Questions

1. (a) Given numerical data and four bins with corresponding frequencies immediately leads to the construction of a histogram. Dotplots and stemplots cannot be constructed when individual values are unknown.
- (b) $\frac{3 + 11}{25} = \frac{14}{25} = 56\%$
- (c) The answer would still be 56%.
2. (a) A complete answer mentions shape, center, spread, and unusual features (such as outliers) while including context within the description.
- Shape:* Unimodal, skewed right, probable outlier at 10.
Center: Around 2 or 3.
Spread: From 0 to 10 goals were scored by each team.
- (b) If the player scored six goals on his own, then his/her team must have scored either 7 or 10 goals. Since the team lost, the only possible final score is 10 to 7.
- (c) No, there were six teams that scored exactly two goals, but there were only five teams that scored less than two goals. So, not all the two-goal teams could have won.
3. (a) The lowest winning percentage over the past 22 years was 46.0%.
- (b) A complete answer mentions shape, center, spread, and unusual features (such as outliers) while including context within the description.
- Shape:* Two clusters, each somewhat bell-shaped.
Center: Around 50%.
Spread: The winning percentages were between 46.0% and 55.6%.
- (c) The team had more losing seasons (13) than winning seasons (9).
- (d) The cluster of winning percentages is further away from 50% than the cluster of losing percentages.
4. (a) Is there a relationship between college students' economic backgrounds and leadership position experience?
- (b) Categorical, taking on values that are categories (economic background).

- (c) Relative frequencies are more appropriate in analyzing these data because there are different numbers of students who were surveyed from each economic category.
- (d) Students who enter college with certain advantages seem to accrue additional advantages in college. In this survey, higher-income economic backgrounds were associated with higher levels of leadership position experience.

Quiz 2 (Pages 57–61)

Multiple-Choice

1. (A) The proportion of students choosing “Classic” was 0.23, but the exact number is unknown because the sample size is not given.
2. (D) The number of participating adults from each age group might be different, which rules out choices (A), (B), and (C). The least relative frequency on the bar chart is 67%, corresponding to age group 18–29.
3. (C) The graph gives relative frequencies, not actual numbers of students. There may be many possible explanations other than cheating for self-grading to result in higher scores.
4. (D) In a skewed right distribution, the right tail is longer than the left and the mean is usually greater than the median. There are 25% of the values between the first quartile and median, and there are 25% between the median and the third quartile.
5. (B) There are two clusters, a gap, and an outlier between 78 and 79. The range is at least $78 - 67 = 11$ and at most $79 - 66 = 13$. There are no low outliers, and although the distribution does have a high outlier, it is not skewed.
6. (B) The second stemplot would be shown as:

–2	0 3 5 5 8 9
–1	0 2 2 5
–0	4 6 8
0	1 9
1	3
2	7

Key: 2 | 7 means 2.7 inches

[Follow for extended description](#)

7. (C) The number of rebounds is an integer. It is numerical because we can take an average.
8. (B) The total number of students represented in the histogram will remain at 35.

Free-Response Questions

1. (a) The distribution of quiz scores is skewed left, is centered around 95, and has a range of $100 - 73 = 27$.
(b) With skewed left distributions, we expect the median to be greater than the mean.
2. (a) Is there convincing statistical evidence of a relationship between school level and e-cigarette use in the population of all students in middle school, high school, and college?

- (b) Categorical, because the data are counts.
 - (c) Relative frequencies are more appropriate in analyzing these data because there are different numbers of students who were surveyed from each school level.
 - (d) Middle school students have the lowest relative frequency use of e-cigarettes. High school students appear to have a relative frequency use of e-cigarettes more than double that of middle school students. College students appear to have a relative frequency use of e-cigarettes more than double that of high school students. In this survey, higher e-cigarette use was associated with higher school levels.
3. (a) Given numerical data and three intervals with corresponding frequencies immediately leads to construction of a histogram. Dotplots and stemplots cannot be constructed when individual values are not given.
- (b) $\frac{15+7}{26} \approx 84.6\%$
- (c) The answer is still 84.6%.
4. (a) A negative difference indicates a volunteer who lost more weight taking Medication *B* than taking Medication *A*.
- (b) The distribution of differences in weight loss (under Medication *A* minus under Medication *B*) is skewed left (toward lower values), has a center around 3.1 pounds, and has a range of $8.5 - (-5.0) = 13.5$ pounds.
- (c) Medication *A* appears to be more effective for weight loss. Eleven participants lost more weight with Medication *B*, while 35 participants lost more weight with Medication *A*.

Quiz 3 (Pages 77–81)

Multiple-Choice

1. (D) The distribution is clearly skewed right, so the mean is greater than the median and the ratio is greater than one.
2. (D) Multiplying every value in a set by the same constant (in this case, by $\frac{1}{60}$) multiplies both the mean and the standard deviation by the same constant. Standardized scores (the number of standard deviations from the mean) are unchanged and without units.
3. (D) Outliers are any values below $Q_1 - 1.5(IQR) = 25 - 1.5(38 - 25) = 5.5$ or above $Q_3 + 1.5(IQR) = 38 + 1.5(38 - 25) = 57.5$. With a minimum of $10 > 5.5$, there are no outliers on the low end; however, with a maximum of $60 > 57.5$, the maximum is an outlier and so are any other values falling between 57.5 and 60.
4. (A) Subtracting 10 from one value and adding 5 to two values leaves the sum of the values unchanged, so the mean will be unchanged. Exactly what values the outliers take will not change what value is in the middle, so the median will be unchanged.
5. (D) The standard deviation depends on all values, not just on the min and max and not just on Q_1 and Q_3 . Standard deviation depends on distances from the mean, not on the size of the mean.
6. (A) The standard deviation, $\sigma = \sqrt{\frac{\sum(x - \mu)^2}{n}}$, and the interquartile range, $IQR = Q_3 - Q_1$, can never be negative.
7. (A) The sum of the scores in one class is $20 \times 92 = 1,840$, while the sum in the other is $25 \times 83 = 2,075$. The total sum is $1,840 + 2,075 = 3,915$. There are $20 + 25 = 45$ students, and so the average score is $\frac{3,915}{45} = 87$.

8. (B) Removing a value that is greater than the average will always lower the average, and adding in a value that is less than the average will also always lower the average.
9. (A) There are 5 classes, and their average size is $\frac{150 + 25 + 25 + 25 + 25}{5} = \frac{250}{5} = 50$.
10. (C) Among the 250 students, there are 100 students in classes of size 25 and 150 students in a class of size 150. The average size of their history class is $\frac{100(25) + 150(150)}{250} = \frac{25,000}{250} = 100$.

Free-Response Questions

1. (a) Adding 10 to each value increases the mean by 10 but leaves measures of variability unchanged. So, the new mean is 340 hours while the range stays at 5,835 hours, the standard deviation remains at 245 hours, and the variance remains at $245^2 = 60,025 \text{ hr}^2$.
- (b) Increasing each value by 10% (multiplying by 1.10) will increase the mean to $1.1(330) = 363$ hours, the range to $1.1(5,835) = 6,418.5$ hours, the standard deviation to $1.1(245) = 269.5$ hours, and the variance to $(269.5)^2 = 72,630.25 \text{ hr}^2$. (Note that the variance increases by a multiple of $(1.1)^2$ and not by a multiple of 1.1.)
2. A complete answer compares shape, center, and spread and mentions context in at least one of the responses.
- Shape:* The distribution of times to complete all tasks by females is skewed right (toward the higher values), whereas the distributions of times to complete all tasks by males is roughly bell-shaped.
- Center:* The center of the distribution of female times (at around $2\frac{1}{4}$ minutes) is less than the center of the distribution of male times (at around $3\frac{3}{4}$ minutes).
- Spread:* The spreads of the two distributions are roughly the same; the range of the female times ($5 - 1\frac{1}{2} = 3\frac{1}{2}$ minutes) equals the range of the male times ($5\frac{1}{2} - 2 = 3\frac{1}{2}$ minutes).
3. z-scores give the number of standard deviations from the mean, so $Q_1 = 300 - 0.7(25) = 282.5$ and $Q_3 = 300 + 0.7(25) = 317.5$.
The interquartile range is $IQR = 317.5 - 282.5 = 35$, and $1.5(IQR) = 1.5(35) = 52.5$.
The standard definition of outliers encompasses all values less than $Q_1 - 52.5 = 230$ and all values greater than $Q_3 + 52.5 = 370$.
4. (a) The Liberian median age lies in the 15–19 age interval because roughly 50% of the total bar lengths is above and below the 15–19 interval.
- (b) Numerical values are needed to calculate means. Histograms such as these give numbers or proportions of values in specific intervals but do not give the actual values.
- (c) Canada has more children younger than 10 years of age. There are about 1.2 million children younger than age 10 in Liberia (boys and girls) and roughly 3.5 million in Canada. (Note the difference in scales!)
- (d) The population pyramid indicates that Canadian women live longer than men because all the higher age intervals show greater numbers of women than men.
- (e) In the Liberian graph, the smaller 15–19 age group shows a definite break with the overall pattern. A plausible explanation is that a great number of children died in the civil war.

Quiz 4 (Pages 82–87)

Multiple-Choice

1. (A) Both distributions are skewed toward the lower values—that is, skewed left.
2. (C) The interquartile range, $Q_3 - Q_1 = 14 - 10$, accounts for about 50% of the data.

3. (C) The range of the Area 1 distribution is a little greater than 50, while the range of the Area 2 distribution is a little greater than 100. The Area 1 distribution is roughly bell-shaped, and its mean is close to its median, while the Area 2 distribution is skewed left, so its mean is less than its median.
4. (A) Multiplying every value in a set by the same constant (in this case by 1.1) multiplies both the mean and the standard deviation by the same constant and leaves the standardized score unchanged.

$$z = \frac{1.1x - 1.1\mu}{1.1\sigma} = \frac{1.1(x - \mu)}{1.1\sigma} = \frac{x - \mu}{\sigma}$$

5. (D) The new temperature is an outlier since it is far below the rest of the data. The median is resistant to outliers, so it is unlikely to change. The mean is not resistant to outliers, so it is likely to change, in this case decrease. The range is not resistant to outliers, so it is likely to change.
6. (B) Both teams have a median of .255, with 50% of the averages to either side. So on both teams at least half their players have an average over .250. There is no way of knowing how many players are on either team, and there is no way of concluding “roughly bell-shaped” from a boxplot.
7. (D) The weight distribution from Site X is much more spread out from roughly where the mean would be than the weight distribution from Site Y, leading to a greater standard deviation of the distribution from Site X. Values may be anywhere in the bins, so there is no way of exactly knowing the median or mean.
8. (D) The median should be a point with half the values on either side; thus, it is in the interval from 100 to 200. With a right-skewed distribution, the mean will be greater than the median.
9. (B) For the men, $Q_1 - 1.5(IQR) = 47.6 - 1.5(47.9 - 47.6) = 47.15$ and $Q_3 + 1.5(IQR) = 47.9 + 1.5(47.9 - 47.6) = 48.35$. So there is at least one low outlier for the men. For the women, $Q_1 - 1.5(IQR) = 52.8 - 1.5(53.6 - 52.8) = 51.6$ and $Q_3 + 1.5(IQR) = 53.6 + 1.5(53.6 - 52.8) = 54.8$. So there are no outliers for the women.
10. (A) Adding the same number to every value will increase Q_1 and Q_3 by the same value. So their difference, which is the IQR, will remain the same. Similarly, the mean will also increase by the same value but differences from the mean will remain the same. So the SD will remain unchanged.

Free-Response Questions

1. (a) The distribution of daily price fluctuations of Stock F has a greater center than the distribution of daily price fluctuations of Stock E (a median of 80 versus a median of 60). Stock E distribution has the greater spread than Stock F distribution (a range of 70 versus a range of 25). Stock E distribution has 3 outliers, while Stock F distribution has none.
- (b) The 3 high outliers in the Stock E distribution are greater than any values in the Stock F distribution, showing a greater potential for a high price. So choose Stock E.
- (c) The lowest value in the Stock F distribution is higher than at least 50% of the Stock E values. Stock F promises a higher minimum value, so choose Stock F.
2. (a) The distribution of casual dining times is roughly right skewed, while the distribution of fine dining times is roughly bell-shaped. Both the center and the spread are greater for the fine dining times.
- (b) There were $12 + 3 = 15$ dining times under 90 minutes.
- (c) All 14 casual dining times were under 105 minutes as were 5 of the fine dining times. A total of 19 dining times were under 105 minutes, and a total of $19 + 4 = 23$ were under 120 minutes. With a total of 42 dining times, the median is between the 21st and 22nd times, so it must be somewhere between 105 and 120 minutes.
3. (a) The distribution of first serve speeds for the male players was roughly bell-shaped, while the distribution of first serve speeds for the female players was skewed right. The center of the male

distribution was greater than that of the female distribution (a median of 127 versus 114). The spread of the male distribution was greater than that of the female distribution (a range of 32 versus 28).

- (b) There were $8 + 1 = 9$ first serves over 125 mph.
- (c) There were $4 + 10 = 14$ first serves under 118 mph and $11 + 3 = 14$ first serves over 118. The median first serve speed among the 30 players was 118 mph.
- 4. (a) In a skewed right distribution, the mean is pulled to the right. So the median, which is resistant to outliers, is a more typical value of the weights of the textbooks.
- (b) For the 397 weights, the median will be the 199th weight in order. Add weights from the left, $18 + 68 + 85 = 171$. It is seen that the 199th weight will be in the 2.5 to 3.0 bin. The median weight is between 2.5 and 3.0 pounds.
- (c) The smaller IQR would identify those weights below $Q_1 - 1.5(0.5) = Q_1 - 0.75$ or above $Q_3 + 1.5(0.5) = Q_3 + 0.75$ as outliers. The larger IQR would identify those weights below $Q_1 - 1.5(1.5) = Q_1 - 2.25$ or above $Q_3 + 1.5(1.5) = Q_3 + 2.25$ as outliers. Therefore, the smaller IQR would potentially identify more textbook weights as outliers.

Quiz 5 (Pages 99–103)

Multiple-Choice

- 1. (A) This was an observational study because no treatment was imposed. It was retrospective because existing data from the past were examined.
- 2. (D) The wording “creating a level playing field” and “a right to express their individuality” are nonneutral and clearly leading phrasings.
- 3. (C) If there is bias, taking a larger sample just magnifies the bias on a larger scale. Even when the subjects are chosen randomly, there can be bias due, for example, to nonresponse or to the wording of the questions. Convenience samples, like shopping mall surveys, are based on choosing individuals who are easy to reach, and they typically miss a large segment of the population. Voluntary response samples, like radio call-in surveys, are based on individuals who offer to participate, and they typically overrepresent persons with strong opinions.
- 4. (D) Each of the 50 states, with its own longtime past state standards and different regional culture, is considered a homogeneous stratum.
- 5. (D) Increasing sample size increases accuracy but has no effect on any bias that is present.
- 6. (D) In a simple random sample, every possible group of the given size has to be equally likely to be selected, and this is not true here. For example, with this procedure it would be impossible for all the players of one team to be together in the final sample. This procedure is an example of stratified random sampling, but stratified random sampling does not result in simple random samples.
- 7. (C) Surveying people coming out of any church results in a very unrepresentative sample of the adult population, especially given the question under consideration. Using chance and obtaining a high response rate will not change the undercoverage bias and make this into a well-designed survey.
- 8. (B) Cluster random samples work when each cluster roughly looks like the population as a whole. However, there is no reason to believe that each conference has similar graduation rates as any other conference. Some conferences may have different academic standards and different support levels for their student athletes. Picking two conferences may not at all be representative of the population of Division 1 college football programs. In general, cluster random samples are known for ease, convenience, and less expense than other sampling methods. Homogeneity among schools in given conferences would relate to picking a stratified random sample with the conferences as strata.

9. (B) Voluntary response bias results from surveys based on individuals who choose to participate. Examples include when people choose to call in to a radio station, to respond to a newspaper ad, to approach and answer a surveyor's questions, or to respond to an online website. The example in which people are randomly chosen from a voter registration list and all answer a survey would not involve voluntary response bias.
10. (B) Nonresponse bias is a potential problem whenever a survey does not obtain responses from everyone surveyed. Choice (A) has potential voluntary response bias, and choice (D) has potential undercoverage bias. However, choices (A), (C), and (D) all achieved responses from everyone who was surveyed, so they do not have nonresponse bias.

Free-Response Questions

1. (a) This is an example of *stratified random sampling*, where the chapters are strata. The advantage is that the student is ensuring that the final sample will represent the 12 different authors, who may well use different average word lengths.
- (b) This is an example of *cluster random sampling*, where for each chapter the three chosen pages are clusters. It is reasonable to assume that each page (cluster) resembles the author's overall pattern. The advantage is that using these clusters is much more practical than trying to sample from among all an author's words.
- (c) This is an example of *systematic random sampling*, which is quicker and easier than many other procedures. A possible disadvantage is that if ordering is related to the variable under consideration, this procedure will likely result in an unrepresentative sample. For example, in this study if an author's word length is related to word order in sentences, the student could end up with words of particular lengths.
2. (a) Method I results in cluster random sampling (the rows are clusters) and can be implemented by numbering the rows 1 through 10, using a random number generator to randomly pick a number between 1 and 10, and using all 15 plots in that row. Method II results in stratified random sampling (the columns are strata) and can be implemented by numbering the rows 1 through 10, using a random number generator to pick a number between 1 and 10 to pick a plot in the first column, repeating this process to pick a random plot in each of the 15 columns, and using the 15 plots resulting from picking one plot from each column.
- (b) Method II, a stratified random sample (with the columns as strata), would be preferable here as each column is somewhat different, so picking a plot from each is meaningful. Method I, an attempt at cluster random sampling with the rows as clusters, would work only if each of the rows was heterogeneous (representative of the whole area); this is clearly not true as, for example, the top and bottom rows have far fewer nests than the middle rows.
3. (a) People who attend the school's basketball game are probably more interested in supporting the athletic program than those who do not attend. This study will probably show that a higher proportion of people favor the increased spending than what is the true proportion.
- (b) Subscribers to a health magazine are more likely to believe in the benefits of eating organic than the general population, so the resulting proportion of believers in this study will be greater than it is in the general population.
- (c) The distributor could put any moldy strawberries on the bottom, out of sight! The proportion of accepted boxes will be higher than it should be.
- (d) Many patients will be reluctant to admit to their doctor that they are not following their diet instructions. So, the proportion who say they are following the instructions will be higher than the true proportion following the instructions.
- (e) Parents who are satisfied with the school system are probably less likely to take the time to respond to this voluntary response survey, while those who are unhappy with the education their kids are

receiving are more likely to respond. So, the proportion of “unhappy” responses will probably be higher than the true proportion of parents who are unhappy.

4. (a) Methods *A* and *C* indicate high precision because they result in statistics that are tightly clustered.
(b) Methods *C* and *D* indicate probable bias because they result in statistics that are not centered around the true population parameter. Note that shape and variability are irrelevant to the issue of bias. Bias is focused on the *center* of the distribution.
5. (a) No, a sample obtained using Method II will not be representative of the 1,000 counties. It will only be representative of either the low- or the high-income counties, whichever was randomly chosen.
(b) Method II will result in less variability in AQI because all the scores will come from the one randomly selected income group of 500 similar-income counties.
(c) Method I was most likely used to select this sample. The bimodal shape of the sample distribution indicates that counties were likely selected from both high- and low-income groups, which happens with Method I. Method II would most likely result in a unimodal distribution centered around either 50 or 100.

Quiz 6 (Pages 104–107)

Multiple-Choice

1. (B) This matched pairs experiment is a special type of randomized block experiment. In this study, each volunteer is a block in which the order of the treatment (watching the television episode or listening to heavy metal music) is randomly assigned.
2. (C) Response bias occurs when the question itself can lead to misleading results because people might not answer truthfully if they don't want to be perceived as having unpopular or unsavory views or don't want to admit to not following instructions. Careful wording of the questions can decrease the likelihood of response bias.
3. (B) Stratified random sampling involves dividing the population into homogeneous groups called strata, then picking random samples from each of the strata, and finally combining these individual samples into what is called a stratified random sample. For example, we can stratify by age, gender, income level, or race/ethnicity.
4. (D) Causation can be concluded only from well-designed experiments, not from observational studies.
5. (B) Random sampling, which is the use of chance in selecting a sample from a population, is critical in being able to generalize from a sample to a population. Random assignment of subjects to treatments in experiments is critical in minimizing the effect of possible confounding variables.
6. (A) Because the students self-selected whether to live in a dormitory, the sample was not representative of all students attending the college. Not including students living off campus can result in undercoverage bias. The true proportion of students happy with the dormitories is probably less than 93%.
7. (C) Driving while intoxicated is both socially unacceptable and illegal. So even if they do this, most people will lie rather than admit to doing this.
8. (B) Cluster random sampling involves dividing the population into heterogeneous groups called clusters and then picking everyone or everything in a random selection of one or more of the clusters. In this situation, the teams were the clusters. Choice (A) was an SRS, choice (C) was a stratified random sample with the conferences as strata, and choice (D) was a systematic random sample.
9. (A) No treatment was applied, so this was an observational study, not an experiment. Measurements were taken of the concentration of the contaminant, so concentration was the response variable.
10. (D) Nonresponse bias occurs when a substantial number of contacted people choose not to respond. People choosing to respond and people not choosing to respond might have very different opinions, which

can lead to unreliable conclusions if only one group participates.

Free-Response Questions

1. (a) Number the employees 1 through 50. Use a random number generator to pick integers between 1 and 50, throwing out repeats, until 5 unique integers are picked. The committee will consist of the 5 employees with numbers corresponding to the 5 integers.
(b) With stratified random sampling, you could guarantee both blue-collar and white-collar employees on the committee. With an SRS, you could end up with only blue-collar or only white-collar employees on the committee. This is important because blue-collar and white-collar employees might have different concerns with regard to the issues under negotiation.
2. (a) Treatments (supplemental protein) were applied, so this was an experiment.
(b) The purpose of the experiment was to see if ingesting supplemental protein is more effective in gaining lean body mass, muscle strength, and physical test performance than ingesting no protein supplement. To do a comparison, there needed to be a group that received no supplement. The study also allows for investigating two different levels of supplemental protein.
(c) These 30 student-athletes were volunteers, who may not represent the population of student-athletes. For example, the volunteers might be more interested in muscle mass and do more weight training than other student-athletes. Ingesting supplemental protein and weight training would then be confounded. So it is not reasonable to generalize conclusions from this study beyond the 30 participants.
3. (a) Surveying the first customers to arrive introduces undercoverage bias because the survey deals with offering a reward for early arrivals. A better method would be to choose a random sample from among all the customers shopping at that store because early arrivals and later arrivals will probably have different views on giving a reward to early arrivals.
(b) Wording such as “deserve a reward” may lead early arrivals to say that is the alternative they prefer. It would be more neutral to simply ask, “Would you rather receive a free piece of candy for arriving early or would you rather the store offer a storewide sale on candy?”
4. (a) Method II results in a stratified random sample where the students attending public colleges are one strata and the students attending private colleges are a second strata.
(b) Method I results in a cluster random sample where the 5,800 colleges are the clusters.
(c) An advantage of Method I (cluster random sampling) is that only 25 colleges need to be surveyed, while a disadvantage is that the students surveyed might be all or mostly from either the private colleges or the public colleges (undercoverage bias).
(d) An advantage of Method II (stratified random sampling) is that it guarantees that students from both private and public colleges will be surveyed, while a disadvantage is that it necessitates listing a very large number of students from which to be able to pick randomly.
5. (a) If the friends are all in the same department and if something happened in the department to change productivity, this would be confounded with the program being implemented.
(b) If something new occurred in the whole company that affected productivity, without the control group it would not be possible to know what increased productivity, one of the programs or what occurred in the whole company.
(c) It would not be reasonable to generalize any conclusion because the study was with volunteers. People who volunteer might be more motivated and would increase productivity simply because they were being observed.

Quiz 7 (Pages 116–119)

Multiple-Choice

1. (A) Random sampling can be used in experimental design but is not a primary element like it is with observational studies.
2. (C) Double-blinding is often important so that those who measure responses are not influenced by the knowledge of which subjects are receiving which treatments.
3. (C) The placebo effect refers to the fact that some people will respond to a perceived treatment even though their treatment is actually an inactive treatment—for example, receiving a sugar pill rather than a medication.
4. (D) Experiments aim to discover whether a treatment of interest (explanatory variable) has a proposed effect (response variable).
5. (A) The main office at your school should be able to give you the class sizes of every math and English class. If need be, you can check with every math and English teacher.
6. (C) In the first study, the families were already in the housing units, while in the second study, one of two treatments was applied to each family.
7. (D) Octane is the only explanatory variable, and it is being tested at four levels. Miles per gallon is the single response variable.
8. (D) Using only a sample from the observations gives less information. It may well be that very bright students are the same ones who both choose to take AP Statistics and have high college GPAs. If students could be randomly assigned to take or not take AP Statistics, the results would be more meaningful. Of course, ethical considerations might make it impossible to isolate the confounding variable in this way.
9. (D) Association does not imply causation. It is likely that regular exercise and a person's age are both associated with greater heart strength, and either one could be the cause of the greater heart strength. Regular exercise and age are confounded. An experiment in which people are randomly assigned to exercise or not can prove a causal relationship between exercise and heart strength.
10. (B) Blocking divides the subjects into groups, such as men and women, or political affiliations, and thus reduces variation. That is, when we group similar individuals together into blocks and then randomize within the blocks, much of the variability due to the differences between the blocks is accounted for and so comparison of the treatment groups is clearer.

Free-Response Questions

1. (a) This cannot involve random assignment as it makes no sense to randomly assign teenagers to go to public or private schools. However, it would be good to use random sampling to select teenagers from each type of school to be asked about *Back to the Future* movies so as to be able to generalize to the population of public and private schools.
(b) It is important for the company CEO to use random assignment to receive hourly pay raises or end-of-the-year bonuses because a cause-and-effect conclusion is of interest.
(c) The sportscaster would probably want to include all the interleague games played by the Cubs and White Sox during the shortened 2020 season, and thus use neither random sampling nor random assignment.
2. (a) The explanatory variable is the frequency of substance abuse. The response variable is the age at death.
(b) This is a prospective observational study. No treatments were assigned, and the researchers asked the rock and pop stars to report their frequency of substance abuse and later noted their ages at time of death.

- (c) No, this is an observational study, so no cause-and-effect conclusion is possible. It is possible that other variables are influencing the response. For example, rock and pop stars might engage in other forms of risk taking, such as poor eating and poor sleep patterns, which also might negatively affect life expectancy.
3. (a) The explanatory variable is flashing or not flashing “BUY POPCORN” on the screen for a fraction of a second. The response variable is the quantity of popcorn sold.
- (b) This is an experiment because a treatment, flashing “BUY POPCORN,” is performed on certain days. There is a control group, the days when no flashing takes place.
 - (c) Double-blinding should be incorporated. It is important that customers don’t know they are seeing or not seeing the subliminal message as that knowledge might influence whether or not they buy popcorn. It is also important that the clerks selling the popcorn don’t know which days the message will be flashing as that might impact how they react to customers.
 - (d) Random assignment should be incorporated. For example, every day for some specified period of time, the manager could flip a coin. If heads, flash the subliminal message all day on the screen. If tails, though, don’t flash the message that day.
 - (e) If the subliminal message always took place on certain days, for example, on weekends, that could be a confounding variable. It would not be clear if observed higher popcorn sales were due to the subliminal messages or to the weekend. Random assignment of days for subliminal messages is a control over this variable.
 - (f) Compare the quantity of popcorn bought by the treatment group—that is, by the people who receive the subliminal message—to the quantity bought by the control group, which is the people who don’t receive the message. This is a well-designed experiment incorporating comparison, replication, random assignment, and control. So if a difference is noted, a cause-and-effect conclusion is justified.
4. (a) This was an experiment because treatments, which involved fitness trackers and incentives, were applied.
- (b) Treatments: (1) tracker and personal cash incentive, (2) tracker and charity cash incentive, (3) tracker alone, and (4) no tracker or cash incentive. Experimental units: the 800 volunteers. Response variable: change in blood pressure over a six-month period.
 - (c) The “no fitness tracker or incentive” group was a *control group* whose purpose is to provide a baseline to determine if the activity tracker or cash incentive treatments had any effect on blood pressure change.
 - (d) Assign each participant an integer from 1 to 800. Use a random integer generator to choose 600 unique integers between 1 and 800, throwing away repeats. The participants corresponding to the first 200 of these selected integers will be put in the first group, the participants corresponding to the second 200 of these selected integers will be put in the second group, the participants corresponding to the last 200 of these selected integers will be put in the third group, and the remaining 200 participants will be put into the fourth group.
 - (e) These 800 participants were volunteers, who may not be representative of a larger population with regard to the effects of wearing fitness trackers or receiving cash incentives on changes in blood pressure. So it is not reasonable to generalize the results of this study to a larger population.

Quiz 8 (Pages 120–124)

Multiple-Choice

- 1. (C) The experimental units are the people who participated in the study.
- 2. (B) If the variation in the THC content within each group is small compared with the variation in the THC content between the two groups, then the THC content data would look like two clusters with a gap

between. This would provide convincing evidence that the average THC content of the flower is different when dried or fresh.

3. (D) Picking names out of a hat gives random assignment. There was a comparison of the two treatments, standing with eyes open or with eyes closed, and there were 15 subjects in each treatment group. Note that replication refers to having more than one experimental unit in treatment groups, not to repeating the experiment multiple times.
4. (C) The treatments, which are the three musical genres piped into the headphones, will be randomly assigned to the volunteers.
5. (A) Using volunteers makes it difficult to generalize to a population. However, the main problem here is the lack of a control group. Without a control group, there could well have been confounding variables, including the placebo effect.
6. (B) The experiment is to determine the effect of having a parent sitting beside a child, and clearly a child will know if a parent is beside him or her.
7. (D) Combining two different populations each possibly subject to several confounding variables will not allow generalizing to any population.
8. (A) The students using the first review book average 1470 on the SAT exam. This says nothing about the spread of exam scores among the 110 students using the book. The spread could be dramatic with nothing unusual about a single score of 1255.
9. (C) What is being manipulated on all the preparations is whether they are baked immediately, chilled for 1 hour before baking, or chilled for 24 hours before baking.
10. (A) In a randomized block design, the subjects are grouped into blocks based on similar characteristics and then randomly assigned treatments within each block. For example, the researchers in this example might want to test the three different teaching methods separately on students with high IQs, on students with average IQs, and on students with low IQs.

Free-Response Questions

1. (a) Treatments: New medication, placebo Experimental units: 50 teens with ADHD Response variable: Lessening of ADHD symptoms
(b) Pairing similar teens (same age, same gender, same ADHD diagnosis, and same severity of symptoms) allows for variation in ADHD symptoms due to the new medication, which could be distinguished from the listed common variables. This would not be possible in a completely randomized experiment. By using the matched pairs design, the difference in symptom lessening between the new medication and a placebo is easier to measure. It is easier to find convincing evidence that the new medication is better than a placebo.
(c) Label each pair of similar teens as *A* and *B*. For each pair, toss a coin. If heads, teen *A* gets the new medication and teen *B* gets the placebo. If tails, teen *B* gets the new medication and teen *A* gets the placebo.
2. (a) This was an experiment because two treatments were applied: 1 lesson followed by using training wheels and 2 lessons followed by using training wheels.
(b) No, they would not be justified. The experiment did not compare 2 lessons followed by using training wheels with just using lessons with training wheels.
(c) It would be important to use random assignment for the two groups. For example, if all the older children were put in one of the groups, then age would have been a confounding variable.
3. (a) It is important for the teacher to use random assignment for choosing which students are offered bonus points and which are offered a chocolate chip cookie because a cause-and-effect conclusion is

of interest.

- (b) This cannot be random assignment because you don't want to assign students to go to a game. Instead, you want to know their preferences. However, it would be good to use random sampling to increase the chance of getting a representative sample and being able to generalize to the whole population.
 - (c) The student would probably want to interview all the administrators and teachers at the school. So neither sampling method is appropriate.
4. (a) The explanatory variable is the correct or improper placing of the marble. This is a categorical variable.
- (b) The response variable is self-reported nausea level. This is a numerical variable as it is on a 1 to 10 scale.
- (c) This is an experiment with two treatments: correct and improper placement of a marble to a pressure point.
- (d) Yes, blinding should be incorporated because of a possible placebo effect. If patients knew they were receiving the correct marble placement, they might psychologically feel less nauseous even if the marble was ineffective. Similarly, if patients knew they were receiving the improper marble placement, they might psychologically feel more nausea than expected.
- (e) Random assignment should be incorporated. Each patient is a block and so should receive both treatments, one each of the two days, randomly chosen. For example, number the patients 1 to 50. Use a random number generator to pick integers between 1 and 50, throwing out repeats, until 25 unique integers are picked. The patients with numbers corresponding to the 25 picked integers should receive correct marble placement the first day, while the remaining 25 patients should receive correct marble placement the second day.
- (f) For example, nausea levels might naturally be less on the second day. If all the patients received the correct pressure point placement on the second day and seemed to feel relief, it would be unclear whether the marble or the day was the cause of the relief. Marble and day would be confounded. This variable is controlled by the random assignment of treatments to days.
- (g) This is a well-designed experiment incorporating comparison, replication, random assignment, and control. So if patients rate nausea relief differently depending upon correct or improper placement of the marble, a cause-and-effect conclusion is justified.

Quiz 9 (Pages 136–140)

Multiple-Choice

1. (D) Of the 500 people surveyed, 250 were students, and $\frac{250}{500} = 0.5$, or 50%.
2. (A) Of the 500 people surveyed, 125 picked *challenging* as most important and were *also* teachers: $\frac{125}{500} = 0.25$, or 25%.
3. (D) There were 50 administrators, and 25 of them picked *strict* as most important: $\frac{25}{50} = 0.5$ or 50%.
4. (D) There were 210 people picking *enthusiastic* as most important, and 150 of them were students: $\frac{150}{210} = 0.714$, or 71.4%.
5. (C) The percentages of students, teachers, and administrators picking *strict* as most important were 20%, 12.5%, and 50%, respectively.
6. (B) Based on lengths of indicated segments, the percentage from the West who prefer country is the greatest.
7. (D) The given bar chart shows percentages, not actual numbers.

8. (D) In mosaic plots, the area of a box is proportional to the count corresponding to that box. Choice (D) is true because the three boxes corresponding to students with GPAs under 3.0 have a total area greater than the three boxes corresponding to students with GPAs 3.0 or higher, or note that along the vertical axis, “GPA under 3.0” has a greater length than “GPA 3.0 or higher.”
9. (B) If there was no association, the conditional relative frequency for the receiver moving toward the corner should be about the same if the server aims for the corner or aims down the line. However, the conditional relative frequencies of the receiver moving toward the corner are $\frac{300}{400} = 0.75$ if the server aims for the corner and $\frac{50}{100} = 0.5$ if the server aims for down the line. So there does appear to be an association.
10. (A) If there was no association, the conditional relative frequency of preferring dogs should be about the same for middle school students and for high school students. The conditional relative frequency of preferring dogs for middle school students is $\frac{0.20}{0.20 + 0.10 + 0.05} = 0.571$, while the conditional relative frequency of preferring dogs for high school students is $\frac{0.30}{0.30 + 0.25 + 0.10} = 0.462$. So there does appear to be an association.

Free-Response Questions

1. (a) Of the speeding pullovers who were given a ticket, 27 out of 93, or $\frac{27}{93}$, were Latino.
- (b) Of the speeding pullovers who were White, 84 out of 120, or $\frac{84}{120} = 0.70 = 70\%$, were given a warning.
- (c) Based upon the mosaic plot, there is an association between race/ethnicity and whether or not a ticket or warning is given to drivers pulled over for speeding at night. Whites are more likely to receive a warning, Blacks are equally as likely to receive a warning or a ticket, while Latinos are more likely to receive a ticket. Latinos were twice as likely to receive a ticket as Whites.

2. (a)

Program	Percentage of Men Accepted (%)	Percentage of Women Accepted (%)
A	62	82
E	28	24

Women seem to be favored in program A, while men seem to be slightly favored in program E.

- (b) Overall, 564 out of 1,016 male applicants were accepted, for a 55.5% acceptance rate, while 184 out of 501 female applicants were accepted, for a 36.7% acceptance rate. This appears to contradict the results from part (a).
- (c) You should tell the reporter that although it is true that the overall acceptance rate in these two programs for women is 36.7% compared with the 55.5% acceptance rate for men, in one program women have a much higher acceptance rate than men, while in the other program women have only a slightly lower acceptance rate than men. The reason behind this apparent paradox is that between these two programs, most men applied to program A, which had a high acceptance rate. However, most women applied to program E, which had a low acceptance rate.

3. (a)

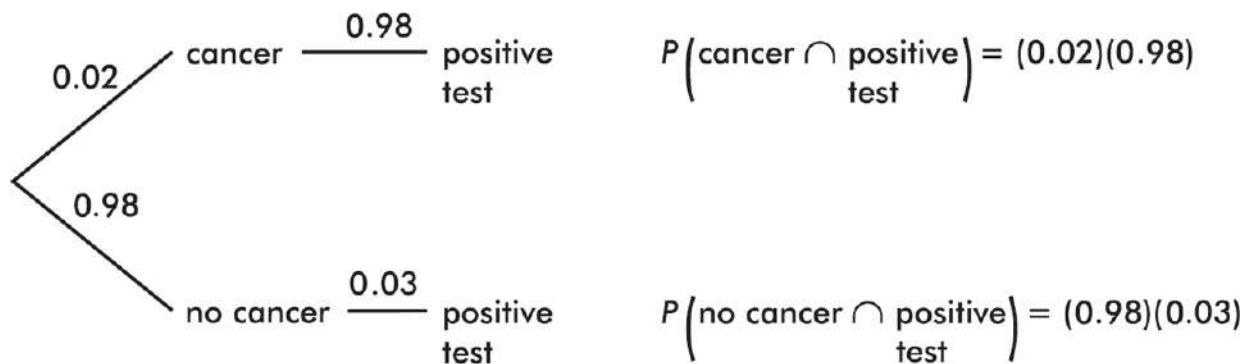
	Review book A	Review book B
Scored 3 or higher	0.638	0.522
Scored below 3	0.362	0.478

- (b) Book A appears to lead to better results with 63.8% of purchasers scoring 3 or higher, while only 52.2% of book B purchasers scored 3 or higher.
- (c) No, this was an observational study, so cause and effect cannot be concluded.
- (d) Among stronger students, $\frac{55}{60} = 91.7\%$ of book B purchasers scored 3 or higher, while only $\frac{400}{530} = 75.5\%$ of book A purchasers scored 3 or higher. Among weaker students, $\frac{125}{285} = 43.9\%$ of book B purchasers scored 3 or higher, while only $\frac{50}{175} = 28.6\%$ of book A purchasers scored 3 or higher. Both stronger and weaker students who purchased book B scored higher than those who purchased book A.
- (e) A high percentage of book A purchasers were stronger students, leading to overall higher scores for book A purchasers. A high percentage of book B purchasers were weaker students, leading to overall lower scores for book B purchasers. However, when looking separately at stronger and weaker student groups, book B purchasers scored higher.

Quiz 10 (Pages 147–150)

Multiple-Choice

1. (A) 35 out of the entire population of 500 both have strep throat and tested positive.
2. (A) Of the 460 healthy people, 25 tested positive.
3. (C) Of the 40 people with strep throat, 35 tested positive.
4. (B) Of the 460 healthy people, 435 tested negative.
5. (A) The probability of the next child being a girl is independent of the sex of the previous children. Before she had any children, if the question had been about the probability of having eight girls in a row, then the answer would have been $(0.5)^8$, or about 1 in 256.
6. (A) The probability of a boy is given to be about 0.5. By the law of large numbers, the more births (Hospital B has more births per day), the more the relative frequency tends to become closer to this probability. With fewer births (Hospital A has fewer births per day), there is a greater chance of wide swings in the relative frequency.
7. (C) $P(\text{exactly one}) = P(\text{math: yes}) P(\text{physics: no}) + P(\text{math: no}) P(\text{physics: yes}) = (0.55)(0.7) + (0.3)(0.45)$. Note that the events “math and not physics” and “physics and not math” are mutually exclusive.
8. (D)



$$P(\text{positive test}) = P(\text{cancer} \cap \text{positive test}) + P(\text{no cancer} \cap \text{positive test})$$

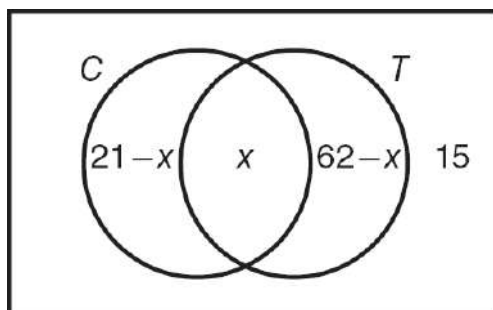
$$= (0.02)(0.98) + (0.98)(0.03)$$

$$P(\text{cancer} | \text{positive test}) = \frac{P(\text{cancer} \cap \text{positive test})}{P(\text{positive test})}$$

$$= \frac{(0.02)(0.98)}{(0.02)(0.98) + (0.98)(0.03)}$$

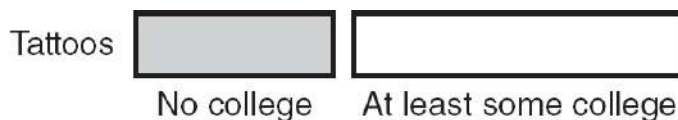
[Follow for extended description](#)

9. (A) We have $(21 - x) + x + (62 - x) + 15 = 90$, which gives $x = 8$. So there are 8 students who have at least two analog clocks and have at least three television sets in their homes. Because the events can occur at the same time, they are not mutually exclusive.



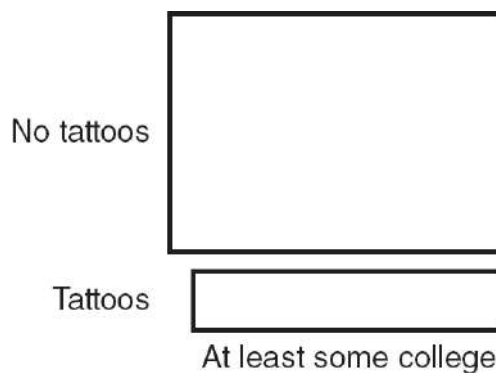
[Follow for extended description](#)

10. (A) Given that a person has a tattoo narrows our interest to



[Follow for extended description](#)

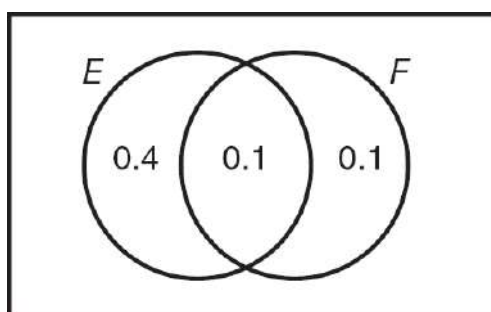
and people with at least some college are clearly over half of this group. Given that a person has taken a college course narrows our interest to



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and people with tattoos are clearly less than half of this group.

11. (D) Events E and F are independent if $P(E \cap F) = P(E)P(F)$. Here, $P(E) = 0.4 + 0.1 = 0.5$ and $P(F) = 0.1 + 0.1 = 0.2$. Then $P(E)P(F) = (0.5)(0.2) = 0.1 = P(E \cap F)$.



[Follow for extended description](#)

12. (D) $P(I) = P(\text{breakfast} \mid \text{senior}) = \frac{P(\text{breakfast} \cap \text{senior})}{P(\text{senior})}$
 $P(II) = P(\text{breakfast} \cap \text{senior})$, and
 $P(III) = P(\text{senior} \mid \text{breakfast}) = \frac{P(\text{breakfast} \cap \text{senior})}{P(\text{breakfast})}$

$P(I) > P(III)$ because the denominator in (I) is smaller than the denominator in (III); that is, it is reasonable to assume that the set of all teenagers who are high school seniors is smaller than the set of all teenagers who eat breakfast.

$P(II)$ is the smallest because (I) and (III) have denominators that are less than one.

Free-Response Questions

1. It's easiest to first sum the rows and columns:

		Years of experience			
		0–5	6–10	>10	
Less than 50 years old		80	125	20	225
More than 50 years old		10	75	50	135
		90	200	70	360

[Follow for extended description](#)

- (a) $P(\text{age} < 50) = \frac{225}{360} = 0.625$
 $P(\text{experience} > 10) = \frac{70}{360} = 0.194$
 $P(\text{age} > 50 \cap \text{experience } 0-5) = \frac{10}{360} = 0.028$
- (b) $P(\text{age} < 50 \mid \text{experience } 6-10) = \frac{125}{200} = 0.625$
- (c) $P(\text{age} < 50) = \frac{225}{360} = 0.625$; however, $P(\text{age} < 50 \mid \text{experience} > 10) = \frac{20}{70} = 0.286$ and so they are not independent. $P(\text{age} > 50) = \frac{135}{360} = 0.375$ and also $P(\text{age} > 50 \mid \text{experience } 6-10) = \frac{75}{200} = 0.375$, and so these two are independent.
2. (a) She is assuming that acceptances to the two colleges are independent. This does not seem reasonable as most colleges use similar acceptance criteria.
- (b) She is assuming that acceptances to the two colleges are mutually exclusive. This does not seem reasonable as students can simultaneously be accepted at both of the colleges.
3. (a) $P((\text{smartphone} \geq 3 \text{ yrs}) \cap \text{Gen } X) = (0.26)(0.09) = 0.0234$
- (b) $P(\text{smartphone} \geq 3 \text{ yrs}) = P(\text{Gen } X \cap (\text{smartphone} \geq 3 \text{ yrs})) + P(\text{Gen } Y \cap (\text{smartphone} \geq 3 \text{ yrs})) + P(\text{Gen } Z \cap (\text{smartphone} \geq 3 \text{ yrs})) = 0.0234 + (0.38)(0.12) + (0.36)(0.15) = 0.123$
- (c) $P(\text{Gen } X \mid (\text{smartphone} \geq 3 \text{ yrs})) = \frac{P(\text{Gen } X \cap (\text{smartphone} \geq 3 \text{ yrs}))}{P(\text{smartphone} \geq 3 \text{ yrs})} = \frac{0.0234}{0.123} = 0.190$

Quiz 11 (Pages 154–155)

Multiple-Choice

- (D) The expected value or mean is a long-term average. It is not a median. This value may not be the most likely value, and in this example, it can never occur. For any given sample, anything might result.
- (C) The standard deviation gives the average or typical amount of variability from the mean for the population.
- (A) The expected value or mean of a discrete random variable is given by $\sum xP(x)$. From the table, $\sum xP(x) = 0(0.16) + 1(0.24) + 2(0.37) + 3(0.15) + 4(0.08)$.
- (C) $1,000(0.2) + 5,000(0.05) = 450$, and $800 - 450 = 350$
- (B) Your expected winnings are only

$$\frac{1}{7,500,000}(1,000,000) + \frac{5}{7,500,000}(10,000) + \frac{20}{7,500,000}(100) = 0.14$$

Free-Response Questions

1. $\sum xP(x) = 10(0.50) + 20(0.25) + 30(0.15) + 50(0.05) + 100(0.05) = 22$. This player will average 22 points per game in the long run. (Note that 22 points can never be the result of a single game.)
2. (a) The probability of 1 or 6 on the first roll is $\frac{2}{6} = \frac{1}{3}$. The probability of 2, 3, 4, or 5 followed by a heads is $\frac{4}{6} \cdot \frac{1}{2} = \frac{1}{3}$. Similarly, the probability of 2, 3, 4, or 5 followed by a tails is $4 \cdot \frac{4}{6} \cdot \frac{1}{2} = \frac{1}{3}$.

	1 or 6 on first roll	2, 3, 4, or 5 and heads	2, 3, 4, or 5 and tails
X (money won)	\$5	\$1	\$2
Probability	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

(b) $E(X) = \sum xP(x) = 5\left(\frac{1}{3}\right) + 1\left(\frac{1}{3}\right) + 2\left(\frac{1}{3}\right) = \frac{8}{3} \approx \2.67

$$SD(X) = \sqrt{\left(5 - \frac{8}{3}\right)^2 \left(\frac{1}{3}\right) + \left(1 - \frac{8}{3}\right)^2 \left(\frac{1}{3}\right) + \left(2 - \frac{8}{3}\right)^2 \left(\frac{1}{3}\right)} = \sqrt{\frac{26}{9}} \approx \$1.70$$

In the long run, a player will average winning about \$2.67 per game. The player winnings for a randomly selected game will generally vary about \$1.70 from the mean of \$2.67.

- (c) To be a fair game, you should be willing to pay about \$2.67.
3. (a) If the ball lands on 13, the player is ahead $\$175 - \$5 = \$170$. If the ball lands on anything other than 13, the player loses \$5.

x	\$170	-\$5
$P(x)$	$\frac{1}{38}$	$\frac{37}{38}$

(b) $E(X) = \sum xP(x) = 170\left(\frac{1}{38}\right) + (-5)\left(\frac{37}{38}\right) = -\frac{15}{38} \approx -\0.39

$$SD(X) = \sqrt{\left(170 - \frac{-15}{38}\right)^2 \left(\frac{1}{38}\right) + \left(-5 - \frac{-15}{38}\right)^2 \left(\frac{37}{38}\right)} \approx \$28.01$$

In the long run, you will average losing about \$0.39 per game. Your winnings for a randomly selected game will generally vary about \$28.01 from the mean of -\$0.39.

- (c) To be over \$100 ahead, you most likely played one game and won, rather than losing some number of games before winning. $P(\text{winning first game}) = \frac{1}{38}$, and losing some number of games before winning would multiply this by $\frac{37}{38}$ some number of times.

Quiz 12 (Pages 160–163)

Multiple-Choice

- (A) This is a binomial with $n = 3$ and $p = 0.90$. Then $P(X \geq 2) = (0.9)^3 + 3(0.9)^2(0.1)$.
- (C) This is a binomial with $n = 10$ and $p = 0.37$, and so the mean is $np = 10(0.37) = 3.7$.
- (B) The sample has five adults, so $n = 5$, and the probability any of these adults experienced cyberbullying is given to be $\frac{1}{6}$.
- (C) This is a binomial distribution with $n = 5$ and $p = \frac{2}{5} = 0.4$. Then, $P(X = 2) = \binom{5}{2} (0.4)^2 (0.6)^3$.
- (B) This is a geometric distribution with $p = 0.4$. Then, $P(X = 3) = (0.6)^2 (0.4)$.
- (D) $P(I) = P(II) = (0.5)^5$, while $P(III) = {}_5C_2 (0.5)^3 (0.5)^2 = 10(0.5)^5$.

7. (A) The mean is 5, so $np = 5$. The standard deviation is 2 so $\sqrt{np(1-p)} = \sqrt{5(1-p)} = 2$. Solving for p results in $5(1-p) = 4$, $1-p = 0.8$, and $p = 0.2$.
8. (B) Random variable F follows a binomial distribution with $n = 10$ and $p = 0.7$. Distribution F has mean $\mu_F = np = (10)(0.7) = 7$ and standard deviation $\sigma_F = \sqrt{np(1-p)} = \sqrt{10(0.7)(0.3)} = 1.45$. Answer choice (C) is the most reasonable simulation result.
9. (C) This is a binomial distribution, so the mean is $\mu = np = (50)(0.34) = 17$. The mean of a probability distribution is the long-run average of the values selected from many repetitions. Choices (A) and (B) refer to particular samples, not a long-run average. Choice (B) refers to a geometric probability; however, this is a binomial. Choice (D) is the interpretation of the standard deviation, not the mean.

Free-Response Questions

1. (a) We have a binomial with $n = 20$ and $p = 0.15$, so $P(X \geq 10) = 1 - \text{binomcdf}(20, 0.15, 9) = 0.00025$.
[Or we can calculate $P(X \geq 10) = \binom{20}{10}(0.15)^{10}(0.85)^{10} + \binom{20}{11}(0.15)^{11}(0.85)^9 + \dots + (0.15) = 0.00025$.]
- (b) If the probability of *Legionella* bacteria growing in an automatic faucet is 0.15, the probability of a result as extreme as or more extreme than what was obtained in the Johns Hopkins study is only 0.00025. With such a low probability, there is strong evidence to conclude that the probability of *Legionella* bacteria growing in automatic faucets is greater than 0.15. That is, there is strong evidence that automatic faucets actually house more bacteria than the manual kind.
2. (a) $\frac{0.02}{0.11 + 0.02 + 0.014 + 0.006} = \frac{0.02}{0.15} = \frac{2}{15} = 0.133$
- (b) $P(\text{at least 2 tattoos}) = 0.02 + 0.014 + 0.006 = 0.04$. The student with tattoos can be the first, second, or third student, so we have $3(0.04)(0.85)^2 = 0.0867$.
- (c) $E(X) = 0(0.85) + 1(0.11) + 2(0.02) + 3(0.0014) + 4(0.006) = 0.216$. $\sigma = \sqrt{\sum(x - 0.216)^2 P(x)} = \sqrt{0.365344} = 0.604$. The mean number of tattoos per college student is 0.216, and for any randomly selected college student, the number of tattoos will vary by about 0.604 from the mean of 0.216 tattoos.
3. (a) This is a binomial distribution with $n = 4$ and $p = 0.35$. Then $P(X = 2) = \text{binomialcdf}(4, 0.35, 1) = 0.3105$. [Or $P(X = 2) = \binom{4}{2}(0.35)^2(0.65)^2 = 0.3105$.]
- (b) Binomial, $n = 4$, $p = 0.35$, and $P(X \geq 2) = 1 - P(X \leq 1) = 1 - \text{binomialcdf}(4, 0.35, 1) = 0.4370$. [Or $1 - [(0.65)^4 + 4(0.35)(0.65)^3] = 0.4370$.]
- (c) $(0.65)(0.35) = 0.2275$.

Quiz 13 (Pages 167–168)

Multiple-Choice

1. (B) Curve a appears to have a mean of 6 and a standard deviation of 2, while curve b appears to have a mean of 21 and also a standard deviation of 2. (It is important to look at the scale of each graph, not just the visual width.)
2. (C) Point E appears to be one standard deviation above the mean. $74.3 + 9.7 = 84.0$. Note that point E (and point C) are points where the slope is steepest.
3. (C) Points B and F are two standard deviations below and above the mean, respectively. By the empirical rule, 95% of the data are in this interval.
4. (A) Point C is one standard deviation below the mean. By the empirical rule, 68% of the data is between one standard deviation below and above the mean. This leaves 32% outside this interval and 16% in each tail.
5. (D) With a normal distribution, 16% in the tail to the right corresponds to one standard deviation above the mean. Then $\mu + 1(2.5\%) = 25\%$ gives $\mu = 22.5\%$.

Free-Response Questions

1. (a) $z = \frac{720 - 600}{120} = 1$, and for the normal model, the proportion of z -scores below 1 is $0.5 + 0.34 = 0.84$, or the 84th percentile.
(b) Both \$360 and \$840 are \$240 from the mean of \$600, which makes \$360 two standard deviations below the mean and \$840 two standard deviations above. For the normal model, the proportion of z -scores between -2 and $+2$ is approximately 0.95 or 95%.
(c) Under a normal curve, 99.7% of the values are within ± 3 standard deviations of the mean: $600 - 3(120) = 240$ and $600 + 3(120) = 960$. So, 99.7% of American teens spend between \$240 and \$960 on food per year.
2. No. In a normal distribution the mean and median should be equal or nearly equal, but here the mean, 6.4, is significantly greater than the median, 5. Furthermore, in a normal distribution the distance from the minimum to the median should be about the same as the distance from the median to the maximum. In this data set, however, the distance median $-$ min $= 5 - 0 = 5$ is different from the distance max $-$ median $= 15 - 5 = 10$. Both these facts suggest the distribution is skewed right.

Quiz 14 (Pages 176–178)

Multiple-Choice

1. (A) The area under any probability distribution is equal to 1. Many bell-shaped curves are *not* normal curves. The smaller the standard deviation of a normal curve, the higher and narrower the graph is. The mean determines the value around which the curve is centered; different means give different centers. Because of symmetry, the mean and median are identical for normal distributions.
2. (A) Probability for a normal model is calculated by the area under the normal curve within a given interval of X values. Since the point $X = 22$ does not represent an interval, $P(X = 22) = 0$.
3. (D) All normal distributions have the same percentage (about 95%) of their observations within two standard deviations of the mean.
4. (A) $P(x > 30) = P\left(z > \frac{30 - 20}{5}\right) = P(z > 2)$
5. (D) The z -scores of 3 and 4 are $\frac{3 - 3.4}{0.5}$ and $\frac{4 - 3.4}{0.5}$, respectively. Then $P(3 < X < 4) = P\left(\frac{3 - 3.4}{0.5} < z < \frac{4 - 3.4}{0.5}\right)$.
6. (C) The critical z -score associated with 99% to the left is $\text{invNorm}(0.99) = 2.326$.
7. (B) The critical z -scores associated with the middle 95% are $\pm \text{invNorm}(0.975) = \pm 1.96$. Converting to raw scores gives $9,500 \pm 1.96(1,750)$.
8. (B) The critical z -score is $\text{invNorm}(0.85) = 1.036$. Then $\mu + 1.036(2) = 16$ gives $\mu = 16 - 1.036(2)$.
9. (C) Based on the table, 68% of the values are within one standard deviation of the mean, 95% are within two standard deviations of the mean, and 100% are within three standard deviations of the mean. These percentages are very close to the 68%-95%-99.7% of the empirical rule.

Free-Response Questions

1. (a) $P\left(z > \frac{x - 150}{25}\right) = 0.90$, so $\frac{x - 150}{25} = -1.282$ and $x = 117.95$ for a cutoff score of 118.
(b) $P(x > 118 | x > 100) = \frac{P(x > 118)}{P(x > 100)} = \frac{0.900}{0.9772} = 0.921$
(c) $1 - (0.921)^3 = 0.219$
2. (a) $P(X \geq 8.5) = 0.11 + 0.07 + 0.02 + 0.01 + 0.01 = 0.22$

(b) $z = \frac{8.5 - 8.03}{0.49} = 0.9592$ and $P(X \geq 8.5) = \text{normalcdf}(0.9592, \infty) = 0.1687$

- (c) The answers to (a) and (b) are different because the discrete model calculation sums the areas of the bars with a left boundary of 8.375 (8.5 is found between 8.375 and 8.625), whereas the continuous model calculation has a left boundary at exactly 8.5. Therefore, the lesser probability is found by using the continuous model from part (b).
- (d) Use the discrete model. The discrete model shows a bar centered directly over $X = 7.25$ and thus gives a probability of 0.06 for $X = 7.25$. However, the continuous model gives a probability of zero since there is no area above the single point $X = 7.25$.
- (e) Use the continuous model. The continuous model gives a probability of $X \geq 7.1$, which corresponds to the area under the curve to the right of 7.1. However, the relative frequency of 7.1 is not clear when using the discrete model because 7.1 does not correspond to any of the bars.

Quiz 15 (Pages 183–187)

Multiple-Choice

1. (D) Distributions describe variability, and variability is the most fundamental concept in statistics. A sampling distribution describes variability among samples. The sampling distribution of a sample mean is the distribution of the statistic $\bar{x}$ for all possible samples from the population.
2. (B) By the central limit theorem, the larger the sample size n , the closer the sampling distribution of $\bar{x}$ will be to a normal distribution.
3. (B) The mean of any sampling distribution of the sample mean will be the population mean. A larger sample size will give less variability in the sampling distribution than will a smaller sample size.
4. (C) With $p = 0.1$ and $n = 10$, the distribution should be centered around $np = 10(0.1) = 1$. With 30 trials, there should be 30 dots in the plot.
5. (A) The larger the sample size n , the tighter the clustering of the sample means. (The sampling distribution of $\bar{x}$ has a standard deviation of $\frac{\sigma}{\sqrt{n}}$. With larger n , the standard deviation $\frac{\sigma}{\sqrt{n}}$ will be smaller.)
6. (C) The larger the population standard deviation σ , the more spread out is the sampling distribution of $\bar{x}$. (The sampling distribution of $\bar{x}$ has a standard deviation of $\frac{\sigma}{\sqrt{n}}$. With larger σ , the standard deviation $\frac{\sigma}{\sqrt{n}}$ will be larger.)
7. (D) The sampling distribution of sample means will be approximately normal if the original population is approximately normal or if the sample size is large enough. Neither of those conditions appears to be true.
8. (C) The distribution of sample means will be normal if the population is normal and will be approximately normal if the sample size is large enough ($n \geq 30$). In this case, we are given that $n = 50 \geq 30$.
9. (D) With samples of size 2, we expect about 25% of the samples will include 2 high school runners, 25% of the samples will include 2 middle school runners, and 50% of the samples will include 1 runner at each grade level.

Free-Response Questions

1. (a) Because $n = 5 < 30$, the sampling distribution of the sample mean will also be skewed left but not quite as strongly as the population distribution.
 (b) Because $n = 50 > 30$, the sampling distribution of the sample mean will be approximately normal by the central limit theorem.
2. (a) When the mean is substantially greater than the median, the distribution is probably skewed right.

- (b) The distribution of a large sample should be approximately the same as that of the population, so the distribution of salaries from a random sample of 50 players should also probably be skewed right.
- (c) By the central limit theorem, with large n , the sampling distribution of $\bar{x}$ will be approximately normal.
- (d) With a larger sample size, the sampling distribution of $\bar{x}$ will be even closer to a normal distribution and will have a tighter clustering (a smaller standard deviation).
- (e) With a larger number of samples, the approximate normality of the sampling distribution of $\bar{x}$ should be clearer, but the standard deviation of the sampling distribution of $\bar{x}$ will not change as it depends upon the sample size, not the number of samples.

Quiz 16 (Pages 195–197)

Multiple-Choice

1. (B) Proportions come from binomial distributions, and a binomial distribution will be approximately normal if n is large enough. We check that np and $n(1-p)$ are both ≥ 10 . A sample size of at least 30 is a condition to be checked with regard to the sampling distribution of a sample mean. np and $\sqrt{np(1-p)}$ refer to the mean and standard deviation of a binomial distribution.
2. (C) A biased estimator is one for which its mean is not equal to the value of the parameter being estimated.
3. (C) The 10% condition says that the sample size should be less than 10% of the population size. Here, the sample size of 75 is greater than 10% of the population size of 560.
4. (C) The sampling distribution of $\hat{p}$ is approximately normal with mean $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.07)(0.93)}{400}}$. The z -scores of 0.05 and 0.10 are $\frac{0.05 - 0.07}{\sigma_{\hat{p}}}$ and $\frac{0.10 - 0.07}{\sigma_{\hat{p}}}$, respectively. The probability of a sample proportion between 0.05 and 0.10 is $P\left(\frac{0.05 - 0.07}{\sqrt{\frac{(0.07)(0.93)}{400}}} < z < \frac{0.10 - 0.07}{\sqrt{\frac{(0.07)(0.93)}{400}}}\right)$.
5. (C) The variance for the sampling distribution of $\hat{p}$ equals $\frac{p(1-p)}{n}$. A larger n in the denominator results in a smaller quotient, and $(0.1)(0.9) < (0.5)(0.5)$.
6. (B) The center of a sampling distribution for a population proportion equals the center of the distribution being sampled, so both distributions should be roughly centered at 0.14. The larger the sample size, the lesser the variability in the sampling distribution, so distribution B with a sample size of 200 will have less variability than distribution A with a sample size of 100.

Free-Response Questions

1. (a) $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.16)(0.84)}{80}} = 0.041$
 - (b) Yes, the conditions for inference are met. First, it is given that a random sample is planned. Second, the sample size, $n = 80$, is clearly less than 10% of all game players. Third, $np = (80)(0.16) = 12.8 \geq 10$ and $nq = (80)(0.84) = 67.2 \geq 10$.
 - (c) The sample size would be larger than 80, because $0.037 < 0.041$. In order for $\sigma_{\hat{p}}$ to be smaller, the denominator in $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ must be larger.
2. (a) 0.58
 - (b) This is a binomial with $n = 3$ and $p = 0.58$.

$$P(X \geq 3) = \binom{5}{3} (0.58)^3 (0.42)^2 + \binom{5}{4} (0.58)^4 (0.42) + (0.58)^5 = 0.647$$

or $1 - \text{binomcdf}(5, .58, 2) = 0.647$.

- (c) The sample is given to be random, both $np = (350)(0.58) = 203 \geq 10$ and $n(1 - p) = (350)(0.42) = 147 \geq 10$, and our sample is clearly less than 10% of all Americans. So, the sampling distribution of $\hat{p}$ is approximately normal with mean 0.58 and standard deviation $\sigma_{\hat{p}} = \sqrt{\frac{(0.58)(0.42)}{350}} = 0.0264$. With a z-score of $\frac{0.5 - 0.58}{0.0264} = -3.030$, the probability that the sample proportion is greater than 0.5 is $\text{normalcdf}(-3.030, 1000) = 0.9988$. [Or $\text{normalcdf}(.5, 1, .58, .0264) = 0.9988$.]

Quiz 17 (Pages 204–206)

Multiple-Choice

- (C) The relevant condition is that both $n\hat{p}$ and $n\hat{q} \geq 10$, and only choice (C) with $n\hat{p} = (75)(0.15) = 11.25$ and $n\hat{q} = (75)(0.85) = 63.75$ satisfies this.
- (C) The critical z-scores with 0.01 in the tails are $\pm \text{invNorm}(0.99) = \pm 2.326$ and $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{(2/3)(1/3)}{60}}$. The confidence interval is $\hat{p} \pm z^* SE(\hat{p}) = \frac{2}{3} \pm 2.326 \sqrt{\frac{(2/3)(1/3)}{60}}$.
- (D) The sample size of 30 is greater than ten percent of the population; that is, $30 > (0.10)(250) = 25$. This violates the 10% condition. Note that $n\hat{p} = 14 \geq 10$, and $n(1 - \hat{p}) = 16 \geq 10$, which ensures that the sampling distribution of $\hat{p}$ is approximately normal.
- (D) A confidence level of 95 percent gives the percent of all possible confidence intervals of the given size that will capture the population parameter. Choice (A) is a correct interpretation of the confidence interval.
- (D) The margin of error is $\pm z^* SE(\hat{p})$, where $SE(\hat{p}) = \sqrt{\frac{(0.29)(0.71)}{1470}} = 0.011835$. Then $z^*(0.011835) = 0.03$, $z^* = 2.535$, and $\text{normalcdf}(-2.535, 2.535) = 0.9888$, or about 99%.
- (B) $1.96 \sqrt{\frac{(0.5)(0.5)}{n}} \leq 0.025$ gives $\sqrt{n} \geq \frac{1.96(0.5)}{0.025} = 39.2$ and $n \geq 1,537$. Note that the population size is not relevant here.
- (B) The interval is a statement about how confident we are that the interval has captured the population parameter. The confidence interval is about the population, not the sample. The probability the true percentage is between 40% and 46% is either 1 or 0 depending upon whether it is or is not in the interval. Choice (D) is a correct interpretation of the confidence level, not the confidence interval.

Free-Response Questions

- (a) The figure 93% comes from a sample, so it is a statistic.
 - (b) The margin of error is proportional to $\frac{1}{\sqrt{n}}$. With such a very large sample size n , the margin of error will be very small.
 - (c) This is a voluntary response survey, and so we should have very little confidence in the conclusion.
- (a) *Parameter:* Let p represent the proportion of the population of American women between the ages of 45 and 64 who would say that breast cancer is the medical condition they fear most.
Procedure: One-sample z-interval for a population proportion.
Checks: We check *independence* and *normality*. For independence, we are given that this is a random sample, and clearly 500 is less than 10% of all American women between the ages of 45 and 64. For normality, $n\hat{p} = 305 \geq 10$ and $n(1 - \hat{p}) = 195 \geq 10$.
Mechanics: Calculator software gives (0.554, 0.666).
[For instructional purposes, we note that $\hat{p} = \frac{305}{500} = 0.61$ and $0.61 \pm 2.576 \sqrt{\frac{(0.61)(0.39)}{500}} = 0.61 \pm 0.056$ or (0.554, 0.666).]

Conclusion in context: We are 99% confident that the true proportion of all American women between the ages of 45 and 64 who would say that breast cancer is the medical condition they most fear is between 0.554 and 0.666.

- (b) If all possible random samples of 500 women between the ages of 45 and 64 were surveyed, we would expect about 99% of the resulting confidence intervals to contain the true proportion of all American women between the ages of 45 and 64 who would say that breast cancer is the medical condition they fear most.
- (c) No, the 3% value is not consistent with the confidence interval because 0.03 is not in the interval (0.554, 0.666).

Quiz 18 (Pages 207–209)

Multiple-Choice

1. (D) Both np and $n(1 - \hat{p})$ have to be ≥ 10 , but $n\hat{p}$, the number of “successes,” is only 8 in this example, so the normality condition is violated. The population size is immaterial because it is more than ten times the sample size.
2. (B) A confidence *level* of 90% gives the percent of all possible confidence intervals of the given size that will capture the population parameter. Choice (A) is a correct interpretation of the confidence *interval*.
3. (C) Because of sampling variability, samples and the resulting sample proportions differ. Thus, using a single $\hat{p}$ to estimate the true population proportion cannot give a one-point answer, but rather encompasses an interval of plausible values. The margin of error helps define this interval of plausible values around $\hat{p}$.
4. (D) There is no guarantee that 13.4 is anywhere near the interval, so none of the statements are true.
5. (A) The margin of error has to do with measuring chance variation but has nothing to do with faulty survey design. Sampling variability (also called sampling error) refers to the natural variation among samples and is quantified by the margin of error. As long as n is large, s is a reasonable estimate of σ ; however, again this is not measured by the margin of error. (As seen in Unit 4, with t -scores there is a correction for using s as an estimate of σ .)
6. (C) The interval is a statement about how confident we are that the interval has captured the population parameter. The confidence interval is about the population, not the sample. The probability the true percentage is between 85% and 89% is either 1 or 0 depending upon whether it is or is not in the interval. Choice (D) is a correct interpretation of the confidence *level*, not the confidence *interval*.
7. (C) $\text{invNorm}(0.95) = 1.645$, $1.645 \left(\frac{0.5}{\sqrt{n}} \right) \leq 0.04$, $\sqrt{n} \geq 20.563$, $n \geq 422.8$, so choose $n = 423$.

Free-Response Questions

1. (a) *Parameter:* Let p represent the proportion of the population of family members who contract H1N1 after an initial family member does in this state.
Procedure: One-sample z -interval for a population proportion.
Checks: We check *independence* and *normality*. For independence, we are given this is a random sample, and we assume that 876 is less than 10% of the number of all family members of H1N1 patients in the state. For normality, $n\hat{p} = 129 \geq 10$ and $n(1 - \hat{p}) = 747 \geq 10$.
Mechanics: Calculator software gives (0.12757, 0.16695).
[For instructional purposes, we note that $\hat{p} = \frac{129}{876} = 0.147$ and $0.147 \pm 1.645 \sqrt{\frac{(0.147)(0.853)}{876}} = 0.147 \pm 0.020$ or (0.127, 0.167).]
Conclusion in context: We are 90% confident that the true proportion of all family members who contract H1N1 after an initial family member does in this state is between 0.127 and 0.167.
- (b) Because $\frac{1}{8} = 0.125$ is not in the interval of plausible values for the population proportion, there is sufficient evidence that the true proportion of all family members who contract H1N1 after an initial family member does in this state is different from the 1 in 8 chance concluded in the published study.

- (c) $0.147 \pm 2.576 \sqrt{\frac{(0.147)(0.853)}{876}} = 0.147 \pm 0.031$ or $(0.116, 0.178)$ which does include 0.125. So, using a 99% confidence interval, there is *not* sufficient evidence that the true proportion of all family members who contract H1N1 after an initial family member does in this state is different from the 1 in 8 chance concluded in the published study.
2. (a) $z^* SE(\hat{p}) = 1.96 \sqrt{\frac{(0.74)(0.26)}{1050}} = 0.0265$. So, the margin of error is $\pm 2.65\%$.
- (b) $0.74 \pm 0.0265 = (0.7135, 0.7265)$. We are 95% confident that between 71.35% and 76.65% of all smokers would like to give up smoking.
- (c) If all possible random samples of 1,050 smokers were surveyed, we would expect about 95% of the resulting confidence intervals to contain the true proportion of all smokers who would like to give up smoking.
- (d) Smoking is becoming more undesirable in society as a whole, so some smokers may untruthfully say they would like to stop.
- (e) To be more confident, we must accept a *wider* interval (the critical z-score would go from 1.96 to 2.576).
- (f) All other things being equal, the greater the sample size, the *smaller* the margin of error is (the standard error is inversely proportional to $\sqrt{n}$).

Quiz 19 (Pages 215–217)

Multiple-Choice

- (A) Both the null hypothesis H_0 and the alternative hypothesis H_a are always about a population parameter, never about a sample statistic.
- (C) If the null hypothesis is true, the p -value represents the probability of obtaining a sample proportion as extreme as or more extreme than the one obtained for the hypothesis test, in the direction of the alternative hypothesis. In this case, the null hypothesis is that 70 percent of teens in the city pass their driving test on the first attempt; given the direction of the alternative hypothesis, “extreme as or more extreme” means “as large as or larger.”
- (C) A Type II error is a mistaken failure to reject a false null hypothesis or, in this case, a failure to realize that a person really does have ESP.
- (A) This is a one-sided z -test, $H_0: p = 0.5$, $H_a: p > 0.5$, with $\sigma_{\hat{p}} = \sqrt{(0.5)(0.5)/565}$ and $\hat{p} = \frac{305}{565} = 0.540$. The p -value equals $P\left(z > \frac{\hat{p} - p_0}{\sigma_{\hat{p}}}\right) = P\left(z > \frac{0.54 - 0.50}{\sqrt{(0.5)(0.5)/565}}\right)$.
- (A) With $H_0: p = \frac{2}{3}$ and $H_a: p > \frac{2}{3}$, 1PropZTest gives $P = 0.0062$. [For instructional purposes, we note that $\hat{p} = \frac{150}{200} = 0.75$ and $z = \frac{0.75 - \frac{2}{3}}{\sqrt{\frac{(\frac{2}{3})(\frac{1}{3})}{200}}} = 2.5$ and $\text{normalcdf}(2.5, 100) = 0.0062$.] With this small of a p -value, $0.0062 < 0.01$, there is *sufficient* evidence to reject the null hypothesis; that is, there is convincing evidence in support of the sociologist’s claim that over two-thirds of adult Whites have entirely White social networks without any minority presence.
- (B) If the null hypothesis is far off from the true parameter value, there is a greater chance of rejecting the false null hypothesis and thus a smaller risk of a Type II error. Power is the probability that a Type II error is not committed, so a lower Type II error results in higher power.
- (D) The one-sample z -test for a population proportion results in a p -value (the probability of observing a value as extreme as or more extreme than $\hat{p}$ given that p_0 is true), which is a measure of the inconsistency between the claimed p_0 and the observed $\hat{p}$. It can be used to make a decision as to whether or not there is

convincing evidence to reject the claimed p_0 . Choice (C) concerns confidence intervals, not hypothesis tests.

Free-Response Questions

1. (a) The parameter of interest is p = the proportion of the population of animal and vegetable protein products that contain measurable amounts of microplastics.
 (b) $H_0: p = 0.90$ and $H_a: p > 0.90$
 (c) $np_0 = (500)(0.90) = 450 \geq 10$ and $n(1 - p_0) = (500)(0.10) = 50 \geq 10$, so the condition that the sampling distribution of $\hat{p}$ is approximately normal is met.
 (d) $SE(\hat{p}) = \sqrt{\frac{p_0(1 - p_0)}{n}} = \sqrt{\frac{(0.90)(0.10)}{500}} = 0.0134$
 (e) $\hat{p} = \frac{462}{500} = 0.924$ and the test statistic $z = \frac{0.924 - 0.90}{0.0134} = 1.79$
2. (a) The p -value of 0.138 gives the probability of observing a sample proportion of GBS complications as great as (or greater than) the proportion found in the study if, in fact, the proportion of GBS complications is 0.000001.
 (b) Since $0.138 > 0.10$, there is not sufficient evidence to reject H_0 ; that is, there is not convincing evidence that under the new vaccine, the true proportion of GBS complications is greater than 0.000001 (one in a million).
 (c) The null hypothesis is not rejected, so there is the possibility of a Type II error—that is, of mistakenly failing to reject a false null hypothesis. A possible consequence is continued use of the vaccine with a higher rate of GBS complications than is acceptable.
3. (a) *Parameter:* Let p represent the proportion of the population of AAUP members who own a Roth IRA.
Hypotheses: $H_0: p = 0.174$ and $H_a: p > 0.174$, where p is the proportion of all AAUP members who own Roth IRAs.
Procedure: A one-sample z -test for a population proportion.
Checks: Random sample (given), $np_0 = 750(0.174) = 130.5 \geq 10$ and $n(1 - p_0) = 750(0.826) = 619.5 \geq 10$ (where p_0 is the claimed 0.174) and the sample size, $n = 750$, is assumed to be less than 10% of all AAUP members.
Mechanics: Calculator software gives $z = 1.878$ and $p = 0.030$.
 [For instructional purposes, we note that:

$$\hat{p} = \frac{150}{750} = 0.2, z = \frac{0.2 - 0.174}{\sqrt{\frac{(0.174)(0.826)}{750}}} = 1.878, \text{ and the } p\text{-value} = P(z > 1.878) = 0.030$$

Conclusion in context with linkage to the p -value: With a p -value this small, $0.030 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is sufficient evidence that more than 17.4% of all AAUP members own Roth IRAs.
- (b) There was convincing evidence to reject the null hypothesis, so a Type I error might have been committed—that is, rejecting a true null hypothesis. If so, the AAUP would incorrectly think that more than 17.4% of their members own Roth IRAs and would not feel the need to recommend the Roth IRAs to their members when perhaps they should.
- (c) Assuming that 17.4% of all AAUP members own a Roth IRA, there is a 0.030 probability of getting a sample proportion of 0.2 or higher by chance alone in a random sample of 750 AAUP members.

Quiz 20 (Pages 224–226)

Multiple-Choice

1. (B) The critical z-scores are $\pm \text{invNorm}(0.995) = \pm 2.576$.

$$n_1 = 300 \quad n_2 = 400$$

$$\hat{p}_1 = 0.65 \quad \hat{p}_2 = 0.48$$

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} = \sqrt{\frac{(0.65)(0.35)}{300} + \frac{(0.48)(0.52)}{400}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* SE(\hat{p}_1 - \hat{p}_2) = (0.65 - 0.48) \pm 2.576 \sqrt{\frac{(0.65)(0.35)}{300} + \frac{(0.48)(0.52)}{400}}$$

2. (D) $\hat{p}_1 = \frac{151}{500} = 0.302$, $\hat{p}_2 = \frac{345}{1,000} = 0.345$, $\hat{p}_c = \frac{151 + 345}{500 + 1,000} = 0.331$. This is a two-sided test ($H_0: p_1 - p_2 = 0$, $H_a: p_1 - p_2 \neq 0$) with $SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = \sqrt{(0.331)(0.669)\left(\frac{1}{500} + \frac{1}{1,000}\right)}$. The p -value equals twice the tail probability (because this is a two-sided test), so the p -value equals $2P\left(z < \frac{\hat{p}_1 - \hat{p}_2}{SE(\hat{p}_1 - \hat{p}_2)}\right) = 2P\left(z < \frac{0.302 - 0.345}{\sqrt{(0.331)(0.669)\left(\frac{1}{500} + \frac{1}{1,000}\right)}}\right)$.

3. (D) The p -value is a conditional probability; in this case, there is a 0.032 probability of an observed difference in sample proportions as extreme as (or more extreme than) the one obtained if the null hypothesis is assumed to be true.
4. (B) A confidence *interval* of 95% means that we are 95% confident that our interval has captured the true population parameter. Choice (C) is a correct interpretation of confidence *level*.
5. (A) The 99% confidence interval does not contain 0. So there is convincing evidence at the 1% significance level that the proportion of teens experiencing cyberbullying is different for the 15–17 and 13–14 age groups. If there is convincing evidence at the 1% level, there will be convincing evidence at the 5% level and, in fact, at any level greater than 1%.

Free-Response Questions

1. (a) *Parameters:* Let p_{US} represent the proportion of the population of new births in the United States that are to unmarried women. Let p_{UK} represent the proportion of the population of new births in the United Kingdom that are to unmarried women.

Procedure: Two-sample z -interval for a difference between population proportions, $p_{US} - p_{UK}$, the difference in population proportions of new births to unmarried women in the United States and the United Kingdom.

Checks: We are given these are random samples, they are clearly independent, and the sample sizes, 500 and 400, are less than 10% of new births in the United States and the United Kingdom, respectively. For normality, $n_{US}\hat{p}_{US} = (500)(0.412) = 206 \geq 10$, $n_{US}(1 - \hat{p}_{US}) = (500)(0.588) = 294 \geq 10$, $n_{UK}\hat{p}_{UK} = (400)(0.465) = 186 \geq 10$, and $n_{UK}(1 - \hat{p}_{UK}) = (400)(0.535) = 214 \geq 10$.

Mechanics: Calculator software (such as 2-PropZInt on the TI-84) gives $(-0.1182, 0.01219)$.

[For instructional purposes, we note that:

$$(0.412 - 0.465) \pm 1.96 \sqrt{\frac{(0.412)(0.588)}{500} + \frac{(0.465)(0.535)}{400}} = -0.053 \pm 0.065 \text{ or } (-0.118, 0.012)$$

Conclusion in context: We are 95% confident that the true difference in proportions, $p_{US} - p_{UK}$, of new births to unmarried women in the United States and the United Kingdom is between -0.118 and 0.012 .

- (b) Because 0 is in the interval of plausible values for the difference of population proportions, this confidence interval does not support the belief by the UN health care statistician that the true proportions of new births to unmarried women are different in the United States and the United Kingdom.

2. (a) The sample was said to be selected randomly. There was no random assignment into ages or into who answered with what response.
- (b) $H_0: p_{35} = p_{65}$ and $H_a: p_{35} \neq p_{65}$, where p_{35} is the proportion of all voters under age 35 who cannot name a single Supreme Court justice, and p_{65} is the proportion of all voters over age 65 who cannot name a single Supreme Court justice.
- (c) The two sample proportions (voters under 35 and over 65 who cannot name a single Supreme Court justice) are 4.43 standard deviations (or standard errors) apart.
- (d) With this large of a z -score, 4.43, the p -value will be very small, so there is sufficient evidence to reject the null hypothesis. That is, there is convincing evidence that the proportion of all voters under 35 years old who cannot name a single Supreme Court justice is different from the proportion of all voters over 65 years old who cannot name a single Supreme Court justice.
- (e) We are 99% confident that the proportion of all voters under age 35 who cannot name a single Supreme Court justice is greater than the proportion of all voters over age 65 who cannot name a single Supreme Court justice by between 0.04612 and 0.17388.
- (f) Since the confidence interval contains only positive values—that is, does not include zero as a plausible value for the difference in population proportions—the two procedures give consistent results.
3. (a) The purpose of random assignment is to create two groups of subjects that are roughly equivalent except for the treatment, hydroxychloroquine or placebo, received. This allows for a cause-and-effect conclusion about hydroxychloroquine and a lowered susceptibility to Covid-19 if the difference between the two groups is statistically significant.
- (b) $H_0: p_{hydro} = p_{placebo}$ and $H_a: p_{hydro} < p_{placebo}$, where p_{hydro} is the true probability of contracting Covid-19 after taking hydroxychloroquine as a postexposure prophylaxis for asymptomatic participants reporting a high-risk exposure to a confirmed Covid-19 contact, and $p_{placebo}$ is the true probability of contracting Covid-19 after taking a placebo for asymptomatic participants reporting a high-risk exposure to a confirmed Covid-19 contact.
- (c) Two-sample z -test for a difference of two population proportions.
- The treatments were randomly assigned.
 - With $\hat{p}_c = \frac{49 + 58}{414 + 407} = 0.130$, we have $n_{hydro} \hat{p}_c = (414)(0.130) = 53.8$, $n_{hydro} (1 - \hat{p}_c) = (414)(0.870) = 360.2$, $n_{placebo} \hat{p}_c = (407)(0.130) = 52.9$, and $n_{hydro} (1 - \hat{p}_c) = (414)(0.870) = 360.2$ are all at least 10.
- (d) With this large of a p -value, $0.152 > 0.05$, there is not sufficient evidence to reject the null hypothesis; that is, there is not convincing evidence that hydroxychloroquine is any better than a placebo for preventing Covid-19 when used as a postexposure prophylaxis for asymptomatic participants reporting a high-risk exposure to a confirmed Covid-19 contact.
- (e) There was not convincing evidence to reject the null hypothesis, so a Type II error might have been committed—that is, failing to reject a false null hypothesis. If so, the researchers would not have recommended using hydroxychloroquine as a postexposure prophylaxis for asymptomatic participants reporting a high-risk exposure to a confirmed Covid-19 contact when, in reality, hydroxychloroquine is effective and should have been recommended.

Quiz 21 (Pages 227–229)

Multiple-Choice

1. (C) A confidence *level* of 90% gives the percent of all possible confidence intervals that will capture the true difference in proportions. Choice (A) is a correct interpretation of confidence *level*. Note that choice

- (B) incorrectly refers to a conclusion about sample statistics rather than the population parameter.
- (A) The 99% confidence interval does not contain 0. Unfortunately, since the interval is entirely in the positive region, this provides convincing evidence that e-cigarette usage among NYC high school students increased between pre-T21 and post-T21 despite the passage of T21.
 - (C) Look for the biggest difference between the proportions in the two groups (choices (A) and (C), compare 10% versus 30%), and when the difference is the same, the larger sample sizes will produce stronger evidence (no choice has larger sample sizes than (C)).
 - (B) With a p -value greater than 0.05, the difference is not statistically significant at the 0.05 significance level, so a 95% confidence interval for the difference in population proportions will include both positive and negative values.
 - (D) The 95% confidence interval for the difference in proportions is (0.0162, 0.0438). The confidence interval for the difference in numbers is found by multiplying by 5,000.

Free-Response Questions

- Parameters:* Let p_B represent the proportion of the population of high school boys who meet the recommended level of physical activity. Let p_G represent the proportion of the population of high school girls who meet the recommended level of physical activity.

Hypotheses: $H_0: p_B - p_G = 0$ (or $p_B = p_G$) and $H_a: p_B - p_G > 0$ (or $p_B > p_G$).

Procedure: Two-sample z -test for a difference of two proportions, $p_B - p_G$.

Checks: We are given independent random samples, and the sample sizes, 850 and 580, are clearly less than 10% of the respective populations. For normality, with

$$\hat{p}_c = \frac{370 + 218}{850 + 580} = \frac{588}{1,430} = 0.411, n_B \hat{p}_c = (850)(0.411) = 349 \geq 10, n_B(1 - \hat{p}_c) = (850)(0.589) = 501 \geq 10, n_G \hat{p}_c = (580)(0.411) = 238 \geq 10, \text{ and } n_G(1 - \hat{p}_c) = (580)(0.589) = 342 \geq 10.$$

Mechanics: Calculator software (such as 2-PropZTest on the TI-84) gives $z = 2.2427$ and $p = 0.012$. [For instructional purposes, we note that

$$\begin{aligned} \hat{p}_B &= \frac{370}{850} = 0.4353, \hat{p}_G = \frac{218}{580} = 0.3759, \hat{p}_c = \frac{370 + 218}{850 + 580} = \frac{588}{1,430} = 0.4112, \\ z &= \frac{0.4353 - 0.3759}{\sqrt{(0.4112)(0.5888)\left(\frac{1}{850} + \frac{1}{580}\right)}} = 2.2427, \text{ and} \\ p &= P(z > 2.2427) = 0.012. \end{aligned}$$

Conclusion in context with linkage to the p -value: With a p -value this small ($0.012 < 0.05$), there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that the proportion of all high school boys who meet the recommended level of physical activity is greater than the proportion of all high school girls who meet the recommended level of physical activity.

- The sample was said to be selected randomly. There was no random assignment into ages or into the responded answers.
 - $H_0: p_{18-25} = p_{65}$ and $H_a: p_{18-25} \neq p_{65}$ where p_{18-25} is the proportion of all voters 18 to 25 years old who believe the false political claim, and p_{65} is the proportion of all voters over age 65 who believe the false political claim.
 - The two sample proportions (voters 18 to 25 years old and voters over 65 who believe the false political claim) are 1.834 standard deviations (or standard errors) apart.
 - This is a two-sided test with a p -value of $2 \times \text{normalcdf}(1.834, 1000) = 2(0.0333) = 0.0666$. With this large of a p -value, $0.0666 > 0.05$, there is not sufficient evidence to reject the null hypothesis. That is, there is not convincing evidence that the proportion of all voters ages 18 to 25 who believe the

false political claim is different from the proportion of all voters over 65 years old who believe the false political claim.

- (e) We are 95% confident that the proportion of all voters ages 18 to 25 who believe the false political claim is between 0.0012 less than and 0.06123 greater than the proportion of all voters over age 65 who believe the false political claim. We are 95% confident that the true difference in the proportions of 18- to 25-year-old voters and voters over 65 who believe the false political claim is between -0.0012 and 0.06123.
 - (f) Since the confidence interval contains both negative and positive values—that is, it includes zero as a plausible value for the difference in population proportions—the two procedures give consistent results.
3. (a) Due to random sampling, the result can be generalized to the population from which the sample was drawn: All adults in Boulder, Colorado.
- (b) $P(\text{exercise addiction}) = \frac{151 + 48}{1,000} = 0.199$
- (c) $P(\text{exercise addiction} \mid \text{negative body image}) = \frac{68 + 26}{395} = \frac{94}{395} = 0.238$
- (d) The confidence interval calculation relies on the fact that the sampling distribution of $p_{neg} - p_{pos}$ is approximately normal. This will be the case if $n_{neg}\hat{p}_{neg}$, $n_{neg}(1 - \hat{p}_{neg})$, $n_{pos}\hat{p}_{pos}$, and $n_{pos}(1 - \hat{p}_{pos})$ are all at least 10.
- (e) We are 95% confident that the true difference between the proportion of people with severe exercise addiction among those in the population who have negative body image and the proportion of people with severe exercise addiction among those in the population who do not have negative body image is in the interval from 0.000815 to 0.0581. That is, we are 95% confident that the proportion of people with severe exercise addiction is between 0.0815 percentage points higher and 5.81 percentage points higher for those who have negative body image than those who do not have negative body image.
- (f) An example of a confounding variable could be a certain chemical imbalance in adults. This imbalance would cause both negativity in general (including negative thoughts about one's body) and a craving for the euphoria of endorphins (which are released during exercise).

Quiz 22 (Pages 235–238)

Multiple-Choice

1. (B) Picking separate samples from each of the schools and classifying according to one variable (perception of quality education) is a survey design that is most appropriately analyzed using a chi-square test of homogeneity of proportions.
2. (D) Tests for independence involve a single sample cross-classified on two variables, while tests for homogeneity involve samples from each of two or more populations to compare the distribution of some variable.
3. (C) With $df = (3 - 1)(5 - 1) = 8$, $P(\chi^2 > 13.95) = \chi^2_{cdf}(13.95, 1000, 8) = 0.083$. Since $0.05 < 0.083 < 0.10$, there is sufficient evidence at the 10% significance level, but not at the 5% significance level, of a relationship between education level and sports interest.
4. (D) With $df = 3$, the p -value is $\chi^2_{cdf}(5.998, 1000, 3) = 0.1117$. With a p -value this large, $0.1117 > 0.10$, there is not sufficient evidence of a difference in cafeteria food satisfaction among the class levels.
5. (A) The degrees of freedom are $df = (\text{rows} - 1)(\text{columns} - 1) = (3 - 1)(4 - 1) = 6$, and the p -value is $\chi^2_{cdf}(12.7, 1000, 6) = 0.048$. With this small of a p -value, $0.048 < 0.05$, there is sufficient evidence to reject the null hypothesis. When the null hypothesis is rejected, there is the possibility of a Type I error—that is, the possibility of mistakenly rejecting a true null hypothesis.

Free-Response Questions

1. (a) The explanatory variable is the amount of lighting in the child's room before age two. The response variable is the child's eyesight diagnosis.
- (b) This is an observational study. Researchers did not assign the children to the amount of light in their rooms; they merely recorded this information.
- (c)
$$P(\text{farsighted} \mid \text{darkness}) = \frac{P(\text{farsighted} \cap \text{darkness})}{P(\text{darkness})} = \frac{40}{172}$$
$$P(\text{darkness} \mid \text{farsighted}) = \frac{P(\text{darkness} \cap \text{farsighted})}{P(\text{farsighted})} = \frac{40}{91}$$
- (d) *Hypotheses:* H_0 : A child's eyesight is independent of bedroom lighting condition, and H_a : A child's eyesight is not independent of bedroom lighting condition.
Procedure: A chi-square test of independence.
Checks: It is given that there was a random sample, the sample size of 479 is less than 10% of all children, the data are measured as "counts," and the expected counts are all at least 5:

21.45	49.19	66.36
14.25	32.68	44.08
39.30	90.13	121.57

Mechanics: Calculator software gives $\chi^2 = 56.51$ with a p -value ≈ 0.000 and $df = 4$.

Conclusion in context with linkage to the p-value: With this small of a p -value, $0.000 < 0.05$, there is strong evidence to reject H_0 ; that is, there is convincing evidence of a relationship between a child's eyesight and the bedroom lighting condition.

- (e) Eyesight is hereditary, so children tend to have eyesight like that of their parents. Furthermore, nearsighted parents may tend to use more light in the child's room than other parents, perhaps so they can more easily check on the child during the night.
2. (a) This was an observational study because no treatment was imposed. It was retrospective because existing data from the past were examined.
- (b)
$$P(4 \text{ or more}) = \frac{584 + 104 + 135}{1,500 + 250 + 300} = \frac{823}{2,050} = 0.401$$
- (c)
$$P(\text{exactly } 3 \mid \text{White}) = \frac{309}{1,500} = 0.206$$
- (d)
$$P(\text{Hispanic} \mid \text{exactly } 2) = \frac{35}{210 + 36 + 35} = \frac{35}{281} = 0.125$$
- (e) *Hypotheses:* H_0 : There is no difference in the distributions of core social network sizes across White, Black, and Hispanic populations. H_a : There is a difference in the distributions of core social network sizes across White, Black, and Hispanic populations.
Procedure: A chi-square test for homogeneity
Checks: i. We are given three independent random samples.
 ii. The three samples are clearly each less than 10% of their respective populations.
 iii. The expected cells (calculated below) are all greater than 5.
- Mechanics:* Using matrix and χ^2 -Test on a calculator, we find the expected cells, the test statistic χ^2 , the p -value, and the degrees of freedom.

410.5	205.6	281.7	602.2
68.4	34.3	47.0	100.4
82.1	41.1	56.3	120.4

$$\chi^2 = 15.13, p\text{-value} = 0.019, df = 6$$

Conclusion in context with linkage to the p-value: With this small a p -value, $0.019 < 0.05$, there is sufficient evidence to reject the null hypothesis. There is convincing evidence that there is a difference in the distributions of core social network sizes across White, Black, and Hispanic populations.

Quiz 23 (Pages 245–247)

Multiple-Choice

- (C)** Because the population is approximately normal, normality can be assumed for both sampling distributions. Both will have mean 0.81. The standard deviation for the $n = 10$ sampling distribution is $\frac{0.24}{\sqrt{10}}$, which is greater than the standard deviation for the $n = 100$ sampling distribution, which is $\frac{0.24}{\sqrt{100}}$.
- (D)** We have a random sample that is less than 10% of the high school football population. With a sample size of 48, the central limit theorem applies, and the sampling distribution of $\bar{x}$ is approximately normal with mean $\mu_{\bar{x}} = 355$ and standard deviation $\sigma_{\bar{x}} = \frac{80}{\sqrt{48}}$. The z -scores of 340 and 360 are $\frac{340 - 355}{\frac{80}{\sqrt{48}}}$ and $\frac{360 - 355}{\frac{80}{\sqrt{48}}}$, respectively, and the probability of a sample mean between 340 and 360 is $P(340 < X < 360) = P\left(\frac{340 - 355}{\frac{80}{\sqrt{48}}} < z < \frac{360 - 355}{\frac{80}{\sqrt{48}}}\right)$.
- (A)** The sampling distribution of the sample mean will be roughly normal if the population is roughly normal or if the sample size is large. Because this sampling distribution is skewed, it is likely that neither of those conditions exist.
- (D)** The mean of the sampling distribution is equal to the population mean. The standard deviation of the sampling distribution is the population standard deviation divided by the square root of the sample size. In this case, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ gives $\frac{\sigma}{\sqrt{25}} = 4$ and thus the population standard deviation equals $\sigma = 4\sqrt{25} = 20$.
- (A)** The sampling distribution of $\bar{x}$ has mean $\mu_{\bar{x}} = \mu = 50$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$. With this large a sample size, $n = 100$, the CLT gives that the shape of the sampling distribution will be roughly normal regardless of what the original distribution is.

Free-Response Questions

- $P(X > 4,250) = P\left(z > \frac{4,250 - 4,000}{125}\right) = P(z > 2) = 0.0228$
[Or `normalcdf(4250, 10000, 4000, 125) = 0.0228`.]
 - The original population is approximately normal, so the sampling distribution of $\bar{x}$ is approximately normal with mean $\mu_{\bar{x}} = \mu = 4,000$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{125}{\sqrt{40}} = 19.764$.
 - $P(X < 3,950) = P\left(z < \frac{3,950 - 4,000}{19.764}\right) = P(z < -2.530) = 0.0057$.
[Or `normalcdf(0, 3950, 4000, 19.764) = 0.0057`.]
 - The answer to part (a) would be affected because it assumes an approximately normal population. The other answers would not be affected because for large enough n , the central limit theorem gives that the sampling distribution of $\bar{x}$ is roughly normal regardless of the distribution of the original population.

2. (a) $\text{invNorm}(0.90) = 1.282$, and $386.3 + 1.282(21.5) = 413.9$. [Or $\text{invNorm}(0.90, 386.3, 21.5) = 413.9$.]
 (b) This is a binomial distribution with $n = 5$ and $p = 0.10$. Then $P(x \geq 3) = 1 - \text{binomcdf}(5, 0.10, 2) = 0.00856$. [Or $10 (0.10)^3 (0.90)^2 + 5 (0.10)^4 (0.90) + (0.10)^5 = 0.00856$.]
 (c) This is a normal distribution with $\sigma = \frac{21.5}{\sqrt{5}} = 9.615$. Then

$$P(x \geq 413.9) = P\left(z \geq \frac{413.9 - 386.3}{9.615}\right) = P(z \geq 2.871) = 0.002..$$

Quiz 24 (Pages 253–256)

Multiple-Choice

1. (A) The t -distribution is bell-shaped (unimodal and symmetric) but lower at the mean, higher at the tails, and more spread out than the normal distribution. The larger the df , the closer the t -distribution is to the normal distribution, so as df increases, the area in the tails decreases.
2. (C) The women working in the mayor's office are a convenience sample, and the randomization condition cannot be verified. The sample size, $n = 38$, is large enough for the central limit theorem to apply and given that the sampling distribution of $\bar{x}$ is approximately normal. The sample size of 38 can reasonably be assumed to be less than 10% of the women in the small city.
3. (A) Random assignment doesn't make sense here. The sample size, $n = 95$, is large enough for the central limit theorem to apply, so the population distribution does not have to be approximately normal. It is clear that the sample size of 95 students is less than 10% of all high school students. Given a random sample and less than 10% of the population, the independence condition is satisfied.
4. (D) The 95% refers to the method: 95% of all intervals obtained by this method will capture the true population parameter. Nothing is certain about any particular set of 20 intervals. For any particular interval, the probability that it captures the true parameter is 1 or 0 depending upon whether the parameter is or isn't in it.
5. (A) If you are willing to accept less confidence—that is, a smaller confidence level—you will have a smaller margin of error and a smaller confidence interval. The margin of error varies directly with the critical value (z^* or t^*) and directly with the standard deviation of the sample but inversely with the square root of the sample size. The value of the sample mean and the population size do not affect the margin of error.
6. (D) $df = n - 1 = 30 - 1 = 29$, $\text{invT}(0.975, 29) = 2.045$, and

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{1.2}{\sqrt{30}}$$
, so the confidence interval is $\bar{x} \pm t_{n-1}^* SE(\bar{x}) = 28.5 \pm 2.045\left(\frac{1.2}{\sqrt{30}}\right)$.
7. (B) Although both methods will give estimates of the total quantity of sugary beverages consumed daily by students at this high school, Method II is better as it draws data from 30 students consuming sugary beverages. Method I could result in choosing a very small number, possibly even none at all, of students consuming sugary beverages to use for the overall estimation.
8. (D) The percent refers to how confident we are that the interval has captured the population mean. The confidence interval is about the population, not the sample. The probability the true mean difference is between 4.6 and 7.3 is either 1 or 0 depending on whether it is or is not in the interval. The confidence interval is about a population mean, not about individual values.
9. (D) To divide the interval estimate by d without affecting the confidence level, multiply the sample size by a multiple of d^2 . In this case, $4(50) = 200$.
10. (C) The formula for margin of error is $t^* \left(\frac{s}{\sqrt{n}}\right)$. Since s is given to be the same for both samples, the width of the confidence interval from X will be less than from Y if the sample sizes are the same but t^* is less (which will occur if the confidence level is less).

Free-Response Questions

1. (a) *Parameter:* the mean weight of all “11 oz” bags of tortilla chips.
Procedure: A one-sample t -interval for a population mean.
Checks: We check *independence* and *normality*. For independence, we are given a simple random sample, and $n = 10$ is less than 10% of all “11 oz” bags of tortilla chips. For normality, a dotplot of the sample data is roughly unimodal and symmetric and is free from strong skewness and outliers.
Mechanics: Calculator software gives (11.08, 11.38).
[For instructional purposes, we note that with $df = 10 - 1 = 9$, the critical t -score is $\text{invT}(0.975, 9) = 2.262$. The confidence interval is $11.23 \pm 2.262\left(\frac{0.2058}{\sqrt{10}}\right) = 11.23 \pm 0.15$.]
Conclusion in context: We are 95% confident that the mean weight of the population of all “11 oz” bags of tortilla chips is between 11.08 and 11.38 ounces.

(b) Five of the bags in the sample have weights {11.1, 11.2, 11.2, 11.2, 11.3} in this confidence interval. Five out of ten gives an answer of 50%.

(c) 50% is not close to 95%, but nothing went wrong. The confidence interval is estimating the mean weight, not weights of individual bags. There is no reason to expect that 95% of the individual weights to be within the confidence interval.
2. (a) The point estimate is the sample mean, \$2,900,000.

(b) $SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{850,000}{\sqrt{30}} = 155,000$.

(c) Yes, the conditions for inference are met: *independence* and *normality*. For independence, the sample is given to be random, and it is reasonable to assume that 30 is less than 10% of all white-collar crimes. For normality, the sample size, $n = 30$, is large enough for the central limit theorem to apply, thus giving the sampling distribution of the sample mean is approximately normal.

(d) We are 95% confident that the mean amount involved in all white-collar crimes is between \$2,580,000 and \$3,222,000.

(e) The confidence interval says nothing about the plausibility of a white-collar crime with a monetary amount of \$2,000,000. A confidence interval provides an estimate of the population mean value but does not provide information about an individual value.

(f) It is not plausible that the population mean is \$2,000,000 because 2,000,000 is not in the interval from 2,580,000 to 3,222,000.
3. (a) *Parameter:* Let μ_{diff} represent the true mean difference (STEM minus non-STEM) between weekly wages of recent graduates.
Procedure: Use a one-sample t -interval for a population mean difference.
Checks: We are given a random sample of paired data, and the sample size is less than 10% of all recent college graduates. For normality, $n = 30$ is large enough for the central limit theorem to apply. Calculation of a t -interval is appropriate.
Mechanics: With an unknown population SD, it's necessary to find a t -interval. Using $\bar{x}_{\text{diff}} = 976$ and $s_{\text{diff}} = 41$, calculator software (TInterval) gives (963.3, 988.7).
[For instructional purposes, we note that $SE(\bar{x}_{\text{diff}}) = \frac{41}{\sqrt{30}} = 7.486$. With $df = 30 - 1 = 29$ and 5% in each tail, the appropriate t -scores are $\pm \text{invT}(0.95, 29) = \pm 1.699$. Then $976 \pm 1.699(7.486) = 976 \pm 12.7$.]
Conclusion in context: We are 90% confident that the true mean difference (STEM minus non-STEM) between weekly wages of recent college graduates is between \$963.30 and \$988.70.

(b) No, there is not significant evidence that the true mean difference is different from \$985 because 985 is in the interval of plausible values from 963.30 to 988.70.

(c) If all possible random samples are taken of 30 pairs of recent graduates where each pair consists of two subjects with the same GPA, one having a STEM degree and the other having a non-STEM

degree, and if a confidence interval for the mean difference of weekly salaries is constructed from each sample, then 90% of all these intervals would capture the true mean difference of weekly salaries.

Quiz 25 (Pages 261–264)

Multiple-Choice

1. (A) Hypotheses are always about population parameters, like μ , and never about sample statistics, like $\bar{x}$. You must look at the context to decide if the test is one-sided or two-sided. If the test is one-sided, you must determine the direction. In this case, the sociologist is interested in whether the true mean is greater than \$13,000.
2. (C) Since the sample size $n = 22 < 30$, the central limit theorem does not apply. In order to conclude approximate normality of the sampling distribution of $\bar{x}$, we either have to be told that the population is approximately normal, or failing that, that the sample distribution is roughly unimodal and symmetric so that it would not be unreasonable to assume the population is approximately normal. However, we are not given that the population is approximately normal, and we are told that the sample distribution is skewed right. Thus, the normality condition is violated.
3. (A) The numerator of the test statistic is the sample mean minus the assumed population mean, while the denominator is the sample standard deviation divided by the square root of the sample size. That is,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{103.2 - 98}{4.8/\sqrt{100}} = \frac{103.2 - 98}{4.8/10}.$$
4. (C) The population standard deviation is unknown, so this is a t -test. Medications having an effect shorter or longer than claimed should be of concern, so this is a two-sided t -test: $H_a: \mu \neq 58.4$, and $df = n - 1 = 40 - 1 = 39$.
5. (D) The test statistic in a hypothesis test for a population mean is $t = \frac{(\bar{x} - \mu_0)}{SE(\bar{x})}$; that is, it is the number of standard errors the sample mean is away from the hypothesized mean. Be careful not to confuse the test statistic with the p -value.
6. (C) The p -value is the probability that, if many samples of size 50 were selected from the population, we would observe a sample mean of 29.4 gallons or more, given that the population mean is 28.2 gallons.
7. (C) This is a hypothesis test with H_0 : breaking strength is within specifications, and H_a : breaking strength is below specifications. A Type I error is committed when a true null hypothesis is mistakenly rejected. In this context, a Type I error would be committed if the breaking strength is within specifications, but because the sample wrongly indicates a drop below the specified level, the production process is unnecessarily halted.
8. (B) $P(\text{at least one Type I error}) = 1 - P(\text{no Type I errors}) = 1 - (0.99)^5 = 0.049$
9. (C) A larger sample size n reduces the standard deviation of the sampling distribution, resulting in a narrower sampling distribution. So, for the given sample statistic, the p -value is smaller, and the probabilities of mistakenly rejecting a true null hypothesis or mistakenly failing to reject a false null hypothesis are both decreased. Furthermore, a lower Type II error results in higher power.
10. (D) This is a hypothesis test of a set of differences (LP rating minus CD rating) with $H_0: \mu_d = 0$ and $H_a: \mu_d > 0$. Typical levels of significance are 0.01, 0.05, and 0.10. When the p -value is greater than the significance level, there is not sufficient evidence to reject H_0 ; that is, there is not convincing evidence of H_a .

Free-Response Questions

1. (a) The p -value of 0.0197 gives the probability of observing a sample mean of 1,002.4 or greater if, in fact, this system results in a mean *E. coli* concentration of 1,000 MPN per 100 mL.

- (b) Since $0.0197 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that the mean *E. coli* concentration is greater than 1,000 MPN per 100 mL.
- (c) With rejection of the null hypothesis, there is the possibility of a Type I error—that is, of mistakenly rejecting a true null hypothesis. Possible consequences are that sales of the system drop even though the system is doing what it claims or that the company performs an overhaul to fix the system even though the system is properly reducing the mean concentration of *E. coli* to 1,000 MPN per 100 mL.

2. (a) The population of interest is all students at the counselor's college.

- (b) Number the students at the college from 1 to 6,000. Use a random number generator to generate 50 unique numbers between 1 and 6,000. The students whose numbers correspond to the generated numbers will be the sample.
- (c) *Parameter*: Let μ represent the mean number of hours worked over the summer break by all students at the college.

Hypotheses: $H_0: \mu = 25$, $H_a: \mu > 25$

Procedure: A one-sample *t*-test of a population mean

Checks: We are given a simple random sample, and the sample size $n = 50$ is less than 10% of the population of $N = 6,000$. The normality condition is satisfied because the sample size $n = 50 \geq 30$, so the central limit theorem applies. Conducting a *t*-test is appropriate.

Mechanics: Calculator software gives $t = 1.394$ and a *p*-value of 0.085.

Conclusion in context with linkage to the p-value: With this large a *p*-value, $0.085 > 0.05$, there is not sufficient evidence to reject the null hypothesis. There is not convincing evidence to support the counselor's claim that the true mean number of hours per week worked over summer break by students at her college is greater than 25.

3. Because the data come in pairs, the proper procedure is to perform a single-sample hypothesis test on the set of differences.

Parameter: Let μ_d represent the mean difference (union minus non-union) in weekly salaries.

Hypotheses: $H_0: \mu_d = 0$, $H_a: \mu_d > 0$

Procedure: A one-sample *t*-test for a population mean difference (union minus nonunion).

Checks: We are given a random sample, and the sample size is less than 10% of the population of nonmanagerial employees in the restaurant industry. The normality condition is satisfied because the sample data (the dotplot of differences) does not show strong skew or outliers. Conducting a *t*-test is appropriate.

Mechanics: Calculator software with $\bar{x} = 126.4$ and $s = 66.4$ gives $t = 6.02$ and a *p*-value of 0.000099.

Conclusion in context with linkage to the p-value: With this small a *p*-value, $0.000099 < 0.01$, there is sufficient evidence to reject the null hypothesis. There is convincing evidence that the mean weekly salary of the nonmanagerial union employees is greater than the mean weekly salary of the nonmanagerial nonunion employees.

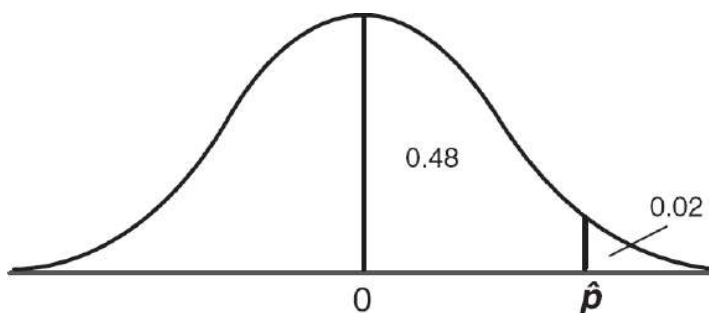
Quiz 26 (Pages 270–272)

Multiple-Choice

1. (D) Since couples are chosen, the samples of grandmothers and grandfathers are not independent, so a two-sample *t*-test cannot be performed. If the procedure would be to perform a one-sample *t*-test on the set of differences, then the conditions for inference would be met. Random assignment does not make sense here. The sample size, $n = 100$, is large enough for the central limit theorem to apply, so the population distribution need not be approximately normal. One hundred grandparents is clearly less than 10% of all grandparents.

2. (D) There were two independent groups, and the data were collected on a single quantitative variable (number of head injuries). The correct procedure is a two-sample t -test for a difference in population means.
3. (A) $SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}} = \sqrt{\frac{(19.1)^2}{347} + \frac{(19.9)^2}{561}}$ and with such large sample sizes, the critical t -scores should be just slightly greater than the critical z -scores of ± 1.96 associated with 95% confidence. The confidence interval is $(\bar{x}_1 - \bar{x}_2) \pm t^* SE(\bar{x}_1 - \bar{x}_2) = (93.5 - 84.5) \pm 1.97 \sqrt{\frac{(19.1)^2}{347} + \frac{(19.9)^2}{561}}$.
4. (A) With unknown population standard deviations, the t -distribution must be used, and $SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{(0.3)^2}{10} + \frac{(0.2)^2}{10}}$. The critical t -score is $t = \frac{\bar{x}_1 - \bar{x}_2}{SE(\bar{x}_1 - \bar{x}_2)} = \frac{(4 - 4.8)}{\sqrt{\frac{(0.3)^2}{10} + \frac{(0.2)^2}{10}}}$.

5. (B)



[Follow for extended description](#)

With 0.02 in the tail, a confidence interval over $0.48 + 0.48 = 0.96 = 96\%$ will contain 0.

Free-Response Questions

1. (a) *Procedure:* A two-sample t -interval for $\mu_{NRW} - \mu_{RW}$, the difference in population means of BPA body concentrations in nonretail workers and retail workers.
Checks: We are given random samples. It is reasonable to assume the samples are independent, and we assume the sample sizes are less than 10% of the populations. For normality, both sample sizes, $528 \geq 30$ and $197 \geq 30$, are large enough so that the CLT applies.
Mechanics: Calculator software gives $(-0.9521, -0.7479)$ with $df = 332.3$.
Conclusion in context: We are 99% confident that the difference in true means of BPA body concentrations in all nonretail and retail workers (nonretail mean minus retail mean) is between -0.75 and $-0.95 \mu\text{g/L}$.
- (b) Because 0 is not in the interval of plausible values for the difference of population means and the entire interval is negative, the interval does support the belief that retail workers carry higher amounts of BPA in their bodies than nonretail workers.
2. (a) Since couples are chosen, the samples of husbands' ages and wives' ages are not independent. So a two-sample t -interval cannot be found. If the procedure would be to calculate a one-sample t -interval on the set of differences, then the conditions would be met.
- (b) Because there is a clear relationship between husbands and wives in couples, the proper procedure is to form the set of 50 differences and to perform a one-sample t -test as follows.
Parameter: Let μ_{diff} = the true mean difference in ages at wedding between husbands and wives (husbands minus wives).
Hypotheses: $H_0: \mu_{\text{diff}} = 0$ and $H_a: \mu_{\text{diff}} > 0$.
Procedure: A paired t -test—that is, a one-sample t -test on the set of differences.
Checks: We are given a random sample, and $n = 50$ is less than 10% of all married couples. For normality, $n = 50$ is large enough for the CLT to apply.

Mechanics: A t -test gives $t = 3.780$ with a p -value of 0.000213.

[Or we can calculate $SE(\bar{x}_{\text{diff}}) = \frac{3.742}{\sqrt{50}} = 0.5292$ and $t = \frac{\bar{x}_{\text{diff}} - 0}{0.5292} = \frac{2}{0.5292} = 3.780$. With $df = n - 1 = 49$, the p -value is $P(t > 3.780) = \text{tcdf}(3.780, 1,000, 49) = 0.000213$.]

Conclusion in context with linkage to the p -value: With this small of a p -value, $0.000213 < 0.05$, there is sufficient evidence to reject H_0 . In other words, there is convincing evidence that the true mean age at wedding for husbands is greater than the true mean age at wedding for wives.

- (c) There was sufficient evidence to reject the null hypothesis. If the null hypothesis were true, we would be committing a Type I error—that is, mistakenly rejecting a true null hypothesis. A possible consequence is that the sociologist publishes a false conclusion and loses credibility in the academic world.

Quiz 27 (Pages 275–279)

Multiple-Choice

1. (B) For this day's sample, $G = \frac{\text{Max} + \text{Min}}{2} = \frac{35.55 + 33.51}{2} = 34.53$. A value of 34.53 or less occurred only twice in the 100 samples. Thus, the estimated p -value is $\frac{2}{100} = 0.02$. If the machinery was operating properly, a G measurement of 34.53 would be very unusual. The conclusion should be to recalibrate the machine.
2. (D) The p -value is a conditional probability, the probability of as extreme as (or more extreme) a result as the one obtained, given that the null hypothesis is true. This is a two-tailed test. In the situation, there were $12 + 7 + 1 = 20$ values on the lower end and $5 + 2 = 7$ values on the upper end that are at least as extreme as the $W = -0.7$ value. Thus, the estimated p -value is $\frac{20 + 7}{1,000} = 0.027$.

Free-Response Questions

1. (a) Although shape cannot accurately be gauged from boxplots, a complete answer here must address center, spread, and outliers, and mention context. The distribution of study hours per week for athletes at this college has a greater center (median) and a smaller spread (range) than the distribution of study hours per week for nonathletes at this college. The nonathlete distribution has three low end outliers while the athlete distribution has no outliers.
- (b) The value of +3 or above occurred in $7 + 10 + 4 + 1 = 22$ of the trials.
- (c) The observed difference in median study hours was $21 - 18 = +3$. From (b), this occurred in 22 of the 110 trials giving a probability of $\frac{22}{110} = 0.2$.
- (d) The observed difference in study hours was three hours, and this value or more will occur an estimated 20% of the time just by chance. Thus, there is not convincing statistical evidence that athletes participating in varsity sports at this college spend more hours per week with academic studies outside the classroom than do nonathletes.
2. (a) $H_0: \mu = 350$ and $H_a: \mu < 350$
- (b) The Normal condition is not met. There is no indication that the population of all battery cranking powers have a roughly normal distribution; the sample size, $n = 5$, is not large enough for the central limit theorem to apply; and since the sample data are not given, there is no way to determine if it is reasonable to assume that the sample data come from a roughly normal distribution.
- (c) The consumer group's random sample of 5 batteries had an average cranking power of only 320 CCA. The simulation indicates that the chance of picking 5 batteries with an average of 320 CCA or lower from a group of new batteries manufactured to have 350 CCA power is 4 out of 160 or 0.025. So, the estimated p -value is 0.025.

- (d) With a p -value this small, $0.025 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that the true mean cranking power of these batteries is less than the company's claim of 350 CCA.
3. (a) $H_0: p_1 = p_2$ and $H_a: p_1 > p_2$, where p_1 is the proportion of all addicts taking desipramine who would not have relapsed after one year and p_2 is the proportion of all addicts taking lithium who would not have relapsed after one year.
- (b) $\frac{14}{25} - \frac{6}{25} = 0.32$
- (c) There are 3 values out of 500 that are 0.32 or greater, so the estimated p -value is $\frac{3}{500} = 0.006$.
- (d) With this small of a p -value, $0.006 < 0.05$, there is sufficient evidence to reject the null hypothesis; that is, there is convincing evidence that the antidepressant desipramine is more effective than the mood stabilizer lithium in treating cocaine addiction.

Quiz 28 (Pages 282–284)

Multiple-Choice

1. (A) The power is the probability of not committing a Type II error—that is, the power is the probability of correctly rejecting a false null hypothesis. In this context, a false null hypothesis means the air quality is poor, and thus, power is the probability of correctly giving a warning when the air quality is poor.
2. (C) The α -risk, also called the significance level of the test, is the probability of committing a Type I error and mistakenly rejecting a true null hypothesis.
3. (C) Power is the probability of not committing a Type II error. Here, $1 - 0.4 = 0.6$.
4. (B) Changing from a significance level of 0.05 to one of 0.01 makes it more difficult to reject the null hypothesis. Thus, there is a smaller probability of being able to reject a false null hypothesis. This gives a decrease in power and an increase in the probability of a Type II error.
5. (A) The null hypothesis is typically that no effect exists. Power, the probability of correctly rejecting a false null hypothesis, is thus the probability of correctly detecting an effect that exists. Choice (B) is a Type I error, and choice (C) is a Type II error.
6. (A) Power is the probability of correctly rejecting a false null hypothesis. All other things being equal, power will be greatest if the true mean is furthest from the mean claimed in the null hypothesis in the direction of the alternative.
7. (D) A larger sample size gives a smaller standard error and thus a greater probability of picking up that the null hypothesis is incorrect. Also, the further the true parameter is from the claimed null hypothesis about the parameter, the greater the probability of picking up that the null hypothesis is incorrect.
8. (D) With a greater significance level, 0.05 rather than 0.01, there is a greater chance of rejecting the null hypothesis. Also, a larger sample size gives a smaller standard error and thus a greater probability of picking up that the null hypothesis is incorrect.

Free-Response Questions

1. (a) Given $N(81, 0.3)$, $P(80.5 < X < 81.5) = 0.904$.
- (b) The sampling distribution of $\bar{x}$ is approximately normal with mean $\mu_{\bar{x}} = \mu = 81$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.3}{\sqrt{5}} = 0.1342$.
- (c) A Type I error will be committed if the null hypothesis is true but is mistakenly rejected. Given $N(81, 0.1342)$, $P(X < 80.5) + P(X > 81.5) = 0.0002$.

- (d) A Type II error will be committed if the null hypothesis is false, but is mistakenly not rejected. Given $N(80.4, 0.1342)$, $P(80.5 < X < 81.5) = 0.228$.

Quiz 29 (Pages 303–309)

Multiple-Choice

1. (A) Residual = Measured – Predicted, so if the residual is negative, the predicted must be greater than the measured (observed).
2. (A) The correlation r is unit-free.
3. (B) The “Predictor” column indicates the independent variable, x , with its coefficient, b , to the right.
4. (D) $r = \sqrt{0.986} = 0.993$. (The sign is the same as the sign of the slope, which in this case is positive.)
5. (C) $\widehat{\text{back}} = 0.056 + 0.920(0.55) = 0.562$, and so the residual = $0.59 - 0.562 = 0.028$.
6. (C) The coefficient of determination r^2 gives the proportion of the y -variance that is accountable from a knowledge of the variability of x . In this case, $r^2 = (0.632)^2 = 0.399$ or 39.9%.
7. (A) The correlation is not changed by adding the same number to every value of one of the variables, by multiplying every value of one of the variables by the same positive number, or by interchanging the x - and y -variables.
8. (B) The word “negative” in the phrase “strong negative linear association” means that generally as one variable increases, the other variable decreases. Thus, regions with a higher percentage of seat belt usage tend to have lower numbers of highway deaths due to failure to wear seat belts. Choice (A) is an interpretation of the y -intercept. Correlation is a measure of the strength of a linear relationship but does not by itself explain the meaning of “positive” or “negative.” While the overall pattern is of a negative association, anything can be true about two points on the scatterplot. Choice (C) is an interpretation of the slope.
9. (D) A scatterplot readily shows that while the first three points lie on a straight line, the fourth point does not lie on this line. Thus, no matter what the fifth point is, all the points cannot lie on a straight line, and so r cannot be 1.
10. (B) Note that the smallest and largest x -values in X have y -values below the regression line—that is, with negative residuals—and so X corresponds to 2. Note that the three largest x -values in Z have y -values with positive residuals, and so Z corresponds to 3.

Free-Response Questions

1. (a) The correlation $r = \sqrt{0.8428} = 0.918$. It is positive because the slope of the regression line is positive.
(b) The slope is 0.283, signifying that each unit increase in strength is associated with a predicted increase of 0.283 in speed.
(c) The linear regression line passes through $(\bar{x}, \bar{y})$ so the mean speed = $4.626 + 0.283(30.47) = 13.25$.
(d) 84.28% of the variation in speed is accounted for by the linear model relating speed and strength.
(e) The linear model predicts Thanos’s speed to be $4.626 + 0.283(45) = 17.36$, so the model underestimates Thanos’s speed by $18 - 17.36 = 0.64$.
(f) $4.626 + 0.283(42) = 16.512$, so the point (42, 16.5) is almost exactly on the regression line. Removing a point that follows the linear pattern will lower the correlation.
(g) Removing the points corresponding to both Superman (with its high x -value) and the Penguin (with its low x -value) swings the line away from these points, in both cases leading to a decrease in the positive slope value.

2. (a) In scatterplot I, the points fall exactly on a downward sloping straight line, so $r = -1$. In scatterplot II, the isolated point is an influential point, and r is close to $+1$. In scatterplot III, the isolated point is also influential, and r is close to 0.
- (b) Inserting a point that lies directly on a straight line drawn through the three points will not change the correlation of $r = -1$.
3. (a) The association between stress level and debt for this sample of 25 college graduates is linear, moderate, and positive.
- (b) $19.376 + 0.42640(60) = 44.96$ and $\text{Observed} - \text{Predicted} = 25 - 44.96 = -19.96$. The residual value means that the stress level is 19.96 points less than what would be predicted for a student with a \$60,000 debt.
- (c) Health is a better choice than income for including with debt in a regression model for predicting stress. The relatively strong association between health and residuals from regression of stress on debt shows that health will help explain the unexplained variation (the residuals) between stress and debt. The random scatter in the plot with income versus residuals from regression of stress on debt shows that income will not help explain the unexplained variation (the residuals) between stress and debt.
4. (a) There is a weak, negative, linear relationship between the difference in House membership (Republicans minus Democrats) in midterm elections and the difference in House membership (Republicans minus Democrats) in the following presidential elections.
- (b) Graph 2 would be more helpful. For each point in the scatterplot, $x + y > 0$ (Republican majority) if the point is above the line $y = -x$, $x + y = 0$ (balanced) if the point is on the line $y = -x$, and $x + y < 0$ (Democrat majority) if the point is below the line $y = -x$.
- (c) In the scatterplot, 10 points would be classified as “Republican majority,” no points would be classified as “Balanced,” and 8 points would be classified as “Democrat majority.”
- (d) The predicted difference in number of House members (Republicans minus Democrats) for the following presidential election would be $\hat{y} = 0.257 - 0.546(7) = -3.565$.

Quiz 30 (Pages 310–315)

Multiple-Choice

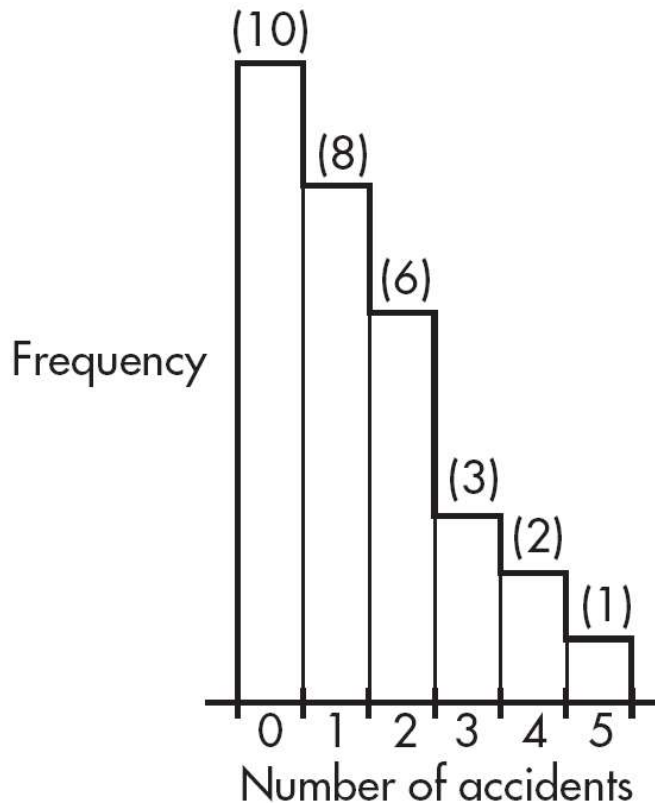
1. (D) The negative value of the slope (-2.84276) gives that, on average, the predicted average total SAT score of a school is 2.84 points lower for each one unit higher in the percentage of students taking the exam. Choices (A) through (C) are incorrect for the following reasons. The variable column indicates the independent (explanatory) variable. The sign of the correlation is the same as the sign of the slope (negative here). In this example, the y -intercept is meaningless (predicted SAT result if no students take the exam). There can be a strong linear relation with high r^2 -value but still a distinct pattern in the residual plot indicating that a nonlinear fit may be even stronger.
2. (D) A negative correlation shows a tendency for higher values of one variable to be associated with lower values of the other; however, given any two points, anything is possible.
3. (A) Since $(2, 5)$ is on the line $y = a + 3x$, we have $5 = a + 3(2)$ and $a = -1$. Thus, the regression line is $y = -1 + 3x$. The point $(\bar{x}, \bar{y})$ is always on the regression line, and so we have $\bar{y} = -1 + 3\bar{x}$.
4. (C) If the points lie on a straight line, $r = \pm 1$. Correlation has the formula $r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$, so x and y are interchangeable, and r does not depend on which variable is called x or y . However, since means and standard deviations can be strongly influenced by outliers, r too can be strongly affected by extreme values. Although $r = 0.75$ indicates a better fit with a linear model than $r = 0.25$ does, we cannot say that the linearity is threefold.

5. **(B)** Residuals can be any sign. In a good straight line fit, the residuals show a random pattern. The linear regression line minimizes, not maximizes, the sum of the squares of the residuals.
6. **(C)** Predicted winning percentage = $44 + 0.0003(34,000) = 54.2$, and Residual = Observed – Predicted = $55 - 54.2 = 0.8$.
7. **(B)** The slope and the correlation coefficient have the same sign. Multiplying every y -value by -1 changes this sign.
8. **(C)** A linear association means that as the explanatory variable (home size here) changes by a constant amount, the response variable (selling price here) also changes by a constant amount, on average. Unless there was perfect linear correlation, the points will not line up on a straight line. No distinct pattern in the residual plot simply means that there is not an obviously better model to be found, but it does not necessarily suggest that the data are linear. The coefficient of determination indicates something about the strength of the relationship but does not define linearity. Choice (D) is an interpretation of the slope, but again, not a definition of linearity.
9. **(A)** “Strong” means that the points in the related scatterplot fall close to the least squares regression line. In context, this means that the actual incidence of skin cancer at a given latitude will be very close to what is predicted by the least squares model. Association does not imply causation. *Although* (C) refers to the existence of an association, and (B) and (D) further refer to the direction of the association, none of these statements refer to the strength of the association.
10. **(B)** The mean equals the sum divided by the number of values; thus, if the sum is zero, so is the mean. However, if the sum of values equals zero, it does not follow that the middle value is zero.

Free-Response Questions

1. (a) The correlation coefficient is $r = \sqrt{0.746} = 0.864$. It is positive because the slope of the regression line is positive.
- (b) The slope is 1.106, signifying that each additional page raises a grade by 1.106, on average.
- (c) Including Mary’s paper will lower the correlation coefficient because her result seems far off the regression line through the other points.
- (d) Including Mary’s paper will swing the regression line down and lower the value of the slope.
- (e) 74.6% of the variation in grades is accounted for by the linear model between grades and page lengths.

2. (a)



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- (b) There is a roughly linear trend with daily accidents increasing during the month.
 - (c) The daily number of accidents is strongly skewed to the right.
3. (a) The association between exam score and hours of sleep is *nonlinear* (or *curved*), weak (or moderate), and positive.
- (b) There appears to be very little, if any, association between exam score and hours of sleep for the 8 students who received at least 6 hours of sleep the night before the exam. (Because of the one very low score that also corresponds to hours of sleep just over 6, one could say that there is a very weak *positive* association.)
 - (c) The student who scored 75 and slept for just over 2 hours the night before the exam has such a strong influence that that particular scatterplot point taken together with all the other points suggests a nonlinear form.
4. (a) This was an observational study as no treatments were applied. Participants were not instructed to consume quantities of ultra-processed foods.
- (b)
 - i. A strong relationship means that the data points fell close to a line or curve.
 - ii. A linear relationship means that as consumption of ultra-processed foods increases by 1 gram, the dementia score tends to change by a constant amount on average.
 - iii. A positive relationship means that adults 55 years of age or older who consume more ultra-processed foods tend to have a higher dementia score in 10 years.
 - (c) No, it is not reasonable to conclude causation. Causation cannot be concluded from observational studies, and linear regression can indicate association but not causation.

- (d) A positive residual would indicate that the linear model underestimated the dementia score for that adult.

Quiz 31 (Pages 322–323)

1. *Procedure:* Chi-square test for homogeneity (“homogeneity” rather than “independence” because there are two independent samples)
Checks: Independent random samples; all expected counts at least 5; $n \leq 0.10N$
Hypotheses: H_0 : There are no differences between the true distributions of reasons for trying out for varsity sports (social, health, or status) for men and women.
 H_a : There is at least one difference between the true distributions of reasons for trying out for varsity sports (social, health, or status) for men and women.
2. *Procedure:* One-sample z -interval for population proportion p
Parameter: Let p represent the true proportion of the population of AP Statistics students who graduate in the top ten percent of their senior class.
Checks: Random sample; at least 10 successes and 10 failures, that is, $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$; $n \leq 0.10N$
3. *Procedure:* One-sample t -interval for population mean μ
Parameter: Let μ represent the true mean number of books read by all high school students during their four years in secondary school.
Checks: Random sample; population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; $n \leq 0.10N$
4. *Procedure:* One-sample z -test for a population proportion p
Parameter: Let p represent the true proportion of the population of racially motivated hate crimes that are motivated by anti-Black bias.
Checks: Random sample; $np_0 \geq 10$ and $n(1 - p_0) \geq 10$; $n \leq 0.10N$
Hypotheses: $H_0: p = 0.70$, $H_a: p > 0.70$
5. *Procedure:* One-sample t -test for population mean μ
Parameter: Let μ represent the true mean speed of all fastballs thrown by a particular pitcher when participating in 10-minute yoga sessions.
Checks: Random sample; population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric
Hypotheses: $H_0: \mu = 95$, $H_a: \mu > 95$
6. *Procedure:* Linear regression
Parameter: Let β represent the true slope of the regression line relating school performance (GPA) to family income.
Checks: Random sample; scatterplot is roughly linear; no pattern in the residual plot;
The resulting coefficient of determination gives the percent of variability in the dependent variable (school performance) that can be explained by the independent variable (family income).
7. *Procedure:* Two-sample z -interval for difference in population proportions, $p_1 - p_2$
Parameters: Let p_1 represent the true proportion of the population of patients with warts who are cured with a treatment of cryotherapy. Let p_2 represent the true proportion of the population of patients with warts who are cured with a treatment of duct tape occlusion.
Checks: Independent random samples; at least 10 successes and 10 failures in each sample, that is, $n_1\hat{p}_1 \geq 10$, $n_1(1 - \hat{p}_1) \geq 10$, $n_2\hat{p}_2 \geq 10$, $n_2(1 - \hat{p}_2) \geq 10$; $n_1 \leq 0.10N_1$ and $n_2 \leq 0.10N_2$
8. *Procedure:* Two-sample t -test for difference in population means, $\mu_1 - \mu_2$
Parameters: Let μ_1 represent the true mean cholesterol level (measured in milligrams per deciliter) for all people living in “Western” countries. Let μ_2 represent the true mean cholesterol level (measured in

milligrams per deciliter) for all people living in “non-Western” countries.

Checks: Independent random samples; for each sample, population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; and for each sample $n \leq 0.10N$

Hypotheses: $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 > 0$

9. Procedure: Linear regression

Parameter: Let β represent the true slope of the regression line relating average increase in CO₂ emissions (in tons per person) to income level (in GDP per capita) among all countries of the world.

Checks: Random sample; scatterplot is roughly linear; no pattern in the residual plot;

The resulting slope gives the predicted change in emissions for each unit increase in income level.]

10. Procedure: Chi-square test for independence

Checks: Random sample; all expected counts at least 5; $n \leq 0.10N$

Hypotheses: H_0 : There is no association between race/ethnicity and whether or not an adult is unbanked.

H_a : There is an association between race/ethnicity and whether or not an adult is unbanked.

11. Procedure: Two-sample z -test for difference in population proportions, $p_1 - p_2$

Parameters: Let p_1 represent the true proportion of the population of at-risk patients receiving rivaroxaban who still have strokes. Let p_2 represent the true proportion of the population of at-risk patients receiving warfarin who still have strokes.

Checks: Independent random samples; $n_1\hat{p}_C \geq 10$, $n_1(1 - \hat{p}_C) \geq 10$, $n_2\hat{p}_C \geq 10$, and $n_2(1 - \hat{p}_C) \geq 10$, where

$$\hat{p}_C = \frac{x_1 + x_2}{n_1 + n_2}; n_1 \leq 0.10N_1 \text{ and } n_2 \leq 0.10N_2$$

Hypotheses: $H_0: p_1 - p_2 = 0$, $H_a: p_1 - p_2 \neq 0$

12. Procedure: Matched pair t -test for mean difference in population means, μ_{diff}

Parameter: Let μ_{diff} represent the true mean difference (with pill – with placebo) in malaria episodes for the population of adults when they take the new pill and when they take the placebo.

Checks: Random sample; set of differences is from a normal population or set of differences is large, at least 30, or distribution of set of differences is unimodal and symmetric

Hypotheses: $H_0: \mu_{\text{diff}} = 0$, $H_a: \mu_{\text{diff}} < 0$

13. Procedure: Two-sample t -interval for difference in population means, $\mu_1 - \mu_2$

Parameters: Let μ_1 represent the true mean infant mortality rate in the population of live births in the U.S. (mean infant deaths for every 1,000 live births).

Let μ_2 represent the true mean infant mortality rate in the population of live births in Japan (mean infant deaths for every 1,000 live births).

Checks: Independent random samples; for each sample, population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; for each sample, $n \leq 0.10N$

14. Procedure: Matched pair t -interval for mean difference in population means, μ_{diff}

Parameter: Let μ_{diff} represent the true mean difference (second score – first score) for the population of students taking a national standardized exam twice.

Checks: Random sample; set of differences is from a normal population or set of differences is large, at least 30, or distribution of set of differences is unimodal and symmetric; $n \leq 0.10N$

15. Procedure: One-sample z -interval for true population mean μ . (Note that we use a normal distribution rather than a t -distribution because the population standard deviation is given.)

Parameter: Let μ represent the mean waist size of the population of male high school math teachers.

Checks: Random sample; population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; $n \leq 0.10N$

Quiz 32 (Pages 324–325)

1. *Procedure:* Two-sample t -test for difference in population means, $\mu_1 - \mu_2$
Parameters: Let μ_1 represent the true mean number of hours per week that first-year students study. Let μ_2 represent the true mean number of hours per week that sophomores study.
Checks: Independent random samples; for each sample, population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; for each sample, $n \leq 0.10N$
Hypotheses: $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 > 0$
2. *Procedure:* One-sample z -interval for population proportion p
Parameter: Let p represent the true proportion of the population of students who would report cheating by other students.
Checks: Random sample; at least 10 successes and 10 failures, that is, $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$; $n \leq 0.10N$
3. *Procedure:* Chi-square test for independence
Checks: Random sample; all expected counts at least 5; $n \leq 0.10N$
Hypotheses: H_0 : There is no association between blood pressure level (low, average, or high) and personality type (high-strung or easygoing).
 H_a : There is an association between blood pressure level (low, average, or high) and personality type (high-strung or easygoing).
4. *Procedure:* One-sample z -test for a population proportion p
Parameter: Let p represent the true proportion of 3-point shot attempts successfully made by the population of Division 1 college basketball players this year.
Checks: Random sample; $np_0 \geq 10$ and $n(1 - p_0) \geq 10$; $n \leq 0.10N$
Hypotheses: $H_0: p = 0.34$, $H_a: p < 0.34$
5. *Procedure:* Matched pair t -test for mean difference in population means μ_{diff}
Parameter: Let μ_{diff} represent the true mean difference (after program – before program) in the number of push-ups a clinic member can complete in 90 seconds.
Checks: Random sample; set of differences is from a normal population or set of differences is large, at least 30, or distribution of set of differences is unimodal and symmetric
Hypotheses: $H_0: \mu_{\text{diff}} = 0$, $H_a: \mu_{\text{diff}} > 0$
6. *Procedure:* One-sample t -test for population mean μ
Parameter: Let μ represent the true mean body temperature in the population.
Checks: Random sample; population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; $n \leq 0.10N$
Hypotheses: $H_0: \mu = 98.6$, $H_a: \mu < 98.6$
7. *Procedure:* Matched pair t -interval for mean difference in population means μ_{diff}
Parameter: Let μ_{diff} represent the true mean difference in resting heart rates before and after an extensive exercise program for teenagers.
Checks: Random sample; set of differences is from a normal population or set of differences is large, at least 30, or distribution of set of differences is unimodal and symmetric
8. *Procedure:* Two-sample z -interval for difference in population proportions, $p_1 - p_2$
Parameters: Let p_1 represent the true proportion of the population of job applications with stereotypically “White” names who receive callbacks. Let p_2 represent the true proportion of the population of job applications with stereotypically “Black” names who receive callbacks.
Checks: Independent random samples; at least 10 successes and 10 failures in each sample, that is, $n_1\hat{p}_1 \geq 10$, $n_1(1 - \hat{p}_1) \geq 10$, $n_2\hat{p}_2 \geq 10$, $n_2(1 - \hat{p}_2) \geq 10$
9. *Procedure:* Chi-square test for homogeneity
Checks: Independent random samples; all expected counts at least 5; $n \leq 0.10N$

Hypotheses: H_0 : There is no difference in the true distributions of highest school levels attained between Whites, Blacks, Asians, and Hispanics (non-White).

H_a : There is a difference in the true distributions of highest school level attained between Whites, Blacks, Asians, and Hispanics (non-White).

10. *Procedure:* Linear regression

Parameter: Let β represent the true slope of the regression line relating selling price to miles noted on the odometer for all used cars of a particular model.

Checks: Random sample; scatterplot is roughly linear; no pattern in the residual plot; The resulting coefficient of determination gives the percent of variability in the dependent variable (selling price) that can be explained by the independent variable (miles on odometer).

11. *Procedure:* One-sample t -interval for population mean μ

Parameter: Let μ represent the true mean pause length in the population of telemarketing calls.

Checks: Random sample; population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; $n \leq 0.10N$

12. *Procedure:* Two-sample z -test for difference in population proportions, $p_1 - p_2$

Parameters: Let p_1 represent the true proportion of the population of White students who report that, at times, they feel unsafe to go to school. Let p_2 represent the true proportion of the population of non-White students who report that, at times, they feel unsafe to go to school.

Checks: Independent random samples; $n_1\hat{p}_C \geq 10$, $n_1(1 - \hat{p}_C) \geq 10$, $n_2\hat{p}_C \geq 10$, and $n_2(1 - \hat{p}_C) \geq 10$, where $\hat{p}_C = \frac{x_1 + x_2}{n_1 + n_2}$; $n_1 \leq 0.10N_1$ and $n_2 \leq 0.10N_2$

Hypotheses: $H_0: p_1 - p_2 = 0$, $H_a: p_1 - p_2 < 0$

13. *Procedure:* Two-sample t -interval for difference in population means, $\mu_1 - \mu_2$

Parameters: Let μ_1 represent the true mean weight of the population of elephants living in captivity. Let μ_2 represent the true mean weight of the population of elephants living in the wild.

Checks: Independent random samples; for each sample, population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; for each sample, $n \leq 0.10N$

14. *Procedure:* Linear regression

Parameter: Let β represent the true slope of the regression line relating pulse rate to quantity of caffeine consumed in a sitting.

Checks: Random sample; scatterplot is roughly linear; no pattern in the residual plot; The resulting slope gives the predicted change in pulse rate per additional ounce of coffee consumed.

15. *Procedure:* One-sample z -test for population mean μ . (Note that we use a normal distribution rather than a t -distribution because the population standard deviation is given.)

Parameter: Let μ represent the true mean hourly wage paid for summer work to the population of high school students.

Checks: Random sample; population is normal or sample is large, at least 30, or sample distribution is unimodal and symmetric; $n \leq 0.10N$

Hypotheses: $H_0: \mu = 10.50$, $H_a: \mu < 10.50$

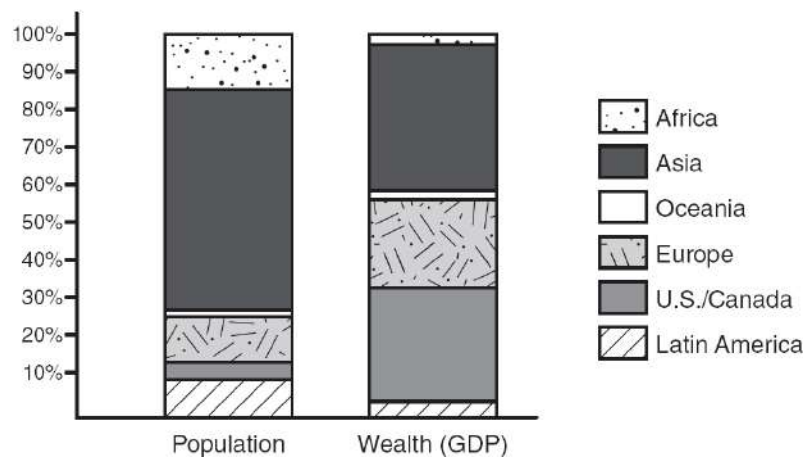
Quiz 33 (Pages 328–336)

1. (a)

Region	Population (in millions)	% of World Population	Wealth (GDP) (in billions)	% of World GDP
Africa	1,341	17.2	2,995	2.8

Region	Population (in millions)	% of World Population	Wealth (GDP) (in billions)	% of World GDP
Asia	4,641	59.5	40,248	38.1
Oceania	43	0.6	2,002	1.9
Europe	748	9.6	24,881	23.6
U.S./Canada	369	4.7	31,319	29.7
Latin America	654	8.4	4,068	3.9
World Total	7,796	100.0	105,513	100.0

- (b) The segmented bar chart treats each bar as a whole and divides each bar proportionally into segments corresponding to the percentage in each group.



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The segmented bar chart shows that both Europe and the U.S./Canada have a hugely disproportionate share of the wealth of the world when compared with their populations, while Asia, Africa, and Latin America have a low share of the wealth of the world when compared with their populations.

- (c) The slopes give each region's wealth in dollars per person. The U.S./Canada region has the steepest slope and thus the highest \$/person, followed by Oceania, Europe, Asia, Latin America, and Africa.
- (d) There are many possible answers. For example, switching to GPI takes into account industrial pollution, which is likely much higher in the U.S./Canada than in the islands of the South Pacific. With GDP per capita as the index, the U.S./Canada ranked slightly higher than Oceania. With CPI per capita as the index, the value for Oceania might go up and the value for U.S./Canada might go down sufficiently for the ranking to be reversed.
2. (a) Fifteen percent of all U.S. Black men are likely to go to prison for the first time by age 25.
- (b) No U.S. White males are likely to go to prison for the first time between ages 60 and 70.
- (c) Age 30. The graph indicates that 16% of all U.S. Hispanic males are likely to go to prison, and half of them (corresponding to 8% on the vertical axis) will go to prison for the first time by age 30.
- (d) Answers should explain how addressing the disproportionate suspensions among students of color should lower the initial slopes of the Black and Hispanic graphs. An answer might also conclude that

dismantling the school-to-prison pipeline would move the x -intercept for all 3 curves to the right—that is, raise the ages for first admissions to prison for all young people.

3. (a) Predicted total SAT = $903.36 + 0.00116(\text{Income})$
 - (b) Every \$1,000 of additional family income is associated with an average of 1.16 points higher on a student's total SAT score.
 - (c) The coefficient of determination $R^2 = 94.0\%$. That is, 94.0% of the variation in total SAT scores is explained by a linear relationship between total SAT and Family Income.
 - (d) Predicted Total SAT = $903.36 + 0.00116(63,500) = 977.02$
Residual = Actual – Predicted = $970 - 977.02 = -7.02$; that is, the actual total SAT score of this student is 7.02 *less* than what was predicted by the family income.
 - (e) $S = 20.936$ is the standard deviation of the residuals. In other words, it is a measure of how the points are vertically spread around the regression line.
 - (f) Correlation does not imply causation. A possible confounding variable, for example, could be instructional money spent per student. Leveling per student expenditures across school districts might lead to higher SAT scores in economically disadvantaged areas.
4. (a) There is a moderate, negative, nonlinear relationship between GNI and percentage of GNI spent on development aid.
 - (b) While the United States has the greatest GNI of any country in the sample, it spends one of the lowest percentages of GNI on development aid of any country in the sample.
 - (c) Because the United States GNI is so large, even a small percentage of it would be large. Thus, the United States will show the highest, or one of the highest, actual amounts spent on development aid.
 - (d) While Belgium, Denmark, Netherlands, Norway, and Sweden have the lowest GNIs compared with the other countries in this sample, they invest the highest percent of GNI in soft power (development aid) compared with the other countries.
 - (e) There is a moderate to strong, negative, linear relationship between $\ln(\text{GNI})$ and $\ln(\text{percentage of GNI spent on development aid})$.
 - (f) The scatterplot of $\ln(\text{GNI})$ versus $\ln(\text{percentage of GNI spent on development aid})$ shows a linear form and there are no leftover patterns in the residual plot.
 - (g) $r = -\sqrt{0.600} = -0.7746$. (Note that r is negative as it takes the same sign as the slope.)
 - (h) For each one unit increase in the natural logarithm of GNI, the natural logarithm of the percent of GNI spent on development aid *decreases* on average by 0.4179.
 - (i) $\ln(\text{percentage GNI}) = 2.4135 + (-0.4179)\ln(\text{GNI})$
 - (j) $\ln(\text{percentage GNI}) = 2.4135 + (-0.4179)\ln(23,400) = 2.4135 - 0.4179(10.0605) = -1.7908$ and so $\text{percentage GNI} = e_{-1.7908} = 0.167$. The predicted percentage of GNI spent on development aid for the U.S. is 0.167%, which is an overestimate of the actual value of 0.15%.

5. (a)

	SOHR			
UN		Reported Deaths	Unreported Deaths	Totals
	Reported Deaths	2,635	992	3,627

	SOHR			
	Unreported Deaths	425	$D - 4,052$	$D - 3,627$
	Totals	3,060	$D - 3,060$	D

- (b) Relative frequencies should be approximately equal if there is independence. The expected cell formula $\text{Exp} = \frac{(\text{Row sum}) \times (\text{Column sum})}{N}$ gives $2,635$ $2,635 = \frac{(3,627)(3,060)}{D}$. (If you look at rows, you get instead $\frac{2,635}{3,627} = \frac{425}{D - 3,627}$. If you look at columns, you get $\frac{2,635}{3,060} = \frac{992}{D - 3,060}$.) Solving any of these equations gives $D = 4,212$.
- (c) *Convenience samples* are based on choosing individuals who are easy to reach and tend to produce data highly unrepresentative of the entire population. In this case, only very readily obtainable reports of deaths are included, violating the assumption of a random sample of deaths for that month.
- (d) Both samples will tend to report the same large casualty events, thus violating the assumption of independence.
- (e) The calculated value of D is almost certainly an underestimate. As shown in parts (c) and (d), the reported deaths tend to be convenience samples of more widely reported deaths and with different samples tending to report the same deaths.
6. (a) This is an example of cluster random sampling, which allows for a quicker, less expensive method of sampling than a simple random sample. It assumes that each cluster, in this case each county, is representative of the whole state.
- (b) Number the counties 1 through 67. Use a random number generator to pick two unique integers between 1 and 67 (ignoring repeats). The two counties to be used will consist of the two counties with numbers corresponding to the two unique numbers.
- (c) The purpose of randomization in picking the 90 jurors is to obtain a sample that is representative of all eligible White jurors in the state so that any conclusions can be generalized to all eligible White jurors in the state.
- (d) Random assignment here is used to minimize the effect of possible confounding variables. For example, if self-selection is allowed, perhaps some White jurors are uncomfortable with people of color and would choose to be on the all-White juries. Because it is likely that these same jurors are more likely biased against a defendant of color, it might be impossible to say whether any conclusions can be made based on the racial makeup of a jury.
7. (a) “Should books with depictions of slavery be banned from school libraries?”
- (b) There are many possible answers. For example, “By restricting information and discouraging freedom of thought, censors undermine one of the primary functions of education: teaching students how to think for themselves. Should books with depictions of slavery be banned from school libraries?” This wording bias will lead to an underestimate of the true proportion of parents in favor of banning books with depictions of slavery.
- (c) There are many possible answers. For example, “It is potentially dangerous to expose schoolchildren to some controversial topics. Should books with depictions of slavery be banned from school libraries?” This wording bias will lead to an overestimate of the true proportion of parents in favor of banning books with depictions of slavery.
- (d) $P(\text{liberal} \cap \text{favor banning}) = P(\text{liberal}) \times P(\text{favor banning} | \text{liberal}) = (0.26)(0.09) = 0.0234$
- (e) $P(\text{favor banning}) = P(\text{liberal} \cap \text{favor banning}) + P(\text{moderate} \cap \text{favor banning}) + P(\text{conservative} \cap \text{favor banning}) = (0.26)(0.09) + (0.38)(0.12) + (0.36)(0.15) = 0.123$

$$(f) P(\text{liberal} | \text{favor banning}) = \frac{P(\text{liberal} \cap \text{favor banning})}{P(\text{favor banning})} = \frac{0.0234}{0.123} = 0.190$$

8. (a) $P(\text{Blast zone}) = \frac{5.5}{38.8} = 0.14175$ and $P(\text{Latino} \cap \text{Blast zone}) = P(\text{Blast zone})P(\text{Latino} | \text{Blast zone}) = (0.14175)(0.75) = 0.10631$

$$(b) P(\text{Blast zone} | \text{Latino}) = \frac{P(\text{Latino} \cap \text{Blast zone})}{P(\text{Latino})} = \frac{0.10631}{0.390} = 0.27259$$

(c) Is $P(\text{Blast zone} | \text{Latino}) = P(\text{Blast zone})$? No, $0.27259 \neq 0.14175$. So, in California, being Latino and living within a blast zone are not independent events.

(d) This is a binomial with $n = 5$ and $p = 0.14175$. Then $P(X \geq 2) = 1 - [(0.85825)^5 + 5(0.14175)(0.85825)^4] = 0.14979$. Calculator software gives $1 - \text{binomcdf}(5, 0.14175, 1) = 0.14979$.

(e) Another possible explanatory variable is poverty. It could be that blast zones tend to be associated with low-income housing and that a higher percentage of people of color than White people are low income.

9. (a) Roughly the same percentage of victims were White and were non-White. Most murders with White victims were committed by White offenders, and most murders with non-White victims were committed by non-White offenders.

$$(b) P(\text{White victim} | \text{execution}) = \frac{P(\text{White victim} \cap \text{execution})}{P(\text{execution})}, \text{ which leads to } 0.76 = \frac{P(\text{White victim} \cap \text{execution})}{0.012} \text{ and } P(\text{White victim} \cap \text{execution}) = (0.76)(0.012) = 0.00912$$

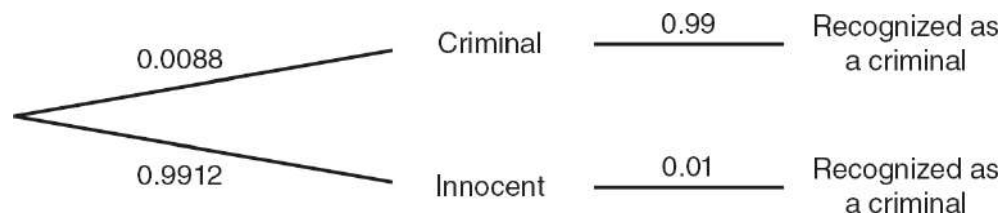
$$P(\text{non-White victim} | \text{execution}) = \frac{P(\text{non-White victim} \cap \text{execution})}{P(\text{execution})}, \text{ which leads to } 0.24 = \frac{P(\text{non-White victim} \cap \text{execution})}{0.012} \text{ and } P(\text{non-White victim} \cap \text{execution}) = (0.24)(0.012) = 0.00288$$

$$(c) P(\text{execution} | \text{White victim}) = \frac{P(\text{White victim} \cap \text{execution})}{P(\text{White victim})} = \frac{0.00912}{0.505} = 0.0181$$

$$P(\text{execution} | \text{non-White victim}) = \frac{P(\text{non-White victim} \cap \text{execution})}{P(\text{non-White victim})} = \frac{0.00288}{0.495} = 0.00582$$

(d) $\frac{0.0181}{0.00582} = 3.11$, so when the defendant is executed, it is more than three times as likely that the victim was White than non-White.

10. (a) $P(\text{criminal}) = 0.0088$. $P(\text{innocent}) = 1 - 0.0088 = 0.9912$.



[Follow for extended description](#)

$$P(\text{criminal} \cap \text{identified as criminal}) = (0.0088)(0.99) = 0.008712$$

$$P(\text{innocent} \cap \text{identified as criminal}) = (0.9912)(0.01) = 0.009912$$

$$P(\text{identified as criminal}) = 0.008712 + 0.009912 = 0.018624$$

$$(b) P(\text{criminal} | \text{identified as criminal}) = \frac{P(\text{criminal} \cap \text{identified as criminal})}{P(\text{identified as criminal})} = \frac{0.008712}{0.018624} = 0.468$$

Alternatively, a table of counts can be constructed for a hypothetical population. In this case, let's assume that the original given percentages hold exactly for a population of size 1,000,000.

	Identified as criminal	Identified as innocent	Total
Criminal			
Innocent			
Total			1,000,000

Numbers in parentheses indicate the order in which the cells can be calculated.

	Identified as criminal	Identified as innocent	Total
Criminal	(3) 8,712	(4) 88	(1) 8,800
Innocent	(5) 9,912	(6) 981,288	(2) 991,200
Total	(7) 18,624	(8) 981,376	1,000,000

Then, the conditional probability in which we are interested can be calculated straight from the first column of the table: $P(\text{criminal} \mid \text{identified as criminal}) = \frac{8,712}{18,624} = 0.468$.

- (c) There is a high risk of police misidentifying and acting against innocent people. By using this “99 percent accurate” facial recognition software, the probability that someone identified as criminal is actually a criminal is only 0.468.
- (d) Yes, a serious civil rights concern is that facial recognition is creating de facto algorithm-based racial profiling.
11. (a) This study looked backward by examining existing data and, therefore, was a retrospective study. Since no treatment was applied—that is, no one was told to use this forensic evidence—this study was not an experiment.
- (b) This is a binomial distribution with $n = 2,500$ and $p = \frac{257}{268} = 0.959$.
Thus, $E(X) = np = 2,500(0.959) = 2,397.4$.
The standard deviation $= \sqrt{np(1-p)} = \sqrt{2,500(0.959)(0.041)} = 9.915$.
- (c) The probability of a case resulting in the death penalty, given it involves flawed forensic testimony, is $\frac{32}{257} = 0.1245$. Thus, the probability of the case not resulting in the death penalty is $1 - 0.1245 = 0.8755$. The probabilities of a death penalty occurring on the first or second case examined, respectively, are 0.1245 and $(0.8755)(0.1245) = 0.1090$. So, the probability that the first case with a resulting death penalty doesn’t occur until at least the third case examined is

$$1 - (0.1245 + 0.1090) = 0.7665$$

[Or recognize that this is a geometric distribution with $p = 0.1245$. Then $P(X \geq 3) = 1 - P(X \leq 2) = 1 - \text{geometcdf}(0.1245, 2) = 1 - 0.2335 = 0.7665$.]

- (d) “The misapplication of forensic science” and “eyewitness misidentification” are not mutually exclusive because, if so, their intersection would be the empty set. Then the probability of their union would be $0.50 + 0.75 - 0 = 1.25 > 1$. It is possible that they are independent because, if so, the probability of their intersection would be $(0.50)(0.75)$ and the probability of their union would be

$$0.50 + 0.75 - (0.5)(0.75) = 0.875 \leq 1$$

12. (a) We have a normal distribution with mean 3,287 and standard deviation 250. Then $P(X > 3,500) = P\left(z > \frac{3,500 - 3,287}{250}\right) = 0.1971$

(b) We have a binomial distribution with $n = 5$ and $p = 0.1971$. Then $P(X \geq 3) = 1 - P(X \leq 2) = 1 - \text{binomcdf}(5, 0.1971, 2) = 0.05572$.

(c) We have a normal distribution with mean 3,287 and standard deviation $\frac{250}{\sqrt{5}} = 111.80$. Then

$$P(\bar{x} > 3,500) = P\left(z > \frac{3,500 - 3,287}{111.80}\right) = 0.0284.$$

(d) Possible answers might range from average yearly proceeds (about \$35 billion) to average number of slaves in the world on any given day (21 to 29 million), from numbers of child slaves to numbers of women in sexual slavery.

13. (a) A complete answer addresses center, spread, shape, and unusual features such as outliers, and mentions context. The median age of hate site members is approximately 23. The ages go from 14 to 65 with a range of $65 - 14 = 51$ and an IQR = $30 - 17 = 13$.

There are 6 outliers between 50 and 65. Although it is not usually possible to describe distribution shape from just a boxplot, in this case the distribution of ages does appear to be skewed to the right (to the larger values).

- (b) In a skewed right distribution, the mean is usually larger than the median. So, we would conclude that the quotient $\frac{\text{Mean age}}{\text{Median age}}$ is greater than 1.
- (c) If mostly the younger, computer savvy members of the hate groups are the ones who are online, the median age of all hate group members is probably greater than the median age of those self-reporting on the site, 23.
- (d) This is a binomial with $n = 3$ and $p = 0.186$. Then $P(X \geq 2) = 0.0909$ or

$$3(0.186)^2(0.814) + (0.186)^3 = 0.0909$$

$$[\text{Or } 1 - P(x \leq 1) = 1 - \text{binomcdf}(n = 3, p = 0.186, x = 1) = 1 - 0.9091 = 0.0909.]$$

Quiz 34 (Pages 337–343)

1. (a) $0.19 + 0.17 - 0.06 = 0.30$

(b) *Parameter:* Let p represent the true proportion of the population of students in grades six through ten who are victims of moderate to frequent bullying.

Procedure: One-sample z -interval for a population proportion

Checks: We must assume this was a random sample, $n\hat{p} = (15,686)(0.17) = 2,667 \geq 10$, $n\hat{q} = (15,686)(0.83) = 13,019 \geq 10$, and $n = 15,686$ is less than 10% of all students in grades six through ten.

Mechanics: Calculator software gives (0.16415, 0.1759).

Conclusion in context: We are 95% confident that the proportion of all students in grades six through ten who are victims of moderate to frequent bullying is between 0.16415 and 0.1759.

- (c) Because 0.22 is not in the interval of plausible values for the population proportion, there is convincing evidence that the proportion of all students who are victims of moderate to frequent bullying is different from the 22% claimed in the one study.
- (d) Issues that might be addressed in planning a survey include stratification among the grade levels, anonymity, and timing (early enough in the school year to allow implementation of a plan and again later in the year to assess the effectiveness of any prevention and intervention program).
2. (a) *Parameter:* Let μ represent the true mean number of firearm suicides in the U.S. each year among the population of all years.
- Procedure:* A one-sample t -interval
- Checks:* We are given a random sample. The nearly normal condition seems reasonable because the sample data show no outliers or strong skewness. We assume that 7 years are less than 10% of all

years under consideration. Under these conditions, the mean firearm suicides per year can be modeled by a t -distribution with $n - 1 = 7 - 1 = 6$ degrees of freedom.

Mechanics: Calculator software gives the interval (20408, 21850). [We can instead calculate $22,128.6 \pm 2.447 \frac{779.6}{\sqrt{7}} = 22,128.6 \pm 721.0$.]

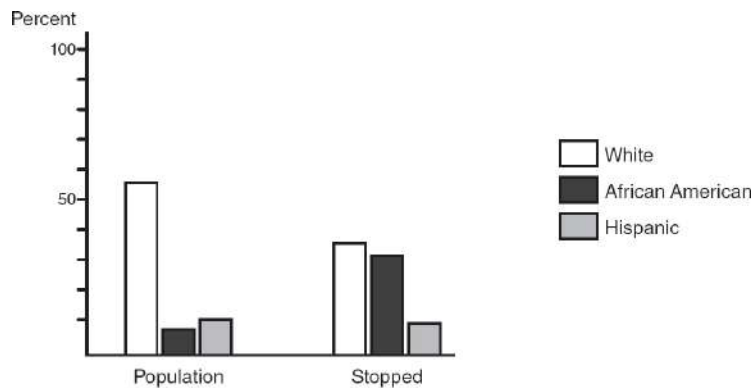
Conclusion in context: We are 95% confident that the true mean number of firearm suicides in the U.S. each year is between 21,408 and 22,850.

- (b) It could well be true (actually, it is true) that most guns in Israel are technically owned by the government (by the IDF—the Israel Defense Forces). The 6.7 guns per 100 people statistic is a significant underestimate of guns in the population.
 - (c) The conversation would change from number of guns in the population to strictness of gun laws pertaining to gun ownership.
 - (d) The U.S. has 120.5 guns per 100 people but only $\frac{120.5}{3} = 40.2$ gun owners per 100 people. Thus, the “firearm suicides per gun owner” rate in the U.S. will be three times as great as the “firearm suicides per gun” rate. How “rates” are calculated can lead to different conclusions.
3. (a) A cumulative frequency plot that rises quickly at first and then rises at a much slower rate later corresponds to a histogram with a large area under the curve early and much less area later. Thus, it is *skewed right* (skewed to the higher values of lead). In skewed right distributions, we expect the *mean* to be greater than the median.
- (b) These figures indicate an extremely serious lead problem in the Flint water supply. The research team found that 25% of the homes had a lead water supply level of at least 15 ppb. Because $25\% > 10\%$ and $15 \text{ ppb} > 5 \text{ ppb}$, the percent of homes with at least 5 ppb is larger than 10%.
- (c) With $H_0: \mu = 10$ and $H_a: \mu > 10$, a Type I error (mistaken rejection of a true null hypothesis) means that we think the lead level is greater than 10 ppb when, in fact, it is 10 ppb. So, we might institute costly and unnecessary cleanup measures. A Type II error (mistaken failure to reject a false null hypothesis) means that the mean lead level is greater than 10 ppb but we don’t have enough evidence to think so. So, we might fail to institute a necessary cleanup. Clearly, with people’s, and especially children’s, health at stake, a Type II error should be more of a concern.
4. (a) The sample proportion is in the middle of the confidence interval, so $\hat{p} = \frac{(0.21882 + 0.28258)}{2} = 0.2507$. The margin of error is $0.28258 - 0.2507 = 0.03188$. We have $1.96 \sqrt{\frac{0.2507(1 - 0.2507)}{n}} = 0.03188$, which gives $\sqrt{n} = 26.64666$, and n is about 710.
- (b) The mosaic plot shows (1) that the sample contained roughly twice as many resumes with “White” names than “non-White” names, and (2) in the sample, a greater proportion of resumes with “White” names than “non-White” names resulted in positive responses (offered interviews).
- (c) With such a small p -value, $0.00486 < 0.05$, there is sufficient evidence to reject H_0 ; that is, there is convincing evidence that resumes with “White” names result in a greater proportion of positive responses (offered interviews) than resumes with “non-White” names.
- (d) Because 0 is in the confidence interval—that is, the interval contains both negative and positive plausible values—there is not convincing evidence to conclude that discrimination against African-Americans is trending higher.
5. (a) This is an example of random assignment applied to a block design. Each pair of children is a block, and in each block one child is randomly assigned to receive the gun safety “treatment” while the other child receives the car safety “treatment.”
- (b) $H_0: p_G = p_C$ and $H_a: p_G > p_C$ where p_G = proportion of all youths 8 to 12 years old who will tell an adult about a gun in the room after being shown a gun safety video and where p_C = proportion of all

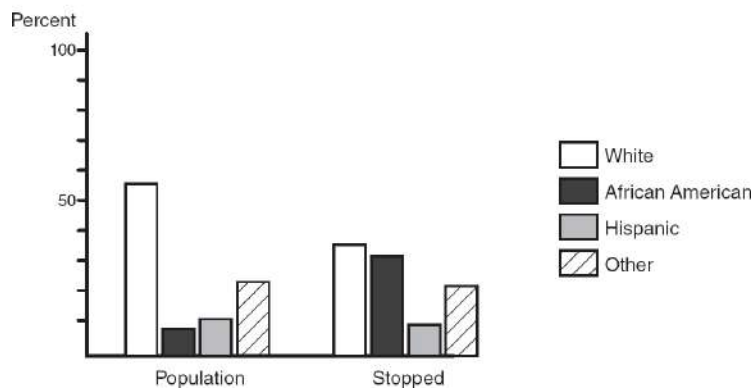
youths 8 to 12 years old who will tell an adult about a gun in the room after being shown a car safety video.

- (c) The two sample proportions, $\hat{p}_G$ and $\hat{p}_C$, are 4.167 standard errors apart.
 - (d) This is a one-sided test with a p -value of $\text{normalcdf}(4.167, 1000) = 0.000$. With this small a p -value, $0.000 < 0.01$, there is sufficient evidence to reject the null hypothesis. In other words, there is convincing evidence that the proportion of all youths 8 to 12 years old who will tell an adult about a gun in the room after being shown a gun safety video is greater than the proportion of all youths 8 to 12 years old who will tell an adult about a gun in the room after being shown a car safety video.
 - (e) We are 99% confident that the proportion of all youths 8 to 12 years old who will tell an adult about a gun in the room after being shown a gun safety video is between 0.09342 and 0.36676 greater than the proportion of all youths 8 to 12 years old who will tell an adult about a gun in the room after being shown a car safety video.
 - (f) Since the confidence interval is entirely greater than 0, the two procedures give consistent results.
 - (g) Younger children find authority figures in uniforms to be especially persuasive. This finding suggests that we can use law enforcement to develop interventions aimed at promoting safe behavior among young children.
 - (h) Perhaps parents with guns are more likely to talk to their children about gun safety than parents without guns.
6. (a) *Parameters:* Let μ_{LEAD} represent the true mean yearly criminal and legal system costs of the population of low-level drug and prostitution offenders who are participating in the LEAD program. Let μ_{control} represent the true mean yearly criminal and legal system costs of the population of low-level drug and prostitution offenders who are not participating in the LEAD program.
Hypotheses: $H_0: \mu_{\text{LEAD}} = \mu_{\text{control}}$ and $H_a: \mu_{\text{LEAD}} < \mu_{\text{control}}$.
Procedure: This is a two-sample t -test for means.
Checks: We are given samples that, by design, are assumed representative and independent. The sample sizes, 203 and 115, are large enough so that by the CLT, the distribution of sample means is approximately normal and a t -test may be run. The sample sizes are assumed to be less than 10% of the population of all low-level drug and prostitution offenders.
Mechanics: Calculator software gives $t = -33.41$ with a p -value of $\gg 0.000$.
Conclusion in context with linkage to the p -value: With this small of a p -value, $0.000 < 0.05$, there is very strong evidence to reject H_0 . In other words, there is convincing evidence that for those adults eligible for the LEAD program, the mean yearly criminal and legal system costs of those participating in the program are less than that for those not participating.
- (b) The null hypothesis was rejected. So, the possibility is of a Type I error—that is, of rejecting a true null hypothesis. A Type I error would mean that in reality, after participating in the program, the LEAD group would have the same mean days per year in jail as the control group. However, in error, we think the LEAD group has significantly fewer mean days per year in jail than the control group. A possible consequence is that the LEAD program is expanded, thinking it will reduce the future mean days per year in jail when, in reality, it won't.
 - (c) It is very difficult to “choose” representative samples. Using randomization has been shown to be the most effective technique to minimize the effect of possible confounding variables.
 - (d) “We ran a large sample statistical study and noted that for our sample, the average yearly criminal and legal costs spent on those offenders participating in the LEAD program were significantly lower than the average spent on those not participating in the program. The difference was so great that it is highly unlikely that it could have occurred by chance.”

7. (a) $\frac{1,710}{4,658} = 36.7\%$, $\frac{1,423}{4,658} = 30.5\%$, and $\frac{543}{4,658} = 11.7\%$
 $\frac{64,412}{116,774} = 55.2\%$, $\frac{10,076}{116,774} = 8.6\%$, and $\frac{11,600}{116,774} = 9.9\%$



or



[Follow for extended description](#)

- (b) *Parameter:* Let p represent the true proportion of the population of all stopped drivers who are African American.
Hypotheses: $H_0: p = 0.086$ and $H_a: p > 0.086$
Procedure: One proportion z-test
Checks: A representative (random) sample is assumed; $np_0 = 4,658(0.086) = 400.6 \geq 10$ and $n(1 - p_0) = 4,658(0.914) = 4,257.4 \geq 10$.
Mechanics: Software such as 1-PropZTest on the TI-84 gives $z = 53.4$ with a p -value of $\gg 0.000$.
Conclusion in context with linkage to the p -value: With this low of a p -value, $0.000 < 0.001$, there is convincing evidence that African Americans are being stopped at a higher rate than what would be indicated by their population percentage.
- (c) *Hypotheses:* H_0 : Race/ethnicity (White, African American, Hispanic) and disposition (arrest/citation or not) are independent.
 H_a : There is a relationship between race/ethnicity (White, African American, Hispanic) and disposition (arrest/citation or not).
Procedure: A chi-square test of independence
Checks: Representative (random sample assumed) and all expected cells as shown below are ≥ 5 .

OBS	White	African American	Hispanic
Citation	1,058	481	237
None	652	942	306

EXP	White	African American	Hispanic
Citation	826.2	687.5	262.3
None	883.8	735.5	280.7

Mechanics: Software (such as χ^2 -Test on the TI-84) gives the above expected values and $\chi^2 = 250.61$ with a p -value of ≈ 0.000 .

Conclusion in context with linkage to the p -value: With this low of a p -value, $0.000 < 0.001$, there is convincing evidence of a relationship between race/ethnicity (White, African American, Hispanic) and disposition (arrest/citation or not).

8. (a) *Parameters:* Let μ_{college} represent the mean lifetime earnings of all workers with some college experience. Let $\mu_{\text{high school}}$ represent the mean lifetime earnings of all workers with a high school degree.

Procedure: A two-sample t -interval for the difference of two population means, $\mu_{\text{college}} - \mu_{\text{high school}}$

Checks: We are given independent random samples, and the sample sizes, 250 and 200, are less than 10% of their respective populations. For normality, both sample sizes, 250 and 200, are greater than 30, so the central limit theorem applies. Calculating a t -interval is appropriate.

Mechanics: Calculator software (2-SampTInt) gives (227,000, 373,000).

Conclusion in context: We are 95% confident that the true difference in mean lifetime earnings (college experience minus high school degree) between all workers with some college experience and all workers with a high school degree is between \$227,000 and \$373,000.

- (b) No, there is not convincing evidence that the mean lifetime earnings of workers with some college experience is more than \$250,000 greater than the mean lifetime earnings of workers with a high school degree because the entire interval is not above \$250,000.
- (c) There are many possible answers. One would be the average debt with which people incur for each year of college. A second would be that lifetime earnings for workers with some college experience are probably different depending upon what courses were taken.

9. (a) *Parameters:* Let p_{2024} represent the proportion of U.S. adults in 2024 who would say they see discrimination against Jews. Let p_{2021} represent the proportion of U.S. adults in 2021 who would say they see discrimination against Jews.

Procedure: Two-sample z -interval for the difference between population proportions, $p_{2024} - p_{2021}$

Checks: We are given independent random samples, and the sample sizes, 12,055 and 12,693, are less than 10% of the U.S. adult population in the two years 2021 and 2024. For normality, $n_{2021} \hat{p}_{2021} = 2,411 \geq 10$, $n_{2021} (1 - \hat{p}_{2021}) = 12,055 - 2,411 = 9,644 \geq 10$, $n_{2024} \hat{p}_{2024} = 5,077 \geq 10$, and $n_{2024} (1 - \hat{p}_{2024}) = 12,693 - 5,077 = 7,616 \geq 10$.

Mechanics: Computer software (2-PropZInt) gives (0.185, 0.215).

Conclusion in context: We are 99% confident that the true difference in the proportions (2024 minus 2021) of U.S. adults who would say they see discrimination against Jews is between 0.185 and 0.215.

- (b) Yes; because the entire confidence interval from 0.185 to 0.215 is above 0.15, there is convincing evidence that the difference (2024 minus 2021) in the proportions of U.S. adults who would say they see discrimination against Jews is greater than 15%.
- (c) If the surveys had just been from among the population of Jewish U.S. adults, the entire interval would have probably had greater proportions as a particular religious or racial group is probably more aware of discrimination directed against them than is the general population.

10. (a) The sample proportion is in the middle of the interval (0.30, 0.36), so $\hat{p} = 0.33$. The margin of error is $0.36 - 0.33 = 0.03$. The critical z -scores associated with 95% confidence are ± 1.96 . We have $1.96\sqrt{\frac{(0.33)(0.67)}{n}} \leq 0.03$, or $\sqrt{n} \geq \sqrt{\frac{1.96(0.33)(0.67)}{0.03}} = 30.72$ and $n \geq 943.7$. Conclude that the sample size was $n = 944$.

(b) With a very large p -value, there is little or no evidence to reject the null hypothesis. In other words, there is little or no evidence of an association between autism and vaccines.

- (c) Given $p = 0.10$ and $n = 500$, we have $\mu_{\hat{p}} = 0.10$ and $\sigma_{\hat{p}} = \sqrt{\frac{(0.10)(0.90)}{500}} = 0.01342$. Then $P(\tilde{p} > 0.12) = P\left(z > \frac{0.12 - 0.10}{0.01342}\right) = P(z > 1.49) = 0.068$.

(d) $R^2 = (>0.58)^2 = 0.336$. Conclude that 33.6% of the variation in Covid-19 severity is accounted for by the linear regression model between mumps IgG titer and Covid-19 severity.

11. (a) There is a positive, linear, moderate to strong relationship between Temperature Anomaly and Time.

(b) *Parameters:* Let p_{Ind} represent the true proportion of the population of Indians who express a great deal of worry about global warming. Let p_{Chi} represent the true proportion of the population of Chinese who express a great deal of worry about global warming.

Hypotheses: $H_0: p_{\text{Ind}} = p_{\text{Chi}}$, $H_a: p_{\text{Ind}} \neq p_{\text{Chi}}$

Procedure: Hypothesis test for a difference between two proportions

Checks: Independent random samples; $n_{\text{Ind}}\tilde{p}_C = (2,029)(0.417) = 846 \geq 10$, $n_{\text{Ind}}(1 - \tilde{p}_C) = (2,029)(0.583) = 1,183 \geq 10$, $n_{\text{Chi}}\tilde{p}_C = (2,180)(0.417) = 909 \geq 10$, and $n_{\text{Chi}}(1 - \tilde{p}_C) = (2,180)(0.583) = 1,271 \geq 10$, where $\hat{p}_C = \frac{x_1 + x_2}{n_1 + n_2} = \frac{1,319 + 436}{2,029 + 2,180} = 0.417$; $n_{\text{Ind}} \leq 0.10N_{\text{Ind}}$ and $n_{\text{Chi}} \leq 0.10N_{\text{Chi}}$

Mechanics: $z = 29.59$, with a p -value ≈ 0.000

Conclusion in context with linkage to the p -value: With this small of a p -value, $0.000 < 0.05$, there is sufficient evidence to reject H_0 . In other words, there is convincing evidence of a difference between the proportion of Indians who express a great deal of worry about global warming and the proportion of Chinese who express a great deal of worry about global warming.

- (c) Phrases like “climate change deniers,” “corporate lobbyists,” and “a dying planet” are clearly nonneutral and leading. The calculated proportion of people who express a great deal of worry about global warming will most likely be greater than the true proportion.

Formulas Given on the AP Statistics Exam

I. Descriptive Statistics

$$\bar{x} = \frac{1}{n} \sum x_i = \frac{\sum x_i}{n}$$

$$s_x = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$\hat{y} = a + bx$$

II. Probability and Distributions

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability Distribution	Mean	Standard Deviation
Discrete random variable X	$\mu_x = E(X) = \sum x_i \cdot P(x_i)$	$\sigma_x = \sqrt{\sum (x_i - \mu_x)^2 \cdot P(x_i)}$
If X has a binomial distribution with parameters n and p , then: $P(X = x) = \binom{n}{x} p^x (1 - p)^{(n-x)}$, where $x = 0, 1, 2, 3, \dots, n$	$\mu_x = np$	$\sigma_x = \sqrt{np(1 - p)}$

[Follow for extended description](#)

III. Sampling Distributions and Inferential Statistics

Standardized test statistic: $\frac{\text{statistic} - \text{parameter}}{\text{standard error of the statistic}}$

Confidence interval: $\text{statistic} \pm (\text{critical value})(\text{standard error of statistic})$

Chi-square statistic: $\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$

Sampling distributions for proportions:

Random Variable	Parameters of Sampling Distribution		Standard Error * of Sample Statistic
For one proportion: $\hat{p}$	$\mu_{\hat{p}} = p$	$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$	$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
For two proportions: $\hat{p}_1 - \hat{p}_2$	$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$	$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	$SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ When $p_1 = p_2$ is assumed: $SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ where $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$

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Sampling distributions for means:

Random Variable	Parameters of Sampling Distribution		Standard Error* of Sample Statistic
For one population: $\bar{X}$	$\mu_{\bar{x}} = \mu$	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$	$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$
For two populations: $\bar{X}_1 - \bar{X}_2$	$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2$	$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

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*Standard deviation is a measurement of variability from the theoretical population. Standard error is the estimate of the standard deviation. If the standard deviation of the statistic is assumed to be known, the standard deviation should be used instead of the standard error.

AP Scoring Guide

The multiple-choice and free-response sections are weighted equally.

There is no penalty for guessing in the multiple-choice section.

To find your score use the following guide:

Multiple-choice section (42 questions)

Number correct $\times 1.19 =$ _____

Free-response section (4 open-ended questions)

Question 1 _____ $\times 1.25 =$ _____
out of 10

Question 2 _____ $\times 1.25 =$ _____
out of 10

Question 3 _____ $\times 1.25 =$ _____
out of 10

Question 4 _____ $\times 1.25 =$ _____
out of 10

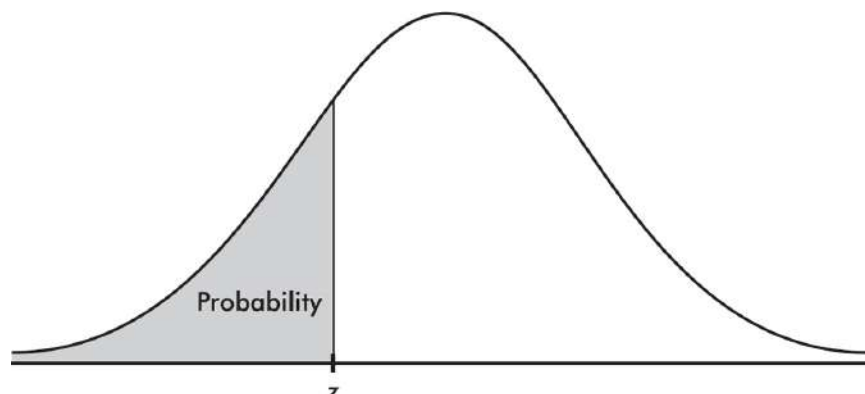
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Total points from multiple-choice and free-response sections = _____

Conversion chart based on average cutoff scores from past exams:

Total Points	AP Score
70–100	5
55–69	4
44–54	3
30–39	2
0–29	1

In the past, roughly 15% of students scored 5, 20% scored 4, 25% scored 3, 20% scored 2, and 20% scored 1 on the AP Statistics exam. Colleges generally require a score of at least 3 for a student to receive college credit.



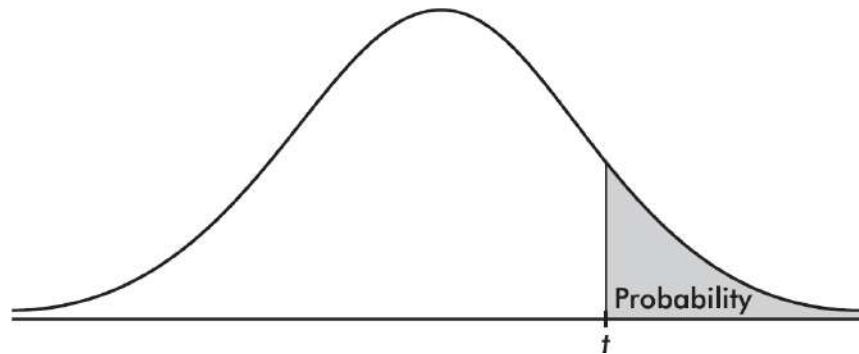
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TABLE A: STANDARD NORMAL PROBABILITIES

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6643	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

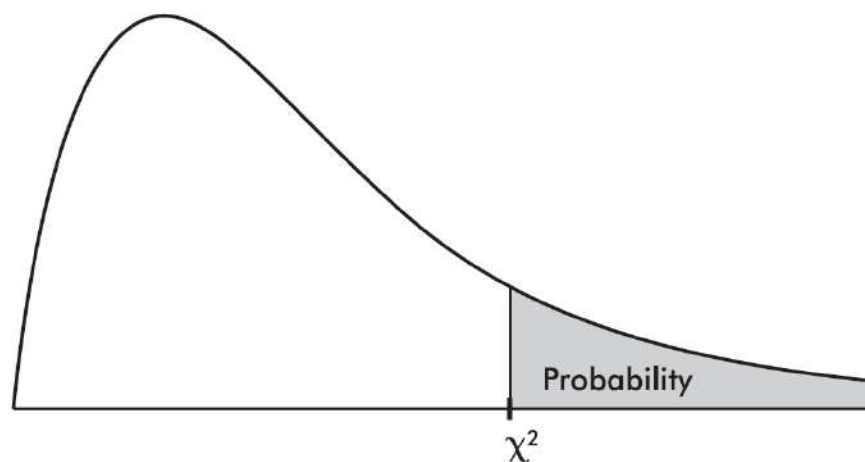


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TABLE B: *t*-DISTRIBUTION CRITICAL VALUES

<i>df</i>	Tail probability <i>p</i>											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792

Tail probability p												
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%
Confidence level C												



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TABLE C: χ^2 CRITICAL VALUES

Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47

Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.51
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61
80	88.13	90.41	93.11	96.58	101.9	106.6	108.1	112.3	116.3	120.1	124.8
100	109.1	111.7	114.7	118.5	124.3	129.6	131.1	135.8	140.2	144.3	149.4

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Procedure	Parameter	Statistic	Conditions	Formula	Calculators*
One-sample z-interval for a proportion	p	$\hat{p}$	- Random sample $n \leq 10\%N$ $n\hat{p} \geq 10, n(1 - \hat{p}) \geq 10$	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$	1-PropZInt 1-Props Z Interval Z-Int: 1π
Two-sample z-interval for a difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ $n_1\hat{p}_1 \geq 10, n_1(1 - \hat{p}_1) \geq 10$ $n_2\hat{p}_2 \geq 10, n_2(1 - \hat{p}_2) \geq 10$	$\hat{p}_1 - \hat{p}_2 \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$	2-PropZInt 2-Prop Z Interval Z-Int: $\pi_1 - \pi_2$

The first row explains a one-sample z-interval for a proportion, where the parameter is p and the statistic is $\hat{p}$. Conditions include a random sample, sample size less than or equal to 10 percent of the population, and at least 10 expected successes and failures. The formula shown is $\hat{p}$ plus or minus z -star times the square root of $\hat{p}$ multiplied by 1 minus $\hat{p}$ divided by n . Calculator options listed include 1-Prop Z Int and 1-Prop Z Interval, Z Interval equals 1π . The second row explains a two-sample z-interval for the difference in proportions, with parameter $p_1 - p_2$ and statistic $\hat{p}_1 - \hat{p}_2$. Conditions include independent random samples, each sample size less than or equal to 10 percent of its population, and at least 10 expected successes and failures in both groups. The formula shown is $\hat{p}_1 - \hat{p}_2$ plus or minus z -star multiplied by the square root of the sum of $\hat{p}_1$ times 1 minus $\hat{p}_1$ over n_1 and $\hat{p}_2$ times 1 minus $\hat{p}_2$ over n_2 . Calculator options include 2-Prop Z Int and 2-Prop Z Interval, Z-Int equals $\pi_1 - \pi_2$.

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Procedure	Parameter	Statistic	Conditions	Formula	Calculators*
One-sample t-interval for a mean	μ	$\bar{x}$	- Random sample $n \leq 10\%N$ - Population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$ with $df = n - 1$	TInterval 1-Sample t Interval T-Int: 1μ
Two-sample t-interval for a difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ - For each sample, population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}$ with df from calculator	2-SampTInt 2-Sample t Interval T-Int: $\mu_1 - \mu_2$
Paired data t-interval for a mean difference	μ_{diff}	$\bar{x}_{diff}$	- Random sample of paired data $n_{diff} \leq 10\%N_{diff}$ - Population of differences $\approx$ normal OR $n_{diff} \geq 30$ OR sample differences shows no strong skewness or outliers	$\bar{x}_{diff} \pm t^* \frac{s_{diff}}{\sqrt{n_{diff}}}$ with $df = n_{diff} - 1$	TInterval 1-Sample t Interval T-Int: 1μ

The table shows six columns labeled Procedure, Parameter, Statistic, Conditions, Formula, and Calculators. The first row describes a one-sample t-interval for a mean. The parameter is μ and the statistic is $\bar{x}$. Conditions include a random sample, sample size less than or equal to 10 percent of the population, and either a population approximately normal or a sample size of at least 30 with no strong skewness or outliers. The formula shown is $\bar{x}$ plus or minus t^* multiplied by s over the square root of n , with degrees of freedom equal to n minus 1. Calculator options include T Interval and 1-Sample t Interval. The second row describes a two-sample t-interval for a difference in means. The parameter is $\mu_1 - \mu_2$ and the statistic is $\bar{x}_1 - \bar{x}_2$. Conditions include independent random samples, sample sizes less than or equal to 10 percent of their populations, and for each sample either a population approximately normal or sample size at least 30 with no strong skewness or outliers. The formula shown is the difference between sample means plus or minus t^* multiplied by the square root of s_1 squared over n_1 plus s_2 squared over n_2 , with degrees of freedom obtained from a calculator. Calculator options include 2-SampTInt and 2-Sample t Interval. The third row describes a paired-data t-interval for a mean difference. The parameter is μ_{diff} and

the statistic is $\bar{x}_{\text{diff}}$. Conditions include a random sample of paired data, paired sample size less than or equal to 10 percent of the population, and either approximately normal population differences or at least 30 paired differences with no strong skewness or outliers. The formula shown is $\bar{x}_{\text{diff}} \pm t^* \frac{s_{\text{diff}}}{\sqrt{n_{\text{diff}}}}$, with degrees of freedom equal to $n_{\text{diff}} - 1$. Calculator options include T Interval and 1-Sample t Interval.

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Procedure	Null Hypothesis	Conditions	Formula	Calculators*
One-sample z-test for a proportion	$H_0: p = p_0$	- Random sample $n \leq 10\%N$ $np_0 \geq 10, n(1 - p_0) \geq 10$	$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$	1-PropZTest 1-Prop Z Test Z-Test: 1π
Two-sample z-test for a difference in proportions	$H_0: p_1 - p_2 = 0$ OR $H_0: p_1 = p_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ $n_1\hat{p}_c \geq 10, n_1(1 - \hat{p}_c) \geq 10$ $n_2\hat{p}_c \geq 10, n_2(1 - \hat{p}_c) \geq 10$ where $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$	$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_c(1 - \hat{p}_c)}{n_1} + \frac{\hat{p}_c(1 - \hat{p}_c)}{n_2}}}$	2-PropZTest 2-Prop Z Test Z-Test: $\pi_1 - \pi_2$

The table consists of six columns, including Procedure, Null Hypothesis, Conditions, Formula, and Calculators. The first row describes a one-sample z-test for a proportion. The null hypothesis is $H_0: p = p_0$. Conditions include a random sample, sample size less than or equal to 10 percent of the population, and at least 10 expected successes and failures using p_0 . The formula shown is $z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$. Calculator options include 1-Prop Z Test and 1-Prop Z Test. The second row describes a two-sample z-test for a difference in proportions. The null hypothesis is either $p_1 - p_2 = 0$ or $p_1 = p_2$. Conditions include independent random samples, each sample size less than or equal to 10 percent of its population, and at least 10 expected successes and failures in both samples using the pooled proportion $\hat{p}_c$. The pooled proportion is defined as $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$. The formula shown is $z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_c(1 - \hat{p}_c)}{n_1} + \frac{\hat{p}_c(1 - \hat{p}_c)}{n_2}}}$. Calculator options include 2-Prop Z Test and 2-Prop Z Test.

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Procedure	Null Hypothesis	Conditions	Formula	Calculators*
One-sample t-test for a mean	$H_0: \mu = \mu_0$	- Random sample $n \leq 10\%N$ - Population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$ with $df = n - 1$	T-Test 1-Sample tTest T-Test: 1μ
Two-sample t-test for a difference in means	$H_0: \mu_1 - \mu_2 = 0$ OR $H_0: \mu_1 = \mu_2$	- Independent random samples $n_1 \leq 10\%N_1, n_2 \leq 10\%N_2$ - For each sample, population $\approx$ normal OR $n \geq 30$ OR sample shows no strong skewness or outliers	$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}}$ with df from calculator	2-SampTest 2-Sample tTest T-Test: $\mu_1 - \mu_2$
Paired data t-test for a mean difference	$H_0: \mu_{diff} = 0$	- Random sample of paired data $n_{diff} \leq 10\%N_{diff}$ - Population of differences $\approx$ normal OR $n_{diff} \geq 30$ OR sample differences shows no strong skewness or outliers	$t = \frac{\bar{x}_{diff} - 0}{\frac{s_{diff}}{\sqrt{n_{diff}}}}$ with $df = n_{diff} - 1$	T-Test 1-Sample tTest T-Test: 1μ

The table consists of six columns, including Procedure, Null Hypothesis, Conditions, Formula, and Calculators. The first row describes a one-sample t-test for a mean. The null hypothesis is $H_0: \mu = \mu_0$. Conditions include a random sample, sample size less than or equal to 10 percent of the population, and either a population approximately normal or a sample size of at least 30 with no strong skewness or outliers. The formula shown is $t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$, with degrees of freedom equal to $n - 1$. Calculator options include T-Test and 1-Sample t-Test. The second row describes a two-sample t-test for a difference in means. The null hypothesis is either $\mu_1 - \mu_2 = 0$ or $\mu_1 = \mu_2$. Conditions include independent random samples, sample sizes less than or equal to 10 percent of their populations, and for each sample either a population approximately normal or a sample size of at least 30 with no strong skewness or outliers. The formula shown is $t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}}$, with degrees of freedom obtained from a calculator. Calculator options include 2-SampTest and 2-Sample t Test. The third row describes a paired-data t-test for a mean difference. The null hypothesis is $H_0: \mu_{diff} = 0$. Conditions include a random sample of paired data, paired sample size less than or equal to 10 percent of the population of differences, and either a population of differences approximately normal or a sample size of at least 30 with no strong skewness or outliers. The formula shown is $t = \frac{\bar{x}_{diff} - 0}{\frac{s_{diff}}{\sqrt{n_{diff}}}}$, with degrees of freedom equal to $n_{diff} - 1$. Calculator options include T-Test and 1-Sample t-Test.

than or equal to 10 percent of the population, and either approximately normal population differences or at least 30 paired differences with no strong skewness or outliers. The formula shown is $t = \frac{\bar{x}_{\text{diff}} - 0}{s_{\text{diff}} / \sqrt{n_{\text{diff}}}}$, with degrees of freedom equal to $n_{\text{diff}} - 1$. Calculator options include T-Test and 1-Sample t Test.

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Procedure	Hypotheses	Conditions	Formula	Calculators*
χ^2 -test for independence	H_0 : Two variables are independent. H_a : Two variables are not independent.	- Random sample $n \leq 10\%N$ - All expected counts > 5	$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$ $df = (\# \text{ of rows} - 1)(\# \text{ of columns} - 1)$	χ^2 -Test χ^2 two-way Test χ^2 Test two-way test
χ^2 -test for homogeneity	H_0 : A distribution is the same across 2 or more populations. H_a : A distribution differs across 2 or more populations.	- Random samples from each population $n_i \leq 10\%N_i$ each sample - All expected counts > 5	$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$ $df = (\# \text{ of rows} - 1)(\# \text{ of columns} - 1)$	χ^2 -Test χ^2 two-way Test χ^2 Test two-way test

The table consists of six columns, including Procedure, Hypotheses, Conditions, Formula, and Calculators. The first row describes a chi-square test for independence. The null hypothesis states that two variables are independent, while the alternative hypothesis states that the variables are not independent. Conditions include a random sample, sample size less than or equal to 10 percent of the population, and all expected counts greater than 5. The formula shown is chi-square equals the sum of observed minus expected squared divided by expected. Degrees of freedom are calculated as the number of rows minus 1 multiplied by the number of columns minus 1. Calculator options include chi-square Test and chi-square two-way Test. The second row describes a chi-square test for homogeneity. The null hypothesis states that a distribution is the same across two or more populations, while the alternative hypothesis states that the distribution differs across two or more populations. Conditions include random samples from each population, each sample size less than or equal to 10 percent of its population, and all expected counts greater than 5. The same chi-square formula is shown, along with the same degrees of freedom calculation based on rows and columns. Calculator options again include chi-square Test and chi-square two-way Test.

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$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability Distribution	Mean	Standard Deviation
Discrete random variable X	$\mu_x = E(X) = \sum x_i \cdot P(x_i)$	$\sigma_x = \sqrt{\sum (x_i - \mu_x)^2 \cdot P(x_i)}$
If X has a binomial distribution with parameters n and p , then: $P(X = x) = \binom{n}{x} p^x (1 - p)^{(n-x)}$, where $x = 0, 1, 2, 3, \dots, n$	$\mu_x = np$	$\sigma_x = \sqrt{np(1 - p)}$

A probability reference table displays formulas for probability rules and probability distributions. At the top, the addition rule is shown as P of A union B equals P of A plus P of B minus P of A intersection B. Beside it, the conditional probability formula is shown as P of A given B equals P of A intersection B divided by P of B. Below, a table with columns labeled Probability Distribution, Mean, and Standard Deviation summarizes formulas for random variables. The first row describes a discrete random variable X. The mean formula is mu-sub-x equals the expected value E of X, calculated as the sum of x-sub-i multiplied by P of x-sub-i. The standard deviation formula is sigma-sub-x equals the square root of the sum of x-sub-i minus mu-sub-x squared multiplied by P of x-sub-i. The second row describes a binomial distribution with parameters n and p. The probability formula is P of X equals x equals n choose x multiplied by p to the x power times 1 minus p to the n minus x power, where x ranges from 0 to n. The mean is mu-sub-x equals n multiplied by p, and the standard deviation is sigma-sub-x equals the square root of n multiplied by p multiplied by 1 minus p.

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Random Variable	Parameters of Sampling Distribution		Standard Error * of Sample Statistic
For one proportion: $\hat{p}$	$\mu_{\hat{p}} = p$	$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$	$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
For two proportions: $\hat{p}_1 - \hat{p}_2$	$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$	$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	$SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ When $p_1 = p_2$ is assumed: $SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ where $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$

The table has columns labeled Random Variable, Parameters of Sampling Distribution, and Standard Error of Sample Statistic. The first row describes one proportion using $\hat{p}$. The mean of the sampling distribution is $\mu_{\hat{p}} = p$. The standard deviation is $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$. The standard error formula is $SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$. The second row describes two proportions using $\hat{p}_1 - \hat{p}_2$. The mean of the sampling distribution is $\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$. The standard deviation formula is $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$. The standard error formula for two proportions is $SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$. When $p_1 = p_2$ is assumed, the pooled standard error formula is shown using $\hat{p}_c$ multiplied by $1 - \hat{p}_c$ times the quantity $\frac{1}{n_1} + \frac{1}{n_2}$. The pooled proportion $\hat{p}_c$ is defined as $\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$.

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Random Variable	Parameters of Sampling Distribution		Standard Error* of Sample Statistic
For one population: $\bar{X}$	$\mu_{\bar{x}} = \mu$	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$	$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$
For two populations: $\bar{X}_1 - \bar{X}_2$	$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2$	$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

The table has columns labeled Random Variable, Parameters of Sampling Distribution, and Standard Error of Sample Statistic. The first row describes one population using $\bar{X}$. The mean of the sampling distribution is $\mu_{\bar{x}} = \mu$. The standard deviation is $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. The standard error formula is $SE_{\bar{x}} = \frac{s}{\sqrt{n}}$. The second row describes two populations using $\bar{X}_1 - \bar{X}_2$. The mean of the sampling distribution is $\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2$. The standard deviation formula is $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$. The standard error formula is $SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$.

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Question 1 _____ $\times 1.25 =$ _____
out of 10

Question 2 _____ $\times 1.25 =$ _____
out of 10

Question 3 _____ $\times 1.25 =$ _____
out of 10

Question 4 _____ $\times 1.25 =$ _____
out of 10

Each line contains a blank space for a score, followed by multiplication by 1.25, an equals sign, and another blank line for the calculated result. Below each question line, the phrase out of 10 is centered.

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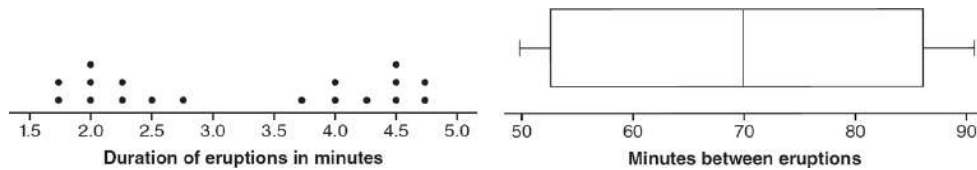
Extended image description for Figure

Dependent variable is: Sodium				
Predictor	Coef	SE Coef	T	P
Constant	-228.33	77.97	-2.93	0.009
Calories	4.0133	0.4922	8.15	0.000
S = 48.5799 R-Sq = 78.7% R-Sq(adj) = 77.5%				

Regression output table for a model where the dependent variable is Sodium. Predictors shown are Constant, Calories, and Fat, with columns for coefficient, standard error, t statistic, and p value. The intercept is -220.33, Calories has a positive coefficient of 4.0133 with $p < .001$, and Fat has a positive coefficient of 7.95 with $p = .000$. Summary statistics at the bottom show $S = 48.5799$, $R\text{-sq} = 78.7\%$, and adjusted $R\text{-sq} = 77.5\%$.

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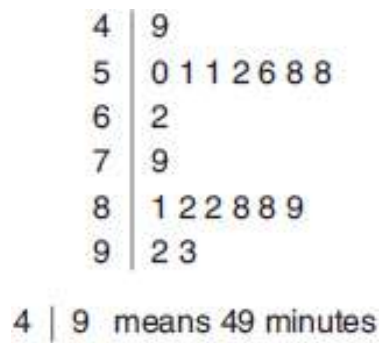
Extended image description for Figure



A pair of horizontal plots comparing sodium levels. On the left, a dot plot labeled “Sodium of doghouse hotdogs” shows values clustered mostly from about 180 to 410, with several observations around 200–280 and a smaller group around 340–400. On the right, a horizontal boxplot labeled “Sodium boxplot” spans roughly 230 to 540, with the median near 420 and the upper whisker extending to about 540.

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A stem-and-leaf plot showing times in minutes. The stems are 4, 5, 6, 7, 8, and 9. Values include 49; 50, 51, 51, 52, 56, 58, 58; 62; 79; 81, 82, 82, 88, 89; and 92, 93. A key below the plot states that “4 | 9 means 49 minutes.”

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Extended image description for Figure

Question 1 _____ $\times 1.25 =$ _____
out of 10

Question 2 _____ $\times 1.25 =$ _____
out of 10

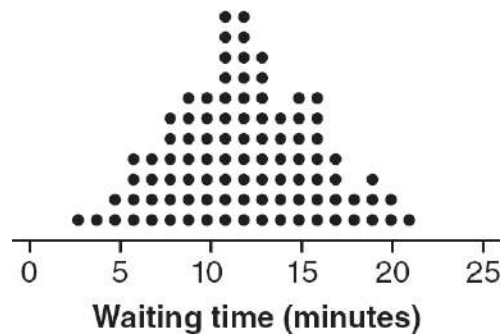
Question 3 _____ $\times 1.25 =$ _____
out of 10

Question 4 _____ $\times 1.25 =$ _____
out of 10

A worksheet scoring template with four lines labeled Question 1 through Question 4. Each question line includes a blank for the score, the phrase “out of 10,” followed by “ $\times 1.25 =$ ” and another blank for the converted score.

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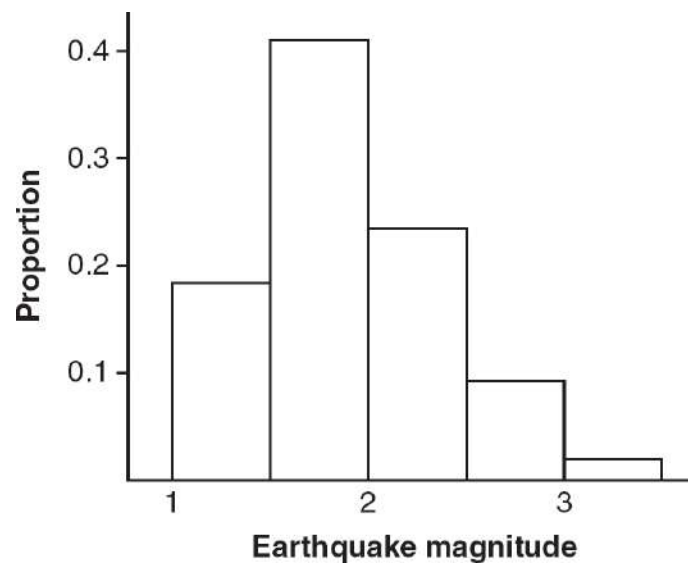
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Alt text: A dot plot titled “Waiting time (minutes)” showing a roughly symmetric, mound-shaped distribution. The data range from about 4 to 19 minutes, with the tallest stack near 11 or 12 minutes. Most waiting times fall between about 7 and 16 minutes, with fewer values at the low and high ends.

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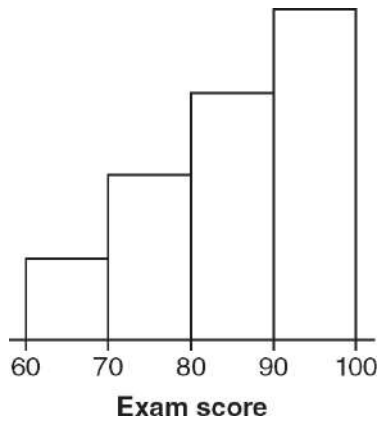
Extended image description for Figure



Alt text: A histogram showing the distribution of earthquake magnitudes. The x-axis is labeled “Earthquake magnitude,” and the y-axis is labeled “Proportion.” Most earthquakes fall between about magnitude 1.5 and 2.5, with the tallest bar around magnitude 1.5 to 2.0. The distribution tapers off for larger magnitudes, with fewer earthquakes between about magnitude 2.5 and 3.5.

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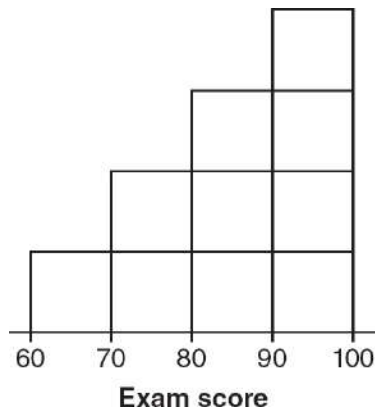
Extended image description for Figure



A histogram of exam scores. The x-axis is labeled “Exam score” and ranges from 60 to 100. The bars increase in height from left to right, with the fewest scores between 60 and 70 and the most scores between 90 and 100. The distribution is concentrated at higher exam scores and is skewed to the left.

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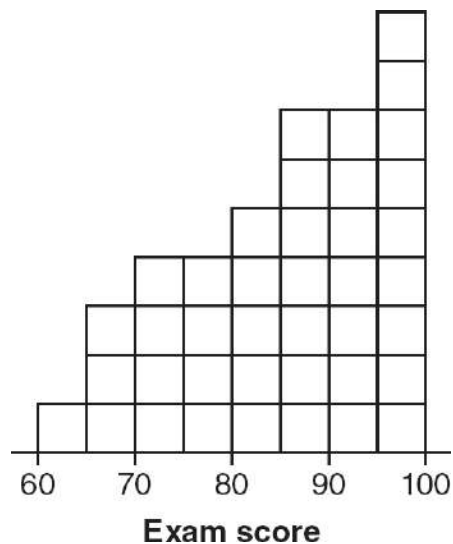
Extended image description for Figure



Alt text: A histogram of exam scores with the x-axis labeled “Exam score,” ranging from 60 to 100. The bars increase in height from left to right: the 60–70 interval has the fewest scores, followed by 70–80, 80–90, and 90–100 with the most scores. The distribution is concentrated at higher scores and is skewed to the left.

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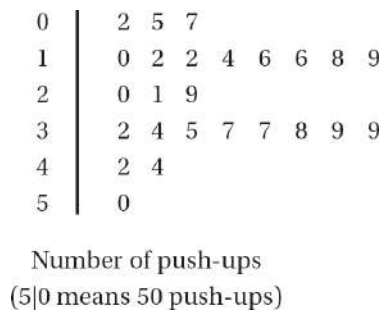
Extended image description for Figure



Alt text: A histogram of exam scores with the x-axis labeled “Exam score,” ranging from 60 to 100. The bars increase steadily from left to right, with the smallest bar in the 60–65 range and the tallest bars near 95–100. The distribution is concentrated at higher scores and is skewed to the left.

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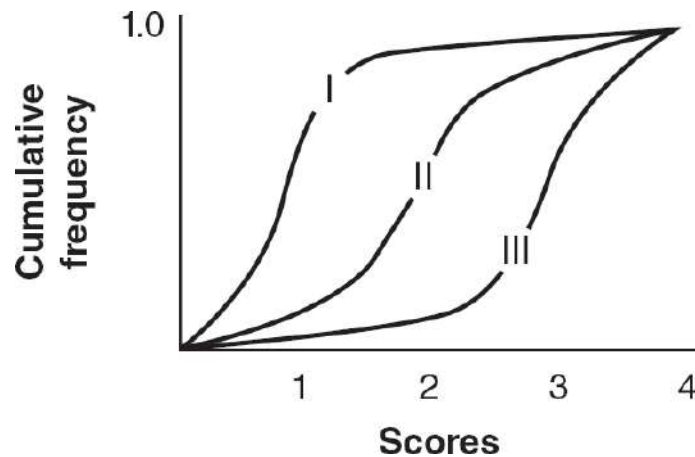
Extended image description for Figure



Alt text: A stem-and-leaf plot for number of push-ups. The stems are 0, 1, 2, 3, 4, and 5, with leaves showing values from 2 to 50 push-ups. Most values are in the 10s and 30s, with fewer values in the 0s, 20s, 40s, and 50s. A key below the plot states that “5|0 means 50 push-ups.”

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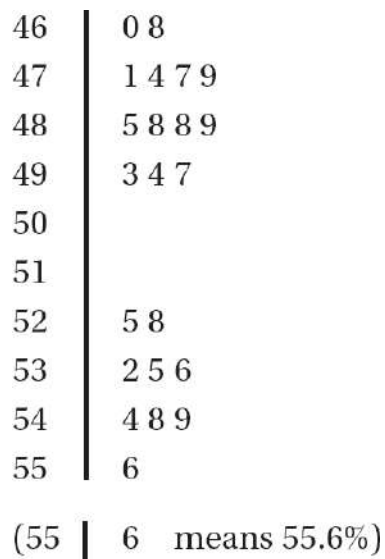
Extended image description for Figure



A line graph showing three cumulative frequency curves labeled I, II, and III. The x-axis is labeled “Scores” and ranges from 1 to 4, while the y-axis is labeled “Cumulative frequency” and ranges from 0 to 10. Curve I rises steeply near the lower scores and levels off early, curve II rises most sharply around score 2, and curve III rises steeply closer to the higher scores. All three curves end near a cumulative frequency of 10 at score 4.

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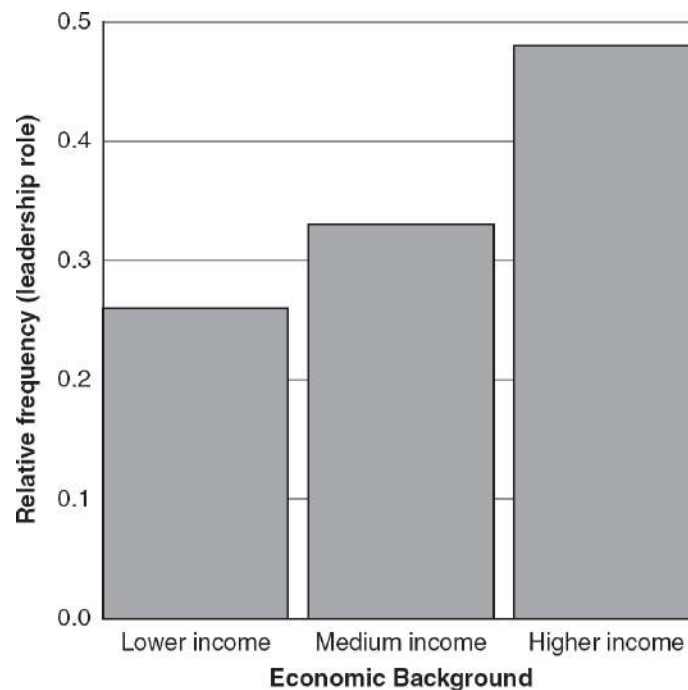
Extended image description for Figure



A stem-and-leaf plot showing values from 46.0 to 55.6. The stems run from 46 to 55, and the leaves represent tenths. Values include 46.0, 46.8, several values in the 47s through 49s, no values in the 50s or 51s, and additional values from 52.5 to 55.6. A key below the plot states that “55 | 6 means 55.6.”

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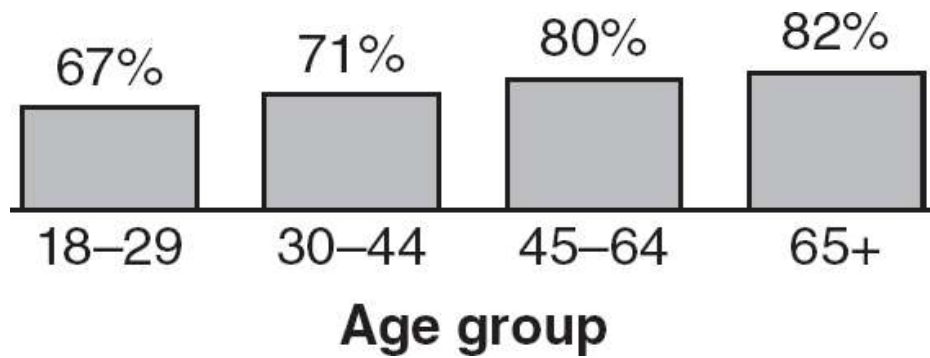
Extended image description for Figure



A bar chart comparing proportions across three economic background groups. The x-axis is labeled “Economic Background,” with categories “Lower income,” “Medium income,” and “Higher income.” The y-axis is labeled “Proportion” and ranges from 0 to 0.5. The bars increase from left to right, with lower-income at about 0.26, medium-income at about 0.33, and higher-income at about 0.48.

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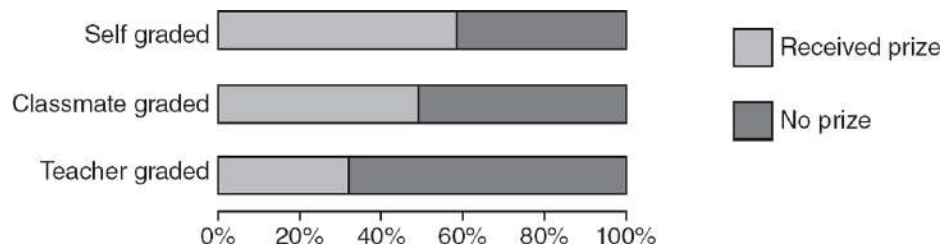
Extended image description for Figure



A bar chart comparing percentages by age group. The x-axis is labeled “Age group” and includes 18–29, 30–44, 45–64, and 65+. The bars increase with age: 67% for ages 18–29, 71% for ages 30–44, 80% for ages 45–64, and 82% for ages 65+.

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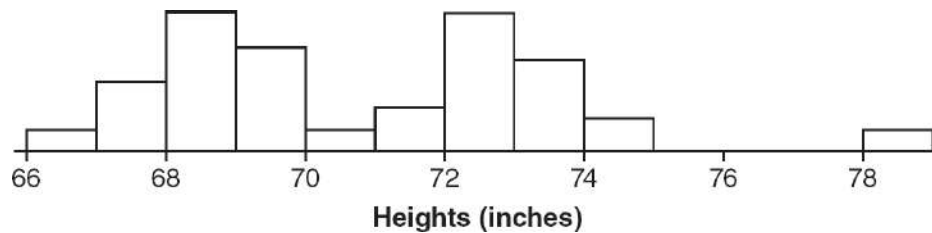
Extended image description for Figure



Alt text: A horizontal stacked bar chart comparing three groups. Each group is shown with a light gray segment on the left and a dark gray segment on the right, representing two response categories. The top group has a somewhat longer dark segment than light segment, while the middle and bottom groups have slightly longer dark segments than light segments. The image is low resolution, so the exact labels and percentages are difficult to read.

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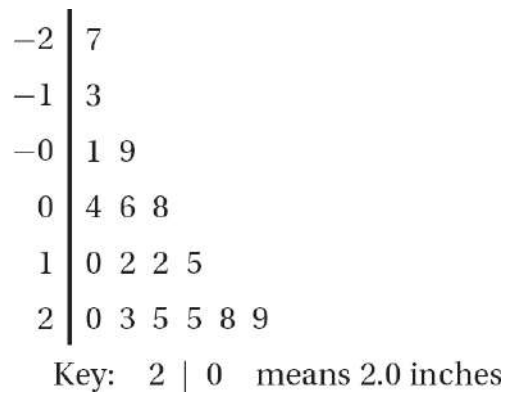
Extended image description for Figure



A histogram showing a distribution with two main clusters. The x-axis is labeled “Weight (tons)” and ranges from about 0 to 11. Several bars appear between about 0 and 5 tons, with the tallest bars around 2 and 5 tons. There is a gap in the middle of the distribution, followed by a single bar near 10 tons.

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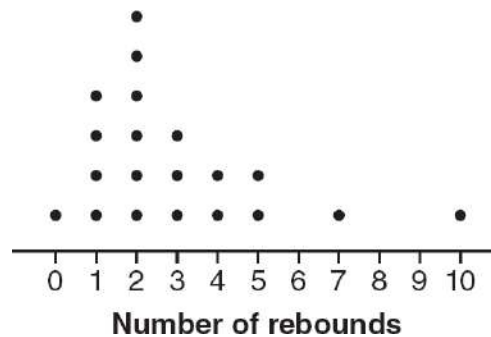
Extended image description for Figure



Alt text: A stem-and-leaf plot showing values in inches, including negative and positive values. The stems range from -2 to 2, with leaves such as -2|7, -1|3, 0|1 9, 0|4 6 8, 1|0 2 2 5, and 2|0 3 5 5 8 9. A key at the bottom states that “2 | 0 means 2.0 inches.”

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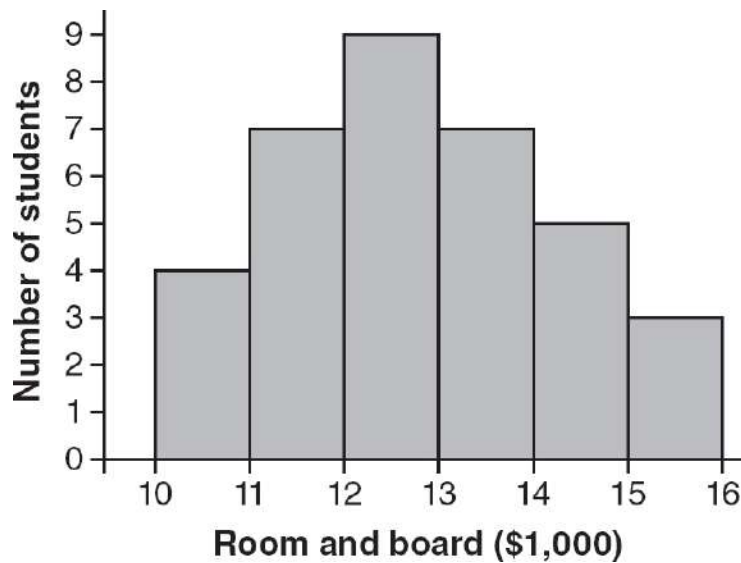
Extended image description for Figure



A dot plot showing the number of rebounds. The x-axis is labeled “Number of rebounds” and ranges from 0 to 10. Most values are clustered between 0 and 5 rebounds, with the highest stack around 2 rebounds. There are isolated values at about 7 and 10 rebounds, suggesting a right-skewed distribution.

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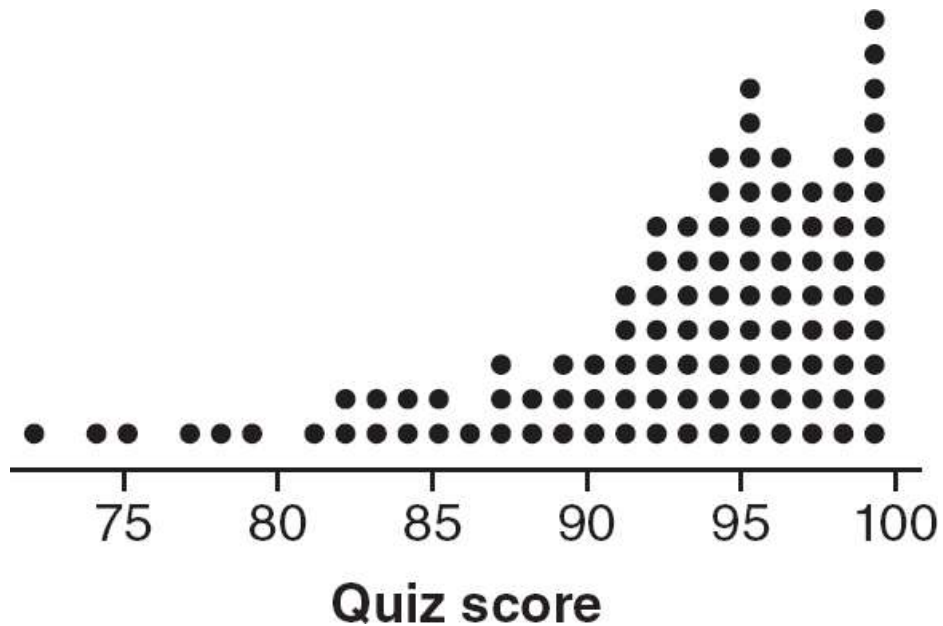
Extended image description for Figure



A histogram showing room and board costs in thousands of dollars. The x-axis is labeled “Room and board (\$1,000)” and ranges from about 10 to 16. The y-axis is labeled “Number of colleges.” The distribution is roughly mound-shaped, with the highest bar around \$13,000. Most colleges fall between about \$11,000 and \$15,000, with fewer colleges at the lower and higher ends.

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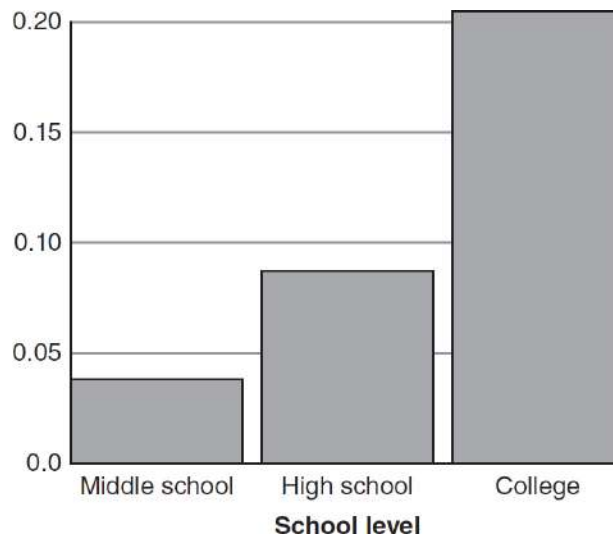
Extended image description for Figure



Alt text: A dot plot showing quiz scores. The x-axis is labeled “Quiz score” and ranges from about 75 to 100. Most scores are clustered at the high end, especially between about 90 and 100, with the tallest stacks near 96 and 100. A few lower scores appear between about 72 and 85. The distribution is skewed to the left, with most students earning high quiz scores.

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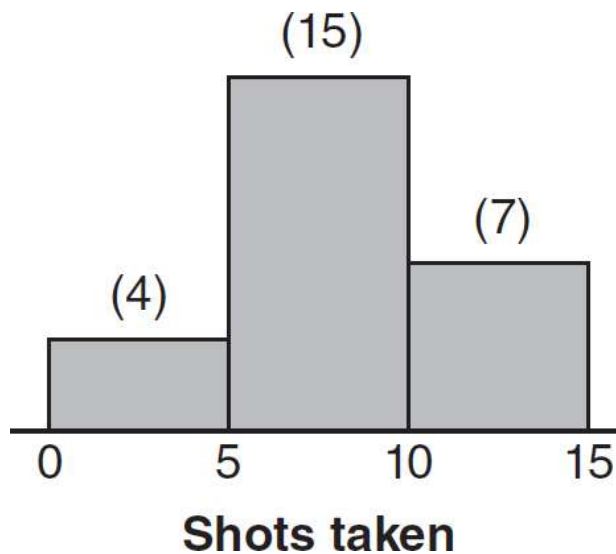
Extended image description for Figure



Alt text: A bar chart showing proportions by school level. The x-axis is labeled “School level” and includes Middle school, High school, and College. The y-axis is labeled “Proportion” and ranges from 0 to 0.20. The bars increase from middle school to college, with middle school at about 0.04, high school at about 0.09, and college at about 0.20.

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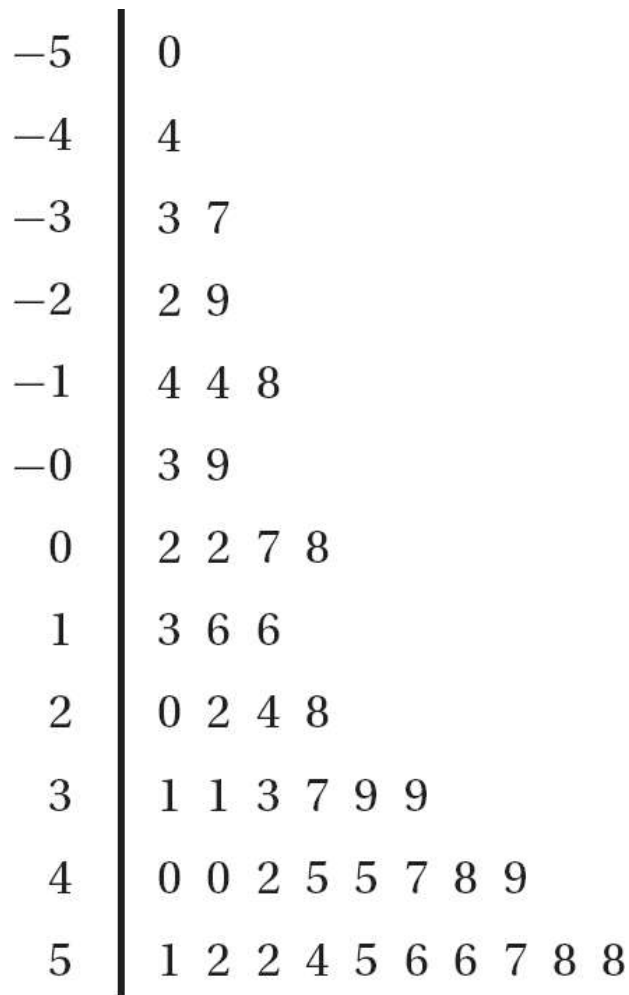
Extended image description for Figure



A histogram showing the number of shots taken. The x-axis is labeled “Shots taken” and ranges from 0 to 15. There are three bins: 0–5 shots with a frequency of 4, 5–10 shots with a frequency of 15, and 10–15 shots with a frequency of 7. Most players took between 5 and 10 shots.

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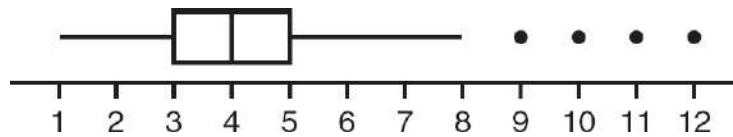


Key: 5|1 = 5.1 pounds

A stem-and-leaf plot showing values in pounds. The stems range from -5 to 5, with leaves listed beside each stem. The plot includes one value at -5, several negative values from about -3 to -1, values around 0, and many positive values from 1 to 5. The greatest concentration appears in the positive range, especially from 3 to 5 pounds. A key below the plot states that “5 | 1 = 5.1 pounds.”

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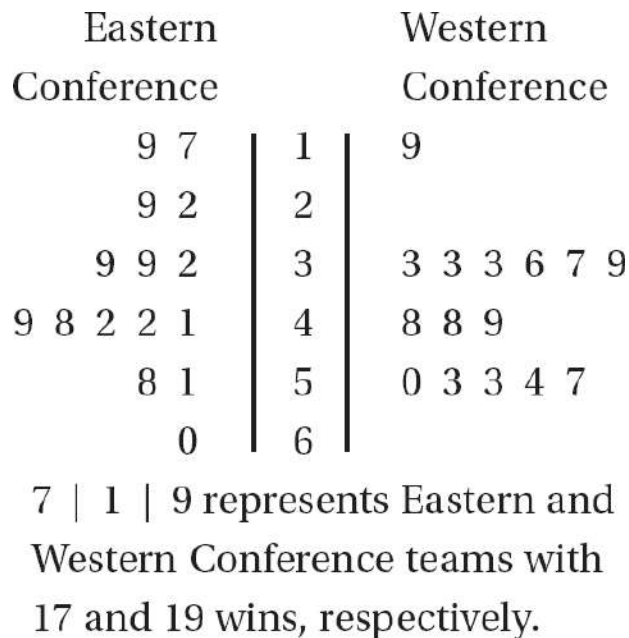
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A horizontal boxplot shown above a number line from about 1 to 14. The box spans roughly 3 to 5, with a median near 4. The left whisker extends to about 1, and the right whisker extends to about 7. Four individual points appear to the right around 9, 10, 11, and 12, suggesting high outliers.

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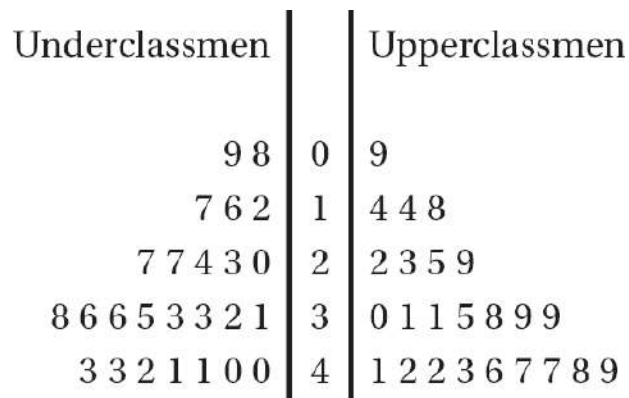
Extended image description for Figure



Alt text: A back-to-back stem-and-leaf plot comparing Eastern Conference and Western Conference points. The stems are 1 through 6, and the key states that “7 | 1 | 9 represents Eastern and Western Conference teams with 17 and 19 wins, respectively.” Eastern Conference values appear on the left side of the stems, and Western Conference values appear on the right. Both conferences have teams ranging from the teens to the 60s, with many values clustered in the 30s, 40s, and 50s.

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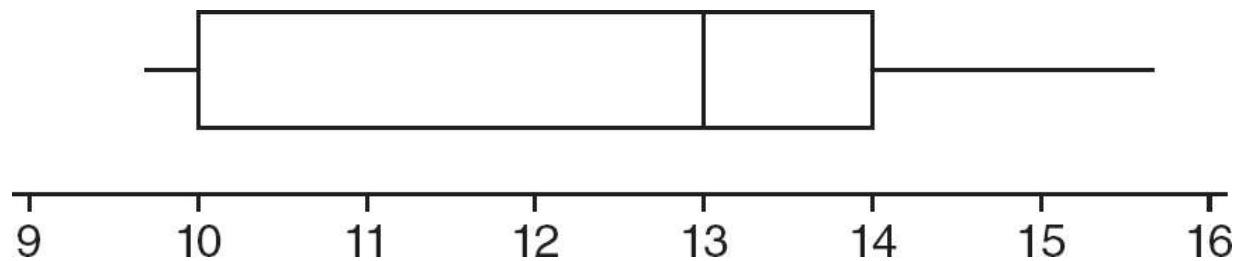


Key: 0|9 = 9 minutes

A back-to-back stem-and-leaf plot comparing waiting times, in minutes, for underclassmen and upperclassmen. Underclassmen values are shown on the left of the stems, and upperclassmen values are shown on the right. The stems range from 0 to 4. Upperclassmen have many values in the 10s, 20s, 30s, and 40s, while underclassmen values are more concentrated in the 20s through 40s. A key below states that “0 | 9 = 9 minutes.”

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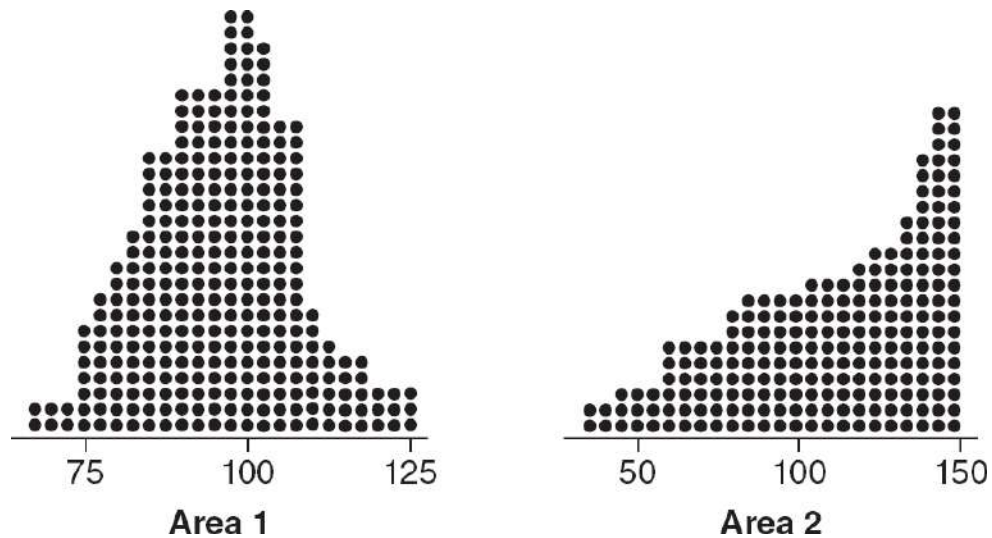
Extended image description for Figure



A horizontal boxplot above a number line. The box spans roughly 2 to 4, with the median close to 4. The left whisker extends slightly below 1, and the right whisker extends to about 5. The distribution appears skewed to the left, with more spread on the lower-value side.

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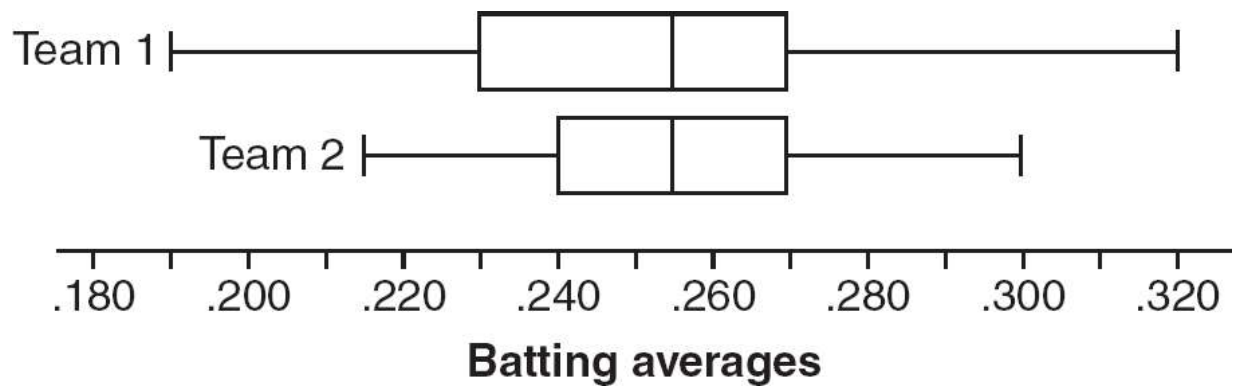
Extended image description for Figure



Two side-by-side dot plots comparing ages. The plot on the left, labeled “Age 1,” shows most values clustered between about 75 and 90, with the highest stack near the low 80s and a tail extending toward older ages. The plot on the right, labeled “Age 2,” shows values spread more widely, with many ages clustered toward the higher end, especially around 95 to 105.

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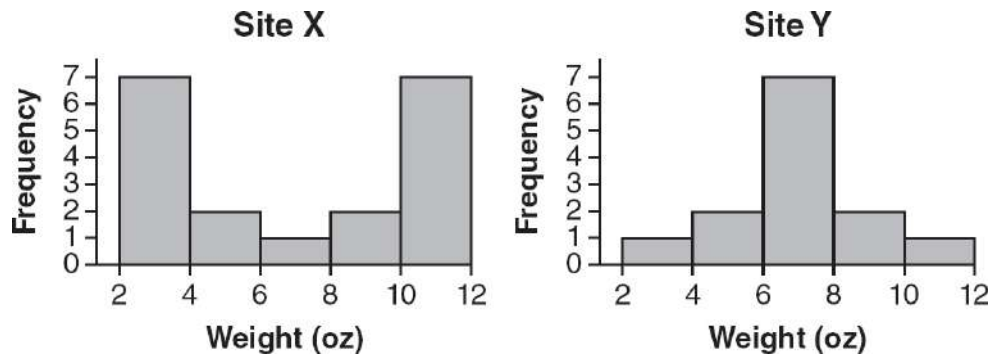
Extended image description for Figure



Two horizontal boxplots comparing batting averages for Team 1 and Team 2. The x-axis is labeled “Batting average” and ranges from about .100 to .500. Team 1 has a wider spread, with values extending from about .150 to .480, while Team 2 has a smaller spread, from about .250 to .420. Both teams have medians around .300.

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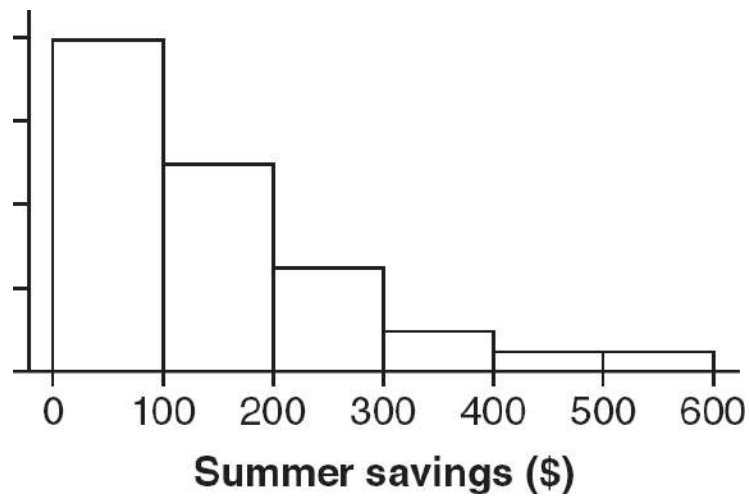
Extended image description for Figure



Two side-by-side histograms comparing two distributions. Both plots have the x-axis labeled “Variable (y)” and show values from about 2 to 10. The left histogram has tall bars at the low and high ends with smaller bars in the middle, giving it a U-shaped distribution. The right histogram has the tallest bar at the high end and smaller bars spread across the lower and middle values.

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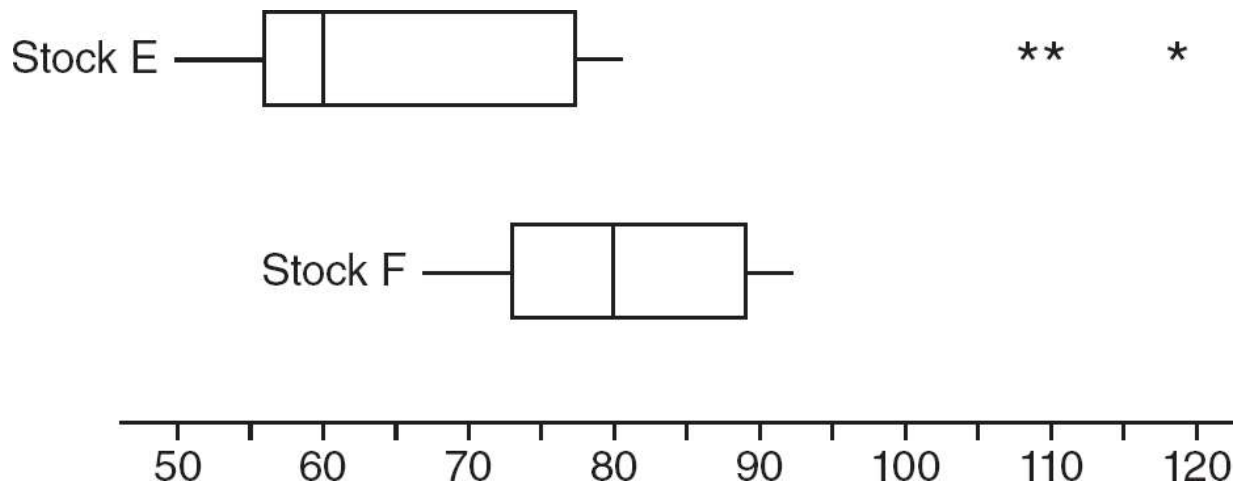
Extended image description for Figure



A histogram showing summer savings in dollars. The x-axis is labeled “Summer savings (\$)” and ranges from 0 to 600. The distribution is strongly skewed to the right, with the tallest bars near the lowest savings amounts and gradually decreasing frequencies as savings increase. Most values are below about \$200, with a long tail extending toward \$600.

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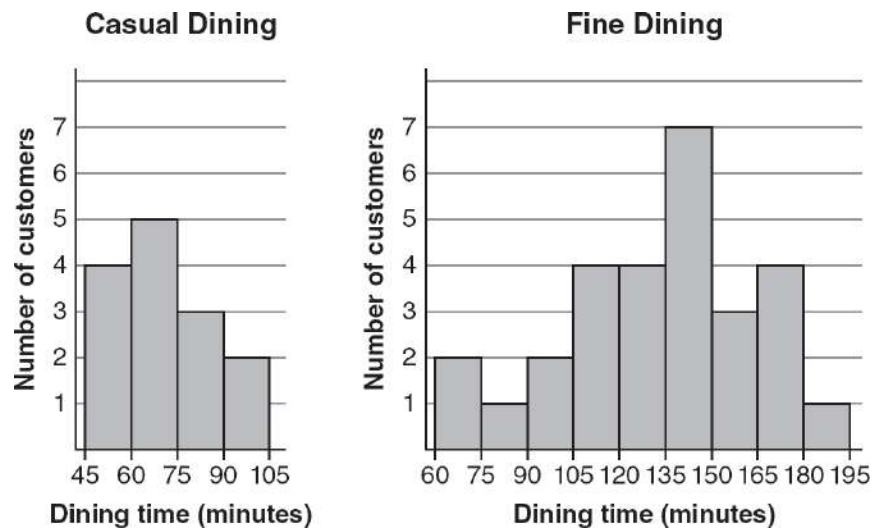
Extended image description for Figure



Alt text: Two horizontal boxplots comparing scores for School A and School B. The x-axis ranges from about 50 to 120. School A has a wider spread, with a box from roughly 60 to 80, a median near 70, and two high outliers around 110 and 120. School B has scores clustered higher, with a box from roughly 75 to 90 and a median in the mid-80s.

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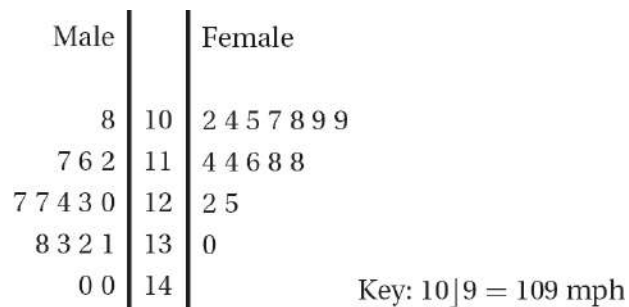
Extended image description for Figure



Two side-by-side histograms comparing dining time in minutes for casual dining and fine dining. The casual dining histogram shows most dining times clustered between about 45 and 65 minutes, with fewer meals lasting longer. The fine dining histogram shows generally longer dining times, with most values between about 75 and 105 minutes and the tallest bar around 85 to 95 minutes.

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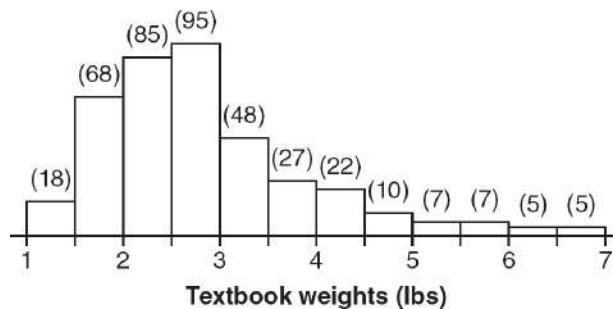
Extended image description for Figure



A back-to-back stem-and-leaf plot comparing speeds, in miles per hour, for males and females. Males are shown on the left of the stems, and females are shown on the right. The stems are 10 through 14, and the key states that “10 | 9 = 109 mph.” Female values are mostly in the 100s and 110s, while male values extend from about 108 to 140 mph, with many values in the 120s and 130s.

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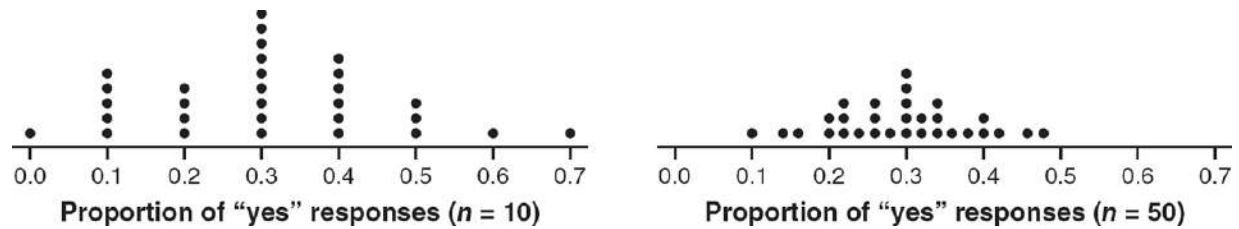
Extended image description for Figure



A histogram showing textbook weights in pounds. The x-axis is labeled "Textbook weights (lbs)" and ranges from 1 to 7. The distribution is strongly skewed to the right, with the highest frequencies around 2 pounds. Bar labels show frequencies such as 65, 95, 90, 48, 27, 22, 10, 7, 7, 5, and 3, with fewer textbooks at heavier weights.

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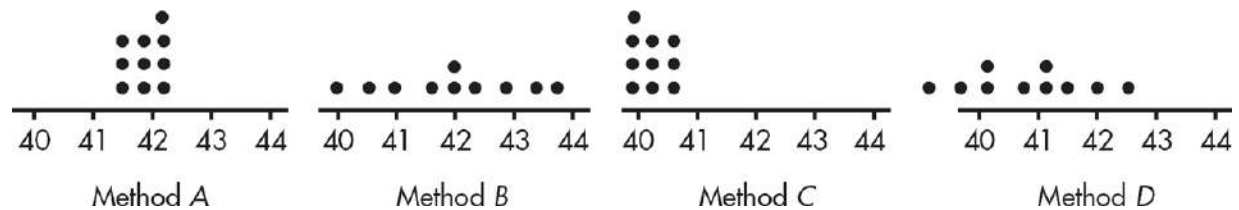
Extended image description for Figure



Two side-by-side dot plots comparing weekly television viewing, with the x-axes labeled “Popular TV shows (hours).” The left dot plot has values spread across a wider range with two tall clusters, while the right dot plot is more tightly clustered around the middle, with one taller stack and shorter stacks nearby.

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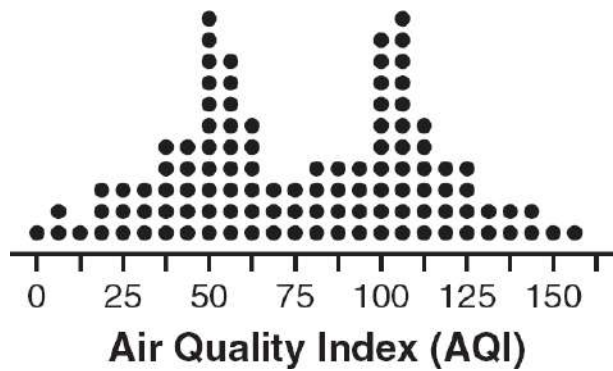
Extended image description for Figure



Two side-by-side dot plots comparing two groups or methods. The left dot plot shows values spread across a wider range with several tall stacks at the low and high ends and fewer values in the middle. The right dot plot shows values clustered more tightly around the center, with several similar-height stacks. The image is low resolution, so the exact axis labels and values are difficult to read.

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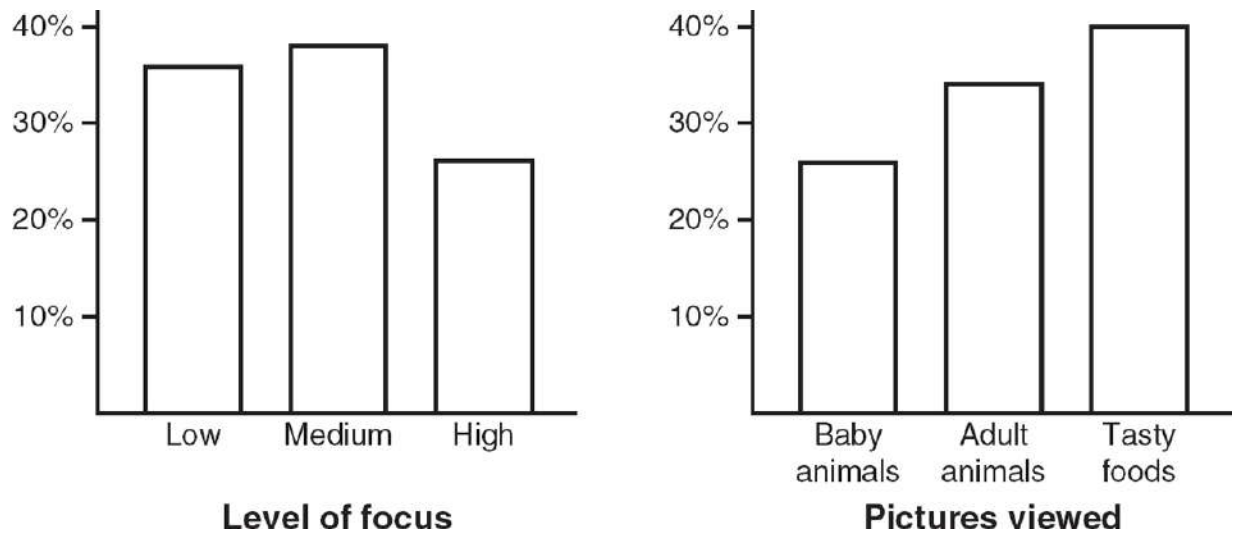
Extended image description for Figure



A dot plot showing Air Quality Index (AQI) values. The x-axis is labeled "Air Quality Index (AQI)" and ranges from 0 to 150. The distribution is roughly symmetric with two tall peaks, one around AQI 50 and another around AQI 100. Most values fall between about 25 and 125, with fewer values at the low and high ends.

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Two bar charts comparing app use and work habit by homework level. The left chart shows students who use the app, with high percentages for low and medium homework levels and a lower percentage for high homework. The right chart shows students who do not use the app, with percentages increasing from low to medium to high homework levels.

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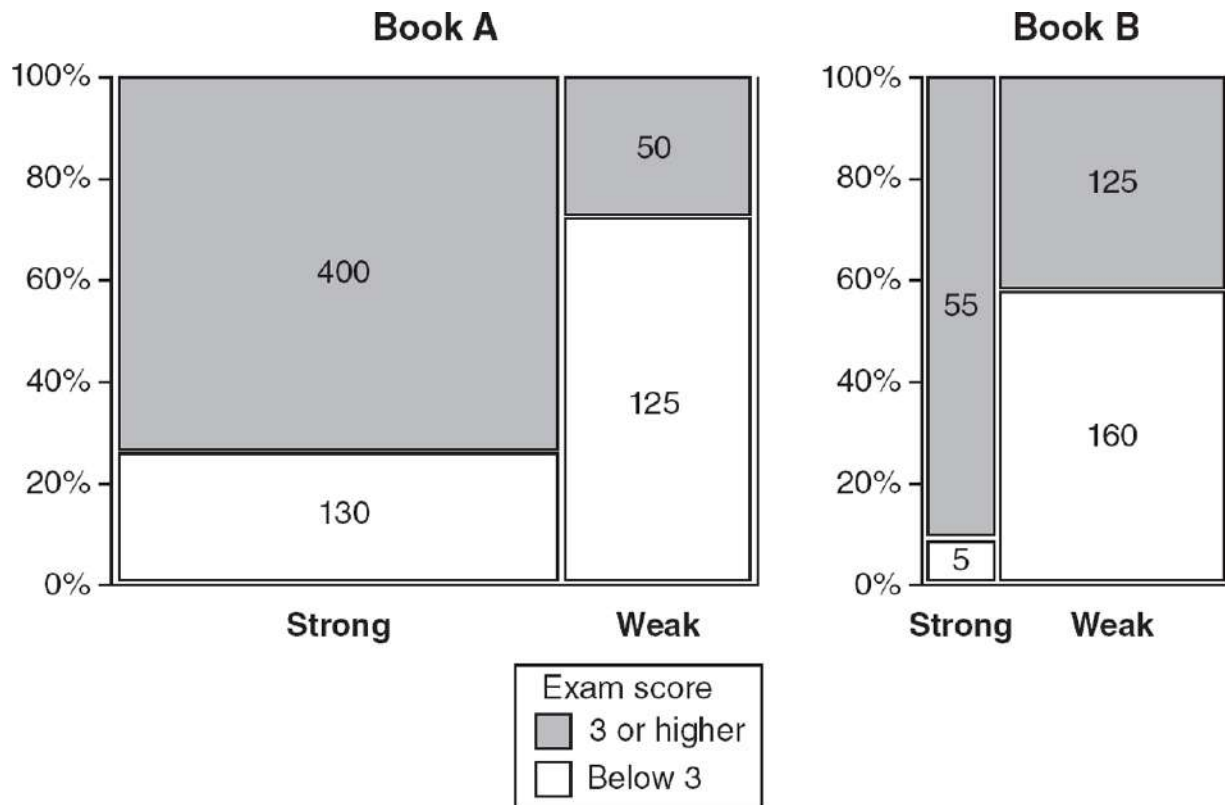
Extended image description for Figure

Good condition			Poor condition		
	Dr. F	Dr. P		Dr. F	Dr. P
Died	8	17	Died	52	33
Survived	60	120	Survived	130	80

Two two-way tables comparing class status and AP class enrollment. The left table is for good school, and the right table is for poor school. In each table, columns show AP and no AP, and rows show junior and senior. The counts are higher for seniors than juniors in both schools, and the no-AP counts are higher than AP counts for both class levels.

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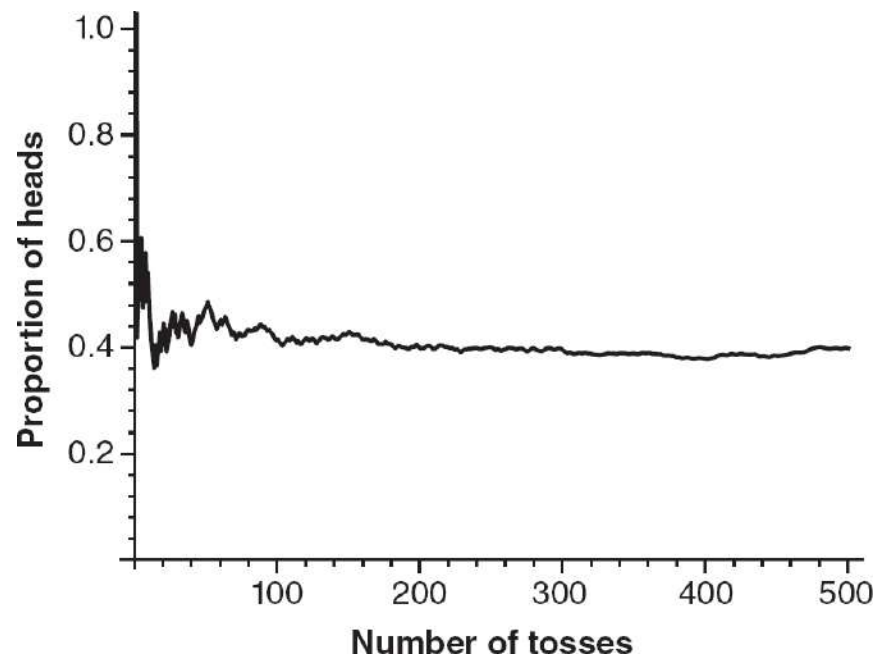
Extended image description for Figure



Two segmented bar charts comparing stress levels by pacing group for Book A and Book B. Each chart has bars for pacing and no pacing, divided into low stress, moderate stress, and high stress. For Book A, both pacing groups show mostly low stress, with smaller moderate-stress and high-stress sections. For Book B, the pacing group shows a mix of low, moderate, and high stress, while the no-pacing group shows a larger low-stress section and a smaller moderate-stress section.

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A line graph showing the proportion of heads versus the number of tosses. The x-axis is labeled "Number of tosses" and ranges from 0 to 500. The y-axis is labeled "Proportion of heads" and ranges from 0 to 1.0. The plotted proportion fluctuates widely at first, then becomes more stable as the number of tosses increases, leveling off near 0.5.

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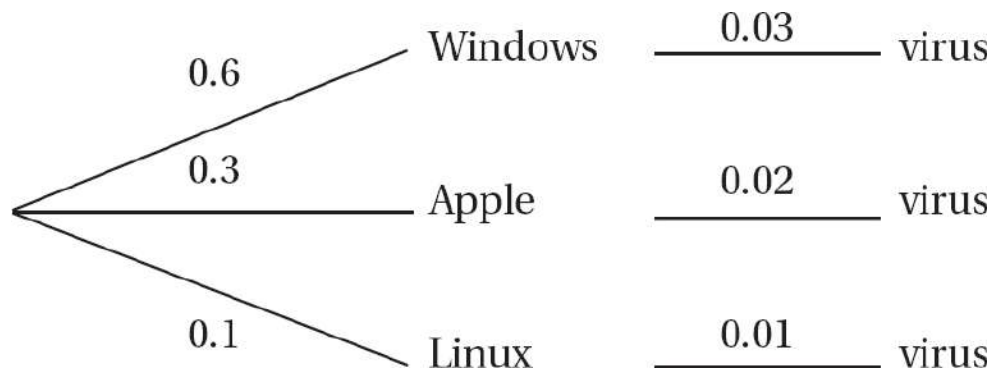
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		Score			<i>Total</i>
		0-59	60-79	80-100	
<i>Strategy</i>	Guess	40	160	100	300
	Choose answer (c)	35	170	115	320
	Choose longest answer	45	200	135	380
<i>Total</i>		120	530	350	1,000

A two-way table showing counts by rowing category and year in school. The columns are labeled “Year” and include 9th, 10th, 11th, and 12th. The rows include College, Club, and a combined category, with counts listed for each grade level.

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Alt text: A tree diagram showing probabilities for choosing among three computer types and then choosing a color. The first split shows Windows with probability 0.6, Apple with probability 0.3, and Linux with probability 0.1. Each computer type then branches to color outcomes labeled White and another color, with conditional probabilities shown on the branches.

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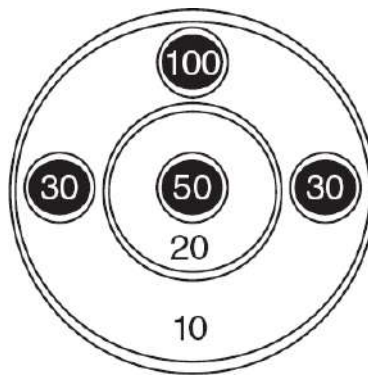
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	Years of experience		
	0–5	6–10	>10
Less than 50 years old	80	125	20
More than 50 years old	10	75	50

A two-way table showing counts by years of experience and age group. Columns are labeled “Years of experience”: 0–5, 6–10, and greater than 10. Rows compare people less than 50 years old and more than 50 years old. The counts are 80, 125, and 20 for the less-than-50 group, and 10, 75, and 50 for the more-than-50 group.

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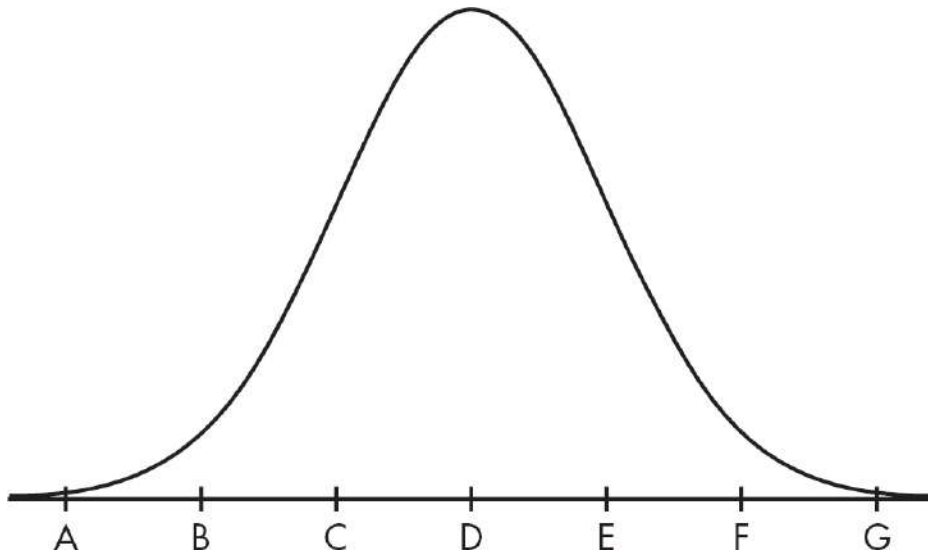
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A circular target with four scoring regions. The center circle is labeled 50, the surrounding inner ring is labeled 20, and the outer ring is labeled 10. Three marked shots are shown: one at the top labeled 100, one on the left labeled 30, and one on the right labeled 30.

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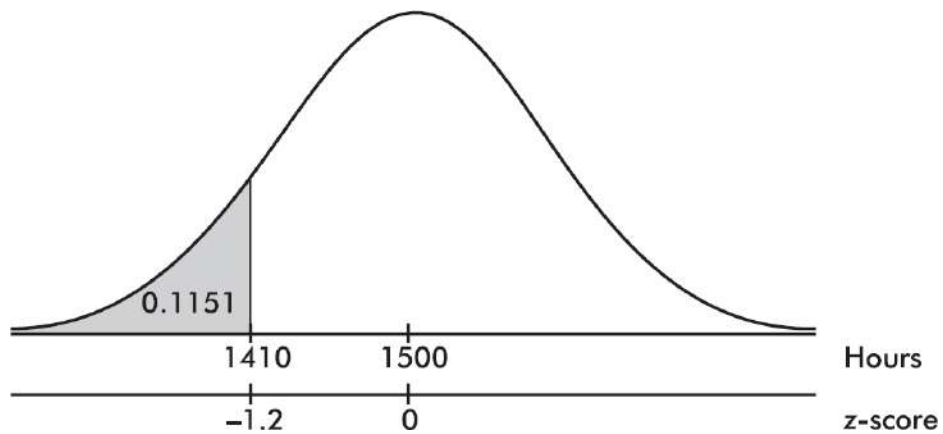
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A bell-shaped normal curve above a horizontal axis labeled A through G. The peak of the curve is centered at D, with the curve tapering toward A on the left and G on the right.

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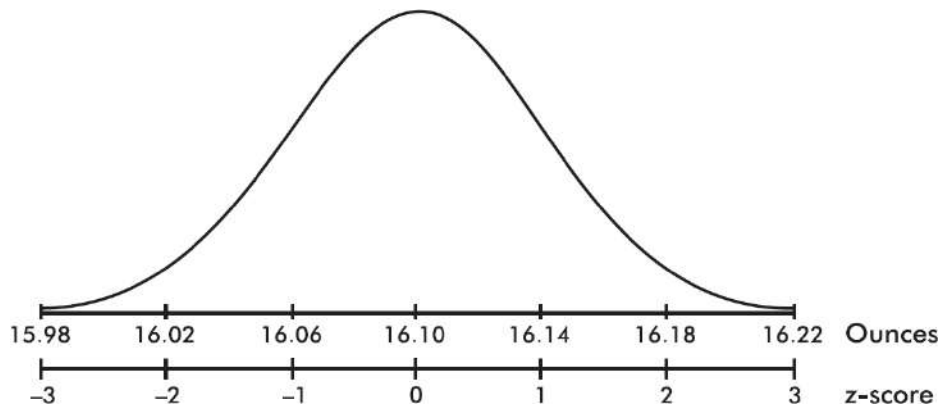
Extended image description for Figure



Alt text: A normal distribution curve with the left tail shaded. The x-axis is labeled with values including 110 and 150, and a lower axis shows corresponding z-scores from about -3 to 3. The shaded region represents the probability of values less than a cutoff on the left side of the distribution.

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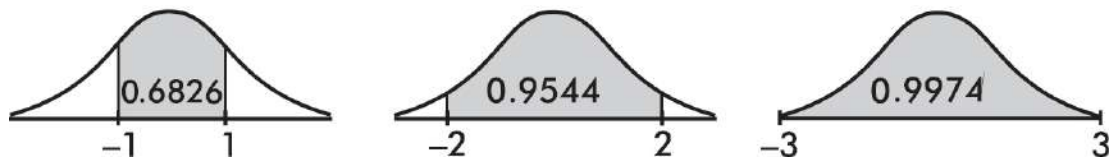
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A bell-shaped normal distribution curve centered at about 16.0. The horizontal axis ranges from about 14.0 to 18.0, with evenly spaced tick marks.

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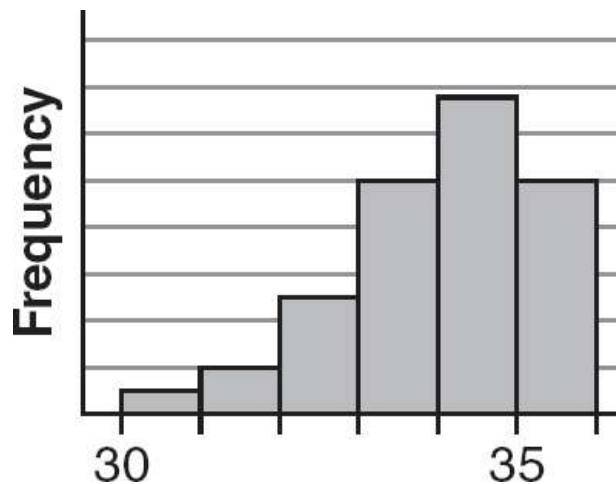
Extended image description for Figure



Three bell-shaped distributions shown side by side on the same scale. Each distribution is centered at a different x-value, with narrow spreads and values clustered closely around each center. The three centers appear near ? 8, ?4, and 0.

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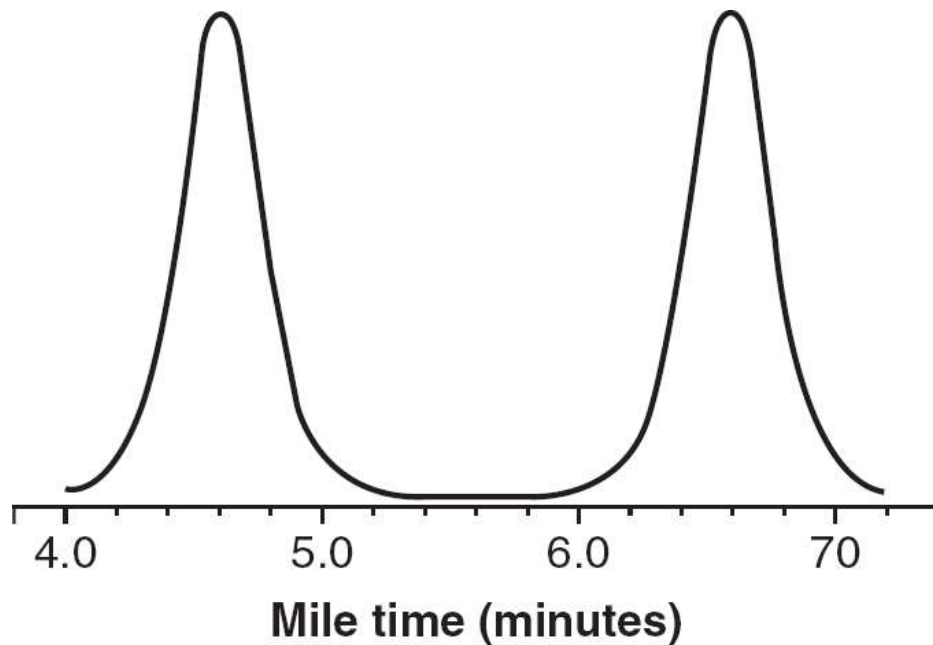
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A histogram showing a distribution of values from about 30 to 36. The bars generally increase from left to right, with the highest frequency around 34 to 35. The distribution is concentrated at the higher end and is skewed to the left.

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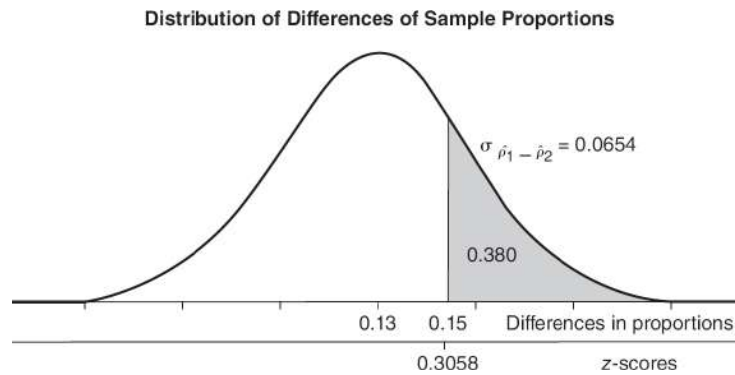
Extended image description for Figure



A bimodal distribution of mile times, in minutes. The x-axis is labeled “Mile time (minutes)” and ranges from 4.0 to 7.0. The distribution has two peaks, one near 4.2 minutes and one near 6.8 minutes, with very few values around the middle of the range.

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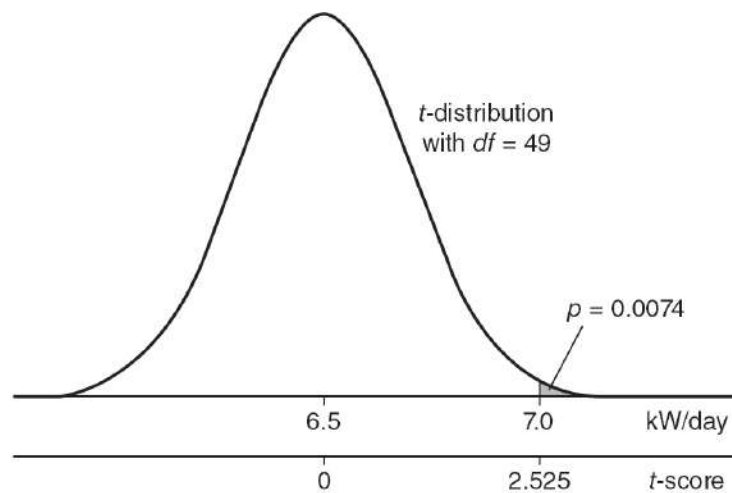
Extended image description for Figure



A normal distribution curve for the difference in sample proportions. The curve is centered near 0.13, and a vertical cutoff line marks 0.098. The area to the right of 0.098 is shaded and labeled as the probability for sample-proportion differences greater than 0.098.

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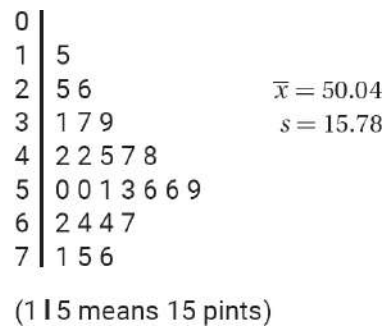
Extended image description for Figure



Alt text: A normal distribution curve centered at 65.5, labeled as an $\bar{x}$ -bar distribution with standard deviation 0.8. A vertical cutoff line at 70 marks the start of the right tail. The right-tail area beyond 70 is shaded and labeled $p = 0.0037$.

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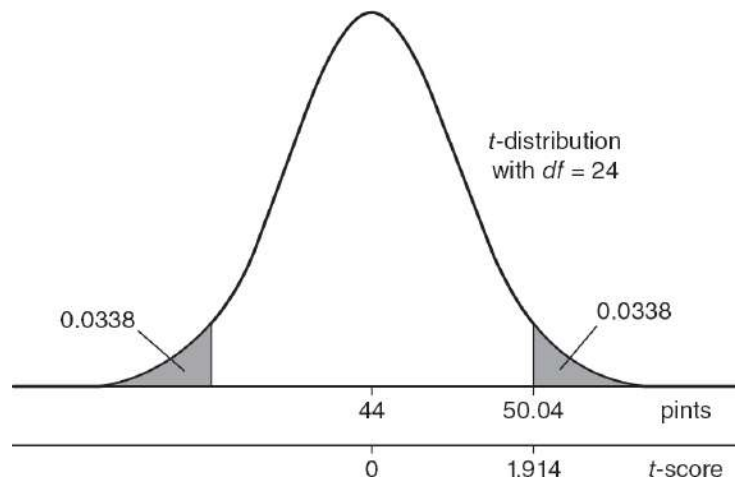
Extended image description for Figure



Alt text: A stem-and-leaf plot showing test scores in points. The stems run from 0 to 7, and the leaves represent ones digits. Values include 15; 25, 26; 31, 37, 39; 42, 42, 45, 47, 48; 50, 50, 51, 53, 56, 56, 59; 62, 64, 67; and 71, 75, 76. Summary statistics show $\bar{x} = 50.04$ and $s = 15.78$. A key states that "1 | 5 means 15 points."

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A normal distribution centered at 50.04, labeled as a distribution with standard deviation 2.4. Vertical cutoff lines mark 46 and 54. The area between 46 and 54 is shaded and labeled 0.9114, while each tail outside the cutoff values is labeled 0.0388.

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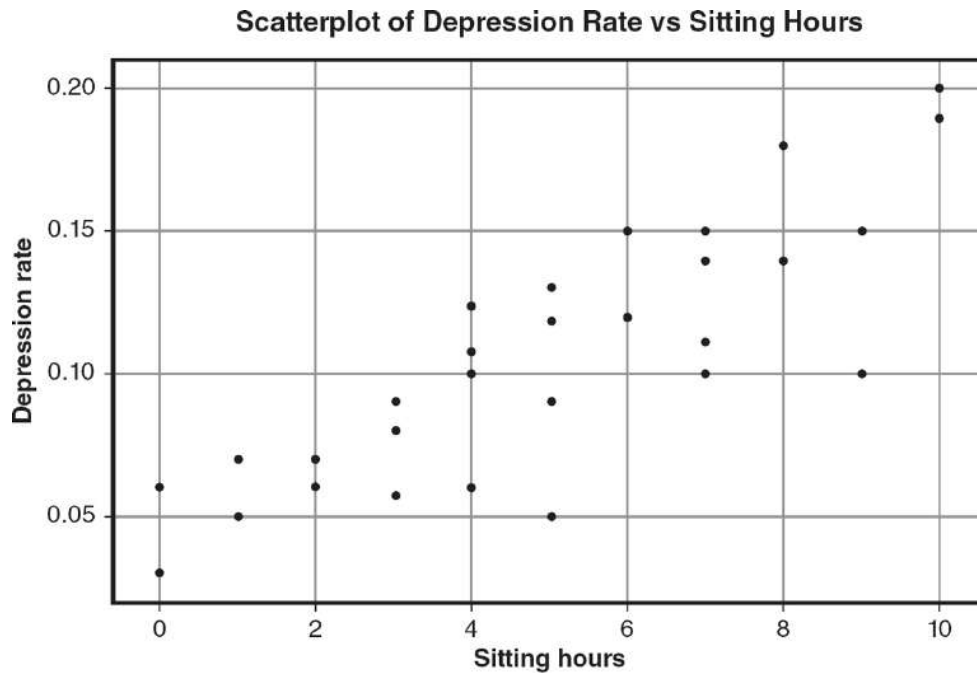
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Union	
Nonunion	
Differences	

Three horizontal dot plots comparing the distributions for Wolf, Rabbit, and Skunk. Each row shows individual data values as vertical marks along the same horizontal scale. Wolf and Rabbit have values spread across the scale, with several values clustered toward the higher end. Skunk has values spread across a similar range, with many values clustered in the middle to higher end.

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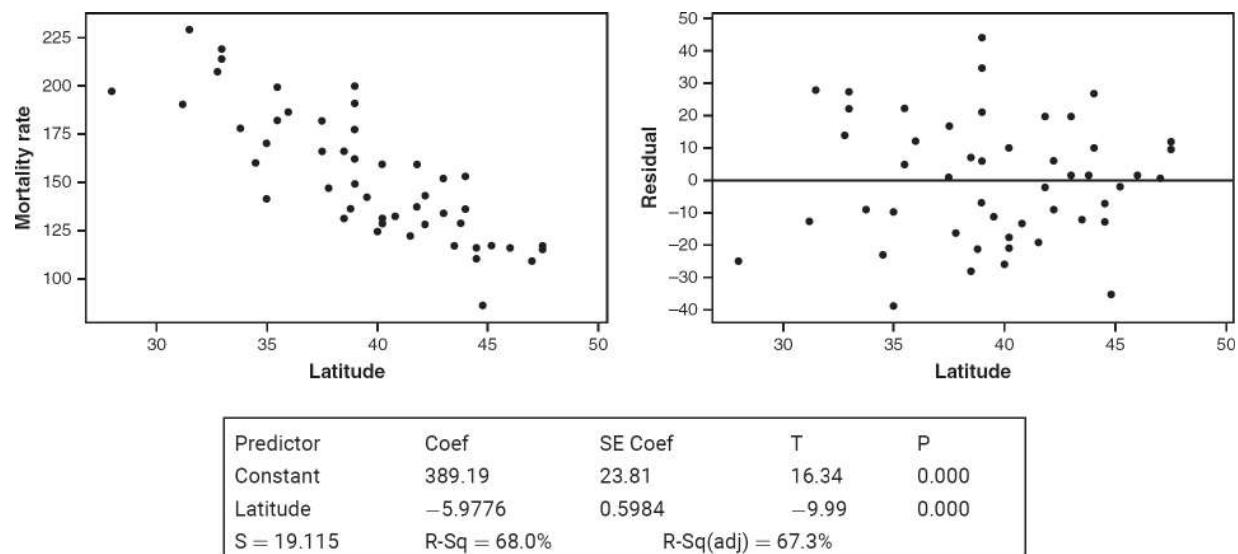
Extended image description for Figure



A scatterplot titled “Scatterplot of Opposite Arm vs. Sitting Burn.” The x-axis is labeled “Sitting burn,” and the y-axis is labeled “Opposite arm.” The points show a positive association: as sitting burn values increase, opposite arm values tend to increase.

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Extended image description for Figure



A regression analysis display with two scatterplots and a regression output table. The left scatterplot shows a negative association between the explanatory variable and response variable, with points decreasing as x increases. The right plot shows residuals scattered around a horizontal line at 0, with no clear curved pattern. The regression output table lists the constant and x-variable terms with coefficient, standard error, t statistic, and p-value, along with model summary statistics including S, R-sq, and adjusted R-sq.

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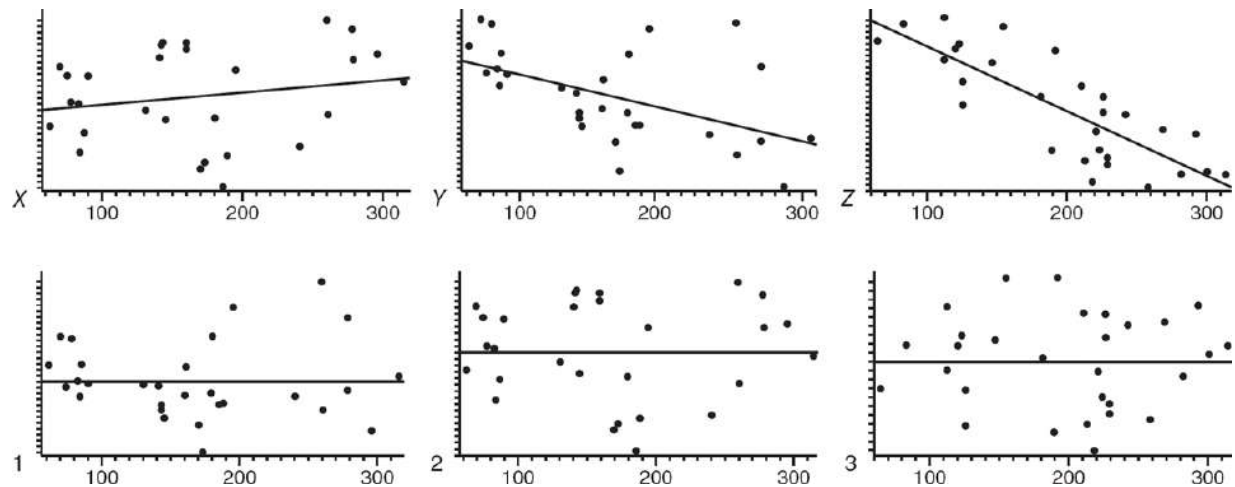
Extended image description for Figure

Predictor	Coef	SE Coef	T	P
Constant	0.05590	0.02368	2.36	0.046
Facing	0.92003	0.03902	23.58	0.000
S = 0.0242922 R-Sq = 98.6% R-Sq(adj) = 98.4%				

A regression output table with columns for Predictor, Coefficient, Standard Error of the Coefficient, t statistic, and p-value. The model includes a constant term and two predictor variables. The output also includes summary statistics at the bottom, including S, R-sq, and adjusted R-sq.

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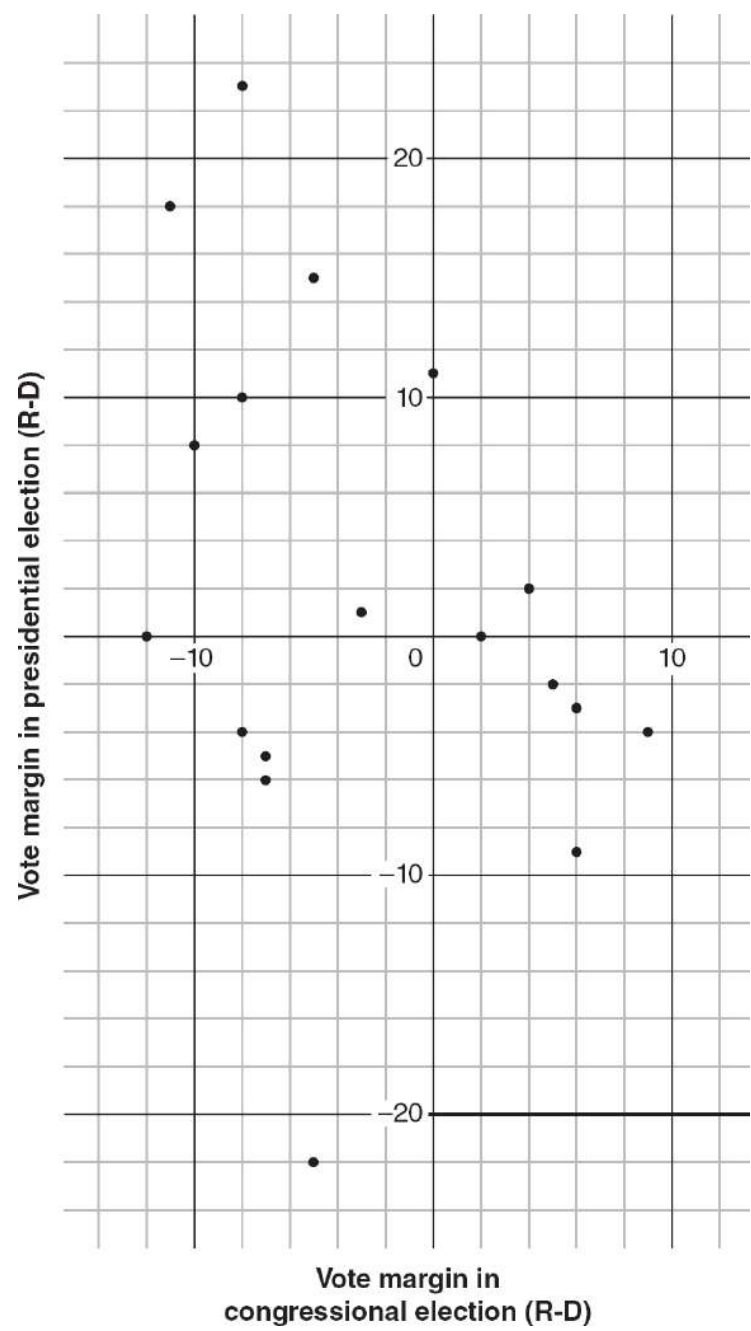
Extended image description for Figure



A set of six small scatterplots labeled A through F, each showing a different pattern of association. Plot A shows a positive linear association. Plot B shows a negative linear association with moderate scatter. Plot C shows a strong negative linear association. Plot D shows little to no association, with points scattered around a horizontal band. Plot E shows a curved or nonlinear pattern. Plot F shows little to no association, with points scattered in a roughly horizontal pattern.

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Extended image description for Figure

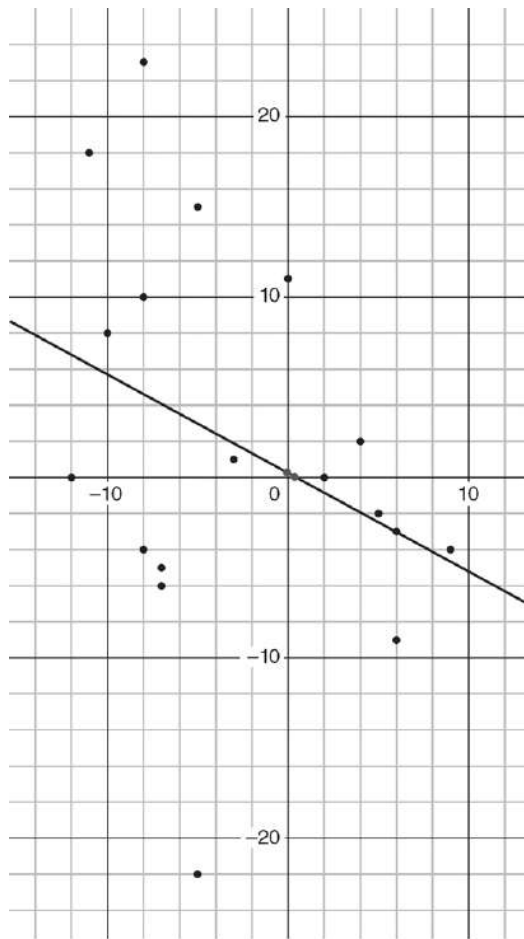


A scatterplot showing the relationship between two quantitative variables. The x-axis shows average time watching TV or videos, and the y-axis shows average time reading. The points are widely scattered, with a general

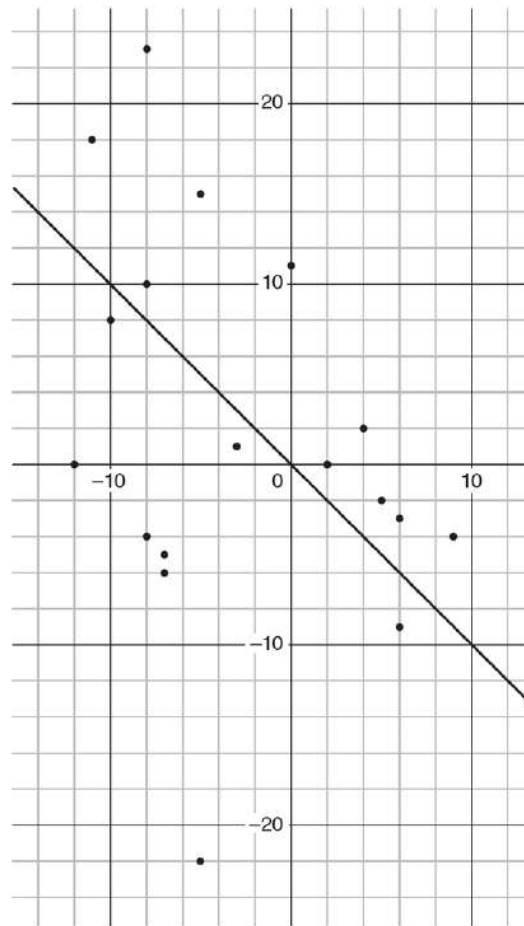
downward trend: higher values on the x-axis tend to be associated with lower values on the y-axis.

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Graph 1

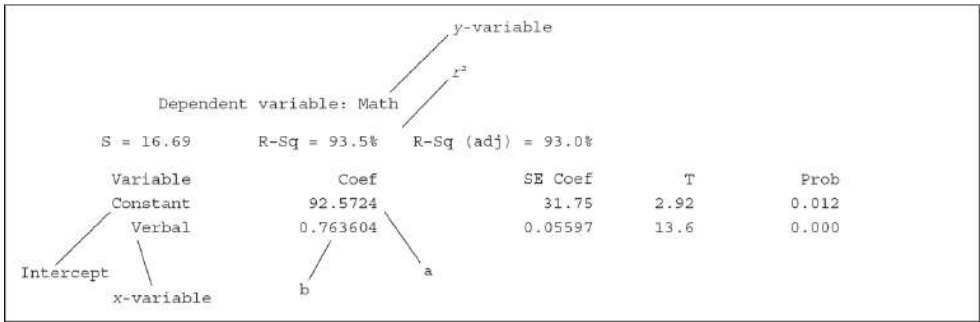


Graph 2

Two coordinate-plane graphs labeled Graph 1 and Graph 2, each showing a linear function with a negative slope. In Graph 1, the line decreases more gradually from left to right. In Graph 2, the line decreases more steeply from left to right.

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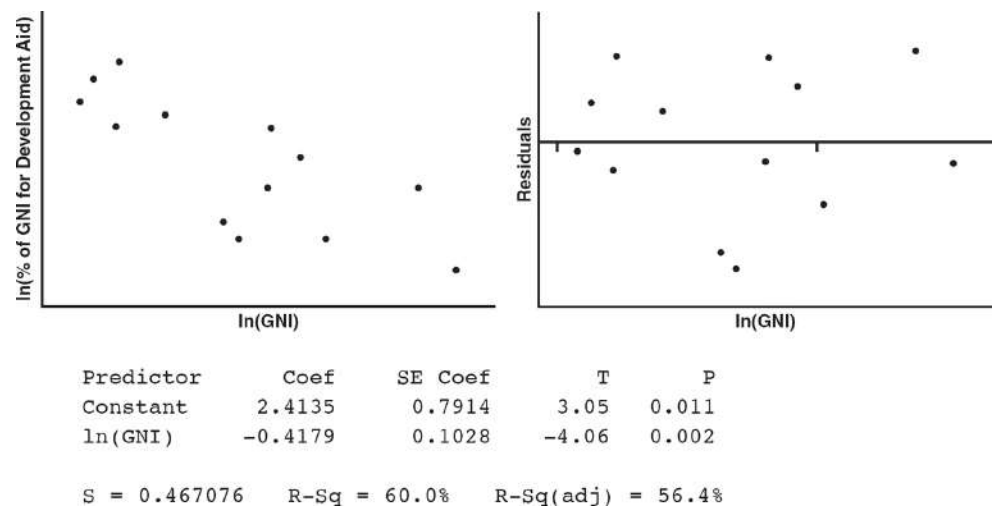
Extended image description for Figure



A statistical output showing a linear regression equation, a table of regression results, and a small regression diagram. The table lists terms for the model, including the constant and a predictor, with columns for coefficient, standard error, test statistic, and p-value. The diagram represents the fitted regression line and the variation around the model.

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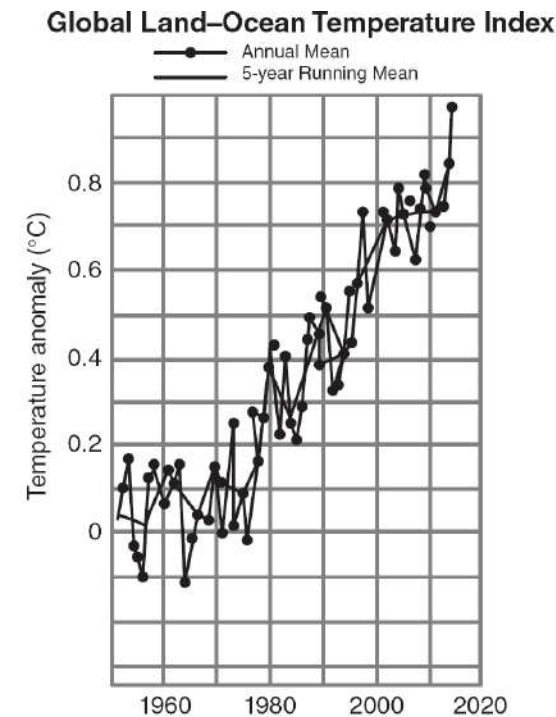
Extended image description for Figure



Alt text: Two small diagnostic plots with a table of numerical results underneath. The left plot is a scatterplot of individual points showing a generally downward trend. The right plot shows several points scattered around a horizontal reference line, suggesting a residual or diagnostic plot. The text and table values below the graphs are difficult to read because the image is low resolution.

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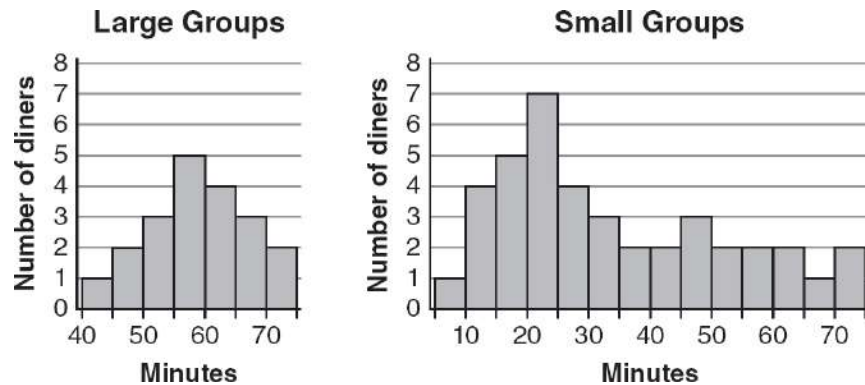


Source: National Aeronautics and Space Administration,
Goddard Institute for Space Studies

Alt text: A line graph titled “Global Land–Ocean Temperature Index” showing global temperature anomalies over time. The x-axis shows years from about 1880 to 2020, and the y-axis shows temperature anomaly in degrees Celsius. The graph includes yearly data points and a smoothed trend line. Temperatures fluctuate around or below 0°C anomaly in the late 1800s and early 1900s, then rise steadily after the mid-20th century, reaching about 1.0°C above the baseline by the 2000s and 2010s.

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Two side-by-side histograms comparing ages for large dogs and small dogs. The large-dogs histogram shows ages clustered mostly around 4 to 8 years, with the tallest bars near 6 to 7 years. The small-dogs histogram is more spread out, with many dogs around 2 to 5 years and a long right tail extending to about 12 years.

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Extended image description for Figure

Dependent variable: Sales

Variable	Coefficient	SE Coef	t-ratio	P
Constant	123.800	1.798	68.84	0.000
Advertising	12.633	0.378	33.44	0.000

R-Sq = 99.4% R-Sq (adj) = 99.3%

s = 2.926 with 9-2 = 7 degrees of freedom

statistical output table, likely showing results for a two-sample test or confidence interval. The table includes columns for two groups, sample statistics such as mean and standard deviation, and a final line with a test statistic or confidence interval. The exact labels and values are difficult to read because the image is low resolution.

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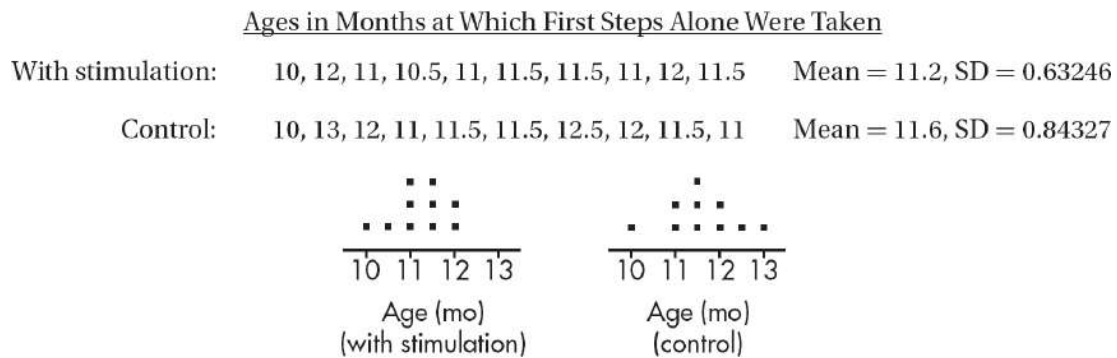
Extended image description for Figure

		College Plans	
		Public	Private
Taking AP Statistics:		18	27
Not taking AP Statistics:		26	40

A two-way table comparing AP Statistics enrollment with college plans. Columns are labeled “College Plans,” with categories “Public” and “Private.” Rows are labeled “Taking AP Statistics” and “Not taking AP Statistics.” The counts are 18 public and 27 private for students taking AP Statistics, and 26 public and 40 private for students not taking AP Statistics.

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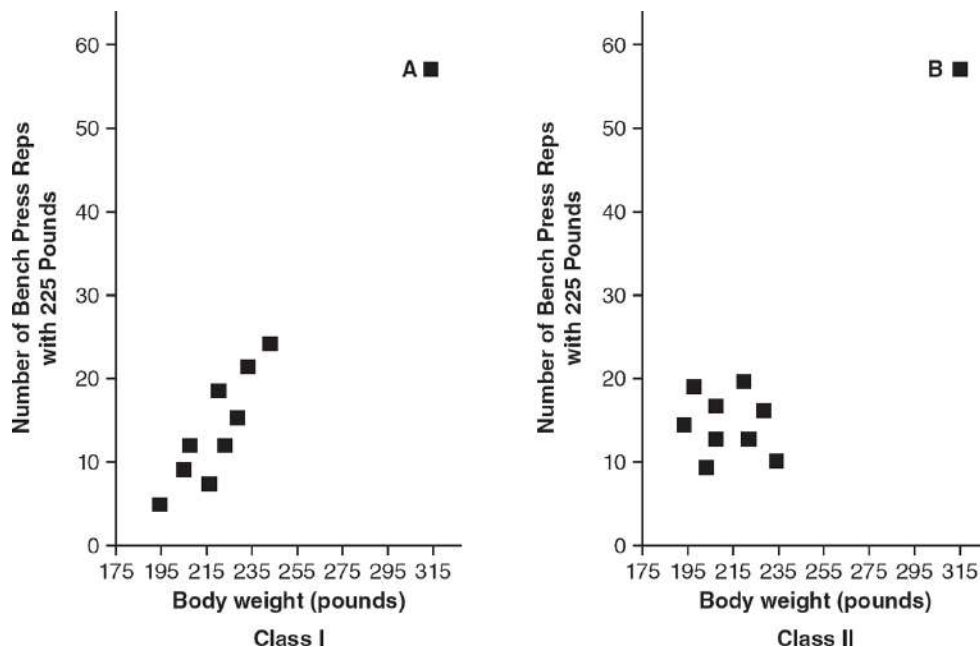
Extended image description for Figure



A diagram showing a hypothesis test or randomization setup. It includes two rows of grouped symbols or letters representing two groups, followed by arrows leading to two possible assignments labeled with “Group 1” and “Group 2.” The exact title, labels, and values are difficult to read because the image is blurry.

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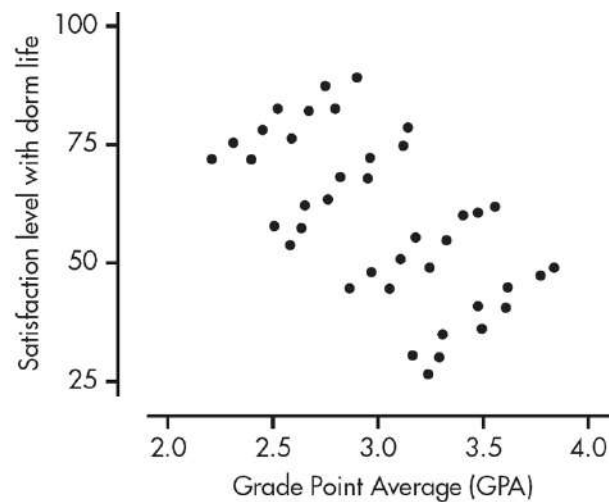
Extended image description for Figure



Two side-by-side scatterplots showing the relationship between two quantitative variables. In both plots, points are grouped into clusters, with one set of points concentrated on the left side and another set on the right. The right-hand cluster appears lower on the vertical axis than the left-hand cluster, suggesting a negative association overall, though the exact axis labels and values are difficult to read because the image is low resolution.

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Extended image description for Figure



$$\text{Predicted satisfaction level} = 117.5 - 19.93(\text{GPA})$$

$$S = 14.80$$

$$df = 38$$

$$t = -3.559$$

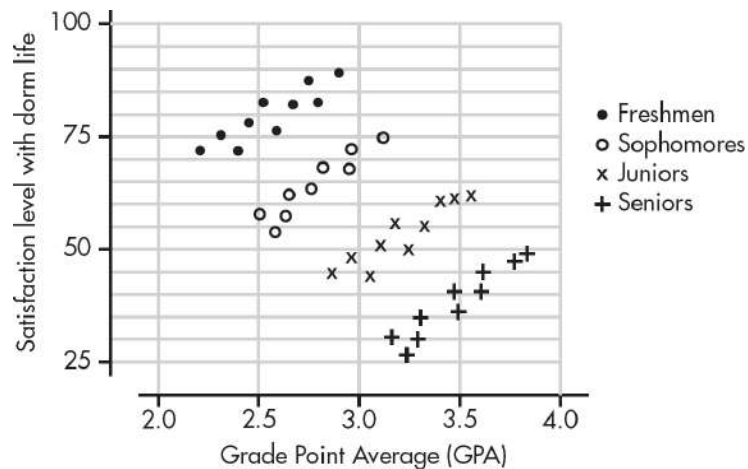
$$P = 0.001$$

$$R\text{-Sq} = 0.250$$

A scatterplot showing a relationship between two quantitative variables. The x-axis appears to show predicted values on a scale from about 0.6 to 1.3, and the y-axis ranges from about 5.0 to 6.6. The points form two main clusters: one cluster in the upper-left portion of the plot and another in the lower-right portion. Within each cluster, the points show a generally positive trend.

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Extended image description for Figure



Alt text: A scatterplot comparing grade point average (GPA) on the x-axis and a second academic score on the y-axis. Points are grouped by school and shown with different symbols for Highview, Springfield, Ashton, and Fairview. The points show several school-based clusters, with some schools concentrated at higher GPA and lower scores and others at lower GPA and higher scores.

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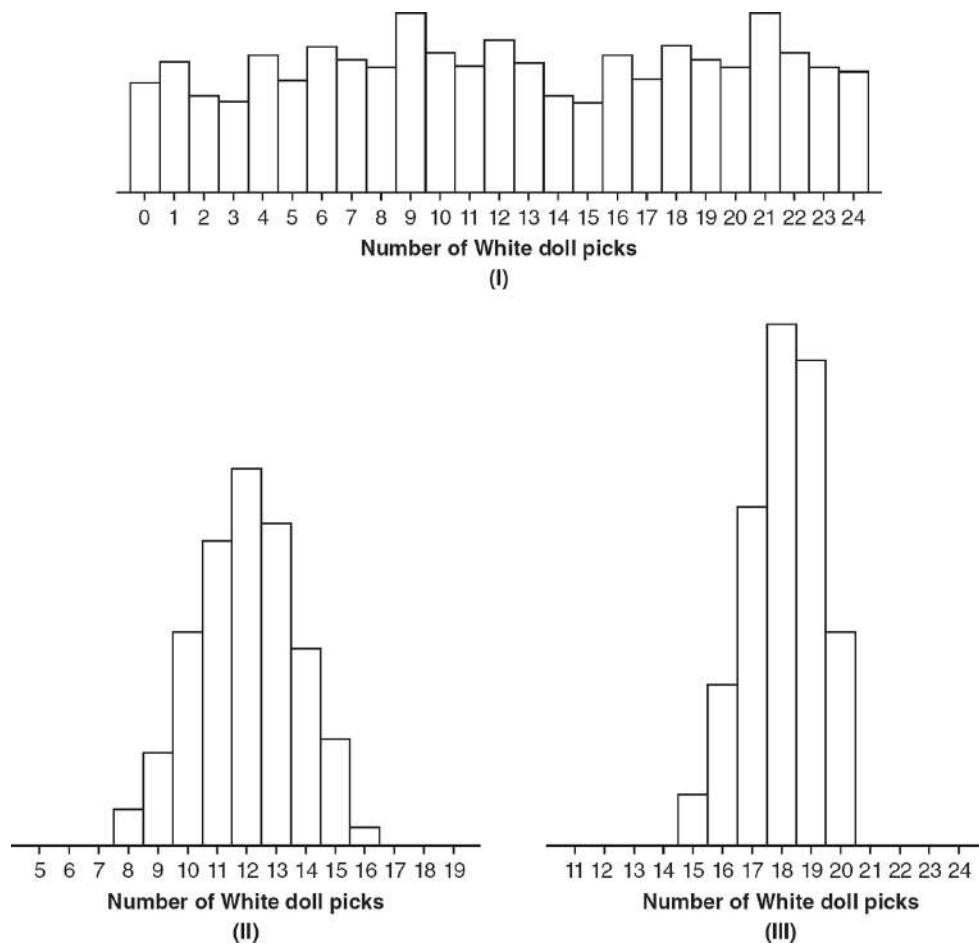
Extended image description for Figure

```
2-SampTTest
 $\mu_1 < \mu_2$ 
t=-3.939503313
df=43.43159286
 $\bar{x}_1=19.5$ 
 $\bar{x}_2=25.6$ 
Sx1=4.5
Sx2=6.3
n1=25
n2=25
```

Output from a two-sample t-test, labeled “2-SampTTest,” testing the alternative hypothesis $\mu_1 < \mu_2$. The test statistic is $t = -3.9395$, with $df = 43.4316$ and $p = 1.9 \times 10^{-5}$. The sample means are $\bar{x}_1 = 19.5$ and $\bar{x}_2 = 25.6$, with sample standard deviations $Sx1 = 4.5$ and $Sx2 = 6.3$. The sample sizes are $n1 = 25$ and $n2 = 25$.

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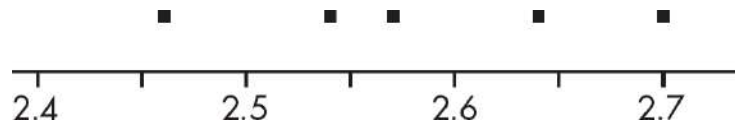
Extended image description for Figure



Three histograms comparing sampling distributions. The top histogram shows many sample proportions or means spread fairly evenly across values from about 0.25 to 0.75. The two bottom histograms show distributions for the number of successes in samples: both are mound-shaped, with the left distribution centered around about 40 and the right distribution centered around about 80. The bottom-right distribution is taller and more tightly clustered than the bottom-left distribution.

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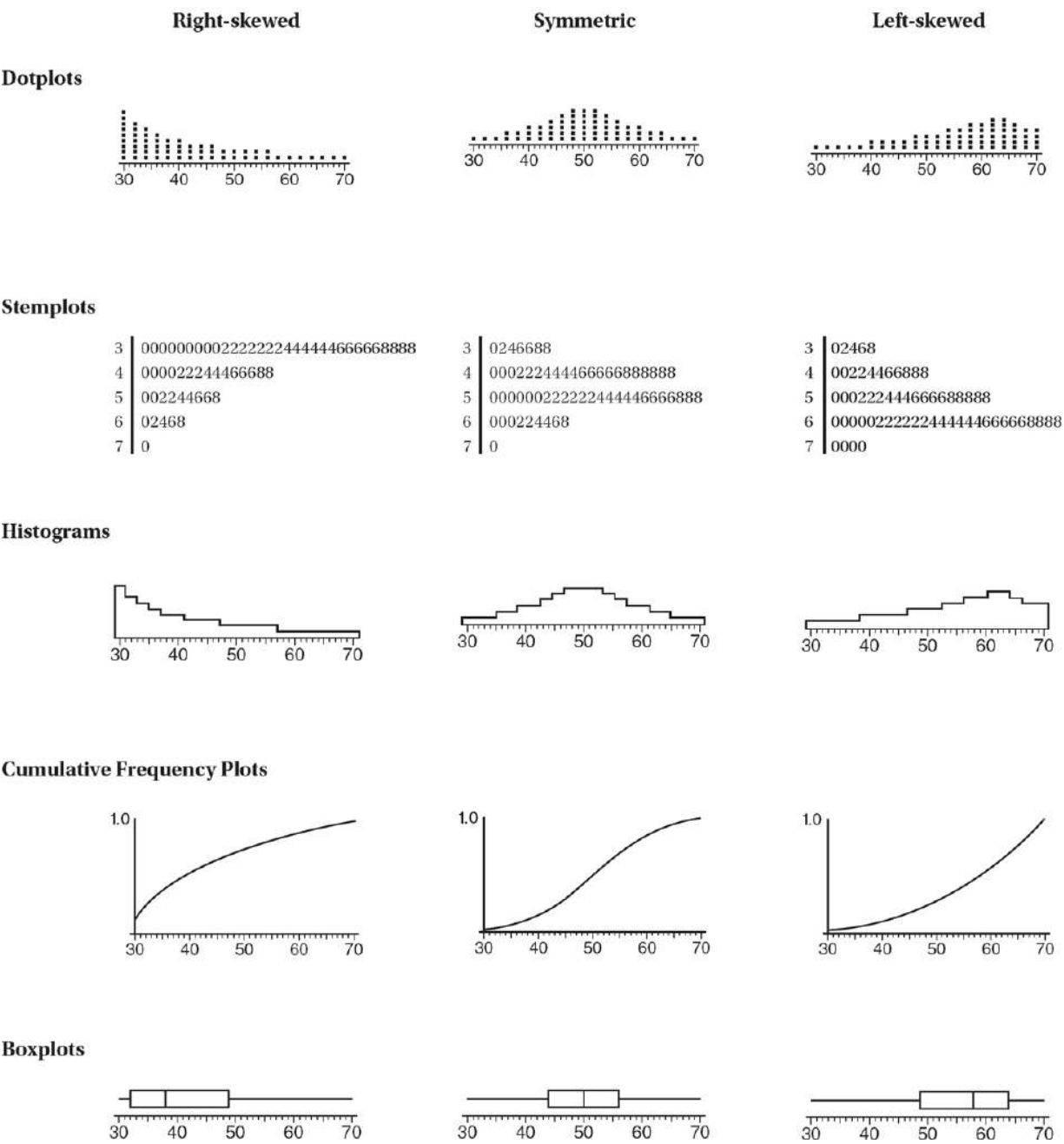
Extended image description for Figure



A small plot shown above a horizontal number line. Several vertical marks appear above values around 24.4 to 24.7, indicating individual data points or a very simple dot/line plot. The image is cropped and low resolution, so the full title, axis label, and exact values are difficult to read.

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Extended image description for Figure

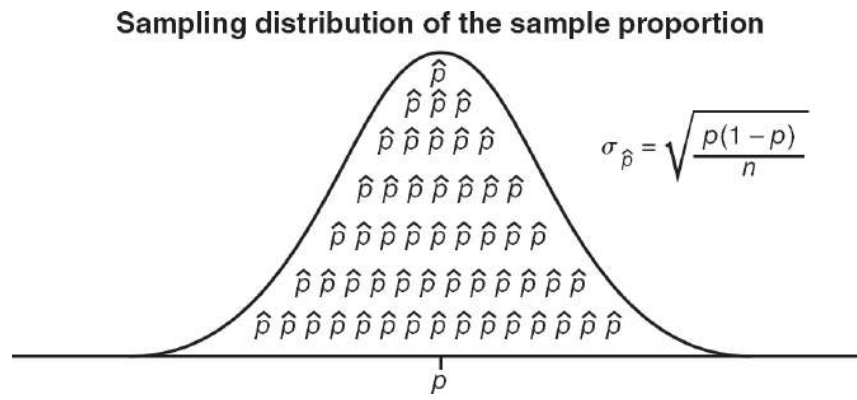


Alt text: A collection of small statistical graphs arranged in a grid. The image includes multiple dot plots, histograms, line graphs, and scatterplots, each showing different distribution shapes and data patterns, such as

skewed distributions, symmetric distributions, clusters, gaps, and trends. The image is low resolution, so the individual graph labels and values are difficult to read.

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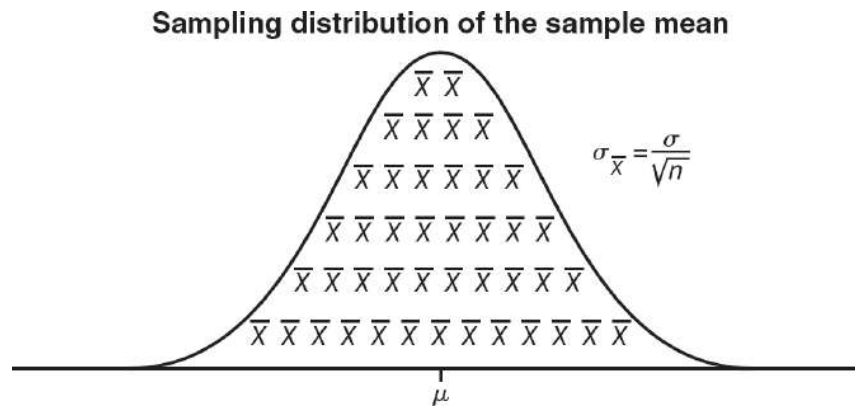
Extended image description for Figure



A diagram titled “Sampling distribution of the sample proportion” showing a bell-shaped curve. The curve is centered at the population proportion, p , and the spread is labeled with the standard error formula, the square root of p times 1 minus p divided by n . Repeated sample proportions are represented inside the curve, illustrating that sample proportions cluster around the population proportion.

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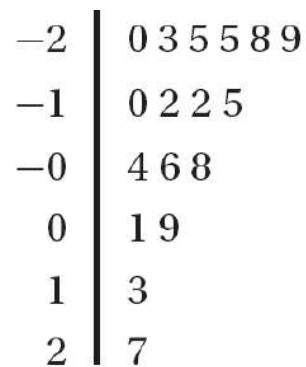
Extended image description for Figure



A diagram titled “Sampling distribution of the sample mean” showing a bell-shaped curve. The curve is centered at the population mean, μ , and the spread is labeled with the standard error formula, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. Repeated sample means are represented inside the curve, illustrating that sample means cluster around the population mean.

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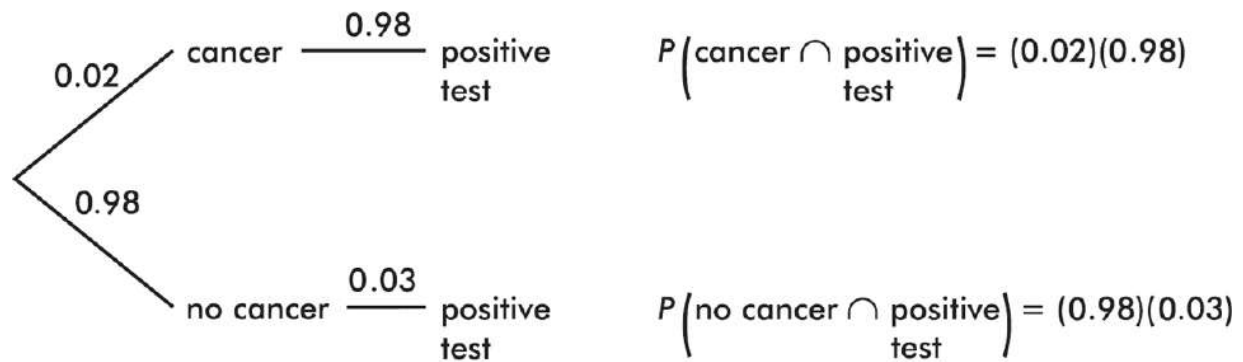


Key: 2 | 7 means 2.7 inches

A stem-and-leaf plot showing values in inches, including negative and positive values. The stems are -2, -1, -0, 0, 1, and 2. Values include -2.0, -2.3, -2.5, -2.5, -2.8, -2.9; -1.0, -1.2, -1.2, -1.5; -0.4, -0.6, -0.8; 0.1, 0.9; 1.3; and 2.7. A key below states that “2 | 7 means 2.7 inches.”

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$$P(\text{positive test}) = P(\text{cancer} \cap \text{positive test}) + P(\text{no cancer} \cap \text{positive test})$$

$$= (0.02)(0.98) + (0.98)(0.03)$$

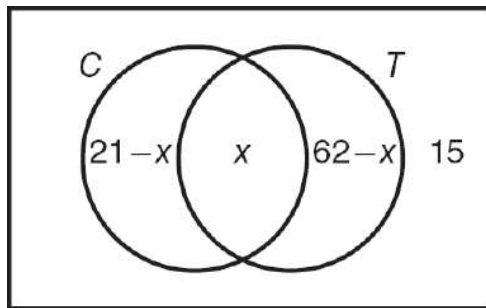
$$P(\text{cancer} | \text{positive test}) = \frac{P(\text{cancer} \cap \text{positive test})}{P(\text{positive test})}$$

$$= \frac{(0.02)(0.98)}{(0.02)(0.98) + (0.98)(0.03)}$$

A probability tree diagram and formula for using Bayes' theorem. The tree begins with two branches labeled for having HIV and not having HIV, with probabilities 0.001 and 0.999. Each branch then splits into positive and negative test results. The diagram highlights the probability of having HIV given a positive test result and shows the Bayes' theorem calculation as a fraction: the probability of HIV and testing positive divided by the total probability of testing positive.

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Extended image description for Figure



A Venn diagram showing two overlapping sets labeled C and T. The left-only section of C is labeled $21 - x$, the overlap between C and T is labeled x , and the right-only section of T is labeled $62 - x$. The area outside both circles is labeled 15.

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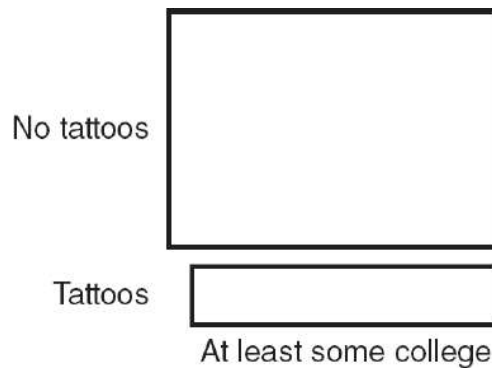
Extended image description for Figure



A segmented bar chart comparing education level by tattoo status. The horizontal axis shows two groups: “No college” and “At least some college.” The vertical axis is labeled “Tattoos.” The “No college” bar is divided into a large lower section and a small upper section, while the “At least some college” bar is mostly one large unshaded section, indicating the proportions differ by education group.

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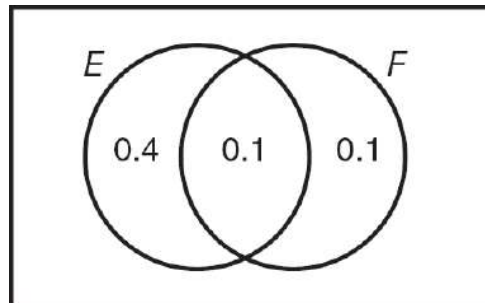
Extended image description for Figure



Alt text: A segmented bar chart comparing tattoo status by education level. The vertical axis has two categories, “No tattoos” and “Tattoos,” and the horizontal axis is labeled “At least some college.” The chart shows a much larger proportion of people with no tattoos than with tattoos, with a smaller shaded segment along the bottom representing people with tattoos.

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A Venn diagram showing two overlapping events labeled E and F . The left-only part of event E is labeled 0.4, the overlap between E and F is labeled 0.1, and the right-only part of event F is labeled 0.1.

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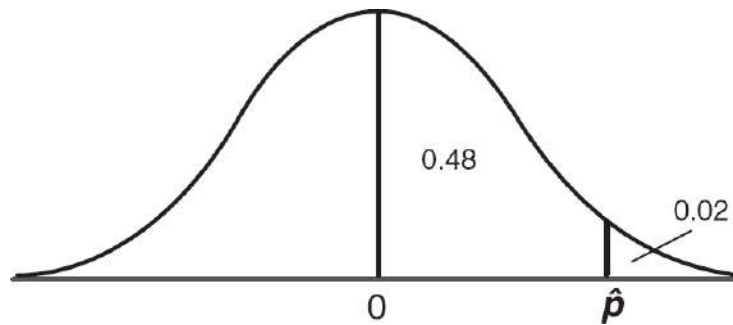
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	Years of experience			
	0–5	6–10	>10	
Less than 50 years old	80	125	20	225
More than 50 years old	10	75	50	135
	90	200	70	360

A two-way table showing counts by years of experience and age group. Columns are labeled “Years of experience”: 0–5, 6–10, and greater than 10. Rows compare people less than 50 years old and more than 50 years old. The counts are 80, 125, 20, and 225 for the less-than-50 group, and 10, 75, 50, and 135 for the more-than-50 group. Column totals are 90, 200, 70, and the overall total is 360.

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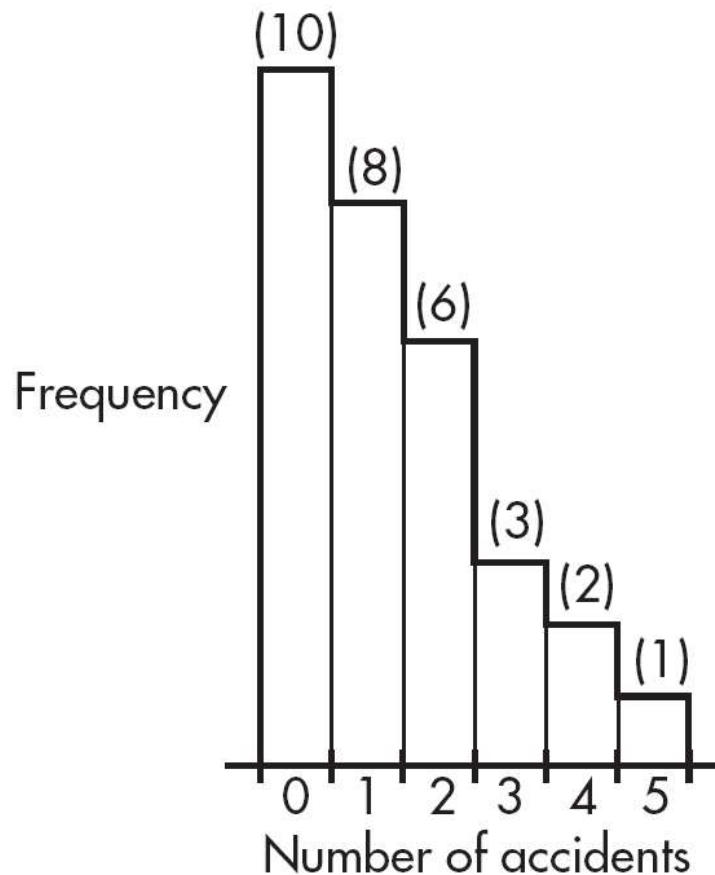
Extended image description for Figure



A bell-shaped normal curve centered at 0 with a vertical line at the center and another vertical cutoff line on the right labeled $\hat{p}$. The area between 0 and $\hat{p}$ is labeled 0.48, and the small right-tail area beyond $\hat{p}$ is labeled 0.02.

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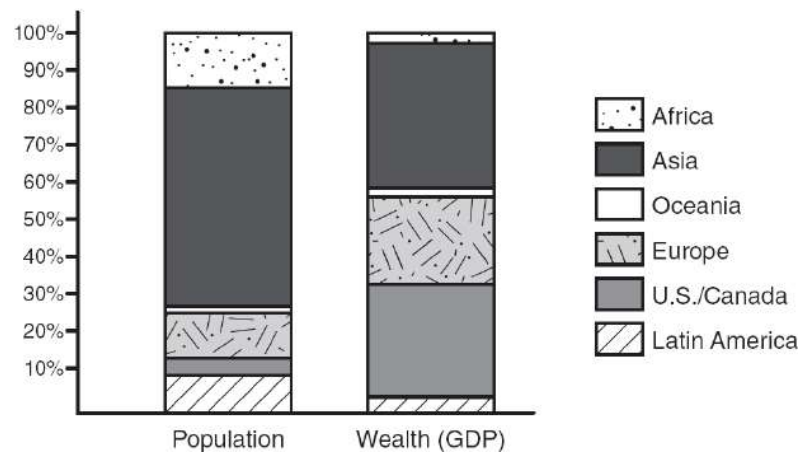
Extended image description for Figure



A histogram showing the number of accidents. The x-axis is labeled “Number of accidents,” with values from 0 to 5, and the y-axis is labeled “Frequency.” The bar labels show frequencies of 10 for 0 accidents, 8 for 1 accident, 6 for 2 accidents, 3 for 3 accidents, 2 for 4 accidents, and 1 for 5 accidents. Frequencies decrease as the number of accidents increases.

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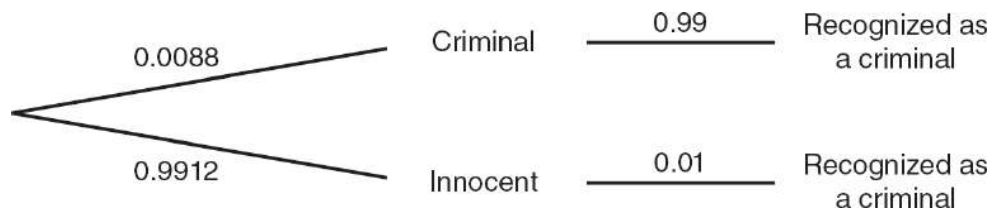
Extended image description for Figure



A stacked bar chart comparing age-group percentages for the population and a sample. The y-axis is labeled Percent and ranges from 0% to 100%. Each bar is divided into age categories shown in a legend: 90+, 80 to under 90, 70 to under 80, 60 to under 70, and less than 60. The population bar is dominated by the 60 to under 70 category, while the sample bar has a larger share of younger age groups and a smaller share in the 60 to under 70 category.

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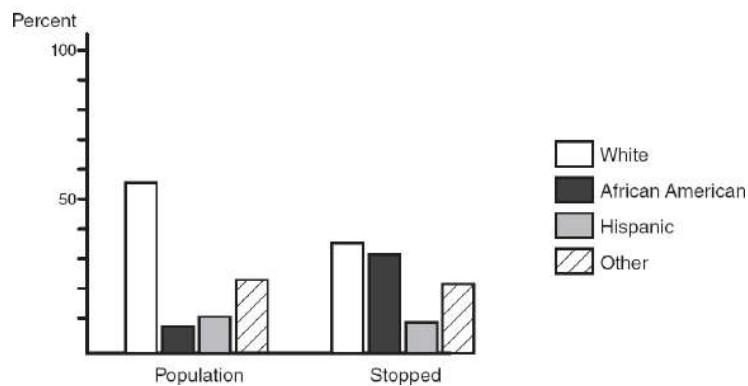
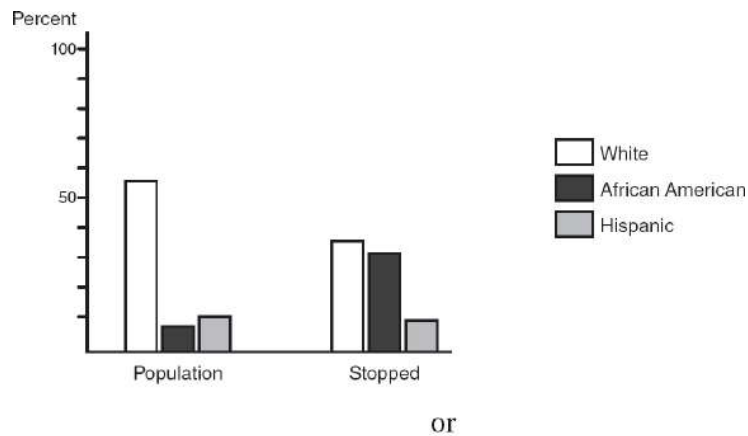
Extended image description for Figure



A tree diagram showing a first split into two outcomes labeled Correct and Incorrect. The Correct branch has probability 0.0108, and the Incorrect branch has probability 0.9892. Each of those branches then splits into two more outcomes, labeled Recognizes and Does not recognize. The image is low resolution, so the smaller probability labels on the final branches are difficult to read.

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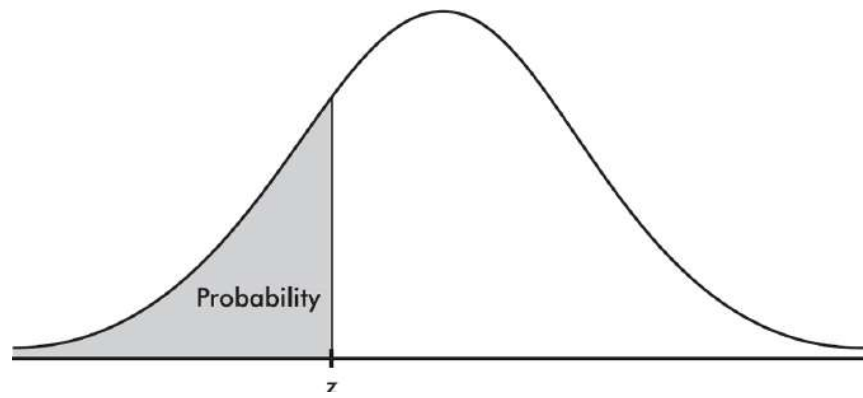
Extended image description for Figure



Two small stacked or grouped bar charts comparing response percentages for two groups, labeled “Prepared” and “Not prepared.” Each chart includes a legend with four response categories shown in different shades. In both charts, the “Prepared” group has one response category that is much higher than the others, while the “Not prepared” group has a more mixed distribution across categories. The image is low resolution, so the exact legend labels and percentages are difficult to read.

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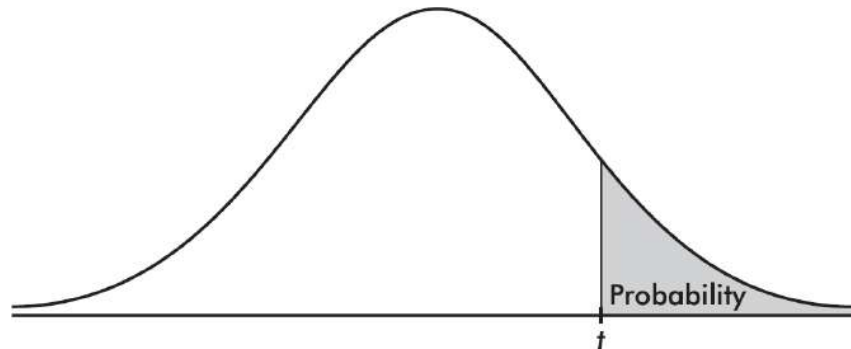
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A bell-shaped normal curve with a vertical cutoff line on the left side of the distribution. The area to the left of the cutoff is shaded and labeled “Probability,” indicating the probability of a value less than the marked cutoff.

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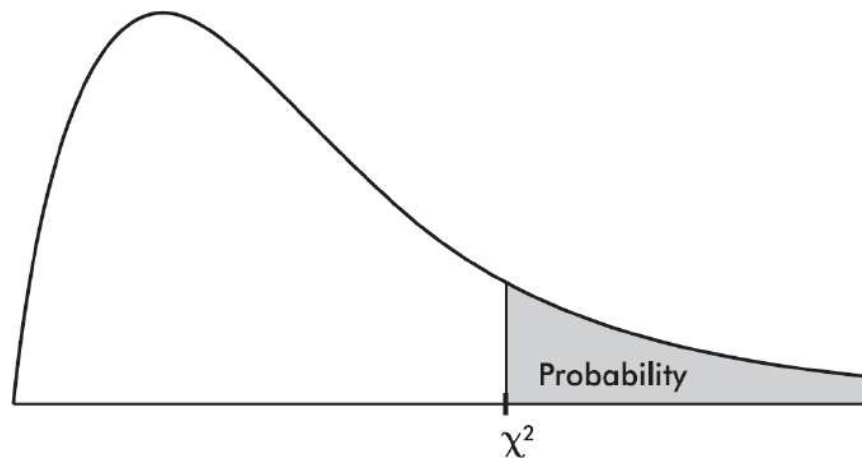
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bell-shaped normal curve with a vertical cutoff line on the right side of the distribution. The area to the right of the cutoff is shaded and labeled “Probability,” indicating the probability of a value greater than the marked cutoff.

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A right-skewed probability distribution curve with the shaded area in the right tail. The x-axis marks a cutoff value labeled " χ^2 ," and the shaded region to the right of that value is labeled "Probability."

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